PRE-DECISIONAL DRAFT

FUTURE OF FLOOD RISK DATA—STANDARD OPERATING PROCEDURES FOR INLAND DATA DEVELOPMENT

VERSION 1

December 2026

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PREFACE

This draft standard operating procedure (SOP) is pre-decisional and documents initial methods and approaches identified to produce probabilistic flood risk information in inland areas of the United States. The U.S. Army Corps of Engineers (USACE) produced this SOP for the Federal Emergency Management Agency (FEMA) as part of the Future of Flood Risk Data (FFRD) initiative.

The four objectives of FFRD are:

1. Shift from binary to probabilistic analysis
2. Modernize data management and delivery
3. Empower both private and state, local, tribal, and territorial (SLTT) stakeholders
4. Drive risk-informed actions

The resulting information would be a more comprehensive dataset with greater consideration for more storm and flood hazard types with the aim of providing people with actionable information to inform their decisions in areas susceptible to flooding.

This SOP draft is shared for informational purposes to demonstrate the scope and scale of the effort. It is not intended for public use for modeling and mapping purposes. The second version of this SOP will incorporate improvements and lessons learned from pilots, also called validation basin projects, to be ready for implementation in late 2027.

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FUTURE STANDARD OPERATING PROCEDURE UPDATES LIST

This document summarizes the proposed or planned methods and procedures that will be advanced in future versions of the Future of Flood Risk Data (FFRD) Standard Operating Procedure (SOP).

Version 1 Schedule: Delivery in late 2026. This version will be applied to validation basin projects.

Future Version Schedule: Delivery of Version 2 planned for October 2027, with subsequent annual update releases.

1.1 STOCHASTIC SIMULATION STRUCTURE

1.1.1 COMPUTATIONAL EFFICIENCIES

Version 1: A clearly defined process for stochastic simulation structure is provided, and limitations are recognized.

Future Versions: Improvements to computational efficiencies will be investigated and considered for incorporation.

1.1.2 PRODUCTION SIMULATION STRUCTURE

Version 1: For stochastic simulations, events are filtered so that only events that contribute to the annual maximum series (AMS) and are rarer than the 1/10 AEP event, based on HEC-HMS and HEC-ResSim results, and events with potential breaches, based on HEC-HMS/HEC-ResSim results with HEC-RAS ratings and system response results, are computed in HEC-RAS.

Future Version: Consider incorporation of other methods to decrease computational burden or provide alternatives to the event filter. Targeted investigations related to importance sampling, subsampling, or Gaussian process regression methods will provide recommendations that will be considered for future SOP inclusion.

1.1.3 UNCERTAINTY QUANTIFICATION

Version 1: The SOP calls for a simplistic 5 realizations to be computed for each project to establish team workflows. Multiple realizations allow quantification of uncertainty but is not rigorous or defensible. The estimates of uncertainty around variable-frequency relationships converge with many additional realizations but would require additional compute costs.

Future Version: Future versions of the SOP will consider alternative methods for quantification of uncertainty based on recommendations of focused investigations.

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1.2 SST

1.2.1 STORM ARRIVAL RATE/STORM CATALOG THRESHOLD

Version 1: The storm catalog is produced using a threshold that provides 10 storms per year of record.

Future Version: It is recognized that the selection of this threshold may influence overall modeling results for frequent events. Future updates to this SOP will consider procedures for selection of threshold to ensure bias is avoided.

1.2.2 NON-PRECIPITATION DRIVEN FLOODING

Version 1: Non-precipitation-driven flooding, such as snowmelt-only and ice-driven flooding, is not addressed within the SOP, but can be a significant contribution to flood risk in specific watersheds.

Future Versions: Investigation into these non-precipitation-driven flood mechanisms is required and will be implemented into future versions of the SOP.

1.2.3 STORM TYPING

Version 1: Storm typing is utilized to categorize storm seasonality and to isolate tropical storms.

Future Versions: Additional investigation into the influence of storm typing on FFRD results is required and will inform how storm typing is used in future iterations. Simplified approaches to storm typing could be implemented to achieve the two purposes currently used for the typing (seasonality and tropical storm isolation).

1.2.4 STORM DURATION

Version 1: A fixed 72-hour storm duration is used for storm catalog generation and synthetic events.

Future Version: Additional study to assess the validity of the selection of this fixed storm duration in a wider range of hydrometeorology is planned.

1.2.5 DATASET FOR STORM CATALOG CONSTRUCTION

Version 1: The Analysis of Record for Calibration (AORC) dataset is specified as the data source for SST storm catalog development in the SOP.

Future Version: Other sources such as the CONUS404 dataset will be evaluated and considered for use in future versions of the SOP.

1.2.6 AUTOMATED TRANSPOSITION REGION DELINEATION

Version 1: Subjective manual transposition region development is allowed, with reference to future methods.

Future Version: Targeted investigations through partnerships with academia will focus on topics such as automated transposition region development based on extremal statistics of rainfall maxima and the method will become the standard process. Recommendations from these investigations will be considered for incorporation into future versions of the SOP.

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1.2.7 SST EXTRAPOLATION LIMITS

Version 1: Includes reference to a heuristic formula to determine the smallest probability the SST method can support ($AEP_{min} \approx p/Y$). Basins where the minimum AEP is more frequent than 1/2,000 are to be flagged in supporting documentation.

Future Versions: Investigation into the limits of SST methodology for very rare precipitation probabilities will be considered for incorporation into future versions of the SOP.

Special consideration may be required for basins where this formula indicates the minimum AEP supported by SST methods to be less than 5E-4 AEP (1/2,000 years). For these basins, this potential theoretical limit should be flagged and documented. Note that this heuristic formula is approximate and conservative based on simplifying assumptions. Additional study is required to better understand the limits of SST methodology regarding very rare precipitation probability.

1.2.8 STOCHASTIC STORM TRANSPOSITION PLACEMENTS

Version 1: Storms from the catalog are placed randomly for the quasi-continuous realizations. The valid storm placements ensure that the study basin is fully contained in the transposition region. For this version of the SOP all valid locations are equally likely.

Future Version: Future versions of the SOP may incorporate improved techniques to account for spatial patterns of precipitation within the Transposition Region.

1.2.9 HISTORICAL STORMS INCLUSION IN SST CATALOG

Version 1: The SST catalog consists of storms included in the observed storm catalog, which is composed of storms dating back to 1979 from the AORC dataset. Historical storms (i.e., those occurring prior to the observed storm catalog) are not included within the SST catalog. This potentially excludes the largest precipitation events from analysis which may bias flood frequency curves.

Future Versions: Investigation may be considered to develop means of incorporation of historical storm events into the SST catalog. This requires research into depth-area-duration relationships of those storms, statistical reanalysis (e.g., downscaling) or use of numerical weather models to generate high-fidelity space-time fields which can be transposed, and selection of sampling probability for historic storms. This approach has not been implemented previously and is complicated by several factors which must be overcome prior to implementation in the FFRD process.

1.3 CALIBRATION

1.3.1 CALIBRATION GUIDANCE

Version 1: Event-based calibration requirements are well defined, and overarching requirements for comparison to long-term historical data in frequency space are provided.

Future Versions: Improved calibration techniques based on recommendations from targeted investigations will be considered.

1.3.2 CONFORMANCE VALIDATION AGAINST OBSERVED DATA

Version 1: SOP recommends probabilistic results from the *conformance phase* be compared against observed frequency data, which includes published stage- and flow-frequency data. Other relationships that *conformance phase* results must be evaluated against are listed in Section 12.4, subsection *Conformance Metrics Evaluation*.

Future Versions: Language will be included in a *Calibration* job aid regarding the use of stage- and flow-frequency estimates. This includes published frequency curves from risk assessments. Future versions will include metrics and procedures for evaluation of frequency-based results. Comparison against current effective models is also planned for inclusion into the SOP. Specific guidance on appropriate evaluation against observed data will be developed for all items listed in Section 12.4, subsection *Conformance Metrics Evaluation*.

1.3.3 MODEL QUALITY RATING SYSTEM

Version 1: Model quality is indicated using standardized calibration metrics compared against published values.

Future Versions: The rating system should be expanded to improve communication and confidence in results at each observed data location.

1.3.4 UNGAGED WATERSHEDS

Version 1: Ungaged watershed model parameters are developed using guidelines from literature, data from other hydrologically similar modeling units within the basin, and professional engineering judgment. Probabilistic results from the conformance phase can be compared with published regional regression estimates where available.

Future Versions: Additional investigation and best-practices will be documented in SOP Version 2 regarding calibration to ungaged model units and subbasins.

1.3.5 UTILIZATION OF ALL AVAILABLE HYDROLOGIC DATA

Version 1: Regional regression equations and traditional flow-frequency analyses are used to informally verify appropriateness of conformance phase outputs, including at-site information.

Future Versions: All available regional regression equations and flood frequency quantile estimates will be considered for future studies. This data requires QA/QC checks prior to comparison with computed probabilistic results. Incorporation of Atlas 14/15 data may also be possible, but requires areal reduction factors prior to comparison with SST outputs. Global Historical Climatology Network (GHCN) data is another source of information which may be incorporated.

1.4 SOFTWARE

1.4.1 SUPPORTED SOFTWARE

Version 1: SOP references existing, stable, accepted software for FFRD analysis.

Future Version: Developing software and new techniques will be explored for potential incorporation into the FFRD process. Some examples of these software and techniques include: HEC-RAS 2025, Storm Typing, SLAM, Structure Survey Tool public deployment, GovCloud-only versus commercial cloud deployment, etc.

1.4.2 HEC-RAS VERSION

Version 1: The SOP Version 1 calls for use of HEC-RAS version 6.6, which is the current release fully supported by cloud compute.

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Version 2: HEC-RAS 7.0.x includes notable advancements including enhanced capabilities for assessing model performance using Linux compute engines within a desktop environment, as well as addressing potential concerns regarding numerical reproducibility in parallelized computations. A future SOP update will leverage these new functionalities.

Version 2+: An SOP update to utilize the HEC-RAS planned release currently referred to as HEC-RAS 2025 will be considered when core capabilities to support FFRD become available and may be paired with more significant workflow and tool updates. Note that in the interim, Version 1 of the SOP does not specify how users must create two-dimensional (2D) meshes and early HEC-RAS 2025 processes may be utilized.

1.4.3 PLUG-IN TO CREATE MANIFESTS FOR CLOUD COMPUTE

Version 1: Relies on the Model Linking Register document to plan modeling linkages and for the integration lead to deliver linking information for cloud compute. The compute-manifests control the actual computes, which are owned by the cloud compute lead. Job Aid 12, *Model Delivery for Cloud Compute*, includes specifics of the inputs used to create the compute manifests.

Future Version: A system for extracting information from the delivered models to automatically build manifests for cloud-compute is envisioned. Testing of prototypes for HEC-RAS models has been performed as represented in "ffrd-manifest-generator".

1.4.4 AUTOMATION OF HYDRAULIC MODEL CONSTRUCTION

Version 1: The SOP provides procedures for manual development and calibration of hydraulic models, following traditional approaches.

Future Version: Automated workflows may be considered for direct incorporation into the SOP.

1.5 MODELING STANDARDS AND WORKFLOWS

1.5.1 TIME WINDOW FOR STOCHASTIC STORM TRANSPOSITION EVENT SIMULATION

Version 1: The model integration lead will guide event time window duration selection, which will be long enough to ensure rainfall in the upper basin has sufficient travel time so that the downstream boundary of the model reaches a peak. HEC-RAS runs will compute for the full duration for each event.

Future Version: For large basins the time window could be tailored for each HEC-RAS model-unit and each event, and significant savings in compute time could be achieved. This is referred to as "Sliding time windows".

1.5.2 DATA PREPARATION: OUTREACH TO DATA HOLDERS

Version 1: The outreach to data holders is described as a key part of the scoping and data preparation phase, but limited details regarding process and specific communication standards are included.

Future Version: Data-holder-specific communication standards will be outlined, including messaging and communication paths for specific federal agencies such as USACE.

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1.5.3 DATA MANIFEST

Version 1: A data manifest that tracks the source and other metadata associated with all datasets for the project is manually created and delivered as a table.

Future Version: Integration of an automated tracking of modern metadata for all data associated with the FFRD project using a cloud-based system will be considered.

1.5.4 ADDITIONAL DATA SOURCES FOR DATA PREPARATION

Version 1: Many data types are identified for collection during the *data preparation* phase.

Future Versions: Additional data types that are not currently identified, but could be included in future versions of the SOP include: geologic features, imagery, high water marks, satellite passes during flood events, on-the-ground flood photos, historic flood-related newspaper articles, hydro-connector lines, StreamStats, additional gridded soils data, storm drains, coastal tide gage references, published stage/flow frequency curves.

1.5.5 INTERNATIONAL DATA SOURCES AND COORDINATION

Version 1: International data sources are not discussed within the SOP. Coordination with international agencies is not discussed within the SOP.

Future Versions: Coordination with Canadian and Mexican agencies and data sharing will be covered in future versions of the SOP. This is required prior to FFRD execution of international watersheds.

1.5.6 HYDRAULIC MODEL-UNIT DELINEATION

Version 1: Hydraulic model-unit delineation is described throughout several locations of the SOP but does not cover all cases.

Future Versions: Additional guidance will be included related to model-unit delineation. These include expansion on delineation best-practices, shared boundaries with other HUC-4 boundaries, discrepancies between high-resolution DEM data and published HUC-4 boundaries, splitting model-units in highly developed areas, water diversion structures, and interaction/coordination between model-unit boundaries, hydraulic model perimeters, and hydrologic subbasin boundaries.

1.5.7 NAMING CONVENTION CONTROL

Version 1: The SOP includes various review steps and checklists to address naming convention issues; there is opportunity for human error.

Future Version: Processes to check for adherence to naming conventions using scripts have the potential to reliably catch issues and ensure consistency across the FFRD program. Standardize and enforce a process to ensure consistency in naming conventions. Additionally, software field character limitations will be addressed in the standard naming conventions.

1.5.8 HYDRAULIC MODEL INITIAL CONDITIONS

Version 1: Allows warm-up time periods when appropriate and varies reservoir starting elevations through initial condition point settings.

Future Version: Consider restart files as a similar approach to vary basin conditions at beginning of simulation without utilizing warm-up. This saves runtime compared with warm-up and also addresses the

fact that depression storage starts dry and all channels upstream of baseflow locations are also dry. It would additionally address the fact that initial flow in baseflow hydrographs is an inadequate representation of desired initial state for HEC-RAS.

1.5.9 HYDRAULIC MODEL PARAMETER UNCERTAINTIES

Version 1: A single geometry is calibrated and validated against 6 historic events and applied to the long-term simulations as a single model with deterministic parameters.

Future Version: Consider incorporating uncertainty in parameters such as hydraulic roughness into the long-term simulations to account for additional knowledge uncertainties.

1.5.10 STARTING RESERVOIR ELEVATIONS IN HYDRAULIC MODEL WHEN ALSO MODELED IN RESERVOIR OPERATIONS MODEL

Version 1: HEC-ResSim period of record (POR) runs are not leveraged for Initial Pool Elevations in HEC-RAS within the current cloud-compute toolset.

Future Version: Leverage HEC-ResSim POR for initial pool elevations for event-based sampling.

1.5.11 URBAN STORMWATER DRAINAGE

Version 1: Overarching guidance to use best practices and engineering judgment to represent urban stormwater drainage networks within areas scoped for additional mesh detail.

Future Version: Will consider incorporation of recommendations from focused investigations targeting simplified techniques for stormwater drainage representation.

1.5.12 HYDRAULIC MODELING OF BRIDGES

Version 1: Bridges are approximated using breaklines and vector terrain modifications (where needed) to represent the bridge approach embankments which act as hydraulic constrictions. Bridge decks and piers are not explicitly represented within the hydraulic modeling unless engineering judgment warrants their inclusion. The SOP recommends against representation of the bridges as hydraulic structures (i.e., SA/2D connections) due to computation and stability costs, as well as data availability.

Future Versions: Additional procedures will be developed to screen where bridge features have a significant influence on flood characteristics. In instances where bridge decks and piers do significantly impact inundation, and where modeling these bridge features is supported by available data, these features will be included explicitly in the hydraulic modeling. Additional investigation is required to better understand what implications modeling bridges as hydraulic structures has on computational cost and model stability. A decision tree, or similar tool, to identify and guide treatment of bridges will be developed.

1.5.13 REPLICATION OF HISTORIC EVENTS WITH INTEGRATED WATERSHED MODEL VIA CLOUD COMPUTE

Version 1: Historic events are replicated after integration of the watershed model in the desktop environment, early in the conformance phase.

Future Version: Add cloud-compute processes and tools to seamlessly perform this task via cloud-based computes.

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1.5.14 STRESS TESTING OF INTEGRATED WATERSHED MODEL VIA CLOUD COMPUTE

Version 1: Stress testing of the linked watershed model occurs after integration in the desktop environment, early in the conformance phase.

Future Version: Add cloud-compute processes and tools to seamlessly perform this task via cloud runs.

1.5.15 FFRD BASIN EXTERNAL BOUNDARY CONDITIONS

Version 1: Upstream and downstream boundary conditions for the HUC-4 FFRD watershed are not defined.

Future Versions: Investigation and documentation will describe external boundary conditions for the FFRD project watershed. This would include how coincident flows and boundary conditions are passed from previously studied FFRD basins to other basins. Reservoir operations which have inter-basin dependencies also require investigation and implementation into the FFRD SOP.

1.5.16 DOWNSTREAM BOUNDARY AT COASTAL SITES

Version 1: While calibration of the hydraulic models is performed using observed tide timeseries, a simplified fixed boundary condition at Mean High Water (MHW) as the coastal boundary condition is applied for all synthetic stochastic storm transposition (SST) events.

Future Version: Improved techniques for selecting tidal boundary condition series for synthetic events will be considered. An early proposed method that uses the Coastal Ocean Reanalysis (CORA) dataset for sampling downstream timeseries boundary conditions for SST events has been discussed.

1.5.17 BREACH SAMPLING PROCEDURES

Version 1: Workflows for delivering models that support breaches and SRPs for each breach location are clearly defined in the SOP. Technical specifications for breach sampling during stochastic simulations are documented within cloud-compute technical manuals outside of the SOP.

Future Versions: Breach sampling methods will be refined based on lessons learned during early basin studies and recommendations from focused investigations.

1.5.18 DAM BREACH TRIGGERS ON RESERVOIR SIMULATION DAMS

Version 1: Trigger elevations are sampled for each event and shared between HEC-ResSim and HEC-RAS. When HEC-ResSim sees a pool above that trigger elevation, it will breach by switching outflows to equal inflows. In the HEC-RAS model during that same event, the pool elevation will track the HEC-ResSim pool and trigger a breach at the same instant and release a larger volume of water through the breach calculated in HEC-RAS. The actual outflow in the HEC-RAS model is the sum of the HEC-ResSim releases and the HEC-RAS-computed breach flows.

There are a few imperfections in this method:

- HEC-RAS and HEC-ResSim Simulated Pool Hydrographs may not track perfectly due to the timing of inflows. HEC-ResSim receives inflows routed via HEC-HMS, while HEC-RAS simulates 2D flow over the mesh using HEC-HMS excess precipitation and baseflow outputs.
    - If pool hydrographs do not track well together, breaches will not trigger at the same time.

- Breaches at upstream dams may trigger breaches at downstream dams. HEC-ResSim will not be able to account for this fact since the large volume in a reservoir is not actually released after a breach.
- After a breach occurs, HEC-RAS may release too much flow, especially if inflows are high at the time of breach.
- It is possible HEC-RAS numerical stability issues will occur if HEC-ResSim inflows are forced out of a reservoir that is dry after a breach has released all water.

Future Version: Investigate an approach that allows for a coupled simulation.

1.5.19 DAM DATA REFINEMENT

Version 1: SOP calls for the National Inventory of Dams (NID) data to be refined prior to FFRD study and is considered separate from the study itself.

Future Version: Develop a workflow to fully incorporate a dam data refinement process into NID data management processes, to remain outside of FFRD SOP procedures.

1.5.20 FLOOD RISK SOURCE ATTRIBUTION

Version 1: Hazard and consequence summaries are reported at all locations and combine all sources of flooding without differentiation of the flood source or driver.

Future Version: Develop additional methods and standard reporting products that differentiate the source of flooding between fluvial, pluvial, dam/levee overtopping, and dam/levee breach.

1.5.21 CRITERIA FOR HOT FIXES

Version 1: Hot fixes to models are applied only to specific events where they were deemed necessary. The team should consider the overall impact of failures of a limited number of events, and may judge that fully addressing every single failed event may not be necessary if they do not have a significant impact on overall hazard frequency results.

Future Versions: Guidelines and specific criteria can be developed and implemented which describe when hot fixes are required, and when they may be ignored.

1.5.22 LONG TERM STORAGE AND ARCHIVAL OF DATASETS

Version 1: A listing of event-level outputs and summary-level outputs to be stored is provided in SOP Volume II, Section 11. These include HEC-RAS Hierarchical Data Format (HDF) output files and gridded outputs. No specific guidance regarding subsets of the data to archive indefinitely is provided.

Future Version: Develop a cohesive system to ensure key inputs and outputs from the study can be accessed indefinitely. This may include a system to be able to reproduce a complete set of event-level outputs on a one-by-one basis, which would reduce the burden of long-term storage of massive datasets.

1.5.23 MODIFICATION OF MODELS FOLLOWING BASIN PRODUCTION

Version 1: FFRD models are developed through the workflows described in SOP Volume II Section 2.1, and do not describe methods to revise or modify models following the *production and final reporting* phases.

Future Versions: Investigation will be required into methods to allow for model refinement and modification after an FFRD basin has been fully produced. Findings of this investigation would be implemented into future versions of the SOP. These methods could allow for implementation of higher

level-of-detail modeling than is currently required by SOP Version 1. The approach would ideally accommodate revisions to modeling based on changed conditions within the watershed.

1.6 CONSEQUENCES

1.6.1 CONSEQUENCE CATEGORIES TO CONSIDER

Version 1: For FFRD base-level runs, the consequence categories are limited to direct structure and content losses. While additional consequence categories exist (such as life loss) and are capable of being computed with the hazard data generated by FFRD, they are not included in the initial FFRD effort.

Future Version: Consider incorporation of additional consequence categories.

1.6.2 CONSEQUENCE COMPUTATION METHOD AND DAMAGE FUNCTIONS

Version 1: Single-parameter deterministic (depth) damage functions are applied to depth-frequency summary results at each structure.

Future Version: Consider multi-parameter damage functions provided by FEMA including those widely accepted within USACE. Also consider event-based damage calculation at each structure, which allows for implementation of multi-parameter damage functions and for summation of regional risk beyond the structure-by-structure summaries.

1.6.3 REFINEMENTS TO STRUCTURE INVENTORY

Version 1: The SOP includes tasks for manually improving the placement of structures and using the placements to inform improvements to the hydraulic models. No workflow to provide feedback from the manual update process back to the NSI database is specified.

Future Version: Structure inventory sourcing, structure inventory improvement tasks, consequence computation procedures, and systematic feedback from project refinements to the structure inventories will be considered.

1.6.4 STRUCTURE SURVEYS

Version 1: No tasks associated with defining the uncertainty of the structure inventory are included in the SOP procedures.

Future Version: Consider incorporation of first floor elevation uncertainty into consequence computations. This may include leveraging the existing National Structure Inventory (NSI) survey tool after it is publicly released via complementary efforts outside of FFRD, pending related conversations connected to consequence computations and structure inventory source.

1.6.5 CONSEQUENCE COMPARISON TO HISTORICAL EVENTS

Version 1: No provision for comparing simulated consequence results to data from historical events.

Future Version: Consider including practical and appropriate workflows that allow comparisons of simulated consequence results to historical data. These workflows must address data sharing and information security requirements and recognize the quality and resolution of the existing historical loss data.

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ABBREVIATIONS

1D	one dimensional
2D	two dimensional
3DEP	3D Elevation Program
AAL	average annualized losses
AEP	annual exceedance probability
AORC	Analysis of Record for Calibration
AMS	Annual Maximum Series
ASCII	American Standard Code for Information Interchange
bc	Boundary Condition
BE-PWP	basin execution project work plan
BLE	base-level engineering
br-hw	Breach Headwater
br-tw	Breach Tailwater
CC	Cold Content
CCP	common computation point
CONUS	contiguous United States
CPU	central processing unit
CRS	Coordinate Reference System
CSV	Comma-Separated Values
CTP	Cooperating Technical Partners
CWBI	Civil Works Business Intelligence
DEM	Digital Elevation Model
ds	Downstream
DSS	Data Storage System
DST	Dam Screening Tool
EAD	expected annualized damages
EAP	Emergency Action Plan
EDH	elevation derived hydrography
EM	Engineer Manual
ESA	elevation-storage-area
FEMA	Federal Emergency Management Agency
FFRD	Future of Flood Risk Data
FGMRES-SOR	Flexible Generalized Minimal Residual with Successive Over-Relaxation
FIRM	Flood Insurance Rate Map
FIS	Flood Insurance Study

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FRM	flood risk management
GAO	Government Accountability Office
GB	gigabyte
GeoTIFF	Georeferenced Tagged Image File Format
GIS	Geographic Information System
GW	groundwater
H&H	hydraulics and hydrology
HEC	Hydrologic Engineering Center
HEC-DSSVue	Hydrologic Engineering Center Data Storage System Visual Utility Engine
HEC-HMS	Hydrologic Engineering Center Hydrologic Modeling System
HEC-RAS	Hydrologic Engineering Center River Analysis System
HEC-ResSim	Hydrologic Engineering Center Reservoir System Simulation
HEC-SSP	Hydrologic Engineering Center Statistical Software Package
HUC	Hydrologic Unit Code
hw	Headwater
IBTrACS	International Best Track Archive for Climate Stewardship
IC	Initial Condition
IRWA	Interagency Reimbursable Work Agreement
lb	Left Bank
LCL	Levee Control Location
LiDAR	Light Detection and Ranging
LULC	Land Use/Land Cover
LWASS	Liquid Water at Soil Surface
MDI	mapping data integration
MIP	Mapping Information Platform
MRMS	multi-radar multi-sensor
ms	Midstream
NAS	National Academy of Sciences
NAVD 88	North American Vertical Datum of 1988
NFHL	National Flood Hazard Layer
NFIP	National Flood Insurance Program
NHD	National Hydrography Dataset
NHDPlus HR	National Hydrography Dataset Plus High Resolution
NGVD 29	National Geodetic Vertical Datum of 1929
NID	National Inventory of Dams
NLCD	National Land Cover Database
NLD	National Levee Database

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NOAA	National Oceanic and Atmospheric Administration
NSI	National Structure Inventory
NT	Non-Tropical
NWS	National Weather Service
PBIAS	percent bias
PC	personal computer
PFRA	Probabilistic Flood Risk Analysis
PMF	Probable Maximum Flood
PMP	probable maximum precipitation
POR	Period of Record
PRISM	Parameter-elevation Regressions on Independent Slopes Model
PWP	Project Work Plan (general)
QC	Quality Control
QL2	quality level 2
QRA	Quantitative Risk Assessment
R	storage coefficient
R2	coefficient of determination
RAM	random access memory
rb	Right Bank
RFC	River Forecast Center
RMSE	root-mean-square error
RSR	root-mean-square-error–observations standard deviation ratio
SNODAS	Snow Data Assimilation System
SNOTEL	Snow Telemetry
SOP	Standard Operating Procedure
SOR	Successive Over-Relaxation Matrix Solver
SPF	Standard Project Flood
S-PWP	Scoping Project Work Plan
SQRA	Semi-Quantitative Risk Assessment
SRP	System Response Probability
SST	Stochastic Storm Transposition
SSURGO	Soil Survey Geographic Database
STATSGO	State Soil Geographic Database
SWE	Snow Water Equivalent
SWE-ELM	Shallow Water Equations, Eulerian-Lagrangian Method
SWE-EM	Shallow Water Equations, Eulerian Method
SWE-LIA	Shallow Water Equations, Local Inertial Approximation

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TC	Tropical Cyclone
TMAC	Technical Mapping Advisory Council
TOD	top of dam
TOG	Technical Oversight Group
TP	technical paper
us	Upstream
U.S.	United States
USACE	U.S. Army Corps of Engineers
USBR	U.S. Bureau of Reclamation
USGS	U.S. Geological Survey
UTC	Coordinated Universal Time
UTF-8	Unicode Transformation Format–8-bit
VDatum	Vertical Datum Transformation tool/service
WBD	Watershed Boundary Dataset
WCM	water control manual
WSEL	water surface elevation
XYZM	X, Y, Z, and Measure coordinates

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GLOSSARY

Area/Reach/Location of Interest—Locations within the watershed where additional model resolution is warranted to support the objective of the FFRD initiative (e.g., urban areas, major rivers, transportation corridors, existing FEMA flood hazard designations).

Background Mesh—The initial computational mesh used to create a 2D flow area. References to the background mesh in this document refer to the lowest resolution 2D cells (largest cells) in the mesh employed in areas where more detailed modeling resolution is not needed. Commonly, the background mesh resolution is used in rural areas in the basin that are not near population centers or other key streams.

Bathymetry—The measurement of depth under the water surface in reservoirs, rivers, oceans or other water bodies. In hydraulic modeling, it is common to leverage a model of the bare-earth land surface that includes land elevations assuming a completely dry landscape including underwater elevations.

Bucket—Specific cloud storage location.

Directed Acyclic Graph—Term used to define the order of model execution that accounts for dependencies between upstream models and downstream models that require inputs from previously computed models.

Directory—Specific file location including path name within larger hierarchy of file storage system.

Event—An event represents a unique set of parameters derived from antecedent conditions, storm selection and placement, and sampled failure elevations for breachable infrastructure. Simulations generally span a time window allowing the sampled precipitation to cycle through the watershed, i.e., time of concentration.

Flood Hazard—Hazard caused by flooding is described by specific parameters such as water depth or velocity in specific locations.

Key Streams—FFRD modeling term to define scoped Mesh Development Criteria for use during model development.

Levee Control Location (LCL)—The location on a levee system expected to overtop first, also called the initial or incipient overtopping location. The levee data lead approximates LCLs during the scoping and data preparation phase, and the hydraulic modeler reviews and confirms them during the calibration phase. An LCL is commonly used as a breach location.

Model Order—Referring to the sequential order in which each component model must be executed to properly pass outputs from certain models as inputs to other models.

Modeling Unit—The 2D modeling extent used for hydraulic model development and calibration. Each HUC-4 basin evaluated in the FFRD initiative is subdivided into modeling units to reduce the computational burden of the model and decrease the model development time. Modeling units are based on major topographic and drainage control features, optimized according to run time. Features considered are location of major reservoirs, stream gages, levees, and population centers or due to the size of the basin. Previous modeling efforts used modeling units varying from 300 to 2,400 square miles with an average of 875 square miles.

Other Streams—FFRD modeling term to define scoped Mesh Development Criteria for use during model development.

Realization—A realization is composed of a set of synthetic years and is used to construct an estimate of a variable-frequency relationship.

Repository—Specific shared file location.

Simulation—A simulation is composed of a set of realizations and is used to describe the knowledge uncertainty around a variable-frequency relationship.

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Stochastic Storm Transposition—Meteorologic technique that applies synthetic rainfall on a study basin leveraging historic storm observations from a larger region with similar meteorologic characteristics.

Synthetic Year (block maxima)—A synthetic year is composed of a set of independent events and is used to determine the annual maximum values for any variable of interest at any location in the watershed. The construction of the storm database determines the number of events randomly selected. The term quasi-continuous is used to describe the independence of events in that each simulated event does not persist state across events.

System Response Probability (SRP) curves—SRP curves provide a relationship of probability of failure (response) to a flood stage elevation.

Transposition Region—A large region that can be assumed to have similar meteorologic characteristics to the basin being studied.

Urban Areas— FFRD modeling term to define scoped Mesh Development Criteria for use during model development.

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PRE-DECISIONAL DRAFT

FUTURE OF FLOOD RISK DATA—STANDARD OPERATING PROCEDURES FOR INLAND DATA DEVELOPMENT

Volume I, Overview

VERSION 1

December 2026

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EXECUTIVE SUMMARY

This draft Standard Operating Procedure (SOP) outlines a modern approach for producing consistent, efficient, and high-quality flood hazard and risk information for inland areas across the United States. Leveraging advancements in data, modeling techniques, and computational tools, the SOP provides a look into the potential framework for delivering scalable, science-based information to improve decision-making and risk communication at all levels of government. The second version of this SOP should incorporate improvements and lessons learned from pilots to be ready for implementation.

EX.1 GOAL

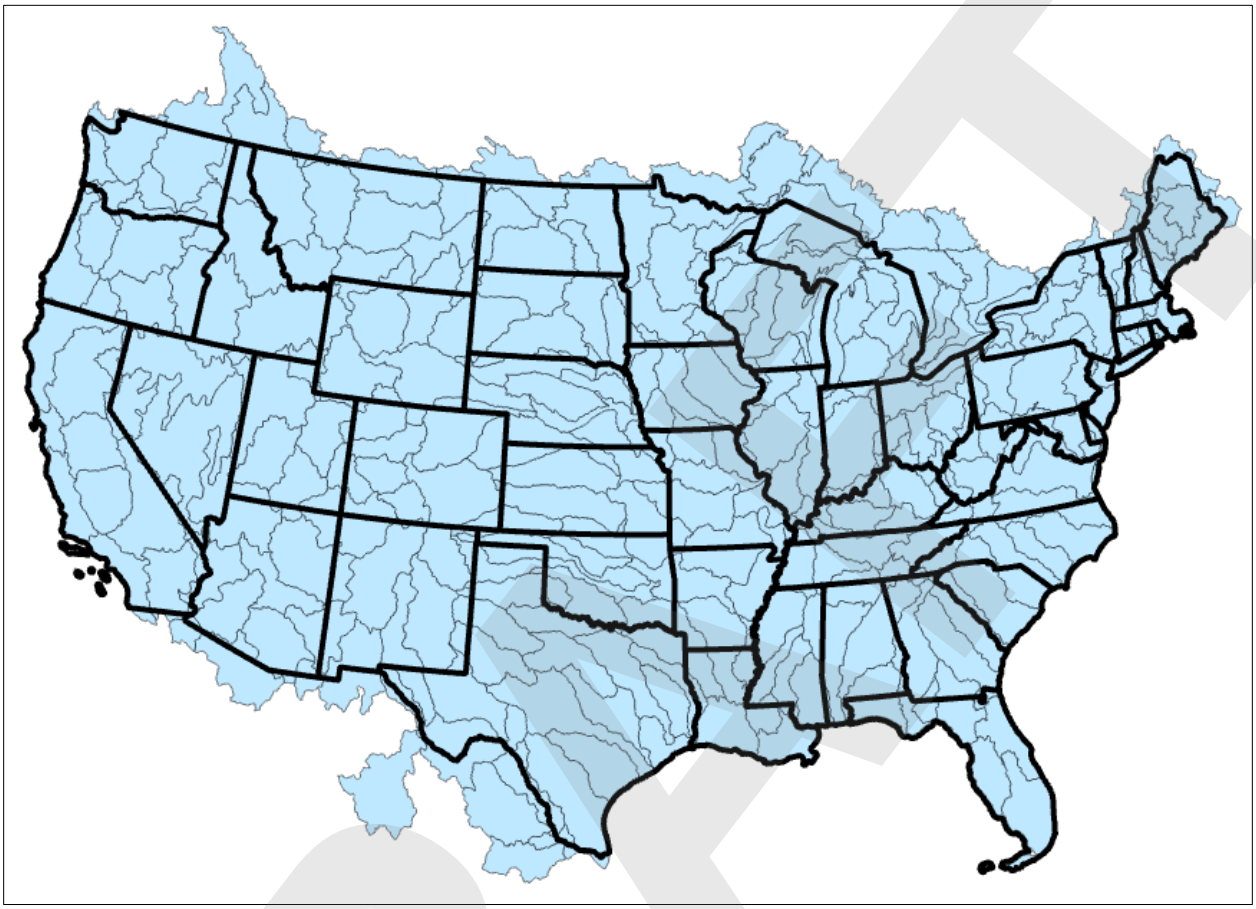
The goal is to create foundational datasets produced through standardized, nationwide procedures to serve as high-quality, foundational flood risk information with consistent coverage across the country. This dataset is intended to provide robust and reliable baseline information that supports a wide range of applications and may be leveraged for more detailed local studies. All models and datasets will be stored using consistent data management protocols to ensure accessibility, transparency, and long-term usability.

EX.2 WATERSHED APPROACH

The modeling approach will be based on the Watershed Boundary Dataset (WBD), with Hydrologic Unit Code 4 (HUC-4) watershed basins serving as the standard unit for modeling. Watershed (or basin) is hereafter used to refer to the HUC-4 subregion level. For major rivers, upstream probabilistic inputs are essential; therefore, an upstream-to-downstream sequence is encouraged, allowing outputs from earlier studies to directly inform downstream analyses and increasing modeling efficiency and consistency.

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Figure EX.1. Map of Hydrologic Unit Code 4 Basins from the Contiguous United States Watershed Boundary Dataset



EX.3 MODELING FRAMEWORK

The methodology outlined in this SOP employs an integrated modeling framework that spans the full hydrologic cycle and the spatial extent of a watershed. This approach is designed to characterize the movement of water through atmospheric processes, overland flow, built infrastructure, and ultimately to damageable assets within the floodplain.

To achieve this, the methodology links a series of individually calibrated models, enabling seamless data transfer from one component to the next. This end-to-end modeling system ensures that outputs from meteorological models feed directly into hydrologic and hydraulic models, maintaining consistency and accuracy throughout the simulation process.

A central component of this methodology is the use of Stochastic Storm Transposition (SST), which generates synthetic rainfall events from an existing observed precipitation dataset. These synthetic events drive both hydrologic and hydraulic simulations. The resulting outputs support the quantification of flood frequency and enable the development of both event-based and graduated flood hazard and risk products.

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SECTION 1—STANDARD OPERATING PROCEDURE DEVELOPMENT

1.1 COLLABORATION WITH THE U.S. ARMY CORPS OF ENGINEERS

Because the U.S. Army Corps of Engineers (USACE) has extensive hydrologic and hydraulic modeling and flood risk management infrastructure expertise, the Federal Emergency Management Agency (FEMA) engaged USACE to develop a recommended methodology and framework to meet the goals of this effort, including development of this draft SOP.

1.2 REQUIREMENTS FOR METHODOLOGY

The procedures detailed throughout this SOP are intended to meet the following requirements:

- Methods must be applicable for all watersheds within the United States and its territories, regardless of differences in geography, watershed characteristics, or local conditions.
- Resulting flood risk products should be accurate and clearly communicate inherent uncertainties and limitations without implying a false sense of precision.
- Provide model and data storage standards that allow for development of a repository that ensures all datasets are findable, accessible, interoperable, and reusable.
- Models, procedures, and results must be traceable and reproducible, allowing them to be leveraged as the foundation for a wide array of possible uses, such as alternative analysis or local refinement.
- Leverage data from partners where possible and contribute to national-level datasets with newly generated data.
- Account for water management systems where they appreciably impact flood hazard.
- Account for the contribution of possible dam and levee breaches to flood risk.
- Provide flood risk assessment for all locations with 2D hydraulic modeling throughout the watershed, accounting for both pluvial and fluvial flood sources.
- Procedures must provide guidance on modeling approaches for stormwater drainage systems, including areas with sparse information to enable reasonable flood risk estimates.
- Procedures must provide guidance on calibrating models and subsequent conformance procedures in regions with and without historical records.
- Provide both pluvial and fluvial flood hazard data across the range of natural variability from the 1/10 to the 1/2,000 annual exceedance probability (AEP).
- Portray the uncertainty around the natural variability of flooding in a watershed.
- Support consequences modeling as part of the modeling sequence or as a post process.

1.3 PRINCIPLES FOR METHODOLOGY SELECTION

Several high-level principles emerged as part of the methodology selection process and are described below.

1.3.1 LEVERAGE EXISTING DATA

Many necessary datasets exist supporting the modeling effort. Data leveraged to support this effort includes terrain data, hydrologic data, levee data, dam data, economic data, and data regarding other critical infrastructure. Some necessary datasets do not exist while others are duplicated across agencies. Coordination to collect missing data and to strategically reduce duplicative datasets is central to the

development of the national water resources model. Centralizing foundational datasets collectively will improve the national ability to model flood related hazards and risks.

1.3.2 IMPROVE EXISTING MODELS

Existing models were designed to meet objectives of a given water resource authority or a specific study. In many cases, existing models may not meet the diverse user requirements spanning authorities across multiple water resources agencies. Developing tools to support all event frequencies as opposed to creating tools that focus on specific frequencies (e.g., floods or low water conditions) is important in developing a cohesive risk management framework. This requires improvements to existing models or integration of existing models to meet additional requirements.

1.3.3 SUSTAINABLY SCALE TO THE CLOUD

FEMA intends to adopt a cloud-first design for the framework, data access, and software-as-a-service capabilities and encourages the advancement of tools towards cloud-native architectures to modernize the management and delivery of flood risk data. Many existing tools require substantial computational resources and were created for use on a personal computer (PC). Shifting PC-based applications into the cloud is an inefficient use of cloud resources. Moving forward, building sustainable, scalable, and certified tools targeted for the cloud environment to leverage the collective knowledge of the water resources community will improve the computational capability of the entire community.

1.3.4 FOSTER COLLABORATION AND COORDINATION

Interagency partnerships encourage and enable engineers and scientists to review, revise, and re-envision how water resource modeling moves into the future, which improves the state of the science and provides coordinated delivery into the state of the practice. Efforts such as collaboration between agencies on developing integrated models and providing consistent underlying information for water resource risk data support coordination of overall integrated water resources management.

1.4 SELECTED METHODOLOGY

The selected methodology described in this SOP can be characterized as follows:

- Utilize sequential meteorologic, hydrologic, reservoir simulation, hydraulic, and consequence models by linking inputs to outputs.
- Perform individual model calibration and validation, utilizing multiple observed historic events.
- Utilize SST method to generate synthetic storm events.
- Perform hydrologic modeling to provide baseflow and gridded excess precipitation outputs.
- Perform reservoir simulation modeling to provide outflows from key flood control dams.
- Account for dam and levee breaches by leveraging available system response probability curves from existing risk assessment studies or other baseline estimates.
- Perform hydraulic modeling with fully 2D coverage to provide stage, flow, and velocity information at all locations throughout the study basin.
- Vary the resolution of hydraulic modeling outputs based on clearly defined basin characteristics. Higher resolution is provided along key streams and urban areas.
- Utilize consequence modeling that leverages actuarial books, user-provided inventories, or nationally available structure inventories.
- Apply a stochastic approach by modeling multiple synthetic events per year of a long-term, quasi-continuous simulation.

- Perform multiple realizations of quasi-continuous simulations to provide quantified uncertainty around all flood risk results.
- Produce a complete picture of flood risk with uncertainty over certain ranges (e.g., from 1/10 to 1/2,000 AEP) and over all flood risk metrics.
- Centralize storage and delivery of all inputs, models, and output data through the Model Library system, which utilizes cloud storage and role-based access control for all users.
- Standardize the use of geographic projection, vertical datum, time zones, naming conventions, and other important parameters.
- Standardize the workflow, including well-defined project development phases, each with a clear review process and clear deliverables for each step within the phases.
- Provide multiple job aids, checklists, and templates to assist in production of products for a basin.
- Develop tools with standard software development practices on open-source libraries.
- Develop tools that are extensible through a plugin-based architecture, which allows for future development of tools and procedures around alternative models or added capabilities as models evolve and mature.

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SECTION 2—TYPICAL INDIVIDUAL BASIN TEAM STRUCTURE

Teams developing individual basin projects will have access to training and technical support. Multiple team members and skill sets are needed to achieve success in the production of a basin. The following sections describe the recommended best practices regarding team composition.

The *scoping and data preparation* phase and the *basin execution* phases (calibration, conformance, production, and final reporting) may be completed by either one team or two separate teams, which may share overlapping personnel or be entirely distinct. If two teams are assigned, the first team develops and completes the scoping and data preparation. Once the first team's work is completed, approved, and advanced, the second team begins the modeling and reporting phases. If only one team is assigned, it must fulfill all roles outlined in both sections.

2.1 SCOPING AND DATA PREPARATION TEAM STRUCTURE

Recommended team member roles for the scoping and data preparation phase are provided in Table 2. below.

Table 2.11. Basin Scoping and Data Preparation Team Members

Role	Description
Technical Team Lead	Experienced co-leader responsible for overseeing the data collection and preparation phases. Coordinates with key stakeholders to verify the successful completion of these phases.
Project Manager	Co-leader of team; focuses on project scheduling, budget tracking, and upward reporting during the pre-model data collection phase.
Technical Team	Technical staff, including Geographic Information System (GIS) specialists, meteorologists, hydrologists, hydraulic engineers, dam and levee fragility analysts, economists, software engineers, and consequences modelers, are responsible for assessing the quality of existing datasets, preparing the most relevant and necessary data, and developing the technical scoping elements for the basin execution phases.
Technical Editor	Technical team member, compiling and editing documentation for technical consistency, spelling, grammar, and formatting.
Dam and Levee Technical Team Members	Responsible for data collection, data refinement, and data analysis needed for developing dam and levee characteristics including location, alignment, and fragility. Team members are part of the conformance review team, verifying statistical appropriateness of work products produced in the data preparation phase.

2.2 BASIN EXECUTION TEAM STRUCTURE

Tables 2.2 and 2.3 outline the recommended roles and describe the basin execution team leads and team members, building on the previously defined pre-model data preparation team roles in Table 2.1. Individuals may fulfill roles on both teams, provided they meet the requirements for each role and are available to effectively support the initiative.

Table 2.22. Basin Execution Project Delivery Team Leads

Role	Description
Technical Team Lead	The technical team lead provides technical leadership and streamlines effective communication within the team and with key stakeholders. The lead oversees the development and delivery of technically accurate work products and ensures coordination and outreach efforts are aligned with project objectives. The lead supports hydrologic, reservoir operations, and hydraulic modelers, guiding them through the calibration phase and ensuring alignment with project goals through to basin closeout. The lead is responsible for scheduling and hosting meetings or delegating as appropriate and ensuring that data is consistently uploaded in a format suitable for future use.
Project Manager	The project manager oversees project scheduling and budget tracking, ensuring alignment with project timelines and budget requirements from project start to closeout. This manager provides upward reporting updates and maintains clear communication with leadership and stakeholders. Working closely with the technical team lead, the project manager schedules key meetings and facilitates communication among team members. Additionally, the project manager oversees the delivery of all study basin deliverables, including models and documentation.
Conformance and Production Lead	The conformance and production lead is a technical professional with a strong background in statistical analysis, hydrology, hydraulics, reservoir operations, and consequences modeling. The lead coordinates with the cloud compute team to verify the successful completion of the conformance and production phases.

Table 2.33. Basin Execution Project Delivery Team Members

Role	Description
Meteorologist/ SST Team Member(s)	Technical staff works through all basin execution phases possessing the necessary technical experience and knowledge to review meteorological data, create homogeneous transposition domains, and develop storm catalogs.
Hydrologic Modeler(s)	Technical staff responsible for using appropriately acquired data to develop hydrologic models specific to the region (e.g., rainfall-runoff, snowmelt). Ensures models are properly calibrated and validated to achieve accuracy and reliability. Completes tasks in the calibration phase and contributes to overall model refinement and assists as needed for model updates within conformance and production phases.
Hydraulic Modeler(s)	Technical staff works through all phases of study and is responsible for developing and refining 2D hydraulic models using appropriately acquired data. Ensures models are properly calibrated and validated for accuracy and reliability. Completes tasks in the calibration phase and contributes to overall model development and improvement and assists as needed for model updates within conformance and production phases.
Consequences Modeler(s)	Technical staff members work through all basin execution phases with an understanding of structure inventory data and standard consequences modeling techniques.

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Role	Description
Conformance and Production Team Member(s)	Team of experienced modelers from the calibration modeling team. Collectively, the team must understand statistical, hydrologic, hydraulic, reservoir operations, and consequences modeling to complete work in the conformance and production phases.
Technical Editor	Technical editors compile and edit documentation for technical consistency, spelling, grammar, and formatting. The editors work in all phases, with a focus on the reporting phase.
Cloud Compute Team Member	Technical staff member working through conformance and production phases with the appropriate security credentials and authorization to process computes.
Levee Data Team Member	Technical team member responsible for data collection, data refinement, and data analysis needed for developing levee characteristics including fragility. Engages in all phases of work.
Dam Data Team Member	Technical team member responsible for data collection, data refinement, and data analysis needed for developing dam characteristics including fragility. Engages in all phases of work.
Reviewers	Individuals with expertise in hydrology, hydraulics, reservoir operations, consequences modeling, statistics, meteorology, geospatial analysis, and documentation will conduct reviews of pre-model, calibration, conformance, and production work products. Reviewers must possess a thorough understanding of the model specialty they are reviewing. Modelers are prohibited from reviewing work on basins they are actively working on but may serve as modelers for other basins while also performing review duties.

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PRE-DECISIONAL DRAFT

FUTURE OF FLOOD RISK DATA—STANDARD OPERATING PROCEDURES FOR INLAND DATA DEVELOPMENT

Volume II, Technical Standards and Procedures

VERSION 1

December 2026

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SECTION 1—ORIENTATION

This SOP volume outlines the authoritative technical standards and procedures for executing and delivering Future of Flood Risk Data (FFRD) projects.

1.1 WORKFLOW

A standardized phased workflow greatly improves the ability of large FFRD teams to coordinate and communicate throughout the development of deliverables for individual basins. Furthermore, this standardized process benefits data management and enables national-level tracking of progress and the delivery of flood risk data on a national scale.

1.2 TECHNICAL STANDARDS

The technical standards enforce a consistent and achievable national approach for developing flood risk information that is founded on rigorous science and leverages modern tools and state-of-the-practice techniques. The standards described in this volume deliver the requirements.

1.3 JOB AIDS AND PROCESSES

Multiple job aid documents are available that represent specific detailed processes and workflows necessary for execution of FFRD projects. In some cases, alternative processes may be allowed and may be utilized to meet the requirements specifically stated in the main body of this volume.

1.4 TOOLS AND TEMPLATES

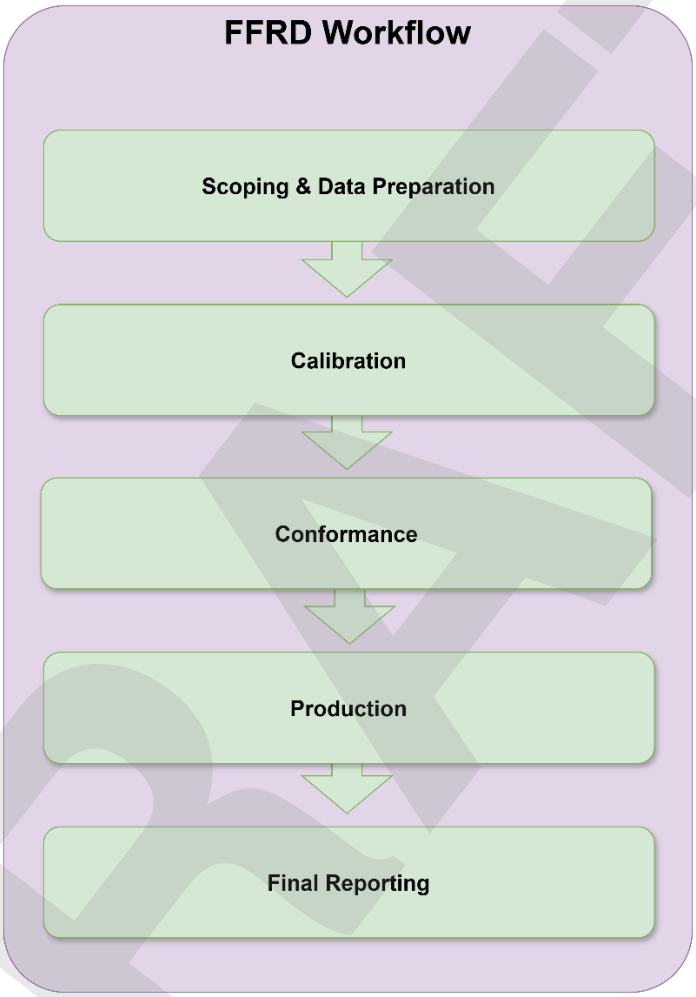
The main body of this document relies on core models and tools that are publicly available and license-free. Specifically, cloud-compute, the Hydrologic Engineering Center's Hydrologic Modeling System (HEC-HMS), Reservoir System Simulation (HEC-ResSim), and River Analysis System (HEC-RAS) are embedded in the processes. Additionally, there are several helpful tools provided in the FFRD Templates and Resources GitHub site that may optionally be used to achieve the requirements described in the main body of this volume. Templates are similarly provided to enforce standardization of documentation in the form of delivered reports, quality reviews, and certain aspects of delivered datasets.

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SECTION 2—PROJECT WORKFLOW AND REVIEW

This section defines the required workflow, deliveries and reviews. Resources for national-level technical support for teams executing FFRD projects are also described.

Figure 2.1. High Level Workflow Diagram for an FFRD Basin Project

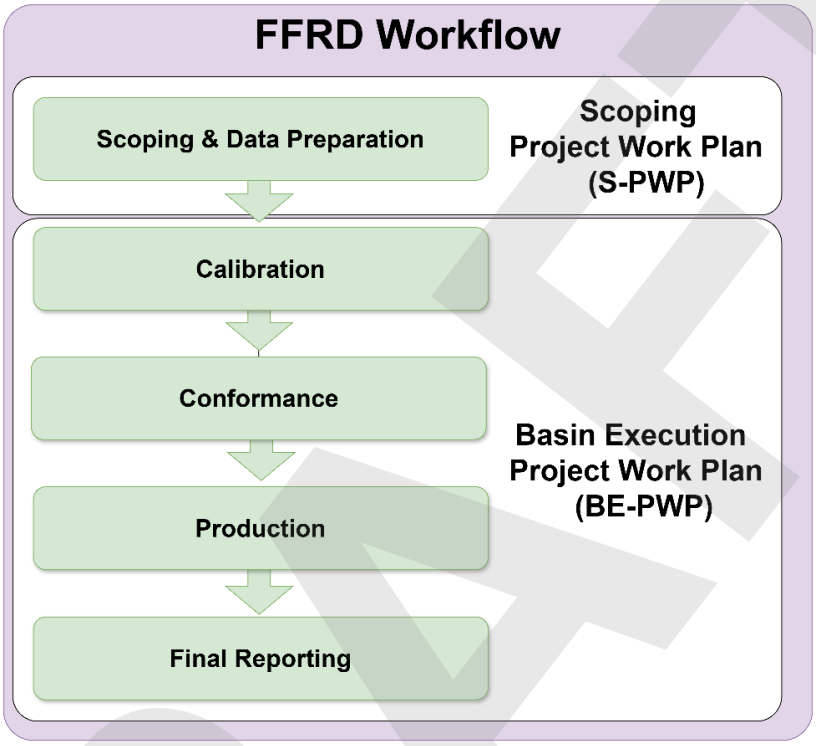


2.1 STUDY WORKFLOW

The FFRD study workflow described in this document is separated into five distinct phases (see Figure 2.1). The **Scoping and Data Preparation** phase includes analysis by a skilled technical team to prepare the shared datasets and define the specifics of scope. Once the basin is scoped and necessary data are developed, the basin study continues with the subsequent four phases (collectively referred to as Basin Execution). The **Calibration** phase includes the bulk of model development for the project and brings in a much larger team of modelers to develop, calibrate, review, and deliver each of the constituent models for the study. During the **Conformance** phase, the models are fully linked together by a smaller modeling team and tested in a stochastic environment to ensure frequency-based results reasonably align with long-term observed data. The **Production** phase produces many model simulations for the entire model suite for multiple realizations. These simulations are the basis for all final delivered results. The **Final Reporting** phase communicates and documents the results and refers to the detailed documentation that

is created throughout the phases. High-level tasks associated with each phase of the process are provided as a graphic in the FFRD Templates and Resources site (View Task Matrix button). Various quality control (QC) and quality assurance (QA) checks of the products are built into each of these phases.

Figure 2.2. Major Phases of FFRD Workflow



2.1.1 BASIN SELECTION

The process of FFRD basin study selection is the responsibility of the National Implementation Management Group in coordination with the Steering Committee. The FEMA Program Manager collates the list of prospective basins with support from National Implementation Team members. Basin selection may be derived from a variety of sources including HQ or FEMA region priority. Prospective basins are presented by FEMA to the Steering Committee where FEMA HQ representatives and steering committee members confirm basins meet selection criteria, the SOP is applicable, and required data are available. After a basin is selected for study and funding is available, scoping and data preparation can begin.

2.1.2 PROJECT WORK PLANS

All phases of FFRD work will be guided by signed and approved project work plan (PWP) documents which define project goals and team member roles. There are two distinct PWP documents: The scoping project work plan (S-PWP) and the basin execution project work plan (BE-PWP). The team lead supplies the technical aspects, while the project manager develops the detailed budget and schedule.

2.1.3 SCOPING AND DATA PREPARATION PHASE

This phase includes a significant amount of technical analysis to verify the future overall modeling effort can proceed with reliable and consistent data that will be coordinated across a large team of modelers. A data preparation team lead, data preparation project manager, FEMA representative, and other roles are

assigned to a selected basin. The team lead assigns other roles as needed and initiates the data collection process. In project leadership roles, the team lead has the primary responsibility for development of technical data and scoping elements for the basin study in direct coordination with the project manager, who is responsible for project schedule, budget, tracking, and communication. Key stakeholder meetings are held to aid in data collection and preparation. The scoping and data preparation phase is considered complete once all review checklists have been posted, all items have been closed, and the team lead has deemed the data suitable for advancing to the basin study. The basin is then presented to the Technical Oversight Group (TOG) for high-level technical review and approval. Once this step is completed, the calibration phase can begin.

2.1.4 CALIBRATION, CONFORMANCE, AND PRODUCTION PHASES—BASIN EXECUTION

Upon completion of the scoping and data preparation phase, a basin study team lead, basin study project manager, and a FEMA representative are assigned to the basin. These individuals may remain the same as those in the scoping and data preparation phase. The basin study project manager and team lead develop the BE-PWP, which includes budget and schedule details and identifies key team members. The basin technical team lead assigns additional roles within the team. The project manager and team lead then schedule and facilitate the basin kickoff meeting, after which modeling begins.

Modeling continues in accordance with the methods and procedures defined in this SOP. The steps, processes, and guidelines outlined in the following sections are followed to maintain consistency, accuracy, and compliance with established protocols throughout the modeling phases. Throughout the basin study, the TOG and assigned subject matter expert (SME) reviewers conduct formal evaluations and technical reviews confirming each study phase is complete and ready for promotion to the next phase through project completion. The high-level technical TOG reviews occur at the end of the calibration, conformance, and production phases.

2.1.5 FINAL REPORTING PHASE—BASIN EXECUTION

After the final TOG meeting and acceptance of the final basin results and reports, the completion of all result summaries, final reports, and review documentation close-outs can take place, following established documentation standards, practices, and templates. The study technical team lead verifies that all data is uploaded, archived for future use, and that file maintenance is performed in accordance with these standards as part of the close-out process.

2.2 COLLABORATION BEST PRACTICES

FFRD teams must operate in close collaboration throughout the process to be successful, as many aspects of modeling require interaction between models being developed by separate individuals. It is recommended that the team lead facilitate weekly team meetings during the model development phases of the project. The regular calls should follow a format that encourages team interaction and discussion of the relevant issues for upcoming key decisions.

In addition, it is recommended that the program manager hold regular meetings with all active team leaders to facilitate technical discussion of issues and concerns occurring across multiple teams.

2.2.1 EARLY MODEL CALIBRATION—OUTPUT DATA EXCHANGE

A critical phase exists during development of initial calibrated models where outputs from the hydrologic model are required for the calibration of the reservoir operations and hydraulic models. Coordination may be required when the hydraulic model has difficulty achieving calibration and may point out a technical consideration that leads to an edit to the hydrologic model. Similar exchanges and coordination are required as models are updated in response to formal review comments. This type of exchange does not

constitute a formal review but is absolutely required for project success. It cannot be done without regular team collaboration.

2.2.2 AUTHORITATIVE INPUT DATASET CHANGES

While the data prepared in the scoping and data preparation phase is carefully planned and thoroughly reviewed, the need for changes may be discovered during the basin execution phases. Any changes to the authoritative input dataset require coordination across the different sub-model teams. When updates are required, the centrally stored dataset in the Model Library will be updated and corresponding updates to the data manifest will be made.

2.2.3 COMMUNICATION AND COLLABORATION TOOLS

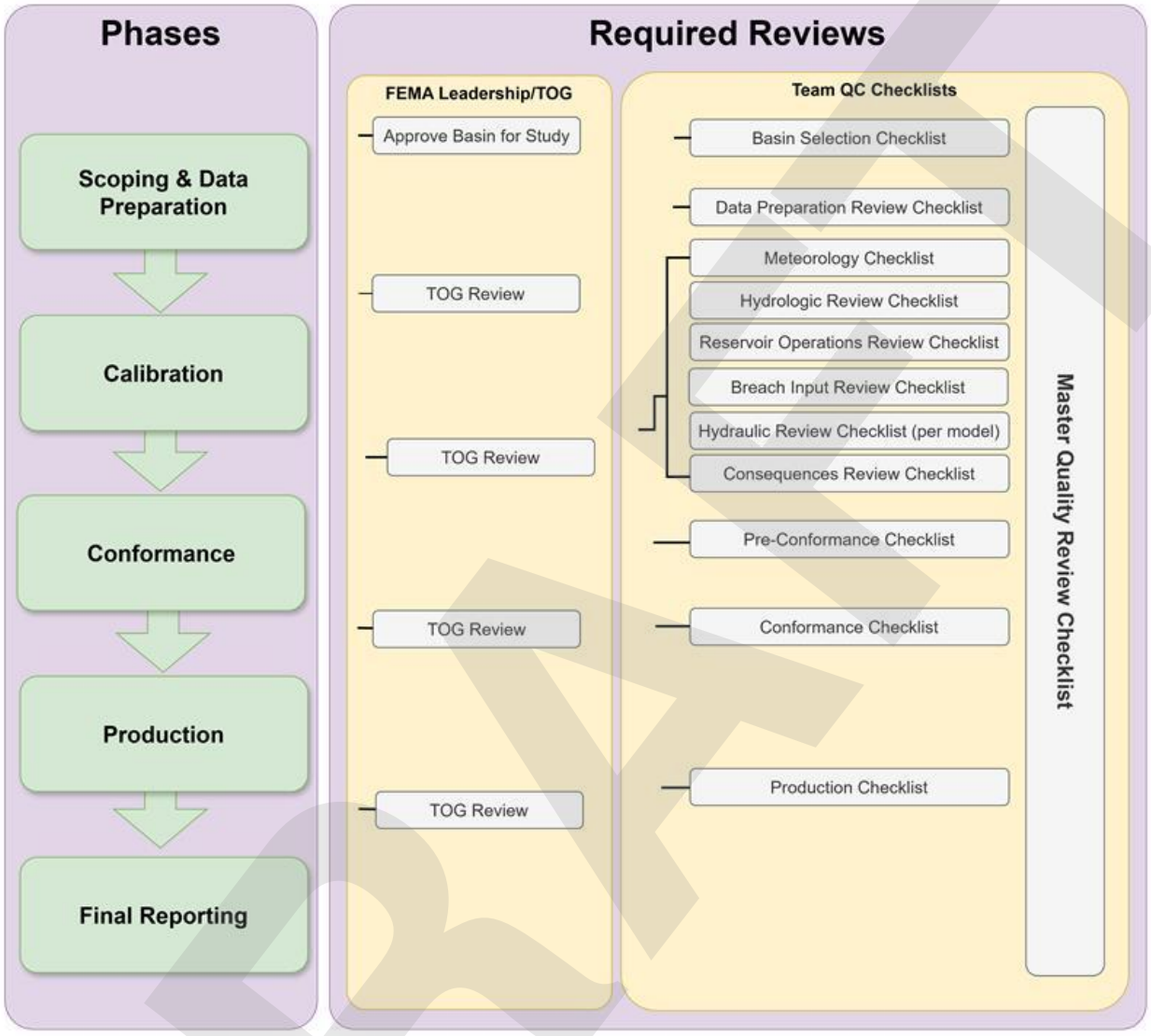
Modeling efforts for the FFRD program are a large effort that requires many team members working together in a cohesive manner. Use of modern tools for full team communication and collaboration is encouraged. For example, centralized discussion forums that log communications among team members have the potential to create a searchable record of the resolution of common issues or to distribute team-wide guidance more efficiently than regular meetings or email distribution lists.

2.3 REVIEW PROCESSES

Reviews across all levels of work are necessary for project success. The review manager assigns reviews and ensures reviewers complete the review on each work package. The technical team lead confirms review comments are sufficiently addressed. After comments are sufficiently addressed, the review manager elevates basins to the TOG for review. Figure 2.3 shows specific reviews that must be documented via review checklists. Any technical issues at the team level that require resolution may be elevated to the TOG. Issues that cannot be resolved at the TOG are elevated to the Steering Committee for final resolution.

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Figure 2.3. Major Phases of FFRD Workflow and Associated Reviews.



2.3.1 TECHNICAL INTERNAL REVIEWS

Several technical QC reviews are conducted throughout the project. Each review is documented in the form of a checklist. Template checklists with pre-specified review prompts are provided as a supplement to this SOP. These checklists include fields for:

- Basin Name
- Team Member Name (producer of the work to be reviewed)
- Review Name
- Review Dates
- Comment for each item (required entry, even if it is confirmation that the item was checked and the standard was met)
- Backcheck and Closeout Discussion
- Status Tracking of each item (Open/Closed)
- Allowance for additional comments that reviewers determine may not fit within the pre-defined checklist items.

An overview of the specific review checklists is listed below.

- Master Quality Review—Primary tracking checklist to confirm major project actions are taken prior to advancing to subsequent phases. For example, the Master Quality Review checklist forces the review manager to confirm that the other technical review checklists have been completed.
- Basin Selection Review—Checklist to confirm the proposed basin has adequate data available to successfully execute an FFRD project prior to starting the scoping and data preparation phase.
- Breach Input Review—Checklist to verify proper delivery of system response curves for infrastructure.
- Data Preparation Review—Checklist to confirm completeness and accuracy of the data preparation processes.
- Meteorology Review—Checklist to confirm technical accuracy of the meteorologic data inputs as well as completeness of documentation.
- Hydrologic Review—Checklist to verify technical accuracy of the HEC-HMS model as well as completeness of documentation.
- Reservoir Operations Review—Checklist to validate technical accuracy of the HEC-ResSim model as well as completeness of documentation.
- Hydraulic Review—Checklist to verify technical accuracy of the HEC-RAS model as well as completeness of documentation.
- Consequences Review—Checklist to confirm technical accuracy of the structure inventory as well as completeness of documentation.
- Pre-Conformance Review—Checklist to verify the full system of models is ready to accept generalized synthetic storms, re-confirm naming conventions were followed, check if model linkages are clearly documented, and confirm that results from cloud compute can be replicated in a desktop environment.
- Conformance Review—Checklist to verify that the system of models meets metrics for comparison to long-term observed data after cloud compute for a single realization of synthetic events.
- Production Review—Checklist to assess if all results from the full set of realizations are acceptable.

2.3.2 TECHNICAL OVERSIGHT GROUP REVIEWS

The TOG provides technical advice, oversight, and project review at a programmatic level. This panel is charged with managing the risks to program execution and credibility. The technical team lead for each basin prepares a presentation and presents results up to the current level of study for the four TOG review engagements.

The TOG reviews each basin at four progress points—after scoping and data preparation phase, after model calibration phase, after model conformance phase, and at the conclusion of the production phase. The TOG meets when reviews are available on up to a monthly basis. Effort will be made to group reviews into presentation meetings with at least three basins presented.

Phase documentation will be provided to the TOG at least 2 weeks prior to a presentation meeting. Presentations are expected to be a high-level technical overview of completed work, and last from 1 to 2 hours.

The TOG provides high-level QA and consistency review comments on the technical aspects of the calibration, conformance, and production work products. Basin modeling teams address comments that fall within the scope of the SOP. Comments outside of the scope of the SOP will be evaluated by the FEMA representative to determine risk to the project and will either be scoped as an additional effort to the ongoing project or flagged as a future update to the SOP.

TOG comments will typically be backchecked by the review manager for minor comments; however, the TOG may request presentation of a basin again, if significant quality concerns need to be addressed.

After TOG comments are satisfied and closed, the FFRD team may proceed to the next phase of the project.

2.3.3 DOCUMENTATION OF REVIEWS

All reviews must be documented with saved comment and comment resolution information. Each FFRD project uses standardized review checklist templates provided as part of the SOP and used to document reviews. Review checklists are maintained within the documentation folder within the basin repository in the Model Library. All review templates allow for free-form commentary on key aspects of the review that may not be directly included in the checklist. All review documentation shall be uploaded within the documentation folder for each discipline.

2.4 CHANGE MANAGEMENT DURING BASIN STUDIES

When issues arise that prohibit production of FFRD results, deviations from the SOP may be justified. Production teams should submit deviation requests that describe the constraints and proposed remedy to FFRD program leadership, and any approved deviations will be recorded in a signed memo.

2.5 NATIONAL-LEVEL RESOURCES FOR SUPPORT AND REVIEW

A comprehensive organizational and governance structure for the FFRD program serves to clarify the roles, responsibilities, and authorities of individuals/agencies needed to promote program success. This support structure is described in Volume I of this SOP.

2.5.1 BASIN LIAISONS

Basin liaisons from USACE with experience developing early FFRD pilot projects will be available to support basin production teams. This role is expected to be critical for early production activities and may diminish over time as collective experience with SOP procedures is gained.

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SECTION 3—STANDARDS

3.1 MODELING SYSTEM AND PLUGINS

The FFRD program is executed in a cloud-based environment requiring each model to have a plug-in to connect with the environment. This SOP references specific plugins that are available for programs built by HEC and tailored to operate in a cloud environment.

There are various alternative options for modeling systems that simulate the hydrologic, hydraulic, reservoir operations, system performance, and consequence response to storm events that could be possible to apply to an FFRD project. It is recognized that building standards and procedures for alternative systems would be a significant effort and would require coordination and approval from FFRD leadership.

3.2 PROJECTION

The Albers Equal Area Projection will be used for the FFRD program. See Table 3.1 for specific parameters associated with the standard projection.

Table 3.1. Parameters for Standard Albers Equal Area Projection for Future of Flood Risk Data Study Watersheds in the Contiguous United States

PROJCS	USA_Contiguous_Albers_Equal_Area_Conic_FFRD
GEOGCS	GCS_North_American_1983
DATUM	D_North_American_1983
SPHEROID	GRS_1980, 6378137.0, 298.257222101
PRIMEM	Greenwich, 0.0
UNIT	Degree, 0.0174532925199433
PROJECTION	Albers
PARAMETER	False_Easting, 0.0
PARAMETER	False_Northing, 0.0
PARAMETER	Central_Meridian, -96.0
PARAMETER	Standard_Parallel_1, 29.5
PARAMETER	Standard_Parallel_2, 45.5
PARAMETER	Latitude_Of_Origin, 23.0
UNIT	Foot, 0.3048

Certain exceptions may apply for studies outside of the contiguous U.S. In those cases, state plane projection for the state where the majority of the modeling unit resides may be used. Any deviations from the standard Albers Projection must be approved by the FFRD National Implementation Group.

3.3 VERTICAL DATUM

The North American Vertical Datum of 1988 (NAVD 88) will be used for all FFRD products. Elevation conversion from local project information may be necessary. When conversions from other standard vertical datums are required, the National Oceanic and Atmospheric Administration (NOAA) Vertical Datum Transformation Tools may be applied but must be documented in FFRD reports and models. When federal dam owners' official vertical datum conversion values are available, they will be used directly.

Certain exceptions may apply for lands that do not coincide with the NAVD 88 footprint (e.g., Hawaii and Puerto Rico). Any deviations from the standard NAVD 88 datum must be approved by the project technical lead.

3.4 DIGITAL ELEVATION MODELS

- Digital Elevation Model (DEM) Quality Standard: FFRD projects will use a DEM based on light detection and ranging (LiDAR) data of Topographic Data Quality Level 2 (QL2) or better, as defined by U.S. Geological Survey's (USGS's) Topographic Data Quality Levels (QLs). If QL2 data is not available, a lower quality DEM may be acceptable after coordination with the technical team lead during the scoping and data preparation phase.
- DEM Minimum Quality: If QL2 data is not available for the study, the minimum requirement for the initiation of an FFRD study is basin-wide availability of the USGS 3D Elevation Program (3DEP) 1/3 arc-second DEM. This dataset is currently available for the entire contiguous United States (CONUS), Alaska, Hawaii, and Puerto Rico.
- DEM Scoping Consideration: A lack of availability of a high-quality DEM may be considered in the prioritization and schedule of basins to receive FFRD study. Procurements of QL2 data may be justifiable as part of the study and must be coordinated with FEMA FFRD leadership during the development of the S-PWP for the scoping and data preparation phase.
- Bare Earth: The official DEM for the study will represent bare earth elevations. The DEM does not represent elevations associated with buildings or vegetation.
- Hydro-connections, Hydro-corrections, and Reconditioning: The official DEM for the project may include adjustments to allow for drainage paths when readily available. This is not a requirement for the official delivered DEM and should not be considered an additional scope item during the data preparation phase.
- DEM Resolution: Standards for DEM Resolutions are provided in the table below.
- Nationwide Reference Grid Standard: To maintain consistency across FFRD deliverables, all DEMs and derivative gridded data must align to a single, nationwide reference grid. This grid is defined as follows:
    - Projection
        - ♦ USA_Contiguous_Albers_Equal_Area_Conic_FFRD
    - Base Cell Size
        - ♦ 4 feet or an exact multiple thereof
    - Reference Origin
        - False Easting: 0.0 feet
        - False Northing: 0.0 feet
        - Central Meridian: -96.0 degrees
        - Latitude of origin: 23.0 degrees
    - Snapping and Corner Matching Rule
        - ♦ All DEMs and derivative grids must be generated such that their origin coordinates and cell boundary coordinates in the standard projection fall exactly on integer multiples of the base cell size.
- DEM Tiling: The full-resolution, official DEM will be delivered in a format that is practical for hydraulic model development purposes. A 3 gigabyte (GB) file size is considered the practical upper limit for hydraulic model development, and DEMs are delivered in a tiled format such that individual model units are at or below that file size limit. Delivery of tiled full resolution DEMs also allows for variation of cell size across the study basin.

- DEM Extent: The DEM must include a large buffer beyond the study extent boundary, which minimizes possible re-work associated with any changes that may occur during scoping.
- DEM File Format: The base-terrain DEMs are delivered in cloud optimized GeoTIFF format.
- DEM Official Delivery: The highest quality available DEM for the entire basin will be prepared and used as the foundation of the study. The DEM will be delivered during the scoping and data preparation phase and be used as the official base DEM for the FFRD project. An alternate lower resolution version will also be prepared for the project and will be based on a down-sampling of the official full resolution DEM. Team members will not use other DEM data sources. Job Aid 06, *Standard Naming Conventions*, describes the DEM file name standard. Procedures for searching for DEM source, FFRD DEM development processes, and FFRD format of delivery are elaborated in Job Aid 02, *Data Preparation Processes*.

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Table 3.2. Base Digital Elevation Model Terrain Delivery Versions

Official DEM Delivery	Storage Location	Description and Purpose	Pixel Size and Origin
Full Resolution DEM	terrain-base directory	The highest quality available bare earth DEM for the entire basin. This full resolution DEM will be used as the basis for: • Hydraulic Model Geometry Development and Pre-processing • Delineation of Hydraulic Model Units and Hydrologic Model subbasins. • Development of Elevation Derived Hydrography (stream centerlines)	Highest resolution DEM supported by LiDAR quality level, rounded to an even value in feet. Typically, a 4-foot resolution DEM based on QL2 LiDAR data. If alternative base DEM resolutions are proposed, careful consideration must be given to the impact of edge-snapping with other products. Resolution may vary across the basin if tiling is used. The reference origin must be used.
100-foot DEM	terrain-base directory	Down-sampled version of full resolution DEM. Lower resolution terrain used for hydrologic model parameterization and general basin overview maps.	100-foot pixels The reference origin must be used.

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3.5 HYDRAULIC MODEL TERRAINS

The hydraulic modeling process involves the conversion of the official source DEM into HEC-RAS-formatted terrains. These terrains can accept additional RAS Mapper terrain modifications for a variety of purposes and may incorporate other data sources such as bathymetry layers. Guidance on development of RAS Mapper terrains for FFRD studies is described in Section 8.3.2 and in Job Aid 12, *Model Delivery for Cloud Compute*. Table 3.3 summarizes required deliveries of final RAS Mapper terrain datasets that are developed during the calibration phase. The file name standard for the hydraulic model terrains is shown in Job Aid 06, *Standard Naming Conventions*.

Table 3.3. Hydraulic Model Terrain Versions

RAS Mapper Terrain Delivery	Storage Location	Description and Purpose	Pixel Size and Origin
Modeling Terrains	terrain-modeling directory	Central storage location for RAS Mapper terrain (including vector modifications) used as the input for each model geometry. A single terrain is required for each HEC-RAS model unit (i.e., multiple versions of terrains are not allowed). These terrains are intentionally stored at a higher-level directory to allow them to be accessed centrally during various phases of the workflow without redundancy.	Use full resolution terrain for model development, typically 4-foot cells. Consistent origin should be maintained as best practice to maintain snapped edges.
Mapping Terrains	terrain-mapping directory	Final HEC-RAS terrain (with any modifications merged into the single tif) associated with each model unit, which is used specifically in development of mapped output grids such as depth.	Use 12-foot cell resolution for mapping terrain. Reference origin must be maintained.

3.6 TIME ZONE

Coordinated Universal Time (UTC) is the time standard used for all FFRD projects.

3.7 OBSERVED GRIDDED DATASETS

Observed temperature and precipitation gridded datasets are used as boundary conditions for modeling efforts for FFRD projects.

For single storm event models, the minimum gridded dataset resolution required is 4-kilometer cells with an hourly timestep.

For long-term simulations (i.e., period of record [POR]), the minimum gridded dataset resolution required is 4-kilometer cells with a daily timestep.

Increased resolution (i.e., smaller cells or shorter timesteps) may be scoped for the study when the data is available and can be applied practically in terms of model development and computing times.

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3.8 NAMING CONVENTIONS

For successful integration of the models, all models and directories must use consistent naming conventions. Job Aid 06, *Standard Naming Conventions*, defines the standard naming conventions for watershed level model data as well as for each constituent model.

3.9 DATA STORAGE LOCATION

The FFRD datasets will be centrally stored in an S3 cloud repository.

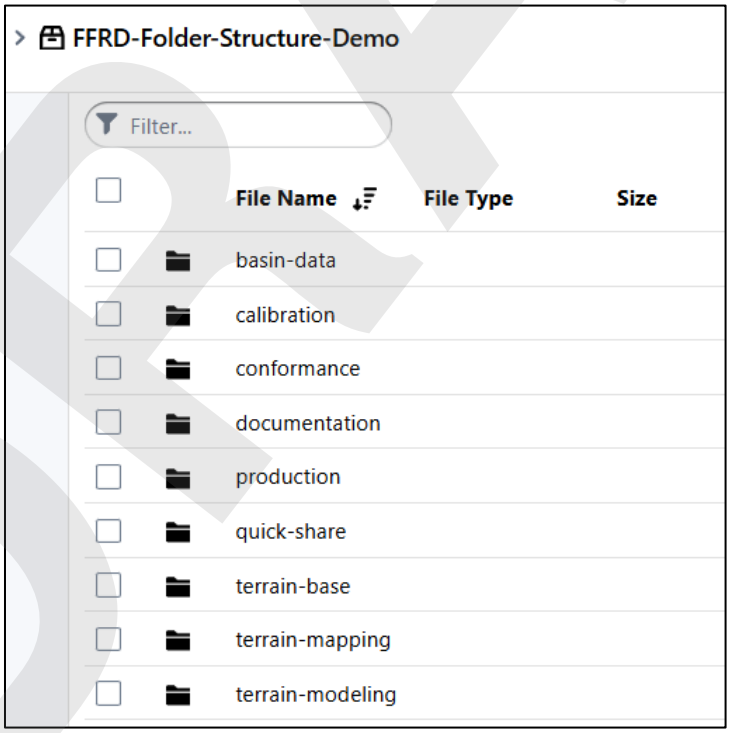
3.10 MODEL LIBRARY INTERFACE

The Model Library will be used as the interface for cloud-based storage that can be accessed by all team members based on user-based access roles. Job Aid 07, *Model Library Use*, contains details on Model Library access and use.

3.11 STANDARD DIRECTORY STRUCTURE

The standard directory structure and Model Library serve as the central cloud storage locations and are accessible by all team members and all organizations throughout the life cycle of the project. The standard directory is available within FFRD Templates. The following subsections describe the purpose of each high-level directory. Job Aid 07, *Model Library Use*, provides additional details on the use of the directory including standard subdirectories.

Figure 3.1. Top-Level Standard Directory for an FFRD Project



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3.11.1 BASIN-DATA DIRECTORY

The basin-data directory stores all official shared datasets for the project, which are delivered in the scoping and data preparation phase. Ad-hoc sharing of unofficial data shall not be posted to the basin-data directory.

3.11.2 QUICK-SHARE DIRECTORY

Temporary ad-hoc shared data can be shared among team members through the quick-share directory. This allows team members to share files that are temporary in nature, but useful for collaboration or technical discussions.

3.11.3 DOCUMENTATION DIRECTORY

The documentation directory is the centralized storage of all work plans, reports, and reviews for the entire project. The documentation will advance from draft stages to final products within this location.

3.11.4 TERRAIN-BASE DIRECTORY

This directory contains the original delivered base DEM for use in the project that was gathered as part of pre-model data preparation. There should be multiple resolutions as defined in Section 3.4.

3.11.5 TERRAIN-MODELING DIRECTORY

This directory contains RAS Mapper-formatted terrains used for HEC-RAS model development. Section 3.5 provides more information.

3.11.6 TERRAIN-MAPPING DIRECTORY

This directory contains modified, RAS Mapper-formatted terrains used specifically in development of mapped output grids. Section 3.5 provides more information.

3.11.7 CALIBRATION DIRECTORY

This directory contains individual models through event-based calibration phase. These models and associated datasets undergo review and backcheck. If model files are updated due to a review, the previous version is first moved to the archive directory of individual model folders. This update and archival process may continue as models progress from draft to final delivery. Note that documentation and terrain files associated with each hydraulic model are posted to separate directories (*documentation* and *terrain-modeling*) and should not be included here.

3.11.8 CONFORMANCE DIRECTORY

Model files for the conformance phase are stored in the Model Library watershed repository under *conformance*. Final individual models from the calibration phase are promoted to the conformance directory, integrated, executed in cloud compute, and adjusted to reasonably align with long-term observed data in frequency space. When adjustments are necessary to individual models, outdated models are moved to the archive subdirectory and replaced with the current version. Integrated models run through many simulations in cloud compute and results are stored here. Note that documentation and terrain files associated with each hydraulic model are posted to separate directories (*documentation* and *terrain-modeling*) and should not be included here.

3.11.9 PRODUCTION DIRECTORY

Model files for the production phase are stored in the Model Library watershed repository under *production*. Final models from the conformance phase are promoted to the production directory and production simulations are executed in cloud compute and output data is stored here. When hot fix model edits are needed for individual events, a *hotfix* subdirectory will be created to store the model versions. Note that documentation and terrain files associated with each hydraulic model are posted to separate directories (*documentation* and *terrain-modeling*) and should not be included here.

3.12 TEMPLATES

Each FFRD project uses standardized documentation templates, review processes, and calculation tools for calibration and other purposes. Various templates supporting FFRD model and report production are available at the FFRD Templates and Resources GitHub website. These template resources include the following categories:

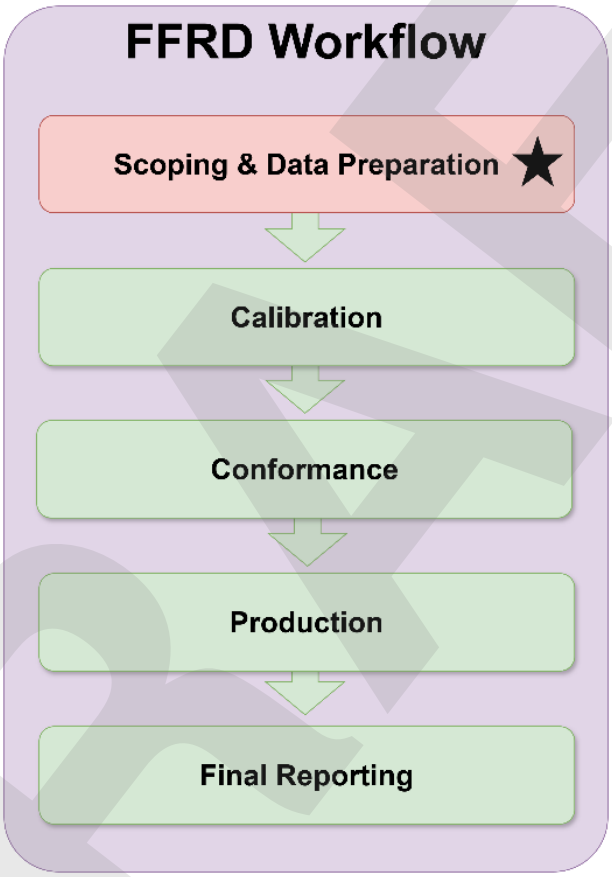
- Data Delivery Templates: Example files for data delivery of calibration event selection, the Dam Scoping Table, model linking, system response tables, and several other items.
- Documentation Templates: Report templates for each phase of the FFRD process, overall project delivery report, project work plans, and suggested language for data-request communication.
- Quality Management Templates: Review checklist templates for each pre-defined review stage.
- Tool Templates: Analytical spreadsheet tools used for computing calibration statistics, and breach parameters using both the regression equation approach and simplified-physical breach methodology.

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SECTION 4—SCOPING AND DATA PREPARATION

Watershed studies for FFRD require a large team working in a coordinated manner. The scoping and data preparation phase is a significant effort executed by a relatively small team to define the specifics of the project scope and provide an authoritative common input dataset that is in place to be used by the entire team throughout the subsequent phases. In addition to data collection and coordination efforts, this phase includes modeling and analysis tasks to confirm that the scope of individual model efforts (to be initiated by a larger team in the subsequent calibration phase) is well defined.

Figure 4.1. Context of Scoping and Data Preparation Phase in Overall Workflow



4.1 PROJECT WORK PLAN—SCOPING AND DATA PREPARATION PHASE

A S-PWP will be developed to outline the team, tasks, schedule, and budget for the scoping and data preparation phase of an FFRD project.

4.2 STAKEHOLDER AND DATA PROVIDER ENGAGEMENT

During the scoping phase, a list of agency stakeholders and data providers is developed, and engagement meetings are held. Stakeholders are provided with an overview of the planned work and the planned schedule for the FFRD project, which will focus on flood concerns in the basin. During engagements, the project delivery team surveys stakeholders regarding the most important aspects of flooding in the basin. Data providers are encouraged to provide information or data related to existing

studies or models that may be useful for the FFRD study and offer general feedback. The team provides periodic progress updates to the stakeholder group throughout the project.

4.3 SCOPING

Several overall modeling approach decisions are made during the scoping phase that impact the proposed cost, schedule, and budget for the subsequent basin execution phases of the project. A decision on the ability to move forward with currently available data must also be made. The following subsections describe several of the high-level considerations for scoping.

4.3.1 WATERSHED STUDY EXTENT

FFRD studies are typically scoped based on USGS Watershed Boundary Dataset (WBD) HUC-4 boundaries. Existing WBD boundaries will be reviewed during project scoping for possible adjustments due to hydrologic interactions with adjacent basins or other considerations. Adjustments to the boundary for the modeling effort will be clearly documented, justified, and described in the BE-PWP and project reports. Note that the HUC-4 boundary is used for project planning and early scoping, but higher resolution boundaries based on the FFRD high-quality DEM will be used to create more detailed and accurate boundaries used in the models as described in other sections of this document.

4.3.2 EXTERNAL BOUNDARY CONDITION DATA

The FFRD methodology requires external boundary conditions (e.g., upstream inflow, downstream stage) as key inputs to a basin study. The general strategy is to develop models sequentially from headwaters to outlets; however, this may not always be possible or advantageous. Large and/or complex modeling domains may require additional boundary conditions to accurately estimate flood frequency at critical locations. The availability of external boundary condition information will be confirmed as part of the FFRD project scoping phase. There may be more complex interactions across basins requiring adjustments to the methodology, and they will also be defined during this phase.

4.3.3 ELEVATION DATA AVAILABILITY

The FFRD standard for elevation data quality is defined in Sections 3.3 and 3.4, along with a description of cases where deviation from the QL2 standard may be accepted or cases where field data collection may be scoped as part of the FFRD basin.

4.3.4 STOCHASTIC STORM TRANSPOSITION APPLICATION

The methodology described in this SOP relies on Stochastic Storm Transposition (SST) techniques. The method is driven by a catalog of historic observed storms within a transposition domain that is larger than the study basin. The ability to identify a transposition domain, an assessment of applicable storm types, and the availability of long-term observed temperature and precipitation data of sufficient spatial and temporal resolution for the domain will be assessed during the scoping phase of the project.

4.3.5 OBSERVED STREAM GAGE DATA AVAILABILITY

The availability of observed stream gage data is assessed as part of the FFRD project scoping phase. This data is cross-referenced with the available observed gridded precipitation and temperature data. The scoping team must confirm a full calibration dataset exists to support the subsequent basin execution phases of the project. If adequate data is not available, the approach to the overall effort should be coordinated with FEMA FFRD leadership.

4.3.6 OBSERVED GRIDDED DATA AVAILABILITY

A substantial record of gridded precipitation and temperature data is a requirement for the methodology described in this SOP. Availability of these datasets will be confirmed during the scoping phase of the project. Three categories of observed gridded data are required:

- Several significant historic events over the study basin
- Long-term POR over the study basin
- Storm catalog of large events within the larger transposition domain

4.3.7 LOCATIONS FOR CALIBRATION

All stream gages with a significant period of observed data are used for calibration of the FFRD models. Each gage is categorized as either primary, secondary, or tertiary gages to describe the hierarchy for use in model calibration. More rigorous calibration metrics (as described in separate sections of this SOP) are used for primary gages than for secondary or tertiary gages. The gages are categorized during the scoping phase to maintain consistency in calibration targets across all FFRD modeling efforts for the basin. Guidance for categorizing gages is defined in Job Aid 01, *Scoping Processes*.

4.3.8 IDENTIFY CALIBRATION EVENTS

Historic storm events to be used for calibration of hydrologic and hydraulic models are selected during scoping. Generally, the largest observed historic events that have occurred in the basin are selected. The availability of both reliable observed stream gage data and gridded precipitation and temperature data must be verified for each selected event. Consideration is also given to events that have occurred in varied spatial domains within the study watershed. More recent historic events may be given preference as they may better reflect current basin conditions and may have higher quality observed data. Specific time windows for the events will be clearly defined in the scope. The guidelines for selecting events for calibration are described in Job Aid 01, *Scoping Processes*.

4.3.9 HYDRAULIC MODELING UNIT DELINEATION

FFRD studies are typically scoped for large-scale basins, and developing full basin two-dimensional (2D) hydraulic models at this scale is generally deemed impractical using the computing capability available with current technology. Therefore, smaller hydraulic modeling units are delineated during the scoping phase. Model units will be delineated according to guidance contained in Job Aid 01, *Scoping Processes*, which includes considerations for total drainage area, calibration locations, model linking strategy, dam locations, and levee locations among other factors.

4.3.10 HYDRAULIC MESH DEVELOPMENT SCOPE

FFRD hydraulic models use a relatively coarse background mesh, with a higher-resolution, more detailed mesh for specific streams, floodplains, and populated areas. Three geospatial layers that define the scope for mesh development (urban areas, key streams, and other streams) are created during the scoping phase according to the guidance and procedures provided in Job Aid 01, *Scoping Processes*. All areas outside these layers are represented by the background mesh. Table 4.1 shows the selection criteria and intended use for each mesh development category, and Section 8.3.1.3 describes the model treatment for each category.

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Table 4.1. Mesh Development Categories, Selection Criteria, and Intended Use

Mesh Development Category	Definition and Criteria for Selection	Intended Use of Output
Background Mesh	All areas not covered by the urban areas, key streams, or other streams layers.	Produce basin-wide H&H data at a coarse resolution that may be leveraged for separate detailed modeling efforts. Provide approximate pluvial and fluvial flood information. Collect and route flows to downstream locations.
Urban Areas (polygons)	Polygons sourced from the U.S. Census *urban areas* layer, which accounts for density of development, land cover, development type, and other factors. Any modifications made during scoping are documented and justified in the scoping report.	Produce reliable pluvial and fluvial flood risk mapping results in populated areas.
Key Streams (lines)	Streams that meet any of the following criteria: a contributing drainage area greater than 100 square miles; an existing Flood Insurance Study (FIS) supporting Zone AE, AH, or AO delineation; or an active stream gage with observed data useful for calibration.	Accurate routing of flood water through larger streams. Accurate flood risk mapping from fluvial sources for the most critical streams in the basin.
Other Streams (lines)	All streams not designated as key streams that have a contributing drainage area greater than 0.25 square miles.	Improve the accuracy of flood water routing through smaller streams. Moderately improve the accuracy of fluvial flood risk mapping along these streams.

4.3.11 LAND COVER DATA

The National Land Cover Database (NLCD) will be obtained during the scoping and data preparation phase. This dataset meets the minimum standard for the base roughness layer for use in FFRD models.

Other sources of land cover mapping that exceed the current NLCD quality and resolution may be scoped for use on specific projects if they are readily available and coordinated with the technical lead and program lead early in the scoping and data preparation phase.

4.3.12 LEVEE DATA

The National Levee Database (NLD) will be leveraged as the source for all levee alignment and crest profile data in FFRD models. Prior to or during the scoping phase, the NLD will undergo a data refinement process to verify sufficient accuracy and completeness of the data for the entire study basin. Processes for levee data refinement are documented in the Reference Documents stored on the NLD.

4.3.13 LEVEE BREACH LOCATIONS

The USACE levee data lead will specify draft breach locations for each levee in the basin during the scoping phase. The locations will be selected according to the guidelines in Job Aid 11, *Levee System Response Development*, which considers the availability of existing risk assessment studies and levee length among other factors.

Note

Levee breach locations may be adjusted later during the calibration phase based on information gained related to the most likely overtopping location from early model results.

4.3.14 DAM DATA

The National Inventory of Dams (NID) will be leveraged as the official source of inventory of dams for FFRD studies. Prior to or during the scoping phase, the NID will undergo a data refinement process to verify sufficient accuracy and completeness of the data for the entire study basin.

4.3.15 DAM AND RESERVOIR MODELING SCOPE

During the scoping phase, the model treatment for each dam in the basin will be clearly specified in the *Dam Scoping Table* (as described in Table 4.2). There are many combinations of how a dam can be modeled across FFRD models, and these treatments are covered in relevant model-specific sections below. Modeling treatments for each dam are selected according to the guidelines in Job Aid 01, *Scoping Process*es, which balances modeling efficiency, accuracy, and the magnitude of impact to overall flood characteristics among other factors. Job Aid 08, *Dam Physical Representation*, provides additional details on modeling and coordination of dams across the FFRD models.

The Dam Scoping Table will provide an authoritative selection of modeling methods throughout the project. During the calibration and conformance phases, computed results within and downstream of the dams of interest may indicate a need for adjustments to the model treatment for certain dams. If the computed results are not adequate, revisions to the Dam Scoping Table may be warranted. Any proposed revisions to the techniques must be approved by the technical lead and coordinated with the full modeling team.

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Table 4.2. Dam Scoping Table Attributes

NIDID
Dam Name
Alternate Dam Name
Max Storage
Hazard Class
Scoping Justification / Discussion
Include in Reservoir Operations Model (Y/N)?
Number of Unique Outflow Features? ("ResSim Dams")
Hydrologic Model Treatment - Historic Events
Hydrologic Model Treatment - Synthetic Events
Hydraulic Modeling Treatment
Include Hydraulic Initial Pool Elevation (IC Point)?
Breach? (Yes/No)
Vary Initial Pool Elevation During Stochastic Events?

4.3.16 UNIQUE FEATURES

Certain unique hydrologic features or infrastructure may play an important role in the flooding characteristics of a basin. Any major features that are not otherwise described will be assessed for possible inclusion in the scope for basin execution.

4.3.17 MODEL ORDER DESIGN

The full set of integrated FFRD models relies on passing model outputs from one model to be used as inputs for other models. The selection of plugins and the strategy for passing data between models is referred to as the model order and will be clearly defined during scoping.

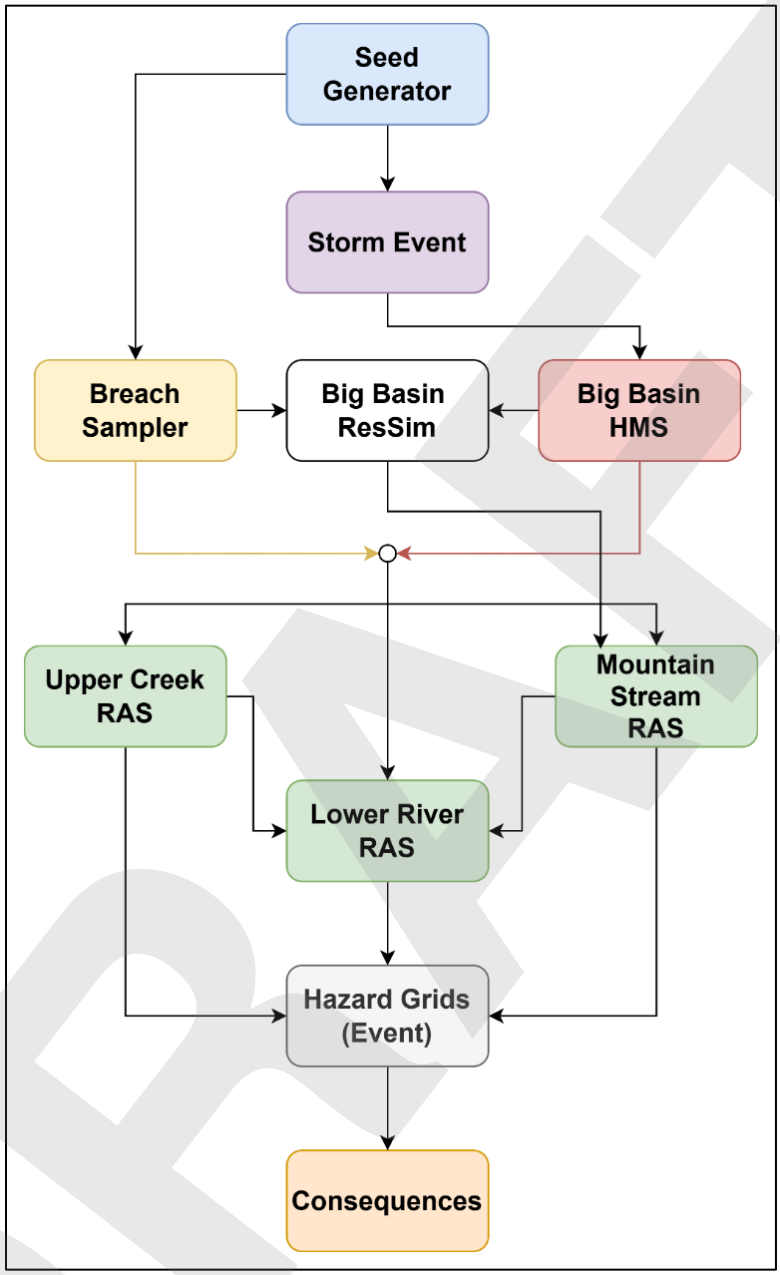
4.3.17.1 CONCEPTUAL MODEL ORDER GRAPHIC

A conceptual model order graphic, such as the example in Figure 4.2, is created along with any additional detail or discussion and included in the BE-PWP.

Separate model orders will be necessary for model calibration (observed events) versus the production of synthetic stochastic simulations of the models. For example, historic event modeling will often utilize observed outflows from large dams while synthetic events will receive outflows from a reservoir operations model.

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Figure 4.2. Example Conceptual Model Order Graphic



4.3.17.2 NAME TABLE

The conceptual model order graphic is supported by an authoritative *Name Table* of all planned model file names and internal model element names which follows the naming standards in Job Aid 06, *Standard Naming Conventions*, and is developed during the scoping and data preparation phase of the project.

4.3.17.3 MODEL LINKING REGISTER

An authoritative *Model Linking Register* lists the specific data that will be passed between models including path names and element names that are developed during the scoping and data preparation

phase of the project. The register will include the plan for linking models during the calibration phase (which leverages historic observation data for some boundary conditions) and a modified plan for linking during the conformance/production phases (which leverages simulated conditions in all cases).

4.3.18 ENUMERATION OF CLOUD SIMULATIONS

Six observed historic events will be selected for model calibration efforts.

The stated FFRD description objective of simulating a full spectrum of frequencies from routine (10-year) to rare (2,000-year or rarer) requires a minimum set of synthetic years per realization of 2,000.

Ten events per synthetic year will be modeled for projects conducted according to this SOP.

The number of realizations impacts scoping decisions as compute costs represent a significant cost within the overall study. Projects executed under the current version of this SOP will include 5 realizations, which will provide a basic understanding of uncertainty around variable-frequency relationships. Table 4.3 shows the number of modeled events in an FFRD study.

The uncertainty estimates around variable-frequency relationships may improve with additional realizations. For statistically appropriate confidence intervals, the number of realizations would be determined based on a set of convergence criteria. Future versions of this SOP may consider allowing flexibility to adjust the number of realizations based on statistical criteria and other factors.

The number of realizations specified is not sufficient to accurately describe knowledge uncertainty; however, it does allow for a full demonstration of the envisioned result datasets for validation of the workflows and procedures contained in this SOP in 2026. Future versions of the SOP will consider advanced techniques to optimize the simulation strategy to meet the requirements related to quantification of uncertainty.

Table 4.3. Total Number of Modeled Events for an FFRD Study

Phase	Number of Synthetic Years	Number of Events per Synthetic Year	Number of Realizations	Total Events
Calibration	N/A	N/A	N/A	6 observed events
Conformance	2,000	10	1	20,000
Production	2,000	10	5	100,000

4.4 DATA PREPARATION

Consolidation of various datasets is integral to the scoping effort and establishes a foundation that allows each individual model to be produced in a consistent manner. Several of the datasets collected during the scoping phase simply obtain data from other sources, but many others are produced through engineering analysis and assessments that are made through the scoping procedures.

The complete prepared dataset is a key delivery from the scoping and data preparation phase of the project. This dataset provides the base data and scope definition data that is needed to execute the project during the subsequent basin execution phases of the project.

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4.4.1 DATA MANIFEST

A data manifest will be created to document and provide metadata for all data created for the project as part of the data preparation phase. Job Aid 02, *Data Preparation Processes*, provides the formatting and delivery method requirements for the data manifest.

4.4.2 STORAGE OF PREPARED DATASETS

All prepared datasets will be posted to the central repository using the standard directory structure. The standard directory structure is provided in the demonstration project directory on the FFRD Templates and Resources GitHub site. The process for establishing a new repository for a new project is provided in Job Aid 07, *Model Library Use*.

4.5 PREPARED DATASET LIST AND DESCRIPTIONS

Tables 4.4 through 4.9 show a complete list of datasets to be prepared as part of the scoping and data preparation phase, including descriptions, intended purpose, and guidance for development and delivery of each item.

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Table 4.4. Prepared Manifest Dataset and Description

Dataset	Description	Purpose	Data Preparation Guidance
Data Manifest	Document listing each of the datasets provided, which fully describes the source, purpose, description, and other metadata.	Key reference to document and track all input data used for FFRD project.	Job Aid 02, *Data Preparation Processes*, provides guidance for the format and critical fields to include in the Data Manifest.

Table 4.5. List of Scoping Data Datasets and Descriptions

Dataset	Description	Purpose	Data Preparation Guidance
Study Area Polygon	Polygon defining the study area for the project	Official study area polygon.	Scoping of the study area is described in Section 4.3.
Stream Gages for Calibration	The point layer for gages that will be used for calibration of the models. Each gage is attributed with its assigned importance category (i.e., primary, secondary, or tertiary).	Key reference layer showing the locations where observed data exists and its treatment for calibration and validation.	Guidance for the assignment of gage categories (i.e. primary, secondary or tertiary) is provided in Job Aid 01, *Scoping Processes*.
Storm Events for Calibration	This tabular report of storm events selected for calibration of models includes time windows for each event.	Clearly defines the basin-wide storm events and time windows used for evaluation of historic event-based calibration metrics, which validates all model efforts using a consistent approach.	Guidance for the selection of historic events for calibration is provided in Job Aid 01, *Scoping Processes*.
Hydraulic Modeling Units	Polygons for each hydraulic modeling unit for the study, each with a defined basin name.	Each modeling unit polygon defines a scope for individual hydraulic model unit development.	Guidance for delineation of hydraulic model units is provided in Job Aid 01, *Scoping Processes*.
Uncalibrated Hydrologic Model	Uncalibrated hydrologic model that includes all planned HEC-HMS model elements including subbasins established during scoping and data preparation phase of the FFRD project.	Establish model elements, naming, and hydrologic subbasins prior to start of major model development efforts. Early identification of the subbasins allows for consistent naming conventions to be developed and for the development of hydraulic model units that are aligned with the subbasins.	Hydrologic model is developed according to general best practices and following guidance in Section 6.
Name Table	Table of adopted names for all elements of FFRD models.	Establish adopted consistent names prior to start of major model development efforts, which facilitates model linking and other uses throughout the project.	Naming convention requirements are defined in Section 3. Guidance on delivery and format of the Name Table is included in Job Aid 01, *Scoping Processes*.
Model Linking Register	Graphic and tabular versions of the Model Linking Register for the project that clearly show the source of all required inputs for each model. Includes a version for the calibration phase (when observed flows for historic events are commonly used as inputs) and for the conformance phase (when all inputs are based on modeled outputs).	In combination with the Name Table document, the Model Linking Register provides a clear definition of the data structure to be used for all modeling efforts for the basin.	Guidance on delivery and format of the Model Linking Register is included in Job Aid 01, *Scoping Processes*.
Dam Scoping Table	Table of all dams in the study area attributed with the scope of modeling treatment in each of the relevant models (Hydrologic Model, Reservoir Operations Model, Breach Model, and Hydraulic Model).	Provide clear definition of the scope of work for the FFRD modeling efforts before major model development begins in the calibration phase.	Guidance on development of the Dam Scoping Table is provided in Job Aid 01, *Scoping Processes*.
Levee Data Tracking Table	Table of all levees in the study area along with key attributes including confirmation of data availability (alignment, profile, existing risk assessment).	The primary function of the Levee Data Tracking Table is to inform the scope of Levee Data Refinement Effort completed prior to the FFRD study; identifying which levees need crest profiles. It assembles available fields on levees within the study basin to identify critical levee data gaps or out-of-date information. The table can also be used to manage the levee data refinement effort by assigning tasks to teams or team members and tracking them to completion.	Guidance on development of the Levee Data Tracking Table is provided in separate procedures associated with Levee Data Refinement efforts.

Dataset	Description	Purpose	Data Preparation Guidance
Levee Breach Locations	Geospatial point data table including various attributes that are populated during the scoping and data preparation phase.	Geospatial file of georeferenced breach locations for the study basin. The associated attribute table will include information that documents levee breach schema (quantity and location selection), as well as critical information needed to develop system response curves (risk assessment type and levee crest and toe elevations). Table will identify draft Levee Control Locations (LCL) to be verified by the RAS team and will be updated if any adjustments are made in later phases. This product informs the hydraulic model development during the calibration phase.	Guidance on development of the levee breach locations file is provided in Job Aid 11, *Levee System Response Development*.
Mesh Development Scoping-Key Streams	A set of stream centerlines considered the dominant and most important streams to characterize flooding in the basin. *Key streams* are selected during the scoping phase according to guidelines in Section 4.3.10 and provided as a shared dataset.	This layer conveys the scope for hydraulic model development in terms of *key stream* centerlines. The hydraulic modeling level of detail for FFRD modeling will be increased along *key streams* and adjacent floodplains as described in Section 8.	The guidelines for scoped key streams layer development are shown in Table 4.1 and procedures for their development are provided in Job Aid 01, *Scoping Processes*.
Mesh Development Scoping-Other Streams	A set of stream centerlines helping to characterize flooding in the basin. The *other streams* are selected during the scoping phase according to guidelines in Section 4.3.10 and provided as a shared dataset.	This layer conveys the scope for hydraulic model development in terms of *other stream* centerlines. The hydraulic modeling level of detail for FFRD modeling will be increased along these streams as described in Section 8.	The guidelines for scoped other streams layer development are shown in Table 4.1 and procedures for their development are provided in Job Aid 01, *Scoping Processes*.
Mesh Development Scoping-Urban Areas	Scoped *urban areas* polygons within the study basin.	The scoped urban areas will receive increased level of hydraulic model detail for the project as described in Section 8.	The scoped urban area polygons are sourced directly from the U.S. Census Urban Areas layer. Any modifications to the areas layers during the scoping phase will be documented and justified in the scoping report.
Full Resolution DEM	The best available DEM for the entire watershed	Consistent DEM to be used as the baseline for all aspects of FFRD modeling for the basin.	Standards for DEM delivery for FFRD are outlined in Section 3.4. Guidance on the creation of the DEM is provided in Job Aid 02, *Data Preparation Processes*.
100-ft DEM	The full basin DEM is down-sampled to a 100-ft by 100-ft resolution.	Smaller file size elevation layer, used for hydrologic model parameterization and for general basin overview maps.	Standards for DEM delivery for FFRD are outlined in Section 3.4. Guidance on the creation of the DEM is provided in Job Aid 02, *Data Preparation Processes*.
National Structure Inventory	The NSI provides an unedited base national dataset of structure points and characteristics.	The NSI locations layer is a useful reference. Also is used to aid in the review of hydraulic model results and as the starting point for structures that will be used to compute consequences for the project (after refinement).	The process for acquisition of the NSI base layer is described in Job Aid 02, *Data Preparation Processes*.
Elevation Derived Hydrography (EDH)-Stream Centerlines	Stream centerlines delineated based on the full resolution DEM for the study.	This comprehensive layer can be incorporated into hydraulic modeling with minimal edits for uses such as terrain modification alignment for bathymetry estimates or 2D mesh cell alignments. EDH lines generally match very well with the project DEM, but some artifacts will remain, and modeling best practices must be considered.	Guidance for the development and acquisition of Elevation Derived Hydrography is included in Job Aid 02, *Data Preparation Processes*.
Dam Inventory Data	Geospatial dataset with attributes and includes all dams listed in the NID within the study basin.	This is a reference dataset for use in several scoping and modeling processes.	Guidance for the acquisition of the NID dataset is included in Job Aid 02, *Data Preparation Processes*.

Table 4.6. List of Base Geospatial Data Datasets and Descriptions

Dataset	Description	Purpose	Data Preparation Guidance
Levee Inventory Data	Geospatial dataset of levee features for all levees in the NLD within the primary basin. Alignment and crest elevation profile is also included in tabular format (XYZM).	This reference dataset is to be used in the development of hydraulic models and an important reference for several aspects of the project. The alignment and crest elevation profile data are used directly in hydraulic modeling levee elements.	The NLD will also go through quality review and refinement efforts related to the FFRD study, which are described in Job Aid 11, *Levee System Response Development*. The activity is called Levee Data Refinement and precedes the other data preparation activities, with work typically completed in the prior fiscal year. Guidance for the acquisition of the NLD dataset is included in Job Aid 02, *Data Preparation Processes*.
Bathymetry	Bathymetry data for reservoirs, navigation channels, coastal areas, or other unique features based on observed measurements.	Bathymetry data is a key component to improve accuracy of channel conveyance to improve flow and water surface elevation results. When available, best available bathymetry should be directly applied to all models in the calibration phase.	Guidance for the acquisition of any available bathymetric data is included in Job Aid 02, *Data Preparation Processes*.
Soil Coverage	The Soil Survey Geographic Database (SSURGO) contains soil data at the county level, and the State Soil Geographic Database (STATSGO) contains soil data at the state level.	Soil coverage is useful for estimating infiltration rates and other hydrologic basin model parameters.	Guidance for the acquisition of the soils data is included in Job Aid 02, *Data Preparation Processes*.
NLCD	The NLCD provides spatial reference and descriptive data for characteristics of the land surface such as thematic class (e.g., urban, agriculture, and forest), percent impervious surface, and percent tree canopy cover.	NLCD serves as the minimum standard for roughness parameters for the hydraulic models.	Guidance for the acquisition of the NLCD data is included in Job Aid 02, *Data Preparation Processes*.
Stream Gage Network	Geospatial dataset of all relevant stream gages for the study basin including basic attributes.	Stream gages in the basin are the basis for model calibration and validation and are used as a key reference layer.	Guidance for the acquisition of the stream gage locations and attributes is included in Job Aid 02, *Data Preparation Processes*.
U.S. Census Topologically Integrated Geographic Encoding and Referencing (TIGER) Data	Geospatial dataset of key features and boundaries within the study basin.	The U.S. Census Urban Areas layer is used directly in the FFRD scoping of the hydraulic model effort. Other U.S. Census datasets are useful reference layers.	Guidance for the acquisition of the U.S. Census data is included in Job Aid 02, *Data Preparation Processes*.
National Bridge Inventory	Geospatial dataset of bridges that includes basic attributes.	Bridge location and attributes are a useful reference layer and may be used in development of certain elements of the hydraulic model.	Guidance for the acquisition of the National Bridge Inventory data is included in Job Aid 02, *Data Preparation Processes*.

Table 4.7. List of Observed Data Prepared Datasets and Descriptions

Dataset	Description	Purpose	Data Preparation Guidance
Observed Stream Gage Hydrographs (curated)	Complete records of all available observed elevation and observed flow stream gage data for the basin, curated and adjusted to serve as official dataset for the FFRD project.	The observed stream gage data is the basis for evaluation of model performance for all aspects of the modeling efforts. Establishment of an official curated and adjusted observed dataset during the scoping and data preparation phase allows for consistency among all model efforts.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring observed gage data and includes FFRD guidance for QC, treatment of missing data, and vertical datum adjustments.
Observed Stream Gage Field Measurements (curated)	Complete records of field measurement data at each stream gage location (which include stage and discharge at the time of measurement) curated and adjusted to serve as the official dataset for the FFRD project.	Measurements at stream gages are the basis for understanding of the stage-discharge relationships. The foundational measurement values are used as a guide for model adjustments during the calibration and conformance phases of the FFRD project.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring field observations at gages and includes FFRD guidance for QC and vertical datum adjustments.
Observed Stream Gage Annual Maximum Series (curated)	Complete record of annual peak discharge and stage for the POR for each stream gage, which is curated and adjusted to serve as the official dataset for the FFRD project.	Annual peaks at stream gages provide the basis for flow and stage frequency relationships and are used for evaluation of metrics during the conformance phase of the FFRD project.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring annual peak records at gages and includes FFRD guidance for QC and vertical datum adjustments.
Stream Gage Stage-Discharge Relationships (Ratings)	Published stage-discharge relationships at stream gages.	Published stage-discharge relationships are used similarly to observed stream gage field measurements. They can be used as a guide for model adjustments during the calibration and conformance phases of the FFRD project.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring published stage-discharge relationships at gages and includes FFRD guidance for vertical datum adjustments.

Dataset	Description	Purpose	Data Preparation Guidance
Observed Frequency Relationships	Computed frequency relationships with uncertainty for curated observed flow, stage and precipitation data.	Provide official observed frequency relationships with uncertainty for benchmarking model performance throughout the project.	Job Aid 05, *Observed Gridded Data Preparation*, and Job Aid 04, *Observed Gage Data Preparation*, provide guidance for developing frequency relationships with uncertainty based on curated observed data.
High Water Mark Data	Records of historic flood high water mark measurements in the basin that may exist in non-gage locations for historic flood events.	When measured high-water marks during floods are available, they are used as supplemental information to check and improve the hydraulic model calibration in locations that do not have stream gage data.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring observed high water mark data in the basin when it is available.
Flood Imagery	Images that document historical flooding that may include aerial photos from aircraft, satellite images, or on-the-ground photos.	When flood imagery is available, it is used as supplemental information to check and improve hydraulic model calibration in locations that do not have stream gage data.	Job Aid 04, *Observed Gage Data Preparation*, provides guidance for acquiring historic flood imagery for the basin when it is available.
Period of Record Gridded Data	Precipitation, temperature, and snow-water-equivalent timeseries grids for the POR for the basin.	Observed gridded data is a key input for the development and calibration of the H&H models.	Job Aid 05, *Observed Gridded Data Preparation*, provides guidance for acquiring gridded data for the POR, which includes QC checks and evaluation of possible sources.
Observed Storm Event Gridded Data	Precipitation, temperature, and snow-water-equivalent timeseries for each storm event selected as a calibration event during the scoping phase.	Observed gridded data truncated to include specific historic events as a key input for the development of the hydrologic model.	Job Aid 05, *Observed Gridded Data Preparation*, provides guidance for acquiring gridded data for the calibration events, which includes QC checks and evaluation of possible sources.
Observed Precipitation	Observed precipitation record based on the gridded data over the POR. Summarized as an annual maximum series at the subbasin level for multiple durations (1 hour, 24 hours, and 72 hours).	Annual maximum series of observed precipitation for each subbasin provides a basis for evaluation of performance of SST processes to be applied for the project.	Job Aid 05, *Observed Gridded Data Preparation*, provides guidance for summarizing observed precipitation.
Observed Snow Water Equivalent (SWE)	Observed SWE record based on the gridded data over the POR. Summarized as an annual maximum series at the subbasin level.	Annual maximum series of observed SWE for each subbasin provides a basis for evaluation of performance of SST processes to be applied for the project.	Job Aid 05, *Observed Gridded Data Preparation*, provides guidance for summarizing observed SWE.
Storm Catalog	Catalog of significant storm events that have occurred in the transposition region within the POR. Each significant event includes observed gridded precipitation data.	Identification of the storm catalog is a critical input for the SST method that drives meteorologic inputs for the project.	Guidance for development of the storm catalog is included in Job Aid 03, *Meteorologic Data Processes*.
Normalization Grid	Grid that captures the spatial variability of precipitation patterns across the transposition region.	The SST method described in this SOP accounts for spatial variability of precipitation through the Normalized Transposition technique which utilizes the Normalization Grid as the key input.	Job Aid 03, *Meteorologic Data Processes*, provides guidance for creation and delivery of the Normalization Grid.

Table 4.8. List of Reference Data and Reports Datasets and Descriptions

Dataset	Description	Purpose	Data Preparation Guidance
Vertical Datum Conversions	Register of any official vertical datum conversion values for infrastructure in the basin.	Reference of official data conversion values supports consistent application of adjustments across all FFRD project efforts.	Job Aid 02, *Data Preparation Processes*, provides guidance for creation and delivery of the register of vertical datum conversions.
Dam Structure and Operation Information	Dam structure information, capacity and operation data is assembled for reference whenever available, respecting all required information security requirements (e.g., USACE water control manuals [WCMs], Federal Energy Regulatory Commission [FERC] permits, or NRCS dam information [DamWatch]).	Dam structure and operation data is directly used in hydraulic and hydrologic modeling and is an important reference throughout the project.	Job Aid 02, Data Preparation Processes, provides guidance for acquisition and delivery of dam structure and operation information.
Other Locally Available Data	Additional local datasets or models that may be available within the project extent.	Certain locally available data may allow for improved resolution or accuracy of FFRD modeling.	Discussion of the search for some specific datasets is included in Job Aid 02, *Data Preparation Processes*. Available data is also sought in data provider meetings. The inclusion of additional local datasets as inputs or reference material for FFRD efforts is at the discretion of the FFRD team.

Table 4.9. List of FEMA Data Datasets and Descriptions

Dataset	Description	Purpose	Data Preparation Guidance
FEMA Effective FIS Reports	All existing effective FIS reports containing full documentation of the FIS model approach.	Background information to supplement FIS model data used to inform model development.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.
FEMA Effective FIS Models	Existing FIS study effective hydrologic and hydraulic models.	Used either as a starting point for the FFRD models or as a reference to previous approaches, depending on the age and quality of the effective models.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.
FEMA Effective Layers	Geospatial dataset of existing FEMA flood hazard including flood zones and other layers. Dataset will include regulatory layers and may also include supplementary flood risk database information for some communities.	Existing flood hazard data is a useful reference for the FFRD study and is used in determination of certain scoping items.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.
FEMA BLE Reports	Reports that provide detailed information on the methodology used for each BLE study.	Background information to supplement BLE model data used to inform model development.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.
FEMA BLE Models	BLE models.	Used either as a starting point for the FFRD models or as a reference to previous approaches, depending on the age and quality of the models.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.
FEMA BLE Data	Geospatial layers from FEMA BLE results that may include regulatory or non-regulatory information.	Existing flood hazard data is a useful reference for the FFRD study and is used in determination of certain scoping items.	Job Aid 02, *Data Preparation Processes*, provides guidance for delivery of FEMA data reference in FFRD efforts.

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4.6 REPORT

A report documenting the findings from the scoping and data preparation phase will be developed, reviewed, and included in the delivery.

4.7 PROJECT WORK PLAN—BASIN EXECUTION

The S-PWP is closed out at the end of the phase. In preparation for the next project phases, a draft BE-PWP will be developed that incorporates the scoping effort results and serves as the basis for all post-scoping activities related to the project. It includes the basin background, a data manifest of prepared datasets, model approach strategy, team member roles, a timeline, and budget. The BE-PWP will be finalized and signed before the start of the basin execution phases.

4.8 REVIEW OF SCOPING AND DATA PREPARATION PHASE PRODUCTS

The QC review for the scoping and data preparation products will be completed after all defined tasks are complete. The QC review will be conducted using the data preparation checklist that requires review input from multiple reviewers, led by the review manager. The template for the checklist includes specific items for review and closeout and is included within FFRD Templates and will be posted to the documentation directory for the project.

The TOG review of the scoping and data preparation phase work products is conducted after the QC review is closed.

The calibration phase shall not be initiated until the scoping tasks are complete, QC review is closed and documented, and TOG has approved the effort to move forward to the calibration phase.

4.9 DELIVERABLES

Table 4.10 contains a list of deliverables that must be delivered to complete the scoping and data preparation phase.

Table 4.10. Final Scoping and Data Preparation Phase Deliveries

Deliverable	Location to Post
Complete Data Preparation Dataset Posted to Model Library	basin-data
Data Manifest	basin-data
Scoping and Data Preparation Report *Signed by FFRD Leadership and TOG	documentation
Technical Review Checklist *All items closed out	documentation
Master Quality Review Checklist *All items closed out for the Phase	documentation
Basin Execution Project Work Plan (Draft)	documentation

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4.10 TRANSITION TO CALIBRATION PHASE

After the scoping and data preparation phase is fully closed out including TOG review and the BE-PWP is finalized and signed, the calibration phase can begin. It is important that all scoping and data preparation steps are complete and well understood by the team leaders. This ensures that information regarding the available data and scope can be clearly conveyed to the larger team. Team member assignments are made, the start date and timeline are finalized, and a kickoff meeting is held. This kickoff brings together the entire team and communicates the project scope, process, schedule, budget, collaboration forums, schedule tracking systems, data sources, and any other technical details to be considered as the larger team embarks on the FFRD basin execution effort.

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SECTION 5—METEOROLOGY

5.1 METEOROLOGIC CONTEXT AND STANDARDS

The FFRD SOP employs the SST technique to apply storm events on the study basin for quasi-continuous simulations. The method uses a catalog of observed storm events from a region of similar meteorology to the study watershed. Storm observations are taken from a high-resolution precipitation dataset with a long POR and many events per year, spread out through a region much larger than the watershed. The region contains a large sample of significant rainfall events that quantify the variability in the intensity, spatial, and temporal patterns of storms in the area. The uncertainty around the interaction of these storms with the study watershed is quantified by selecting storms and moving them around the region of homogeneous meteorology at random. Each time this process is repeated, it creates one plausible alternate realization of the observed record and can be used to drive a hydrology model to estimate flood response throughout the watershed.

5.1.1 PHASES OF METEOROLOGIC MODELING

The meteorologic data and modeling are produced in the context of the FFRD project in phases according to the workflows described in Section 2 of this SOP, and high-level meteorologic modeling tasks in each phase are described below:

- Scoping and Data Preparation Phase: The transposition region is developed, the regional storm catalog and normalization grid are created. The observed gridded data over the study basin is also acquired, adjusted as needed, and summarized.
- Calibration Phase: Observed meteorologic data from historic events are applied in the development and calibration of H&H models.
- Conformance Phase: Development of valid random storm placements (fishnets) is performed. SST performance for single realization is evaluated and adjustments are applied as needed.
- Production Phase: Application of SST to a multi-realization framework.
- Reporting Phase: Compilation and delivery of results from the project.

5.1.2 SOFTWARE AND VERSION

The FFRD meteorologic data development and application efforts are accomplished with the aid of actions available within the following cloud-compute plugins.

Table *5.1*. Software for FFRD Meteorologic Processes

Software	Version
storm-cloud	version 1.0.6
seed-generator-plugin	version 1.0.5
basin-record-gen-plugin	version 1.0.1
hms-mutator	version 1.0.4

5.1.3 DATA PROVIDED TO METEOROLOGIC TEAM

The key input for the meteorologic data and modeling effort is the study basin polygon, which is identified prior to project kickoff.

5.1.4 NAMING CONVENTIONS

Standard naming conventions for all project model files and data including meteorology data are provided in Job Aid 06, *Standard Naming Conventions*, and must be followed.

5.2 REGIONAL PRECIPITATION DATA

High-resolution precipitation data with a long POR are crucial for implementing SST. The Analysis of Record for Calibration (AORC) dataset is utilized as the primary source of gridded data for FFRD studies, which is available as both a 1-kilometer and a 4-kilometer resolution product with a 1-hour timestep and a POR beginning in 1979. The first and last year in the data POR should be checked for completeness, as they may be missing data at the beginning or end of the record. Storm data should only be taken from years of the precipitation POR that are complete (i.e., have data for all 365 or 366 days). The data should be taken from all complete years in the POR.

5.3 TRANSPOSITION REGION

The fundamental assumption SST makes is that a region of homogeneous meteorology containing the study watershed can be constructed. The domain is used to identify and select storm events to use in the catalog and to define the limits of where events can be moved during stochastic simulations.

Development of a transposition domain requires expert judgment and remains a subjective process until a repeatable, objective process is developed. The region must contain the study watershed in its entirety and be several times larger with a recommended lower limit of five times the watershed area. In particularly complex regions that have many mechanisms that generate heavy rainfall, such as tropical storms, orographic enhancement, rain shadows, atmospheric rivers, and other meteorological complexities, extra care must be taken in drawing the region.

5.4 STORM CATALOG

An SST storm catalog is unique for the study watershed. It is a function of the meteorology of the watershed, as well as the watershed's size and shape. For each study, a storm catalog will be developed. The storm catalog lists all storm events severe enough to be used in the study, some metadata about those storms, and precipitation data for each event. The following subsections describe the process for developing the storm data and metadata and the formats they should be stored in. Collectively, these items are referred to as the *storm catalog*.

A storm is selected for inclusion in the catalog based on a value called its *potential precipitation*, which measures how much total rainfall the storm would bring to the study watershed if its position were optimized by transposition.

5.4.1 STORM DURATION

The duration for storm analysis and simulation is 72 hours for FFRD studies. This duration is meant to capture the most intense portion of synoptic-scale storms and the entirety of mesoscale storms and smaller. It is the typical duration associated with extreme precipitation analyses such as the probable maximum precipitation (PMP). The duration affects the accumulation map used for measuring potential precipitation and the storm data included in the catalog. Shorter durations may not capture the full extent of widespread rainfall due to large-scale storms, and longer durations may include multiple events that are not independent of one another. Deviation from this duration, especially shortening it, is not recommended and should not occur without a compelling technical reason.

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5.4.2 STORM ARRIVAL RATE

All storms with a potential precipitation over a certain threshold are included. The threshold is chosen to achieve a specific catalog that averages 10 storms per year. For example, if 45 years of record are available, the threshold will be chosen such that the storm catalog includes 450 storms.

It is recognized that the selection of this threshold may influence overall modeling results for frequent events. Future updates to this SOP will consider procedures for selection of a threshold to minimize bias.

5.4.3 STORM RANKING: POTENTIAL PRECIPITATION

Storms are extracted from the POR precipitation dataset for the transposition region. Ranking is based on potential precipitation defined as the storm's area-averaged, 72-hour accumulation over an area the size and shape of the study watershed. Extraction of independent storm events requires significant computational effort and can be accomplished via geospatial analysis of the datasets.

Tools are available through the cloud-compute storm-cloud plugin to automate the process of generating catalogs that meet the above requirements, with additional details described in Job Aid 03, *Meteorologic Data Processes*.

5.4.4 STORM CATALOG DELIVERY SPECIFICATIONS

Each storm will be stored in its own HEC-DSS version 7 file with a filename and internal pathnames following the standard provided in Job Aid 06, *Standard Naming Conventions*. The catalog will be stored in the project directory specified within Job Aid 03, *Meteorologic Data Processes*.

5.5 TEMPERATURE DATA

In watersheds that require temperature data to accurately model events (e.g., where snow processes are being modeled), temperature data needs to be included in the storm .dss files as well. The temperature data should come from the same data source as the precipitation. Temperature fields only need to cover the watershed extents because they are not being transposed. They must overlap the duration of the storms and include any additional simulation period needed to capture attenuation and translation of the flood wave to the outlet of the watershed.

5.6 STORM TYPING

Storm typing is a novel addition to the traditional SST recipe that improves the hydrologic modeling of flood responses and allows for interfacing with the coastal hazards modeling process. The categories used for this approach are intentionally broad since the distinction is important for two uses: storm seasonality and isolation of tropical storms. The seasonality of storm events is used to determine initial watershed conditions during the stochastic simulations so that implausible combinations of meteorology and watershed state cannot be created. Isolation of tropical cyclones is important for handling additional coastal modeling that may be required to characterize compound hazards caused by the combination of rainfall and storm surge.

5.6.1 STORM CATEGORIES

After the catalog is developed, each storm will be classified into two broad storm types:

- Tropical Cyclones
- Non-Tropical

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Most rainfall events that occur on a sufficiently widespread scale for inclusion in a HUC-4 scale watershed catalog will fall into the non-tropical category. Each storm in the storm catalog will be manually classified following guidelines in Job Aid 03, *Meteorologic Data Processes*.

5.6.2 STORM SEASONALITY OUTPUT

For each type of storm, a seasonality analysis will be produced. The seasonality analysis will describe the daily frequency of occurrence for each storm type based on the start date of the events in the storm catalog. Each day of the year from 01 January through 31 December will have a storm occurrence count computed as the count of storms of that type which begin on that day of the year. The sum of the counts should equal the total number of storms of that type, and the sum of all tables should total the number of storms in the catalog.

5.7 NORMALIZED TRANSPOSITION

The use of the *normalized transposition* technique is recommended for FFRD studies. The use of *normalized transposition* accounts for spatial variability in precipitation patterns within the transposition region and study basin. This technique has been demonstrated to improve performance of the SST at multiple scales within a watershed.

HEC-HMS natively handles transposition normalization using a bias grid in the Gridded Precipitation method in the Meteorologic Model. The modeler needs to create a *Precipitation-Normal Grid* in their HEC-HMS project and then apply it in the *Bias Grid* option in their meteorologic model.

5.7.1 NORMALIZATION BASED ON NOAA ATLAS 14

The preferred source for a normalization field is NOAA Atlas 14. GIS Grids provided via the NOAA Atlas 14 Precipitation Frequency Data Server should be used where they are available. Use the 1/100 annual exceedance probability (AEP) grid for 3-day duration based on annual maximum series estimates. The grids must cover the entire transposition domain. If the transposition domain spans several volumes of NOAA Atlas 14, a mosaic of the grids will need to be created.

5.7.2 NORMALIZATION BASED ON PERIOD OF RECORD PRECIPITATION DATASET

When NOAA Atlas 14 is not available, a suitable alternative is to create a mean annual maximum 72-hour precipitation field from the precipitation data used to generate the storm catalog. To generate this field, compute the annual maximum 72-hour precipitation value for each year at each grid cell, creating an annual maximum series for each cell. Then, take the average of these values and generate the raster. To check that the result makes sense, compare it to the spatial patterns of rainfall present in the Parameter-elevation Regressions on Independent Slopes Model (PRISM) 30-year precipitation normal grid for the transposition domain. PRISM should not be used for normalization.

5.8 STOCHASTIC STORM TRANSPOSITION PLACEMENTS

Storms from the catalog are placed randomly for the quasi-continuous realizations. The valid storm placements ensure that the study basin is fully contained in the transposition region. For Version 1 of the SOP, all valid locations are equally likely.

The valid storm placements are created with the aid of the cloud-compute hms-mutator plugin and are termed *fishnets*.

Future versions of the SOP may incorporate improved techniques to account for precipitation spatial patterns within the Transposition Region, storm-type specific spatial patterns, and other techniques to improve computational efficiency.

5.9 SST UPPER LIMITS

The SST methods described in the SOP leverage a limited period of record of observed gridded datasets, and the substitution of space (from a larger transposition region) for record length allows extrapolation to less frequent events. However, there is an upper limit to this approach due to limited information content of the underlying transposition region and storm catalog. The theoretical upper limit of extrapolation for the SST method is easily calculated and must be reported as part of the FFRD study. The return period ceiling is approximated by the following formulas:

$$AEP_{\min} \approx \frac{\left({}^{A_B}/_{A_D}\right)}{Y} \approx \frac{p}{Y} \qquad T_{cap} = \frac{1}{AEP_{min}}$$

Where AEP_{min} is the theoretical smallest exceedance probability supported by the SST method, T_{cap} is the corresponding return period limit, p is the ratio of basin area (A_B) to transposition domain area (A_D), and Y is the record length in years.

5.10 METEOROLOGICAL MODEL REVIEW PROCESS AND DELIVERABLES

Meteorological data and model review is a critical part of the FFRD production process. The meteorology reviews will examine the transposition domain, the storm catalog, and the storm typing classifications.

Review checklists are used to track comments regarding pre-defined aspects checked throughout the review process. FFRD Templates and Resources contains *Quality Management Templates*.

The following subsections describe specific review steps for meteorologic products and associated deliveries.

5.10.1 SCOPING AND DATA PREPARATION PHASE REVIEWS

The following reviews are completed during the scoping and data preparation phase.

5.10.1.1 INTERIM DISCUSSIONS

The meteorological team members may seek out collaboration and advice on general direction from national technical experts regarding items such as selection of the transposition region, storm catalog development, and the normalization grid during development. These are considered informal discussions and are not required to be documented.

5.10.1.2 METEOROLOGY DATA REVIEW

The draft final meteorology deliveries from the scoping and data preparation phase will undergo a complete technical review. The required deliverables for the review and the scope of the review are described below.

The review will be documented using the Meteorology Review checklist (template included as part of the SOP package).

Deliverables for review:

- Transposition Region Polygon

- Storm Catalog
- Normalization Grid
- Completed Meteorological sections of the Scoping and Data Preparation Report

5.10.1.2.1 SCOPE OF TRANSPOSITION REGION REVIEW

The transposition domain review must determine if the delineated domain is reasonably homogeneous. The transposition domain must completely contain the study watershed and must be at least five times larger than the target watershed by area. The domain needs to have the same types of heavy rainfall mechanisms as the target watershed, even if the watershed spans a large area with a variety of storm and flood types. It will follow large natural breaks in climatic regime, such as prominent mountain ranges, that cause rain shadow effects or other weather blocking. Frequency of tropical storms should always be considered when drawing the domain. Data supporting the delineation should express the climatic regime inside the watershed and be able to show areas surrounding the watershed that are similar to it when measured by several climatic variables (e.g., elevation, annual and maximum precipitation, annual and seasonal temperature, humidity, and prevailing winds).

Inherently, the delineation of a transposition domain involves subjectivity and judgment. If the choices are well-documented and supported by available data, the transposition domain is defensible.

Automated techniques for domain generation are allowed, but if the delineation is only based on one aspect of extreme rainfall such as maximum rainfall statistics, the domain needs to be compared to other datasets for consistency. The review will consider the feasibility of the method used for automated delineation for the target watershed's climate.

5.10.1.2.2 SCOPE OF STORM CATALOG REVIEW

The catalog will be reviewed for completeness and compliance with the data standards outlined in the previous sections. The proper count of storms overall based on the length of the POR is the most important check. Each storm's center coordinates must be correctly specified in the HEC-HMS .grid file and all centers must be inside the transposition domain. All file naming and DSS path conventions must be observed.

5.10.1.2.3 SCOPE OF STORM TYPING REVIEW

The primary goal of the storm typing process is to confirm storm precipitation characteristics are paired with the correct watershed conditions to produce credible flood responses. The key linkage is in the seasonality of the storm categories. The storm typing review will check the classification procedures, data assumptions, and verify that values and thresholds used for partitioning make meteorological sense for the type of storm being classified. Determining if the resulting seasonality distributions for each storm type make sense for the storm type being evaluated is the final check.

5.10.1.3 METEOROLOGICAL REPORTING REVIEW

Reporting must address the following topics:

- Development of the transposition domain
- Precipitation and temperature data source
- Description of criteria for inclusion of storms in catalog
- Delineation of storm types

Documentation of the data sources and delineation choices for the transposition domain must support the assumption that the domain is meteorologically homogeneous. It must show how the data was used to reach this conclusion and address any uncertainty or sensitivity in the data or interpretation of it. If an

automated technique was used to develop the domain, an explanation of the assumptions and methods the technique uses must be included.

The precipitation and temperature data used for developing the storm catalog should be the same as used in the hydrologic model calibration and validation process. The report should detail the data source, its spatial and temporal resolution, the range of dates of the dataset used to create a POR, and document any completeness issues with the data.

Severity criteria determine whether a storm will be included in the catalog or not and will be documented in the meteorology report. This is a threshold value of potential precipitation and all storms with potential precipitation above this value will be included. The threshold is chosen to achieve the 10 storms per year average occurrence rate. The report will also include an analysis of the number of storms for each year in the POR and the distribution of storm counts per year.

A detailed explanation of the storm typing procedure should include, but not be limited to, documentation of datasets, any algorithms or flowcharts developed and applied, references to storm or flood reports, and validation of automated storm typing procedures. Manual storm typing results will show examples of storm classifications for each type, including surface analyses, precipitation maps, and other data used to make the judgment. The validation should include performance statistics for the classifier including a confusion matrix, precision, and recall statistics for each of the storm classifications.

5.10.2 CONFORMANCE PHASE REVIEWS

During the conformance phase of the FFRD project, the stochastic storms are evaluated against observed data from the POR with the expectation that they will reasonably align.

The comprehensive conformance phase review is documented via the conformance review checklist, provided within FFRD Templates and Resources, which includes several prompts for evaluation of specific SST performance aspects.

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SECTION 6—HYDROLOGIC MODELING

6.1 HYDROLOGIC MODELING CONTEXT AND STANDARDS

This section outlines the hydrologic modeling process and standards necessary to meet the overall FFRD objectives. The hydrologic modeling domain is the entire FFRD study watershed as a single model. The objectives of FFRD will be met via basin-level study, and it is acknowledged that other needs and use cases (e.g., local regulatory needs) may require additional detailed studies.

Hydrologic models produced following this guidance will have the following characteristics:

- Single model extent for entire study basin.
- Coordinated for seamless integration into system of models in terms of subbasin alignments, naming conventions, and other modeling strategies.
- Capable of accurately estimating spatial and temporal runoff volume for both discrete storm events and long-term continuous simulations within a reasonable amount of computation time.
- Capable of accurately estimating baseflow response for each subbasin for both discrete storm events and long-term continuous simulations within a reasonable amount of computation time.
- Capable of providing basis for an array of antecedent conditions based on historic POR data.
- Capable of providing basis for quantification of uncertainty in basin parameters.

6.1.1 PHASES OF HYDROLOGIC MODELING

The hydrologic model is developed for an FFRD project in phases according to the workflows described in Section 2. This hydrologic modeling section applies to multiple phases, and high-level modeling tasks in each phase are described below:

- Scoping and Data Preparation Phase: The hydrologic subbasins, reaches, junctions, reservoirs, and other model elements are built into a base model. No effort to parametrize or calibrate the model is made. This includes no effort to incorporate geometry or operational features at reservoirs. Standard naming conventions are applied. Subbasin delineation is leveraged for scoping and delineation of hydraulic modeling units. Meteorologic data is acquired and adjusted. Construction of the base model is performed concurrently with development of the Name Table, Model Linking Register, and Dam Scoping Table, which are all key deliverables for this phase.
- Calibration Phase: Calibration of the hydrologic model to historic observed events. Hydrologic model outputs for each historic event are provided to hydraulic models for use in calibration efforts. Calibration of a POR hydrologic simulation is also performed.
- Conformance Phase: Integration of the hydrologic model into the system of models, including delivery of a combination of initial states and valid parameter datasets to be used for randomized synthetic events. Computation of synthetic events in cloud compute, and adjustments to all models' performance to verify it conforms to defined quality metrics for a single long-term realization.
- Production Phase: Computation of multi-realization events. Hydrologic models may require hotfixes or updates to address specific issues.
- Reporting Phase: Compilation and delivery of results from the project.

6.1.2 SOFTWARE VERSION

The hydrologic model will be developed using HEC-HMS version 4.13.

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6.1.3 DATA PROVIDED TO HYDROLOGIC MODELER

A comprehensive shared dataset for the project is developed during the scoping and data preparation phase of the project. The development of an uncalibrated hydrologic base model occurs during the scoping and data preparation phase concurrently with development of several key scoping deliverables such as the hydraulic model unit delineation, Name Table, and Model Linking Register.

Key prepared datasets provided to the hydrologic modeler follow.

- Full Resolution and 100-foot DEM terrain data for consistent use for the entire project
- Key streams (see Section 4 for definition of this project scoping layer)
- Other streams (see Section 4 for definition of this project scoping layer)
- Elevation derived hydrography (stream centerlines aligned with DEM flow paths)
- POR gridded data (precipitation, temperature, and snow grids)
- Observed storm event gridded data (precipitation, temperature, and snow grids)
- Observed flow and water surface elevation data
- Stream gages for calibration including assigned categories for calibration criteria
- Selected historic storm events for calibration
- Dam and reservoir physical data
- Dam operations manuals when available
- Scope for modeling treatment of each dam in the basin
- All available bathymetry for rivers, lakes and reservoirs
- Land Cover/Land Use dataset
- Urban Imperviousness dataset
- Soils Datasets (SSURGO, gridded SSURGO [gSSURGO], and STATSGO)
- Existing hydrologic models
- Various other reference layers

6.1.4 CUSTOM REFERENCE DATA

This SOP includes directions to utilize a common set of shared basin data that is created during the scoping and data preparation phase, which are stored in the *ffrd_[basin-name]/basin-data/* directory. The hydrologic model will reference this official data for many aspects of the modeling process.

However, there may be datasets that will be developed during model development specifically to support an individual model. For calibration phase deliverables, these datasets typically should be saved within a reference directory inside the *model-unit* directory (*ffrd_[basin-name]/calibration/hydrology/*).

6.1.5 MODEL LINKING STRATEGY

A comprehensive Model Linking Register and Name Table will be developed during the scoping and data preparation phase concurrent with development of the uncalibrated hydrologic base model.

During the calibration phase of the project, the model is calibrated against observed historic events and for the POR. For model development during the calibration phase, the hydrologic model will apply the following boundary conditions:

- **Gridded Precipitation, Temperature, and SWE Data—**These are observed gridded timeseries datasets of precipitation, temperature, and SWE from observed events and for the POR.
- **Other Boundary Conditions—**When appropriate, other boundary conditions will be used in certain situations such as observed outflow timeseries at dams.

During the conformance phase of the project, the models are adapted to fully accept modeled output from synthetic storm events as boundary conditions rather than observed data which may require certain adjustments that will be accounted for in the Model Linking Register.

6.1.5.1 NAMING FOR MODEL LINKING ELEMENTS

All hydrologic model files and model elements will be named according to the comprehensive Name Table and Model Linking Register, which are developed during the scoping and data preparation phase.

6.1.6 COMPREHENSIVE NAMING CONVENTIONS

Model naming conventions for all model elements are an important aspect of FFRD, as standard naming and descriptions may be also used to support metadata that can be queried through the Model Library. Consolidated requirements for naming conventions and project descriptions for HEC-HMS and other models are defined in Job Aid 06, *Standard Naming Conventions*.

6.1.7 COORDINATION WITH OTHER MODELS

Throughout model development and calibration, the hydrologic modeler coordinates with the technical lead and entire modeling team to exchange input and output datasets as appropriate. Some typical points of this coordination are described below.

6.1.7.1 HYDRAULIC MODEL

Early iterations of hydrologic model results may be insufficient to achieve calibration targets when applied to the hydraulic model. The hydraulic and hydrologic modelers will coordinate to determine the most appropriate model adjustments to achieve calibration targets for both models. Also note that hydrologic subbasin delineation will be aligned with the hydraulic model unit delineation during the scoping and data preparation phase, such that each subbasin is fully contained within a single hydraulic model unit.

6.1.7.2 RESERVOIR OPERATIONS MODEL

The reservoir operations modeler and hydrologic modeler must verify the reservoir storage datasets being applied in each model are consistent. Although the Dam Scoping Table will guide the modeling approach for each dam, coordination on certain details of outflow modeling approach between the models may be needed.

6.2 MODEL DEVELOPMENT

The following sections describe the process and standards for hydrologic model development for FFRD projects.

6.2.1 MODEL SETUP

The hydrologic model elements are defined during the scoping and data preparation phase and provide key scoping information for the study. Guidelines for hydrologic modeling activities during this phase are described in the sections below.

6.2.1.1 STUDY AREA DELINEATION

Initial development of the hydrologic model occurs during the scoping and data preparation phase. Leveraging the high-resolution DEM for the project, the hydrologic modeler will provide critical input for the scoping team regarding the accuracy of the study extent polygon used for early project planning. Updates to the study area may be considered by the scoping team when appropriate.

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6.2.1.2 BASIN ELEMENT DELINEATION

The hydrologic modeler identifies the outlet of the basin. The modeler may choose to add additional break points within the watershed at this point, but doing so is not necessary to continue the delineation process. Once the modeler defines the outlet, the modeler delineates the subbasin and reach elements within the basin using the provided 100-foot DEM.

Delineation and placement of hydrologic model basin elements (e.g., subbasins, reaches, reservoirs) is an iterative process. After initial delineation, the modeler should check if the delineation process is working properly and that no major errors exist with the terrain data. This subbasin delineation must be examined relative to the high-resolution DEM to confirm agreement. Significant variation in subbasin and/or reach delineation from published data may indicate a need to refine the terrain data further.

The USGS WBD HUC subbasin boundaries and National Hydrography Dataset (NHD) stream alignments provide a baseline against which the HEC-HMS delineations are compared. The EDH produced during the scoping and data preparation phase also provides a high-quality comparison for auto-generated hydrologic model reaches. These data sources are described in greater detail within Job Aid 02, *Data Preparation Processes*.

The hydrologic model subbasins must align with the hydraulic modeling units with no significant disagreements. This can be accomplished by burning streams and/or building walls. The EDH, WBD, and NHD layers should all be used within this process. Refer to HEC-HMS Tutorials and Guides for more information.

After subbasins are delineated, they are used by the scoping team as a key input into the hydraulic model unit delineation described in Job Aid 01, *Scoping Processes*. Iterations on both the subbasin and hydraulic model unit delineation may be necessary as a collaboration between hydrologic modeling, hydraulic modeling, and GIS team members.

6.2.1.3 REFINING DELINEATION

The initial modeling unit and stream delineation are reviewed and modified to verify alignment with the overall modeling plan. Subbasins and routing reaches can be merged or split to obtain the desired configuration using the GIS tools in HEC-HMS, which allow actions such as building walls and burning streams to enforce desired alignments. Things to consider when refining the initial delineations are the locations of stream and/or reservoir gages, locations that are critical to reservoir operations, HEC-RAS basin boundary condition locations, calibration targets, and desired output locations. Also, basin element delineations should be made at major stream confluences, hydraulic structures, abrupt changes in channel or overland slope, soil, land use, or physical characteristics affecting runoff generation. Minimum or maximum modeling unit sizes are dependent upon the modeling domain. Coordination of modeling unit delineations is made with the hydraulic, reservoir simulation, and consequence models to ensure the hydrologic model aligns with project purposes.

The FFRD HEC-HMS model must not be overly complex to run efficiently within the FFRD cloud compute framework with numerous stochastically sampled scenarios.

6.2.1.4 ELEMENT NAMING

All element names included in the hydrologic base model will follow the FFRD naming conventions established in Job Aid 06, *Standard Naming Conventions*. These names will be utilized in the development of the Model Linking Register and Name Table during the scoping and data preparation phase.

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6.2.1.5 TEST SIMULATION

Once the modeling unit delineation is approved by the team and elements are named appropriately, a meteorologic model and corresponding control specification are developed to produce a fully functional HEC-HMS project. The model is not calibrated, but incorporates all major data linkages, contains reasonable initial parameter estimates, and runs without errors and significant warnings.

This uncalibrated base model is a core delivery for the scoping and data preparation phase of the FFRD project.

6.2.2 COMPUTING WATERSHED CHARACTERISTICS

Watershed characteristics are applied to the model during the calibration phase of the project. If adapting an existing HEC-HMS model for an FFRD study, watershed characteristics may already be incorporated into the model. However, HEC-HMS modelers must review (and, if necessary, recreate) watershed characteristics to verify they are appropriate for the watershed.

If a new HEC-HMS model is developed, once the stream and modeling unit delineation are finalized, the modeler computes physical characteristics of the watershed using automated tools within HEC-HMS. Physical characteristics for a stream include the length, slope, relief, and sinuosity. Similarly, physical characteristics for subbasins include the longest flowpath length and slope, centroidal flowpath length and slope, and overall basin slope and relief, among others. These physical characteristics are then used to estimate the initial model parameters along with commonly available GIS data.

6.2.2.1 FFRD STANDARD HYDROLOGIC MODELING METHODS

Unless the watershed is unique enough to justify otherwise, the following HEC-HMS modeling methods are recommended for use in FFRD HEC-HMS models:

- Discretization: Structured
- Canopy: Simple
- Transform: ModClark
- Loss: Deficit and Constant
- Baseflow: Linear Reservoir
- Routing: Muskingum-Cunge, Muskingum, and/or Modified Puls
- Reservoirs: Specified Release, defined Elevation-Storage-Discharge relationships, and/or Rule-Based Operations
- Snowmelt: Gridded Temperature Index
- Basin Model Local flow option: '*yes*'
- Precipitation: Gridded Precipitation
- Temperature: Gridded Temperature
- Evapotranspiration: Gridded Hamon

In general, initial model parameters are estimated from GIS datasets, regional regression equations, and existing models and then adjusted to achieve calibration targets.

6.2.2.2 ESTIMATING MODEL PARAMETERS

If adapting an existing HEC-HMS model, watershed parameters are already included within the model. However, as part of the FFRD effort, the model parameters must be verified and tested using FFRD event-based calibration data.

If a new HEC-HMS model is developed, physical characteristics of the watershed elements are used along with additional GIS datasets (soil data, land use and land cover [LULC] data) to estimate the initial

model parameters. Modelers compute hydrologic parameters, such as the percent impervious area and time of concentration, using the Expression Calculator in HEC-HMS or within the GIS software. More information is available in the HEC-HMS tutorials and guides.

6.2.2.2.1 DISCRETIZATION METHOD

The Structured Discretization method is recommended for FFRD HEC-HMS models. This method allows for the distributed application of gridded boundary conditions as well as pertinent modeling methods (e.g., losses and transformation).

The chosen Coordinate Reference System (CRS) utilized for Structured Discretization method will utilize the Standard Hydrologic Grid (SHG), which is consistent with the standard FFRD projection defined in Section 3.2. The chosen horizontal resolution of the discretization must be equal to or greater than the smallest resolution of the meteorologic boundary conditions (e.g., 2,000 meters by 2,000 meters). Modelers should refer to HEC-HMS online tutorials when applying the Structured Discretization method.

6.2.2.2.2 TRANSFORM METHODS AND PARAMETERS

The ModClark transform method is recommended for FFRD HEC-HMS models. This method is mature, well established, well documented, and simple to set up and use. It has also been used successfully within thousands of studies throughout the United States. Also, parameters can be regionalized, related to measurable basin characteristics, and varied with excess-precipitation rates, which are important traits for methods used in extreme event simulations. Modelers should refer to Engineer Manual (EM) 1110-2-1417 and HEC-HMS online tutorials when estimating initial transform parameters.

6.2.2.2.3 LOSS METHODS AND PARAMETERS

The Simple Canopy method is recommended for FFRD HEC-HMS models. This method is mature, utilized throughout the United States, simple to set up and use, and scalable in complexity. Within FFRD implementations, losses should be assumed to occur solely within the chosen infiltration routine (e.g., Deficit and Constant). However, to accurately simulate infiltration capacity recovery during prolonged periods of little to no precipitation, a Canopy method is required within HEC-HMS. For this reason, the Initial and Maximum Storage parameters that are required as part of the Simple Canopy method should be set to a small value (e.g., 0.01 inches) and remain unmodified during model calibration. However, the Crop Coefficient parameter, which controls how potential evapotranspiration is converted to actual, may be modified during model calibration to accurately estimate moisture deficits following prolonged periods of little to no precipitation.

The Deficit and Constant loss method is recommended for FFRD HEC-HMS models. This method is mature, utilized throughout the United States, simple to set up and use, and scalable in complexity. Using this method, parameters can be related to predominant soil textures and estimated using multiple literature sources, and the method avoids complexity compared to other methods. Loss parameters are often estimated for each subbasin utilizing readily available GIS data. For example, constant loss rates may be estimated by spatially averaging hydraulic conductivity values from soils databases (e.g., gSSURGO). Directly connected impervious areas may be estimated using NLCD impervious datasets. Care must be taken to estimate directly connected impervious areas in lieu of total impervious areas. Additionally, large bodies of water must be accounted for within the directly connected impervious area estimates.

Modelers should refer to EM 1110-2-1417 and HEC-HMS online tutorials for additional background and examples when estimating loss parameters.

6.2.2.2.4 BASEFLOW METHODS AND PARAMETERS

The linear reservoir baseflow method is recommended for FFRD HEC-HMS models. Using this method, mass is automatically conserved, baseflow generation is linked to infiltration, inflow can be partitioned to multiple layers or lost to deep aquifer storage, and parameters can be regionalized and estimated using watershed physical characteristics. Also, this method is scalable because it can be used for a single event or continuous simulations with little to no differences in parameterization. Modelers should refer to HEC-HMS online tutorials when estimating initial baseflow parameters.

Within FFRD HEC-HMS models, the linear reservoir baseflow method is typically defined by using two groundwater (GW) layers. However, a third layer may be required depending upon physical considerations of the modeling domain in question. The GW 1 layer must be conceptualized as conveying interflow or quick-responding subsurface water while GW 2 layer must be conceptualized as conveying slow-responding baseflow. As such, GW 1 parameters must never be used that result in GW 1 outflow preceding direct runoff provided by the transform method. Similarly, GW 2 parameters must never be used that result in GW 2 outflow preceding outflow from GW 1.

Initial discharge within each GW layer may be initially estimated using any value but should be adjusted based on observable baseflow at the start of each event during calibration. Initially, infiltrated water may be evenly split between the GW layers (i.e., 50 percent of infiltrated water sent to GW 1 and GW 2). The GW coefficients may also be initially estimated using the rule of thumb that the GW 1 coefficient is typically three times the transform method's storage coefficient while the GW 2 coefficient is typically ten times the transform method's storage coefficient. All linear reservoir parameters must be adjusted as needed throughout calibration.

6.2.2.2.5 REACH ROUTING METHODS AND PARAMETERS

The Muskingum-Cunge routing method is recommended for FFRD HEC-HMS models. This method is mature, allows parameters to be efficiently estimated from measurable channel characteristics, and provides accurate results for a wide variety of physical settings. The HEC-HMS modeler estimates initial parameters from GIS datasets, regional regression equations, and/or existing models, if available. If reliable storage-discharge relationships are available throughout the modeling domain, the Modified Puls routing method can also be utilized for FFRD HEC-HMS models. This method provides accurate results for a wide variety of physical settings including very flat slopes and areas influenced by complex hydraulic interactions. Reliable storage-discharge relationships can be derived from a calibrated HEC-RAS project. When input data is not readily available for use within the Muskingum-Cunge and/or Modified Puls methods, the Muskingum method may also be utilized for FFRD HEC-HMS models. Modelers should refer to EM 1110-2-1417 and HEC-HMS online tutorials for further guidance when estimating initial reach routing parameters.

Routing methods and parameters in the HEC-HMS model need to match those in the reservoir operations model; if they do not, there can be discrepancies in computed hydrographs from the two models. HEC-HMS calibrated routing reach parameters are generally adopted directly into the reservoir operations model since the HEC-HMS model focuses more on hydrologic calibration and routing while the reservoir operations model focuses more on calibrating reservoir rules and operations.

6.2.2.2.6 RESERVOIR MODELING METHODS

A comprehensive strategy for dam and reservoir modeling within the full system of models for the FFRD project is developed during the scoping and data preparation phase and documented in the Dam Scoping Table (see Table 4.2). This table may be developed concurrently with the development of the uncalibrated base model and will clearly indicate which dams will be included as reservoirs in HEC-HMS and the selected methods for modeling of releases during both the calibration phase and conformance/production phases.

For historic event simulations, the specified release method is recommended for use when observed outflow data is available. When observed data is not available for use, the Outflow Curve and/or Rule-Based Operations methods may be used instead. Regardless of the chosen modeling method, reservoir elements must include elevation-storage information and initial conditions so the observed pool elevation can be monitored during calibration (since it is generally more reliable than estimated reservoir inflow data). Nearby USGS stream gages may serve as additional calibration resources. The elevation-storage information modeled within the hydrologic model must match those elevation-storage relationships modeled within the reservoir operations and hydraulic models.

For stochastic event simulations, the specified release method is not suitable for use due to a lack of observed data. Therefore, the Outflow Curve and/or Rule-Based Operations methods must be used. Modelers should refer to HEC-HMS online tutorials when estimating initial reservoir parameters.

6.2.2.2.7 SNOWMELT METHODS AND PARAMETERS

If the watershed is hydrologically impacted by snowmelt, the Gridded Temperature Index method is recommended for FFRD HEC-HMS models. This method is mature, simple to set up and use, and avoids complexity compared to other methods. Modelers should refer to HEC-HMS online tutorials when estimating initial snowmelt parameters.

The Snow Data Assimilation System (SNODAS) or University of Arizona SWE datasets are recommended for use within watersheds requiring snowmelt simulations. Selection of a snow dataset may vary by its accuracy within the watershed and POR.

6.2.2.2.8 ADDING OBSERVED FLOW

During the calibration phase, observed flow data from historic events is added to the model through the Time-series Data Manager. The modeler creates a new timeseries data entry for each gage location and populates the name and description with the corresponding information for the gage. A timeseries gage entry for each gage location is also created. Gage names are chosen so they are easily identifiable, and the actual gage name and USGS ID (if applicable) are included in the description field. Observed data for each gage is entered in the Component Editor and is associated with the corresponding basin model element. Once the observed data is associated with the proper basin element, the element is set as a computation point so the element can be easily identified and used to perform more efficient computations during calibration runs. Computation points are used to speed up calibration efforts as HEC-HMS computes only the modified elements upstream of the computation point.

6.2.2.2.9 CREATING METEOROLOGIC MODELS OF HISTORIC EVENTS

The modeler creates meteorologic models for historic events using precipitation and temperature data, which is created as part of the scoping and data preparation phase. This meteorologic data is used later in the calibration process but also allows for creation of uncalibrated base simulation runs to verify the HEC-HMS project runs and is stable. Gridded meteorologic data are required for use within FFRD implementations. In general, the AORC gridded precipitation and temperature product is recommended, as it is available for CONUS from 1979 to the present.

The AORC data is available at an hourly timestep for a relatively long POR. However, the accuracy of AORC precipitation volumes may vary regionally. During the scoping and data preparation phase the accuracy of the AORC data will be evaluated and may be normalized using the PRISM daily precipitation data (as described in Job Aid 05 – Observed Gridded Data Preparation), which generally contains more accurate precipitation volumes within a daily timestep. More information and examples on normalizing gridded datasets can be found within the HEC-HMS online tutorials and guides website.

Information on additional readily available gridded data sources is found online at the HEC-HMS tutorials and guides website.

FFRD project workflows for acquisition of gridded data, adjustments to the data, summarization of the data, naming conventions, and storage locations during the scoping and data preparation phase are described in Job Aid 05, *Observed Gridded Data Preparation*.

Gridded data is added by creating a new precipitation gridset in the Grid Data Component Manager. Once created, the appropriate filename and path for the gridded data is entered in the Component Editor. Once a precipitation gridset is created and added to the HEC-HMS project, a meteorologic model for that gridset is created using the Meteorologic Model Manager. The modeler selects the appropriate precipitation gridset and enters a description for the meteorologic model in the Component Editor. All .dss files with observed data should be saved within the HEC-HMS project folder. When the .dss files are in the HEC-HMS project folder, HEC-HMS uses a relative pathname for the .dss file.

The Gridded Hamon potential evapotranspiration method is recommended for FFRD HEC-HMS models. This method is mature and provides reliable results for a wide variety of physical settings.

6.2.2.2.10 CREATING SIMULATIONS OF HISTORIC EVENTS

Once the basin model and meteorologic model are created, a control specification is created to perform a model simulation. The control specification file identifies the dates and times (in UTC) for which the simulation is run and should match the precipitation event of interest in the meteorologic model. If there is more than one rainfall event of interest in the meteorologic data, a control specification is created for each event.

The simulation timestep is also defined in the control specification. For most models, a timestep of one hour is the preferred standard. Ideally, the timestep is based on modeling unit hydrograph and routing reach parameters, precipitation inputs, and reservoir response. The simulation timestep determines the number of ordinates that define the runoff hydrograph. A general rule of thumb is at least four points (ordinates) on the rising limb of the hydrograph are required to adequately define the shape. For example, a subbasin with a concentration time of four hours requires a simulation timestep of approximately one hour.

Once the basin model, meteorologic model, and control specifications are set up, a simulation run is created for each historic calibration event. Refer to Job Aid 06, *Standard Naming Conventions*, for guidance on naming of the simulation.

6.2.2.2.11 RUNNING THE UNCALIBRATED MODEL

Once the simulation runs are created, the HEC-HMS modeler computes each simulation and reviews the message logs to determine if any major warnings need to be addressed. If no major warnings are found, output from the model at the basin outlet and locations of interest (gage locations, stream junctions, and local inflows) are examined to determine if the model is functioning properly. The modeler verifies the observed flows are plotting correctly in the results for all gage locations with observed data.

6.3 EVENT-BASED MODEL CALIBRATION

The objective of the event-based model calibration is to compute incremental excess precipitation in gridded format and baseflow timeseries for the HEC-RAS model calibration and provide a range of calibration parameters for use in the POR simulation. Once the HEC-HMS model is developed and running, the basin model is further refined to obtain a calibrated model. The modeler calibrates the model parameters for multiple elements upstream of each gage by adjusting them in parallel until the simulated results match observed gage records. The goal is to adjust model parameters such that model output matches the peak, timing, and/or runoff volume of historic events. Section 4.3.8 discusses identification of calibration events.

6.3.1 CALIBRATION OF MODEL USING HISTORIC EVENTS

Observed streamflow and precipitation data are necessary inputs when calibrating a hydrologic model. At least one basin model and simulation event were created during the previous stage of model development, but other events are added to the HEC-HMS project to aid in calibration. If calibrating for numerous events, multiple basin models are created to track parameters during calibration. It is recommended that the model be calibrated to historic events with both wet and dry antecedent conditions. This allows the modeler to determine reasonable bounds for many model parameters.

FFRD HEC-HMS models are calibrated for six historic events that are selected during the scoping and data preparation phase. The primary focus of the hydrologic model development and calibration effort during the calibration phase of the project is achieving the calibration criteria for these events. If the watershed is hydrologically impacted by snowmelt, at least two snowmelt-influenced events are calibrated in addition to at least four rainfall events.

Care should be taken to use physically justifiable parameters given the hydrologic characteristics of the modeling domain. Also, parameters must be relatively consistent across subbasins for hydrologically similar regions. If the need for significant parameter inconsistencies or unrealistic parameters is identified, they must be explained and justified in accompanying reports.

Model performance during extreme event simulations must be given priority during this phase. Parameter modifications for flood events and/or locations where surface runoff dominates (i.e., high magnitude floods) should be the primary focus. Conversely, parameter modifications for events and/or locations where small to moderate flooding is being simulated should be performed afterwards. Over-fitting should also be avoided. Instead, inter-event and regionally consistent model parameterizations should be identified and preferred.

Table 6.1 highlights parameters to investigate and vary during model calibration.

Table 6.1. Calibration Parameters and Approach

Process	Parameter	Calibration Approach
Canopy	Initial Storage	In general, significant canopy interception is not included within FFRD implementations. Instead, this parameter is not varied during model calibration except when necessary to correct for a lack of standing water or large changes in impervious area throughout time. Physical justification must be provided if using varying values for multiple historic events.
Infiltration	Initial Deficit	This parameter is varied to match the observed initiation of runoff.
Infiltration	Maximum Deficit	This parameter varies to match the initiation of runoff after prolonged periods with little to no precipitation during continuous simulations. Physical justification must be provided if using varying values for multiple historic events.
Infiltration	Constant Loss Rate	This parameter is varied to match the observed runoff peak and volume.
Infiltration	Percent Impervious Cover	In general, this parameter is not varied during model calibration except when necessary to correct for a lack of standing water or large changes in impervious area throughout time. Physical justification must be provided if using varying values for multiple historic events.

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Process	Parameter	Calibration Approach
Runoff Transform	Time of Concentration (Tc)	This parameter varies to match the observed hydrograph shape, primarily on the rising limb. This parameter is also varied to match the observed time of peak flow.
Runoff Transform	Storage Coefficient (R)	This parameter varies to match the observed hydrograph shape, primarily on the receding limb. This parameter is also varied to match the observed peak flow magnitude.
Baseflow	GW 1 Initial Discharge	This parameter varies to match the observed discharge at the beginning of the simulation.
Baseflow	GW 1 Fraction	This parameter is varied to match the observed runoff volume.
Baseflow	GW 1 Coefficient	GW 1 is conceptualized to represent interflow (i.e., fast-responding portion of baseflow). Therefore, this coefficient must be set to a smaller value than the GW 2 coefficient. This parameter varies to match the observed hydrograph shape.
Baseflow	GW 1 Number of Steps	This parameter varies to match the observed hydrograph shape.
Baseflow	GW 2 Initial Discharge	This parameter varies to match the observed discharge at the beginning of the simulation.
Baseflow	GW 2 Fraction	This parameter is varied to match the observed runoff volume.
Baseflow	GW 2 Coefficient	GW 2 is conceptualized to represent the slow-responding portion of baseflow. Therefore, this coefficient must be set to a larger value than the GW 1 storage coefficient. This value varies to match the observed hydrograph shape.
Baseflow	GW 2 Number of Steps	This parameter varies to match the observed hydrograph shape.
Streamflow Routing	Length	This parameter should be estimated using physically measurable characteristics and should not be varied during model calibration within Muskingum-Cunge routing reaches. Physical justification must be provided if using varying values for multiple historic events.
Streamflow Routing	Slope	This parameter should be estimated using physically measurable characteristics and should not be varied during model calibration within Muskingum-Cunge routing reaches. Physical justification must be provided if using varying values for multiple historic events.
Streamflow Routing	Manning's "n" values	This parameter is varied to match the observed hydrograph translation and attenuation within Muskingum-Cunge routing reaches.
Streamflow Routing	Storage-Outflow Relationship	This parameter is varied to match the observed hydrograph translation and attenuation within Modified Puls routing reaches.

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Process	Parameter	Calibration Approach
Streamflow Routing	Cross Section Shape	This parameter should be estimated using physically measurable characteristics and should not be varied during model calibration within Muskingum-Cunge routing reaches. Physical justification must be provided if using varying values for multiple historic events.
Streamflow Routing	Number of Steps	This parameter is varied to match the observed hydrograph translation and attenuation within Modified Puls routing reaches.
Streamflow Routing	K	This parameter is varied to match the observed hydrograph translation within Muskingum routing reaches.
Streamflow Routing	X	This parameter is varied to match the observed hydrograph attenuation within Muskingum routing reaches.
Streamflow Routing	Number of Subreaches	This parameter is varied to match the observed hydrograph attenuation within Muskingum routing reaches.
Snowmelt	PX Temperature	This parameter is varied to match the observed SWE accumulation.
Snowmelt	Base Temperature	This parameter is varied to match the observed SWE depletion during snowmelt.
Snowmelt	Wet Melt rate	This parameter is varied to match the observed SWE depletion during rainfall.
Snowmelt	Rain Rate Limit	This parameter is not varied during model calibration.
Snowmelt	Dry Melt rate	This parameter is varied to match the observed rate of SWE depletion during times when rainfall rates were less than the Rain Rate Limit.
Snowmelt	ATI-Cold rate Function	This function varies to match the observed timing of the start of SWE melt.
Snowmelt	Snowpack Water Capacity	This value varies to match the observed runoff generation from the melting snowpack.
Snowmelt	Ground melt	This value is set to zero and is not varied during model calibration.

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6.3.2 CALIBRATION PERFORMANCE RATINGS

Model performance statistics are used to quantitatively assess the HEC-HMS calibration performance throughout the watershed. Table 6.2 provides the industry-standard performance ratings (Moriasi et al., 2007). Tools provided in FFRD Templates and Resources may be applied to aid in the calculation of these calibration statistics.

Calibration performance rating goals are set by the primary, secondary, and tertiary gage categories:

- Primary Locations: Statistical comparisons of computed and observed flow/stage at all primary locations are in the satisfactory or higher range.
- Secondary Locations: Statistical comparisons of computed and observed flow/stage at all secondary locations are in the satisfactory or higher range but are considered more leniently than for primary locations.
- Tertiary Locations: Statistical comparisons of computed and observed flow/stage at all tertiary locations are in the satisfactory or higher range but considered more leniently than for secondary locations.

Table 6.2. Performance Ratings for Evaluation Metrics

Performance Rating	Color Code	R^2	NSE	RSR	PBIAS
Very Good	Dark Green	$0.65<R^2$	$0.65<NSE\leq1.00$	$0.00<RSR\leq0.60$	$PBIAS\leq\pm15$
Good	Light Green	$0.55<R^2\leq0.65$	$0.55<NSE\leq0.65$	$0.60<RSR\leq0.70$	$\pm15<PBIAS\leq\pm20$
Satisfactory	Orange	$0.40<R^2\leq0.55$	$0.40<NSE\leq0.55$	$0.70<RSR\leq0.80$	$\pm20<PBIAS\leq\pm30$
Unsatisfactory	Red	$R^2\leq0.40$	$NSE\leq0.40$	$RSR>0.80$	$PBIAS>\pm30$

Based on (Moriasi, et al., 2007).

6.3.3 PEAK FLOW PERFORMANCE

Modeled peak flow for each calibration event must be compared to observed peak flow for each location. Adjustments to the parameters will be made to match peak flows within acceptable tolerances with consideration of possible uncertainties in observed values.

6.3.4 CALIBRATION MODEL OUTPUT

Once a calibration run is complete, the HEC-HMS modeler provides calibrated flow information to the HEC-RAS modeler for calibration of the hydraulic model. Information includes:

- Baseflow timeseries from each modeling unit
- Incremental excess precipitation grids covering the entire modeling domain.

As the HEC-RAS modeler utilizes the output from HEC-HMS within the HEC-RAS model, they examine results with the HEC-HMS modeler to identify any discrepancies significant enough to warrant another calibration run through the HEC-HMS model. This process is repeated until the HEC-RAS and HEC-HMS modelers are both confident the models are calibrated for the historic event being simulated.

6.4 CONTINUOUS SIMULATION

A continuous simulation is also developed during the calibration phase of the project. The objective of the continuous simulation is to obtain a single set of parameters within the range of event-based calibration

parameters that perform well for longer durations. It is the first step in development of the POR simulation which is leveraged during the conformance and production phases of the FFRD modeling process.

Once the HEC-HMS model is calibrated to historic events, a continuous simulation calibration of the model is run, testing the model accuracy and performance over longer periods of time. The continuous simulation calibration event includes at least two water years and at least one high flow event. If the watershed hydrology is influenced by snowmelt, the continuous simulation period also includes at least one year where the SWE within the basin was high and contributed significantly to the annual runoff.

6.4.1 CONTINUOUS SIMULATION PARAMETERS

The range of event-based calibration parameters is considered when selecting an initial parameter set for the continuous simulation calibration. The HEC-HMS modeler uses engineering judgment to determine which initial parameters both best represent the watershed over long periods of time and reasonably recreate the available observed data with a particular emphasis on accurately capturing annual maximum peak flows.

6.4.2 CONTINUOUS SIMULATION CALIBRATION

The HEC-HMS modeler compares output from the continuous simulation run to observed streamflow data. If the simulation does not adequately match the observed data, volume, and annual maximum peaks, further adjustment to the continuous simulation parameters is needed using the same procedure described for the calibration process in Section 6.3. This may be an iterative process until reasonable calibration targets are met for the continuous simulation event. Final continuous simulation calibration parameters should fall within the range of event-based calibration parameters.

6.5 PERIOD OF RECORD SIMULATION

A simulation of the full POR of available gridded data is also developed during the calibration phase of the project. The key purpose of the POR simulation is to provide a distribution of reasonable starting states for every day of the year across the entire historic POR to support Knowledge Uncertainty estimation, which is described in greater detail in Sections 11 and 12.

Once the HEC-HMS model is successfully calibrated with a reasonably long continuous simulation (recommended minimum of two water years), the model is tested over the full POR. The simulation is run at an hourly timestep. The length of the POR simulation may vary due to the boundary condition (precipitation, temperature) and observed data (streamflow, reservoir) available within the watershed.

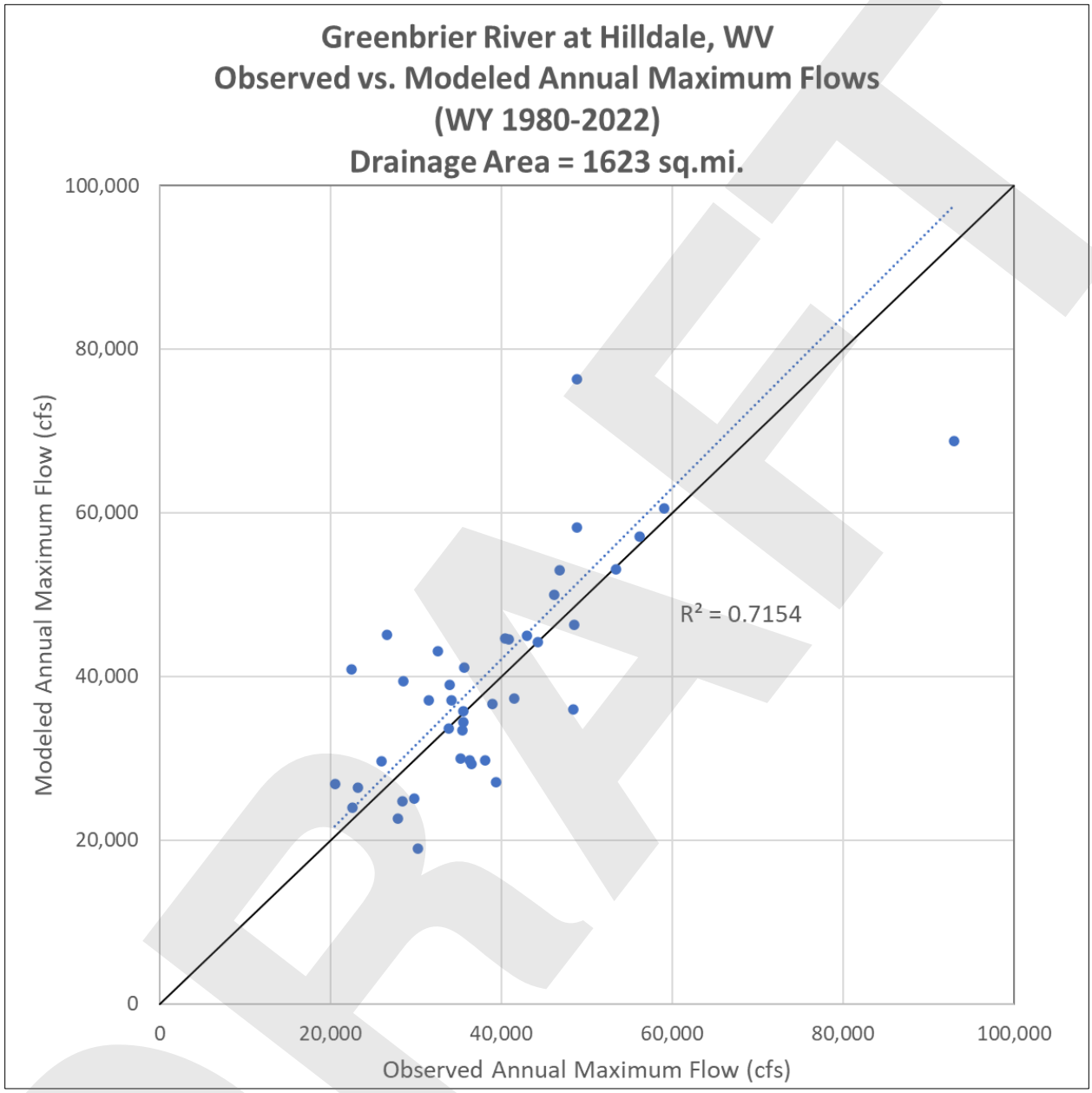
6.5.1 PERIOD OF RECORD SIMULATION PARAMETERS

Initial POR simulation parameters are adopted from the successful continuous simulation calibration and adjusted as necessary to reasonably reproduce annual maximum flows throughout the watershed.

6.5.2 PERIOD OF RECORD CALIBRATION

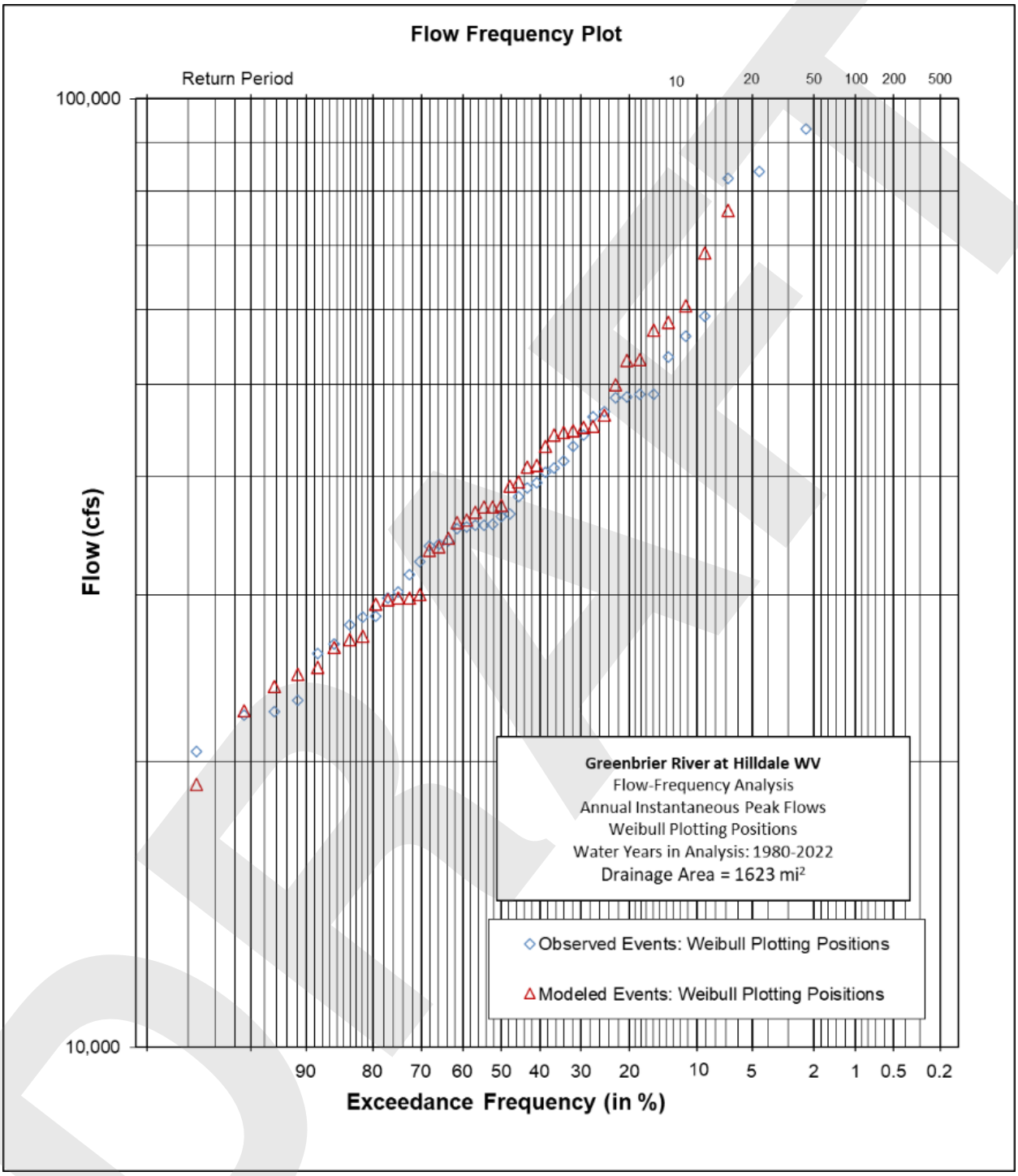
The HEC-HMS modeler extracts and compares modeled to observed annual maximum peak flows at primary gage locations throughout each full water year of the POR simulation. If the simulation does not adequately match the observed annual maximum peaks, further adjustment to the POR simulation parameters may be needed. This may be an iterative process until the FFRD team feels the annual maximum peaks are reasonably replicated within the POR simulation. Final POR simulation parameters should still fall within the range of event-based calibration parameters. Figures 6.1 and 6.2 provide examples of comparison between modeled and observed annual maximum peak flows.

Figure 6.1. Direct Comparison Plots between Modeled and Observed Data



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Figure 6.2. Using Plotting Positions to Plot Empirical Flow Frequency Curves for Both Modeled and Observed Peaks



Ungaged subbasin units should use hydrologic parameters developed using guidelines from literature, data from other hydrologically similar modeling units within the basin, and professional engineering judgment. Probabilistic results from the conformance phase can be compared with published regional regression estimates where available.

6.5.3 OBSERVED LIQUID WATER AT SOIL SURFACE ANNUAL MAXIMUM SERIES

After review of the POR Simulation is complete, subbasin-averaged annual maximum series of the Liquid Water at Soil Surface (LWASS) must be produced for multiple durations (e.g., 1-hour, 1-, 2-, and 3-day), if snow accumulation/melt processes were simulated. This is an important dataset that will be leveraged for the evaluation of stochastic model performance during the conformance phase of the project. Procedures for development are included within Job Aid 05, *Observed Gridded Data Preparation.*

6.6 DOCUMENTATION OF HYDROLOGIC MODELING

A set of standard reports is written for each phase in the FFRD project. FFRD Templates contains report templates that include hydrologic modeling sections.

Table 6.3 summarizes locations of key hydrologic modeling documentation included in each of the reports.

Table 6.3. Locations of Key Hydrologic Modeling Documentation

Report Title	Description
Scoping and Data Preparation Phase Report	Documentation of hydrologic model setup, delineations of model elements and contributions to the Model Linking Register and Name Table.
Calibration Phase Report	Primary documentation of the hydrologic model development including watershed characteristics, methods, parameters, and calibration. Includes documentation of the calibration of both the event-based simulation and POR simulation.
Conformance Phase Report	Documentation of calibrated model results used to create a range of antecedent conditions and model parameters for use in stochastic simulations. Description of any hydrologic model adjustments made to reasonably align with observed frequency data.
Production Phase Report	Descriptions of any hot fixes or patches required in the production phase.

6.7 HYDROLOGIC MODEL REVIEW PROCESS AND DELIVERABLES

Model review is a critical part of the FFRD production process. Several review steps are required to verify the final model delivered can effectively run in a stochastic environment and is capable of accurately depicting basin-wide flood risk on a broad scale. Review checklists are used to track comments regarding pre-defined aspects checked throughout the review process. FFRD Templates contains checklist templates.

The following subsections describe specific review steps and associated deliveries.

6.7.1 SCOPING AND DATA PREPARATION PHASE REVIEW

The following hydrologic model review steps are completed during the scoping and data preparation phase.

6.7.1.1 COORDINATION AND TEAM DISCUSSIONS

Although not documented as a formal review, team coordination is required throughout the scoping and data preparation phase. When the draft HEC-HMS model element delineations are complete, the full FFRD team reviews model layout. The HEC-HMS modeler meets with the FFRD team during the scoping and data preparation phase to review the layout and verify the HEC-HMS subbasins and planned HEC-RAS modeling unit polygons are compatible and consistent with the Model Linking Register, Name Table, and Dam Scoping Table. The goal of this team coordination is the development and delivery of a comprehensive and cohesive scope for the project that will be leveraged later during the calibration phase.

6.7.1.2 SCOPING AND DATA PREPARATION PHASE REVIEW

Final draft deliverables for all aspects of the scoping and data preparation phase are fully and formally reviewed prior to completion of the phase of work. The review is documented via the Scoping and Data Preparation Checklist, which requires multiple contributors.

The checklist includes specific prompts for review of initial hydrologic model setup, naming conventions, meteorologic data preparation, and observed data preparation among other items. The hydrologic model setup, hydraulic model unit delineation, Model Linking Register, Name Table, and all other scoping and data preparation phase items are reviewed and documented.

The hydrologic modeler's sole deliverable responsibilities from the scoping and data preparation phase are the following:

- Model: Unzipped uncalibrated base HEC-HMS model posted to *basin-data/project-scoping/uncalibrated-hydrologic-model/*
- Report: Completion of the Hydrologic Modeling section of the Scoping and Data Preparation Report
- Checklist: Population of pre-review commentary for checklist prompts related to hydrologic modeler deliveries. The checklist is centrally stored in the *documentation* directory.

6.7.1.3 BACKCHECK OF SCOPING AND DATA PREPARATION PHASE REVIEW

The hydrologic modeler is responsible for responding to all comments related to hydrologic products generated during the scoping and data preparation review. Updates to the draft delivered products may be necessary to achieve close-out of the review.

6.7.2 CALIBRATION PHASE REVIEWS

The hydrology review checklist is used to track comments regarding aspects checked throughout the calibration phase review process. This review checklist remains with the model and is used throughout the calibration phase review process.

The hydrologic model shall go through the following review steps during the calibration phase.

6.7.2.1 INITIAL PROGRESS REVIEW

The purpose of this early review is to confirm progress is being made toward delivery for review and for reviewers and team leads to provide support to modeling staff to verify that full guidance has been provided and that model development adheres to FFRD standards. Model data is typically not transferred during this review, but webinars are held to show progress and discuss possible issues.

Additional check-ins during model development may be useful and are encouraged as best practice.

6.7.2.2 QUALITY CONTROL REVIEW

The deliverable will be posted to the Model Library *calibration* directory and will include:

- Model: Unzipped model files posted to the *calibration/hydrology/* directory as described below:
    - Working calibrated model files for all historical events and continuous simulation events/time periods, including all files and data required to run and replicate the results.
    - Non-final model or data files associated with out-of-date runs should not be included.
- Report: Draft calibration phase hydrologic modeling section.
- Checklist: Hydrologic model review checklist for historical events and continuous simulation calibration with modeler comments/notes included for every review item. This confirms all aspects of the model to be reviewed have been addressed by the modeler. The checklist is centrally stored in the *documentation* directory.
- Reference material: As needed.

6.7.2.3 BACKCHECK OF QUALITY CONTROL COMMENTS

After QC comments have been provided by the reviewer and addressed by the modeler, the updated deliverable will be posted to the Model Library *calibration* directory, and the previous version will be moved to the archive subdirectory. This delivery will include the following:

- Model: Unzipped model files posted to the *calibration/hydrology/* directory as described below:
    - Calibrated model files for all historical events and continuous simulation events/time periods, including all files and data required to run and replicate the results.
    - Non-final model or data files associated with out-of-date runs should not be included.
- Report: Final calibration phase hydrologic modeling section.
- Checklist: Hydrologic Model review checklist with Resolution of Reviewer Comments by Modeler section completed for all comments. The checklist is centrally stored in the *documentation* directory.
- Reference material: As needed.

6.7.2.4 FINALIZATION OF CALIBRATION PHASE DELIVERY

Any necessary additional iterations of delivery updates to fully close the review will be similarly posted to the *calibration* directory and the previous version moved to archive. The final hydrologic modeling report section should also be posted.

6.7.3 CONFORMANCE PHASE REVIEWS

Two reviews are documented during the conformance phase, each with its own review checklist to document the close-out of all items. Both checklists require input from the multiple team members.

6.7.3.1 PRE-CONFORMANCE REVIEW

After the calibration phase is complete, the ownership of the model will transition to a smaller model integration team who will promote a copy of the calibrated models to the *conformance* directory. Certain changes to the model will be made to enable successful execution of synthetic stochastic events.

The pre-conformance review checklist acknowledges that calibrated models are fully prepared for cloud-compute which occurs during the conformance phase. This checklist includes items related to model integration, linkages, and naming conventions. This checklist also includes items related to the delivery of multiple hydrologic model files for cloud computing which are described in more detail in Sections 11 and 12.

6.7.3.2 CONFORMANCE PHASE REVIEW

The conformance review checklist focuses on ensuring that results of a full realization of synthetic events meet conformance metrics established in Section 12. Items required to fully close the conformance phase of the project are also included in terms of documentation and data management.

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SECTION 7—RESERVOIR OPERATIONS MODELING

7.1 RESERVOIR OPERATIONS MODELING CONTEXT AND STANDARDS

The reservoir operations model for an FFRD watershed should be capable of simulating the regulation of a flood event over a time window that includes the duration of a typical storm and the recession limb of resulting hydrographs. This section guides HEC-ResSim modelers on systems where a reservoir operations-specific model was deemed necessary to represent flood risk within a study basin and included in the Dam Scoping Table during the scoping and data preparation phase of the project.

Operations must include the changed behavior of the system of reservoirs if one or more reservoirs are interoperable.

The reservoir operations models developed for FFRD will have the following characteristics:

- Able to assess the level of modeling necessary to reasonably reflect the reservoir operations impacts to flood risks.
- Capable of modeling operational behavior of any reservoir with impacts significant to the study flood risk assessment.
- Capture reservoir system operations that are significant to the flood risk assessment.
- Coordinated for seamless integration into system of models in terms of alignments, naming conventions, and other modeling strategies
- Not overly complex to run efficiently within the FFRD cloud compute framework with numerous statistically sampled scenarios.
- Capable of accurately modeling reservoir response to flood events ranging from a 1/10 to 1/2,000 AEP in an event-based Monte Carlo simulation.

7.1.1 ROLE OF USACE IN RESERVOIR OPERATIONS MODELING

USACE has a unique role and responsibility regarding reservoirs that have flood risk management objectives (e.g., USACE projects, projects under Section 7 of the 1944 Flood Control Act, and potentially others). USACE also has a particular role regarding breach considerations (Section 9.1.1). USACE is developing reservoir modeling training to guide the reservoir modeling approach, particularly in reflecting those flood risk management operations and dam breach scenarios. The FFRD National Implementation Management Group contains USACE reservoir modeling experts who support and review work completed during the flood operations portion of the study for all reservoirs with flood risk management objectives.

7.1.2 SCOPE OF RESERVOIR OPERATIONS MODEL

The determination of which reservoirs require a reservoir operations model is done during the scoping and data preparation phase (Section 4.3.15). See Job Aid 01, *Scoping Process*es, for details on the decision-making considerations and options for how dams and reservoirs shall be modeled.

7.1.3 PHASES OF RESERVOIR OPERATIONS MODELING

The reservoir operations model is developed for an FFRD project in distinct phases according to the workflows described in Section 2. This reservoir operations modeling section applies to multiple phases, and high-level modeling tasks in each phase are described below:

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- Scoping and Data Preparation Phase: High-level scoping and selection of which reservoirs will be included in the reservoir operations model. Scoping of all Naming Conventions and a Model Linking Register is performed.
- Calibration Phase: Development, calibration, and stress testing of the reservoir operations model is performed. POR reservoir operations simulation is also performed.
- Conformance Phase: Integration of the reservoir operations model into the system of models, computation of synthetic events in cloud-compute, and adjustments to all models to confirm performance conforms to defined quality metrics for a single long-term realization.
- Production Phase: Computation of multi-realization events. Reservoir operations model may require hotfixes or updates to address specific issues.
- Reporting Phase: Compilation and delivery of results from the project.

7.1.4 SOFTWARE VERSION

The reservoir operations model will be developed using HEC-ResSim version 3.5.1.

7.1.5 DATA PROVIDED TO THE RESERVOIR OPERATIONS MODELER

A comprehensive shared dataset for the project is developed during the scoping and data preparation phase. The reservoir operations model development begins during the calibration phase and will leverage this shared dataset. Key prepared datasets provided to the reservoir operations modeler include:

- Dam Scoping Table that describes the modeling treatment of each dam in the basin
- List of reservoirs to be included in the reservoir operations model
- Model Linking Register
- Name Table for all elements in the full system of models
- Existing H&H models and their documentation, if available
    - A single model selected as most appropriate for conversion to the FFRD purpose
    - Other models, only if useful to the effort (e.g., if they have some data or operations that are important but were not present in the selected model)
    - Associated documentation including information on the model's metadata such as vertical datum and time zone.
- Dam physical data and existing operations manuals including physical data
- Reservoir physical data for each reservoir that needs to be modeled
    - Elevation-flow capacity curves for hydraulic outlets
    - Top of dam (TOD) length, elevation, and the overtopping elevation-flow curve, if available
    - Pool elevation-storage-area (ESA) curve
    - Net evaporation rates
- Operations guidance
    - WCMs and water control plans
    - SOPs
    - Emergency Action Plans (EAPs)
    - FERC licenses for non-federal facilities
- Observed flow and water surface elevation data
- Available historical timeseries data to include reservoir pool elevation, reservoir inflow and outflow, and observed flow data for any gages within the model extents

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- A HEC-ResSim modeler may need to review and identify history of change points in reservoir conditions (e.g., construction of new facilities in the watershed, new outlets at existing facilities, major operational changes).

7.1.6 MODEL LINKING STRATEGY

A comprehensive Model Linking Register and Name Table will be developed during the scoping and data preparation phase prior to reservoir operations model development and will guide model development in terms of element names and common computation points (CCPs).

7.1.7 COMPREHENSIVE NAMING CONVENTIONS

Model naming conventions for all model elements are an important aspect of FFRD, as standard naming and descriptions may also be used to support queryable metadata through the Model Library. Consolidated requirements for naming conventions and project descriptions for HEC-ResSim and other models are defined in Job Aid 06, *Standard Naming Conventions*.

7.1.8 COORDINATION WITH OTHER MODELS

Throughout model development, the reservoir operations modeler coordinates with the technical lead, hydrologic modeler, and hydraulic modelers to exchange input and output datasets as appropriate. Some typical points of this coordination are described below.

The Model Linking Register and Name Table developed during the scoping phase of the project will guide model development and all boundary conditions setup. The reservoir operations modeler and technical lead will communicate any concerns regarding model linking that may arise during the model development process with the full modeling team.

7.1.8.1 HYDROLOGIC MODEL

As communicated through the Model Linking Register, careful linking of hydrologic model outputs to reservoir operations inputs is an important aspect of the modeling system. The Model Linking Register will be developed during the scoping and data preparation phase, and any issues identified during the calibration or conformance phase regarding the Model Linking Register must be coordinated with the technical lead and hydrologic modeler. The HEC-ResSim modeler will need to coordinate closely with the HEC-HMS modeler to make certain CCPs are included where HEC-ResSim will need data from HEC-HMS.

7.1.8.2 HYDRAULIC MODEL

As communicated through the Model Linking Register, careful linking of reservoir operations outputs to hydraulic model inputs is an important aspect of FFRD modeling. The Model Linking Register will be developed during the scoping and data preparation phase, and any issues identified during the calibration or conformance phase regarding this mapping must be coordinated with the technical lead and hydraulic modeler.

The outflow assumptions must be carefully coordinated between the hydraulic and reservoir operations model particularly for large events that overtop dams or for dam breach events.

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7.2 MODEL SETUP

The following sections describe the process and standards for reservoir operations model development for FFRD projects and guidelines for reservoir operations modeling activities during this phase.

7.2.1 VERTICAL DATUM

The modeler needs to confirm that any necessary datum shift has been applied to all physical properties, as well as any rules or scripts that reference an elevation. All historic records also need to be in NAVD 88. The user should also note that the model uses NAVD 88 in the reservoir editor description and any other description fields where there is potential for uncertainty.

Note

Do not use the *Vertical Datum* feature in HEC-ResSim, as it does not work sufficiently in HEC-ResSim version 3.5 and has potential to corrupt the data.

7.2.2 TIME ZONE

UTC should be used when applying any rules that reference time, and this effort should be noted in the reservoir editor description field.

7.2.3 DEFINITION OF MODEL EXTENTS

Reservoirs to include in the reservoir operations model are determined during the scoping and data preparation phase as described in previous sections.

The HEC-ResSim model network must cover the most upstream CCPs needed to determine reservoir operations—typically, this is the inflow to the most upstream reservoir or an upstream gage used to determine operations—down to the outlet of the watershed. To minimize computing time and size, the extents of the reservoir operations model are limited to elements required for the computation of reservoir operations.

For the portions of the HEC-ResSim model that overlay with the HEC-HMS models, the HEC-ResSim model must include all the junctions within that extent. These junctions are CCPs, and they must be placed at the same locations to enable data handoff between the models. Reach routing methods and parameters must also match those used by the HEC-HMS and HEC-RAS models.

Generally, a single HEC-ResSim model (one network and one ResSim alternative) should be used for the entire HUC-4 watershed, covering all the reservoirs that have been selected for modeling. However, if a headwaters basin requires a smaller timestep than the rest of the basin, it may be necessary to split the basins into separate HEC-ResSim models so that each can use a different run timestep. Should this approach be needed, it is important to note that there can be no downstream dependency represented in the headwaters model due to the nature of the directed acyclic graph.

7.2.4 WATERSHED CONFIGURATION

The HEC-ResSim modeler will need to coordinate closely with the HEC-HMS modeler to verify CCPs where HEC-ResSim will need data from HEC-HMS are included. The HEC-ResSim model should also use the shared basin maps and stream centerlines established for the project during the scoping and data preparation phase. While it would be ideal for the stream centerlines to be identical across all models, it is not necessary for HEC-ResSim. If there are changes that are made to the centerlines during HEC-RAS model development, the updated streams do not need to be incorporated into the HEC-ResSim model.

7.2.4.1 COMMON COMPUTATION POINTS OR JUNCTIONS

Model Linking Register and *Name Table* are developed during the scoping and data preparation phase and provided prior to the start of reservoir operations model development during the calibration phase. They guide the naming of network elements to enable naming consistency and linking compatibility across all models.

CCPs in the watershed configuration are shared across models as hand off points for data. These are represented in the HEC-ResSim network as junctions joining routing reaches and reservoirs. Junction locations include confluences, the downstream and upstream ends of reservoirs, anywhere that an inflow is to be added to the river, stream gage locations, and at control points where flow or stage is used to determine reservoir operations.

7.2.4.2 RESERVOIRS AND DIVERSIONS

The reservoirs and diversion elements are included in the reservoir operations model as determined during the scoping and data preparation phase.

7.2.5 LINKING FOR HISTORIC EVENTS VERSUS STOCHASTIC EVENTS

During the calibration phase, the HEC-ResSim model is developed, tested, and calibrated leveraging observed data. For stochastic events to be computed during the conformance phase and production phase of the FFRD project, the model will utilize the HEC-HMS simulated inflows.

7.2.5.1 LOCAL INFLOWS

Local inflows are developed in coordination with the HEC-HMS models from their calibration dataset but are not derived from the HEC-HMS model results until the HEC-ResSim model validation is complete. Gage data used for local inflows is converted to the appropriate timestep for the model by period averaging. Incremental or local inflows are estimated as the difference between downstream and upstream gages. Gap-filling methods may be applied on inflow timeseries.

7.2.5.2 HISTORICAL OBSERVED DATA FOR CALIBRATION

Observed reservoir outflows and pool elevations do not have significant processing applied unless being used to control the Set Q/Set Z historical data alternatives described in Section 7.4.

7.3 RESERVOIR NETWORK DEVELOPMENT

The HEC-ResSim modeler creates a reservoir network based on the watershed configuration. For FFRD, only one network is necessary since the physical arrangement of elements should not change.

7.3.1 NETWORK CONFIGURATION FOR FFRD PROJECT SCOPING

The configuration of Reservoirs, Dams and Outlets in the model must be fully accounted for in the Name Table and Model Linking Register documents. These documents are prepared during the scoping and data preparation phase and provide a clear point of communication as the team confirms model linkages can be made seamlessly.

Some reservoirs have simple operations but are scoped for inclusion in the HEC-ResSim model because they impact the operational decisions made by other reservoirs within the system. These simple

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operations may be modeled using simple rules or run of river/guide curve operations, in which outflow typically equals inflow.

7.3.2 PHYSICAL DATA GENERAL CONSIDERATIONS

Once the modeler has determined which physical data to incorporate into the model, they must also document the source of the data (along with any data processing that was performed) in the final report. If the modeler digitizes any physical data, a copy of the digitized items must be saved with the report to provide alongside all the deliverables.

To address curve resolution, half- or tenth-foot increments are used for physical capacity curves unless necessary to capture the shape of the curve. Hundredth-foot increments are not necessary and discouraged unless other models use them, and the team agrees this is a necessary level of refinement.

7.3.3 INITIAL POOL ELEVATIONS

For the POR run, the lookback pool elevation should use the observed value. If there is no observed value, the model should set the lookback to start at guide curve or typical normal pool.

For the FFRD event compute, it should be set to use the timeseries of the initial pool elevation from HEC-HMS (which comes from the bootstrapping sampling).

7.3.4 REACH ROUTING METHODS

Routing methods and parameters are derived from the HEC-HMS or HEC-RAS models wherever routing reach model elements overlap. Modified Puls is the preferred HEC-ResSim routing method when parameters are acquired from a high-quality calibrated one-dimensional (1D) hydraulic model. If Modified Puls is not available, Muskingum-Cunge with 8-point cross sections is the secondary preferred routing method. The physical characteristics for this method can be copied from the HEC-HMS model. The number of subreaches is adjusted as appropriate for the reservoir operations model timestep. The HEC-ResSim modeler will need to coordinate with the HEC-HMS modeler on any losses or gains in the network including, but not limited to, irrigation and diversions.

7.3.5 RATING CURVES AT CONTROL POINTS

Rating curves used within the HEC-ResSim model for downstream controls must be validated. The HEC-ResSim modeler must work with the HEC-HMS and HEC-RAS modelers to validate and finalize the curves.

7.3.6 PHYSICAL CAPACITY DEFINITIONS

Capacity curves are provided from the most up-to-date sources available; these are typically found in the WCM. Capacity curves need to be extrapolated to cover the full range of elevations likely to be experienced under the events sampled for the model. ESA curves are defined to extend above the highest pool elevation expected during a probable maximum flood (PMF) event, extrapolating if necessary. Outlet capacity curves are defined over their entire operable range, or for a spillway to the same elevation as the ESA curves.

7.3.6.1 RESERVOIR POOL ELEVATION-STORAGE-AREA CURVES

For model validation, elevation-storage relationships must match the ESA curve used for calculation of storage and inflows in the observed dataset used to calculate reservoir inflows; otherwise, a mass

balance error will occur. Often, the ESA curve is developed from surveyed pool bathymetry and updated as part of sedimentation studies. This may result in several breakpoints in the observed data, making it difficult to validate the model across a full POR. The elevation-storage relationship must be coordinated with the hydrologic and hydraulic modelers to verify models are using compatible datasets.

Area is only necessary if evaporation is modeled.

7.3.6.2 OUTLETS

Outlets may have various names across various sources of documentation, and care should be taken to confirm all outlets are properly accounted for. Reviewing blueprint or as-built drawings for the dam may be useful in identifying outlets requiring modeling and the correct number of each type.

If a facility has a system of gates, or several outlets with identical release capacity curves, the outlet is modeled as a single outlet element with the number of gates set to the appropriate number. Do not use a composite curve that does not indicate the number of gates.

- **Controlled Outlets**
    - **Gated Spillways, Regulating Outlets**—Gated spillways and regulated outlets are modeled with the controlled outlet element type.
    - **Powerplants**—Use controlled outlets for powerplants unless there are operations determining net release from the reservoir as a function of hydropower production required. If only hydraulic capacity is known, or power production expressed as energy or power available is not a significant factor in reservoir operations, the most efficient choice is to model these as a controlled outlet. Modeling these outlets as a powerplant requires additional details about the turbine capacity that may not be available for non-federal projects.

- **Uncontrolled Outlets**—Uncontrolled outlets represent spillways, emergency overflows, and other structures with a known physical capacity curve. Releases through these outlets always take priority over the release allocation and operating rules.
- **Estimating Physical Capacity Curves if Unavailable**—If a physical capacity curve is not available, estimates based on the weir equation and known or estimated physical dimensions may be used, and either verified directly by modeling with observed data, or compared against relationships between observed data, such as reservoir outflow and pool elevation.
- **Implementing Physical Constraints as Rules**—Occasionally, reservoir physical constraints need to be defined as a rule, for example, outlets sharing a common penstock through the dam but releasing to different channels or having different capacities, and, due to the common penstock, are mutually exclusive. A rule in the operations set may be defined for such a constraint and is placed as the highest priority rule in each zone. These rules are clearly labeled as representing a physical capacity, using a prefix such as [Physical]-.
- **TOD as an Uncontrolled Outlet**—If a dam is designed to overtop as part of normal or emergency operations, or if an overtopping capacity curve is available, it is recommended that an uncontrolled outlet be added at the TOD elevation, and the TOD elevation for the dam is ignored. The TOD release used otherwise determines the capacity from the elevation and length of the TOD. If the TOD elevation varies, a capacity curve based on simple weir flow estimates may be developed. Extreme events larger than the dam design flood must be accounted for.
- **Other Outlets**—Reservoir evaporation, leakage, and seepage can be significant factors in reservoir mass balance. These are factored into the mass balance equation used to calculate the new pool storage and elevation at each timestep. Evaporation, leakage, and seepage are the only ways water can be removed from the reservoir below the inactive zone.
    - **Evaporation** represents a net loss of volume to the atmosphere and is removed from the pool storage at each timestep. A seasonal pattern based on average evaporation per day, expressed as a depth over the pool area is the preferred way to enter evaporation for FFRD

models. The use of a historical timeseries based on pan evaporation or other estimates may be valuable for model validation but should not be used for the delivered model.
    - **Leakage** through the dam represents water passed downstream without being lost from the system but not controlled by the dam. It can be defined as a constant value or a value for a given pool elevation. If neither of these is appropriate, or another method to calculate leakage or other miscellaneous releases is needed, a controlled outlet can be defined and governed by a rule labeled as a physical constraint and placed at the top of the rule priority stack in every zone.
    - **Seepage** is water lost from the pool into the groundwater below the reservoir. This water does not continue downstream and is lost from the model. If a documented seepage rate exists for a reservoir, it is entered as a seepage outlet. If the seepage relationship does not follow a typical relationship with pool elevation, it may be modeled with a diverted outlet and governed by a rule labeled as a physical constraint and placed at the top of the rule priority stack in every zone.

- **Breach Outlet**—This outlet is a controlled outlet to circumvent the physical limitations of the reservoir in a breach event. If a breach event were to occur, the model should release run-of-river. Depending on the breach elevation, HEC-ResSim may run into physical limitations with the project gates. The breach outlet needs to have a large enough release capacity to match inflow.

7.3.7 SCRIPTS

The use of scripts for FFRD models should be avoided whenever possible. Scripts should only be used in the HEC-ResSim model when necessary to reflect key operational decisions. Any scripts that are necessary to the model may not depend on any outside information, including external scripts and libraries. HEC-ResSim will not have access to external sources such as websites or network resources. If an operation requires external input, this can be accomplished using HEC-ResSim's Global Variable feature. For scripted operations that depend on random behavior, if other simplifying assumptions cannot be made, modelers should use the random sampling option that depends on a reproducible seed value.

7.4 INITIAL MODEL VALIDATION

Two temporary alternatives are developed to test the reservoir network prior to the development of the final operations sets and alternatives. The intent of these two alternatives is to verify the reservoir network passes flow correctly from upstream to downstream and confirm the physical capacity tables are reasonable, and the inflow dataset used for model development and validation represents all the inflows necessary to replicate historical operations.

In developing these alternatives, a simple operations set is configured, typically starting with a run-of-river operations set containing no rules and holding the reservoir at full pool, followed by the Set Q/Set Z operations sets.

7.4.1 CHECK FOR NETWORK CONNECTIVITY

It is essential to develop a network connectivity alternative where a constant timeseries is mapped to each inflow location and the sum of the unregulated flow at the downstream-most junction (once steady-state has been reached) is confirmed to be equal to the constant value multiplied by the number of inflow locations. If this criterion is not met, there are two possibilities: either there are losses defined on one or more of the reaches in the system, or the network connectivity is not complete and water is generated or removed from the system due to a misconnected element.

To check for any losses defined on the routing reaches, use the Reach Report available in the Reservoir Network module. If losses are present, the modeler must determine whether the losses are a constant or a function of flow. If losses are a function of flow, the modeler must determine the number of inflow nodes

upstream of that reach to determine how much flow is lost. If defined losses are not responsible for the difference, the modeler should find the elements that have caused the break in the network. Working upstream in a binary search helps home in on elements causing the problem. These elements, and potentially the surrounding elements, must be deleted and redrawn. Then the network connectivity test should be rerun to confirm the fix was successful.

7.4.2 HISTORICAL DATA ALTERNATIVE (SET Q/SET Z)

To validate that the reservoirs' physical capacity definitions are reasonable, an alternative is created with simple operations that force the reservoir to follow observed data. This is achieved by creating an operations set at each project—either an operation set with a single rule in all zones that follows an external timeseries of releases (Set Q for flow) or an operation set with a zone that follows an external timeseries of observed pool elevation (Set Z for elevation). The other observed historical timeseries are mapped as observed data and used to validate that the reservoir's releases and pool elevation follow closely. If this simulation does not show a close match, either a component of the reservoir's mass balance is missing or incorrect, or the physical capacity is not correct, such as an outlet that does not have enough hydraulic capacity to pass historical inflow, or a pool elevation-storage relationship that is not the same as the one used to determine reservoir inflow.

7.5 DEVELOPMENT OF RESERVOIR OPERATIONS SETS

The final model should only have one operation set; however, during development the modeler may need to develop several operation sets to test various alternatives. Prior to handoff, all operation sets except for the rule-based single FFRD event operations should be removed. While setting up the operation sets, the description textboxes in the Reservoir Editor should be used to document data sources and any assumptions made.

7.5.1 SCOPE OF RESERVOIR OPERATIONS MODELED

FFRD models include formally documented operations and informal operations consistently used to achieve reservoir objectives. Occasional deviations and rule-of-thumb-type operations are avoided but may be modeled to assist with validation.

7.5.2 REVIEW OF WATER CONTROL MANUALS, STANDARD OPERATING PROCEDURES, OTHER OPERATING DOCUMENTS

The operating documents are reviewed and used to identify rules and operating zones. These include, but are not limited to:

- WCMs for USACE-operated reservoirs and those operated by USACE under Section 7 of the Flood Control Act, or similar agreements. In particular, the water control plan, typically in Chapter 7 of the WCM, documents operations at the reservoir.
- SOPs for U.S. Bureau of Reclamation (USBR)-operated reservoirs.
- Fish Passage Plans, Biological Opinions, other documents used to determine ecosystem operations.
- FERC licenses for hydropower facilities not operated by USACE.

As a last resort, historical observed data is examined, particularly the outflow from the project or at gages immediately downstream and at potential damage centers downstream. Typical high and low flows, if consistent, may indicate a maximum and minimum release target from the reservoir.

Operations are cataloged by priority and applicable operating zone, identifying features such as:

- Objective type: minimum, maximum, or specified release

- Timing: seasonality, within-week, within-day.
- Control points downstream in the system
- Other variables necessary to determine operations
- Assumptions made in determining operations.

The model should be comprehensive and not only focus on flood risk but instead should represent realistic operations throughout the year. Low flow operations including irrigation, water supply, ecosystem, and hydropower are to be included to the extent that reliable and consistent data to support these is available. Drought contingency rules do not need to be modeled in the single event alternative if not already incorporated or if they do not function properly with a shorter simulation window.

7.5.3 DEFINITION OF ZONES

At a minimum, a TOD, guide curve, and inactive zone must be specified. A normal top of flood control pool and conservation zone may also be defined, either of which may be the guide curve. Additional zones spelled out in the operating documents are specified and modeled.

The TOD zone is located at the highest elevation at which any reservoir operations need to be modeled. Rules within this zone are only those used to operate the dam during an extreme flood event where dam safety is paramount. This zone does not include rules to limit flows for downstream locations.

The guide curve zone is the target elevation for the reservoir. This zone may be prescribed as part of the operating documents, or, if that is unavailable due to the reservoir not having a flood risk management purpose, an average of historical values by season may be used.

The inactive zone is the lowest zone and contains no rules. When the pool elevation is in this zone, only evaporation, leakage, and seepage demands are met.

7.5.4 DEFINITION OF RULES

Rules derived from WCM, SOP, and other operating documents should be implemented first. Each objective in the WCM should be modeled separately. For example, do not combine minimum flow rules for ecosystem and navigation requirements, or various ecosystem rules. Documentation should identify which objective in the operating documents is supported by each rule.

Informal rules based on the experience of the operators should be clearly labeled and thoroughly tested before inclusion in the deliverable. Any deviation from documented operations required to help smooth or stabilize the operation of the HEC-ResSim model should be clearly labeled as such.

7.5.5 MODELING FOR DAM BREACH

Each reservoir that has been selected as a potential breach location for modeling (likely all reservoirs represented in the HEC-ResSim model) shall be set up to allow for the triggering of dam breach scenarios based on externally sampled breach elevations. Breach scenarios are triggered in HEC-ResSim based on comparing the pool elevation to the sampled breach elevation. If a breach is triggered in HEC-ResSim, the reservoir operations are shifted to run-of-river and HEC-RAS takes full responsibility for modeling the actual breach conditions and flows. Thus, the primary goal of breach modeling in HEC-ResSim is to determine if and when breaches are triggered and if, after a breach, the HEC-ResSim model is no longer simulating the expected scenario; in particular, if downstream reservoirs do not see the inflows that would occur during and after a breach.

To model a breach event in HEC-ResSim, an additional rule nested within an if/else block should be incorporated into the model and added to all zones. This nested condition would dictate that if a target elevation is met or exceeded, the outflow will match inflow. The model should be developed to allow the breach elevation to vary. To accomplish this, set up the breach elevation using a scalar input global

variable. A state variable will be needed to pull the global variable value into the operation rule set. An additional rule must be created and added to the Else portion that states if a breach is not occurring, releases from the Breach Outlet must be zero.

7.6 DEVELOPMENT OF RESERVOIR OPERATIONS ALTERNATIVES

During model development, the modeler will need to create the following alternatives to calibrate and test the model:

- Network Connectivity Alternative
- Historic Data Alternative
- Single Event Alternative
- Extreme Event Alternative

For handoff, a copy of the model needs to be created where all alternatives, with the exception of the Single Event Alternative, need to be deleted.

7.6.1 SETTING ANTECEDENT CONDITIONS FOR OPERATIONS

HEC-ResSim simulations require a lookback window, which is a period of warmup time to allow water to route through the full network before the simulation period with release decisions begins. Model alternatives require the setting of antecedent conditions via the lookback feature. Typically, it is recommended that the lookback time window should cover a period sufficient for water to be routed from the uppermost headwaters input to the furthest downstream location. For many of the FFRD reservoir models, it is not reasonable to have the HEC-HMS model run each event for such a long period of time that would be necessary to provide sufficient lookback inflows. Instead of getting inflows from HEC-HMS, the HEC-ResSim POR run will be used to obtain lookback data associated with the historic data preceding each sampled event.

To best prepare the model for such lookback sampling, some simplifications should be made to operations that typically depend on institutional antecedent conditions. Institutional antecedent conditions, or operational antecedent conditions are variables not directly or significantly tied to hydrology (e.g., history of gate maintenance, deliveries of water supply or hydropower, and things that inform operations but should not or cannot be modeled). When an operation requires data that begins significantly before the event, the possible approaches, in order of preference, are to simplify that condition out of the model, reduce the operation to the essentials (e.g., to limit the length of lookback time needed), or convert a dependency to a randomly sampled variable that reflects the probability of occurrence if a lookback requires an excessive historic time period. This can be accomplished using the HEC-ResSim Monte Carlo feature or a seeded random generation from a previous plugin in the compute, if necessary.

7.6.2 TIMESTEP AND COMPUTATION METHOD

The HEC-ResSim model uses a timestep appropriate to capture the duration of a flood event with an adequate resolution to represent the rate of change constraints and routing effects to control points. Typically, this is somewhere between hourly and six-hour timesteps but may be shorter for small reservoirs and flashy watersheds. HEC-ResSim does not allow for multi-timestep routing through long reservoir pools. Therefore, pool inflows become available and releasable storage for the following timestep after arrival. If one of these small reservoirs is responsible for increasing the overall model timestep, and can be modeled independently of the others, it should be broken out into a separate reservoir operations network and model alternative.

Period average is the preferred flow computation method for all alternatives.

7.6.3 PERIOD OF RECORD MODEL CONSIDERATIONS

A POR model is run for the watershed to determine a distribution of initial starting conditions for the reservoir during event-based modeling. This POR compute is necessary to apply current operations to the whole POR, during which the operations may not have been constant over time. Antecedent conditions will be selected from the POR output associated with the sampled event storm and date. The model will select a starting point from a 2-week period around the actual date across all historic years and HEC-HMS and HEC-ResSim will select a starting point from the POR that is representative of that time of year and consistent across hydrologic and reservoir antecedent conditions. Long-term operations related to demand rules, water quality, ecosystem operations, hydropower, and drought conditions should be included in the POR run, but they should be simplified to reflect a consistent pattern to the extent possible. For example, do not model the POR using nonstationary demands or rules from specific periods of time. Instead base the POR operations on the current conditions to the extent that they can be considered stationary or use a short-term window to develop high, medium, and low curves.

7.6.4 EVENT-BASED MODEL CONSIDERATIONS

The model is ultimately configured for event-based modeling, using HEC-ResSim POR results to set the initial values for pool elevation, gate releases, and inflow timeseries in the Alternative Editor Lookback tab.

The model does not rely on parameters that may be unavailable within the event, such as an operation that depends on the value of a variable several months prior. If this is the case, a Monte Carlo variable is used to provide a randomly sampled input, with the distribution of the variable derived from the POR compute.

Each event simulation time window should be set up to include a lookback window that meets the minimum time requirement determined as the time necessary for inflow to be routed from the highest decision point to the lowest downstream flow location, or approximately 2 weeks, whichever is less. The HEC-ResSim modeler should work with the HEC-HMS modeler to confirm that the event period extends long enough to capture the recession limb of the event.

7.7 SAVED OUTPUT SETTINGS

HEC-ResSim has default settings in the Alternative Editor that should be adjusted prior to final handoff. These changes will ensure the correct data are written out, will remove superfluous data, and will cut down on the run folder file size. The minimum needed includes anything the HEC-RAS model requires, pool elevations, control point flows, gate flows.

In the HEC-ResSim alternative editor, modify the DSS Output by unchecking *Write all computed Time series* and selecting to only keep reservoir elevation, junctions and key state variables. Reaches, diversions, and all other reservoir timeseries will typically be left as not selected.

The Run Control tab in the Alternative Editor defaults the Flow Computation Method to Program Determined. For HEC-ResSim runs, it is usually recommended to set this to Period Average if the input timeseries is greater than 1 hour; however, this can cause issues for the HEC-RAS model. If the timestep is 1 hour or less, the HEC-ResSim modeler can set the flow computation method to instantaneous. If the timestep is greater than 1 hour, the HEC-ResSim modeler should set the flow computation method to period average, and the HEC-RAS modeler will need to process the data prior to its input to the HEC-RAS model.

The DSS Output tab can also be pared down. A more detailed description on the output settings can be found in Job Aid 12, *Model Delivery for Cloud Compute*.

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7.8 SIMULATIONS FOR VALIDATION AND TESTING

Four types of runs are completed to validate and review the performance of the HEC-ResSim model alternatives: event-based computes of historical floods, single event computes using HEC-HMS outputs, synthetic extreme event computes, and a POR compute.

7.8.1 HISTORICAL EVENT-BASED COMPUTES

The event-based compute for historical floods is chosen for an event within the recent POR in which operations are most similar to the current operations of reservoirs within the basin, and other projects that are in place within the watershed. If none of these are available in the recent record, older events are used, but decisions are validated through closer examination of performance and checking that relevant operating constraints are met.

7.8.2 EXTREME EVENTS STRESS TESTING COMPUTES

Synthetic extreme events are used to evaluate the performance of the HEC-ResSim model for large flows as stress testing in preparation for the distributed cloud compute. The intent of these computes is to show that the operation sets behave appropriately for events larger than observed. This should catch issues with operations, like oscillating zone boundaries around top of flood pool, as well as physical data issues like spillway curves that are not defined above top of the flood control zone.

A sufficiently large event is one that confirms the model operates correctly through any surcharge operations and up to the dam crest. Specifications for large flood events, such as the standard project flood (SPF), design flood, and/or PMF may be found in design documents or other studies for the reservoir and may be accompanied by an estimate of regulated outflow during such an event. Typically, SPF is a good event stress test, although the stress test does not need to target a certain probability or use realistic hydrology. If the extreme events available do not sufficiently stress the reservoir's operations, the use of an inflow multiplier with a synthetic or historical event can be used.

Evaluation of the extreme event runs confirms the flood risk management objectives are met correctly for large events, rules are operating correctly, and outlet capacity curves at high elevations and flows are defined correctly.

7.8.3 HYDROLOGIC MODELING SYSTEM EVENT-BASED COMPUTE

The HEC-HMS event-based computations are single event simulations that are run using timeseries data from the HEC-HMS model. These simulations will validate the models are linking correctly and can successfully run. While the HEC-HMS simulations may not match historic data, the modeler should review the results to confirm that HEC-ResSim makes the right decisions based on its operation set.

7.8.4 HYDROLOGIC MODELING SYSTEM PERIOD OF RECORD

The HEC-ResSim modeler will need to run a simulation using the HEC-HMS POR event. The output from this simulation will be provided back to the HEC-HMS modeler so they can include the reservoir elevations in the run used to create bootstrap save states. This facilitates spatial variation of basin parameters and reservoir contents. The HEC-ResSim model should then be configured to read the initial pool elevation from an external timeseries, which will be linked to the timeseries output by the HEC-HMS model in the FFRD event computes.

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7.9 DOCUMENTATION OF RESERVOIR OPERATIONS MODELING

A set of standard reports is written for each phase in the FFRD project. FFRD Templates contains report templates for the reservoir operations modeling reports.

Table 7.1 summarizes locations of key reservoir operations modeling documentation included in each of the reports.

Table 7.1. Locations of Key Reservoir Operations Modeling Documentation

Report Title	Description
Scoping and Data Preparation Phase Report	Documentation of the selection and rationale of the dam modeling approach for each dam in the basin, including a list of dams that will be included in the reservoir simulations modeling effort. The Model Linking Register that includes details regarding the plan for linking reservoir output data to the hydraulic model is also established and documented in this report.
Calibration Phase Report	Primary documentation of reservoir operations model development, calibration and stress testing.
Conformance Phase Report	Description of all reservoir operations model adjustments made to accept synthetic SST events and any other adjustments to reasonably align with observed frequency data
Production Phase Report	Descriptions of any reservoir operations model hot fixes or patches required in the production phase.

7.10 RESERVOIR OPERATIONS MODEL REVIEW PROCESS AND DELIVERABLES

Model review is an important part of the FFRD production process. Several review steps are required to verify the final model delivered can efficiently run in a stochastic environment and is capable of accurately simulating reservoir operations.

The following subsections describe specific review steps.

7.10.1 SCOPING AND DATA PREPARATION PHASE REVIEW

The Model Linking Register, Name Table, and all other scoping and data preparation phase items are reviewed and documented in the Scoping and Data Preparation Checklist, which requires multiple reviewers and is described in Section 4.

This Scoping and Data Preparation Checklist includes items for review of overall configuration of the reservoir operations model.

7.10.2 CALIBRATION PHASE REVIEWS

The reservoir operations review checklist is used to track comments regarding aspects checked throughout the calibration phase review process. This review checklist remains with the model and is used throughout the calibration phase review process.

The reservoir operations model shall go through the following review steps during the calibration phase.

7.10.2.1 INITIAL PROGRESS REVIEW WEB MEETING—INFORMAL

The purpose of this early review is to confirm progress is being made toward delivery for review and for reviewers and team leads to provide support to modeling staff to verify that full guidance has been provided and that model development adheres to FFRD standards. Model data is typically not transferred during this review, but webinars are held to show progress and discuss possible issues using screen sharing.

Additional check-ins during model development may be useful and are encouraged as best practice.

7.10.2.2 QUALITY CONTROL REVIEW

The deliverable will be posted to the Model Library *calibration/reservoir-operations/* directory and will include:

- Working calibrated model.
    - A draft final model that includes all alternatives and simulations created during calibration of the model must be provided at handoff. The simulation .dss files from the simulation runs can be deleted prior to zipping up the model to help reduce space. In the event this model would be needed for reference, the user would only need to run an extract again and then recompute.
    - All temporary or unnecessary model files or references must be deleted from the delivery dataset

- Reservoir Operations QC review checklist with modeler comments/notes included for every review item. This verifies that all aspects of the model to be reviewed have been addressed by the modeler.
- Reservoir Operations Section of Report Draft Completed
- Reference material.

7.10.2.3 BACKCHECK OF QUALITY CONTROL COMMENTS

The updated deliverable will be posted to the Model Library *calibration/reservoir-operations/* directory (previous version will be moved to archive according to directory structure guidance in Job Aid 07, *Model Library Use*) and will include the following:

- Working, calibrated model for all scenarios.
- Reservoir Operations Section of Report Completed
- Reservoir Operations review checklist with Resolution of Reviewer Comments by Modeler section completed for all comments

7.10.2.4 FINALIZATION OF CALIBRATION PHASE DELIVERY

Any necessary additional iterations of delivery updates to fully close the review will be similarly posted to the *calibration* directory and the previous version moved to archive. The final reservoir operations modeling report section should also be posted.

7.10.3 CONFORMANCE PHASE REVIEWS

Two reviews are documented during the conformance phase, each with its own review checklist to document the close-out of all items. Both checklists require input from the multiple team members.

7.10.3.1 PRE-CONFORMANCE REVIEW

After the calibration phase is complete, the ownership of the model will transition to a smaller model integration team who will promote a copy of the calibrated models to the *conformance* directory. Certain changes to the model will be made to ensure synthetic stochastic events can be executed successfully.

The pre-conformance review checklist verifies that calibrated models are fully prepared for the cloud-compute that occurs during the conformance phase. This checklist includes items related to model integration, linkages, and naming conventions. This checklist also includes items related to the delivery of the reservoir operations model files for cloud computing, which are described in more detail in Sections 11 and 12.

The final reservoir operations model to be provided for FFRD must only have one single network and single event alternative. Refer to Job Aid 12, *Model Delivery for Cloud Compute,* for additional details.

7.10.3.2 CONFORMANCE REVIEW

The conformance review checklist focuses on ensuring that results of a full realization of synthetic events meet conformance metrics established in Section 12. Items required to fully close the conformance phase of the project are also included in terms of documentation and data management.

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SECTION 8—HYDRAULIC MODELING

8.1 HYDRAULIC MODELING CONTEXT AND STANDARDS

The procedures outlined in this section document the process and standards for the development of hydraulic models to support the FFRD program. The FFRD hydraulic models created using this document are basin-level studies developed with the understanding that additional future refinement may be undertaken after the project is delivered. Practical run-time considerations have motivated many of the standards included in this section. Models produced following this guidance will have the following characteristics:

- Foundational model that may be refined for any future detailed studies.
- Coordinated for seamless integration into the system of models in terms of alignments, naming conventions, and other modeling strategies.
- Computationally efficient and stable model capable of producing reasonable results for the entire basin and adequate to run in a stochastic environment for numerous events.
- Capable of assessing both pluvial and fluvial flood hazard data across the range of natural variability from the 10-year to the 2,000-year modeled events (1/10 to 1/2,000 AEP).
- Watershed wide systems-based accounting of water volume and timing during dam or levee breach events.
- Fully 2D geo-referenced, calibrated hydraulic model capable of simulating runoff when applied in conjunction with excess precipitation and baseflow provided by a separate hydrologic model.

8.1.1 PHASES OF HYDRAULIC MODELING

The hydraulic models are developed for an FFRD project in phases according to the workflows described in Section 2. This hydraulic modeling section applies to multiple phases, and high-level hydraulic modeling tasks in each phase are described below:

- Scoping and Data Preparation Phase: High-level scoping of the hydraulic modeling units in terms of model unit extents, mesh development categories, input data preparation, Name Table, and Model Linking Register.
- Calibration Phase: Development and calibration of the hydraulic models to historic observed events.
- Conformance Phase: Integration of the hydraulic model into the system of models, computation of synthetic events in cloud-compute, and adjustments to all models to ensure performance conforms to defined quality metrics based on a single 2,000-year realization.
- Production Phase: Computation of multi-realization events. Hydraulic models may require hotfixes or updates to address specific issues.
- Reporting Phase: Compilation and delivery of results from the project.

8.1.2 SOFTWARE VERSION

The hydraulic models will be developed using HEC-RAS version 6.6.

8.1.3 DATA PROVIDED TO HYDRAULIC MODELER

A comprehensive shared dataset for the project is developed during the scoping and data preparation phase of the project. The hydraulic model development begins during the calibration phase and will leverage the shared dataset. These key prepared datasets are provided to the hydraulic modeler:

- DEM: Full resolution DEM for consistent use for the entire FFRD project

- Hydraulic Model Unit Polygon: The 2D modeling extent used for hydraulic model development and calibration. Each FFRD study basin is subdivided into hydraulic modeling units to reduce the computational burden of the model and decrease the model development time.
- Scope for mesh development
- Urban areas polygons
- Key streams
- Other streams
- Observed flow and water surface elevation data
- Stream Gages for Calibration—including assigned categories for calibration criteria
- Selected Historic Storm Events for calibration
- Elevation derived hydrography (stream centerlines aligned with DEM flow paths)
- Levee alignments and profiles
- Levee physical data
- Levee breach locations for FFRD modeling (overtopping locations to be verified by hydraulic modeler)
- Levee operations manuals when available
- Dam physical data
- Dam modeling scope defining modeling treatment of each dam in the basin
- Dam operations manuals when available
- Comprehensive Name Table
- Model Linking Register
- Bathymetry: All available bathymetry for rivers, lakes and reservoirs
- Land cover dataset
- Existing hydraulic models
- Various other reference layers

The hydraulic modeler should fully utilize the provided shared datasets and limit searches for alternative or competing datasets. However, the modeler may become aware of additional data sources that could be used to support model development. Improved input data can increase the quality of the delivered model and is encouraged if it does not impede model production and delivery. Note that any new data must be coordinated with the technical lead and shared with the full modeling team. Note that some datasets may be deemed too detailed for direct incorporation into the FFRD hydraulic model.

The scope for hydraulic model mesh development is well defined during the scoping and data preparation phase, but the modeler may identify certain areas where a high level of refinement could be efficiently included in the model to increase model accuracy without increasing the scope. Any proposed scope increases must be requested and approved by program leadership prior to implementation.

8.1.4 CUSTOM REFERENCE DATA

This SOP includes directions to utilize a common set of shared basin data that is created during the scoping and data preparation phase, which are stored in the *ffrd_[basin-name]/basin-data/* directory. Each hydraulic modeling unit will reference this official data for many aspects of the modeling process.

However, there are some datasets that will be developed during model development specifically to support an individual model. For calibration phase deliverables, these datasets typically should be saved within a reference directory inside the *model-unit* directory (*ffrd_[basin-name]/calibration/hydraulics/[model-unit]/*).

8.1.5 MODEL LINKING STRATEGY

A Model Linking Register and Name Table will be developed during the scoping and data preparation phase prior to hydraulic model development.

During the calibration phase of the project, the model is developed and calibrated against observed historic events. During the conformance phase of the project, the models are adapted to fully accept modeled output from synthetic storm events as boundary conditions rather than observed data. For model development during the calibration phase the hydraulic model will apply the following boundary conditions:

- **Gridded Excess Precipitation Data (Hydrologic Model Output)—**This is a timeseries gridded dataset of excess precipitation data developed from a calibrated hydrologic model used as a boundary condition in the 2D model.
- **Baseflow Data (Hydrologic Model Output)—**This is a timeseries dataset of baseflow data for each subbasin developed from a calibrated hydrologic model used as inflow boundary conditions in the 2D model.
- **Upstream Inflows—**For non-headwaters basins, upstream inflows will be sourced from either observed gage data or other FFRD model results (upstream hydraulic model, hydrologic model, or reservoir operations model results).
- **Downstream Water Surface—**Downstream water surfaces will be modeled with either a rating curve, normal depth assumption, or an appropriate water surface related to a large water body. In many cases, it will be appropriate to increase the model extent beyond the scoped model unit to avoid errors introduced by the downstream water surface assumption.
- **Other Boundary Conditions—**When appropriate, other boundary conditions will be used in certain situations such as initial water surfaces or internal flow timeseries at dams.

Adjustments to model linkages will be necessary for fully synthetic events modeled in the conformance phase and will be accounted for in the Model Linking Register. Common adjustments include:

- Use of dam outflows modeled by either the reservoir simulations or hydraulic model rather than use of observed data
- Use of modeled inflows from an upstream hydraulic model rather than from a stream gage
- Use of a downstream rating curve or assumed water surface boundary condition rather than from an observed stream gage

8.1.5.1 NAMING FOR MODEL LINKING ELEMENTS

All hydraulic model files and model elements will be named according to the comprehensive Name Table and Model Linking Register provided to the modeler prior to the start of model development.

8.1.6 COMPREHENSIVE NAMING CONVENTIONS

Model naming conventions for all model elements are an important aspect of FFRD, as standard naming and descriptions may also be used to support queryable metadata through the Model Library. Consolidated requirements for naming conventions and project descriptions for HEC-RAS and other models are defined in Job Aid 06, *Standard Naming Conventions*.

8.1.7 COORDINATION WITH OTHER MODELS

Throughout model development and calibration, the hydraulic modeler coordinates with the technical lead, hydrologic modeler, and reservoir operations modeler to exchange input and output datasets as appropriate. Some typical points of this coordination are described below.

The *Model Linking Register* and *Name Table* developed during the scoping phase of the project will guide model development and all boundary condition setup. The hydraulic modeler and technical lead will communicate any concerns regarding model linking that may arise during the model development process with the full modeling team.

Note

Hydrologic subbasin delineation will be aligned with the hydraulic model unit delineation during the scoping and data preparation phase.

8.1.7.1 HYDROLOGIC MODEL

Early iterations of hydrologic model results may be insufficient to achieve calibration targets when they are applied to the hydraulic model. The hydraulic and hydrologic modelers will coordinate to determine the most appropriate model adjustments to achieve calibration targets for both models.

8.1.7.2 RESERVOIR OPERATIONS MODEL

Careful linking of reservoir operations outputs to hydraulic model inputs is an important aspect of FFRD modeling. The Model Linking Register will be developed during the scoping and data preparation phase, but any issues identified during the calibration or conformance phases regarding this mapping must be coordinated with the technical lead and reservoir operations modeler.

8.1.7.3 SYSTEM RESPONSE MODELS

The hydraulic model must include breachable elements (2D Connections) that are unique for each location allowed to breach. In the case of levee breaches, early results from the hydraulic model will inform breach locations associated with overtopping events. Therefore, breach locations impact the hydraulic model setup and must be communicated clearly between the hydraulic modeler and the system response modelers. Section 8.9 describes additional, specific dam and levee deliverables from the hydraulic modeler at each step.

The breach parameters (e.g., erodibility, width, development time) are set in the hydraulic model and should be coordinated between the system response modeler and hydraulic modeler. See Section 8.5.9 for breach parameter guidelines.

The names of the 2D Connections must be consistent with those used to deliver the system response tables. These names will be guided by the Name Table and the Model Linking Register and will require coordination between the system response modelers and the hydraulic modelers.

8.1.7.4 CONSEQUENCES MODEL

Review of hydraulic mapped results in relation to the structure inventory occurs during the calibration and conformance phases. This review fosters a collaboration that aims to improve both structure placement and hydraulic model configuration.

8.2 HYDRAULIC MODEL SCENARIOS

Descriptions of the hydraulic model scenarios are provided below along with some general context and phase occurrence.

8.2.1 HISTORIC CALIBRATION EVENTS

FFRD models are calibrated for six historic events that are selected during the scoping and data preparation phase. The primary focus of the hydraulic model development and calibration effort during the calibration phase of the project is achieving the calibration criteria for these events with a single geometry and parameter set.

Final delivered versions of these scenarios are posted to the */calibration/hydraulics/[model-unit]/* folder.

8.2.2 STRESS TESTS

Stress tests will be included as 1 to 3 plans within the HEC-RAS model and delivered as part of the calibration phase model deliveries. The primary purpose of the stress test scenarios during the calibration phase is to identify and address model instabilities that may occur in certain situations, resulting in a more robust and stable model for use in stochastic simulations.

The hydraulic modeler shall perform stress testing after initial calibration is completed. These stress tests should include simulation of very large events that may exceed the largest event of record. These tests should be performed assuming both a breach and non-breach condition of dams and levees, as breaches can often result in rapidly varied flow conditions, which present a stability challenge for hydraulic models. Stress test scenarios may be artificially created by scaling inflow hydrographs to produce very large flood events, such as those that overtop dams and levees. While overtopping tests are beneficial to evaluate model stability, breach stress testing should be performed when levees and/or dams are nearly fully loaded but not overtopping, which can create a critical scenario from a model stability perspective. An additional stress test should be run in which breach trigger elevations throughout the model are set to the structures' toe elevation. This is to verify breaches occur and can result in a highly submerged breach, which often highlights model stability issues. Scenarios with very low flows should also be stress tested.

Final delivered versions of these scenarios are posted to the */calibration/hydraulics/[model-unit]/* folder along with the historic calibration events.

8.2.3 MODEL INTEGRATION AND HISTORIC EVENT REPLICATION

During the conformance phase, the final calibration phase models are adjusted to function as a fully linked system (hydrologic model, reservoir operations model, and all hydraulic models) and to accept synthetic storm events. As an early step in the conformance phase, the historic events used for calibration are recomputed in the desktop environment through the fully integrated model suite to be sure historic events are adequately replicated. This step combines all hydraulic, hydrologic, and reservoir operations models in the study basin and is coordinated by the model integration lead for the project.

After completing this step on desktop, final delivered versions of these scenarios, with results are posted to the */conformance/event-validation/* folder.

8.2.4 CLOUD COMPUTE SIMULATIONS

During the conformance phase, after the models are integrated on desktop they are delivered for cloud computing simulations. A single set of hydraulic model files is delivered to the *conformance/hydraulics* directory and will be used for all stochastic simulations. There are review steps to verify models are delivered in the appropriate format and to confirm that cloud-compute produces appropriate results based on a few events before larger stochastic simulations with many events are computed. These review steps and processes are described in other sections.

Final delivered versions of hydraulic models that are fully prepared for cloud-compute are posted to */conformance/hydraulics/[model-unit]/*.

8.3 GEOMETRY

The following sections provide the standards and guidelines for development of HEC-RAS geometry for FFRD projects.

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8.3.1 HYDRAULIC MODEL CONSTRUCTION

The requirements, guidelines and general processes for construction of the hydraulic mesh are described in the sections below. Hydraulic models created for the FFRD program will include a single geometry with a single set of parameters.

8.3.1.1 MODEL EXTENT

The hydraulic mesh construction begins with the provided Hydraulic Modeling Unit Extent polygon that defines the study area for each model unit. The hydraulic model will be developed using fully 2D geometry. The mesh boundary conforms to the ridgeline separating adjacent hydraulic modeling units. A buffer should not be applied to the hydraulic modeling unit boundary to extend the boundary beyond the basin boundary, although it is good practice to extend the model further downstream of the study area within the existing floodplain to minimize the impact of approximate boundary condition estimates on results within the study area. Certain situations may require other adjustments to the model extent to follow the Model Linking Register, such as around dams that are near the upstream or downstream limits of the model unit.

8.3.1.2 USE OF ELEVATION DERIVED HYDROGRAPHY DATASET

The EDH dataset is produced during the scoping and data preparation phase and provided to the modeler. The EDH is considered the best source of stream alignment information and although some modifications may be needed, can be applied to certain steps in model development such as terrain modifications or mesh alignment along flow paths.

8.3.1.3 CATEGORIES FOR MESH DEVELOPMENT

The scope for level of detail to add to the hydraulic mesh is set during the scoping and data preparation phase. Layers that delineate the extent for each mesh development category are created during scoping and will be adhered to during model development (see Section 4.3.10). The general model treatment for each of the categories and the intended use of the outputs in each area are described in Table 8.1. More specific mesh development guidance is provided in the following sections.

Table 8.1. Model Treatment and Intended Use for each Mesh Development Category

Mesh Development Category	Modeling Treatment	Intended Use of Output
Background Mesh	No added mesh resolution. Maximum cell size that achieves computational stability and accurately routes excess precipitation over landscape. Terrain modifications when needed to hydraulically connect flow paths.	Produce basin-wide H&H data at a coarse resolution that may be leveraged for separate detailed modeling efforts. Provide approximate pluvial and fluvial flood information. Collect and route flows to downstream locations.
Urban Areas (polygons)	Increased 2D mesh resolution. Various features such as drainage ditches, road embankments, or pump stations may be accounted for directly.	Produce reliable pluvial and fluvial flood risk mapping results in populated areas.

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Mesh Development Category	Modeling Treatment	Intended Use of Output
Key Streams (lines)	Increased modeling detail to accurately route flood waters and produce accurate mapping results for the entire floodplain of key streams. Align cells with flow path of stream and account for bathymetry in terrain.	Accurate routing of flood water through larger streams. Accurate flood risk mapping from fluvial sources for the most critical streams in the basin.
Other Streams (lines)	At minimum, align cells with flow path of stream. Terrain modifications when needed to hydraulically connect flow paths.	Improve the accuracy of flood water routing through smaller streams. Moderately improve the accuracy of fluvial flood risk mapping along these streams.

8.3.1.4 MESH DEVELOPMENT GUIDELINES

This SOP aims to balance the need for comprehensive basin-wide coverage in 2D hydraulic models with the need for computational efficiency in a stochastic modeling framework. The 2D mesh will incorporate varying levels of detail based on the mesh development scoping layers. These layers are designed to strategically allocate higher mesh resolution to areas where increased model precision is needed or where improved accuracy is critical. Table 8.2 provides guidelines for mesh cell sizes corresponding to each mesh development category.

The recommended model development process includes starting with a relatively coarse mesh and iteratively refining as needed to achieve calibration targets and other criteria.

Table 8.2. Recommended Cell Sizes for Each Mesh Development Category

Mesh Development Category	Mesh Size (feet)
Background Mesh	100–900
Urban Areas	25–250
Key Streams **including adjacent floodplains*	25–250
Other Streams	100–900

Cell sizes should be selected to secure acceptable model accuracy (i.e., the model's ability to reproduce observed data), numerical stability (e.g., satisfying the Courant condition, minimizing iteration counts, volume conservation errors, and water surface elevation discrepancies), and computational efficiency (i.e., maintaining a reasonable total run time). Appropriate cell sizes should be informed by the characteristics of the area being modeled and may require iterative testing to optimize.

Modelers should approach mesh development with the goal of minimizing unnecessary detail while still meeting required standards for accuracy, stability, and efficiency. Because each modeling unit has unique features and topography that influence performance, engineering judgment is essential in selecting the background mesh resolution and the resolution within other scoped mesh development categories along streams, floodplains, and urban areas.

8.3.1.4.1 BACKGROUND MESH

The 2D computational mesh consists of a coarse background mesh in all areas not specifically scoped for higher resolution modeling. The primary modeling goal in these areas is to provide boundary conditions that can be leveraged for future more detailed studies and to provide approximate pluvial and fluvial flood information.

Hydraulic structures that are present in areas not otherwise scoped for additional modeling detail will also be present in the background mesh. Breaklines to represent major watershed features such as significant roadway embankments or ridgelines may be incorporated to improve model accuracy.

The coarse background mesh computation point spacing will generally vary from about 400-foot to 600-foot polygonal mesh. In cases of remote areas with relatively flat slopes, it is reasonable to have a larger point spacing, up to 900-foot polygonal mesh. But this is not common and should be used sparingly and results should be verified for stability and accuracy.

8.3.1.4.2 URBAN AREAS

The urban area polygons are obtained during the scoping phase of the project and provided to the modeler to define areas for additional mesh detail. These areas will include smaller cell sizes as well as breaklines representing roadways/hydraulic structures. Various other features such as major channels, drainage ditches, culverts, bridges, hydraulic structures, or pump stations may be included if needed to produce reliable flood mapping results. Terrain modifications will commonly be associated with urban area mesh refinements.

A mesh resolution of 150 feet can be used as a starting point for mesh resolution in populated areas. Significant professional engineering judgment should be used in model development for urban areas. Care is taken to account for these features while balancing model run times with model detail.

8.3.1.4.3 KEY STREAMS

The key streams dataset is identified during the scoping phase of the project and defines the streams to receive a high level of additional modeling detail. The key streams are provided to the modeler prior to the start of model development.

Key streams are modeled with increased cell resolution, including cells that are aligned to the stream channel. Additionally, the floodplain areas adjacent to these key streams—including those exposed to flooding during extreme events—are modeled with enhanced detail, following the same guidelines as those applied to urban areas. Modelers should utilize breaklines along key stream banks to ensure the 2D mesh respects high-ground terrain features. Because these high-ground bank features dictate when river flow overtops into the floodplain, they must be adequately resolved and represented within the 2D computational mesh.

The terrain also will be adjusted along key streams to account for the bathymetric elevations that are not part of the bare-earth terrain data. Terrain modifications to represent bathymetry may be sourced from either surveyed data or estimated using the methods described in Section 8.3.2.8.

8.3.1.4.4 OTHER STREAMS

The other streams dataset is identified during the scoping phase of the project and defines the streams to receive some additional modeling detail. The other streams are provided to the modeler prior to the start of model development.

Other streams are modeled to ensure the mesh follows the alignment of the channel. Additional cell resolution beyond the background mesh is not required. Incorporation of stream bathymetry is not

required. Terrain modifications to ensure the streams are hydraulically connected across embankments and bridges may be necessary.

8.3.1.5 HYDRAULIC STRUCTURES

Various hydraulic structures that affect flow patterns and flood extents may be present in the basin to be modeled. The intent is not to model all hydraulic structures in the basin, but to identify the features that may have a significant influence on the ability of the modeling system to accurately portray the probability of flooding in the basin. The requirements and guidelines for inclusion of specific types of structures are described in the subsections below.

8.3.1.5.1 DAMS

The modeling treatment of each dam for the entire basin is determined during the scoping and data preparation phase of the project and provided to the modeler prior to the start of model development. The Dam Scoping Table lists the dam configuration category for each dam to be applied in the hydraulic model geometry. Job Aid 01, *Scoping Processes*, describes the guidelines applied to develop the Dam Scoping Table. Job Aid 08, *Dam Physical Representation*, provides additional details regarding incorporation of the dams into hydraulic models.

The following sections describe the hydraulic model treatment of each dam configuration category to be applied for each category of dam.

8.3.1.5.1.1 DAM TYPE A—OPERATION MODELED OUTSIDE OF HYDRAULIC MODEL

The hydraulic model for Type A dams will include a 2D connection to represent the dam, but no outlets will be included, and a very high crest elevation will be entered to ensure no water will be allowed to pass.

Outflow boundary conditions will be applied to the hydraulic model and be sourced from either observed flows (for historic calibration events during the calibration phase of the project) or from simulated outputs from other models according to the Model Linking Register developed during the scoping and data preparation phase.

A terrain modification to remove the dam embankment may be necessary to ensure both the pool and tailwater of the dam are able to pass the routed flows without being inhibited.

Bathymetry of the pool upstream of the dam will be included as terrain modifications using the methods described in Section 8.3.2.8 and Job Aid 08, *Dam Physical Representation*. The storage capacity of the modifications will be cross-checked for consistency with the other models.

Additionally, terrain modifications may be needed to improve numerical stability for events that breach the 2D connection.

8.3.1.5.1.2 DAM TYPE B—OPERATION MODELED BY HYDRAULIC MODEL

The hydraulic model for Type B dams will include a 2D connection to represent the dam, with outlets, spillways, and overtopping of the crest to allow water to pass through the dam. Operations of the structures will be limited to basic operations (such as rating curves, elevation-controlled gates, or user-defined outlet curves). No complex rulesets to operate structures will be used.

A terrain modification to remove the dam embankment may be necessary to ensure both the pool and tailwater of the dam are able to pass the routed flows without being inhibited.

Bathymetry of the pool upstream of the dam will be included as terrain modifications using the methods described in Section 8.3.2.8. The storage capacity of the modifications will be cross-checked for consistency with the other models when applicable.

Additionally, terrain modifications may be needed to improve numerical stability for events that breach the 2D connection.

8.3.1.5.1.3 DAM TYPE C—SIMPLIFIED OUTFLOWS MODELED BY HYDRAULIC MODEL

Type C dams will be represented in the hydraulic model as a 2D connection. Outlet features and dam crest will be approximated based on the best available information. A terrain modification to remove the dam embankment may be necessary to ensure both the pool and tailwater of the dam are able to pass the routed flows without being inhibited. If Type C dams are also modeled explicitly in the hydrologic model, operations will be cross-checked for consistency with the hydrologic model.

Bathymetry of the pool upstream of the dam will be included as terrain modifications using the methods described in Section 8.3.2.8. The storage capacity of the modifications will be cross-checked for consistency with the other models when applicable.

Additionally, terrain modifications may be needed to improve numerical stability for events that breach the 2D connection.

8.3.1.5.1.4 DAM TYPE D—BASIC REPRESENTATION IN HYDRAULIC MODEL

Type D dams will be represented in hydraulic models with a breakline to represent the dam such that cell faces will hold the pool and release flows according to the underlying terrain. A relatively narrow notch terrain modification may be included to allow water to pass through the dam before spillway flow occurs.

Bathymetry of the pool upstream of the dam will not be included beyond data already available within the terrain surface.

8.3.1.5.1.5 DAM TYPE E—NOT MODELED

Type E dams will not be modeled with any detail in the hydraulic model.

8.3.1.5.2 LEVEES

FFRD hydraulic models will include a representation of all levees in the NLD. Levee alignments and profiles will be included in the models exactly as represented in the NLD. The NLD data for the entire basin will go through a data refinement effort for the entire study basin as part of the scoping and data preparation phase of the project, which will ensure the alignment and profile of the data is complete, accurate, and aligned with the project DEM. All NLD levees will be modeled as a hydraulic structure (2D connection) and will assume all closures are in place. The levee alignment and profile data will be provided to the modeler in an XYZM format to allow for easy import into HEC-RAS as described in Job Aid 09, *Levee Physical Representation*.

During the modeling development process, NLD levee elevation data should be compared with terrain elevation, aerial photography, street view images or other sources to evaluate and confirm accuracy of the data. Obvious errant data, closure structures, and lack of floodwall elevation data may require professional engineering judgment to establish a levee profile for use in the model. Any adjustments to either the NLD levee alignment or NLD crest profiles must be documented in the applicable report (Calibration Phase Report or Conformance Phase Report), which will include a table for commentary on each levee system in the model.

Increased mesh resolution and detail behind the levee will be included if required for model stability and accuracy around breach locations or if the leveed area is included in areas otherwise scoped for increased resolution (urban areas or floodplains adjacent to key streams).

8.3.1.5.2.1 LEVEE SYSTEM SEGMENTATION

Levees will be split into multiple 2D connections when necessary to allow for multiple breach locations to be assessed in a single system when called for by the levee data lead in the scoped approximate levee breach locations. Coordination with the levee data lead is required in terms of specific breach location confirmation and siting as described in Job Aid 11, *Levee System Response Development*.

8.3.1.5.2.2 DRAINAGE SYSTEMS ACROSS LEVEES

Culverts and pumps that have a significant effect on the inundation within the leveed area will be included in the model geometry. Locations, sizes, and capacities of the culverts or pumps across the levee are approximated using NLD data or other readily available data. The behavior of culverts through levees is typically modeled with flap gates or other appropriate assumptions best representing their function.

8.3.1.5.3 URBAN DRAINAGE SYSTEMS

Urban drainage modeling is limited to open channel drainage within urban areas. Subsurface urban drainage is not included in the 2D mesh.

It is recognized that subsurface urban drainage systems play a major role in pluvial urban flooding. Future techniques may allow for a more direct modeling method to account for these systems.

8.3.1.5.4 TRANSPORTATION FEATURES AND BRIDGES

Key hydraulic structures associated with transportation features significantly impacting inundation in populated areas or features that are needed to accurately reproduce observed data should be included in the 2D mesh. Engineering judgment is needed to identify the key structures to include. Transportation features in the background mesh category are of lower priority. Transportation layers from the U.S. Census TIGER dataset will be provided to the modeler to aid in identifying the key features.

When included, hydraulic structures associated with transportation features (e.g., bridges, roadway embankments) are modeled as breaklines and/or terrain modifications (not internal 2D connections) and best practices for modeling should be applied. Bridge approach embankments that act as significant floodplain constrictions should, at a minimum, be reflected within the 2D mesh using breakline or arc features. If bridge piers are judged to significantly influence flood characteristics, these piers should be approximated using vector terrain modifications and respected within the 2D mesh.

8.3.1.5.5 OTHER HYDRAULIC FEATURES

Modeling may include other unique hydraulic features within the mesh to capture key aspects of the flood characteristics of the area when deemed critical by the hydraulic modeler using professional judgment. Examples of such features that significantly influence flood inundation may include embankments, diversions, canals, weirs, saltwater barriers, bridges, floodway tunnels, or other structures. High-ground features that act as hydraulic controls and significantly impact flood inundation must be accurately represented. Modelers should align 2D computational cell faces along the high-ground ridge lines of these features.

8.3.2 TERRAIN CREATION

The following subsections describe the standards, guidance, and best practices for creation and storage of the RAS Mapper terrain for hydraulic modeling use.

8.3.2.1 DIGITAL ELEVATION MODEL

The full resolution base DEM provided for the modeling project (see Section 3.4) will be used as the basis for the RAS Mapper terrain.

8.3.2.2 TERRAIN VERSIONS

As described in Section 3.5, two distinct versions of the terrain will be produced.

The *modeling terrain* is the key terrain used for the development of the hydraulic model; it will retain the full resolution of the base DEM and follow the guidance in the sections below.

The *mapping terrain* is an exported, lower resolution terrain that will be used for development of hazard gridded outputs. This terrain is created at the end of the calibration phase as a deliverable.

8.3.2.3 TERRAIN STORAGE LOCATION

To allow for collaboration with the full team, all official HEC-RAS terrains for each modeling unit will be stored in the *terrain-modeling* directory. It is encouraged to set up the HEC-RAS model on your desktop with reference to the DEM that is above the HEC-RAS model directory during model development. The final HEC-RAS terrain is required to be posted to the *terrain-modeling* directory as part of each delivery step during the calibration phase (Section 8.9 describes specific delivery steps).

8.3.2.4 MERGED TERRAIN DELIVERY FOR MAPPING

The HEC-RAS modeler is required to deliver an exported reduced-resolution RAS Mapper terrain posted to the *terrain-mapping* directory. See Section 3.5 for standard cell sizes for mapping products. This reduced resolution terrain is necessary for efficient FFRD production of mapped output products that retain acceptable quality. This RAS Mapper terrain must be exported with rasterized modifications due to the fact that cloud-compute tools do not support terrains that include terrain modifications (as of late 2026).

8.3.2.5 USE OF TILED TERRAINS

Tiled terrains may be used for the RAS Mapper terrain, but they are not required.

8.3.2.6 MODIFICATIONS OUTSIDE OF RAS MAPPER

Modifications to the hydraulic model terrain may be made using tools outside of the RAS Mapper system provided they maintain the fidelity of the original source data and the edits are well documented.

8.3.2.7 BUILDINGS

Terrains used for FFRD will represent bare earth. Modification of the terrain to represent building/structures within the terrain is not performed. The hydraulic impacts of these features are included with Manning's "n" roughness values associated with urban areas.

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8.3.2.8 BATHYMETRY

The provided DEM is expected to represent top-of-water surface elevations and therefore will not include a representation of the below-water elevations that can have a significant effect on the hydraulic model results. Existing bathymetry information for rivers, lakes, and reservoirs should be incorporated into the RAS Mapper terrain when it is defined in the scope of the modeling effort or existing measured data is readily available. Various techniques for incorporation of bathymetry exist and are categorized below. For all techniques, care is taken so terrain modification only takes place below the water surface, and the existing above-water high-quality terrain is not modified by an errant slope projection or unrealistic stream bottom width.

8.3.2.8.1 GRIDDED BATHYMETRY FROM SURVEY DATA

In some instances, existing digital bathymetry based on field surveys may be available and will be obtained during the scoping and data preparation phase of the project. Other local sources of digital bathymetry identified during model development may also be considered for inclusion in the terrain. This data is incorporated into the terrain file used in RAS Mapper if it is determined to be a better representation of the terrain below the water surface and is consistent with the scoping document for the modeling project.

8.3.2.8.2 SURVEYED CROSS-SECTION DATA

Available surveyed cross-section bathymetry data should be incorporated if deemed to be of high quality. Although other workflows for incorporation of this data exist, it is commonly represented in existing 1D hydraulic models and accomplished by generating a terrain surface using the cross-section data between bank points and incorporated into the terrain as described in the HEC-RAS Mapper User's Manual.

8.3.2.8.3 ESTIMATE CHANNEL BATHYMETRY

When surveyed bathymetry is not available for incorporation into the terrain, channel bathymetry may be estimated. This should be done if either the stream was scoped to include estimated bathymetry (i.e., it is a *key stream*) or channel bathymetry is needed to achieve accurate flood routing to meet calibration targets.

Bathymetric parameters can be estimated using bankfull channel geometry regression equations as summarized in FEMA SO2 Innovations, *Bathymetry Sensitivity Testing* report (April 2024).

Alternatively, aerial photography and the existing terrain may be used to estimate the side slope and bottom width of the channel. Existing FIS profiles and existing stream gage rating curves may be used to estimate the channel invert. Professional engineering judgment may also be applied to estimate channel bathymetry along a reach. When observed data is available to support it, channel bathymetry (width/depth/slope) parameters should be iterated to achieve model calibration. The modeler may justify the selection of dimensions of estimated channels in the report by comparing known rating curves with model results.

The estimated bathymetry parameters may be incorporated into the terrain using RAS Mapper terrain modification tools. RAS Mapper terrain modification tools are limited to a single trapezoidal section per reach. Multiple reaches may be required to achieve the best estimate channel bathymetry and avoid unrealistic terrain modifications. Multiple terrain modifications can be stacked on top of one another to create more complex cross section shapes.

8.3.2.8.4 ESTIMATE RESERVOIR BATHYMETRY

Terrain modifications are required behind most significant dams in the basin, as defined on a per-dam basis in the Dam Scoping Table provided to the modeler. Although incorporating surveyed bathymetry behind the dam is the preferred modeling technique, in many instances, surveyed bathymetry data is not available and an estimate of the below water elevations within the reservoir is needed.

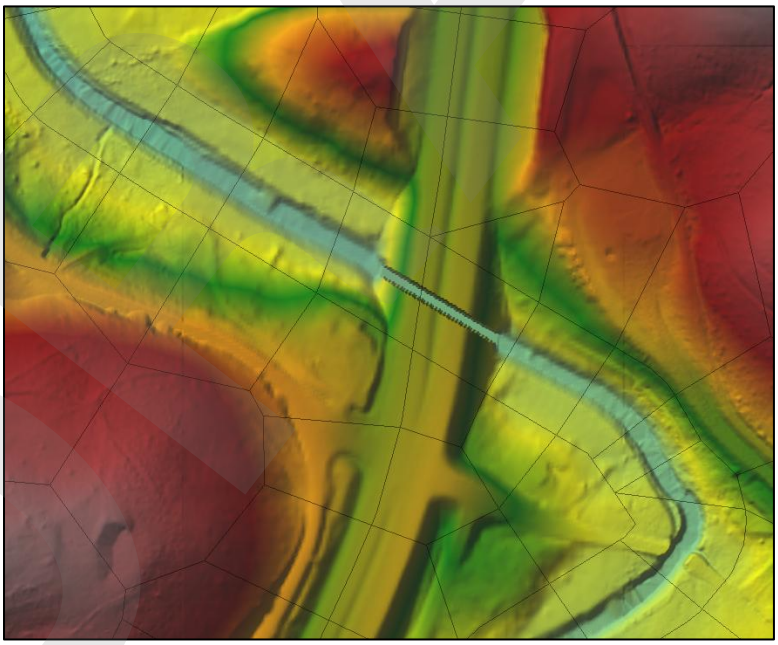
Terrain modifications can be stacked within RAS Mapper to develop a broad floodplain and incorporate a river channel as described in more detail in Job Aid 08, *Dam Physical Representation*. In some instances, terrain modification is required to decrease the footprint of the dam for model stability or to provide the capability within the model to evaluate a dam breach.

Volume comparisons with available elevation-storage relationships or NID data are made to confirm that the terrain modifications are representative of the volume in the reservoir. The resulting elevation-storage relationship within the terrain must be coordinated with the hydrologic model and reservoir simulation model to ensure each model uses identical relationships.

8.3.2.9 STREAM PATH CONDITIONING

Despite efforts to hydro-enforce DEMs provided at the start of the study, unrealistic damming effects at various locations within the watershed may require terrain modifications in HEC-RAS. This can be accomplished with the RAS Mapper vector terrain modification tools to lower the terrain across the feature based on the estimated size of the culvert or bridge opening (Figure 8.1). The result is a channel through the roadway embankment. This terrain modification method uses line vertices to assign an elevation to the terrain modification. Typically, a straight line beginning at the channel invert upstream of hydraulic structure and ending at the channel invert downstream of the hydraulic structure is used.

Figure 8.1. Typical Terrain Adjustment at a Major Road Crossing Preventing Unrealistic Damming Effects

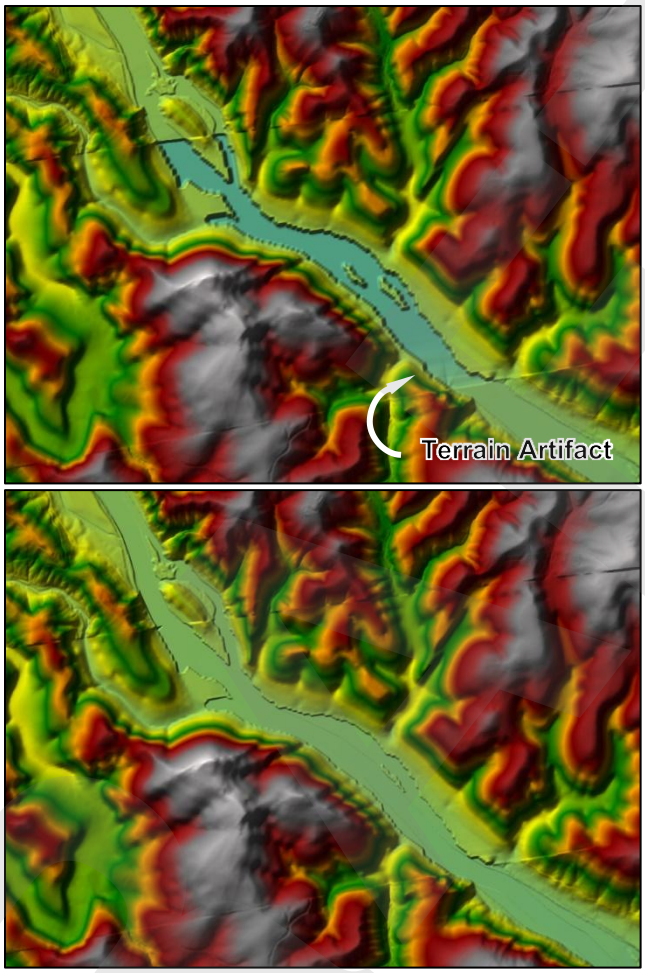


8.3.2.10 TERRAIN ARTIFACTS

In instances where a LiDAR tile is unavailable or corrupt, a coarse DEM tile is required to supplement the high-resolution data. This may result in holes or blockages in the channel that are modified using

professional engineering judgment (Figure 8.2). Additional artifacts may exist in the terrain that are reviewed and corrected using terrain modification tools to meet the objectives of the FFRD model.

Figure 8.2. Terrain Artifacts are Filled in to Match Higher Resolution Data



8.3.3 HYDRAULIC ROUGHNESS

Assignment of hydraulic roughness parameters to the hydraulic model is a key aspect of model calibration. The following subsections describe requirements and guidelines for roughness.

8.3.3.1 ROUGHNESS STORAGE LOCATION

Final hydraulic roughness base layer and any calibration regions will be stored within the hydraulic model unit directory. The path for roughness layer storage is *ffrd_[basin-name]/calibration/hydraulics/[model-unit]/*.

8.3.3.2 ROUGHNESS DATA SOURCES

The NLCD data will be obtained during the scoping and data preparation phase and provided to the hydraulic modeler prior to the start of model development. This dataset meets the minimum standard for the base roughness layer for use in FFRD models, although modifications are required along streams as described in the following sections.

Separate higher-resolution and higher-quality sources for land cover may be applied to specific projects if collection of these data sources was included in the scoping and data preparation phase of the project.

8.3.3.3 ROUGHNESS VALUES

Best practices within the hydraulic modeling community of practice must be used in assignment of roughness values in the model and engineering judgment will be required. Typical ranges for Manning's "n" values by NLCD land cover categories are suggested in the HEC-RAS user's manual and may be used as a guide during model development and calibration. For higher gradient natural channels, *Determination of Roughness Coefficients For Streams in Colorado* by Robert Jarrett should be considered for the assignment of roughness values in the model calibration process.

8.3.3.4 ROUGHNESS ALONG STREAMS

Accurate representation of hydraulic roughness along the streams in the basin is a key aspect of model development and calibration. Due to typical land cover dataset resolution, parts of channels can be categorized as forest or other land covers and be given the higher Manning's "n" value associated with that (incorrect) land cover unless appropriate edits are made.

The hydraulic modeler will incorporate the stream centerline layer into the land cover data layer along with classification polygons/calibration regions to represent all streams with drainage areas larger than 0.25 square miles within the modeling unit. The modeler will utilize the EDH stream centerlines that are provided in the scoping and data preparation phase for this step. Stream widths for each reach must be estimated for this step. Many methods for estimating stream widths or developing continuous bank lines within the hydraulic modeling community of practice are available and may be considered.

8.3.3.5 OTHER ADJUSTMENTS TO ROUGHNESS

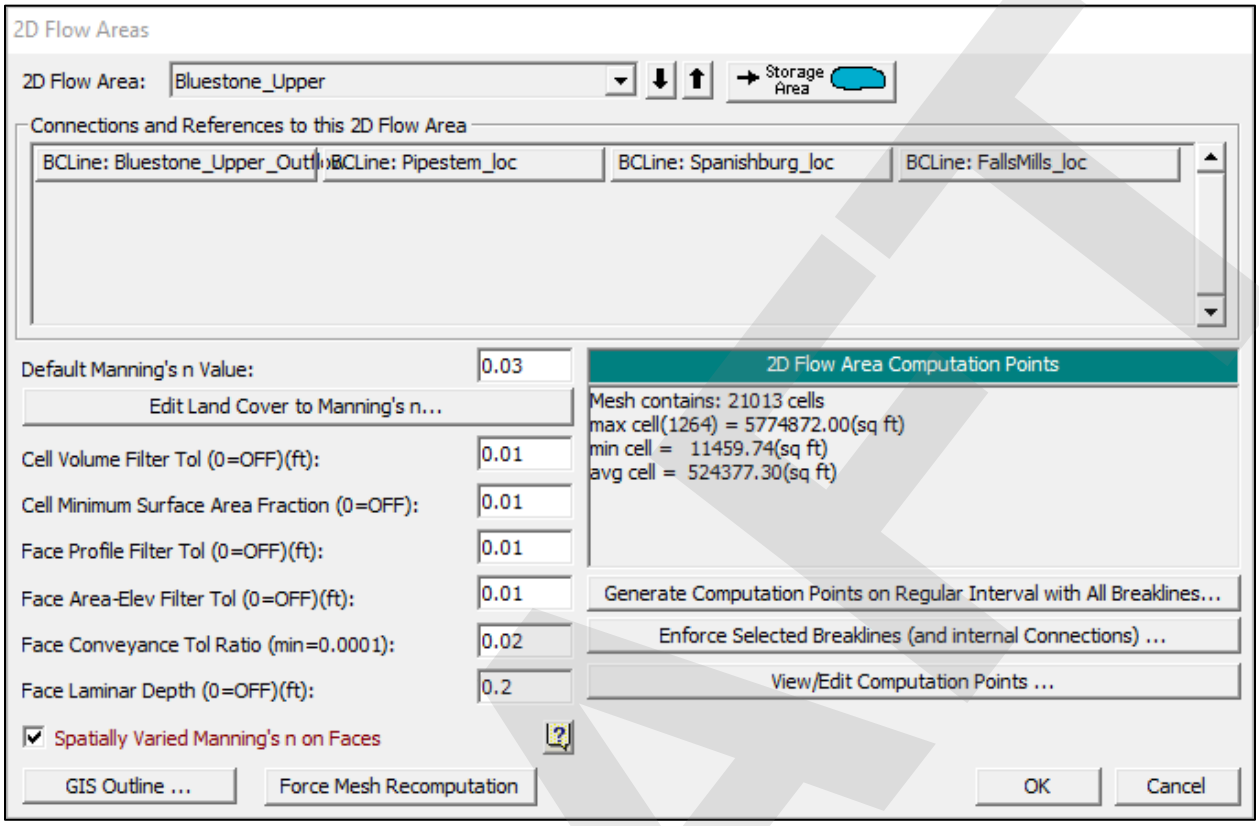
Other adjustments to the hydraulic roughness, such as *Land Classification polygons* or *Calibration Regions,* may be applied when appropriate according to best practices within the hydraulic modeling community of practice. These features are often useful for defining differing regions of the model domain during model calibration.

8.3.3.6 SPATIALLY VARIED ROUGHNESS ON CELL FACES

Manning's "n" values will be spatially varied across each face in the 2D mesh. The default method of selecting a Manning's "n" value in HEC-RAS is to use the "n" value from the land cover data at the face centroid. From the 2D editor (Figure 8.3), spatially varied Manning's "n" on faces can be activated to process depth-varied Manning's "n" values across each face based on the land cover data.

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Figure 8.3. Spatially Varied Manning's "n" Values on Faces in the Two-Dimensional Flow Area Editor



8.3.4 SAVED OUTPUT LOCATIONS

Reference lines and reference points are key model elements that can be used to extract information from the hydraulic HEC-RAS model. Specifically, these elements can extract variables including water surface elevation/stage, flow, and velocity. These elements are defined within the model geometry and must be incorporated prior to a simulation for the model to write output from these defined locations. When reference lines and points are defined, they write timeseries model output to both the project .dss file and plan .HDF file at the specified hydrograph output interval. Timeseries and rating curves of observed data can be assigned to reference lines and reference points in the Unsteady Flow Data editor; this allows for graphical plotting of both modeled and observed data in the Stage and Flow Hydrograph viewer.

Reference locations are discussed in the HEC-RAS Mapper User Manual. Refer to official software documentation for additional discussion and guidance on use of reference features.

The following sections describe the requirements and guidelines for the use of reference locations in FFRD hydraulic models. The saved output locations listed below are a required minimum, but it is acknowledged that other locations not listed may provide useful output and are allowed.

8.3.4.1 NAMING AND DOCUMENTATION OF OUTPUT LOCATIONS

Reference lines and reference points will be named according to the naming convention standards described in Job Aid 06, *Standard Naming Conventions*. Many specific output location names will be provided to the modeler in the Name Table that is developed during the scoping and data preparation phase.

Reference lines and points will be described in the hydraulic modeling documentation report such that what each reference feature is describing is well communicated. The name of a reference feature alone may not be sufficient to communicate what that feature is measuring and its intended purpose. Tables and descriptions of reference features in the report are valuable for model-to-model linking and integration of multiple models.

8.3.4.2 REFERENCE LINES

Reference lines are linear features used to extract modeled discharge and water surface elevation data. They are essential for model calibration, inter-model comparisons, and output summary data extraction.

8.3.4.2.1 REFERENCE LINE CREATION GENERAL GUIDELINES

Full Floodplain Coverage: Reference lines should be drawn such that they span the full width of the floodplain for large events. This ensures that all flow passing a gage location will be recorded by a reference line. If reference lines are drawn shorter, from only left-bank to right-bank over the channel for example, it is possible that flow during large events may flank around the reference line and not be recorded by the line. This can be complicated as USGS may not be measuring flow across the entire floodplain width during observed extreme floods, but best engineering judgment should be used.

Orientation: Where applicable, all reference lines should be drawn perpendicular to the flow direction, from left to right from the perspective of looking downstream. The flow is then appropriately calculated as a positive value when flowing downstream or a negative value when flowing upstream.

Alignment with Mesh: It is recommended that reference lines be enforced in the 2D computational mesh such that 2D cell faces are aligned to the reference line and perpendicular to the flow. This can be accomplished by co-locating and overlapping a 2D breakline, which is used to enforce 2D cell faces. Some manual manipulation of 2D cell points may be required to confirm cell faces are properly snapped to the breakline/reference line feature.

Near Structures: In some instances, there may be a need to measure discharge through an embankment, such as in the case of a dam. In this case, placing a reference line over the embankment will not capture discharge being transferred through the dam as part of a storage area/2D connection's culvert or gate. Therefore, the reference line should be placed downstream of the embankment, so all flows transferred through the structure pass over the line. The storage area/2D connection itself will also provide flow measurements through the embankment as a separate timeseries; however, a reference line is also recommended to provide consistency in how data is extracted from the model.

8.3.4.2.2 REFERENCE LINES AT STREAM GAGES

Reference lines are required at all observed gage locations. For calibration event modeling, these gage reference lines will include observed stage and flow data linked to the observed data .dss file in the Unsteady Flow Data editor. They will be drawn perpendicular to the flow in the stream channel (oriented from left bank to right bank) and be wide enough to capture the entire cross section including overbank flow.

For hydraulic model development in the calibration phase of the project, observed data should be linked to specific reference lines for the calibration events in the Unsteady Flow Data editor. The flow, stage, and rating curve information provided by reference lines at gages is the primary data to be compared with observed data for calibration and validation.

8.3.4.2.3 REFERENCE LINES AT MODEL DOWNSTREAM BOUNDARIES

Reference lines are required at the downstream boundary of each hydraulic model unit. These reference lines may coincide with the downstream boundary condition line or may be interior to the model if the model extends downstream to avoid error associated with an assumed boundary condition. They will be drawn perpendicular to the flow in the stream channel (oriented from left bank to right bank) and wide enough to capture the entire cross section including overbank flow.

8.3.4.2.4 REFERENCE LINES AT HYDROLOGIC MODEL SUBBASIN LOCATIONS

Reference lines are required at the downstream outlet of all hydrologic model subbasins and allow for inter-model comparisons. They will be drawn perpendicular to the flow in the stream channel (oriented from left bank to right bank) and wide enough to capture the entire cross section including overbank flow.

Identifying differences in runoff routing between the two models can provide insight into model performance and aid in calibration when there is difficulty achieving acceptable hydraulic model calibration metrics. Rating curves can also be extracted at these locations, which may be useful in the future to inform other modeling efforts.

8.3.4.2.5 REFERENCE LINES AT HYDROLOGIC MODEL JUNCTION LOCATIONS

Reference lines are required at all hydrologic model junction locations and allow for inter-model comparisons. They will be drawn perpendicular to the flow in the stream channel (oriented from left bank to right bank) and wide enough to capture the entire cross section including overbank flow.

Like the rationale for including reference lines at subbasin locations, identifying differences in runoff routing between the two models can provide insight into model performance and aid in calibration when there is difficulty achieving acceptable hydraulic model calibration metrics. Rating curves can also be extracted at these locations, which may be useful in the future to inform other modeling efforts.

8.3.4.3 REFERENCE POINTS

Reference points are like reference lines except they only provide information on water surface elevation and velocity at a single point, rather than calculating discharge flux across a line. They are essential for model calibration, inter-model comparisons, and output summary data extraction from stochastic simulations.

8.3.4.3.1 REFERENCE POINTS AT GAGE LOCATIONS

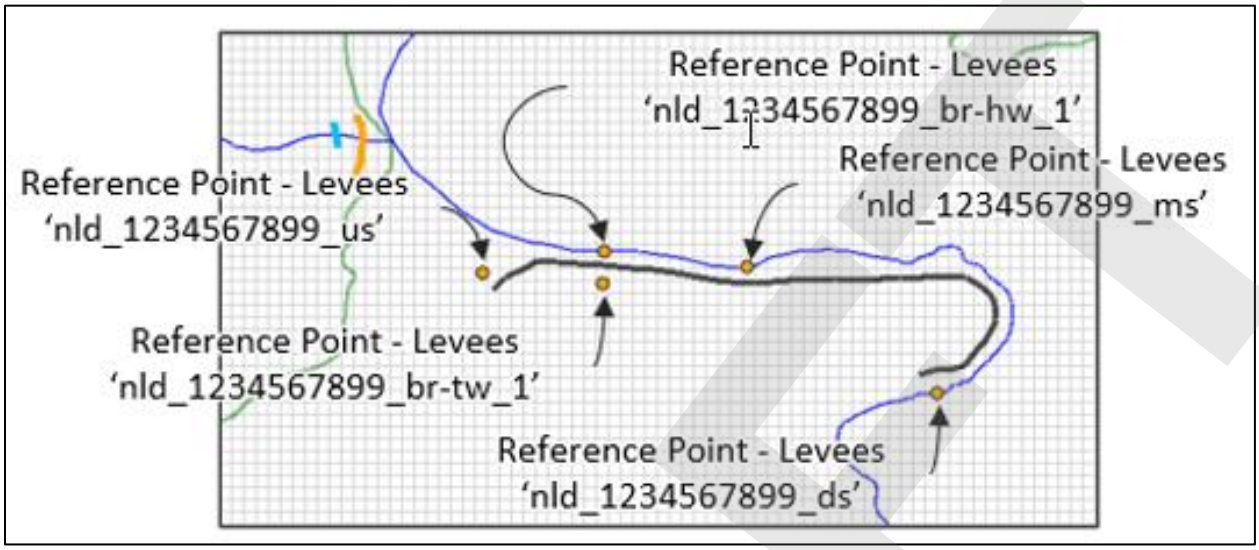
Reference points will be included at each observed data gage location in the basin and be located near the stream centerline. These reference points will overlay the reference line at the same gage location.

8.3.4.3.2 REFERENCE POINTS NEAR LEVEES

Reference points will be included along storage area/2D connections at levees located at the upstream end of the levee system, mid-point of the levee system, and downstream end of the levee system. Additionally, two reference points will be included at each levee breach location (riverside/headwater and landside/tailwater). Figure 8.4 is an example of a typical levee with five associated required reference points.

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Figure 8.4. Typical Reference Point Locations along a Levee with a Single Breach Location



8.3.4.3.3 REFERENCE POINTS NEAR RESERVOIRS

Reference points are recommended within reservoirs to extract water surface information regarding the lake. Specifically, these reference points should be in the deepest portion of the reservoir so that water surface information can be extracted even if the reservoir were to go completely dry. These points are useful in extracting information from the model, particularly in a cloud environment where extracting information from thousands of simulations is required.

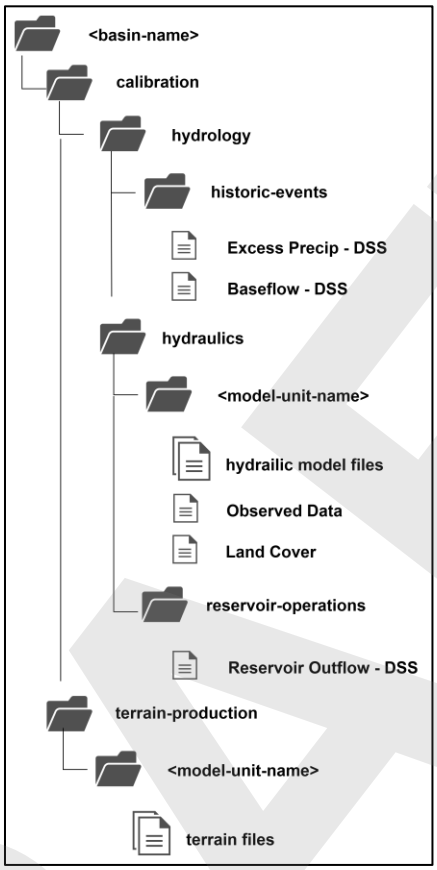
8.4 UNSTEADY FLOW DATA

Guidelines for hydraulic model configuration of all unsteady flow data for FFRD hydraulic models are described in the sections below.

In general, the hydraulic model should use relative references to linked data from their official storage locations in the Model Library according to Job Aid 07, *Model Library Use*, and characterized in Figure 8.5.

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Figure 8.5. Characterization of the Official Directory Structure to be Used for Relative References (Example Shown for Calibration Phase Directory)



8.4.1 BOUNDARY CONDITIONS

The following subsections describe boundary condition standards for FFRD hydraulic models.

8.4.1.1 BASEFLOWS

Baseflow hydrographs from each subbasin in the hydrologic model will be used in the hydraulic model as a flow hydrograph internal boundary condition. The baseflow boundary condition line is added to the HEC-RAS model within the stream channel near the HEC-HMS subbasin outlet from which it was developed.

During the calibration phase, the hydraulic modeler coordinates with the hydrologic modeler to obtain the appropriate baseflow hydrograph output corresponding to the appropriate historic event computed from HEC-HMS. The official storage location of the HEC-HMS DSS output at this stage is *calibration/hydrology/historic-events/*.

The boundary condition lines that represent baseflow locations will be named according to guidance in Job Aid 06, *Standard Naming Conventions*.

Baseflow boundary condition lines must be located near the downstream end of the subbasin. Baseflow data provided from HEC-HMS output is point flow data that is hydrologically routed to the subbasin outlet. Do not introduce baseflow in the middle or upstream portion of the basin; doing so would double-route the flow in the modeling system.

8.4.1.2 UPSTREAM INFLOWS

In non-headwater basins, upstream boundary conditions are required to represent the flow entering the basin. This flow is included in the hydraulic model as a flow hydrograph external boundary condition. The source of inflow (observed data or output from a separate model) will be defined during the scoping and data preparation phase (e.g., upstream hydraulic model output, observed data, other modeled output) and provided to the modeler prior to model development. The upstream inflow boundary condition lines will be named according to guidance in Job Aid 06, *Standard Naming Conventions*.

8.4.1.3 DOWNSTREAM BOUNDARY CONDITION

Downstream boundary conditions are set to appropriately model conditions at the downstream limit of the study area. Boundary conditions are (where possible) located where flow is 1D. In situations where the selection of the downstream water surface is uncertain, it is best practice to extend the model further downstream to be sure the uncertainty does not impact results in the study area.

For modeling of historic events during the calibration phase, observed stage data may be used as the downstream water surface if available, although hydraulic model units will typically be scoped to allow for use of observed data for calibration (i.e., stream gages will be within the modeled area rather than at the downstream boundary). Any observed data utilized will be based on the data provided for the project during the scoping and data preparation phase.

When no observed data is available for boundary condition use, normal depth or an estimated rating curve is generally used for models with riverine characteristics at the downstream boundary. When tidal conditions exist at the downstream boundary of the model, the modeler substitutes an appropriate tidal pattern based on available information. When the downstream conditions are lake-like, an appropriate known water surface may be used.

For conformance and production phase modeling, observed data boundary conditions will be replaced with an appropriate estimate or synthetic timeseries.

When tidal boundary conditions are present, it is acceptable to use Mean High Water (MHW) as a fixed water level. More sophisticated sampling of appropriate tidal boundary conditions for synthetic events is being considered for future versions of this SOP.

The downstream boundary condition lines will be named according to guidance in Job Aid 06, *Standard Naming Conventions*.

8.4.1.4 DAM OPERATIONS

The method for representation of each dam in the basin within the FFRD system of models is clearly defined during the scoping and data preparation phase of the project. The various methods for dam geometry representation and outflows within the hydraulic model are described in Section 8.3.1.5.1.

8.4.2 INITIAL CONDITIONS

The following subsections provide guidance on defining initial conditions for FFRD hydraulic models.

8.4.2.1 INITIAL CONDITION POINTS

The initial conditions definition in FFRD hydraulic models will generally be limited to initial condition points upstream of dams. Initial condition points provide a fixed water surface at the start of the event simulation. For the calibration phase models, initial condition points will be set to an appropriate value based on the

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historic event being modeled. For conformance and production phase models which simulate synthetic events, the initial conditions elevations will be sampled for each event.

Initial elevation points representing dam pools must use the *Fixed During Warmup* option (set in the unsteady flow file) to make sure the initial pool does not vary from the intended value when combined with a warm-up option (set in the plan file).

8.4.2.2 WARM-UP

A warm-up period (configured in the plan file) should be used in FFRD hydraulic models. The warm-up allows major rivers, particularly those downstream of inflow boundary conditions, to fill and align with the initial hydrograph flow values before the simulation starts. See Section 8.5.6 for guidance on warmup settings.

8.4.2.3 RESTART FILES

For models created during the calibration phase, restart files that provide comprehensive stage and flows at the initial timestep may be used to streamline the calibration process. However, restart files will not be applied to the conformance or production phase version of the models at this time due to the complexities associated with sampling of initial water surfaces for stochastic runs.

8.4.2.4 KNOWN WEAKNESS

The hydraulic model stochastic simulations executed following this SOP do not fully prime the landscape with water that may be present prior to an event, which may tend to bias results as some excess precipitation may collect within undrained areas and contribute to filling channels that start dry. The use of restart files has the potential to address this concern but additional tools and methods to effectively sample both reservoir elevations and initial conditions across the landscape and in all channels have not been fully developed as of October 2026. A project to develop a more advanced method for hydraulic model initial conditions sampling has been scoped and would be focused on utilizing a future version of HEC-RAS.

8.4.3 METEOROLOGICAL DATA—EXCESS PRECIPITATION TIMESERIES GRIDS

Runoff is modeled as excess precipitation sourced from HEC-HMS simulations and therefore no infiltration model will be applied to the hydraulic model. A gridded excess precipitation file for each event will be exported from HEC-HMS and included as a boundary condition in the unsteady flow editor as a meteorological variable.

During the calibration phase, the hydraulic modeler coordinates with the hydrologic modeler to obtain the appropriate excess precipitation output corresponding to the appropriate historic event computed from HEC-HMS. The official storage location of the HEC-HMS DSS output at this stage is *calibration/hydrology/historic-events/*.

8.4.4 OBSERVED DATA INPUT

It is recommended that observed flow, stage, and rating data be added to the HEC-RAS flow file to provide a useful quick reference during the calibration phase modeling and review efforts. This reference data must be removed from the flow files for the conformance and production phase models that simulate hypothetical events, although it is acceptable for rating curve observed data to remain if deemed valuable for evaluation of conformance and production phase efforts.

8.5 PLAN FILE—SIMULATION OPTIONS

The following subsections describe standards for the plan files including simulation options for FFRD hydraulic models.

8.5.1 TIME WINDOW

For calibration phase models that replicate historic events, all model simulations for calibration use a simulation start time and simulation end time that match the selected historic events defined during the scoping phase.

For conformance and production phase models that are delivered to cloud compute, an event duration will be determined by the technical team lead that is long enough to capture the peak flood response for all storm events throughout the basin. The specific simulation time window dates are not critical (as they are not used directly in cloud compute), and the technical team lead will define the standard time window for all team members to use for consistency. Note that the time window will be varied by cloud-compute for each modeled event.

8.5.2 COMPUTATION INTERVAL

The computation interval is set to a value that optimizes the model calibration, model stability, and model run-time. The Advanced Time Step Control (ATSC) option is considered during model development. Adjusting the timestep based on the Courant condition may reduce model run-times while maintaining model stability throughout the mesh. Based on the mesh refinement investigation, a starting computation interval of 30 seconds is recommended. When using ATSC, it may be necessary to have at least two halving steps to obtain model stability based on the wide range of cell sizes incorporated.

8.5.3 HYDROGRAPH OUTPUT INTERVAL

The hydrograph output interval is set to ensure the peak stage and flow are captured in the reference lines and internal 2D connections. Typically, this is set to a maximum 1-hour time interval or lower if calibration data warrants a more frequent output interval.

8.5.4 MAPPING OUTPUT INTERVAL

The mapping output interval should match the hydrograph output interval.

8.5.5 DETAILED OUTPUT INTERVAL

The detailed output interval is not critical for 2D FFRD studies. This can vary during the modeling process, which usually means a lower interval is used during calibration for model debugging and refinement, then a larger value is used for final production to reduce the size of the output files.

8.5.6 WARM-UP SETTINGS

The warm-up is a simulation that is performed prior to the model start time and contributes to the initial conditions that are used for the actual model simulation. For FFRD 2D models, the warm-up settings are defined in the *plan editor – options – computational options and tolerances – 2D flow options* window. The Initial Conditions Time and Initial Conditions Ramp Up Fraction variables should be set to values according to modeling best practices.

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Duration and ramp-up options should be set based on their effectiveness in achieving accurate model results. For conformance and production phase models, the additional runtime and compute costs associated with the warm-up option may be considered.

8.5.7 EQUATION SET

The diffusion wave equation set is preferable due to model run times; however, model performance should be compared with the Shallow Water Equations, Eulerian-Lagrangian Method (SWE-ELM) and Shallow Water Equations, Local Inertial Approximation (SWE-LIA) 2D equations. Model equation set should be tested prior to calibration using the initial parameters. If a significant difference is observed between the diffusion wave and SWE equations, model development should focus on developing a calibrated model using the SWE 2D equations. If model run time is excessively impacted by SWE 2D equations use and it is required for model accuracy, escalate to project leadership for guidance.

8.5.8 MATRIX SOLVER

The Parallel Direct Sparse Solver (PARDISO) solver is a robust solver and typically used as the default solver for 2D hydraulic modeling in HEC-RAS. Alternatively, the Successive Over-Relaxation (SOR) and Flexible Generalized Minimal Residual with Successive Over-Relaxation (FGMRES-SOR) solvers may provide computational speed improvements. FFRD models should utilize an appropriate solver that minimizes runtime while maintaining stability. Optimization of the matrix solver has potential to result in significant cost savings during the Production phase of modeling, when very large numbers of simulations will be performed.

8.5.9 BREACHES

Dam and levee breaches are an important aspect of FFRD modeling and must be incorporated into hydraulic models. A list of structures to be breached for the modeling effort is scoped for the entire basin in the scoping and data preparation phase of the project and provided to the modeler prior to the start of model development.

During the calibration phase, hydraulic models are developed to accommodate breaches that will be based on random sampled breach trigger elevations during the conformance and production phases. Specific tasks and general approaches for the hydraulic modeler related to breaching during the calibration phase are listed below.

- **Levee Control Location (LCL) Review and Coordination:** LCLs (also known as initial overtopping locations) are approximated during the scoping and data preparation phase by the levee data lead and are to be reviewed and confirmed by the hydraulic modeler during the calibration phase, with feedback provided to the levee data lead for potential adjustments to specific breach siting.
- **Breach Elevation Data:** Modelers must provide crest and toe elevation data for all breached structures. This is done through careful review of the DEM at each specified location and delivered and coordinated with the dam and levee data leads.
- **Segmentation of Storage Area/2D Connections:** Dam and levee structures are segmented to enable specific breach locations as defined in the scoping phase. Storage area/2D connections for dams and levees are segmented to allow for unique breach locations as scoped.
    - Breaches are entered at fixed locations, and only one breach per storage area/2D connection is allowed in HEC-RAS.
- **Input of Breach Parameters:** Breach parameters are entered into the plan file using the Breach Editor, based on the guidelines provided in subsequent sections.

- For historic event calibrations, breach locations will typically be included in the model but configured to not trigger (unless it is known that breach did occur during the historic event being replicated).

- **Model Stress Testing:** Hydraulic models must be stress-tested to assess numerical stability during breach scenarios. Dam and levee breaches often result in dynamic flow conditions and rapidly varied flow, which can cause hydraulic model instability. Stress testing should be performed in the hydraulic model to verify the model performs acceptably under the full range of expected breach conditions. Section 8.2.2 discusses this testing.
- **Breach Input Documentation:** The modeler must coordinate with the dam and levee data leads to ensure the official breach location listings include accurate information consistent with the final hydraulic model (see Section 9).
- **Structure Categorization for Breach Parameter Method:** Custom breach parameters will be created for each structure based on the structure type and characteristics. Table 8.3 indicates the breach parameter method to apply for various structure categories. The specific breach parameters for each of the breach parameter methods are detailed in the following sections.

Table 8.3. Structure Categorization for Breach Parameter Method

Structure Category	Breach Parameter Method
Earthen Embankment Levees	Simplified Physical
I-wall Levees	Simplified Physical
T-wall and L-wall Levees	Fixed Width
Any Floodwall Type Embedded in Earthen Embankment	Simplified Physical
Levee Closure Structures	Fixed Width
Earthen Dams greater than 30 feet in height	User Defined
Earthen Dams less than 30 feet in height	Simplified Physical
Concrete Dams	Fixed Width
Concrete Gravity Dams with Earthen Embankment Sections	Unique Cases
Other	Unique Cases

Additional information regarding dam and levee breaching can be found in the HEC-RAS user manual.

8.5.9.1 SIMPLIFIED PHYSICAL METHOD

The simplified physical breach method requires an input of breach widening rate versus flow velocity relationships that represent soil parameters (erodibility) and allows the breach development to be simulated based on modeled hydraulics (velocity, time).

Resulting breach dimensions are carefully inspected after the hydraulic simulation to validate if the method produces reasonable results for all scenarios and adjustments are made as necessary. Table 8.4 summarizes breach parameters for this method.

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Table 8.4. Breach Parameters for Simplified Physical Breach Approach

Breach Parameter	Value to Use	Notes and Selection of Values
Breach This Structure (yes or no)	Enabled during model development. Disabled upon delivery from calibration phase	Not all storage area/2D connections are to be breached. This table applies only to dams and levees selected for breach analysis. Additional discussion below.
Breach Method	Simplified Physical	Used for levees and dams less than 30 feet
Center Station	Varies	Additional discussion below
Max Possible Bottom Width	Varies	Additional discussion below
Minimum Possible Bottom Elevation	Varies	Additional discussion below
Left/Right Side Slope	0 (vertical)	Zero unless compelling reason otherwise
Breach Weir Coefficient	Varies	Additional discussion below
Breach Formation Time	N/A	Not required for the simplified physical breach method
Failure Mode	Piping	Standard value for FFRD study
Piping Coefficient	0.5	Default value
Initial Piping Elevation	Same as minimum bottom elevation	Standard value for FFRD study
Initial Piping Diameter	1 foot	Standard value for FFRD study
Mass Wasting Feature	Disabled	Standard value for FFRD study
Trigger Failure At	Water Surface Elevation	Standard value for FFRD study
Starting Water Surface	See discussion below	Additional discussion below
Simplified Physical downcutting velocity vs erosion rate relationship	Varies	Additional discussion below
Simplified Physical widening velocity vs erosion rate relationship	Same as downcutting relationship	Assume symmetrical breach development
Breach Repair	Blank	A blank value disables breach repair

8.5.9.1.1 BREACH CENTER STATION

The breach center station should be selected based on best available data, including risk assessments and levee screening tool analyses. Breach width cannot extend beyond the edge of the storage area/2D connection in which the breach is modeled. Therefore, the breach center station should be far from the end of a storage area/2D connection so that the length of the structure does not artificially limit the potential maximum breach width. Suitable locations for breach center stations include areas of known poor levee performance, incipient levee overtopping locations, and locations of maximum levee height.

8.5.9.1.2 MAXIMUM BREACH WIDTH

The maximum breach width is set to a value large enough such that it does not limit the breach development. A maximum width of 1,500 feet is suggested. If there are physical constraints that would limit breach development (e.g., tie into high ground or a well-founded structure), the maximum breach width is set accordingly. Maximum breach extents cannot extend beyond the storage area/2D connection they are modeled within.

8.5.9.1.3 MINIMUM POSSIBLE BOTTOM ELEVATION

The breach invert for a levee is set at approximately the interior land elevation adjacent to the levee at the specific breach location. The breach invert for a dam that is using the simplified physical method should be set to the elevation of the downstream streambed.

8.5.9.1.4 BREACH SIDE SLOPES

By default, breach side slopes should be set to zero (i.e., vertical side slopes) for levees and for dams less than 30 feet in height.

8.5.9.1.5 BREACH WEIR COEFFICIENT

The breach weir coefficient impacts the velocity through the breach and can also have significant impacts on the breach width when using the simplified physical breaching methodology. The weir coefficient for levees where the river is parallel to the levee and any breach will have an indirect flow through a breach should be set to a value of 2.0, which is a conservative estimate for lateral flows. For dams and for levees where the flow through the breach will be more direct and behaves similarly to a dam, the default weir coefficient of 2.6 should be used.

8.5.9.1.6 BREACH FAILURE MODE

For all scenarios, the failure mode should be set to piping due to complexities of switching failure modes in stochastic computing. This enables all scenarios, whether piping or overtopping, to initiate a breach.

8.5.9.1.7 INITIAL PIPING ELEVATION (PIPING MODE ONLY)

Initial piping elevation is set equal to the minimum possible bottom elevation, also known as the breach invert. If detailed information is available from a risk assessment or other reliable source, this value can be adjusted.

8.5.9.1.8 INITIAL PIPING DIAMETER (PIPING MODE ONLY)

The initial piping diameter will be set to 1 foot by default.

8.5.9.1.9 MASS WASTING FEATURE

The mass wasting feature will not be used for breach models and should be disabled.

8.5.9.1.10 TRIGGER OF BREACH

Breaches will be triggered based on a specified water surface elevation.

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During the calibration phase, historic events will typically be modeled with breaches disabled unless actual breaches were observed.

During the calibration phase, stress testing for model stability is also performed, and triggers may be set to critical conditions, such as the toe of the breach or near the peak stage of the hydrograph used for stress testing.

During the conformance and production phases, the trigger elevation will be set stochastically for each event within the cloud-compute environment. The hydraulic modeler will deliver the model for cloud compute with the trigger elevation set to the toe of the breach, which will be overridden during the cloud compute process.

8.5.9.1.11 EROSION RATE VERSUS VELOCITY RELATIONSHIPS

Erosion rate versus velocity relationships will be estimated using the methods annotated within the spreadsheet tool provided in FFRD Templates and Resources—*Simplified Physical Widening Rates*. Unless additional detailed information is available, an erodibility parameter (k_d) of 25 mm/hr/Pa should be used. When additional erodibility data is readily available (e.g., when levee risk assessment has been completed and includes erodibility information), a custom erodibility parameter can be applied.

Resulting breach widths are to be assessed through stress testing scenarios or other sensitivity tests. Any unrealistic breach widths should be addressed and could be used as rationale to adjust the erodibility parameter, maximum breach width or other assumptions. Engineering judgment is required for any adjustments and any deviation from the standard assumptions must be well documented.

8.5.9.2 USER DEFINED METHOD

Breach parameters for earthen dams with a height of more than 30 feet will be estimated from regression equations and entered in the HEC-RAS file as *user-entered data*.

An annotated breach equation spreadsheet (Tools—*Breach Parameter Regression Equations*) is provided in FFRD Templates and Resources to aid in the application of the regression equations. This tool also includes links for key references that support the methods and a calculator to estimate breach parameters. A variety of breach regression equations are computed and include MacDonald and Langridge-Monopolis, Froehlich (1995a), Froehlich (2008), and Von Thun and Gillette (1990).

A single breach width and development time will be used for all FFRD events. Inputs to the regression equations will assume the pool at top of dam (TOD) and the breach invert is set to the downstream streambed elevation.

Resulting breach dimensions should be carefully inspected and the most appropriate results selected for incorporation into the hydraulic models. Some equations may be easily eliminated from consideration if the breach parameters are larger than the dam embankment or produce too small of an opening given the size of the reservoir. Engineering judgment should be exercised for each dam, which may include knowing the condition of the dam, the core material, or other elements that may provide insight into the selection of the appropriate, realistic parameters. Appropriate breach parameters should produce breach dimensions and formation times such that at the completion of breach formation, flow velocities have dropped below a threshold where significant additional widening of the breach would not occur. If a breach regression equation predicts dimensions and formation times that result in a rapid breach where significantly large head differentials and flow velocities remain following the breach formation, these regression parameters may not be reasonable.

8.5.9.2.1 BREACH CENTER STATION

The center of the breach should generally be in the center of the embankment over the tallest section of the dam. If information is available from prior risk assessment studies, these can be used to inform breach locations.

8.5.9.2.2 BREACH BOTTOM WIDTH

The breach bottom width from the selected regression equation will be used in the hydraulic model. Breach widths will typically be 0.5 to 5 times the height of dam. However, wider breach widths may be reasonable, especially for dams with significant volume.

8.5.9.2.3 BREACH BOTTOM ELEVATION

The breach invert should be set to the elevation of the downstream streambed.

8.5.9.2.4 BREACH SIDE SLOPES

The breach side slopes should be set to the side slopes specified for the regression equation that was selected.

8.5.9.2.5 BREACH WEIR COEFFICIENT AND PIPING COEFFICIENT

A breach weir coefficient will be set to 2.6, and the piping coefficient will be set to 0.5.

8.5.9.2.6 BREACH FORMATION TIME

The breach formation time from the regression equation that was used for the breach bottom width selection should be used in the hydraulic model. Formation time of breach should typically fall in the range of 0.1 to 4 hours. However, longer formation times may be reasonable, especially for dams with significant reservoir volume.

8.5.9.2.7 INITIAL PIPING ELEVATION

Initial piping elevation is set equal to the breach bottom elevation.

8.5.9.2.8 BREACH FAILURE MODE

For all scenarios, the failure mode should be set to piping due to complexities of switching failure modes in stochastic computing.

8.5.9.2.9 TRIGGER OF BREACH

Breaches will be triggered based on a specified water surface elevation.

During the calibration phase, historic events will typically be modeled with breaches disabled unless actual breaches were observed.

During the calibration phase, stress testing for model stability is also performed, and triggers may be set to critical conditions, such as the toe of the breach or near the peak stage of the hydrograph used for stress testing.

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During the conformance and production phases, the trigger elevation will be set stochastically for each event within the cloud-compute environment. The hydraulic modeler will deliver the model for cloud compute with the trigger elevation set to the toe of the breach, which will be overridden during the cloud compute process.

8.5.9.2.10 BREACH PROGRESSION

The breach progression should be based on a sine wave as opposed to linear progression. The breach progression is defined as a fraction of total breach progression as a function of the total time. Breaches tend to start slowly, and finish asymptotically, with the maximum deepening and widening rate occurring in the middle of the breach. The sine wave more realistically approximates the progression of a breach.

8.5.9.3 FIXED WIDTH

For non-earthen structures such as concrete gravity dams, T-wall levees, or arch dams, breach parameters are estimated with a fixed width according to the guidance below.

8.5.9.3.1 BREACH CENTER STATION

The center of the breach should generally be in the middle of the concrete portion of the structure unless there are geometric constraints or detailed available information suggesting a different location.

8.5.9.3.2 BREACH BOTTOM WIDTH

For concrete gravity dams, the breach width is assumed to be half the length of the dam but not to exceed 600 feet, which would be equivalent to six 100-foot monoliths.

For concrete arch dams, it will be assumed the entire dam is breached. The breach width of an arch dam is the distance between the abutments.

For T-wall or L-wall levees, the breach width will be assumed to be 2 monoliths wide. If monolith width is not known, assume a 40-foot monolith width.

For closure structures on levees, the breach width should be set to the width of the closure.

8.5.9.3.3 BREACH BOTTOM ELEVATION

The breach invert should be set to the elevation of the downstream streambed.

8.5.9.3.4 BREACH SIDE SLOPES

The breach side slopes will be set to zero (i.e., vertical). As an exception, the breach side slopes for a concrete arch dam will be set to the slope of the valley walls.

8.5.9.3.5 BREACH WEIR COEFFICIENT AND PIPING COEFFICIENT

The breach weir coefficient will be set to 2.6, and the piping coefficient will be set to 0.5.

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8.5.9.3.6 BREACH FORMATION TIME

Concrete dam breaches typically occur near-instantaneously but for hydraulic model stability, the breach formation for both concrete gravity and concrete arch will be set to 0.1 hours.

8.5.9.3.7 INITIAL PIPING ELEVATION

The initial piping elevation is set equal to the breach bottom elevation.

8.5.9.3.8 BREACH FAILURE MODE

For all scenarios, the failure mode should be set to piping due to complexities of switching failure modes in stochastic computing.

8.5.9.3.9 TRIGGER OF BREACH

Breaches will be triggered based on a specified water surface elevation.

During the calibration phase, historic events will typically be modeled with breaches disabled unless actual breaches were observed.

During the calibration phase, stress testing for model stability is also performed, and triggers may be set to critical conditions, such as the toe of the breach or near the peak stage of the hydrograph used for stress testing.

During the conformance and production phases, the trigger elevation will be set stochastically for each event within the cloud-compute environment. The hydraulic modeler will deliver the model for cloud compute with the trigger elevation set to the toe of the breach, which will be overridden during the cloud compute process.

8.5.9.3.10 BREACH PROGRESSION

The breach progression should use a sine wave. However, given the rapid nature of a concrete breach, the breach progression is not anticipated to have a significant impact on the breach flood wave.

8.5.9.4 UNIQUE CASES

Certain structures to be breached are not easily categorized as earthen dams, earthen levees, or floodwall structures. For example, a dam may include both earthen embankment and concrete gravity dam features. In other cases, the design section at the breach site may be unique.

In these cases, engineering judgment must be used in selection of which section to breach. Some of the factors that may influence the decision could be known risk drivers, condition of sections, overtopping frequency, or other components. A sensitivity analysis may be warranted to determine impacts from a breach of each section. Once the section is selected, the breach parameters should be determined using an appropriate method described in the previous sections.

8.6 RAS MAPPER FILE

The RAS Mapper file should include references to all files that are required to execute the model on desktop. Additional reference layers may be included to assist with review efforts.

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8.7 CALIBRATION AND PERFORMANCE CRITERIA

The primary objective of hydraulic model development during the calibration phase is to ensure the model reasonably replicates observed data for the historic events at stream gage locations identified during the scoping and data preparation phase. This ability to match observed data from historic events provides confidence that the models will accurately predict the response from synthetic flood events.

In addition to replicating observed conditions, models must demonstrate numerical stability, and the resulting mapped outputs must be of sufficient quality. Objective criteria for model calibration, stability, and quality of mapped results are provided in the sections below. Tools available to aid in the evaluation of calibration are also provided in FFRD Templates and Resources.

8.7.1 CALIBRATION TARGETS

The following subsections provide specific targets for calibration to observed data.

8.7.1.1 STAGE AND FLOW TIMESERIES

Objective model calibration criteria for stage and flow timeseries for the duration of the flood event is based on standard model evaluation statistics with corresponding performance ratings. Model evaluation statistics include the coefficient of determination (R^2), Nash-Sutcliffe Efficiency (NSE), Ratio of the Root Mean Square Error to the Standard Deviation of Observations (RSR), and percent bias (PBIAS). Table 8.5 provides model performance ratings to apply to flow and stage for each calibration event for all locations where data is available. Tools provided in FFRD Templates and Resources may be applied to aid in the calculation of these calibration statistics.

All stream gages with observed data are categorized as either primary, secondary, or tertiary during the scoping and data preparation phase. This categorization guides the evaluation of these statistics as follows:

1. Primary gage locations are evaluated first. Statistical comparisons of computed and observed flow/stage at all primary locations must be in satisfactory (per Table 8.5) or higher range.
2. Secondary locations are evaluated second. Statistical comparisons of computed and observed flow/stage at all secondary locations must be in the satisfactory or higher range. If appropriate technical justification is provided and primary locations evaluate favorably, some leeway is granted.
3. Tertiary locations are evaluated last. Statistical comparisons of computed and observed flow/stage at all tertiary locations should be in the satisfactory or higher range. However, if secondary and primary locations evaluate favorably (and adequate hydrologic justification is provided), some leeway is granted.

If calibration metrics are found difficult to achieve, the underlying cause for the poor fit must be explained, discussed with the team, and well documented in the calibration phase report. This includes the pros and cons of the calibration statistics and how the model is reflective of the gage rating curve.

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Table 8.5. Model Calibration Performance Ratings

Performance Rating	Color Code	R^2	NSE	RSR	PBIAS
Very Good	Dark Green	R^2>0.65	0.65<NSE≤1.00	0.00<RSR≤0.60	PBIAS≤±15
Good	Light Green	0.55<R^2≤0.65	0.55<NSE≤0.65	0.60<RSR≤0.70	±15<PBIAS≤±20
Satisfactory	Orange	0.40<R^2≤0.55	0.40<NSE≤0.55	0.70<RSR≤0.80	±20<PBIAS≤±30
Unsatisfactory	Red	R^2≤0.40	NSE≤0.40	RSR>0.80	PBIAS>±30

8.7.1.2 SIMULATED FLOW ACROSS MODELS

During the calibration phase, simulated flow hydrographs from the hydraulic model will be compared to those from the hydrologic model at all coincident locations for each historic event. In areas unaffected by dam regulation, the hydrographs should reasonably align in terms of volume, timing, and peak flow. While no specific quantitative criteria are provided in this SOP, the alignment of simulated results across models should be sufficient to provide confidence in model performance. Engineering judgment must be applied to assess whether the hydrographs are in reasonable agreement.

8.7.1.3 PEAK FLOW

Peak flows at calibration locations must be evaluated versus both observed values and values from the hydrologic model, and significant differences must be addressed. Consideration should be given to local flow patterns and gage locations, as the specific extent of hydraulic model reference lines used to extract flow results may impact reporting of hydraulic model flow results.

8.7.1.4 PEAK STAGE

During the calibration phase, the simulated peak stage for each calibration event must be within 1 foot of the observed peak stage for that event. If the simulated result falls outside this 1-foot threshold, the calibration phase report must include a clear and specific rationale for the exception, supported by engineering judgment and relevant data.

8.7.1.5 DISCHARGE-ELEVATION RELATIONSHIP

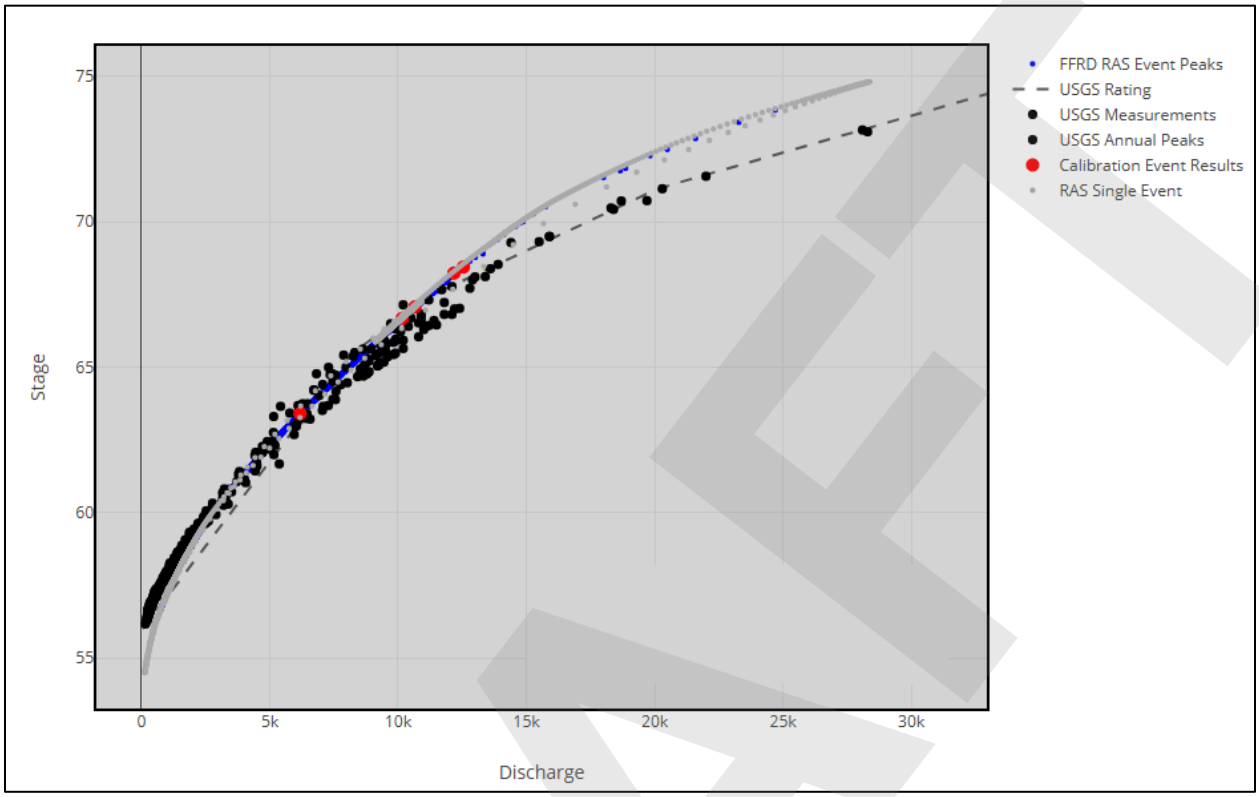
The modeled discharge-elevation relationship must be evaluated against available field observations and annual peak data at all stream gages. Simulated relationships may be extracted from model results at each timestep of the largest modeled event, as well as from peak values across each modeled event. While a statistical comparison is not required, the rating curve provides a qualitative measure of confidence that informs the calibration process.

The simulated relationship should generally fall within the scatter of observed data across the full range of flows and be within 1 foot of the largest observed peak stages. Greater emphasis should be placed on how well the model captures out-of-channel flows and large flood events. During qualitative evaluation, the age of observed data and any significant watershed changes since those measurements may be considered.

If the simulated results deviate from observed data, the calibration phase report must include a clear, specific rationale for the exception, supported by engineering judgment and relevant supporting data.

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Figure 8.6. Graphical Discharge-Elevation Relationship



8.7.1.6 OTHER OBSERVED DATA COMPARISONS

An attempt is made to identify additional observed flood event data that could aid in the model calibration during the scoping and data preparation phase. Additional data may include high-water mark data, satellite imagery during floods, or photographs. When available, qualitative comparisons to this data should be made and may support adjustments to model parameters in some cases.

8.7.1.7 UNGAGED BASINS

Ungaged modeling units should use hydraulic parameters developed using guidelines from literature, data from other hydrologically similar modeling units within the basin, and professional engineering judgment. Probabilistic results from the conformance phase can be compared with published regional regression estimates where available.

8.7.2 QUALITY OF MAPPED RESULTS

The mapped depth and velocity outputs from any given hydraulic modeled event must be of sufficient quality and resolution. The standard for mapping quality is higher in urban areas and along key streams and their floodplains as defined by the mesh development criteria during the scoping and data preparation phase. At a minimum, depth maps along perennial streams should appear connected.

Map rendering mode in RAS Mapper can have a significant impact on the appearance of mapped results for individual events in the desktop environment. However, FFRD risk products are computed based on depth grids generated with the cloud-compute tools that specifically reference the *Sloping (Cell Corners + Face Centers – Depth Weighted Faces 'Precip Mode')*.according the this SOP. Consideration will be given to adjustments to the rendering options in future versions of this SOP.

Mapped results are compared against structure placements in the structure inventory, which can aid in identification of hydraulic model issues or structure placement issues. Addressing issues identified requires coordination with the consequences team member and may result in adjustments to the structure inventory or hydraulic model as appropriate. These steps are also referred to in Section 10 and in Job Aid 14, *Consequence Processes*.

8.7.3 MODEL RUNTIME CRITERIA

Model runtime is an important consideration in hydraulic model development and runtime must be minimized. The recommended model development process starts with a relatively coarse mesh that is iteratively refined as needed to achieve numerical stability and acceptable quality of mapped results. Efforts should be made to increase model stability and reduce runtimes as much as is practical, utilizing best practices within the hydraulic modeling community of practice.

8.7.4 MODEL STABILITY CRITERIA

HEC-RAS models should compute minimal computational errors for all scenarios. The tighter the error tolerance, the more confidence there is in hydrograph timing and peak flow routing. Evaluation of the following stability metrics must be performed.

- Volume Accounting Error: Error should be less than 3 percent and is required to be less than 5 percent for all scenarios. Volume errors within 1 to 2 percent are considered very good for 2D modeling. Values less than 5 percent are reasonable if the spatial mapping aligns with expectations.
- Maximum Water Surface Elevation Error: The error for maximum water surface is generally considered as more of a tolerance. Tolerance of less than 0.05 feet is generally considered high accuracy while a tolerance of less than 0.1 feet is considered reasonable. When water surface elevation tolerance exceeds about 0.1 feet, this often suggests numerical instability and is not considered reasonable. The spatial extent of the maximum water surface errors may also be considered, as isolated instabilities may be tolerable.
- Cell Water Surface Error: Cell water surface error is computed by a comparison of the change implied by the computed water surface. Acceptable levels of error are consistent with the maximum water surface elevation error designation and values less than 0.1 feet are considered reasonable and acceptable.
- Cumulative Maximum Iterations: Maximum iterations are an indication of whether the model is converging. Convergence is measured by continuity and water surface error. Maximum iterations exceeded should be rare and continuity errors less than 5 percent and cell water surface errors less than 0.1 feet are acceptable.

8.8 DOCUMENTATION OF HYDRAULIC MODELING

A set of standard reports is written for each phase in the FFRD project. FFRD Templates contains report templates for the hydraulic modeling reports.

Table 8.6 summarizes locations of key hydraulic modeling documentation included in each of the reports.

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Table 8.6. Locations of Key Hydraulic Modeling Documentation

Report Title	Description
Scoping and Data Preparation Phase Report	Documentation of model unit delineation and many other scoping decisions for model development and calibration
Calibration Phase Report	Primary documentation of hydraulic model development and calibration
Conformance Phase Report	Description of all hydraulic model adjustments made to accept synthetic SST events and any adjustments to reasonably align with observed frequency data
Production Phase Report	Descriptions of any hot fixes or patches required in the production phase.

8.9 HYDRAULIC MODEL REVIEW PROCESS AND DELIVERABLES

Model review is a critical part of the FFRD production process. Several review steps are required to verify the final model delivered can effectively run in a stochastic environment and is capable of accurately depicting basin-wide flood risk on a broad scale. Review checklists are used to track comments regarding pre-defined aspects checked throughout the review process. FFRD Templates contains checklist templates.

The following subsections describe specific review steps.

8.9.1 SCOPING AND DATA PREPARATION PHASE REVIEW

The model unit delineation, Model Linking Register, Name Table, and all other scoping and data preparation phase products are reviewed and documented in the Scoping and Data Preparation Checklist, which requires multiple reviewers and is described in Section 4.

8.9.2 CALIBRATION PHASE REVIEWS

The following reviews are performed during the calibration phase.

8.9.2.1 INITIAL PROGRESS REVIEW WEB MEETING

The purpose of this early review is to confirm progress is being made toward delivery for review and for reviewers and team leads to provide support to modeling staff to verify that full guidance has been provided and that model development adheres to FFRD standards. Model data is typically not transferred during this review, but webinars are held to show progress and discuss possible issues using screen sharing.

Additional check-ins during model development may be useful and are encouraged as best practices.

8.9.2.2 QUALITY CONTROL REVIEW

The deliverable will be posted to the Model Library calibration directory and will include:

- Model:
    - Model file posted to the *calibration/hydraulics/[model-unit]/* directory

- Files must be present in the Model Library unzipped
- Working calibrated model files for all calibration/validation events including all files required to run and replicate the results.
- Do not include non-final extraneous plans in the model or output files or records associated with out-of-date runs.
- Do not include terrain inside the hydraulics directory
- Scenarios (plans) to be included:
    - Each historic calibration and validation event
    - Stress test events
- Output files required:
    - HDF output files
    - DSS output files (single DSS for all scenarios)
    - Do not include saved maps (i.e., depth grids) in the review package.
- Breach Data. Draft table including the model unit name, HEC-RAS 2D connection name, crest, and toe elevation for each breach location that are present in the HEC-RAS model.
- Documentation of the confirmation of the LCLs and coordination with the levee data lead.

- RAS Mapper Terrain:
    - Post draft final RAS Terrain to the *terrain-modeling* directory
    - Do not include the terrain directly in the *hydraulics* directory
    - Do not create duplicate or alternate versions of the terrain used for HEC-RAS model but confirm that the version in *terrain-modeling* is current. Optionally, outdated versions of the terrain may be moved to an *archive* subdirectory within *terrain-modeling*.
    - Do not provide alternate iterations of the terrain. Only the draft final terrain is to be delivered.

- Report: Draft calibration phase report hydraulic modeling section.
- Checklist: Hydraulic model review checklist with modeler comments/notes included for every review item. This verifies that all aspects of the model to be reviewed have been addressed by the modeler prior to review.
- Reference material: As needed.

8.9.2.3 BACKCHECK OF QUALITY CONTROL COMMENTS

After QC comments have been provided by the reviewer and addressed by the modeler, the updated deliverable will be posted to the Model Library *calibration* directory, and the previous version will be moved to the archive subdirectory. This delivery will include the following:

- Model: Working calibrated model for all calibration/validation events
    - Files must be present in the Model Library unzipped
    - Working calibrated model files for all calibration/validation events including all files required to run and replicate the results
    - Do NOT include non-final extraneous plans in the model or output files or records associated with out-of-date runs
    - Scenarios (plans) to be included
        - Each historic calibration and validation event
        - Stress test events
    - Output files required

    - HDF output files
    - DSS output files (single DSS for all scenarios)
    - Do not include saved maps (i.e., depth grids) in the review package.
  - Breach Data. Table including the HEC-RAS 2D connection name, crest, and toe elevation for each breach location. Documentation of the confirmation of the LCLs and coordination with the levee data lead.

- RAS Mapper Terrain:
  - Post draft final HEC-RAS Terrain to the *terrain-modeling* directory
  - Do not include the terrain directly in the *hydraulics* directory
  - Do not create duplicate or alternate versions of the terrain used for HEC-RAS model but confirm that the version in *terrain-modeling* is current. Optionally, outdated versions of the terrain may be moved to an *archive* subdirectory within *terrain-modeling*.
  - Do not provide alternate iterations of the terrain. Only the draft final terrain is to be delivered.

- Mapping RAS Mapper Terrain: Merged exported version of HEC-RAS Terrain at 12-foot resolution with rasterized modifications to the *terrain-mapping* directory.
- Report: Final modeling report.
- Checklist: Hydraulic model review checklist with Resolution of Reviewer Comments by Modeler section completed for all comments.

8.9.2.4 FINALIZATION OF CALIBRATION PHASE DELIVERY

Any necessary additional iterations of delivery updates to fully close the review will be similarly posted to the *calibration* directory, and the previous version moved to archive. This is the final calibration phase model that will not be edited going forward in the FFRD process.

8.9.3 CONFORMANCE PHASE REVIEWS

The following reviews are performed during the conformance phase.

8.9.3.1 PRE-CONFORMANCE REVIEW

Preparation for hydraulic model use in conformance phase requires a review. After the calibration phase hydraulic model review is closed out, it will be saved and remain in its directory location and be the official calibration model. Note that the Calibration Phase Review Checklist includes items that focus on ensuring the transition from historic event-based calibration to stochastic event modeling will be seamless.

All constituent models posted to the *conformance* directory are assessed by the project's model integration lead for interoperability aspects of the project, and comments are provided for the modeler to address.

8.9.3.2 CONFORMANCE PHASE REVIEW

After the calibration phase is complete, the conformance phase begins, and ownership of the model will transition to a smaller model integration team who will promote a copy of the calibrated models to the *conformance* directory. Certain changes to the model will be made to confirm synthetic events can be executed successfully. Section 12 describes these steps in detail.

Once the backcheck review is complete and any model integration items are addressed, the modeling technical lead determines if additional items need to be addressed or if the model is complete and ready to proceed to be applied to the conformance phase of the project.

SECTION 9—DAM AND LEVEE SYSTEM RESPONSE MODELING

This section focuses on dam and levee system response modeling. The breach locations and system response of dams and levees when loaded are important to the modeling effort. Siting specific breach locations for dams and levees, development of system response probability (SRP) data, and sampling approaches for breaches are described here. Other related topics, such as physical representation of dams and levees in the models and selection of breach parameters (size, development time), are covered in other sections.

9.1 DAM AND LEVEE MODELING CONTEXT AND STANDARDS

This section outlines the dam and levee system response process and standards necessary to meet overall FFRD objectives. FFRD seeks to map residual risks behind dams and levees. The objectives of FFRD will be met via basin-level study using available information on the dams and levees in the study area. The approach combines robust statistical methods, practical engineering judgment, and efficient computational strategies to deliver on FFRD goals. It is acknowledged that data gaps exist, and probabilistic dam and levee modeling is a developing research area.

Dam and levee system response models produced following this guidance will have the following characteristics:

- Ability to capture dam and levee performance to reasonably reflect their impact on flood risks. All levees in the NLD and select dams from the NID with significant impact to flood risk will be included.
- Utilization of system responses sourced from available performance data, including the most recent USACE risk assessment.
- Utilization of dam and levee crest elevations from an elevation source with equivalent or higher resolution than the study terrain (Section 4).
- Breach parameters consistent with the type of flood risk management feature (e.g., earthen embankment versus floodwalls for levees). Section 8 contains more detail.
- Selection of modeled dam and levee resistance (trigger elevation) statistically sampled from the developed system response curves for each modeled event.

9.1.1 ROLE OF U.S. ARMY CORPS OF ENGINEERS

USACE maintains reports and data associated with risk assessments for dams and levees within its authorities and has a key role in the management of national inventory databases for all dam and levee infrastructure. USACE personnel are staffed within the FFRD National Implementation Management Group to support basin modeling teams with identification and development of data to support incorporation of dam and levee breaches into FFRD modeling projects.

9.1.2 PHASES OF BREACH MODELING

The system response models are developed for an FFRD project in phases according to the workflows described in Section 2 of this SOP. This system response section applies to multiple phases, and high-level system response modeling tasks in each phase are described below.

- Scoping and Data Preparation Phase: This phase verifies dam and levee inventories are adequately refined to support scoping of the FFRD approach for the basin. In this phase, the team collects and inventories existing risk assessment data. The team develops a comprehensive dam modeling scope, including the list of dams to incorporate in system response modeling.

Levee breach locations are sited, including draft initial overtopping locations. The Name Table and Model Linking Register are documented.

- Calibration Phase: SRP tables are developed. Levee breach locations are verified, including location of initial overtopping. The hydraulic modeler incorporates breaches to prepare for sampling that occurs in future phases. Levee breach sampling scheme is defined and documented. System response curves are developed.
- Conformance Phase: Breach trigger elevations for each stochastic event in a multi-year realization are sampled within the fragility-curve-plugin cloud-compute plugin. Hydraulic model and reservoir operations models are computed for each event. Breach occurrence frequency for the simulated realization is assessed to verify acceptable results.
- Production Phase: Computation of multi-realization events. Frequency of breach occurrence over multiple realizations is assessed to verify acceptable results.
- Reporting Phase: In this phase, results from the project are compiled and delivered.

9.1.3 SOFTWARE VERSION

The breach trigger elevations for stochastic simulations are produced by the *fragility-curve-plugin*, version 1.0.6, which is part of the FFRD cloud-compute toolset that uses the provided SRP tables and breach location names as inputs.

9.1.4 NATIONAL DATABASE REFINEMENTS

Levee data in the NLD and dam data in the NID are validated as part of a data refinement effort early in FFRD's scoping and data preparation phase. This is a critical step to verify that the FFRD project is based on an accurate inventory of dam and levee infrastructure.

Guidelines and processes for these data refinement activities are referred to in Section 4.3.

9.1.5 DATA PROVIDED TO SYSTEM RESPONSE MODELERS

A comprehensive shared dataset for the project is developed during the scoping and data preparation phase of the project. Scoping of dam and levee inclusion in the model, initial breach siting, and assembly of existing risk assessment data are all primarily the responsibilities of the dam and levee data leads and occur during the scoping and data preparation phase concurrently with development of several key scoping deliverables.

The following shared datasets also produced by the scoping and data preparation phase may be relevant to the dam and levee data lead duties:

- DEM: Full resolution DEM for consistent use for the entire FFRD project.
- Model Linking Register
- Comprehensive Name Table.
- Refined NLD inventory of levees in the basin.
- Key levee data in the NLD.
- Refined NLD alignment and profile data.
- Levee operations manuals when available.
- LCLs (first overtopping points for the levee system and/or segments provided by the hydraulic modeler during calibration phase).
- Dam Scoping Table.
- Refined list of NID dams for the basin, including key physical attributes.
- Key dam data in the NID.
- Dam operations manuals when available.

9.1.6 COMPREHENSIVE NAMING CONVENTIONS

The Name Table for the entire modeling effort is established during the scoping and data preparation phase. A Name Table and Model Linking Register will specify the names for dams and levee model elements according to the guidance in Job Aid 06, *Standard Naming Conventions*. System response deliveries will include references to these established names.

9.1.7 COORDINATION WITH OTHER MODELS

Coordination with the hydraulic model, reservoir operations model, and cloud-compute inputs is required to successfully simulate the occurrence of dam and levee breaching in the FFRD modeling process. Some typical points of this coordination are described below.

The Model Linking Register and Name Table developed during the scoping and data preparation phase of the project will guide model development. The dam and levee system response modelers and technical lead will communicate any concerns regarding model linking that may arise during the model development process with the full modeling team.

9.1.7.1 HYDRAULIC MODEL

The hydraulic model must include breachable elements (2D connections) that are unique for each location allowed to breach. In the case of levee breaches, early results from the hydraulic model will inform breach locations associated with overtopping events. Therefore, breach locations impact the hydraulic model setup and must be communicated clearly between the hydraulic modeler and the system response modelers.

The breach parameters (e.g., erodibility, width, development time) are set in the hydraulic model and should be coordinated between the system response modeler and hydraulic modeler. Section 8.5.9 contains breach parameter guidelines.

The names of the 2D connections must be consistent with those used to deliver the system response tables. These names will be guided by the Name Table and Model Linking Register and will require coordination between the system response modelers and the hydraulic modelers.

9.1.7.2 RESERVOIR OPERATIONS MODEL

The reservoir operations model accounts for breaches of dams within its domain by adjusting operational releases when a breach is triggered according to the simulated pool elevation. The system response modeler can review model results for events that include breach occurrences on major dams.

9.1.7.3 *CLOUD COMPUTE* LEAD

The system response modelers provide inputs that are directly utilized by cloud-compute plug-ins. This SOP provides templates and standard locations for data delivery for cloud-compute and review processes to avoid possible pitfalls.

9.2 DAM SYSTEM RESPONSE MODELING

The methodology for development and sampling of dam system responses in the FFRD systems model is established for the project to reflect the best available risk assessment data for all significant dams in the basin.

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9.2.1 SELECTION OF DAMS TO BREACH

The inventory of every dam in the basin will be developed based on the NID database, and the modeling effort for each dam will be defined in the Dam Scoping Table. This table will specify which dams will be allowed to breach, following guidelines in Job Aid 01, *Scoping Processes.*

9.2.2 DAM RISK ASSESSMENT DATA

All existing dam risk assessment data and reports will be assembled during the scoping and data preparation phase. Processes and sources for obtaining this data are included in Job Aid 02, *Data Preparation Processes*.

9.2.3 DAM BREACH LOCATION

Dam breaches for FFRD should be consolidated into a single representative breach location, which is generally considered adequate for flood risk characterization at the basin scale. Unique dam configurations or other circumstances may warrant a more complex treatment of certain dams, which must be coordinated with the full project team.

9.2.4 DAM SYSTEM RESPONSE CURVE DEVELOPMENT

The development of SRPs will follow procedures described in Job Aid 10, *Dam System Response Development*. The SRP development process is adapted based on the type of risk assessment information available.

9.3 LEVEE SYSTEM RESPONSE MODELING

The methodology for incorporating levee system response into the FFRD modeling framework is established reflecting the current state of available levee data. Available information on levee performance is leveraged but will be supplemented with engineering judgment.

9.3.1 LEVEE RISK ASSESSMENT DATA

All existing levee risk assessment data and reports will be assembled during the scoping and data preparation phase. Processes and sources for obtaining this data are included in Job Aid 02, *Data Preparation Processes*.

9.3.2 LEVEE BREACH QUANTITY

The FFRD modeling framework assesses breaches at specific defined locations. Levee breach locations and associated system responses are developed for every levee in the NLD. The levee data lead determines the breach quantity for each levee system. The breach quantity will be set by a combination of many factors including levee mileage, number of levee segments, and risk assessment data availability. Modeled results may be particularly sensitive to breach location and quantity for very long levee systems and/or steep terrain. Job Aid 11, *Levee System Response Development*, provides guidance on selecting breach quantity and describes required documentation of applied judgment.

9.3.3 LEVEE BREACH LOCATION

Once the target quantity of breaches per levee system is determined, the breaches can be located. Breach location selection is generally based on current USACE Mapping, Modeling, and Consequences

(MMC) Production Center's Technical Manual for Levees found in the FFRD Templates and Resources. Locations are identified using the following prioritized considerations:

1. Breach location at the incipient overtopping location, also known as the LCL. Note that selection of localized low spots is completed initially and updated throughout hydraulic modeling in the calibration phase.
2. Documented areas of expected poor performance (e.g., based upon risk assessment).
3. Locations of expected poor performance based on engineering judgment.
4. Location in the upstream third of the levee system.
5. High population or structure density areas adjacent to the levee.

9.3.4 LEVEE SYSTEM RESPONSE CURVE DEVELOPMENT

The development of SRPs for each levee breach location will follow procedures described in Job Aid 11, *Levee System Response Development*. The SRP development process is adapted based on the type of risk assessment information available.

9.4 SYSTEM RESPONSE CURVE DELIVERY FORMAT

The SRPs prepared for FFRD use will be provided as a 2-column table as Elevation versus Breach Probability for each breach location. Additional header information for each location that captures the structure ID (NID or NLD), FFRD HEC-RAS hydraulic model unit name, HEC-RAS 2D connection name, toe elevation, crest elevation, and other attributes is included. All SRPs for the full FFRD basin are provided as a single CSV formatted file that can be created using the *system-response* template provided in FFRD Templates. Separate CSV files will be delivered for all dam SRPs and all levee SRPs.

For levees, a corresponding levee breach locations geospatial file that tracks the point locations and names for each breach location along with the key attributes is also included in the delivery package.

9.5 STOCHASTIC BREACH MODELING

Dam or levee breaches at each location are triggered for individual events in the quasi-continuous simulations when the water surface generated from the sampled storm event exceeds the sampled breach trigger elevation, which are based on provided SRP tables at each breach location. The specifics of the breach trigger sampling algorithm are described in cloud-compute technical documentation.

9.6 HYDRAULIC MODELING OF BREACHES

Guidelines for hydraulic modeling of breaches are provided in Section 8.5.9.

9.7 BREACH PARAMETERS

The parameters determining the breach timing, development rate, and final dimensions are modeled in HEC-RAS and discussed in detail in Section 8.5.9.

9.8 DOCUMENTATION OF SYSTEM RESPONSE MODELING

A set of standard reports will be written for each phase in the FFRD project. Dam and levee data leads have the responsibility to develop the response modeling sections and appendices of the larger reports to fully document the breach modeling efforts. FFRD Templates contains report templates for the breach modeling reports.

Table 9.*1* summarizes key dam and levee response modeling documentation included in each of the reports.

National database refinement efforts are generally not documented as part of the FFRD reports as any data refinement actions on dams and levees will be stored directly in the NID and NLD, respectively. Both databases are managed by USACE. Those actions have distinct SOPs, data management workflows, documentation, and review standards. However, listings of dams and levees within the study basin will be included in the FFRD reporting and any special project-specific adjustments, findings or feedback to the national databases will be included.

*Table 9.*1. Key System Response Modeling Documentation by Phase

Phase	Included Documentation
Scoping and Data Preparation Phase Report	Identification of all dams and levees included in the FFRD study. Listing of all existing and available risk assessment data. Documentation of planned scope of dams selected to breach in FFRD study. Documentation of proposed levee breach locations.
Calibration Phase Report	Documentation of any adjustments to breach locations planned during scoping phase. Documentation of SRP development and sources. Statements regarding breach sampling scheme.
Conformance Phase Report	Documentation of resulting occurrence of breaches in the single realization. Description of any system response modeling adjustments made based on single realization results.
Production Phase Report	Descriptions of any hot fixes or patches required in the production phase.

9.9 DAM AND LEVEE BREACH MODEL REVIEW PROCESS AND DELIVERABLES

Reviews of formal deliveries are a critical part of the FFRD production process, including all aspects of system response modeling inputs. Several review steps are required to verify the final deliveries form part of a modeling system that is capable of accurately depicting basin-wide flood risk. Review checklists are used to track comments related to pre-defined review prompts throughout the review process. FFRD Templates contains checklist templates.

Table 9.2 summarizes key final deliverables by each project phase for dams and levees. The following subsections describe specific review steps along with associated deliveries.

Table 9.2. Overview of Key Dam and Levee System Response Final Deliverables by Phase

	Dams	Levees
Prior to or Concurrent with FFRD Scoping and Data Preparation	NID Data Refinement*	NLD Data Refinement*
Scoping and Data Preparation Phase	Document Confirmation NID Data Refinement Efforts Dam Scoping Table (Breach Scope) Dam Risk Assessment Data Collection	Document Confirmation NLD Data Refinement Efforts Levee Breach Location (Initial) Geospatial file Levee Risk Assessment Data Collection

	Dams	Levees
Calibration Phase	Dam Breach Location Geospatial file Dam SRPs Documentation of SRP development	Levee Breach Location (Final) Geospatial file Levee SRPs Documentation of SRP development
Conformance Phase	Documentation of Breach Results and adjustments during Conformance	Documentation of Breach Results and adjustments during Conformance
Production Phase	Final Reporting of Breach Results	Final Report Review

*Documented by others

9.9.1 SCOPING AND DATA PREPARATION PHASE REVIEW

The system response modeler is responsible for the following deliverables for the scoping and data preparation phase:

- Confirmation of separate NID Data Refinement Efforts, as documented in Scoping and Data Preparation Report
- Confirmation of separate NLD Data Refinement Efforts, as documented in Scoping and Data Preparation Report
- Dam Scoping Table (Breach? Yes/No) with justifications documented in Scoping and Data Preparation Report
- Dam Risk Assessment Data Collection
- Levee Breach Location selection and justification documented in Scoping and Data Preparation Report
- Levee Breach Location (Initial) Geospatial file with attributes per template
- Levee Risk Assessment Data Collection

The Scoping and Data Preparation Checklist includes charge questions related to each of these deliverables.

9.9.2 CALIBRATION PHASE REVIEWS

System response modeling products undergo an informal working review to coordinate and finalize levee breach locations leveraging hydraulic modeler inputs, and a full review of the delivered system response products for the calibration phase.

9.9.2.1 LEVEE BREACH LOCATION ADJUSTMENT INFORMAL WORKING REVIEW

An intermediate review consists of a hydraulic modeler review of draft breach locations intended to be sited at the LCL (i.e., initial overtopping location). Specifically, early hydraulic model results may indicate improved placement of the LCL breach location. Adjustments will be documented by updating the *Location Description and Rationale* field included in the breach location geospatial file. This process may be iterative, and additional updates may be made as hydraulic models are refined during the calibration phase.

The *Breach Input Review Checklist* includes charge questions to verify this working review was performed.

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9.9.2.2 REVIEW OF FINAL CALIBRATION PHASE DELIVERIES

The system response modeler is responsible for the following final deliverables for the Calibration Phase:

- SRP tables (template format)
- SRP Calculation Worksheets to support FFRD SRP deliveries
- Dam Breach Location Geospatial file with attributes per template
- Levee Breach Location (Final) Geospatial file with attributes per template
- SRP development documented in Calibration Phase Report
- Breach sampling schema discussion to be utilized in the Conformance Phase documented in the Calibration Phase Report

The breach input review checklist is used to track comments and document review of these deliverables. This checklist includes charge questions related to review of SRP development computations, adjustments to earlier breach location selection, and breach sampling schema to be utilized in the conformance phase.

9.9.3 CONFORMANCE PHASE REVIEW

The Conformance Phase Review Checklist focuses on ensuring the results of a full realization of synthetic events produce expected trends considering dam and levee response modeling. No separable dam or levee deliverable is provided in this phase unless updates are required. Key dam and levee response modeling areas of review focus include spatial mapping of residual risk, verification of LCL, reasonable representation of annual probability of failure (APF), and reasonable modeled breach parameters. Output from the conformance phase will be reviewed by the levee data lead to verify reasonable levee reliability results that are reflective of expected behavior. Refer to charge questions in the Conformance Checklist.

Updates to dam and levee input data may be needed during the conformance phase as the team makes model adjustments to achieve conformance requirements. If the SRP data is updated, files should replace the original existing data in the standard directory location and filenames. Optionally, superseded data may be copied to an archive subdirectory.

9.9.4 PRODUCTION PHASE

Dam and levee sections of the report are prepared. A standardized narrative of levee input and output is prepared and QC reviewed.

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SECTION 10—CONSEQUENCES

This section outlines the process and requirements for developing and utilizing a structure inventory for FFRD modeling projects. The resulting product is an estimate of average annualized losses (AAL) as well as damage frequency curves with uncertainty at each structure in the United States.

The consequence categories considered are limited to direct structure and content losses. While additional consequence categories exist (such as life loss) and are capable of being computed with the hazard data generated by FFRD, they are not being deployed with the initial effort of FFRD.

10.1 STRUCTURE INVENTORY

The following subsections describe guidelines for development of the structure inventory for FFRD projects.

10.1.1 SOURCE

For FFRD, this SOP outlines the process for computing consequences with the National Structure Inventory (NSI) and a standard damage function library. However, to meet a broader set of needs, the hazard data generated could be applied to a user-provided structure inventory and damage function set.

10.1.2 SCHEMA

To maintain consistency throughout the nation, the normalized schema and data types in the NSI are adopted as the standard for the FFRD initiative. If a structure inventory with a different schema is utilized, a schema crosswalk must be created. The NSI is a public database describing structure locations as points that include attributes required for flood risk consequence assessment. Table 10.1 provides NSI critical attributes. Table 10.2 provides descriptions of the NSI standard occupancy types for each damage category.

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Table 10.1. National Structure Inventory Required Attributes

Field Name	Description	Attribute Type
fd_id	A number that should be unique for all structures.	Integer
cbfips	A census block containing the structure.	String
x	X coordinate of each structure.	Double
y	Y coordinate of each structure.	Double
st_damcat	Damage category of the structure.	String
occtype	Damage function or occupancy type of the structure.	String
val_struct	Value in dollars of the structure*	Double
val_cont	Value in dollars of the contents of the structure*	Double
found_ht	Describes the foundation height of the structure in feet from the ground elevation.	Double
found_type	Describes the type of foundation of the structure (C=Crawl, B=Basement, S=Slab, P=Pier, I=Pile, F=Fill, W=Solid Wall).	String
num_story	The number of stories of the structure.	Integer

*In the NSI, structure value and content value are based on depreciated replacement value.

Table 10.2. Standard Occupancy Types and Descriptions

Damage Category	Occupancy Type	Description
Residential	RES1-1SNB	Single Family Residential, 1 story, no basement
Residential	RES1-1SWB	Single Family Residential, 1 story, with basement
Residential	RES1-2SNB	Single Family Residential, 2 story, no basement
Residential	RES1-2SWB	Single Family Residential, 2 story, with basement
Residential	RES1-3SNB	Single Family Residential, 3 story, no basement
Residential	RES1-3SWB	Single Family Residential, 3 story, with basement
Residential	RES1-SLNB	Single Family Residential, split-level, no basement
Residential	RES1-SLWB	Single Family Residential, split-level, with basement
Residential	RES2	Manufactured Home
Residential	RES3A	Multi-Family housing 2 units
Residential	RES3B	Multi-Family housing 3-4 units
Residential	RES3C	Multi-Family housing 5-10 units
Residential	RES3D	Multi-Family housing 10-19 units
Residential	RES3E	Multi-Family housing 20-50 units
Residential	RES3F	Multi-Family housing 50 plus units
Residential	RES4	Average Hotel
Residential	RES5	Institutional Dormitories

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Damage Category	Occupancy Type	Description
Residential	RES6	Nursing Home
Commercial	COM1	Average Retail
Commercial	COM2	Average Wholesale
Commercial	COM3	Average Personal and Repair Services
Commercial	COM4	Average Professional Technical Services
Commercial	COM5	Bank
Commercial	COM6	Hospital
Commercial	COM7	Average Medical Office
Commercial	COM8	Average Entertainment/Recreation
Commercial	COM9	Average Theater
Commercial	COM10	Garage
Industrial	IND1	Average Heavy Industrial
Industrial	IND2	Average Light Industrial
Industrial	IND3	Average Food/Drug/Chemical
Industrial	IND4	Average Metals/Minerals processing
Industrial	IND5	Average High Technology
Industrial	IND6	Average Construction
Commercial	AGR1	Average Agricultural
Commercial	REL1	Church
Public	GOV1	Average Government Services
Public	GOV2	Average Emergency Response

10.1.3 SCALES

Structure inventory data is generated for the entire project area and broken out into individual inventories for each hydraulic modeling unit.

10.1.4 BASE INVENTORY

During the scoping and data preparation phase, the consequences team member will prepare the base structure inventories and post them to the *basin-data/structure-inventory* directory following guidance provided in Job Aid 14, *Consequence Processes.*

10.1.5 STRUCTURE INVENTORY ADJUSTMENTS

Leveraging the structure inventory through the process of hydraulic calibration can improve both the hydraulic models and the structure inventory. For details on methods to identify hydraulic modeling and structure placement issues see Job Aid 14, *Consequence Processes*.

During the calibration phase, steps are performed by both the hydraulic modeling team members and the consequences team members. Hydraulic modeling team members utilize the structure inventory to

identify potential issues with the hydraulic model as part of the model development and calibration process. When the hydraulic model is submitted for review, the hydraulic results from multiple historic events will be available. The consequences team member will review the hydraulic model output and document the review using the consequences checklist. The review checklists for both the hydraulic model and consequence model include specific items that focus on ensuring the structure placement is checked. Any structure placement issues will be resolved by the consequences team, and an updated structure inventory will be uploaded to the Model Library as directed in Job Aid 14, *Consequence Processes*.

During the conformance phase, consequences are computed over a long-term realization using the hydraulic models and structure inventories that have already been refined during the calibration phase. The ability to view simulated depth-frequency or damage-frequency results is likely to highlight additional hydraulic modeling, levee fragility, or structure placement issues. The Conformance Review Checklist includes items to review consequence results to verify acceptability. Additional adjustments to the models and structure inventory will be made to address any issues identified. Details on methods to identify issues during the conformance phase are included in Job Aid 14, *Consequence Processes*.

10.2 DAMAGE FUNCTIONS

The damage functions for each structure are selected based on the characteristics of the hazard, the occupancy type, and the foundation type.

Within the consequences folder for the appropriate phase, the damage functions JSON file defines the damage functions utilized in the consequences compute. This SOP leverages the FEMA inland damage functions for appropriate residential occupancy types and USACE standardized damage functions for all other occupancy types. Damages computed via this SOP will be based solely on depth at structures based on FEMA guidance. This methodology is embedded in the *compute-multi-frequency-mean-stdev-single-parameter* action of the consequences-runner plug-in in the cloud-compute toolset.

Computing with multi-parameter functions requires event-based consequence calculations, which are not being pursued with this version of the SOP per FEMA guidance.

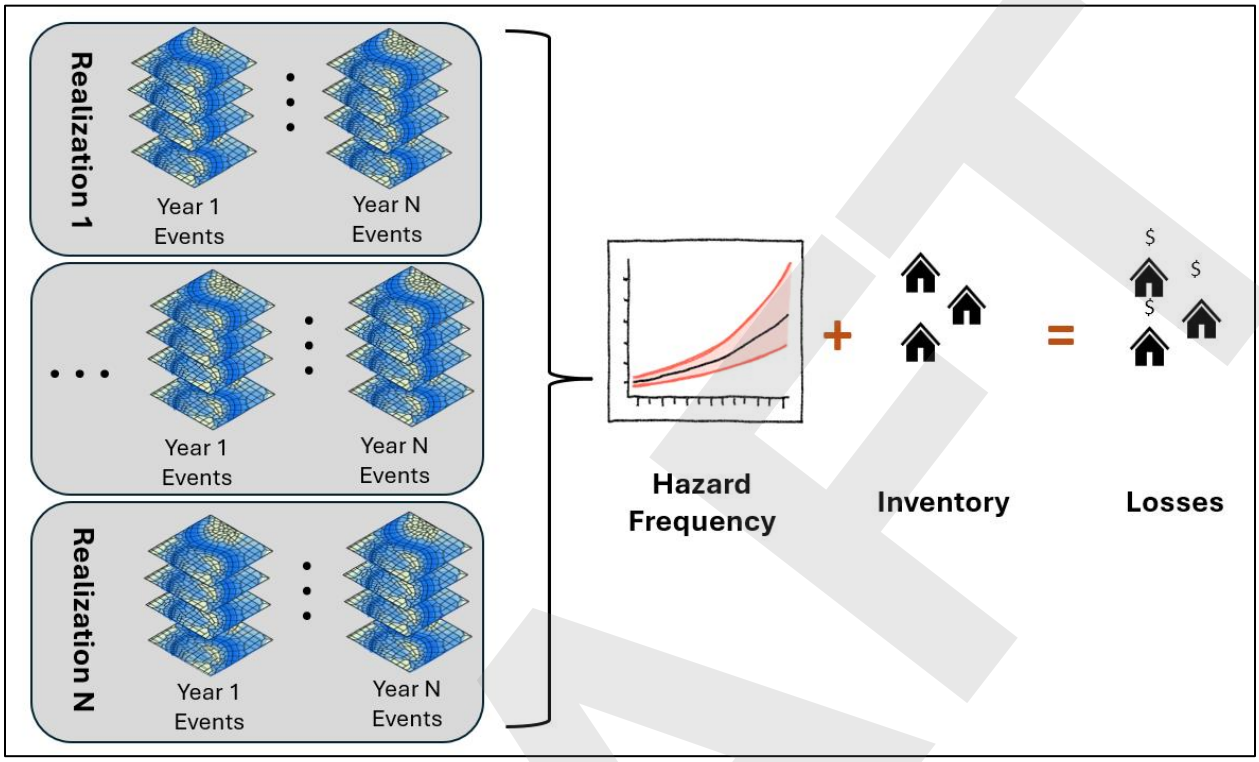
10.3 CONSEQUENCE RESULTS COMPUTATION

During the conformance phase, deterministic depths and damages at each structure for a range of frequency ordinates are computed, and a single value for EAD is computed by integration of the damage-frequency relationship.

During the production phase, multiple realizations provide a mean and standard deviation of depth at each structure at each frequency ordinate. Mean damages and mean EAD are computed based on the mean depth at each structure. The uncertainty in consequences at each structure is represented by sampling various depth-frequency curves based on the standard deviation of depth at each ordinate, and computing damages and EAD. The uncertainty in the structure inventory in terms of foundation heights or other uncertain parameters is not directly considered in this version of the SOP.

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Figure 10.1. Conceptual Diagram of Consequence Computation Method



Consequences are aggregated for relevant geographical units such as the full study basin, individual model units, political boundaries, or leveed areas for the purposes of summary reporting.

Additional details regarding workflow steps to compute and summarize consequences for the project are included in Job Aid 14, *Consequence Processes*.

Future improvements to this SOP will consider computing damages for each event to allow for implementation of multi-parameter damage functions in a defensible way and to provide a more accurate and robust characterization of uncertainty. Incorporation of uncertainty in the structure inventory will also be considered.

10.4 DOCUMENTATION OF CONSEQUENCES MODELING

A set of standard reports will be written for each phase in the FFRD project. Consequence team members have the responsibility to develop the consequence sections and appendices of the larger reports to fully document the consequence modeling efforts. FFRD Templates contains report templates for the consequence modeling reports.

Table 10.3 summarizes locations of key consequences modeling documentation included in each of the reports.

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Table 10.3. Locations of Key Consequences Modeling Documentation

Report Title	Description
Scoping and Data Preparation Phase Report	Documentation of structure inventory source and delivery.
Calibration Phase Report	Primary documentation of structure placement adjustments during calibration. Individual appendices for each model-unit.
Conformance Phase Report	Description of all structure inventory adjustments made based on frequency-based results.
Production Phase Report	Descriptions of any hot fixes or patches required in the production phase.

10.5 CONSEQUENCES REVIEW PROCESS AND DELIVERABLES

Consequence reviews are fully closed out at the end of each study phase. The consequence reviewer and the consequence modeler should verify that documentation and intermediate products are uploaded to the Model Library and that they follow the processes included in this SOP and the *Consequence Processes* job aid. The consequence checklist will provide review documentation. The checklist templates provide major items for the consequence review.

10.5.1 SCOPING AND DATA PREPARATION PHASE REVIEW

Delivery of structure inventories will be performed as described in Section 4 and Job Aid 14, *Consequence Processes*, and review of those products is required. This review will be documented in specific items of the Data Preparation Review Checklist. This shared checklist for the overall scoping and data preparation phase requires multiple reviewers as described in Section 4. The checklist template including review prompts is provided in FFRD Templates and Resources and is delivered to the *basin-name/documentation/review-checklists* directory when complete.

10.5.2 CALIBRATION PHASE QUALITY CONTROL REVIEW

The calibration phase consequence efforts will be reviewed using the consequences review checklist. The deliverables to initiate this review will be posted to the Model Library calibration directory and are listed below:

- Adjusted calibration phase structure inventories. Posted to *calibration/consequences* directory.
- Complete documentation for all calibration phase consequence efforts in report sections and appendices.
    - Final calibration phase reporting will be posted to the *documentation/reports/2-calibration-phase* directory
- Review checklist with modeler comments and assumptions fields populated.

The review checklist will be backchecked by the original reviewer to close each item. The final review checklist will be posted to *documentation/reviews* to complete the delivery.

10.5.3 CONFORMANCE PHASE REVIEW

This review of conformance phase consequence modeling efforts will be documented in specific items of the Conformance review checklist. This shared checklist for the overall phase requires multiple reviewers.

The consequence modeling deliverables for this review are listed below:

- Adjusted conformance phase structure inventories. Posted to conformance/consequences directory
- Completed documentation of all conformance phase consequence efforts in report sections and appendices.
    - Final conformance phase reporting will be posted to the *documentation/reports/3-conformance-phase* directory
- Review checklist with modeler comments and assumptions fields populated for consequence items.

The review checklist will be backchecked by the original reviewer to close each item. The final review checklist will be posted to *documentation/reviews* to complete the delivery.

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SECTION 11—STOCHASTIC SIMULATION STRUCTURE

This section outlines the stochastic simulation structure and related concepts used to generate FFRD results. The simulation structure provides the means to produce graduated hazard and risk estimates across the entire study domain, including direct quantification of uncertainty that preserves the underlying correlations of all variables.

Simulations produced following this guidance will have the following characteristics:

- Quasi-continuous realization within a nested Monte Carlo simulation driven by meteorological events generated through SST.
- Leverage hydrologic model, reservoir operations model, and hydraulic models that are independently calibrated to historic events.
- Account for possible breach of dam and levee infrastructure.
- Provide both pluvial and fluvial flood hazard data across the range of natural variability from the 1/10 to the 1/2,000 AEP.
- Account for natural variability and model uncertainties in terms of storm events, antecedent conditions, breaching, model parameters, and structure inventories.
- Quantify uncertainty in hazard and risk.
- Capable of being leveraged for production of simulations of alternative scenarios or other studies.

11.1 USE OF CLOUD COMPUTE TOOLSET

The FFRD official cloud compute toolset was developed to support the production of hazard and risk products in alignment with this SOP. Maintained as an open-source project by USACE, the toolset allows contributions to future enhancements through its source-controlled repository. Cloud compute tool user documentation and getting-started resources are provided at the *cc-home* repository. The sections below provide an overview of use of these tools for FFRD projects.

11.1.1 CONTAINERIZED PLUGINS AND RUNNERS

The FFRD simulations require the use of cloud-based plugins and runners that have been developed specifically to support the simulation structures described in this SOP. Each plugin is developed in its own container. A container is a package that provides everything necessary to run a program: the executable code, the operating system, and any necessary dependencies.

The tables below provide an overview of the role and purpose of each plugin and the specific actions within the plugins that are needed for execution of this SOP. Additional detailed documentation for the use of these plugins for FFRD is included in the source-controlled repositories.

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Table 11.1. storm-cloud-plugin

Repository	**storm-cloud-plugin**
Description	StormHub cloud plugin: builds storm catalogs from NOAA AORC precipitation data and converts them to HEC-DSS files and an HMS grid file.
Latest release / package	ghcr.io/usace-cloud-compute/storm-cloud-plugin:1.0.6 amd64 image.
Dependencies	**Cloud Compute:** cc-py-sdk 1.1.0 (installed from the GitHub release wheel) **External:** stormhub (Git submodule hosted on Dewberry's GitHub account); hecdss; xarray; zarr; dask; pystac; rasterio; geopandas; dataretrieval **Container base:** Amazon Linux 2023
Action	**Action Description**
download-inputs	Download AORC and other inputs
process-storms	Search for storms and build the storm catalog
convert-to-dss	Convert catalog storms to HEC-DSS
create-grid-file	Write the HMS .grid file
upload-outputs	Upload outputs to the store

Table 11.2. seed-generator-plugin

Repository	**seed-generator-plugin**
Description	Generates reproducible realization, block and event seeds, and the block structure used to organize events into synthetic years.
Latest release / package	ghcr.io/usace-cloud-compute/seed-generator-plugin:1.0.5 amd64 and arm64 images.
Dependencies	**Cloud Compute:** cc-go-sdk (November 10, 2025 commit), including its TileDB store **External:** gonum 0.16; golang.org/x/exp **Container base:** cc-tiledb-base
Action	**Action Description**
block-all-seed-generation	Generate all seeds from blocks.json (the FFRD workflow)

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Table 11.3. hms-mutator-plugin

Repository	**hms-mutator-plugin**
Description	Performs stochastic storm transposition for HEC-HMS gridded meteorology, selecting storms and computing new storm centerings.
Latest release / package	ghcr.io/usace-cloud-compute/hms-mutator-plugin:1.0.4 amd64 and arm64 images.
Dependencies	**Cloud Compute:** cc-go-sdk (November 10, 2025 commit); filesapi **External:** HEC go-statistics (January 2024); dewberry/gdal; google/uuid **Container base:** cc-tiledb-base
Action	**Action Description**
valid_stratified_locations	Generate stratified locations filtered to valid positions ("fishnets")
full_simulation_sst	Run full-realization stochastic storm transposition (best estimate, bootstrap or jackknife sampling)

Table 11.4. basin-record-gen-plugin

Repository	**basin-record-gen-plugin**
Description	Antecedent conditions plugin that processes HMS basin files that account for parameter uncertainty and sampled initial states.
Latest release / package	ghcr.io/usace-cloud-compute/basin-record-gen-plugin:1.0.1 amd64 image.
Dependencies	**Cloud Compute:** cc-py-sdk 1.1.0 (installed from the GitHub release wheel) **External:** hecdss 0.1.28; tiledb; pyarrow; pylance; boto3; numpy **Container base:** cc-tiledb-python
Action	**Action Description**
copy-inputs	Copy input files locally
process-basin-files	Process basin files for parameter uncertainty and sampled initial states
copy-outputs	Copy results to the store

Table 11.5. fragility-curve-plugin

Repository	**fragility-curve-plugin**
Description	Samples dam and levee fragility curves to produce failure elevations for each event, using the knowledge-uncertainty seed for the curve sample and the event seed for the value sample.
Latest release / package	ghcr.io/usace-cloud-compute/fragility-curve-plugin:1.0.6 amd64 and arm64 images.
Dependencies	**Cloud Compute:** cc-go-sdk (November 10, 2025 commit), including its TileDB store **External:** HEC go-statistics (January 2024) **Container base:** cc-tiledb-base
Action	**Action Description**
all-samples	Sample fragility curves for all events

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Table 11.6. hms-runner

Repository	**hms-runner**
Description	Runs HEC-HMS on Linux and provides DSS-to-HDF and DSS-to-CSV conversion utilities.
Latest release / package	ghcr.io/usace-cloud-compute/hms-runner:v1.0.5
Dependencies	**Cloud Compute:** cc-java-sdk 1.0.0 jar (referenced from a local path; current SDK release is 1.1.2) **External:** HEC-HMS 4.13-beta.6 jars (referenced from a local path) **Container base:** eclipse-temurin 17 JDK build stage; Ubuntu 20.04 runtime
Action	**Action Description**
compute_simulation_all_placements_given_storm	Run every storm placement for one storm
dss_to_hdf	Convert DSS output to HDF
dss_to_hdf_tsout	Convert DSS timeseries output to HDF
dss_to_hdf_observations	Convert observed flows to HDF
dss_to_hdf_pool_elevations	Convert pool elevations to HDF
copy_precip_table	Copy the precipitation table
export_excess_precip	Export excess precipitation

Table 11.7. ressim-runner

Repository	**ressim-runner**
Description	Runs HEC-ResSim on Linux.
Latest release / package	ghcr.io/usace-cloud-compute/ressim-runner:v0.0.1
Dependencies	**Cloud Compute:** cc-java-sdk jars from a local /cloud-compute/jar folder (version not recorded) **External:** HEC-ResSim 3.5.1.18 Linux beta jars; AWS SDK for Java v1 (S3); Jackson 2.12 **Container base:** Ubuntu 20.04
Action	**Action Description**
compute_simulation	Run a ResSim simulation
update_timewindow_from_dss	Set the simulation time window from DSS data
dss_to_dss	Copy DSS records
dss_to_dss_merge	Merge DSS records
post_peaks	Post peak values
peaks_merger	Merge peak values

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Table 11.8. ras-runner

Repository	**ras-runner**
Description	Runs HEC-RAS 6.3.1, 6.4.1, 6.5.0 and 6.6.0 on Linux, with actions that link upstream data into RAS models and extract RAS results.
Latest release / package	ghcr.io/usace-cloud-compute/ras-runner-6.6.0:v1.0.0
Dependencies	**Cloud Compute:** cc-go-sdk (November 24, 2025 commit); go-hdf5 **External:** HDF5 1.14.5; HEC-RAS Linux binaries **Container base:** cc-hdf5-ras-base v1.0.0 (Rocky Linux 8)
Action	**Action Description**
unsteady-simulation	Run an unsteady RAS simulation; geometry preprocessing is set by an attribute of this action
column-to-boundary-condition	Link column-oriented HDF data to a model boundary condition (the README calls this column-to-bc)
refline-to-boundary-condition	Link reference-line results from an upstream RAS model to a boundary condition (README: refline-to-bc)
hdf-to-hdf	Copy datasets between HDF files (not documented in the README)
update-boundary-condition	Not documented
update-breach-bfile	Write fragility-curve breach elevations to the RAS B-file (README: update-breach-data)
update-outlet-ts-bfile	Write observed flow data to the RAS B-file (README: update-outletts-data)
update-bfile-skip-dss	Set the B-file flag that suppresses DSS export
ras-extract	Extract user-defined datasets and attributes from RAS HDF output
ras-breach-extract	Extract 2D flow area connection breach data
create-ras-tmp	Create the .tmp file that Linux RAS requires from the plan HDF file
copy-inputs	Bulk copy of model input files
post-outputs	Copy output files to an external store

Table 11.9. RAS Post Processor

Repository	(private)
Description	Generates grids using RAS 2025
Latest release / package	ghcr.io/usace-cloud-compute/ras-postprocess:v1.0.3
Dependencies	**Cloud Compute:** cc-go-sdk; filesapi; **Base Image:** registry-public.hecdev.net/ras-full:master-79
Action	**Action Description**
export-grids	Uses the RAS 2025 API to export a grid.

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Table 11.10. Aggregator

Repository	(private)
Description	Aggregates gridded data at scale
Latest release / package	ghcr.io/usace-cloud-compute/aggregator:v1.0.0
Dependencies	**Cloud Compute:** cc-go-sdk; filesapi; go-hdf5; go-gdal
Action	**Action Description**
ep-grids-action	Create AEP grids from grid data.
freq-grids-action	Create frequency grids from grid data.
ras-summaries-action	Summarize RAS outputs into standard tables
raster-algebra-action	Perform raster algebra on multiple raster grids such as mean, variance, standard deviation, and maximum across a cell across the grids.

Table 11.11. consequences-runner

Repository	**consequences-runner**
Description	Computes economic consequences for event, frequency, coastal and FEMA frequency analyses.
Latest release / package	ghcr.io/usace-cloud-compute/consequences-runner:v1.0.1
Dependencies	**Cloud Compute:** cc-go-sdk (November 24, 2025 commit); filesapi **External:** USACE go-consequences; HEC go-statistics; dewberry/gdal; GDAL 3.8 **Container base:** osgeo/gdal 3.8.3 build stage; Ubuntu 24.04 runtime
Action	**Action Description**
copy-inputs	Stage input files
compute-fema-frequency-single-parameter	Compute FEMA single-parameter frequency damages
post-outputs	Copy outputs to the store

11.1.2 MANIFESTS

All cloud-compute compute jobs require a manifest that contains instructions for the cloud computing plugin or runner. In many cases, the modeling team must provide necessary inputs to build these manifests by leveraging the information in the Model Linking Register.

11.1.3 CLOUD COMPUTE ENVIRONMENT SUPPORT

Support for applying cloud compute tools will be available from the FFRD National Implementation Group staffed by USACE, as described in Section 2 of this SOP.

11.2 SIMULATION STATISTICAL APPROACH

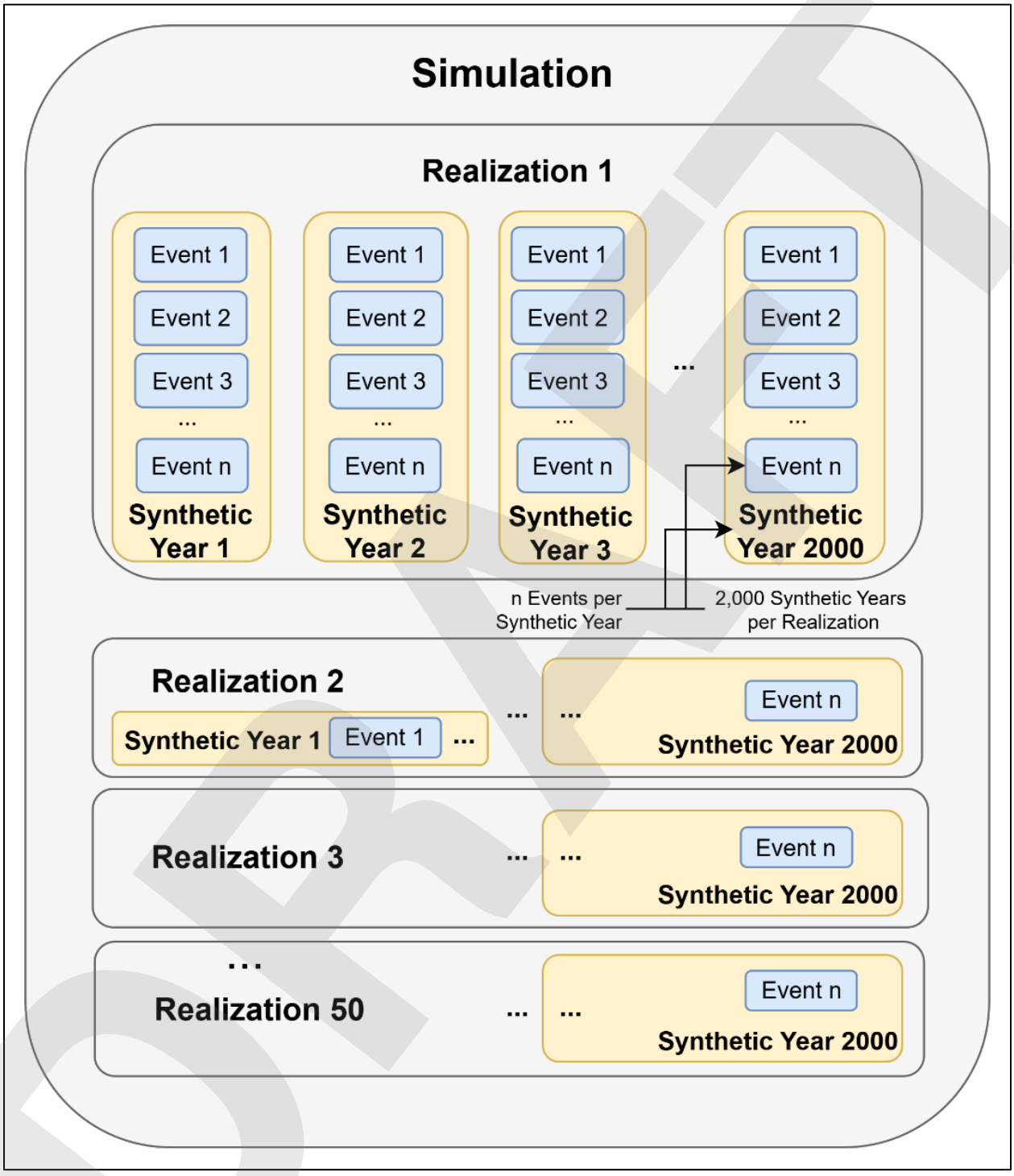
To support a nested Monte Carlo simulation containing quasi-continuous events, the following terminology is used to describe the simulation structure.

- **Event**—An event represents a unique set of parameters derived from antecedent conditions, storm selection and placement, and sampled breach elevations for infrastructure with the capability of breaching. Simulations generally span a time window allowing the sampled precipitation to cycle through the watershed (i.e., time of concentration).
- **Synthetic Year**—A synthetic year is composed of a set of independent events and is used to determine the annual maximum values for any variable of interest at any location in the watershed. The construction of the storm database determines the number of events randomly selected. The term quasi-continuous is used to describe the independence of events in that each simulated event does not persist state across events.
- **Realization**—A realization is composed of a set of synthetic years and is used to construct an estimate of a variable-frequency relationship.
- **Simulation**—A simulation is composed of a set of realizations and is used to describe the knowledge uncertainty around a variable-frequency relationship.

Figure 11.1 illustrates the relationships of the elements within the simulation structure. Realizations are represented as boxes within a simulation, synthetic years are represented by boxes within a realization, and events are represented as boxes within a synthetic year. Figure 11.2 through Figure 11.4 display example outputs for single events, single realizations, and a full simulation. Section 11.6 provides a full inventory of the standard output products provided for an FFRD study.

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Figure 11.1. Relationship of Defined Terms with the Simulation Structure



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Figure 11.2. Example Depth of Inundation Mapped Result from a Single Event

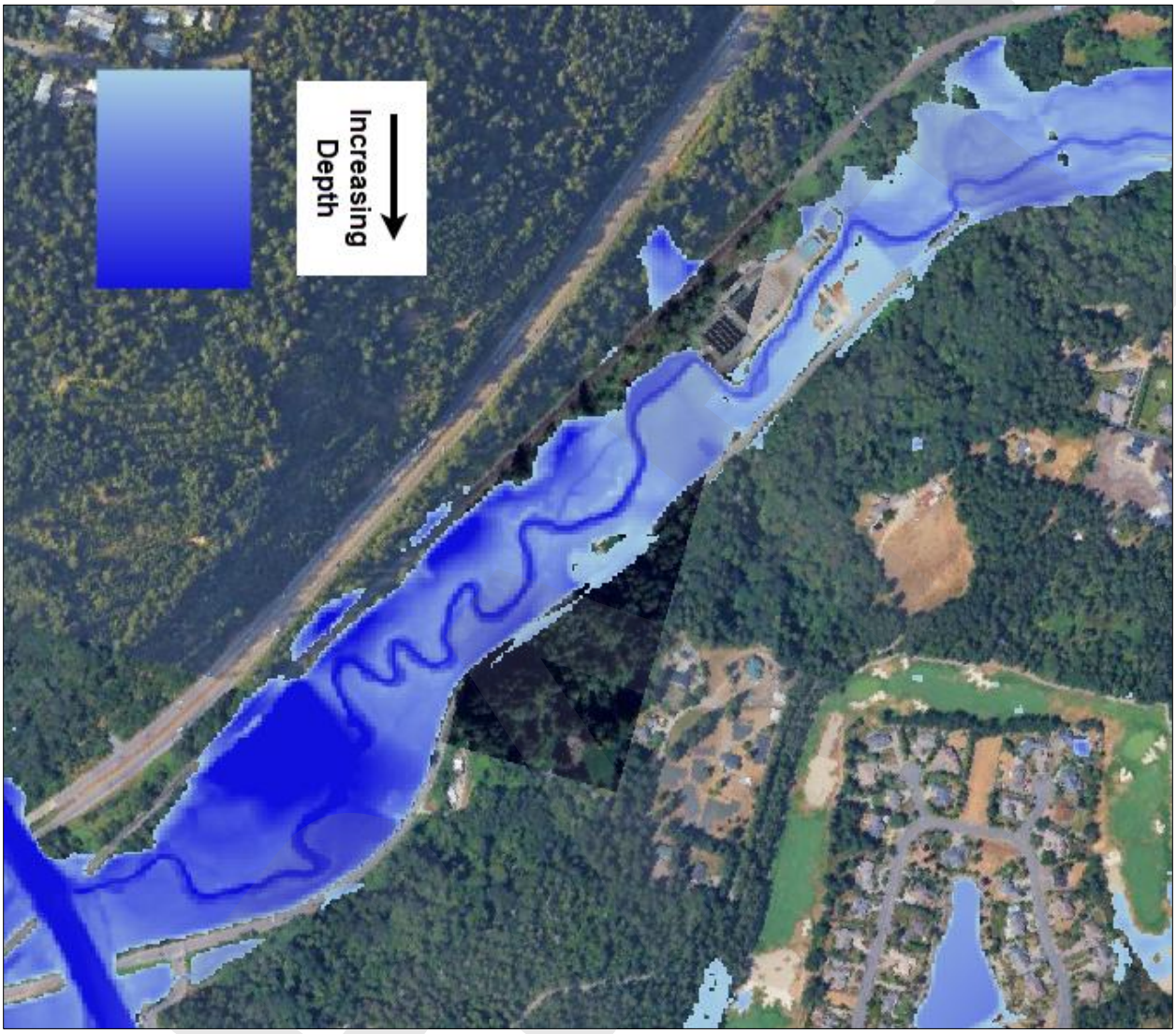


Figure 11.3. Example Annual Probability of Inundation (Depth Greater than Zero) for a Single Realization

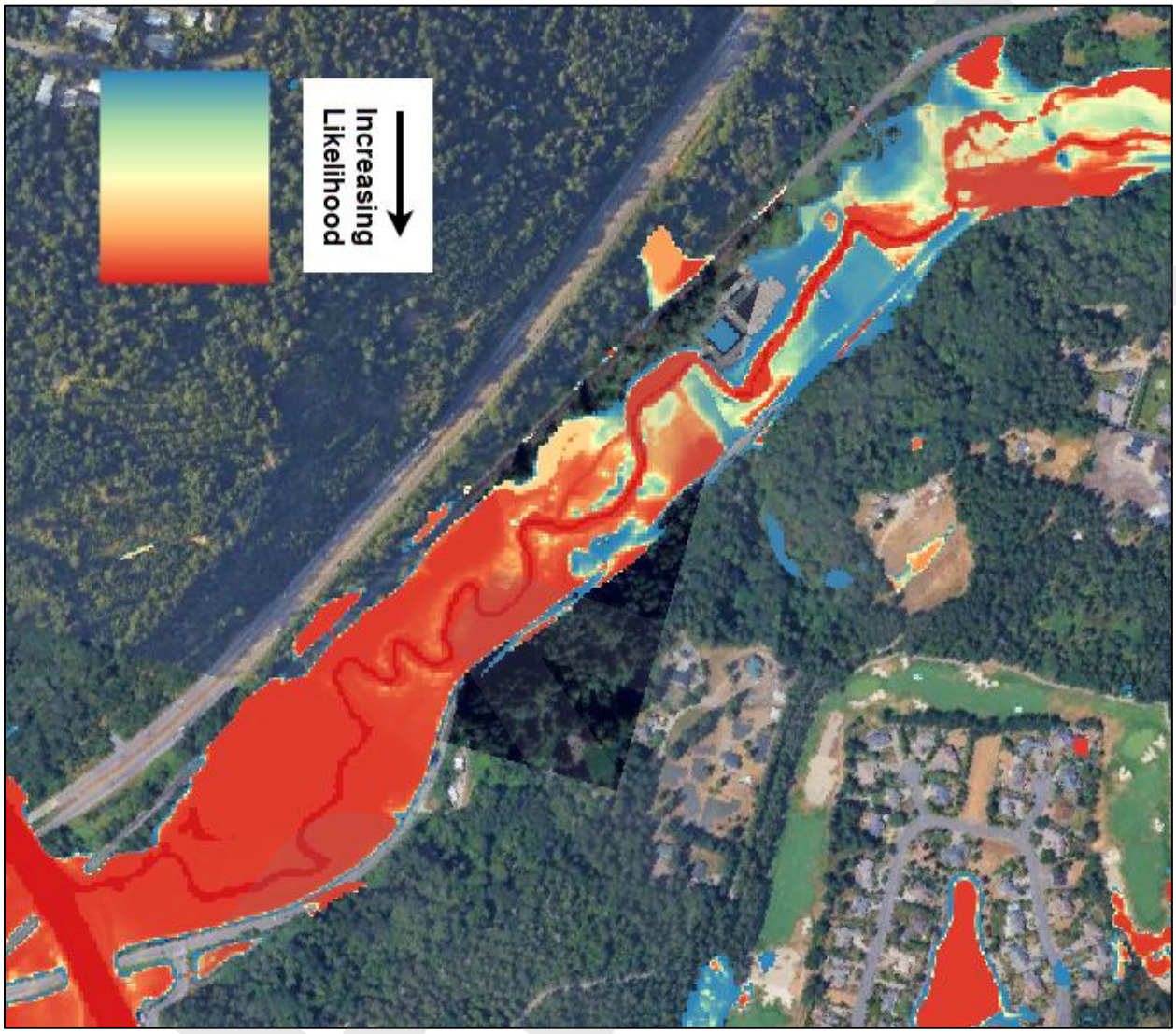
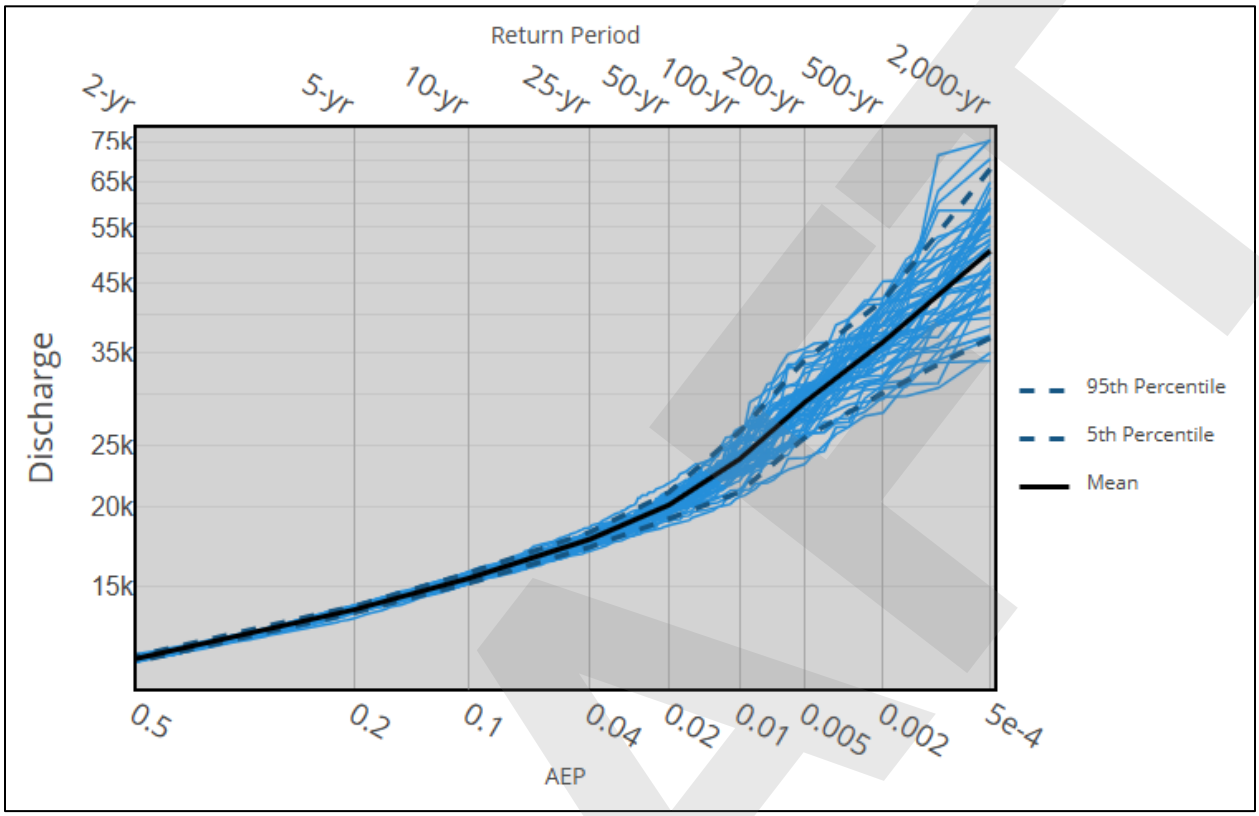


Figure 11.4. Example Computed Variable-Frequency Relationship for Simulation Containing Multiple Realizations



11.3 SAMPLED PARAMETERS DEFINITION

The simulation structure described in Section 11.2 defines the terms realization and event. Within the simulation structure, the representation of uncertainty can be separated into two distinct sources, natural variability and knowledge uncertainty. Natural variability represents the fact that variables, such as the peak flow in a river each year, are random by nature. Knowledge uncertainty represents the fact that there exists limited information of the variable in question. Natural variabilities are sampled with each simulated event. Within the simulation structure, knowledge uncertainties are sampled once per realization.

Table 11.12 lists sampled parameters and their representation as natural variability or knowledge uncertainty within the simulation structure.

Table 11.12. Parameters and Uncertainty Types

Parameter	SOP Version 1 Representation	Sampling by	SOP Tool
Storm Catalog	Knowledge Uncertainty	Pre-process	storm-cloud-plugin
Storm Selection	Natural Variability	Event	hms-mutator-plugin
Storm Location	Natural Variability	Event	hms-mutator-plugin

Parameter	SOP Version 1 Representation	Sampling by	SOP Tool
Hydrologic Parameters (e.g., time of concentration, storage coefficient, constant loss)	Natural Variability	Event	hms-mutator-plugin
Antecedent Conditions (e.g., starting reservoir pool elevations, basin wetness, SWE)	Natural Variability	Event (accounting for seasonality)	hms-mutator-plugin
Breach Trigger Elevation	Knowledge Uncertainty	Block	fragility-curve-plugin
Synthetic Year Event Count	Deterministic	Simulation	storm-cloud-plugin*
Hydraulic Parameter Uncertainty	Deterministic	N/A	N/A
Structure Inventory Uncertainty (foundation heights)	Deterministic	N/A	consequences-runner*
Damage Function Uncertainty	Deterministic	N/A	consequences-runner*

*option for non-deterministic treatment available in plug-in but not called for in SOP version 1.

To provide consistency throughout the simulation of an event, parameters used by more than one model are sampled prior to computation of either model and used by both (e.g., starting pool elevation is sampled prior to hydrology, reservoir operations, and hydraulic modeling and the same value is used by all).

Advancements in the stochastic simulation structure are being considered for future versions of the SOP.

11.4 EVENT SAMPLING PROCEDURE

The sections below provide an overview of the procedures for sampling an individual event.

11.4.1 PRODUCTS FOR SAMPLING SETUP

The products below are established prior to event sampling and are directly applied in the sampling procedure for a single event.

11.4.1.1 STORM CATALOG

The full set of observed storms in the homogeneous transposition domain is assembled into a catalog and classified with the storm type as described in Section 5, which is used as the basis for storm selection.

11.4.1.2 STORM TYPE SEASONAL DISTRIBUTION

The storm type seasonality analysis describes the daily frequency of occurrence for each storm type based on the start date of the events in the storm catalog as described in Section 5.

11.4.1.3 VALID HYDROLOGIC MODEL PARAMETER SETS

For each historic event used for calibration, a unique valid set of model parameters is developed as described in Section 6, which provides the basis for parameter uncertainty.

11.4.1.4 PERIOD OF RECORD HYDROLOGIC MODEL

A POR hydrologic model is calibrated as described in Section 6, which provides daily basin conditions.

11.4.1.5 VALID COMBINATIONS OF HYDROLOGIC MODEL PARAMETER SETS AND ANTECEDENT CONDITIONS

Model files representing a full set of possible combinations of antecedent basin conditions and hydrologic parameter sets are developed as part of the process to deliver the model for cloud-compute according to the procedure described in Job Aid 12, *Model Delivery for Cloud Compute*.

11.4.1.6 EXTERNAL BOUNDARY CONDITION TIMESERIES

If external boundary conditions are being used to represent a related and/or linked (i.e., upstream/downstream) watershed in concert with meteorologic boundary conditions provided by SST, a homogeneous timeseries representing the variable(s) of interest must also be assembled. These homogeneous timeseries must span the same period as the Storm Catalog and can be generated using observed data, POR hydrologic model outputs, or a combination of the two sources. A list of observed data, prepared datasets, and descriptions that are provided upon completion of the scoping and data preparation phase is contained within Section 4. Hydrologic model POR results are described in Section 6.

11.4.1.7 DAM AND LEVEE SYSTEM RESPONSE CURVES

Probability of breach as a function of the loading experienced at each possible breach location is developed as described in Section 9, which provides the basis for selection of breach trigger elevations.

11.4.2 EVENT SAMPLING

Utilizing all products required for sampling setup, the steps in this subsection are followed.

11.4.2.1 STORM SELECTION

Select a storm from the storm catalog with uniform probability and maintain the storm type from the selected storm.

11.4.2.2 STORM PLACEMENT

The *x* and *y* location is selected randomly within constraints for the selected storm.

Storm placements are based on the selected storm name from pre-generated fishnets. The pre-generated fishnets are generated using the transposition domain, original storm centering, and watershed domain. A uniform mesh of points at 4-kilometer spacing was created from the transposition domain; each storm placement within the transposition domain is used to shift the watershed boundary and determine if the watershed boundary contains area outside of the transposition domain. If it does not contain area outside the transposition domain, the placement location is determined to be valid (having

no null precipitation in the storm over the watershed) and the point is recorded into the fishnet. Each storm has a list of valid storm placements spaced at 4-kilometer spacing that are equiprobable for selection. Once a storm is selected, the fishnet is read in, and a random *xy* pair is selected for the storm placement. Each event gets a randomly selected placement from the fishnet for a given storm.

11.4.2.3 ANTECEDENT CONDITION AND HYDROLOGIC PARAMETER SET SELECTION

The day of year of the storm is selected based on a sample that maintains relative occurrence of the storm types for each season. The year of the storm is selected from a uniform distribution of all years in the POR. Antecedent conditions are then selected to match the sampled day and year of the sampled storm. The hydrologic model parameters are selected from the six valid parameter sets (one set for each of the historic events utilized during the model calibration phase).

When snowmelt is included, the observed temperature grids for the sampled event date are utilized without any transposition.

11.4.2.4 EXTERNAL BOUNDARY CONDITION SELECTION

If external boundary conditions are being used to represent a related and/or linked watershed, homogeneous timeseries are sampled for the locations and variables of interest using the sampled day and year of the sampled storm.

11.4.3 BREACH TRIGGER ELEVATIONS

The breach trigger elevations are sampled for the event according to the methods described in Section 9. Breaches will occur for events whenever loads at the breach locations exceed the sampled trigger elevation.

11.4.4 HYDROLOGY AND HYDRAULICS MODEL EXECUTION

The hydrologic, reservoir operations, and hydraulic models are then computed for the event. Consistent antecedent conditions are maintained across each model, although some simplifications are made. The event starting reservoir pool elevation is maintained for dams (according to the Dam Scoping Table developed during the scoping and data preparation phase).

For the current version of the SOP, uncertainty in hydraulic model parameters and initial basin wetness has not been incorporated. Future versions of the SOP may consider incorporation of these additional uncertainties.

11.4.5 EVENT HAZARD GRIDS

After model execution, hazard maps are generated on an event-by-event basis. Note that output grids are generated at a resolution lower than the highest quality source DEM (see Section 3.5 for standards).

Hydraulic model output for a wide range of parameters can be represented as gridded data. For FFRD, the standard saved gridded hazard variables are depth and velocity.

11.4.6 CONSEQUENCE COMPUTATION

For the current version of this SOP, consequences are not calculated on an event-by-event basis and, therefore, sampling of the structure inventory parameters and execution of the consequences are not

included here. For future versions of the SOP, adjustments to the consequence computation method may be considered.

11.5 ENUMERATION OF EVENTS

Section 4.3 states the number of synthetic years, events per year, and realizations for an FFRD study under this SOP.

11.6 DELIVERY OF RESULTS

The use of the FFRD directory structure and the cloud-compute plug-in toolset facilitates a standard and transparent structure of study inputs, indexes, and output results. The use of the standard directory structure is included in Job Aid 07, *Model Library Use*.

Subsections 11.6.1 through 11.6.3 describe results produced for individual event and simulation summary output.

11.6.1 SOURCE MODELS AND INDEX DIRECTORIES

Subsections 11.6.1.1 through 11.6.1.6 provide information on source tracking of models and tracking of the boundary conditions applied to the models for each computed event.

11.6.1.1 SOURCE MODELS

Source model files used for computation of the events are stored in directories above the simulations directory, which includes model results (*/hydrology*, */reservoir-operations*, */system-response*, and */hydraulics*).

11.6.1.2 LINKING INFORMATION

The human-readable Model Linking Register is created during the scoping and data preparation phase of the project and maintained in *basin-name/basin-data/project-scoping/model-linking-register*.

11.6.1.3 MANIFESTS

A *manifests* directory contains information used for sending compute jobs via cloud-compute plugin, which also includes explicit instructions for model linking.

11.6.1.4 SEEDS

A *seeds.json file* in the */simulations* directory stores randomly generated seeds for each plugin that consumes random numbers.

11.6.1.5 EVENT, SYNTHETIC YEAR, AND REALIZATION INDEX

A *blocks.json* is stored in the root of the */simulations* directory and provides an index of all sampled events and their relationship to synthetic years and realizations within the simulation.

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11.6.1.6 ANNUAL MAXIMUM CONTRIBUTING EVENTS

An *ams-contributing-events.csv* list of events that contributed to the annual maximum hazards and were computed through the hydraulic model is saved to */simulations*.

11.6.2 INDIVIDUAL EVENT OUTPUT

Stored output data for each sampled event is saved to the *simulations/event-data/[event-no]* directory, which includes the specific outputs described in subsections below.

11.6.2.1 HYDROLOGY

A *hydrology* directory contains the input to the hydrology model (i.e., data/storm.dss, .met, .grid, .basin, and .control files), the hydrology output (i.e., .dss), and the .tmp.hdf file for each hydraulic model containing the excess precipitation and baseflow from the hydrology run.

11.6.2.2 RESERVOIR OPERATIONS

A *reservoir-operations* directory contains the output from the reservoir operations model (i.e., .dss file).

11.6.2.3 SYSTEM RESPONSE

A *system-response* directory stores randomly generated breach elevations for each breachable structure (i.e., *failure-elevations.json*).

11.6.2.4 HYDRAULICS

A *hydraulics* directory for each modeling unit contains the .hdf file containing the results for the run and a log file.

11.6.2.5 EVENT HAZARD GRIDS

A *grids* directory for each modeling unit contains output maximum depth and maximum velocity .tif files.

11.6.2.6 CONSEQUENCES

A *consequences* directory contains a GeoPackage for each hydraulic modeling unit containing damages for each structure in the modeling unit's footprint (not including overlaps in the mesh).

11.6.3 SIMULATION LEVEL SUMMARY OUTPUTS

Subsection 11.6.3 describes the core simulation level summary outputs generated for an FFRD study. These outputs include tabular hazard summary data, gridded data, spatial point data, and plots. All these simulation level outputs represent data that is embedded in the individual event outputs and are summarized as a set of core standard outputs for easy communication of results and provide a basis for further analysis of results by others. Additionally, a subset of the gridded hazard outputs is required as input for consequence computations.

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11.6.3.1 TABULAR RESULTS SUMMARIES

Tabular summaries of model results for every event in the entire simulation are saved to the *conformance/simulations/frequency-calibration/summary-output* and *production/simulations/current-condition/summary-output* directories.

Table 11.13 lists datasets included in the summaries.

Table 11.13. Listing of Event-level Tabular Results Summaries

Model	Location	Variable(s)
Meteorology	Each Hydrologic Subbasin	Peak Precipitation for 1-hour, 24-hour, 48-hour, and 72-hour durations.
Hydrologic	All Junctions	Peak Flow for 1-hour, 24-hour, 48-hour, and 72-hour durations.
Hydrologic	All Subbasins	Summary parameters including infiltration, interception, LWASS, moisture deficit, percolation, excess precipitation, SWE, groundwater recharge, evapotranspiration, and baseflow.
Reservoir Operations	Modeled Reservoirs	Peak Outflow for 1-hour, 24-hour, 48-hour, and 72-hour durations.
Reservoir Operations	Modeled Reservoirs	Peak pool stage
System Response	All Breach Locations	Sampled breach trigger elevation
Hydraulic	Reference Lines	Peak flow and peak water surface elevation
Hydraulic	Boundary Condition Lines	Peak flow and peak water surface elevation
Hydraulic	2D Area Connections	Peak flow and peak water surface elevation
Hydraulic	Reference Points	Maximum water surface elevation Minimum water surface elevation
Hydraulic	Breach Locations	Breach occurred (Y/N) Maximum water surface loading at breach (headwater and tailwater) Breach initiation time Headwater elevation at breach time Tailwater elevation at breach time Peak flow through breach Maximum breach bottom width Breach progression duration
Hydraulic	Each Hydraulic Model Unit	Volume accounting error percentage Maximum water surface elevation (WSEL) error Computation time Finished successfully (Y/N)

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11.6.3.2 PLOTS

A variety of standard plots are created that leverage these standard simulation level outputs as well as observed data for comparison. Precipitation-frequency, LWASS, flow frequency, elevation-discharge, water surface elevation frequency, breach probability, and damage-frequency plots are all included as core deliveries.

11.6.3.3 SUMMARY HAZARD GRIDS

Grids that summarize hazard results for each synthetic year, each full realization and for the multi-realization simulation are created. These summary hazard grids are saved to the */conformance/simulations/frequency-calibration* and *production/simulations/current-condition/summary-output* directories. Standard frequencies for generation of grids are 1/10, 1/25, 1/50, 1/100, 1/250, 1/500, 1/1,000, and 1/2,000.

11.6.3.3.1 ANNUAL MAXIMUM GRIDS

Synthetic year maximum hazard grids represent the maximum across all event level grids of a specific hazard for a given synthetic year within a given realization. These grids take care to preserve the statistical characteristics of the annual maximums across a synthetic year. Each grid cell contains the maximum value for the critical event during the year for the specific hazard variable. Therefore, these grids are a mosaic of maximum values sourced from multiple events across each synthetic year.

The following annual maximum grids are created and stored in */conformance/simulations/frequency-calibration* and *production/simulations/current-condition/summary-output*.

Table 11.14. Annual Maximum Grids

Hazard Variable	Grid
Depth	Annual maximum depth for each synthetic year in the simulation
Velocity	Annual maximum velocity for each synthetic year in the simulation

11.6.3.3.2 PER REALIZATION SUMMARIES

The following single realization summary grids are created and stored in */conformance/simulations/frequency-calibration* and *production/simulations/current-condition/summary-output.*

Table 11.15. Single Realization Summary Grids

Hazard Variable	Grid
Depth	Annual Exceedance Probability of Depth Exceeding Zero Feet
Depth	Depth at each Standard Frequency (8 grids)
Velocity	Annual Exceedance Probability of Velocity Exceeding 0 Feet per Second
Velocity	Annual Exceedance Probability of Velocity Exceeding 6.6 Feet per Second
Velocity	Annual Exceedance Probability of Velocity Exceeding 13.1 Feet per Second
Velocity	Velocity at each Standard Frequency (8 grids)

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11.6.3.3.3 MULTIPLE REALIZATION SUMMARY

The following summary grids that express mean results with uncertainty derived from multiple realizations are created and stored in */production/simulations/current-condition/summary-output/grids/.*

Table 11.16. Summary Grids of Mean Results with Uncertainty

Hazard Variable	Grid
Depth	Mean depth at each Standard Frequency (8 grids)
Depth	Standard deviation of depth at each Standard Frequency (8 grids)
Depth	Mean Annual Exceedance Probability of Depth Exceeding Zero Feet
Depth	Standard Deviation of Annual Exceedance Probability of Depth Exceeding Zero Feet
Velocity	Mean velocity at each Standard Frequency (8 grids)
Velocity	Standard deviation of velocity at each Standard Frequency (8 grids)
Velocity	Mean Annual Exceedance Probability of Velocity Exceeding 0 Feet per Second
Velocity	Standard Deviation of Annual Exceedance Probability of Velocity Exceeding 0 Feet per Second
Velocity	Mean Annual Exceedance Probability of Velocity Exceeding 6.6 Feet per Second
Velocity	Standard Deviation of Annual Exceedance Probability of Velocity Exceeding 6.6 Feet per Second
Velocity	Mean Annual Exceedance Probability of Velocity Exceeding 13.1 Feet per Second
Velocity	Standard Deviation of Annual Exceedance Probability of Velocity Exceeding 13.1 Feet per Second

11.6.3.4 CONSEQUENCES

A spatial output file for each hydraulic modeling domain describing AAL at each structure is stored in the Model Library under simulations in the */production/simulations/current-condition/summary-output/consequences* directory. The output is delivered in a standard spatial GeoPackage format. In the parameterization of the compute, the user defines the standard frequencies for the compute; those are reproduced in the output demonstrated in Table 11.17. Each frequency will include an estimate of the mean structure damage, mean content damage, standard deviation structure damage, standard deviation content damage, mean hazard, and standard deviation of the hazard.

Table 11.17. Consequences Results Field Names and Descriptions

Field Name	Description
ORIG_ID	Unique Identifier related to the NSI
REPVAL	Depreciated Replacement value
STORY	Number of stories
FOUND_T	Foundation type
FOUND_H	Foundation height

Field Name	Description
x	X coordinate
y	Y coordinate
OccType	Occupancy type
DamCat	Damage category
BASEFIN	Basement finish status
FFH	First floor height
DEMFT	Ground elevation from DEM (feet)
BAAL	Building average annualized loss
CAAL	Content average annualized loss
TAAL	Total average annualized loss
PROB	Probability of the first instance of flooding at a structure
10yr_MS	10-year mean structure damage
10yr_SS	10-year stdev structure damage
10yr_MC	10-year mean content damage
10yr_SC	10-year stdev content damage
10yr_MH	10-year mean hazard
10yr_SH	10-year stdev hazard
25yr_MS	25-year mean structure damage
25yr_SS	25-year stdev structure damage
25yr_MC	25-year mean content damage
25yr_SC	25-year stdev content damage
25yr_MH	25-year mean hazard
25yr_SH	25-year stdev hazard
50yr_MS	50-year mean structure damage
50yr_SS	50-year stdev structure damage
50yr_MC	50-year mean content damage
50yr_SC	50-year stdev content damage
50yr_MH	50-year mean hazard
50yr_SH	50-year stdev hazard
100yr_MS	100-year mean structure damage
100yr_SS	100-year stdev structure damage
100yr_MC	100-year mean content damage
100yr_SC	100-year stdev content damage
100yr_MH	100-year mean hazard

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Field Name	Description
100yr_SH	100-year stdev hazard
250yr_MS	250-year mean structure damage
250yr_SS	250-year stdev structure damage
250yr_MC	250-year mean content damage
250yr_SC	250-year stdev content damage
250yr_MH	250-year mean hazard
250yr_SH	250-year stdev hazard
500yr_MS	500-year mean structure damage
500yr_SS	500-year stdev structure damage
500yr_MC	500-year mean content damage
500yr_SC	500-year stdev content damage
500yr_MH	500-year mean hazard
500yr_SH	500-year stdev hazard
1000yr_MS	1,000-year mean structure damage
1000yr_SS	1,000-year stdev structure damage
1000yr_MC	1,000-year mean content damage
1000yr_SC	1,000-year stdev content damage
1000yr_MH	1,000-year mean hazard
1000yr_SH	1,000-year stdev hazard
2000yr_MS	2,000-year mean structure damage
2000yr_SS	2,000-year stdev structure damage
2000yr_MC	2,000-year mean content damage
2000yr_SC	2,000-year stdev content damage
2000yr_MH	2,000-year mean hazard
2000yr_SH	2,000-year stdev hazard

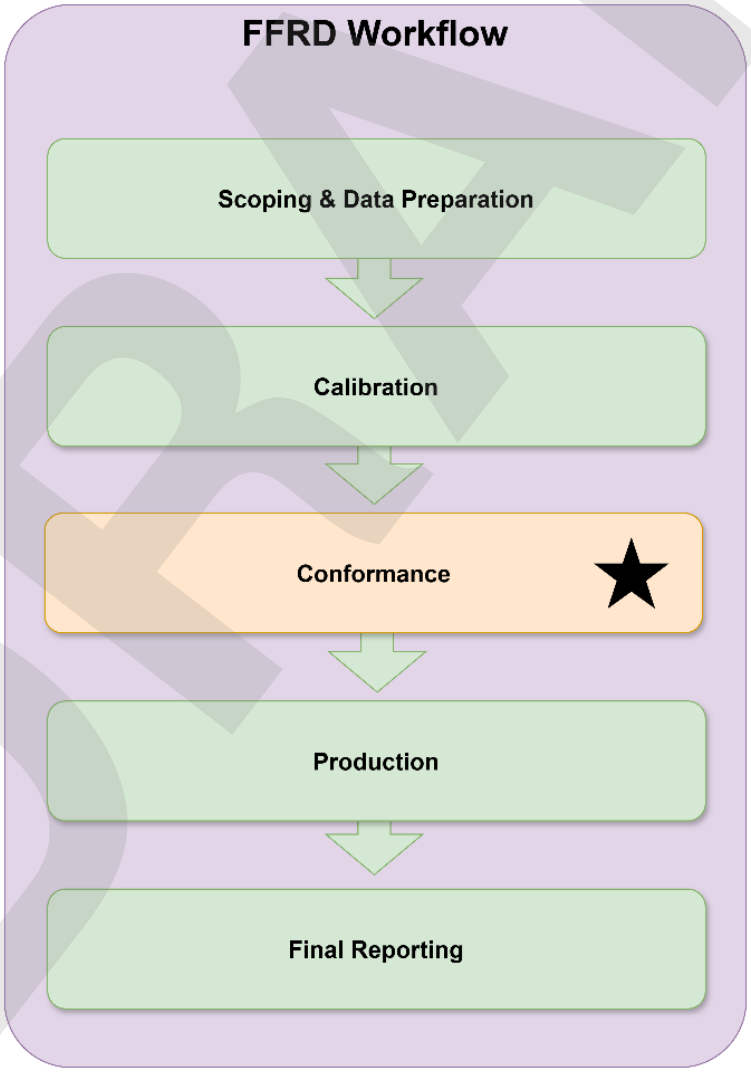
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SECTION 12—CONFORMANCE

The conformance phase of the project integrates the individual calibrated models into a full watershed model and ensures acceptable results can be achieved. The watershed model is first validated for performance against historic events, and then a single long-term realization of storm events is computed. Conformance metrics, which compare simulated frequency curves at key locations to observed frequency curves, are evaluated. Acceptable conformance metrics must be achieved prior to completing the phase and proceeding to production level computations.

During the conformance phase, the individual models will be promoted to the conformance directory, adjusted to represent synthetic events, fully linked, tested for successful execution in the cloud, and checked against desktop results for fidelity. After fidelity is verified, a long-term realization of events will be executed in the cloud, and the suite of models will be adjusted as necessary to meet conformance metrics as defined in the SOP. After conformance metrics are met, reviewed, and approved, the full model suite is considered finalized, and the conformance phase is closed out. After the conformance phase is completed, the production phase begins, in which the model suite is utilized for execution of the full production level simulation, including multiple long-term realizations.

Figure 12.1. Context of the Conformance Phase in the Future of Flood Risk Data Workflow



12.1 STAFFING FOR CONFORMANCE PHASE

The conformance phase is led by the model integration lead and executed by a relatively small team that utilizes the final calibrated models delivered during the calibration phase. The team integrates the models and makes watershed-wide adjustments as necessary to meet the QC and conformance metrics described in the sections below. Significant support from the cloud environment lead is also required to build the compute manifests and to execute models in the cloud through several iterations. Any significant changes to the configuration of the model suite must be coordinated with the appropriate team members involved in the calibration and development of the models (e.g., breach location changes, dam operations changes).

12.2 SELECTION OF TIME WINDOW DURATION FOR STOCHASTIC EVENTS

The time window for stochastic simulations will be selected early in the conformance phase. The selected time window will be of adequate duration, such that the storm event occurs and peak water surfaces are fully routed through the entire basin prior to the end of simulation.

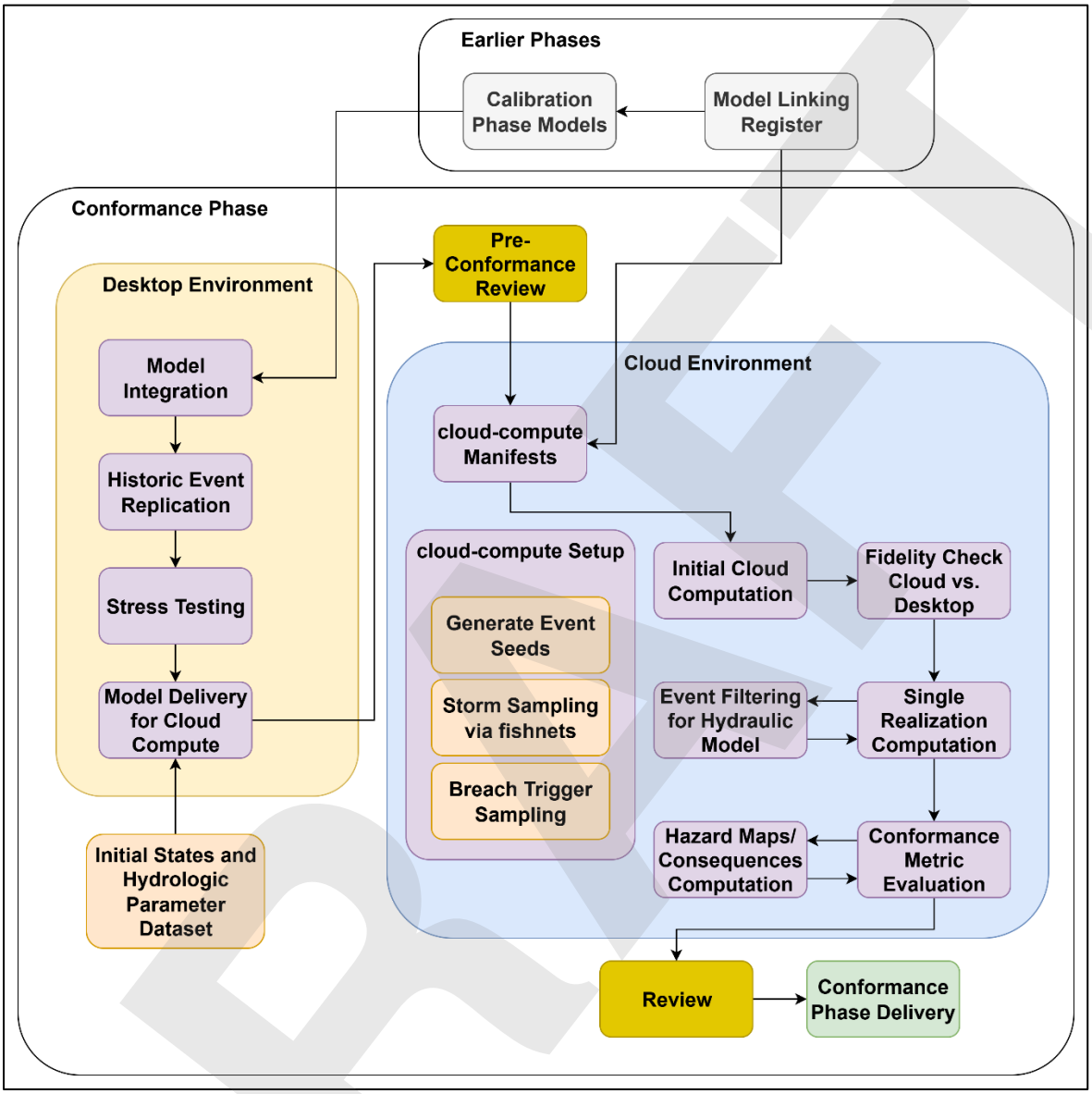
12.3 CONFORMANCE WORKFLOW OVERVIEW

The conformance phase of the project includes several logical major steps to achieve success as briefly described below.

1. Cloud Compute Setup: Setup of cloud compute to accept the fully integrated system of models for computation of stochastic events. Includes pre-processing random seeds, synthetic years, and fishnets for storm placements.
2. Model Integration: Full model linking following the previously scoped Model Linking Register and naming conventions in the desktop environment.
3. Historic Event Replication: QC step to verify that the fully linked system of models still reproduces historic event results, which can be completed in desktop environment.
4. Stress Testing: Desktop execution of model runs that push the limits of model stability and accuracy. Events selected include extreme combinations of storm events and breaches of dams and levees.
5. Model Delivery for Cloud Compute: The full model suite is tailored for cloud compute and delivered to the appropriate location in the *cloud* directory to allow cloud computations.
6. Pre-conformance Review: Review that thoroughly checks the models delivered to confirm all standards are met, documented with review checklist.
7. Cloud-compute Manifests: Development of compute manifests that contain instructions for each cloud-compute plugin to perform the computations.
8. Initial Cloud Computation: Cloud computation of 3 to 5 synthetic events.
9. Fidelity Check Cloud versus Desktop: Direct comparison of desktop results to cloud results for the same storm events to verify fidelity.
10. Single Realization Computation: Cloud computation of a single realization of stochastic storm events.
11. Event Filtering for Hydraulic Model: Selection of a subset of events for computation through the hydraulic model that are relevant for characterizing flood risk. This recognizes that the cost of computing every stochastic event through the hydraulic model is not justified.
12. Hazard Maps and Consequences Computation: Cloud computation of event-based depth grids, summarized hazard maps, and consequences.
13. Conformance Metric Evaluation: The results from the single realization will be evaluated against the established conformance metrics.

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Figure 12.2. Conformance Phase Workflow



12.4 CONFORMANCE TASK DESCRIPTIONS

The following subsections provide additional descriptions of the tasks shown in the workflow overview above.

12.4.1 CLOUD COMPUTE SETUP

There are several important cloud setup activities that can begin immediately at the start of the conformance phase and can occur concurrently while models are being integrated, tested, and prepared for Model Delivery to the cloud.

12.4.1.1 GENERATE EVENT SEEDS

The generated seeds provide the basis for stochastic sampling for the project.

12.4.1.2 STORM SAMPLING AND PLACEMENTS VIA FISHNETS

The storm events including placements are sampled for the conformance realization.

12.4.1.3 BREACH TRIGGER SAMPLING

The breach triggers are sampled for each event within the conformance realization. See Section 9 for criteria for the breach sampling scheme.

12.4.2 MODEL INTEGRATION

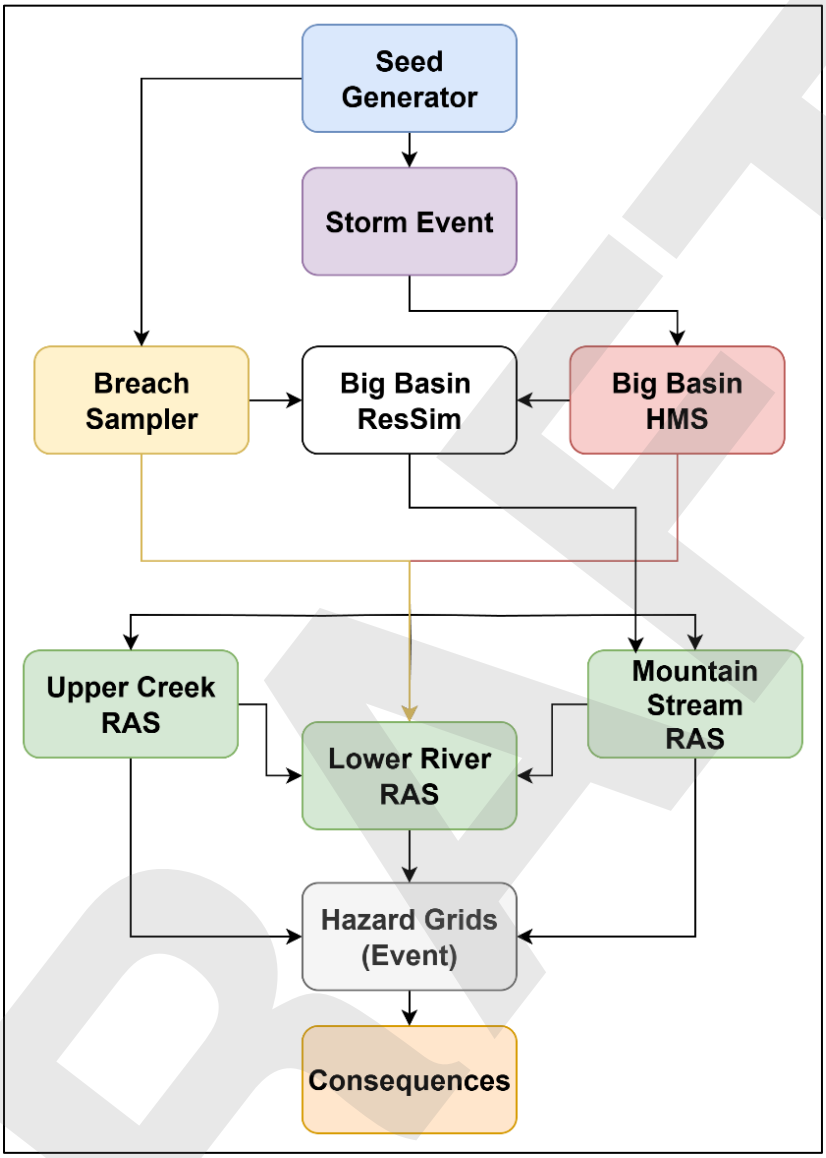
Model integration is the responsibility of the model integration lead and is performed in a desktop environment. This integration includes the hydrologic model, reservoir operations model, and all hydraulic models for the project. The Model Linking Register and Name Table are created during the scoping and data preparation phase and will guide all model integration efforts. Communication of the Model Linking Register can be aided by graphical representations of how output data is used as input for subsequent models as shown in Figure 12.3 below as an example.

During model integration, all models undergo several changes so they can accept synthetic events rather than observed events. This includes various boundary conditions and naming adjustments that are detailed in Job Aid 12, *Model Delivery for Cloud Compute*.

Desktop model integration allows the team members to thoroughly review and verify the configuration and performance of the full watershed model for individual events with easily identifiable and traceable boundary conditions in a traditional computing environment.

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Figure 12.3. Conceptual Model Linking Diagram



12.4.3 HISTORIC EVENT REPLICATION

During the previous calibration phase the individual models were calibrated to match observations from historic storm events. Within the desktop environment, the fully integrated models must be checked against those same historic events to confirm acceptable results are achieved.

12.4.4 STRESS TESTING

The system of models will be subjected to a wide variety of boundary conditions ranging from a dry basin with no rainfall to extreme floods on wet antecedent conditions during the stochastic model runs. Dams and levees will also be subjected to breaches. To minimize model instabilities during the production runs, the models are stress tested by the project's model integration lead in the desktop environment. This effort involves running extreme or unusual storm conditions through the model to check for any model instability or other issues. For instance, the largest storm in the meteorologic region (from the storm

catalog described in Section 5.4) could be centered over a critical location in the basin. All levees and dams could be allowed to breach during the same event. The initial model states may also be adjusted to extreme conditions to pair with the precipitation. Extreme low flows and dry conditions are considered as well as extreme high flows.

12.4.5 MODEL DELIVERY FOR CLOUD COMPUTE

After the models have been integrated, verified, and stress tested, they are nearly ready for delivery for cloud compute. However, there are several steps that must be taken, which are fully detailed in Job Aid 12, *Model Delivery for Cloud Compute*. After models have been posted to the correct locations on the conformance folder along with the final version of the Model Linking Register, the model integration lead is responsible for ensuring the data is accurate and ready for cloud compute via the pre-conformance review (next step).

After the integrated models are initially delivered to the cloud, changes will be needed to address errors or to iterate and achieve the desired model results. The *archive workflow*, as described in Job Aid 07, *Model Library Use*, is designed to minimize mistakes in model versioning. Before any model is updated, the previous version of the model is archived in Model Library through the rename functionality. The model should be renamed with */archive/* prepended to the model's name to support the model archive. After the previous model is archived, the new model is updated where the previous model was stored. This workflow ensures the most current models exist in a static location that does not change, while also maintaining a record of the superseded models.

12.4.6 PRE-CONFORMANCE REVIEW

After the models have been integrated into the desktop environment, verified against historic events, and stress tested, they are nearly ready to be delivered to the cloud storage for cloud computations of synthetic events.

Prior to handing off to the cloud compute team member, a full review of the models to be delivered will be completed and the models will undergo a QC check documented in the Pre-Conformance Review Checklist. Cloud computing does not begin until the pre-conformance review has been fully closed and approved by the model integration lead.

12.4.7 CLOUD-COMPUTE MANIFESTS

After all models have been reviewed and delivered to the cloud, the cloud-compute manifests are created as described in Job Aid 12, *Model Delivery for Cloud Compute*. These files provide instructions for cloud-compute plug-ins to perform compute jobs. The Model Linking Register and the data in the actual delivered models are utilized as inputs to create the compute-manifests. Strict adherence to naming conventions and scripting to automate certain steps in the production of the computing manifests has the potential to improve efficiency and ability to QC linking data.

12.4.8 INITIAL CLOUD COMPUTATION

Three to five events will be computed as a starting point before running any large-scale, cloud-compute simulations for the project. The cloud-compute lead builds the required compute-manifest to execute hydrologic, reservoir operations, and hydraulic models. The cloud-compute lead will verify that inputs and results from the events are being properly varied and then pass the model to the model integration lead for additional review checks.

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12.4.9 FIDELITY CHECK CLOUD VERSUS DESKTOP

The model integration lead will work with the cloud compute team member to perform checks and verify cloud compute produces identical results to a desktop computation with identical boundary conditions and inputs for each event. Any identified issues with the model linkages, desktop models, or cloud compute will be addressed. The cloud results will be evaluated for model linking, antecedent conditions, storm placements, and breach triggers among other items. Iterations will occur until discrepancies are fully resolved. The conformance review checklist includes items to verify that this step was completed.

12.4.10 SINGLE REALIZATION COMPUTATION

A single realization will be executed through cloud computing, which allows model performance over a long-term simulation to be evaluated against observed data. Frequency curves must be compared across the range of observed data with one realization of 2,000 years having a stable mean to the 100-year event.

To inform the event filtering described in the next section, each event in the realization is first computed through the HEC-HMS and HEC-ResSim models. The filtered events to run through the hydraulic model are then selected based on the results from the earlier models.

Each selected event will be executed through the hydraulic model step (i.e., depth grids, hazards grids, and consequences are not evaluated initially) and summary results are output to allow for analysis of model performance. Performance is then evaluated as described in the next step, and models are updated and recomputed until conformance metrics are met.

12.4.11 EVENT FILTERING FOR HYDRAULIC MODEL

It is recognized that not every synthetic event contributes to the overall characterization of the flood risk for the project since some of the sampled events produce only minor runoff compared to all other events in the synthetic year. The fully 2D hydraulic models for the project are somewhat computationally expensive and costs associated with running every event in cloud compute are not justified.

Leveraging the results from computation of all events through hydrology, reservoir operations, and breach sampling, a list of specific events for computation through the hydraulic models is developed. Tabulated summaries of the event-by-event maximums of many variables in many locations for each event are generated.

The cloud compute team member and model integration lead will work together to create a filtered list of sampled events that will be executed through cloud compute based on their importance to flood risk characterization for the project following the process described in Job Aid 13, *Event Selection for Hydraulic Modeling*.

Results from a full realization through the hydrologic and reservoir operations models are needed to inform filtering; therefore, there is a feedback loop with the Single Realization Computation step described in the previous section.

12.4.12 HAZARD MAPS AND CONSEQUENCE COMPUTATION

Cloud computation of event-based depth grids, summarized hazard maps, and consequences are computed based on the results from the hydraulic model. It is recognized that there is a relatively high computation cost associated with generating depth grids for each event and summarized hazards grids. Therefore, the conformance metrics described in the section below that relate model results through the hydraulic model will be evaluated to verify conformance prior to the computation of hazard maps and consequences.

12.4.12.1 CONSEQUENCES

Adjusted structure inventories are updated based on the results of the previously quality checks. Frequency-based consequence estimates are computed using damage function as described in Section 10. Expected annual damages and frequency of first inundated are produced for each structure. Refer to Job Aid 14, *Consequence Processes*, for details on the workflow.

Consequence team members must review outlier results for reasonableness, mapping results for coverage, and verify that consequences appear to be working as intended.

12.4.13 CONFORMANCE METRICS EVALUATION

The model results of the single realization allow for a robust evaluation of model performance over a long-term simulation against observed data. Fundamentally, the FFRD modeled results must reasonably align with observed long-term data and a thorough evaluation of the single realization results during the conformance phase provides credibility that the modeling system can provide flood risk estimates for the watershed.

Table 12.1 lists the relationships that must be evaluated for an FFRD study during the conformance phase. Evaluations must be performed via a graphical comparison to observed data and via computation of objective statistical metrics.

In addition to ensuring the model results match observed data, model stability is also a key criterion for model conformance. A broad review of overall results from the system of models must be completed. In particular, hydraulic model volume errors must be less than 5 percent for all events.

Additional specifics regarding objective pass or fail criteria for acceptable frequency-based model results will be considered for a future update to this SOP.

Table 12.1. List of Relationships of Long-term Modeled Data versus Observed Data for Conformance Evaluation

Relationship
Storm Types
Storm Dates/Seasonality
Storm Locations
Precipitation Frequency
Liquid Water at Soil Surface Frequency
Flow Frequency
Elevation Discharge Relationship
Elevation Frequency
Dam and Levee Breaching Frequency
Flood Extents versus Historical Event Information
Consequence Frequency versus Claims Data

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12.5 ITERATIONS TO ACHIEVE CONFORMANCE

The models are expected to undergo multiple iterations to achieve conformance to long-term observed data. Deliberate and careful model data management is required for successful completion of the project and improves the ability to communicate model updates with the project team. The current version of each model must be maintained as a static directory, which requires an archival of the previous version prior to upload of the update. This archival process and the standard directory names are described in Job Aid 07, *Model Library Use*.

12.6 COMPUTATIONAL RESOURCES

During the earliest cloud-compute efforts, the compute team monitors and records the demand for central processing unit (CPU), random access memory (RAM), disk space, and runtime for computing an event through the full system of models. These resource demand estimates inform the cloud computing resources needed for the full watershed multi-realization compute.

12.6.1 OPTIMIZATION OF COMPUTES

There are several variables that can be adjusted to optimize cloud computations. Shorter compute times for individual events can be achieved by using more powerful cloud resources, but experiences to date have shown that compute costs are optimized when less powerful cloud resources are utilized. Additionally, certain model settings related to the use of compute resources also impact cost and runtime, which must be optimized prior to initiation of large-scale cloud computes for FFRD. The primary objective of optimization is to achieve the lowest compute cost. Additional detail and considerations for optimization are included in cloud-compute documentation.

12.6.2 COMPUTE COSTS

After optimization of the computations, an estimate of resources and cost to compute the full set of events for the study is made.

12.7 CONFORMANCE PHASE REPORTING

A set of standard reports is written for each phase in the FFRD project. FFRD Templates contains the report template for the conformance phase report.

The conformance phase report describes model integration, calibration verification, and conformance outcomes. The report also includes a full description of updates required in the hydrologic, reservoir operations, system response, or hydraulic models that were made to achieve conformance targets. Reports are filed in the documentation folder within the watershed repository in the Model Library.

12.8 CONFORMANCE PHASE REVIEW PROCESS AND DELIVERABLES

Review of the conformance phase deliveries is critical for project success. The review steps during the conformance phase are defined below. Review checklists are used to track comments regarding pre-defined aspects checked throughout the review process. FFRD Templates contains checklist templates.

12.8.1 PRE-CONFORMANCE REVIEW

The pre-conformance review occurs after the system of models has been integrated in the desktop environment, tested, and delivered for cloud compute. The pre-conformance review verifies that the system of models is fully ready for the cloud compute process. A review checklist template with standard

prompts guides the review and documents completion of the review. This pre-conformance review checklist must be closed out prior to hand-off of the model to the cloud compute team member.

12.8.2 CONFORMANCE REVIEW

The conformance review covers all aspects of review of cloud-compute results, and the conformance phase report will be documented via the conformance review checklist template. This checklist includes items related to review of the 3 to 5 initial cloud compute events as well as the review of results from the full realization.

Throughout the conformance phase, iterations to the models may be necessary to achieve acceptable results.

After the initial cloud computations are complete, the team will verify the fidelity of outputs and iterate as necessary to confirm that the desktop models and cloud computing are producing intended results.

After the computation of a single realization in cloud compute, the results will be thoroughly reviewed and assessed for conformance and again iterated as necessary to verify that conformance targets are met.

The final delivery for the conformance phase close-out includes:

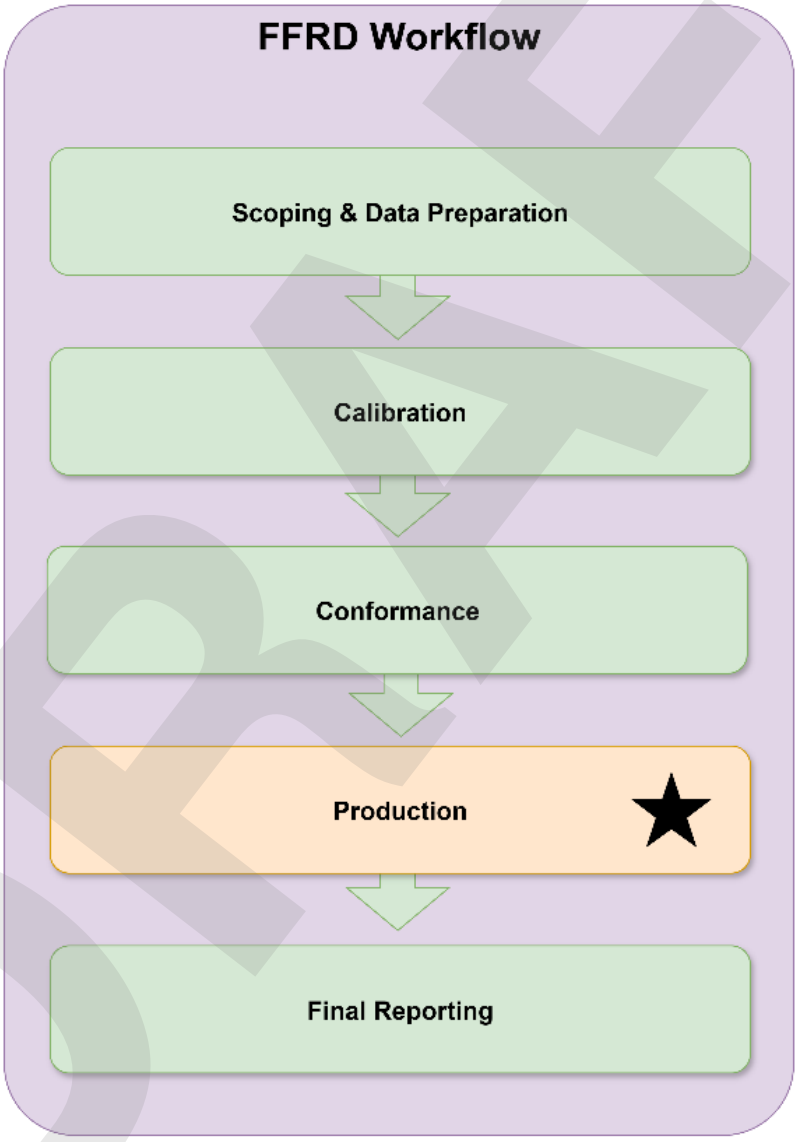
- Conformance phase report
- Final models posted to */conformance/[model]/* directories
- Conformance phase event-level model results posted to *conformance/simulations* directory
- Conformance phase summary model results posted to *conformance/simulations.*

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SECTION 13—PRODUCTION OF STOCHASTIC SIMULATIONS

The production phase of the project begins after demonstration that models achieve the standards set in the conformance phase of the project and the conformance phase review has been closed out. The number of runs transitions from a single realization to a full multi-realization simulation. The dominant characteristic of the production phase is the computation of a very large number of events, which requires significant cloud computing resources to achieve.

Figure 13.1. Context of the Production Phase of the Future of Flood Risk Data Workflow



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13.1 PROMOTION OF MODELS TO PRODUCTION DIRECTORY

All activities for the project leading up to the production phase verify that the system of models meets quality standards and can confidently be executed for a very large number of events. The models will not experience significant changes or updates during the production phase.

At the start of the production phase, the final set of models from the conformance phase are copied to the *production* directory and will be used directly in the production computations.

13.2 CLOUD COMPUTE SETUP

Event seeds, storm selection, storm placement, and breach trigger elevations are all produced for each event for the full stochastic simulation early in the production phase. This is done in a similar manner as described in Section 12.4.1.

13.3 STRATEGIES TO PROMOTE EFFICIENT USE OF CLOUD COMPUTE RESOURCES

Early in the production phase, the models have already been fully prepared for execution of all events in the stochastic simulation using cloud-compute tools. Before submitting the full set of compute jobs for all events and all models, small batches of runs should be made and evaluated to verify that accurate cloud configurations are in place. The following batches of runs are recommended for execution and evaluation in a stepwise fashion to confirm configurations:

- Execute full model suite for 3 to 5 events and evaluate results
- Execute hydrologic model for entire simulation and evaluate results
- Execute reservoir operation model for entire simulation and evaluate results
- Execute hydraulic models for all selected events and evaluate results
- Execute hazard grid generation for all selected events and evaluate results
- Compute consequences and evaluate results

13.4 EVENT FILTERING FOR HYDRAULIC MODEL

The simulation structure for FFRD includes modeling of multiple storm events per year. The goal is to produce annual probabilities of hazard and risk, and not every event contributes to the annual maximum of hazards. Given the computational burden of executing many events through the hydraulic models, it is worthwhile to select a subset of events with the potential to contribute to annual maximum hazards.

No single event necessarily drives the annual maximum hazards everywhere for a given variable of interest. Within each synthetic year, the maximum hazards for unique locations could result from locally intense precipitation events, upstream runoff, breaches, or other complex interactions.

Events will be filtered to execute only a portion of the events through the hydraulic model. Additional description and process for filtering the events is included in Section 12.4.11 and Job Aid 13, *Event Selection for Hydraulic Modeling*.

13.5 HOT FIXES

Running large quantities of numerical models within the stochastic simulation can yield conditions for specific events that cause the model to fail. Very low failure rates are expected due to the rigorous QC and QA steps that take place prior to the start of the production phase. However, since production simulations can produce extreme events or unique combinations of conditions, model failures may occur.

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When model failure occurs, the team will investigate the cause. Typically, failure can be addressed with a hot fix of the models. A hot fix is an alternate version of the input model(s) that addresses a specific accuracy or stability issue that arises in one or more events during the production phase. Hot fix model versions are applied only to specific events where they were deemed necessary. The team should consider the overall impact of failures of a limited number of events, and may judge that fully addressing every failed event may not be necessary if there is no significant impact on overall hazard frequency results.

The standard directory structure for the production phase provides for storage of the hot fix model versions along with metadata files that document the specific events they were utilized for.

The general workflow for implementing hot fixes is shown below.

1. Model integration lead investigates the cause of model failure
2. Model integration lead should take appropriate actions to correct the model.
3. Model integration lead should upload the new model files into Model Library in a hot fix location consistent with the required standard directory structure (see Job Aid 07, *Model Library Use*) and notify the cloud compute team. Also refer to Job Aid 12, *Model Delivery for Cloud Compute*, to verify models are uploaded in the required format for cloud compute.
4. Cloud compute team member reruns the failed events and works with model integration lead to verify the hot fix resolved the issue.
5. Cloud compute team member identifies which events the hot fix resolved and uploads a list of those event numbers to the hot fix model folder.

When hotfixes are applied to the project, the overall results from the full simulation are sourced from slightly different input models but represent a complete and cohesive quantification of hazards and risk for the basin.

13.6 INVESTIGATION OF OUTLIERS

In addition to certain events that may lead to a model failure due to model stability, the team must evaluate the results for possible outlier results that may indicate stability or accuracy issues. Outliers must be reviewed to determine if the results are valid or if a hot fix may be required.

13.7 SIGNIFICANT MODEL UPDATES

If the team determines the required fixes for certain failed events or outlier events would significantly impact the results for the entire basin, this may suggest the need for a more comprehensive update to the input models for all events. The review process leading up to the start of the production phase is designed to verify major model changes during the production phase will be extremely rare. Any decisions related to major model changes during production must be elevated to program leadership.

13.8 PRODUCTION RESULTS COMPARISON TO OBSERVED DATA

Results of the full stochastic simulation will be compared to observed data. This evaluation is like the evaluation done during the conformance phase (Section 12.4.13), although multiple realizations will be available to provide insight into the quantification of knowledge uncertainty and there is no expectation to adjust the models to improve the results.

13.9 PRODUCTION PHASE REPORTING

A set of standard reports is written for each phase in the FFRD project. FFRD Templates contains the report template for the production phase report. This report will contain an overview of the tasks executed

during the production phase, a summary of the results, and documentation of any model hot fixes or patches that occurred during the production phase.

13.10 STOCHASTIC SIMULATIONS REVIEW PROCESS AND DELIVERABLES

A review of the full production level results including precipitation, hydrology, reservoir operations, system response, hydraulics, mapping, and consequences is conducted. Knowledge uncertainty in results as quantified by differences among the multiple realizations is evaluated during this review. Review of the production phase deliveries is documented through a review checklist. FFRD Templates contains the checklist template.

The deliverables from the production phase are:

- Completed Production Phase Review Checklist
- Completed Production Phase Report
- Final models applied to simulation (including any hot fix alternate versions) posted to */production/[model]/* directories
- Production phase event-level model results posted to *production/simulations* directory
- Production phase summary model results posted to *production/simulations* directory

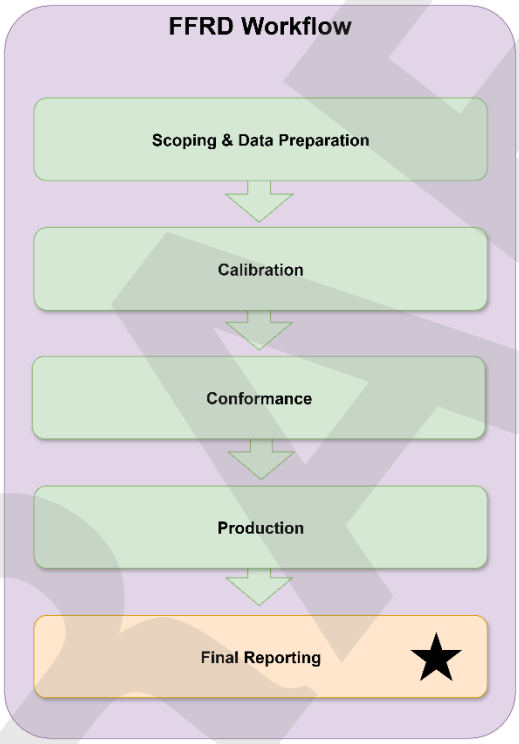
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SECTION 14—FINAL REPORT PHASE

After the production phase has been completed, including close-out of the quality control review checklist and approval from the TOG review, the final reporting phase transfers and communicates the results of the study. The full project directory dataset is archived for long-term storage. The final report is prepared to document the overall effort and to formally close the project.

A set of standard reports is written for each phase in the FFRD project and templates for these reports are included in FFRD Templates. The final report will be a relatively short document that refers to the documentation created at the end of each phase for detailed model development, calibration, and results.

Figure 14.1. Context of the Final Report Phase of the Future of Flood Risk Data Workflow



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JOB AID 1—SCOPING PROCESSES

This job aid describes considerations for the major project scoping decisions that define the overall project scope.

1.1 WATERSHED STUDY EXTENT

The Future of Flood Risk Data (FFRD) initiative will typically be scoped using Hydrologic Unit Code (HUC)-4 watersheds as previously established in the national watershed boundary dataset, but special considerations may apply. The scoping documentation will clearly describe all rationale related to any adjustments to the watershed study extent, and these adjustments should be vetted with FFRD initiative leadership.

An FFRD-specific geospatial polygon will be created for the project; it will be based on the full resolution project digital elevation model (DEM) and will follow FFRD spatial projection standards.

```
Delivery Location: basin-name > basin-data > project-scoping > study-area-polygon
```

1.2 HYDRAULIC MODELING UNIT DELINEATION

FFRD studies at the HUC-4 level are generally too large to efficiently develop and calibrate two-dimensional (2D) hydraulic models with sufficient detail to meet the study objectives. Therefore, the study basin is split into multiple hydraulic model units that define the extents for separate 2D hydraulic models. Considerations for hydraulic model unit delineation for FFRD studies are provided below.

```
Delivery Location: basin-name > basin-data > project-scoping > hydraulic-model-units
```

1.2.1 BASIN SIZE

The FFRD program uses HUC-4 boundaries to define the watershed study area. Developing full basin 2D hydraulic models at the HUC-4 scale following the procedures in this document is generally deemed impractical using the computing capability available with current technology. Therefore, smaller hydraulic modeling units will be defined as part of the FFRD scoping process.

Subbasins or model units with drainage areas of 500 to 1,000 square miles are practical for FFRD hydraulic modeling. The HUC-8 level boundaries provide a rough guide for subbasin delineation. The technical lead reviews the HUC-8 boundaries and adjusts to find more natural breaks from a hydraulic modeling perspective. There are various reasons the HUC-8 boundaries may be adjusted; specifically, consideration is given to the location of significant dams, stream gages, or population centers.

1.2.2 HYDROLOGIC MODEL BOUNDARIES

The boundaries will also align with hydrologic model boundaries; therefore, subbasins of the FFRD hydrologic models must be considered. It is best practice for hydraulic model unit boundaries to align with hydrologic model boundaries, such that hydrologic model output can be more seamlessly used as an input boundary condition for the hydraulic model. Although hydrologic subbasin boundaries may be edited during the FFRD model development process, this is an important planning consideration.

Hydrologic model boundaries and catchments are commonly delineated in a coarser resolution DEM than is generally used for hydraulic modeling. This can result in differences in watershed delineation, which should be resolved for FFRD projects so that the boundaries match.

1.2.3 DIGITAL ELEVATION MODEL ALIGNMENT

Hydraulic model unit boundaries should align with the full resolution DEM for the project. Effort should be made to identify structures, such as culverts and other hydro-connecting features, which can significantly alter the watershed delineation. Local knowledge and judgment may be required for a satisfactory watershed delineation, for example when there are interbasin canals or in watersheds that are very flat or contain significant underground drainage systems.

1.2.4 STREAM GAGES

Stream gage data is critical to successfully developing the runoff, conveyance, and backwater parameters. These gages also serve as the primary means of evaluating model accuracy by providing observed data against which modeled results can be compared. Therefore, model unit boundaries should generally be located away from important stream gage locations to prevent model results at gage locations from being unduly sensitive to boundary condition assumptions. Locating model unit boundaries away from important stream gages allows model results and calibration metrics to be independent of boundary conditions.

If model unit boundaries must be located near stream gages for reasons that cannot be avoided, these gage locations can still be useful for informing boundary condition assumptions and ensuring accurate transition between upstream and downstream model units. In this case, it may be appropriate to use the gage's published stage-discharge rating curve (which may need to be extended to include all flows of interest, even infrequent and extreme flows) as a downstream boundary condition in the hydraulic model.

1.2.5 LEVEES

Model unit boundaries should generally not intersect levee alignments, as splitting the levee limits the ability to evaluate the levee's behavior as a system. There may be occasional cases for extremely long levees where model unit boundaries must split the levee. In this case, careful consideration is required to avoid likely breach and overtopping locations. If a levee breach or overtopping occurs at the upstream end of the levee, separating the leveed area into multiple model units makes it difficult for leveed area inundation to flow through the artificial model unit boundary. The same effect can occur even if the breach/overtopping occurs near the downstream end of the levee. Modelers and team leads should carefully consider representation of inundation in areas where model unit boundaries are adjacent to or split a levee system.

1.2.6 BACKWATER EFFECTS AND RIVER CONFLUENCES

Model unit boundary selection should consider whether a given area is subject to significant backwater effects, which can significantly complicate boundary condition assumptions. River confluences are a common location where backwater impacts can be significant, and for this reason, model unit boundaries should avoid major river confluence areas.

Backwater impacts can be due to many factors other than river confluences. River control structures (such as locks and dams) can produce significant backwater effects, as can valley-wide constructions that limit the conveyance of a river valley. It may not be practicable to avoid all backwater impacts when selecting model unit boundaries. In these cases, boundary condition assumptions must account for these backwater influences. One way to accomplish this is to use a stage-discharge rating curve developed to account for backwater influence as the boundary condition. These rating curves can be developed from downstream model unit results and therefore require additional coordination between model units.

The scoped model unit polygons must not overlap, and their edges must match at all boundaries. Note that the extent of the hydraulic models developed for each model unit will commonly extend beyond the downstream boundary to avoid errors associated with boundary condition assumptions (more guidance included in SOP Volume II, Section 8).

1.2.7 URBAN AREAS

Urban areas should be contained within a single hydraulic model unit whenever possible.

1.2.8 DAMS

When possible, hydraulic model unit boundaries should not be located close to significant dams for the following reasons:

- To minimize complex linking with other models. The linkages within the FFRD system of models can be simplified when each dam is internal to a single hydraulic model unit.
- To better utilize available observed data for calibration. Dams often have associated flow and stage observed data that is best utilized for model calibration if boundary condition assumptions are remote from the observed data.
- To limit impacts to dam breach scenarios. HEC-RAS hydraulic breach scenarios are impacted by the reservoir pool level, and the downstream tailwater can rapidly rise during a breach event. The local hydrodynamics associated with a breach are more easily modeled when contained within a single model unit with minimal interactions with separate models.

1.2.9 DELINEATION PROCESS

Considering the factors included in this section, a geospatial polygon will be developed and coordinated with the larger scoping team to provide workability and to identify any potential modeling issues. When a final delineation is made, it will be uploaded to the directory location listed in Section 1.2 as a key scoping deliverable.

1.3 UNCALIBRATED HYDROLOGIC MODEL

A complete uncalibrated hydrologic model will be developed as part of the scoping phase, in advance of development of other models during the calibration phase. This model will fully define the subbasins. Reaches and reservoir elements are delineated during the scoping phase, which is important baseline information for development of the Name Table and Model Linking Register.

The guidance for development of the HEC-HMS model and the configuration of elements for FFRD studies is included in Volume II of the SOP.

```
Delivery Location: basin-name > basin-data > project-scoping > uncalibrated-hydrologic-model
```

1.4 MESH DEVELOPMENT CRITERIA FOR HYDRAULIC MODEL

The scope for the development of hydraulic 2D meshes for the FFRD initiative is guided by specific layers developed during the scoping phase. The criteria for developing each of the layers are shown in Volume II, Section 4.3.10, of the SOP. All areas not included in these three layers will be part of the *background mesh* in FFRD models.

- Urban Areas: Polygons should be sourced directly from the U.S. Census and provided in a geospatial format to match FFRD projection standards.
- Key Streams: Key streamlines will be selected manually by following the criteria in the SOP. This geospatial layer that defines a portion of the scope of the modeling effort may be based on the existing National Hydrography Dataset (when Elevation Derived Hydrography is not available early in the scoping and data preparation phase), which may not align perfectly with stream centerlines based on the project DEM. This scoping layer will be used together with the Elevation Derived Hydrography (based on project DEM) stream layer during the modeling development process (calibration phase).

- Other Streams: Developed manually based on criteria in the SOP. Similar use caveats apply as for the key stream layer described above.

```
Delivery Location: basin-name > basin-data > project-scoping > mesh-development-criteria
```

1.5 NAME TABLE

The comprehensive Name Table will include all planned element names for all models. This is aided by the HEC-HMS subbasins, reaches, and reservoir elements that are built into the uncalibrated base model during the scoping and data preparation phase. A template document with all necessary attributes is provided as the *name-table* and is available through the FFRD Templates and Resources. The names will all follow the conventions in Job Aid 06, *Standard Naming Conventions*.

```
Delivery Location: basin-name > basin-data > project-scoping > name-table
```

1.6 MODEL LINKING REGISTER

The comprehensive Model Linking Register will include all planned connections for all models. There will be a separate table for historic event scenarios and for synthetic event scenarios (where no historic observation data is leveraged for inputs). A template document is included in FFRD Templates and Resources and includes all necessary attributes. The names will all follow the conventions in Job Aid 06, *Standard Naming Conventions*.

Optionally, a graphical version of the Model Linking Register may be developed to support team communication and understanding of the data linkages.

Special planning for reservoir component linkages is highly recommended during the scoping phase. It may be necessary to link separate components of the dam to separate outflow timeseries independently (for example, one timeseries for conduit releases, another for spillway flows, and a third for dam overtopping flows).

Note that the purpose of the Model Linking Register is to develop a clear plan during scoping and improve team communication with regard to model linking throughout the project. This file is not directly used in cloud compute for applying the linkages. The process for ensuring accurate linkages is applied within cloud-compute events and is described separately.

```
Delivery Location: basin-name > basin-data > project-scoping > model-linking-register
```

1.7 DAM DATA

Dam data refinement and dam data collection are mentioned in the sections below but are not core scoping activities.

1.7.1 DAM DATA REFINEMENT

The National Inventory of Dams (NID) will be leveraged as the official dam inventory source for FFRD studies. Prior to or during the scoping phase, the NID will undergo a data refinement process to provide sufficient accuracy and completeness of the data for the entire study basin with a focus on ensuring data is of adequate quality to make dam scoping decisions for the FFRD study. The guidelines and procedures for this dam data refinement process will be advanced further in future versions of this SOP..

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1.7.2 DAM DATA COLLECTION

Efforts to obtain additional dam information for the purposes of FFRD modeling, such as physical structure data, reservoir storage information, risk assessment information, or operations manuals will be targeted to specific dams and based on the Dam Scoping Table developed according to the guidelines in Section 1.8. Processes for obtaining these datasets are provided in Job Aid 02, *Data Preparation Processes*.

1.8 DAM MODELING SCOPE SELECTION

During the scoping phase, the model treatment for each dam in the basin will be clearly specified in the *Dam Scoping Table*, which provides an authoritative summary of the planned modeling approach for each dam. The FFRD Templates and Resources GitHub Site contains a template for the Dam Scoping Table.

The Dam Scoping Table will include a listing for every dam listed in the NID. The modeling approach in terms of operations modeling, hydrologic modeling, hydraulic modeling, breaching, and treatment of initial pool levels for synthetic events will be specifically defined in the table.

There are many combinations by which to model a dam across FFRD models, and dam scoping decisions must be made with consideration for the function of the full system of models. Guidelines for scoping the treatment of each dam within each component model are provided in the sections below, which balance modeling efficiency, accuracy, and the magnitude of impact to overall flood characteristics among other factors.

```
Delivery Location: basin-name > basin-data > project-scoping > dam-scoping-table
```

1.8.1 RESERVOIR OPERATIONS MODEL SCOPE SELECTION GUIDELINES

Major reservoirs or reservoir systems that are actively operated for a downstream constraint should be considered for possible inclusion in an FFRD reservoir operations model. If a dam operates as part of a system only in rare cases, operates within a small system of dams, or has otherwise uncertain or complicated operations, scoping decisions will require engineering judgment. Reservoirs without a controlled outlet, or those that do not operate for a downstream constraint or in coordination with a system of other reservoirs should not be scoped for inclusion within the reservoir operations model. Reservoirs without controlled outlets should only be modeled in the reservoir operations model if their state (i.e., pool elevation, storage, outflows, etc.) is required to determine the operations at another facility being modeled. Reservoir operations that can be reasonably represented with simplified approaches in either the hydrologic or hydraulic model should not be scoped for inclusion in the reservoir operations model for the FFRD initiative.

Each dam selected for inclusion in the reservoir operations model will be documented in the Dam Scoping Table, and the high-level modeling approach will be specified according to the attributes listed in Tables 1 and 2.

Table 1. Reservoir Operations Model Scoping Options

Include in Reservoir Operations Model (Y/N)?
No
Yes, Include in HEC-ResSim

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Table 2. Scoping Options for Reservoir Operations

Number of Unique Outflow Features? ("ResSim Dams")
N/A
1
2
3
>3

1.8.2 HYDROLOGIC MODEL SCOPE SELECTION GUIDELINES

HEC-HMS reservoir elements should be included for dams that are expected to significantly impact flood characteristics at the basin scale, with special consideration for dams that impact the ability to match downstream observed gage data for flood events. The availability of physical dam data and operational data to support modeling inputs may be considered when scoping the hydrologic modeling approach for individual dams. Watershed size, reservoir storage, operation characteristics, and river slope are among the factors in the evaluation of the impact of individual dams. As a general guideline, dams with over 10,000 acre-feet in storage and a flood control mission are candidates for inclusion as HEC-HMS reservoir elements.

For dams selected for inclusion as reservoir elements in the hydrologic model, the specific modeling method for outflows will also be included in the Dam Scoping Table. Various methods are available depending on the reservoir characteristics. As reservoirs and their operations increase in size and complexity, additional modeling techniques are required. These techniques are described below in Table 3.

During the calibration phase of the project, it is acceptable to use the Specified Release routing method with observed outflows from the reservoir. This allows downstream subbasins to be calibrated without the added uncertainty of reservoir routing. When observed specified releases are leveraged for the calibration phase, the simplest reservoir routing method necessary should be chosen for use in the conformance and production phase models.

Simple reservoir operations include one-to-one storage and discharge relationships, uncontrolled outlets, and ungated spillways. Typically, if the outlets and spillways have simple rating curves not dependent on inflow or seasonality, the dam can be modeled with simple procedures in HEC-HMS. A single outflow curve is the simplest hydraulic routing method available in HEC-HMS. If outlets are gated or controlled, simple operations can still be used by analyzing observed data and estimating an average outflow for a given elevation or storage condition. For more complex reservoir operations, modelers typically rely on HEC-ResSim to model operating schedules, rules, and downstream constraints; however, rule-based operations in HEC-HMS can serve as a close approximation.

Table 3. Hydrologic Model Dam Treatment for Historic Events

Hydrologic Modeling Treatment – Historic Events
Reservoir Element—Rule-Based Operations
Reservoir Element—Outflow Structures
Reservoir Element—Outflow Curve
Reservoir Element—Specified Release
No Reservoir Routing Element

Table 4. Hydrologic Model Dam Treatment for Synthetic Events

Synthetic Event Hydrologic Modeling Treatment
Reservoir Element—Rule-Based Operations
Reservoir Element—Outflow Structures
Reservoir Element—Outflow Curve
Reservoir Element—Specified Release
No Reservoir Routing Element

1.8.3 HYDRAULIC MODEL SCOPE SELECTION GUIDELINES

In the hydraulic model, various treatments and levels of detail are available depending on the reservoir characteristics. As reservoirs and their operations increase in size and complexity, more detailed modeling techniques are needed to account for the impact of dam operations, initial pool levels, and the possibility of breach events to accurately characterize flood risk. The selection of the specific hydraulic modeling treatment for each dam must occur in coordination with selected techniques for the hydrologic and reservoir operations models with the goal of establishing a cohesive plan for the full system of models. Table 5 shows the options for hydraulic modeling techniques and methods. Volume II, Section 8 of the SOP includes specific modeling techniques for each type of model treatment.

Table 5. Hydraulic Model Dam Modeling Treatment

Hydraulic Modeling Treatment
Type A—Operation Modeled outside of Hydraulic Model*
Type B—Operation Modeled by Hydraulic Model*
Type C—Simplified Outflows Modeled by Hydraulic Model*
Type D—Basic Representation in Hydraulic Model
Type E—Not Modeled

*Hydraulic Model Treatment can support breach events

The key consideration for hydraulic model treatment selection is to ensure that the effects of all dams with significant impact on flood risk characteristics, either due to providing flood storage or inducing excess breach flooding, are represented by the model. All dams that meet these criteria shall be scoped to receive Type A, B, or C treatment, which account for reservoir storage and can assess the impacts of breach events.

A storage threshold of 2,500 acre-feet can be considered a guideline for applying Type A, B, or C treatments to dams. The volume of storage relative to drainage area for the dam, hazard classification, and immediate proximity to population centers may also be considered in the selection of hydraulic model treatment.

Type A treatment will be chosen for dams that have been selected for inclusion in either the reservoir operations model or for advanced outflow modeling method in the hydrologic model.

Type B treatment will be chosen for dams when the hydraulic model will provide superior outflow simulation compared to the chosen methods for the hydrologic model, and physical dam information and operational information are available to support a more detailed simulation within the hydraulic model.

Type C treatment will be chosen for dams when the hydraulic model will provide superior outflow simulation compared to the chosen methods for the hydrologic model, but physical and operational data are limited, and outflow features must be approximated.

Type D treatment will be chosen for dams that do not meet the threshold for Type A, B, or C treatments.

Type E treatment is reserved for dams that are deemed to be insignificant. This determination may be related to inability to confirm a dam exists, very small dams, partially breached dams, or other factors determined by professional judgment.

Table 6. Initial Pool Elevation

Include Hydraulic Initial Pool Elevation (Initial Condition [IC] Point)?
Yes
No

IC points must be used in the hydraulic model to simulate events with reservoirs that are not dry at the start of the storm event. All dams selected for Type A, B, or C modeling treatments must include IC points, and all other dams are not required to include IC points.

1.8.4 DAM BREACH MODEL SELECTION GUIDELINES

The Dam Scoping Table will include a clear determination of which dams will be allowed to breach in the FFRD stochastic simulations. The guideline for selection of these dams follows the threshold described for selection of hydraulic model treatments, and therefore all dams that are modeled with Type A, B, or C treatments will also be allowed to breach.

Table 7. Breach Modeling of Dams

Sample Breach
Yes
No

1.8.5 RESERVOIR INITIAL POOL ELEVATIONS FOR STOCHASTIC EVENTS

The variation in initial reservoir pool elevations at the start of simulated events can be an important factor for flood characterization within the context of the quasi-continuous simulation approach for FFRD. A clear determination of which dams will be simulated with a varied initial pool elevation will be included in the Dam Scoping Table. The source for the sampled elevations will also be identified.

The available data to support sampling of varied pool elevations is the key consideration regarding this scoping decision. Calibrated period-of-record simulations from either the reservoir operations or hydrologic models in certain locations may provide a defensible simulated observed record for locations where observed data is not available or reliable.

When no observed or simulated data is available to support sampling, a static water surface for all stochastic events is acceptable, but care should be taken to prevent this assumption from driving the representation of flood characteristics for the basin, and it should be avoided for larger significant dams that may have variable water levels.

Table 8. Options for Initial Pool Elevation for Stochastic Events

Vary Initial Pool Elevation during Stochastic Events
Yes—Sourced from Reservoir Operations Model Period of Record Simulation Results
Yes—Sourced from Hydrologic Model Period of Record Simulation Results
Yes—Sourced from Observed Period of Record
No—Static Initial Pool for all Events
Not Applicable (No IC Point)

1.9 LEVEE DATA

The National Levee Database (NLD) will be leveraged as the official source of inventory, alignments, and crest profiles of levees for FFRD studies. Prior to the study, the NLD will undergo a data refinement process to provide sufficient accuracy and completeness of the data for the entire study basin to adequately support the FFRD study. The guidelines and procedures for this levee data refinement process are documented outside of this SOP.

```
Delivery Location: basin-name > basin-data > levee-data > nld
     Files: levee alignment and profile geospatial data
```

1.10 LEVEE MODELING SCOPE SELECTION

All levees will be represented in the FFRD hydraulic models (according to guidelines in Job Aid 09, *Levee Physical Representation*) and all levees will be scoped for at least one breach location (as described in Job Aid 11, *Levee System Response Development*).

```
Delivery Location: basin-name > basin-data > project-scoping > levee-modeling-scope
     Files: levee breach locations (geospatial with attributes)
```

1.11 CALIBRATION SCOPE

A clear plan for the approach to event-based calibration of FFRD models is established during scoping. This plan consists of both establishing stream gage performance metric target prioritization and selecting specific historic events to be used during model calibration efforts.

```
Delivery Location: basin-name > basin-data > project-scoping > calibration-scope
     Files: Gages (geospatial with attributes), Calibration Event Table
```

1.11.1 CALIBRATION GAGE CATEGORY ASSIGNMENTS

The following sections outline how gages are assigned to the primary, secondary, and tertiary categories.

1.11.1.1 CALIBRATION METRIC HIERARCHY

All active stream gages with a significant period of record of observed data are used for calibration of FFRD models. Each gage is categorized as either primary, secondary, or tertiary gages to describe the hierarchy for use in model calibration. For example, more rigorous calibration metrics are used for primary gages than for secondary gages. The gages are categorized during the scoping phase, ensuring consistency in calibration targets across all FFRD modeling efforts for the basin. The selected gages—

with primary, secondary, and tertiary category attributes—are delivered as part of the scoping and data preparation phase.

Primary gages are typically located on major streams, have plentiful and reliable observed data, and/or are used in reservoir operational decisions. Secondary gages may be sited similarly to primary gages; however, they may be located on smaller streams and not used when operating flood control reservoirs. In general, one primary and one secondary gage should be located within each hydraulic modeling domain with the remaining gages designated as tertiary. However, this categorization is subjective.

1.11.1.2 PRIORITIZATION OF GAGES

Some modeling units may include numerous stream gages, and while information from all these gages would be incorporated into FFRD models in an ideal scenario, it must be acknowledged that there are constraints of time and funding which limit what refinement can be performed. Prioritization of gages should consider factors such as data availability, potential for flood risk near the gage, and many other factors, such as:

- Location of gages within model unit—At least one primary gage within each model unit, if one exists.
- Gage Period of Record—Consider:
    - Does observed data cover the majority of calibration events?
    - Does the data contain significant gaps?
    - Do these gaps occur at times of peak flow/stage?
    - Do large reservoirs or other control features exist?
    - Is this gage used in reservoir operational decisions?
- Population Centers near Gages.
- Gage Data Quality—Consider:
    - Is gage data reliable and consistent over the period of record?
    - Does the gage provide data at a sufficient time interval, such as 15-minute or hourly data, as opposed to daily data?
    - If only annual peak flow data or nearby high-water marks are available, this can still be useful as a peak stage/flow comparison.
- Nearness to other gages—Redundancy of gages representing similar or overlapping areas should be avoided.
- Spatial Coverage throughout Basin—Select gages that provide coverage throughout the basin, even if some have observed data of lesser quality.
- Consider mainstem rivers, major tributaries, then minor tributaries.
- Consider representing both regulated and unregulated areas within the basin.
- Drainage Area Coverage—Assign less priority to areas with small drainages.
- Similarity of Physiographic Features between Nearby Gages (e.g., slope, land-use, land-coverage, soil type, etc.)—Prioritize representing the range of conditions existing within the watershed if a modeling unit contains diverse physiographic properties.

1.11.1.3 EXCLUSION OF GAGES

There may be scenarios where gage data will not be used. Strong justification is required to support the decision to exclude a stream gage from consideration. Examples may include a stage-only gage lacking a reliable elevation datum, a gage representing a very small drainage area (less than one square mile), or a gage where data was not recorded during the calibration events. For some streams and basins where flows are particularly flashy, daily data may not be sufficient for use in calibration. However, there are

scenarios where daily data may be a useful data source; engineering judgment should be used in these cases.

1.11.1.4 OBSERVED DATA CONNECTION TO MODEL ELEMENT NAMES

The observed data will be compared to modeled data throughout the project. During the scoping phase, the connection of observed gage locations to the model element names will be included as attributes in the calibration scope geospatial point file. For each gage location, the HEC-HMS, HEC-ResSim, and HEC-RAS model elements (matching the Name Table) will be included as attributes.

1.11.2 CALIBRATION EVENT SELECTION

Historic storm events to be used for FFRD model calibration will be selected during the scoping and data preparation phase of the FFRD project. Typically, six events are selected. Fewer events may be selected if data is not available in areas with limited gage record or arid regions with few events.

Calibration events are selected with consideration of the factors listed below in priority order:

1. Precipitation data availability
2. Gage data, or other observed data availability
3. Spatial coverage throughout basin (precipitation coverage in such a way that the entire model unit experiences a range of flooding)
4. When considering events such that each portion of the basin experiences at least one flood event, prioritization is given to the largest events of recent record (the largest events with associated high-quality gridded precipitation data)
5. Type of storm event (representation across all event types applicable to the basin, like snowmelt, rain on snow, tropical storm, etc.)
6. Consideration is given to similar events within weather generation—a single event instead of a wet season or long series of smaller flood events, when possible
7. Consideration is given to changes within a watershed that would majorly impact calibration. If possible, events should be selected which best represent the current condition of the basin. Examples of large system changes are long-term cyclical weather patterns, construction of a large dam or levee system, significant mining or urbanization, or significant changes to a system response due to wildfire or sedimentation.
8. Seasonality of impactful events

Each model type (i.e., HEC-RAS, HEC-HMS, HEC-ResSim) should use consistent simulation times (start time and end time) for each calibration event. For example, for each calibration event, all hydraulic models should specify identical simulation windows. These simulation windows may vary between model types, but models with dependencies should fully cover the simulation window of the models that depend on them. For example, the HEC-HMS simulation window should equal or exceed the HEC-RAS simulation window because HEC-RAS depends on output from HEC-HMS.

1.11.2.1 CALIBRATION EVENT LENGTH—RAINFALL

The time window for each selected event will be clearly stated. A rainfall driven calibration event ranges from just prior to the start of precipitation and extends through the hydrograph peak at the downstream model unit boundary. Where possible, these events should extend 1 to 2 weeks, but up to 30 days is acceptable. HEC-HMS model times may start just before the HEC-RAS models (generally 1 day but could extend further) to address ICs. If a downstream boundary includes a tropical surge, HEC-RAS modeling should start prior to both the surge and rainfall.

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1.11.2.2 CALIBRATION EVENT LENGTH—SNOWMELT

A snowmelt driven event should start prior to the rise of the flow and extend through to the peak of the hydrograph at the downstream model unit boundary. If rainfall is a driver of snowmelt, the simulation should start prior to the start of precipitation. Initiating snowmelt may take more than just a single day of warm weather, so the timing of event simulation initiation should consider the flow hydrograph in addition to the start of forcing factors. HEC-HMS model times may start just before the HEC-RAS models (generally 1 day but could extend further) to address these ICs. Snowmelt is typically a slow process; however, a snowmelt flood event is typically associated with a factor that triggers more rapid snowmelt. These events last on the order of one month but may need to extend up to 2 to 3 months to capture the peak of the snowmelt hydrograph and the receding limb. Snowmelt is an annual event, so it can be helpful to model consecutive water years and examine the observed and modeled snow water equivalent (SWE) in a continuous model to understand the snowmelt seasonal patterns and typical snowpack duration.

1.11.2.3 CALIBRATION EVENT LENGTH—COMBINATION EVENT

A rain-on-snow event should range from just prior to the start of precipitation and extend through to the peak of the hydrograph at the downstream model unit boundary. Where possible, these events should extend 1 to 2 weeks, but up to 30 days is acceptable. During a rain-on-snow event that occurs during the snowmelt season, prior to precipitation falling there may already be an underlying baseflow due to snowmelt occurring in the basin. HEC-HMS model times should start at a point to capture any of these ICs contributing to the overall peak flow. Therefore, it is acceptable that HEC-HMS models start just before the HEC-RAS models (generally 1 day but could extend further) to address these initial conditions. If a downstream boundary includes a tropical surge, HEC-RAS modeling should start prior to both the surge and rainfall.

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JOB AID 2—DATA PREPARATION PROCESSES

1.1 GENERAL

The scoping and data preparation phase of an FFRD project includes the preparation of several datasets that will serve as the foundation for model development and production during the subsequent phases of the project. This document details the guidance and workflows for collecting many of the required datasets as listed in Section 4 of SOP Volume II.

Note that other datasets that are not included in this job aid are also prepared during the scoping and data preparation phase, but they require additional coordination with Hydraulics and Hydrology modeling team members and are described in more detailed documents.

1.1.1 DATA DELIVERY LOCATION AND NAMING CONVENTIONS

All shared data prepared and delivered for FFRD project use is stored in the cloud-based project directory accessible via the Model Library according to the standard directory structure. Specific paths for individual datasets are called out throughout this document. Naming conventions (provided in Job Aid 06) must also be followed.

1.1.2 GEOSPATIAL DATA PROJECTION

SOP Volume II discusses the specific parameters associated with the standard projection, which must be adhered to for all geospatial datasets.

1.1.3 GEOSPATIAL DATA FORMATS

The preferred file format for vector data (points, lines, and polygons) is the GeoPackage.

The preferred file format for basin-wide raster data (digital elevation models [DEMs], landcover, etc.) is GeoTIFF.

1.2 DATA MANIFEST

The data manifest is a required deliverable from the scoping and data preparation phase. It serves as a comprehensive record of all datasets delivered during this phase, including metadata for geospatial files, models, reports, and other data types provided. The data manifest must document all pertinent information about the datasets. The manifest enables traceability, consistency, and usability of the data for subsequent phases of the project.

Multiple team members contribute to the preparation of data supplied during the scoping and data preparation phase. Therefore, multiple team members must contribute to populate the data manifest. The prepared data can be broadly categorized into three categories: geospatial data, reference data, and scoping data, which are described below.

1. **Geospatial Data.** The collection of all official input data to be used as a consistent source for the entire FFRD modeling effort. Many of the datasets are data pulls from centralized nationwide databases that are maintained by others and represent a static snapshot of the data, clipped to the project study extent to be applied to the FFRD project consistently. This includes the DEM, National Land Cover Database (NLCD), National Hydrography Dataset (NHD), and several other datasets.

2. **Reference Data.** The collection of all relevant data to be consistently referenced by all efforts for an FFRD project. This includes observed data, existing models, and existing reports, among other items.

3. **Scoping Data.** Datasets that define the scope of the FFRD project. Many of the scoping datasets involve analysis by the technical team to define the scope. This includes the study extent polygon, model unit polygons, mesh development criteria, and calibration event selection, among other items.

A template file for the data manifest is available in FFRD Templates and Resources. The following attributes are included in the data manifest.

Table 1. Attributes for Datasets Included in Data Manifest Template

Attribute	Description
Dataset Name	Name of dataset
Path	Location of dataset in FFRD repository
Description of Content	Content description
Category	Category of dataset: Geospatial, Reference Data, Scoping Data, Observed Data, Models, other.
Purpose	Purpose of the dataset for FFRD Project Use. Refer to Section 4 of SOP Volume II.
Generation Method	Is data direct from source or derived based on the source?
Study Type	Applicable for Flood Insurance Rate Map (FIRM) Database-specific datasets: Effective, Base Level Engineering (BLE), BLE Effective
Source	Publishing organization plus any specific data systems within the organization. (e.g., Federal Emergency Management Agency [FEMA] Mapping Information Platform [MIP], U.S. Geological Survey [USGS] Three-Dimensional Hydrography Program [3DHP], National Structure Inventory [NSI], etc.)
Source Type	National, State, Local, Multiple, Other
Source Detail	Descriptive text regarding source, such as URL used for access.
File size	Size of the dataset
Currency	Last update date of the dataset provided for FFRD project
Geospatial Data Type	Polygon/Point/Polyline/Raster/Image/Not Applicable

At the end of the scoping and data preparation phase, the data manifest must be delivered to the project location below, and it will be reviewed for accuracy and completeness prior to close-out of the phase.

```
Delivery Location: basin-name > basin-data > data-manifest.xlsx
```

1.3 FEMA DATASETS

Applicable FEMA data for the FFRD modeling efforts include regulatory effective data and BLE data. Various spatial vector data, raster data, and reports are available to download from both publicly available and access-restricted sources. The information in the sections below provides comprehensive listings of

available FEMA data and access information. The items listed in Section 4 of SOP Volume II are the minimum required deliverables for the FFRD scoping and data preparation phase.

```
Delivery Location: basin-name > basin-data > fema
     *all FEMA data including FIS reports and effective NFIP flood zone polygons.
Delivery Location: basin-name > basin-data > existing-models
     *all existing H&H models associated with FEMA efforts
```

1.3.1 EFFECTIVE DATA

Relevant effective data include the FIRM database, Flood Insurance Study (FIS) reports, and Flood Risk Database.

1.3.1.1 FLOOD INSURANCE RATE MAP DATABASE GEOSPATIAL LAYERS

The FIRM is the official regulatory map of a community on which FEMA has delineated the Special Flood Hazard Areas (SFHAs) and Base Flood Elevations (BFEs). The effective FIRM data relevant to FFRD modeling efforts include the spatial vector layers and tables noted below.

Metadata, project narratives, and the source citation table may also be available to reference the sources used in the study and other background information. Effective data issued more than 10 years ago may not have as much detailed data available due to changes in the data, software, tools, and modeling methods used over the years. Most hydraulic data for studies newer than 1995 are based on Hydrologic Engineering Center's River Analysis System (HEC-RAS) software, but most pre-1995 studies are based on HEC-2 data. Most hydrologic data for studies newer than 1992 are based on HEC's Hydrologic Modeling System (HEC-HMS) software, but most pre-1992 studies are based on HEC-1 data.

- **Base Flood Elevations:** The S_BFE layer shows the location and attributes for BFE lines and can be used to reference elevations where no effective cross sections are available.
- **Flood Hazard Areas:** The S_Fld_Haz_Ar layer shows the location and attributes for flood risk zones.
- **Gages:** The S_Gage layer is based on hydrologic data and provides the location and attributes for gages in a study area.
- **General Structures:** The S_Gen_Struct layer shows the location and attributes for flood control structures such as dams, culverts, bridges, etc., in the study area.
- **Profile Baselines:** The S_Profil_Basln layer shows the location and attributes for profile baseline and stream centerline features for the Flood Risk Project area.
- **Cross Sections:** The S_XS layer shows the location and attributes for cross section lines for flooding sources. This layer contains all cross sections in a model, not just the lettered cross sections. This layer can be used as a reference for locating relevant cross section data, such as elevation and bathymetric data not captured by LIDAR.
- **Subbasins:** The S_Subbasin layer is based on hydrologic data and provides the extents and attributes for subbasins in the study area.
- **Summary of Discharges:** The L_Summary_Discharges table is based on hydrologic data and provides information about the hydrologic discharge information associated with nodes referenced in the FIS Report Summary of Discharges table. The information on stationing, discharges, and elevations in this table can be used to inform model development and cross-check results.
- **Summary of Elevations:** The L_Summary_Elevations table provides information about SFHAs with static elevations that are referenced in the FIS Report Summary of Elevations table. This table can be used as a reference for cross-checking water surface elevation results.

- **Water Areas:** The S_Wtr_Ar layer shows the location and attributes for hydrography features shown on the FIRM as areas. This layer can be used as a reference for flooding sources not included in the S_Fld_Haz_Ar.
- **Water Lines:** The S_Wtr_Ln layer shows the location and attributes for hydrography features shown on the FIRM as lines. This layer can be used as a reference for flooding sources where profile baselines are not available.

1.3.1.2 FLOOD INSURANCE STUDY REPORTS

The FIS report is a compilation and presentation of flood risk data for specific watercourses, lakes, and coastal flood hazard areas within a community. When a flood study is completed for the National Flood Insurance Program (NFIP), the information and maps are assembled into an FIS. The FIS report contains detailed flood elevation data in flood profiles and data tables and can be used as a reference for elevation data and FIS background information that may be referenced in developing FFRD hydrologic and hydraulic models.

1.3.1.3 FLOOD RISK DATABASE

The Flood Risk Database is a resource that supplements flood hazard information on effective FIRMs, FIS reports, and FIRM database products. This database includes water surface elevation grids, depth grids, and Areas of Mitigation Interest. This database may not be available for all communities.

1.3.2 BASE LEVEL ENGINEERING DATA

There are some instances where effective data is not available for a community, but BLE data is available. The BLE process uses the best available data and combines automated techniques with traditional model development to create flood hazard boundaries for the 1-percent-annual-chance event. Flood hazard boundaries are also typically estimated for other recurrence intervals. The results from BLE can be used to produce both regulatory and non-regulatory products. BLE data relevant to the FFRD modeling efforts include HEC-RAS models, vector spatial data, raster data, and a BLE report. More information on the relevant data associated with BLE studies is listed below:

- **Base Flood Elevations:** S_BFE data is available for BLE studies and follows the same requirements as the S_BFE layer listed in the FIRM Database subsection above.
- **Flood Hazard Areas:** S_Fld_Haz_Ar data is available for BLE studies and follows the same requirements as the S_Fld_Haz_Ar layer listed in the FIRM Database subsection above.
- **Water Lines:** The S_Wtr_Ln data is available for BLE studies and follows the same requirements as the S_Wtr_Ln layer listed in the FIRM Database subsection above.
- **HEC-RAS Models:** Hydraulic models associated with BLE studies are based on HEC-RAS software and typically include multiple events, including the 1- and 0.2-percent-annual-chance events. Types include both steady-state one-dimensional (1D) models and unsteady two-dimensional (2D) models.
- **Flood Depth Grids:** These grids are produced as raster data and typically include the 1- and 0.2-percent-annual-chance events. The grids are typically stored in the Flood Risk Database referenced in the Effective Data section above.
- **Water Surface Elevation Grids:** These grids are produced as raster data and typically include the 1- and 0.2-percent-annual-chance events. The grids are typically stored in the Flood Risk Database referenced in the Effective Data section above.
- **BLE Report:** This report provides detailed information on the methodology used for the study.

1.3.3 DATA GATHERING PROCESSES

A structured data gathering process is essential to ensure accuracy, comprehensiveness, and documentation of information available for FFRD modeling efforts. This section outlines the key sources and steps for obtaining relevant effective and BLE data.

1.3.3.1 SOURCES FOR FEMA DATA

Sources for effective and BLE data from FEMA are listed below:

- **FEMA Flood Map Service Center (MSC):** This is a publicly available source that can be used to obtain effective flood hazard data, including Flood Risk Databases, FIRMs, and FIS reports. Digital effective spatial data is available to download by county or state through the National Flood Hazard Layer (NFHL).
- **Estimated Base Flood Elevation Viewer (eBFE):** This is a publicly available source for Region 6 (Arkansas, Louisiana, New Mexico, Oklahoma, and Texas) and Region 9 (Arizona, California, and Nevada) that can be used to review which Hydrologic Unit Codes (HUCs) are located within the area of interest and to download available BLE data. Data is generally not available for other regions.
- **Mapping Information Platform:** This source requires access to the Risk Analysis and Management (RAM) Access Portal (RAP). The Flood Risk Study Engineering Library (FRiSEL) and MIP File Explorer can be used to obtain hydrologic and hydraulic data.
- **FileTrail:** This source requires access to RAP. It can be used to request data for studies that are too large to download from the MIP.

1.3.3.2 PROCESS TO OBTAIN DATA

By leveraging the available tools and platforms above, users can efficiently collect and validate the necessary datasets. The following subsections provide a comprehensive guide to accessing and utilizing these resources effectively.

While the MSC and eBFE Viewers are valuable resources for effective and BLE data, ensuring an individual with MIP access is involved in the data gathering process can significantly enhance the data available to use. MIP users benefit from advanced search tools and broader data access, which can help overcome the limitations faced by non-MIP users. MIP users have access to a broader range of hydrologic and hydraulic data, including detailed datasets that may not be available through public sources, such as the MSC or eBFE Viewer.

1.3.3.2.1 MAPPING INFORMATION PLATFORM AND FLOOD RISK STUDY ENGINEERING LIBRARY

1. Use keywords, such as the subbasin, watershed, or county name, to search for hydraulic data using FRiSEL on the MIP to get an initial idea of what may be available for the area in question.

2. Use Geographic Information System (GIS) software to determine which counties intersect the watershed or subbasin for your area of interest.

3. Search FRiSEL using the Advanced Search to search by state, county, or community to determine the availability of hydrologic and hydraulic data.

4. For data too large to download from FRiSEL, use File Explorer in the MIP to download hydrologic or hydraulic data in smaller portions where possible.

5. For remaining data too large to download from FRiSEL or File Explorer in the MIP, request data from FileTrail.

1.3.3.2.2 MAP SERVICE CENTER AND ESTIMATED BASE FLOOD ELEVATION VIEWER

1. To obtain data from the MSC, use the MSC Search All Products feature to search by county or community to locate any effective data available for the area in question. If you search for effective NFHL data by state, you may search for any community in the state.

2. To obtain data from the eBFE Viewer, select whether your area of interest is in FEMA Region 6 or Region 9 on the homepage. On the next window that appears, select the *I Want to Download* option. Then select the watershed(s) noted as having data available to download the available datasets and models.

1.4 ACQUISITION OF CORE GEOSPATIAL DATASETS

1.4.1 DIGITAL ELEVATION MODEL

The requirements for the project DEM are provided in SOP Volume II. The delivery location for the DEM is shown below.

```
Delivery Location: basin-name > terrain-base
```

**note full resolution and 100-foot versions are posted to the same directory*

1.4.1.1 SOURCE DATA SEARCH

The primary source for DEMs in the United States is the USGS 3D Elevation Program (3DEP), where the best available data can be accessed from The National Map.

Other local data sources that may provide higher quality DEM data should also be investigated. For example, many states maintain DEM data, as demonstrated in the following examples:

- Connecticut: https://cteco.uconn.edu/data/download/flight2016/index.htm
- Illinois: https://clearinghouse.isgs.illinois.edu/data/elevation/illinois-height-modernization-ilhmp
- Kentucky: https://kyfromabove.ky.gov/maps/kygeonet::kentucky-elevation-data-dem-and-dem-tile-index
- Massachusetts: https://maps.massgis.digital.mass.gov/MassMapper/MassMapper.html
- New Hampshire: https://lidar.unh.edu/map/
- North Carolina: https://sdd.nc.gov/
- Ohio: https://gis1.oit.ohio.gov/geodatadownload/
- West Virginia: https://data.wvgis.wvu.edu/elevation/

1.4.1.2 TIPS

The following considerations and tips are intended to aid in DEM acquisition and preparation for the FFRD project.

1.4.1.2.1 SNAPPING/EDGE MATCHING

As described in SOP Volume II, the origin of the base DEM and all derivative grid products must be consistent, and any lower resolution products must use a cell size that is an integer multiple of the full-resolution base DEM cell size. The reference origin is shown in SOP Volume II.

Methods for meeting this standard vary, but the cell size and origin coordinate of any raster can be checked using the following GDAL command:

Unix shell: $ gdalinfo your-dem.tif | grep -E "Origin|Pixel Size"

Windows Termial: gdalinfo your-dem.tif | findstr "Origin Pixel"

1.4.1.2.2 INTERPOLATION METHOD

When reprojecting or resampling to a lower resolution, DEMs should be processed using bilinear interpolation, and corner matching must be maintained.

1.4.1.2.3 TILING

Tiled GeoTIFFs can be preferable to a single, merged, high-resolution DEM for the entire study basin due to file size challenges associated with transferring and consuming the data. Hydraulic modeling software accepts *.vrt files that are capable of compiling tiled GeoTIFFs.

1.4.1.2.4 DOWNLOAD AND PROCESSING SCRIPTS

Processing of data to produce DEMs for FFRD use will typically be aided by scripts that have been thoroughly reviewed, and several open-source tools may be available to support this effort. Application of any scripts for FFRD DEM creation must be reviewed to confirm that SOP requirements are met.

1.4.1.2.5 METADATA

Metadata associated with the source DEMs should be retained, including metadata that clearly identifies all original sources of the final project DEM and their spatial extent.

1.4.2 ELEVATION DERIVED HYDROGRAPHY

Elevation derived hydrography (EDH) refers to geospatial layers that represent water features—including streams, rivers, shorelines, and catchments—directly from high-resolution elevation data. These layers provide a consistent, topographically accurate hydrography dataset that aligns with terrain representation.

In the context of FFRD, EDH:

- Replaces or enhances traditional hydrography datasets that may be outdated or misaligned with modern elevation surfaces.
- Improves spatial accuracy of flow paths and watershed boundaries, which is important for modeling surface water movement and flood behavior.
- Provides an authoritative hydrographic base layer for hydraulic modeling and floodplain mapping efforts.

If available, USGS 3DHP data is downloaded and used directly. Where 3DHP data is not available, EDH data is produced as part of the data preparation effort for the FFRD project. This centerline data is provided in two versions: short reaches that include attributes such as drainage area and longer reaches with unattributed segments.

Efforts to align with USGS 3DHP program priorities for data development should be made.

Various GIS tools are available to support development of EDH, such as Esri's Arc Hydro toolbox. Edits are commonly necessary at breaks in the channel flowline, such as roadway culvert crossings, where the stream may not align correctly between the upstream and downstream channels.

All identified streams with a drainage area greater than 0.25 square miles should be included in the EDH delivery for FFRD.

The EDH layers provided will be in GeoPackage format and delivered to the location below.

```
Delivery Location: basin-name > basin-data > elevation-derived-hydrography
```

1.4.3 NATIONAL HYDROGRAPHY DATASET AND WATERSHED BOUNDARY DATASET

The legacy National Hydrography Dataset Plus High Resolution (NHDPlus HR) and Watershed Boundary Dataset (WBD) are available for download by Hydrologic Unit Code 4-digit or 8-digit identifiers (HUC-4/HUC-8), by state, or nationally at the following link:

https://www.usgs.gov/national-hydrography/access-national-hydrography-products

The NHDPlus HR was processed using 10-meter 3D Elevation Program (3DEP) data. The dataset was initially released between 2018 and 2021 and was reprocessed for certain regions in 2022 and 2023.

The NHD and WBD layers provided will be in GeoPackage format and delivered to the locations below.

```
Delivery Location: basin-name > basin-data > nhd
Delivery Location: basin-name > basin-data > wbd
```

1.4.4 VERTICAL DATUM CONVERSIONS

An official listing of vertical datum conversions for point locations at critical sites throughout the basin will be established and documented during the scoping and data preparation phase. Conversion values must be included for all stream gages and dam locations where conversions are needed to adjust water surface elevation records or as-built dam elevation data to the vertical datum used for modeling. The source of the conversion values will be documented in the table.

Conversion of any gridded data, such as surveyed bathymetry with no specific XY point location, should also be described with a table entry.

When available, conversion values should be linked directly to official values reported for USGS stream gages, values adopted by dam operators, or other sources that have previously been adopted by others.

When no previous conversion values are available, the NOAA Vertical Datum Transformation (VDatum) tool is recommended.

A template for the listing of conversions is provided in FFRD Templates and Resources, and the completed table will be delivered to the following location.

```
Delivery Location: basin-name > basin-data > vertical-datum-conversions
```

1.4.5 NATIONAL LAND COVER DATABASE

NLCD is used as the minimum standard to aid in the parameterization of roughness coefficients, specifically Manning's "n" values, and percent impervious areas, which are crucial for simulating surface flow.

Data can be downloaded through the Multi-Resolution Land Characteristics Consortium (MRLC) website. Annual data dating to 1985 is available at a 30-meter resolution. This data must be reprojected to the FFRD projection, clipped to the buffered study area, and provided in GeoTIFF format.

```
Delivery Location: basin-name > basin-data > land-cover
```

1.4.6 SUPPLEMENTAL LAND COVER DATASETS

Optionally, additional land cover datasets may improve the ability to calibrate the FFRD models. The NOAA Coastal Change Analysis Program (C-CAP) Land Cover Atlas is available. Aerial imagery may be processed with advanced techniques to directly support modeling efforts.

If these optional supplemental datasets are scoped for inclusion and coordinated with the leadership team for specific projects, they would be stored in the following project directory location.

```
Delivery Location: basin-name > basin-data > land-cover
```

1.4.7 SOIL COVERAGE

The Natural Resources Conservation Service (NRCS) Soil Survey Geographic Database (SSURGO) provides delineations of soils information in shapefile format to which other attributes can be joined. Soil textures or hydrologic soil groups can be used for hydrologic model parameterization.
This dataset will be processed for the study basin and delivered to the following location.

```
Delivery Location: basin-name > basin-data > soils
```

1.4.8 BATHYMETRIC DATA

A full search for available bathymetric data to supplement the above-water terrain information for the study basin is conducted during the scoping and data preparation phase. The availability, type, and format will vary across individual study basins and across the nation. Table 2 provides common sources of data that may be used to support the FFRD study, but it is not intended to be comprehensive. Judgment and coordination with the larger project delivery team (PDT) will be required to determine whether certain available datasets will be of value to the project and if their acquisition is warranted.

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Table 2. Non-comprehensive FFRD Data Source List

Source	Data and Information Link	Description
USACE Hydrographic Surveys Program	Link	Includes links to the USACE Hydrographic Surveys interactive map and downloadable data, the National Channel Framework Dataset, and the Joint Airborne Lidar Bathymetric Technical Center of Expertise.
NOAA Bathymetry	Link	Includes links to various interactive maps and bathymetry products as well as resources for creating gridded data. Interactive maps include the National Centers for Environmental Information (NCEI) Bathymetric Data Viewer and the International Hydrographic Organization (IHO) Data Centre for Digital Bathymetry Viewer (DCDB). Other products include Coastal Elevation Models, Estuarine Bathymetric Digital Elevation Models, Great Lakes Bathymetry, and other bathymetry projects.
Reservoir Sediment Surveys or Tabulated Storage-Elevation Data	N/A	Owners or operators of large reservoirs may have an estimate of storage for a range of pool elevations and may regularly monitor changes in available storage with surveys.
USGS Inland Bathymetry	Link	Inland bathymetry is a component of USGS 3D National Topography Model (3DNTM) and the next generation of the 3D Elevation Program (3DEP), the goal of which is to map all the topography of the nation, including submerged topography under lakes, rivers, and reservoirs, using airborne topobathymetric LiDAR (green wavelength LiDAR).
USGS Maps of America's Submerged Lands	Link	Compilation of USGS published bathymetric information for select locations
Northeast Ocean Data	Link	Library of information along the northeastern coast of the United States
Existing Hydraulic Model Cross Sections	N/A	Legacy hydraulic models may retain valuable cross-section field survey data not preserved in other formats. Integrating this data into current models may be considered, provided its quality and significance warrant the effort associated with its extraction.
Recreational Maps	N/A	Various sources of recreational maps provide bathymetric information, particularly in lakes.

Generally, the data collected should be provided in its original format, which may include point data, raster data, surveyed river cross sections, contour line data, or other formats. Certain formats should be prepared for FFRD use by ensuring that the location is represented in a geospatial format that adheres to the FFRD projection and vertical datum standards. Key metadata of each dataset should be maintained with the dataset and included in the data manifest.

```
Delivery Location: basin-name > basin-data > bathymetry
```

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1.4.9 NATIONAL LEVEE DATABASE

A copy of the National Levee Database (NLD) geospatial data is placed in the FFRD project directory. A data refinement effort for the NLD should be completed prior to beginning an FFRD project, and documentation of that effort is also included as a deliverable.

The layers delivered will include system alignment lines, system routes, centerlines, floodwall lines, closure lines, and point feature data.

Additionally, an XYZM table based on the system route will be produced that captures the XY locations, levee system stationing, and elevation. The XYZM table will apply the FFRD projection, units, and vertical datum standards.

The delivery of NLD datasets is can be automated, and the use of scripts to produce the required deliverables is encouraged.

```
Delivery Location: basin-name > basin-data > levee-data > nld
```

1.4.10 NATIONAL INVENTORY OF DAMS

A copy of the National Inventory of Dams (NID) attributed point data is placed in the FFRD project directory. A data refinement effort for the NID should be completed prior to the start of the FFRD project, and documentation of that effort is also included as a deliverable.

```
Delivery Location: basin-name > basin-data > dams > nid
```

1.4.11 NATIONAL STRUCTURE INVENTORY

The NSI is a comprehensive system of databases managed by USACE. The main purpose of the NSI is to provide a detailed inventory of structures across the United States. This data is crucial for various applications, including flood risk management, emergency response, and infrastructure planning. The NSI databases contain structure inventories of varying quality and spatial coverage. The first version was published in 2011 and has since been updated in 2016, 2019, and 2022.

NSI data is comprehensive and includes information on the location, type, and characteristics of structures. This data is essential for detailed risk assessments and planning. The NSI provides extensive documentation, including technical references and user guides, to help users understand and utilize the data effectively. The NSI data is accessible through various platforms, allowing users to integrate it into their GIS and analysis workflows. An interactive map supports identification and download of available data.

The NSI API, documented in the NSI API reference guide, provides access to NSI structures through query by polygon, bounding box, or Federal Information Processing Standards (FIPS) code. This allows a user to fetch structures in multiple formats, programmatically, to support a variety of workflows.

The naming convention from Job Aid 06, *Standard Naming Conventions,* must be followed, and additional details regarding interaction with the NSI data are provided in Job Aid 14, *Consequence Processes*.

```
Delivery Location: basin-name > basin-data > structure-inventory
```

1.4.12 STREAM GAGE NETWORK

Stream gage data is foundational for calibration of model parameters and validation of results. The USGS stream gage network typically serves as the primary source, although other stream gage networks may be available depending on location.

Separate instructions (Job Aid 04, Observed Gage Data Preparation, and Job Aid 01, Scoping Processes) describe the acquisition of the gage data and the scoping of the use of data at gages for model calibration.

```
Delivery Location: basin-name > basin-data > stream-gages
```

1.4.13 NATIONAL BRIDGE INVENTORY

The National Bridge Inventory includes a shapefile of roadway bridges, including attributes that may be referenced for development of hydraulic model inputs. Background information and a coding guide are available through the U.S. Department of Transportation Federal Highway Administration Bridges and Structures website. An interactive map supports identification and download of available data.

Bridge attributes relevant to hydraulic modeling to retain include:

- Road Name
- Year Built
- Deck Width
- Skew
- Span Type, Number, Length
- Structure Type (Culvert, Bridge, etc.)

```
Delivery Location: basin-name > basin-data > transportation
```

1.4.14 NATIONAL RAILROAD BRIDGES

The National Bridge Inventory does not include railroad bridges. However, the Federal Railroad Administration's Bridge Inventory can be used as a supplement. Its data dictionary and links to an interactive map are available.

Attributes relevant to hydraulic modeling to retain include:

- Owner
- Name and ID
- Number of tracks

```
Delivery Location: basin-name > basin-data > transportation
```

1.4.15 U.S. CENSUS URBAN AREAS

The U.S. Census Urban Area polygons are leveraged directly for certain aspects of scoping the FFRD study. A raw copy of the polygons is placed in the project directory. The data can be acquired at the U.S. Census' TIGER/Line Shapefiles page.

```
Delivery Location: basin-name > basin-data > census > urban-areas
```

1.4.16 U.S. CENSUS INCORPORATED PLACE BOUNDARIES, ROAD AND RAILROAD DATA

Incorporated place boundaries, roads, and railroad data can be useful reference layers. The data can be acquired at the U.S. Census' TIGER/Line Shapefiles page.

```
Delivery Location: basin-name > basin-data > census
```

DRAFT

1.5 RISK ASSESSMENT STUDIES

A search for dam and levee risk assessment data will be conducted. Common sources will include USACE, Bureau of Reclamation, and Federal Energy Regulatory Commission risk assessments. For FFRD projects, the USACE dam and levee data leads will be responsible for assembling the existing risk assessment datasets. Collected reports should be posted to the locations below.

```
Delivery Location: basin-name > basin-data > risk-data
```

1.6 PHYSICAL STRUCTURE AND OPERATION INFORMATION—DAMS AND LEVEES

Physical structure and operation information for dams and levees is a key input to the scoping and modeling efforts for an FFRD study. These datasets include operations manuals, as-built drawings, and reservoir storage data among other information. Additional details specifying the procedures for searching for and acquiring the data are currently being developed and will be elaborated in a future version of this document.

```
Delivery Location: basin-name > basin-data > dams
Delivery Location: basin-name > basin-data > levees-data
```

1.7 ACQUISITION OF OTHER EXISTING MODELS

A full search for additional existing hydraulic and hydrologic models will be conducted. Acquisition of models associated with the FEMA NFIP is described separately. Other common sources of available models include USACE Corps Water Management System (CWMS) models, local flood risk reduction feasibility study models, and dam breach models, among others. Not all models serve as direct inputs into the FFRD project, but they are a useful reference. Judgment may be required to ascertain if model datasets are worth the effort to acquire.

```
Delivery Location: basin-name > basin-data > existing-models
```

1.8 LOCAL DATA

Various other existing datasets may add value to the FFRD process, such as those that are brought forth during stakeholder engagement activities. Judgment will be required to incorporate certain datasets into the project directory. When supplied data do not fit the categories provided, they may be posted to the directory below.

```
Delivery Location: basin-name > basin-data > other
```

DRAFT

JOB AID 3—METEOROLOGIC DATA PROCESSES

This job aid provides supplemental information and details related to the production of meteorologic data for an FFRD study.

1.1 WATERSHED BOUNDARY INPUT

The watershed boundary for an FFRD study is clearly defined during the scoping phase. This boundary is provided as input to the meteorology team member for determining the appropriate transposition domain and assembling the collection of storms for the study.

The watershed domain is stored as a GeoPackage with a single-part polygon at the location below.

```
Delivery Location: basin-name > basin-data > project-scoping > study-area-polygon
```

1.2 TRANSPOSITION REGION SELECTION

Storms do not occur consistently throughout the United States. To ensure storms are only moved within areas where the storm is equally likely to occur, the areas for collecting storms to transpose are limited to a transposition domain. A transposition domain represents a spatial area having homogeneous meteorologic conditions. Developing a good transposition domain requires meteorologic training. Transposition domains are generated by a small team and are delivered to the shared *basin-data* folder of the project directory.

A transposition region is the key assumption and driving factor in the Stochastic Storm Transposition (SST) approach for generating synthetic precipitation events. This domain is a geographic extent that experiences homogeneous storm-generating meteorology and contains the study watershed. The foundational assumption of SST is that a storm that occurred in one location within the transposition domain can freely be transposed to any other location within the domain. The size and shape of the transposition domain extent is informed by meteorological processes.

Rough initial domains can be established using products such as the U.S. Climate Divisions by starting with the division(s) that contain the study watershed and adding adjacent and nearby divisions that are sufficiently similar based on climate. The divisions have several limitations that make them unfit for generating a final transposition domain but can be indicative of transitions between climatic zones within a state. Comparing these boundaries to regions of high gradient in climatic variables (e.g., annual precipitation) or physical characteristics (e.g., elevation) can indicate meteorological driving mechanisms in the region.

An initial investigation of historical flood and storm causal mechanisms for the study watershed is warranted. Storm and flood reports from the largest events on record provide narrative descriptions and data summaries for significant events and an impression of important storms in the region. A transposition domain should contain the area around the study watershed where these kinds of storms are also likely and should not contain physical features that would prevent or drastically alter the behavior of a storm in that area (e.g., mountain ranges).

An initial screening step should be an evaluation of tropical cyclone (TC) activity using a dataset such as International Best Track Archive for Climate Stewardship (IBTrACS). TC activity can be visualized relative to the study watershed, and areas with similar numbers of storm tracks are likely to be included in the transposition domain. The tracks may also reflect important physical features, such as blocking mountain ranges, that would need to be excluded. In areas where few TCs affect the watershed, areas of high TC activity should be excluded from the domain. Coastally adjacent areas are likely to have many TC tracks,

and the domain should contain similar levels of activity throughout. An important boundary delineator in this case is often the distance from the coastline.

A meteorologist will take the input Hydrologic Unit Code (HUC)-4 watershed domain and utilize key datasets to generate a transposition domain.

Promising techniques that use extremal statistics of rainfall maxima to determine transposition regions directly from observed data may also be considered.

The transposition region for the study is developed during the scoping and data preparation phase, stored as a GeoPackage with a single-part polygon, and posted to the location below.

```
Delivery Location: basin-name > basin-data > storm-catalog > transposition-region
```

1.3 STORM CATALOG DEVELOPMENT

Storms are identified from a source of gridded precipitation that has a long period of record. Currently the appropriate data source is the Analysis of Record for Calibration (AORC) dataset. It has an approximately 45 year period of record with hourly precipitation grids, at 4- or 1-kilometer resolution.

Storms are selected based on the target watershed; the objective is to find a storm that optimizes the precipitation over the target watershed for a given duration.

A threshold is leveraged to select storms that exceed a cumulative precipitation total, which helps to define which storms are included or excluded from the catalog. The threshold is set to a number of targeted storms per year over the duration of the input data source. For instance, in a 45-year data source, if the objective count of storms per year is 10, a threshold is set to collect 450 storms from the data source that optimize the storm over the size of the target watershed.

The following inputs are required:

1. A data source of gridded precipitation (AORC dataset used for FFRD studies)
2. A transposition domain (defined according to guidelines and considerations above)
3. A watershed domain (typically HUC-4 watersheds)
4. A user-defined storm duration (72 hours for FFRD studies)
5. A target threshold (average of 10 storms per year for FFRD studies)

1.3.1 STORM CATALOG DEVELOPMENT PROCEDURE

The following algorithmic procedures are employed by the cloud-compute storm-cloud plugin to develop a ranking of storm events within the transposition region and build the storm catalog for use within the SST framework.

Storm events are identified from the aggregated AORC data by looking at rolling 72-hour precipitation totals and finding events with significant 72-hour precipitation over an area the size and shape of the target watershed. Beginning at t=0, which corresponds to the first day of the period of record (01 January 1979), the procedure works as follows:

1. Aggregate the daily product to a three-day total by taking the sum of the three daily total precipitation rasters (t, t+1, and t+2).
2. Optimize the position of the watershed polygon within the transposition domain such that the three-day precipitation is maximized when averaged across the polygon. Limit the movement of the watershed polygon so no portion is ever outside the transposition domain during the search for the optimum position.
3. Compute summary statistics for the precipitation inside the optimized watershed polygon position and associate those with the date t.

4. Increment *t* by 1 and return to Step 1.

This process results in a list of dates with associated precipitation depths creating severe storms in an area the size and shape of the target watershed. The storm dates are separated such that the largest precipitation in each 72-hour window is retained, ensuring the same storm is not double counted. Once there is a complete listing of event dates and the associated maximized watershed precipitation, a threshold is chosen to isolate the most severe events. A threshold that results in an average of *k* events per year is determined to select storms. The threshold is applied to all storms and the *k*×number of years largest storms in the storm data source are selected and added to the catalog.

For each storm event selected to be included in the catalog, additional metadata beyond the rainfall fields are computed and retained.

- The location of the storm center is stored as a coordinate pair, determined by the center of the pixel where the maximum total accumulation of rainfall occurred.
- The beginning date of the storm event.
- The minimum, average, maximum, and total values of the rainfall accumulation pixel values in the maximized position over the watershed.
- The storm type (assignment of type described in a separate section).
- The ratio of the mean accumulation in the maximized position to the average of the 72-hour 2-year rainfall from NOAA Atlas 14 at the same location.

No assumption is made about whether an individual event is an annual maximum with respect to any statistic, and creating an annual maximum series is not necessary. When the transpositions are performed, the storms themselves are never treated as annual maxima but only as one of many possible realizations of a storm event within a year.

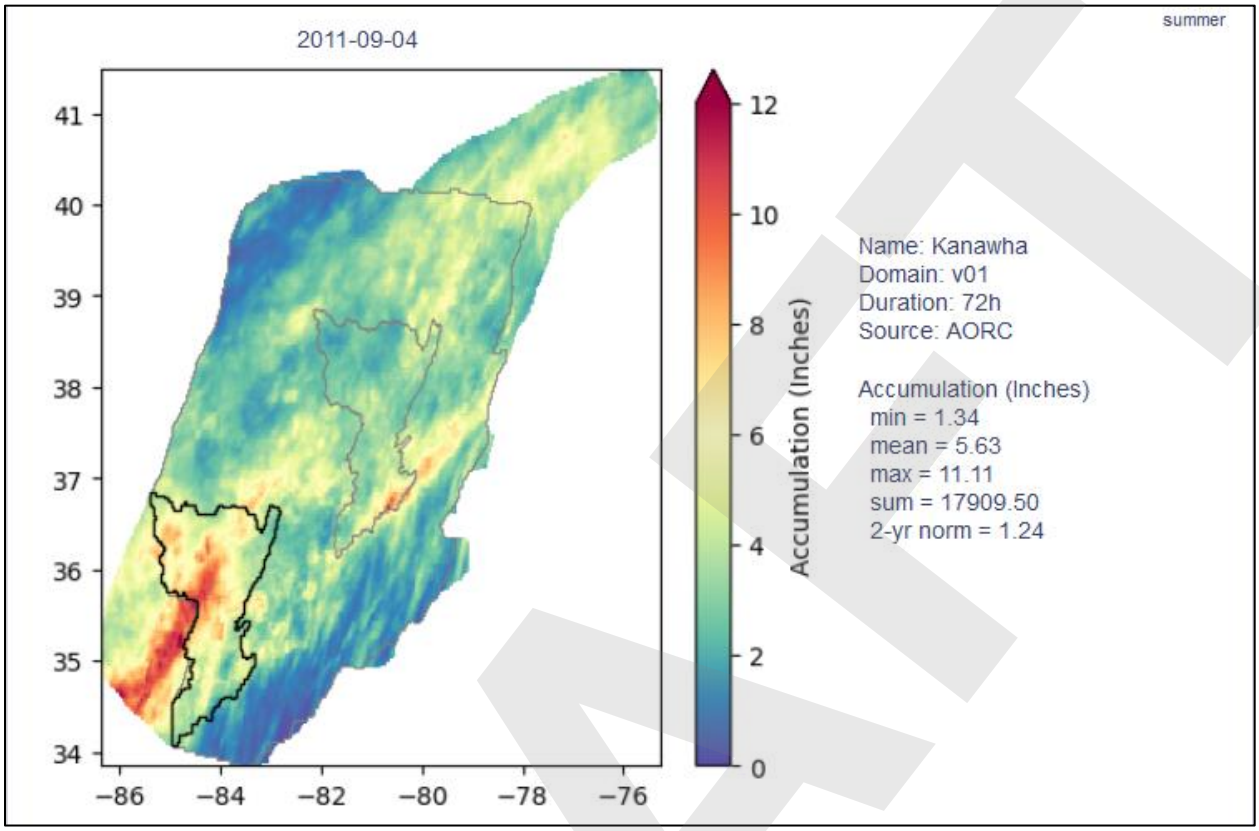
1.3.2 VISUALIZATION AND REVIEW OF IDENTIFIED STORMS

Storm events can be filtered and sorted based on several statistics. Displaying a visualization of the 72-hour precipitation as well as the optimized watershed location within the transposition domain aids in quality control of the extracted storm events by providing a visual check on the behavior of the precipitation field, as well as displaying a number of useful statistics about the event that help determine if the storm event will be added to the catalog.

Figure 1 shows an example of a web-based tool for viewing identified and ranked storm events in the precipitation data. Colored raster extents coincide with the transposition domain. Original watershed position is in grey. Optimized watershed position is in black. This storm event would be included in the catalog because its watershed-averaged precipitation is 5.63 inches exceeds the catalog threshold.

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Figure 1. Example View of Identified Storm



1.3.3 DELIVERY OF STORM CATALOG

The selected storm catalog storms are delivered in named Data Storage System (DSS) files as well as in the form of a full catalog in HEC-HMS .grid format. This allows for easy importing into HEC-HMS for evaluation.

In a subdirectory called seasonality-distributions, a .csv file of the cumulative distribution function of the day of year with the probability of the storm of a given storm type occurring on that day is stored and named by storm type. In a subdirectory called *storms*, the .dss files containing precipitation and temperature are stored.

```
Delivery Location: basin-name > basin-data > storm-catalog
```

1.4 NORMALIZATION GRID FOR TRANSPOSITION REGION

A normalization grid is created to support the Normalized Transposition technique to improve the performance of SST. Section 5.7 of SOP Volume II includes motivation and recommendations for developing and applying a normalization grid as part of the SST procedure. In areas where NOAA Atlas 14 is available and fully covers the transposition domain, the annual maximum series-derived, 72-hour, 100-year precipitation field should be used to normalize. It may be necessary to mosaic together several rasters spanning multiple volumes of Atlas 14. The rasters should be generated in GeoTIFF format. When Atlas 15 becomes available, it should replace any use of Atlas 14. It is anticipated that Atlas 15 will provide a seamless product that is spatially consistent and will not require creating a mosaic.

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Where Atlas 14 or 15 is not available, the preferred normalization field is a 72-hour mean annual maximum field developed from the source precipitation dataset (AORC or similar). Parameter-elevation Regressions on Independent Slopes Model (PRISM) normal or seasonal precipitation should not be used.

```
Delivery Location: basin-name > basin-data > storm-catalog > normalization-grid
```

1.5 STORM TYPING

Storm typing is a novel addition to the traditional SST recipe that improves the hydrologic modeling of flood responses and allows for interfacing with the coastal hazards modeling process. The categories used for this approach are intentionally broad since the distinction is important for two uses: storm seasonality and isolating tropical storms. The seasonality of storm events is used to determine initial watershed conditions during the stochastic simulations so that implausible combinations of meteorology and watershed state cannot be created. Isolation of tropical cyclones is important for handling additional coastal modeling that may be required to characterize compound hazards caused by the combination of rainfall and storm surge. Currently, for FFRD studies, two broad storm categories are used:

- Tropical Cyclones (TC)
- Non-Tropical (NT)

Most rainfall events that occur on a sufficiently widespread scale for inclusion in a HUC-4 scale watershed catalog will fall into the NT category.

Manual storm classification of each storm in the catalog is recommended for FFRD studies, as no cloud-compute software action for automated classification has been fully released. Recommended procedures for automated techniques and tools for storm classification are expected to be incorporated into a future version of the FFRD SOP.

1.5.1 MANUAL STORM CLASSIFICATION

TCs, which include all phases of the lifecycle of a tropical storm (e.g., tropical depression, hurricane, etc.), are most prevalent near the Gulf of Mexico and the Atlantic Coast, although Pacific storms can occur as well. Watersheds that are significantly inland may occasionally be impacted by remnants of tropical storms. Of all storm types, these are typically the easiest to identify due to the availability of datasets like IBTrACS or the Hurricane Database (HURDAT), and their clear impact on surface analysis charts.

A combination of visual information is recommended for the classification process. Daily Weather Maps or other sources of surface analyses can often be used to rapidly identify midlatitude cyclones and TCs. IBTrACS has a spatial product that makes visualization of TC tracks straightforward. Reanalysis datasets such as the contiguous United States for 40 years at 4-kilometer resolution (CONUS404) produce useful visualizations of other key data indicators in space and can be interacted with using Python or other data visualization tools. Some key variables to evaluate include:

- Storm season
- Precipitation
- Convective Available Potential Energy (CAPE)
- Mean Sea-Level Pressure (MSLP)
- Storm Relative Helicity (SRH)
- 500 hPa geopotential height and gradient

Other data and visualizations used at the discretion of the analyst can assist in completing the classifications.

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1.5.2 STORM SEASONALITY OUTPUT

For each type of storm, a seasonality analysis will be produced. The seasonality analysis will describe the daily frequency of occurrence for each storm type based on the start date of the events in the storm catalog. Each day of the year from 01 January through 31 December will have a count of storm occurrence computed as the count of storms of that type beginning on that day of the year. The sum of the counts should equal the total number of storms of that type, and the sum of all tables should total the number of storms in the catalog.

1.6 FISHNETS FOR STORM TRANSPOSITION

Fishnets represent valid storm placements for transposition and are created with the help of the HMS-mutator plugin and are saved as a .csv file. Uniform placements will be applied for studies carried out with version 1 of the SOP. Additional details on the inputs, outputs, and algorithm for fishnet development are included in the cloud-compute documentation.

```
Delivery Location: basin-name > basin-data > storm-catalog > fishnets
```

1.7 STOCHASTIC METEOROLOGIC MODEL RESULTS

Basin-average precipitation frequency data from SST must be extracted from modeled storm placements to evaluate the conformance of results to observed data.

SST is driven by the Uncertainty Analysis, randomly varying the storm selection and storm coordinates, and extracting the precipitation that results from each transposition. An Uncertainty Analysis occurs in HEC-HMS with three parameters: Storm Center X-coordinate, Storm Center Y-coordinate, and grid name (representing an event from the storm catalog). Set up a precipitation-only discretization basin model for the project with a single subbasin of the entire study domain and follow the steps presented in the HEC-HMS Tutorials and Guides, *Fundamentals of Stochastic Storm Transposition* page. The Grid to Point Converter may be useful in this task to extract subbasin-averaged annual maximum series.

The annual maximum series data of subbasin-averaged precipitation must be posted, where it can be programmatically leveraged for analysis, plotting, and evaluation.

```
Single Realization Results from Conformance Phase Delivery Location:
basin-name > conformance > simulations > frequency-calibration
Multi-Realization Results from Production Phase Delivery Location:
basin-name > production > simulations > frequency-calibration
```

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JOB AID 4—OBSERVED GAGE DATA PREPARATION

An authoritative set of observed data will be collected during the data preparation phase and shared among the modeling team. Subsequent modeling efforts must make use of this authoritative set of observed data to avoid using conflicting data. Data that is gathered must also be adjusted prior to use to adhere to vertical datum and time zone requirements (as established in SOP Volume II, Section 3).

The authoritative dataset will be saved to the central *basin-data* directory for the project.

If other data becomes available during later phases (e.g., calibration, conformance, etc.), it must be incorporated into the authoritative set of data and shared with the entire modeling team. Any updates to the authoritative dataset must be coordinated with the technical lead.

This document describes the types and sources of data, data processing tools, and typical modifications that are required for use within FFRD applications.

1.1 TYPES OF DATA TO BE COLLECTED

Several types of data should be collected during an FFRD application. This includes:

- Flow
- Stage
- Stage-Discharge Relationships
- Stream Gage Field Measurement Dataset
- High-Water Marks
- Flood Imagery—Satellite or Airborne
- Flood Photography—Ground Photography

Additionally, data should be collected at numerous temporal resolutions including (but not limited to):

- Instantaneous (e.g., 5, 10, 15, 30, and/or 60 minutes)
- Sub-daily (e.g., 3, 6, and/or 12 hours)
- Daily
- Annual Maximums

1.2 DATA SOURCES

The U.S. Geological Survey (USGS) collects water resources data at over 10,000 active stream gages. This data is disseminated through websites like the USGS Water Data for the Nation site. Additional information including historical observations, statistics, instantaneous peak flow/gage height, and field measurements are also available. In most FFRD applications, data collected and distributed by the USGS will constitute most of the pertinent data. However, there are numerous additional sources of water resources data that may be pertinent to FFRD applications including (but not limited to):

- https://tidesandcurrents.noaa.gov/
- https://www.usbr.gov/pn/hydromet/index.html
- https://cdec.water.ca.gov/ (California Data Exchange Center)
- https://water.usace.army.mil/
- https://apps.usgs.gov/fev/ (high water mark data)
- https://www.usgs.gov/centers/eros (EROS satellite imagery depot)
- https://browser.dataspace.copernicus.eu/ (Copernicus satellite imagery depot)

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An excellent summary of pertinent data sources that are commonly utilized within USACE dam and levee safety studies can be found in RMC-TR-2019-07, *Data Sources for Estimating Hydrologic Hazards for Semi-Quantitative Risk Assessments*.

1.3 VERTICAL DATUMS

Commonly, historical data will be reported in a local/project-specific vertical datum or the National Geodetic Vertical Datum of 1929 (NGVD 29). As established in SOP Volume II, Section 3, all stage and/or elevation data used within FFRD applications must be referenced to the North American Vertical Datum of 1988 (NAVD 88). Utilities such as the NGS Coordinate Conversion and Transformation Tool or NOAA's VDatum can be utilized to convert stages and/or elevations to NAVD 88.

Conversions from local project datums will also be clearly defined during the scoping and data preparation phase when they are available.

1.4 ACQUISITION OF STREAM GAGE DATA

For all stream gages within the modeling domain, download and/or ingest the following data using HEC-SSP or HEC-DSSVue:

- Instantaneous and/or sub-daily stage and flow data
- Daily average flow data
- Annual maximum series of flow
- Annual maximum series of stage
- Stream gage field measurement paired dataset (flow and stage)
- Published stage-discharge relationships

Examples demonstrating these capabilities can be found in HEC's guide on how to Create a new HEC-SSP Study, Import Data, and Visualize Multiple Data Sets.

Additionally, several commonly available scripting languages (e.g., R and Python) have libraries that can be utilized to download this data as well. Examples of these libraries include:

- https://doi-usgs.github.io/dataRetrieval/
- https://github.com/DOI-USGS/dataretrieval-python

1.5 EXTRACTING N-DAY DURATION TIME SERIES

For all stream gages within the modeling domain, calculate *n*-day duration flow timeseries (e.g., 1-, 2-, and 3-day durations) using HEC-SSP or HEC-DSSVue. Examples demonstrating these capabilities within HEC-SSP can be found in HEC's guide to Data Filtering, Extracting an n-Day Duration Time Series.

1.6 QUALITY ASSURANCE/QUALITY CONTROL OF TIMESERIES DATA

Performing quality control on observed data is a critical step in all FFRD applications as the use of incorrect or poor-quality observed data may result in inaccurate results. An excellent summary of quality assurance (QA)/quality control (QC) processes that must be undertaken during flood hazard studies is contained within RMC-TR-2018-03, Hydrologic Hazard Methodology for Semi-Quantitative Risk Assessments: An Inflow Volume-Based Approach to Estimating Stage-Frequency Curves for Dams. The following text was summarized from that document.

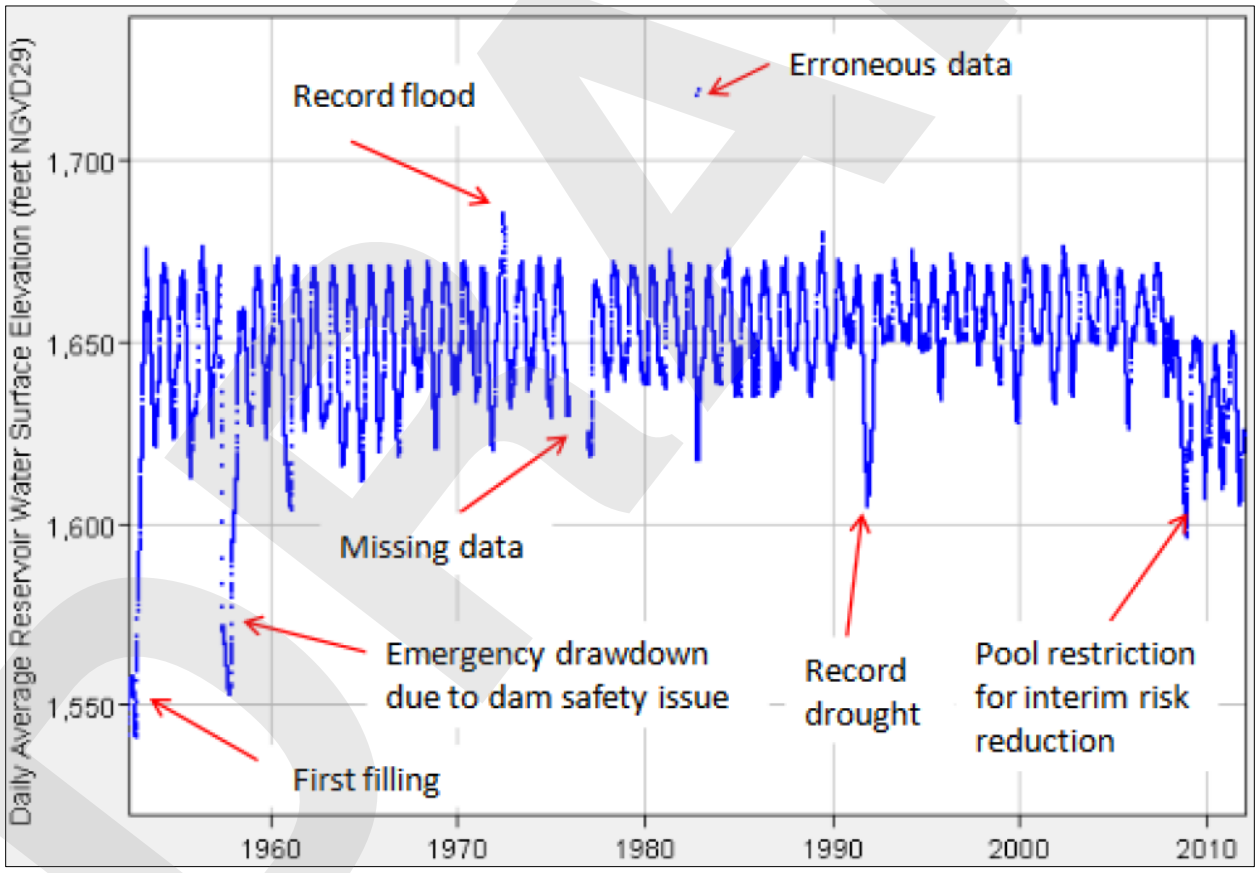
In cases where systematic data is obtained for streamflow gages, information on manual measurements of streamflow and gage height, including indirect measurement, can be obtained from the USGS's Field Measurements site. These measurements can then be used to develop streamflow rating curves, supplement and/or verify the accuracy of the automatically recorded estimates at gages, as well as to

compute streamflow based on gage height. These are valuable for flood frequency studies to aid hydrologists in understanding how the largest flood estimates are made (such as indirect measurements), and in estimating uncertainty. See details on streamflow measurement and computation and discharge ratings from the USGS's Surface Water Technical Procedures page. At most large dams, reservoir inflow is indirectly calculated using measurements of reservoir storage (via stage) and outflow (via rating curves or gage measurements). The water management entity that owns/operates the reservoir in question typically utilizes QA/QC steps on the calculated reservoir inflows, so the data may be used as provided after a cursory review is made to check for any obvious errors.

In addition to potential measurement errors, many datasets have missing data, duplicate values, erroneous data, or other issues that can impact the inflow volume- and stage-frequency assessments. Plotting the flow and stage timeseries data can assist with identifying any potential errors.

Chapter II-1, *Reservoir and River Stage Exceedance Probabilities*, from Best Practices (U.S. Bureau of Reclamation and U.S. Army Corps of Engineers, 2015) provides a useful example data set showing daily average reservoir stages for a dam as presented in Figure 1. The plot reveals several potential data quality issues. The time periods associated with the initial reservoir filling, the dam safety emergency, and the pool restriction for interim risk reduction may not be representative of normal operation. Some of the data also appears to be missing and incorrect based on a visual inspection of the plot. Missing data should be filled, and flawed data should be corrected or removed, as deemed appropriate.

Figure 1. Example of Daily Average Reservoir Stage Data



1.7 PRODUCE PROBABILISTIC SUMMARIES

Probabilistic summaries of timeseries data will be used to ascertain the accuracy of the modeling tools and techniques within calibration, conformance, and production phases.

1.7.1 DETERMINE REGULATED/UNREGULATED TIME WINDOWS

Prior to extracting any probabilistic results, homogeneous time windows for each stream gage must be identified. This must be done in order to avoid violating the independent and identically distributed assumption and align with FFRD modeled outputs. When there is a known source of regulation upstream of a location of interest, but the extent of the impacts is uncertain, further investigation is required. An excellent summarization of steps that can be taken to ascertain the degree of regulation is contained within the technical report titled *Estimating Flood Hazard for Dams and Levees with Upstream Regulation*.

For all locations, consider changes in land use, etc. that may impact the homogeneity of the observed data. Also, for locations that are subject to upstream regulation, consider changes in regulation from upstream storage and diversion projects that may impact the homogeneity of the timeseries (i.e., not all periods of regulation should be considered significant).

1.7.2 ANNUAL MAXIMUM SERIES

Commonly, annual maximum series (AMS) of flow and stage/elevation will be available at most (if not all) USGS gages within the modeling domain. However, these AMS are typically indicative of an instantaneous peak duration (i.e., 1-hour). While the instantaneous peak duration is adequate for use when comparing observed and computed stage/elevation data, additional durations for flow data are needed. Thus, for all stream gages within the modeling domain, flow AMS must be extracted from the observed data for 1 day, 2 days, and 3 days.

1.7.3 PLOTTING POSITIONS AND CONFIDENCE LIMITS

For all stream gages within the modeling domain, 5 and 95 percent confidence limits (i.e., 90 percent confidence interval) must be computed for both flow and stage/elevation for homogeneous time periods.

Confidence intervals for all homogeneous flow AMS should be computed using one of the following approaches:

- For unregulated locations and time periods, use multiple Bulletin 17C analyses (i.e., one analysis per duration) and/or a single volume frequency analysis to compute confidence intervals using Bulletin 17C procedures. Examples demonstrating these capabilities can be found here: Bulletin 17C | Expected Moments Algorithm Examples.
- For regulated locations and time periods, use a general frequency analysis to graphically fit an empirical distribution and estimate confidence intervals using order statistics. Examples demonstrating these capabilities can be found here:
    - Create and Parameterize a General Frequency Analysis
    - General Frequency – Graphical Analysis

Confidence intervals for all homogeneous stage/elevation AMS should be estimated by first graphically fitting an empirical distribution and then computing confidence limits using order statistics. Examples demonstrating these capabilities can be found here:

- Create and Parameterize a General Frequency Analysis
- Example 7. Analyzing Stage Data

1.8 FORMAT FOR DELIVERY

Observed data should be provided in HEC-DSS format for efficient usage by subsequent modeling applications. Probabilistic results must be provided in both .dss and comma-separated values (.csv) formats for use within the calibration, conformance, and production phases. The naming conventions in Job Aid 06, Standard Naming Conventions, must be used.

Authoritative observed data will be posted to the *basin-data/observed-data* central directory for the project at the end of the scoping and data preparation phase.

1.9 DOCUMENTATION OF DATA COLLECTION EFFORTS

All data collection efforts must be summarized and provided to the modeling team and included in the report for the scoping and data preparation phase.

Entries for observed data will be included in the Data Manifest.

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JOB AID 5—OBSERVED GRIDDED DATA PREPARATION

This document describes the processes to develop the authoritative set of observed gridded data for the study basin for use in an FFRD project. This data is collected, processed, quality control (QC) checked, and delivered to the project directory during the scoping and data preparation phase of the FFRD project.

1.1 TOOLS AVAILABLE

Data tools for processing the gridded datasets include the following:

- HEC-Vortex
- HEC-DSSVue
- HEC-SSP
- QGIS or ArcGIS Pro

1.2 DATA TO BE DELIVERED

The observed gridded datasets in the following subsections are to be prepared for the FFRD project. These datasets include available observed gridded data that occurred over the study basin (which is separate from gridded storm catalog data for the larger region, developed separately). The spatial extent of these datasets should be buffered beyond the study basin extent to ensure no data is missing along the edge of the boundary and that the data can accommodate slight adjustments to the study basin boundary if they occur later in the project.

```
Delivery Location: basin-name > basin-data > observed-data
```

1.2.1 PERIOD OF RECORD FOR STUDY BASIN EXTENT

Gridded datasets for the full continuous period of record (POR) of precipitation, temperature, and snow are acquired, adjusted, and delivered to the project directory in DSS format.

Processes for acquisition and considerations for adjustments and QC of the datasets are outlined in the sections below.

1.2.2 HISTORIC CALIBRATION EVENTS

Gridded precipitation, temperature, and snow datasets for the historic event time windows are clipped from the POR data for use in model event-based calibration efforts and delivered to the project directory in DSS format.

The time windows for the events will be defined according to the guidance for calibration event selection provided in Job Aid 01, *Scoping Processes.* Observed gridded datasets should be buffered both before and after the selected events by at least 1 week. For example, if a historic event is expected to span 15 January 1996 through 15 February 1996, create DSS files containing gridded precipitation, temperature, and snow data that spans 01 January 1996 through 28 February 1996 (i.e., buffered slightly in time to avoid reprocessing if the event window is changed).

1.2.3 OBSERVED PRECIPITATION ANNUAL MAXIMUM SERIES

During the scoping and data preparation phase of the project, annual maximum series (AMS) of observed subbasin-averaged precipitation will be created and delivered. The precipitation AMS will be created based on the final gridded dataset after all QC and normalization adjustments.

Using the uncalibrated HEC-HMS model and the POR precipitation data, a precipitation-only period-of-record simulation can provide the data to extract the AMS. All hydrologic processes are turned off in the model, except for discretization. AMS of subbasin-averaged observed precipitation for 1-hour, 24-hour, and 72-hour durations will be calculated and extracted in HEC-DSSVue for the POR simulation. Results will be plotted and assessed for possible input data problems. Reference the Validating Precipitation Grid Sampling and Grid-to-Point Converter HMS resources for additional information on this process.

1.2.4 OBSERVED SNOW WATER EQUIVALENT ANNUAL MAXIMUM SERIES

During the scoping and data preparation phase of the project, the AMS of observed subbasin-averaged snow water equivalent (SWE) will be created and delivered. The SWE AMS will be created based on the final gridded dataset after all QC and normalization adjustments.

Extract subbasin-averaged timeseries using the Grid-to-Point tool within the HEC-Vortex utility in HEC-HMS. Reference the Grid to Point Converter Tutorial for guidance on this process. This should be done separately for both Analysis of Record for Calibration (AORC) and Parameter-elevation Regressions on Independent Slopes Model (PRISM) gridded precipitation datasets as the comparison of the two can be a helpful resource during the QC process. The mean statistic should be selected to get a single observed value averaged over the spatial domain of each subbasin.

With the hourly and/or daily subbasin-averaged timeseries, compute the n-day duration precipitation accumulation timeseries (e.g., 1-, 2-, and 3-day durations) using either HEC-SSP or HEC-DSSVue. Reference the Data Filtering, Extracting an *n*-Day Duration Time Series Guide for guidance on this process. Finally, compute an AMS of precipitation and SWE using either HEC-SSP or HEC-DSSVue. The deliverables from this process are separate .dss files for each dataset containing the subbasin-averaged precipitation and SWE AMS to provide to the HEC-HMS modeler(s).

1.2.5 OBSERVED LIQUID WATER AT SOIL SURFACE ANNUAL MAXIMUM SERIES

During the calibration phase of the FFRD project after the POR HEC-HMS model has been calibrated and completed, the observed AMS of the subbasin-averaged liquid water at the soil surface (LWASS) will be created and delivered.

The Grid-to-Point tool within the HEC-Vortex utility in HEC-HMS is used to extract the subbasin-average timeseries of LWASS from the POR simulation results. Reference the Grid to Point Converter Tutorial for guidance on this process. The mean statistic should be selected to get a single observed value averaged over the spatial domain of each subbasin.

With the hourly subbasin-averaged , compute the n-day duration LWASS accumulationtimeseries timeseries (e.g., 1-, 2-, and 3-day durations) using either HEC-SSP or HEC-DSSVue. Reference the Data Filtering, Extracting an *n*-Day Duration Time Series Guide for guidance on this process. Finally, compute an AMS of LWASS using either HEC-SSP or HEC-DSSVue. Reference the Data Filtering, Extracting an Annual Maximum Series Guide for guidance on this process.

1.3 SPATIAL PROJECTION AND NAMING CONVENTIONS

The spatial projection and naming conventions for these datasets must conform to the standards described in SOP Volume II, Section 3, and Job Aid 06, Standard Naming Conventions.

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1.4 DATA SOURCES

The primary source for short-interval gridded precipitation and air temperature is the National Oceanic and Atmospheric Administration (NOAA) AORC dataset. The AORC dataset contains 1-hour accumulated surface precipitation, and the temperature data consist of instantaneous 2-meter above-ground air temperature. Both precipitation and air temperature are reported at the top of the hour.

Another source for gridded precipitation data is the Oregon State University PRISM dataset. This includes daily, monthly, and long-term averages for precipitation and is typically used if data normalization is deemed necessary.

SWE data is available from the University of Arizona and is based on assimilated Snow Telemetry (SNOTEL) and PRISM data.

1.5 RAW GRIDDED DATA ACQUISITION

Gridded datasets are imported and converted to a DSS file using the HEC-Vortex utility. Raw datasets should be clipped to the buffered modeling domain during the import process. This creates a .dss file containing gridded precipitation, temperature, and snow data for 1979 through the present. This set of files will be used within the HEC-HMS POR simulations as well as Stochastic Storm Transposition (SST) processing. An overview of this process is provided in the Gridded Data Import Guide.

All available datasets should be acquired as inputs. The need for normalization assessment is described in sections below. Only the recommended final (adjusted as needed) gridded datasets for use as observed data for the FFRD project are required for delivery to the shared project directory.

1.6 NORMALIZATION OF GRIDDED PRECIPITATION DATA FOR STUDY BASIN

The need for normalization of raw observed gridded precipitation data over the study basin must be assessed. If large discrepancies in precipitation volumes between datasets exist, normalization may be required.

To assess the need for normalization, the total mass of precipitation over the POR is compared between the AORC and PRISM datasets. If a large difference in total precipitation volume exists, the AORC data should be normalized based on the PRISM dataset. A 50 percent volume difference can be used as a guideline to trigger a need to normalize the data.

1.6.1 COMPUTE MASS OF PRECIPITATION

For AORC and PRISM datasets, extract subbasin-averaged timeseries using the Grid-to-Point tool within the HEC-Vortex utility in HEC-HMS. Reference the Grid to Point Converter Tutorial for guidance on this process. The mean statistic should be selected to get a single observed value averaged over the spatial domain of each subbasin. Using the *Tools* | *Math Functions* | *Accumulation Operator* in HEC-DSSVue, create accumulated mass curves for each of the precipitation timeseries datasets that were extracted for each subbasin.

1.6.2 NORMALIZATION OF PRECIPITATION DATA

When the criteria for normalization of the observed data are met, proceed with normalizing the POR AORC dataset to correct the volume so that it aligns with the PRISM dataset. Note the time window and normalization interval to best target the sections of the dataset with suspected data inaccuracies. Reference the Normalizing Precipitation Data Tutorial for guidance on this process. Produce a new .dss file containing the normalized AORC precipitation dataset, and provide it to the HEC-HMS modeler(s). Because the boundary condition data is the basis for future SST modeling, any normalization of precipitation data applied to the observed data in the study basin would also be needed across the entire

SST domain for consistency, and the storm catalog from the transposition region for the FFRD study must also be normalized (discussed in other areas of the SOP).

1.7 QUALITY CONTROL

QC is an important step before beginning the modeling for both historic event simulation and POR simulations. The gridded datasets should be inspected for errors including, but not limited to, holes in the grids, missing timesteps, far outlying values, time-shifted data, and mismatched values and units. The quality checks safeguard against missing or erroneous data use in the model calibration process. Quality checks also prevent potentially time-consuming model simulation errors caused by erroneous meteorologic data.

Use the *Image-Exporter* tool in the HEC-Vortex utility to convert a .dss record into a .tiff raster that can be inspected in QGIS or ArcGIS Pro to check for missing data. Next, the HEC-Vortex Sanitizer tool can be used to explicitly address missing data, screen for minimum/maximum values, and fix .dss pathname parts, among other functions. If there is a shift needed with the time and date of gridded records or a discrepancy in the start time (Part D) or end time (Part E) of a record, the *HEC-Vortex Time-Shifter* tool can be used to adjust the time of records. For unit conversions, the HEC-Vortex Calculator tool can perform simple operations on all records in the DSS file. Reference the HEC-Vortex Utility Guide for additional details and guidance on each of these tools.

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JOB AID 6—STANDARD NAMING CONVENTIONS

For successful integration of the models, all models and directories must use consistent naming conventions. This Job Aid defines the standard naming conventions for watershed level model data as well as for each constituent model. The more detailed naming conventions of folders and key files in the standard directory structure are provided in the FFRD demo directory, which is referred to in Job Aid 07, *Model Library Use*.

A comprehensive Name Table will be created during the scoping and data preparation phase and provided to modelers prior to the start of the calibration phase and will follow the requirements and guidelines in the sections below. This table will reinforce full-basin consistency, avoid duplication, and serve as the key reference for model linking that will occur during the conformance phase.

1.1 GLOBAL NAMING CONVENTION REQUIREMENTS

- Naming conventions will use only valid American Standard Code for Information Interchange (ASCII) characters.
- Uppercase characters will never be used (i.e., use all lowercase characters).
- Spaces will never be used.
- Hyphens (-) will be used for separating words to improve human readability.
- Field Delimiters: Underscores (_) will be used as a universal field delimiter to separate individual fields contained within folder names, file names, and model element names.
- Pre-defined formulas for the sequence of fields in the names for most model elements are defined in the sections below and shall be followed.

1.2 GENERAL STRUCTURE OF NAMES

Rules for naming conventions are outlined in the sections that follow. For each object included, a formula is provided to instruct users on the proper naming in a format that clearly and consistently defines the fields included in each name.

The rules are presented using the following terminology:

- fields (separable parts of the name)
- delimiters (the character used to separate individual fields)
- formulas (derivation of the object names built from fields and delimiters)
- examples

Literal text that does not vary is shown in the formulas in between single quotations (‘literal-text’). In some cases, optional fields are included in formulas to account for edge cases; these are shown in square brackets.

Table 1 includes terms, abbreviations, and delimiters that are referenced in the detailed sections that follow, and Table 2 provides basin-name examples.

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Table 1. Commonly Used Terms in the Creation of Names for Files and Model Elements

Name/ Abbreviation	Description	Options/Examples
archived-date	Date appended for files, models, or data that have been superseded and retained for reference.	20210115
base	Indicator of modeled baseflow that is produced by the hydrologic model and input as a boundary condition to the hydraulic model	
basin-name	Name of overall study basin	big-horn trinity
bc	Indicator of a boundary condition in the hydraulic model.	
event-name	Name of the event. For historic event used for calibration or validation, format is based on the month and year when it occurred (mmmYYYY). For continuous simulations, descriptive text for the time span is used. For synthetic events or events created for stress testing, another format may be used.	apr1996 2x-apr1996 por wy2009to2011 pmf sst
event-start-date	Start date of simulation in UTC. Format: YYYYMMDD	20210203
data-interval	Field describing the time interval used for hydrologic timeseries in the hydrologic model.	1hr 15min 1day
data-type	Type of observed gridded data	precip (precipitation) temp (temperature) swe (snow water equivalent) cc (cold content)
ds	Downstream	
dem	Digital elevation model (DEM)	
gage-id	ID assigned by gage operator.	12105900 (USGS) 8594900 (NOAA)
gage-operator	Operator of the observed data gage	usgs noaa usace
hms-project-use	Descriptor for HMS project filenames for use when multiple HEC-HMS projects are created.	calibration continuous por synthetic
hw	Headwater	
event-type	Type of hydraulic simulation.	calibration validation stress-test breach synthetic

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Name/ Abbreviation	Description	Options/Examples
hydro-enforced	Indicates that DEM is hydro-enforced to remove artificial impediments to stream flow such as roadways with culverts or bridges.	
ic	Indicator of an initial condition point in the hydraulic model	
lb	Left Bank	
merged	Indicator that a HEC-RAS Terrain has been exported into a single raster such that it does not include RAS Mapper formatted terrain modifications.	
model-unit	Name of the hydraulic modeling unit. Due to character length limitations in certain software, it is necessary that each model-unit string be less than 14 characters.	
nid	National Inventory of Dams	
nid-id	The unique identifier of a dam in the National Inventory of Dams.	tx05966
dam-outlet-descriptor	Short descriptor of a specific outlet from a reservoir.	spill gates rim
nld	National Levee Database	
nld-system-id	The unique identifier of a levee system in the National Levee Database	505000356
nld-segment-id	The unique identifier of a levee segment in the National Levee Database	5604350001
nsi	National Structure Inventory	
nsi-version-year	Version year of the National Structure Inventory (NSI) dataset.	
price-level	Price versioning used for the NSI.	
other--id	A field in the National Inventory of Dams that is used when a single dam includes multiple separable structures.	
other-type-description	Description of structure type used for unique 2D Connections that are not associated with dams or levees	bridge railroad diversion roadway
pt-purpose	Indicator used for hydraulic model Reference Points to convey the purpose of the point (specific to levees).	us (upstream) ms (midstream) ds (downstream) br-hw (breach headwater) br-tw (breach tailwater)
por	Period of Record	

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Name/ Abbreviation	Description	Options/Examples
rb	Right Bank	
reference-name	Short names useful for referencing features to provide an alternative to numeric ID's or provide additional context. May be abbreviations of long names.	lost-creek trinity-river white-rock trinity-bay
resolution	Resolution description of a DEM or Terrain.	4ft 12ft 100ft
storm-duration	Duration of storm from database of events for a given project/duration. Format: 00hr	72hr
storm-rank	Rank of storm from database of events for a given project/duration. Format: r001	r026
storm-type	Type of storm as described in Volume II, section 5.	st2 st3
structure-source	Source of levee, dam, or other structure data.	nld nid
unique-id	Unique ID used for HEC-HMS model elements. Usually paired with an identifier of the element type (s for subbasin, j for junction, r for reach). See naming formulas in the section for application. Increasing values from downstream to upstream.	010 020 030

1.2.1 BASIN-NAME EXAMPLES

Since the FFRD work is based on the Hydrologic Unit Code (HUC)-4 level of the National Hydrography Dataset, or subregions, the *basin-names* referred to within this report generally relate to the subregion names. All delimiters in basin names shall be hyphens. No periods, apostrophes, commas, or other symbols shall be used. Many HUC-4 basins are named with a hyphen to separate different major rivers within the basin. For this convention, those hyphens will remain, and all spaces will also be replaced with hyphens. Table 2 lists example HUC-4 basins and how they shall be named to follow this convention.

Table 2. Example Hydrologic Unit Code 4 Basin Naming

HUC-4 Subregion Code	HUC-4 Subregion Name	FFRD basin-name
0702	St. Croix-Trout-Kettle	st-croix-trout-kettle
1302	Middle Rio Grande	middle-rio-grande
1703	Columbia-Lower Snake	columbia-lower-snake
0307	Altamaha-St. Marys	altamaha-st-marys

1.3 PROJECT REPOSITORY NAME

The FFRD watershed study is organized within the Model Library within a repository named using the FFRD descriptor followed by the basin name text. The basin name will typically match the HUC-4 name in the Watershed Boundary Dataset (WBD).

- Formula: ‘ffrd’ _ basin-name
- Examples: ffrd_big-horn, ffrd_trinity

1.4 DIGITAL ELEVATION MODEL NAMES

The base DEMs created during the scoping and data preparation phase will be named according to the formulas below. Which formula to use depends on whether the delivery is provided as a single DEM for the full study basin or broken out into smaller pieces according to the hydraulic model units.

The naming convention for DEMs provided for the full study basin will follow the formula below.

- Formula: ‘dem’ _ basin-name _ resolution [_ ‘hydro-enforced’]
- Examples: dem_big-horn_100ft, dem_big-horn_4ft

The naming convention for the DEM produced for individual hydraulic model units follows the formula below.

- Formula: ‘dem’ _ model-unit _ resolution [_ ‘hydro-enforced’]
- Examples: dem_west-fork-creek_4ft, dem_east-fork-creek_4ft

1.5 RAS MAPPER TERRAIN NAMES

The terrains created by RAS Mapper produced for each model unit will follow the formula below.

- Formula: ‘terrain’ _ model-unit _ resolution
- Example: terrain_cedar-creek_4ft

1.6 STORM NAMES

Each storm in the storm catalog is stored in an individual HEC-DSS file, which is named based on the storm start date, the duration, data type, storm type, and the total cumulative precipitation ranking of the storm.

Note

r is used in place of t, which is commonly used as a parser for time in ISO formats.

- Formula: event-start-date _ storm-duration _ storm-type _ storm-rank
- Example: 19860102_72hr_st4_r001

1.6.1 STORM DSS PATH NAMES

HEC-DSS files use a path name internally to organize data in a tidy way. There are some required path name parts and some free-form parts to a valid .dss path. A path has six parts labeled A through F. The C, D, and E parts have strict requirements for grid data. A and B are free form but have a small character limit. F is usually the catch-all path part due to its longer character limit.

The convention for the storms catalog data is as follows:

/CRSresolution/$basin/PRECIPITATION/$start-date/$end-date/$data-source/

- A part: The Coordinate Reference System (CRS) should be SHG for the Standard Hydrologic Grid. The resolution should match the source precipitation data resolution (e.g., 4K for 4-kilometer resolution).

- B part: Use the same short basin name that is used in the file name.
- C part: For gridded precipitation, always use PRECIPITATION.
- D part: For gridded data, this part indicates the data's recording start time in ddBBBYYYY:HHMM format, where BBB is the first three letters of the month name, and dd is a zero-padded day.
    - Each 1-hour precipitation grid has its own path name. A storm .DSS file will contain 72 path names. The D part of the first path will match the start date in the filename.
- E part: For gridded data, this part indicates the data's recording end time in the same format as the D part. For hourly data, this should be one hour later than the D part.
    - The E part of the last path will match the end date in the filename.
- F part: Indicates the source of the precipitation data (e.g., AORC).

An example of a valid .dss path is below.

/SHG4K/TRINITY/PRECIPITATION/06OCT1982:0000/06OCT1982:0100/AORC/

1.7 HYDROLOGIC MODEL NAMES

Note that within the HEC-HMS software, certain element name fields are limited to 28 characters, which can conflict with naming convention guidance provided within this document. In these cases where naming convention guidance would exceed allowable field lengths, logical abbreviations are permitted to reduce overall name length. Additionally, duplicate names for multiple elements are not permitted. In cases where standard naming conventions would produce duplicate names, a model unit prefix is suggested to provide a unique identifier of the element. For example, two streams both named "Mill Creek" would use their model unit as a prefix, such as *salt-mill-creek* and *pepper-mill-creek*. It is recommended to document all deviations from standard naming conventions.

1.7.1 PROJECT NAME

The HEC-HMS project name is based on the basin name. In cases where multiple hydrologic models are created for a single study, an optional field may be added using an integer starting with 1, for each model created. Project names and titles must be identical.

- Formula: basin-name [_ integer]
- Examples:
    - trinity
    - rock-river_1
    - rock-river_2

1.7.2 PROJECT COMPONENTS

An HEC-HMS project is defined using several files that contain information pertaining to the land surface processes/parameters, meteorologic conditions, computational settings, and simulations. The various project component names (e.g., basin models, meteorologic models, control specifications, computes, and output .dss file) for conformance/production phase models will be a fixed name, "sst".

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1.7.2.1 BASIN MODELS

Basin models must be named using the historic event time period and/or purpose.

- Formula: event-name
- Examples:
    - apr-1997
    - por
    - sst

1.7.2.1.1 METEOROLOGIC MODELS

Meteorologic models must be named using the historic event time period and/or purpose.

- Formula: event-name
- Examples:
    - apr-1997
    - por
    - sst

1.7.2.1.2 CONTROL SPECIFICATIONS

Control specifications must be named using the historic event time period and/or purpose.

- Formula: event-name
- Examples:
    - apr-1997
    - por
    - sst

1.7.2.1.3 COMPUTES

Computes (i.e., simulation runs, forecast alternatives, etc.) must be named using the historic event time period and/or purpose.

- Formula: event-name
- Examples:
    - apr-1997
    - por
    - sst

1.7.3 SHARED COMPONENT DATA

Hydrologic simulation requires a wealth of data that is used as initial conditions, boundary conditions, or parameters. Timeseries, paired data, gridded data, and terrain data are all examples of shared component data.

1.7.3.1 TIMESERIES DATA

Timeseries data in the hydrologic model must be named to match the location and timestep of the data.

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- Formula: location _ timestep
- Examples:
    - trinity-river-at-dallas_1-day
    - bridgeport-lake-inflow_1-hour
    - shasta-lake-elevation_1-hour
    - sayers-dam-outflow_15-min

1.7.3.2 PAIRED DATA

Paired data in the hydrologic model must be named to match the location and type of data.

- Formula: location _ data-type
- Examples:
    - big-bear-lake_storage-outflow
    - big-bear-lake_elevation-storage
    - delaware-rv_r020_storage-outflow
    - wb-susquehanna-rv_r150_xs

1.7.3.3 GRID DATA

Gridded data in the hydrologic model that is being used during the calibration phase must be named to match the data source and historic event time period.

- Formula: data-source _ data-type
- Example: aorc_apr-1990

1.7.3.4 HYDROLOGIC TERRAIN

Terrain data in the hydrologic model that is being used during the calibration phase will be sourced directly from the 100-ft resolution DEM created during the scoping and data preparation phase and the name will not be altered when loaded in the HEC-HMS project.

- Formula: data-source _ horizontal-resolution
- Example: ned-100ft

1.7.4 BASIN ELEMENTS

Basin elements consist of subbasins, reaches, reservoirs, junctions, diversions, sources, and sinks. These elements are used to describe the physical properties of the watershed and the topology of the stream network. Basin elements are organized within basin models.

1.7.4.1 SUBBASINS

Subbasins must be named using a reference name following the HEC-HMS convention, suffixed with an s, followed by the HEC-HMS unique element ID. The unique ID for subbasins will follow a numbering system that increases from downstream to upstream, starting at 010 and incrementing by 10 (e.g., 010, 020, 030, etc.).

When a subbasin element is used as a comparison to observed data at a gage, the description field in the model should be used to document the authoritative name of the gage.

- Formula: reference-name _ ‘s’ unique-id

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- Example: white-rock-ck_s040

1.7.4.2 REACHES

Reaches must be named using a reference name following the HEC-HMS convention, suffixed with an r, followed by the HEC-HMS unique element ID. The unique ID for reaches will follow a numbering system that increases from downstream to upstream, starting at 010 and incrementing by 10 (e.g., 010, 020, 030, etc.). When multiple reaches are used between two junctions, numbers should be increased by 1 (e.g., 010, 011, etc.).

When a reach element is used as a comparison to observed data at a gage, the description field in the model should be used to document the authoritative name of the gage.

- Formula: reference-name _ 'r' unique-id
- Example: trinity-rv_r070

1.7.4.3 JUNCTIONS

Junctions are complex and require special consideration. In general, junctions should be named using a reference name following the HEC-HMS convention, suffixed with a j and the HEC-HMS unique element ID. In cases when the junction is directly associated with the outlet of a subbasin, the same unique ID should be used for both the subbasin and junction. This will work in many cases where unnamed tributaries join main stems of larger water bodies. In cases where larger water bodies connect, the plus sign (+) can be used to identify the confluence using a second optional reference name. For cases where streams and rivers flow into reservoirs, **'**in' should be used as the second reference name. Likewise, junctions that represent outflow from a reservoir should use **'out'** as the second reference name.

When a junction element is used as a comparison to observed data at a gage, the description field in the model should be used to document the authoritative name of the gage.

- Primary delimiter: underscore (universal for all FFRD models and filenames)
- Secondary delimiter: Optional plus-sign for additional reference names
- Formula: reference-name [+ reference-name] _ 'j' unique-id
- Examples:
    - west-fork_j090
    - elm-fork+clear-ck_j050
    - nid_tx05966_in_j070
    - nid_tx05966_out_j010

1.7.4.4 RESERVOIRS

Reservoirs will be named using *'nid'* followed by the National Inventory of Dams Identifier (NID ID). The name may also include the human-readable dam name and Other Structure ID from the NID as the following fields. These fields are sourced from the NID. Note that guidance for modeling methods for individual dams will be created as part of the scoping and data preparation phase.

The reservoir description field will include the authoritative NID Name field.

- Formula: 'nid' _ nid-id [_ dam-name] [_ other-structure-id]
- Examples:
    - nid_tx05966
    - nid_mn00584_springfield-dam_s002
    - nid_mn00584_springfield-dam

1.7.4.5 DIVERSIONS

Diversions must be named using a reference name following the HEC-HMS convention, suffixed with a d, followed by the HEC-HMS unique element ID.

- Formula: reference-name _ ‘d’ unique-id
- Example: lost-creek-diversion_d001

1.7.4.6 SINKS

Sinks must be named using a reference name following the HEC-HMS convention.

- Formula: reference-name
- Example: trinity-bay

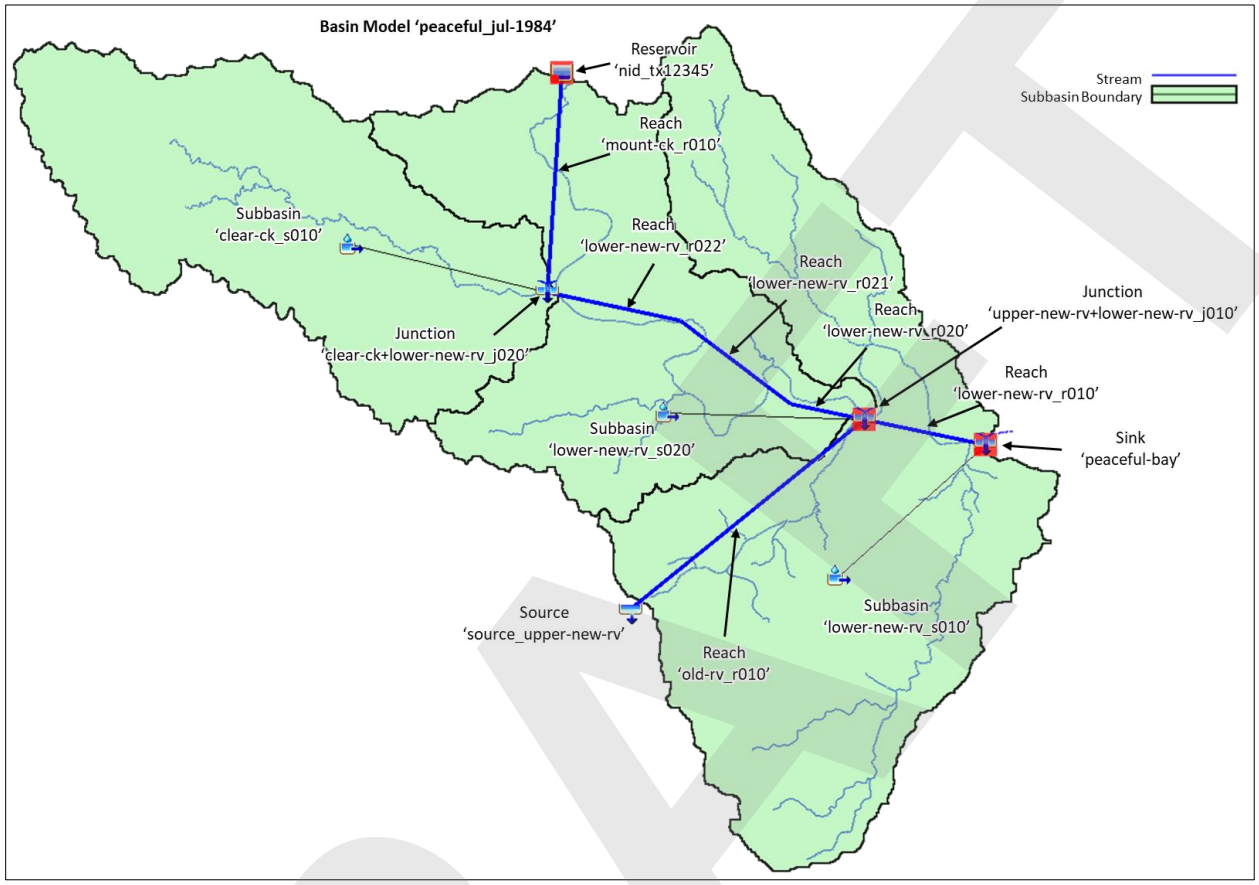
1.7.4.7 SOURCES

Sources must be named using a reference name following the HEC-HMS convention and prefixed with ‘source’.

- Formula: ‘source’ _ reference-name
- Example: source_trinity-bay

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Figure 1. Hydrologic Model Calibration Phase Element Naming Convention Example for HEC-HMS Basin Model.



1.8 RESERVOIR OPERATIONS MODEL NAMES

1.8.1 WATERSHED NAME

The HEC-ResSim watershed name is also the name of the directory containing all elements of the model—it is essentially the model's name. When creating a new HEC-ResSim model for an FFRD project, it shall be named according to the following:

- Formula: 'ffrd' _ basin-name
- Examples:
    - ffrd_trinity
    - ffrd_rock-river
    - ffrd_altamaha-st-marys
    - ffrd_rio-grande-elephant-butte

If an existing watershed is used, the watershed name can be changed within the file system by changing three items in coordination while HEC-ResSim is not running: the main directory name, the project file (.wksp) name, and the first line of the .wksp file.

1.8.2 ALTERNATIVE NAMES

HEC-ResSim alternative names are limited to only ten characters, so the naming must be succinct.

For the calibration phase modeling, several alternatives will be created, including the guide curve operations, the pulse flow test, and the calibration events. Each alternative should include an informative description. The calibration events shall be named using the following formula:

- Formula: [first letter of the month during which the event begins]-[4-digit year]
- Examples:
    - April 1990 = a-1990
    - October-November 2015 = o-2015 (or o-n-2015)

For the Conformance/Production Phase models, the HEC-ResSim alternative name will use a standard name: ‘fema_ffrd’

1.8.3 RUN NAMES

The modeler does not create run names, but scripting with HEC-ResSim and HEC-ResSim output uses run names that are based on the ten-character alternative name appended with a zero for normal runs. HEC-ResSim has the option to create trial versions of runs, which are numbered sequentially; however, trials shall not be used in the FFRD modeling. If the alternative name is fewer than ten characters, HEC-ResSim appends dashes until it gets to ten characters before appending the final zero character.

- Examples:
    - a-1990----0
    - o-2015----0
    - o-n-2015--0

1.8.4 SIMULATION NAMES

Every simulation shall have a short description to indicate its purpose. For example, “Calibration Event for Apr-May 1990”, “conservation of mass test run”, or “FEMA Future of Flood Risk Data cloud compute simulation”.

For the simulations of calibration events in the calibration phase model, the HEC-ResSim simulation name will follow the following formula:

- Formula: ‘c’ _ alternative-name (the “c” stands for calibration)
- Examples:
    - c_a-1990
    - c_o-2015
    - c_o-n-2015

For the conformance/production phase models, the HEC-ResSim simulation name will use a standard name: ‘fema_ffrd’

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1.8.5 RESERVOIR NAMES

Where possible, reservoir names will directly correspond with the reservoir element names applied in the hydrologic model, using the description and formula provided in Section 1.7. If a large, complex pre-existing model is used for the study, the level of effort required to rename the reservoirs is not justified, and instead a lookup table shall be provided in .json form within the main watershed directory.

The reservoir description field shall always include the authoritative NID *Name* field associated with the primary dam on the pool. The description shall also include any alternate, local, or familiar names used to reference the reservoir and dam.

- Examples: nid_tx05966

1.8.6 DAM NAMES

Dams are sub-elements of reservoirs, and their names will matter in watersheds that have reservoirs with diverted outlets or multiple dams. Name these sub-elements “dam”, or in cases with multiple dams at a reservoir, append a short descriptor to dam, e.g., *“dam-aux”*. This descriptor is referred to in other sections of this document as ‘dam-outlet-descriptor’.

1.9 HYDRAULIC MODEL NAMES

Note that within the HEC-RAS software, certain element name fields are limited to 16 characters, which can conflict with naming convention guidance provided within this document. In these cases where naming convention guidance would exceed allowable field lengths, logical abbreviations are permitted to reduce overall name length. Additionally, duplicate names for multiple elements are not permitted. In cases where standard naming conventions would produce duplicate names, a model unit prefix is suggested to provide a unique identifier of the element. For example, two streams both named “Mill Creek” would use their model unit as a prefix, such as *salt-mill-ck* and *pepper-mill-ck*. It is recommended to document all deviations from standard naming conventions.

1.9.1 PROJECT FILE NAME

The modeling unit name should be based on the largest stream in the model unit and will commonly match a HUC name that may approximately be coincident with the model unit. The model unit will be defined during the scoping and data preparation phase.

- Formula: model-unit
- Example: lower-new

1.9.2 PROJECT DESCRIPTION FIELD

All models created for FFRD will include a thorough description field providing users with pertinent information. The following items must be listed in the HEC-RAS model description field with each item on a new line using a colon to separate the item name and response:

- Program and Version: HEC-RAS 6.X.X September 20XX
- Status of Model: Interim Review, Final Review, Final Draft, or Final
- Primary Basin: Primary Basin Name
- Modeling Unit: Modeling Unit Name
- Date of last modification: MM/DD/YYYY
- Model Developed by: Company/Agency Name
- Vertical datum and units: NAVD 88, feet

- Horizontal datum and units: NAD 1983, feet
- Horizontal Projection: USA_Contiguous_Albers_Equal_Area_Conic_FFRD

1.9.3 GEOMETRY NAME

All models utilize a single geometry named using the modeling unit name (identical to the project file name).

- Formula: model-unit
- Example: lower-new

1.9.4 UNSTEADY FLOW NAME

The flow file for calibration phase modeling efforts will be named to match the project name, with a filename extension in the form of .u0X. Each unsteady flow event will be named according to the calibration event name. The format for historic unsteady flow names for historic events is mmmYYYY. If historic events are increased with a multiplier for the purposes of a stress test, the field uses a #x-mmmYYYY format and considered a descriptive name. The unsteady flow file name for conformance/production phase models will be a fixed name, “sst”.

- Formula: event-name
- Examples: apr1997, 2x-apr1997, sst

1.9.5 PLAN NAME

The plan file name for calibration phase modeling efforts will be named to match the project name, with a filename extension in the form of .p0X. Each plan will be named based on the scenario type and the calibration event name. The plan file name for conformance/production phase models will be a fixed name, “sst”.

- Calibration Phase
    - Formula: event-type _ event-name
    - Examples:
        - calibration_apr1997
        - validation_jun1988
        - breach_apr1997
        - stress-test_2x-apr1997
- Conformance and Production Phase
    - Formula: event-type (fixed)
        - sst

1.9.6 PLAN SHORT ID

The short plan name must be identical to the plan name.

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1.9.7 TERRAIN NAMES

The hydraulic model terrains are created using RAS Mapper software using the full resolution DEM created during the scoping and data preparation phase as the basis. The terrains will be named using the model unit and resolution as in the formula below. Note that the formula includes an optional field to indicate terrains that have been exported into a single raster and include no RAS Mapper-based terrain modifications (i.e., *'merged'*). This name is the file name for the .hdf and .vrt files. No standard is set for the underlying .tif files that make up the terrain, but human readability should be considered.

- Formula: model-unit _ 'terrain' _ resolution [_ 'merged']
- Examples:
    - little-fork-creek_terrain_4ft
    - little-fork-creek_terrain_12ft_merged

1.9.8 TWO-DIMENSIONAL FLOW AREA ELEMENT NAMES

Two-dimensional (2D) flow area names will be identical to the project file name established in the Name Table and used directly. Optionally, names may include an integer field, starting with 1, to distinguish when multiple areas are used in the model. If more than one 2D flow area exists in the model, the integer must be used for all elements.

- Formula: model-unit [_ integer]
- Example: lower-new_1

1.9.9 BOUNDARY CONDITION NAMES

Multiple distinct boundary condition line uses are called for in this SOP, and the naming formulas will vary for each based on the categories below.

1.9.9.1 INFLOW BOUNDARY CONDITION FROM UPSTREAM BASINS

The inflow boundary condition (BC) lines that supply inflow from upstream basins have names that descriptively indicate the link to an upstream basin and are intentionally repeated to be identical to the reference line from the upstream source. An optional integer field may be added so that multiple locations for passing flow between models can be distinguished when needed. The formula below will be followed.

- Formula: model-unit (upstream) _ 'to' _ model-unit (downstream) [_ integer]
- Examples:
    - mount-creek_to_lower-new
    - baldhill_to_sunny-valley_1

1.9.9.2 INTERNAL BOUNDARY CONDITION FROM SUBBASIN BASEFLOWS

The inflow BC lines that are sourced from HEC-HMS subbasin baseflows include an indicator that they are BC lines, the reference name, *'s'* and unique ID from the HEC-HMS subbasin that supplies the flow.

- Formula: 'bc' _ reference-name _ 's' unique-id _ 'base'
- Example: bc_clear-crk_s230_base

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1.9.9.3 INTERNAL BOUNDARY CONDITION FOR DAM OUTFLOWS

The flow hydrograph boundary conditions (storage area/2D area connections) that are internal to the HEC-RAS model extent and are sourced from either observed flows, HEC-ResSim simulated outflows, or HEC-HMS simulated outflows are named according to the formula below.

- Formula: 'nid' _ nid-id [_ dam-outlet-descriptor]
- Examples:
    - nid_tx05966
    - nid_ms02541_spill
    - nid_ms02541_gates

1.9.9.4 DOWNSTREAM OUTFLOW BOUNDARY CONDITION

The outflow BC lines include a BC line indicator, the model unit name, an outflow indicator, and an optional integer so that multiple outflows can be distinguished if necessary.

- Formula: 'bc' _ model-unit _ 'out' [_ integer]
- Example: bc_lower-new_out_1

1.9.9.4.1 EXCESS PRECIPITATION (METEOROLOGICAL) BOUNDARY CONDITION

No name is applied to the meteorological data boundary condition in HEC-RAS. In a desktop HEC-RAS model, the path to the source of the data is a simple .dss path link included in the plan file.

1.9.10 INITIAL CONDITION POINT NAMES

Initial condition points can be used to set the initial water surface elevations at the start of simulations and will commonly be associated with reservoir pools. When they are used to set an initial pool level, they will be named with a type identifier, a descriptive name for the reservoir and an optional integer. When initial condition points are used for other purposes, no specific formula is required.

- Formula: 'ic' _ 'nid' _ nid-id [_ reservoir-name] [_ integer]
- Examples:
    - ic_nid_ms58374
    - ic_nid_tx00776_homme-dam_1

1.9.11 STORAGE AREA/TWO-DIMENSIONAL CONNECTION NAMES

Internal storage area connections are used to represent dams and levees as well as other unique features. The names of these structures will use an indicator of the type of structure followed by a numeric identifier according to the naming formulas and descriptions below. Note that a character limit of 32 applies for all storage area/2D connection names. Additionally, requirements for the description fields are also included for each type of structure below.

1.9.11.1 DAMS

Dam names will directly correspond with the element names applied in the hydrologic and/or reservoir operations models, using the description and formula provided in Section 1.7.

The description field will include the authoritative NID Name field.

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- Examples: nid_tx05966, nid_mn00584_s002

1.9.11.2 LEVEES

2D connections representing levees will be named using *'nld'* followed by the National Levee Database System ID (System ID). The name will also include an integer field for cases when the levee system is broken into multiple structures (to allow multiple breaches or other reasons). When the integer field is used, it should be applied starting at 1 at the upstream end and increasing downstream.

The description field will include the authoritative name of the levee system found in the NLD (NLD System Name) on the first line, and a list of all NLD Segment IDs and NLD Segment Names that make up the system.

- Formula: 'nld' _ nld-system-id [_ integer]
- Examples: nld_3005000153, nld_505000356_2
- Description Example:

Prairie du Pont / Fish Lake System 2 Segment System 5604160001 - Fish Lake Drainage & Levee District No. 8 5604350001 - Prairie du Pont Levee and Sanitary District

1.9.11.2.1 OTHER STRUCTURES

Other types of structures modeled using storage area/2D connections will be named using a *'conn'* prefix, followed by a suitable ID, followed by a suitable descriptor of the structure type. The description field should be used to document the authoritative name of the structure and other useful information.

- Formula: 'conn' _ other-id _ other-type-description
- Examples: conn_1_bridge, conn_4_rr-embank, conn_3_diversion, conn_10_roadway

1.9.12 REFERENCE LINE NAMES

Reference lines are used for several distinct purposes. Naming conventions for reference lines for each purpose are detailed below.

1.9.12.1 OBSERVED DATA GAGE

Reference lines that capture modeled output at observed stream gage locations must be named with an identifier indicating the purpose of the reference line (*'gage'*), an indicator of the gage operator, and the gage operator's ID. The gage operator ID will be sourced from the authoritative databases (commonly hosted by USGS or NOAA). Lists of the gages and complete observed data will be documented during the scoping and data preparation phase and provided to the model developer. Optionally, the gage name or an abbreviation can be appended to improve human readability.

- Formula: 'gage' _ gage-operator _ gage-id [_ reference-name]
- Examples:
    - gage_usgs_12105900_gr-at-auburn
    - gage_usgs_12105900
    - gage_noaa_8594900
    - gage_noaa_8594900_washington-dc
    - gage_usace_1290183026_green-peter-outflow

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1.9.12.2 HYDROLOGIC MODEL ELEMENTS

Reference lines are included in the hydraulic model at all relevant hydrologic model elements such as headwater subbasin outlets, junctions, and reservoirs to allow for direct comparison of modeled results across each model. These reference lines will be named identically to the corresponding hydrologic model element.

- Formula: See Subsections 1.7.4.11.7.4.3 through 1.7.4.4.
- Examples:
    - s-grn-50
    - white-rock-ck_s040
    - trinity-river_r040
    - west-fork_j090
    - west-fork_j090
    - elm-fork+clear-ck_j052
    - lewinsville-inflow_j099
    - hhd-reservoir-out_j010
    - trinity-river_r040
    - white-rock-ck_s040
    - nid_tx05966
    - nid_mn00584_s002

1.9.12.3 OUTFLOW LINKING LOCATIONS

Reference lines will be created to capture outflows from the models that will be passed to downstream models (note that outflow boundary condition lines will not be used for this purpose). The reference lines that pass flow to downstream model units will have names that descriptively indicate the link to the downstream model and are intentionally repeated to be identical to the inflow boundary condition name of the downstream model. An optional *'integer'* field may be added so that multiple locations for passing flow between models can be distinguished when needed.

- Formula: model-unit (upstream) _ 'to' _ model-unit (downstream) [_ integer]
- Examples:
    - mount-creek_to_lower-new
    - baldhill_to_sunny-valley_1
    - baldhill_to_sunny-valley_2

1.9.12.4 OTHER REFERENCE LINES

Reference lines for other purposes should use a *'ref-ln'* prefix followed by a location name.

- Formula: 'ref-ln' _ location-name
- Example: ref-ln_sunrise-bridge

1.9.13 REFERENCE POINT NAMES

Reference points are used in the model to capture water surface elevations at stream gages and loading conditions along levees. They also may be used for other miscellaneous reasons.

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1.9.13.1 OBSERVED DATA GAGE

Reference points will be included at each stream gage location and placed near the river centerline. Note that there is a 32-character limit for reference point names in HEC-RAS. The *'reference-name'* field is optional and may be left out if the character limit does not allow it.

- Formula: 'ref-pt' _ 'gage' _ gage-operator _ gage-id [_ reference-name]
- Examples:
    - ref-pt_gage_usgs_12105900_gr-at-auburn

1.9.13.2 LEVEE REFERENCE POINTS

Reference points will be included along storage area/2D connections at levees located at the upstream end of the levee system, mid-point of the levee system, and downstream end of the levee system. Additionally, two reference points will be included at each levee breach location (riverside/headwater and landside/tailwater). An integer value should be appended when more than one breach location is included for the levee system.

- Formula: 'nld' _ nld-system-id _ pt-purpose [_ integer]
- Examples:
    - nld_3005000153_us (Upstream)
    - nld_3005000153_ms (Midstream)
    - nld_3005000153_ds (Downstream)
    - nld_3005000153_br-hw (Breach headwater)
    - nld_3005000153_br-tw (Breach tailwater)
    - nld_5009000634_br-hw_2 (Breach headwater)

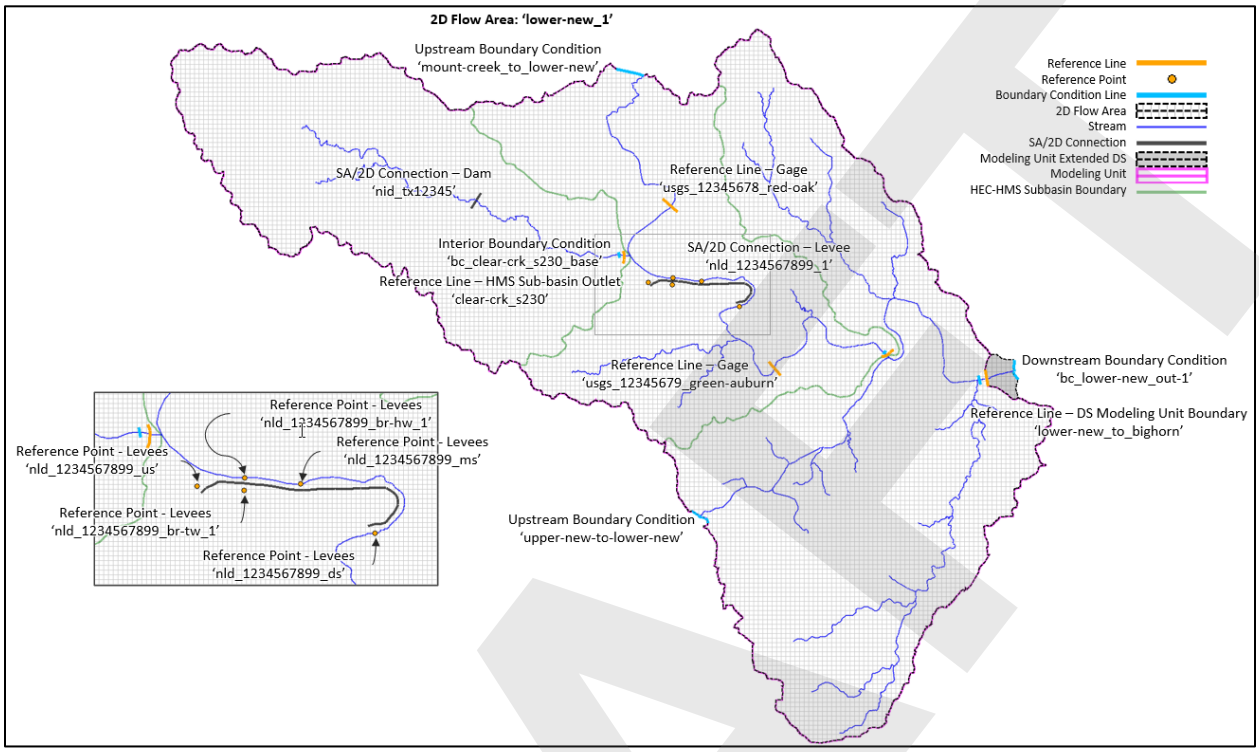
1.9.13.3 OTHER REFERENCE POINTS

Reference points for other purposes should use a *'ref-pt'* prefix followed by a location name.

- Formula: 'ref-pt' _ location-name
- Example: ref-pt_sunrise-bridge

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Figure 2. Hydraulic Model Element Naming Convention Example for Boundary Conditions, Storage Area Connections, Reference Lines/Points, and 2D Flow Area.



1.10 CONSEQUENCES MODEL NAMES

1.10.1 STRUCTURE INVENTORIES

1.10.1.1 FULL BASIN STRUCTURE INVENTORY

The structure inventory file for the entire study basin will be delivered in .gpkg format. The naming convention will include the basin name, NSI version, and the year of the price level, as shown in the formula below.

- Formula:
  - basin-name _ 'nsi' _ nsi-version-year _ price-year _ 'pricelevel' [_ archive-date]
- Examples:
  - trinity_nsi_2022_2024_pricelevel.gpkg

1.10.1.2 MODEL UNIT STRUCTURE INVENTORY

Structure inventories are also provided for each hydraulic model unit, which will be named following the formula below.

- Formula:
  - model-unit _ 'nsi' _ nsi-version-year _ price-year _ 'pricelevel' [_ archive-date]

- Examples:
    - little-creek_nsi_2022_2024_pricelevel.gpkg
    - fast-river_nsi_2022_2023_pricelevel_20240612.gpkg

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JOB AID 7—MODEL LIBRARY USE

This document defines key details regarding the use of the Model Library specifically for the FFRD initiative. The Model Library application allows access to view and interact with datasets stored in the cloud, independent of the specific bucket, and meets all USACE security protocols. The Model Library supports the FFRD initiative by providing a place for team members to store, share, and review models as well as providing a platform for staging models for cloud computing.

1.1 MODEL LIBRARY KEY INFORMATION

The Model Library is a general tool used for projects other than FFRD. Key background information can be found via the links below.

Table 1. Model Library Links

Information Source	Link
Model Library	https://model-library.sec.usace.army.mil/
Model Library User's Manual	https://www.hec.usace.army.mil/confluence/model-library/
Model Library Support	mlhelp@usace.army.mil

1.2 CURRENT MODEL LOCATION

FFRD projects require contribution from many team members throughout the process, and various models and datasets will be updated multiple times throughout their development. Strong control of model versions is critical to avoid potential errors or oversights.

The key tenet of FFRD model data management is that the current version of each model will always remain in a single, stable location.

1.2.1 ARCHIVE PROCESS FOR SUPERSEDED MODEL DATA

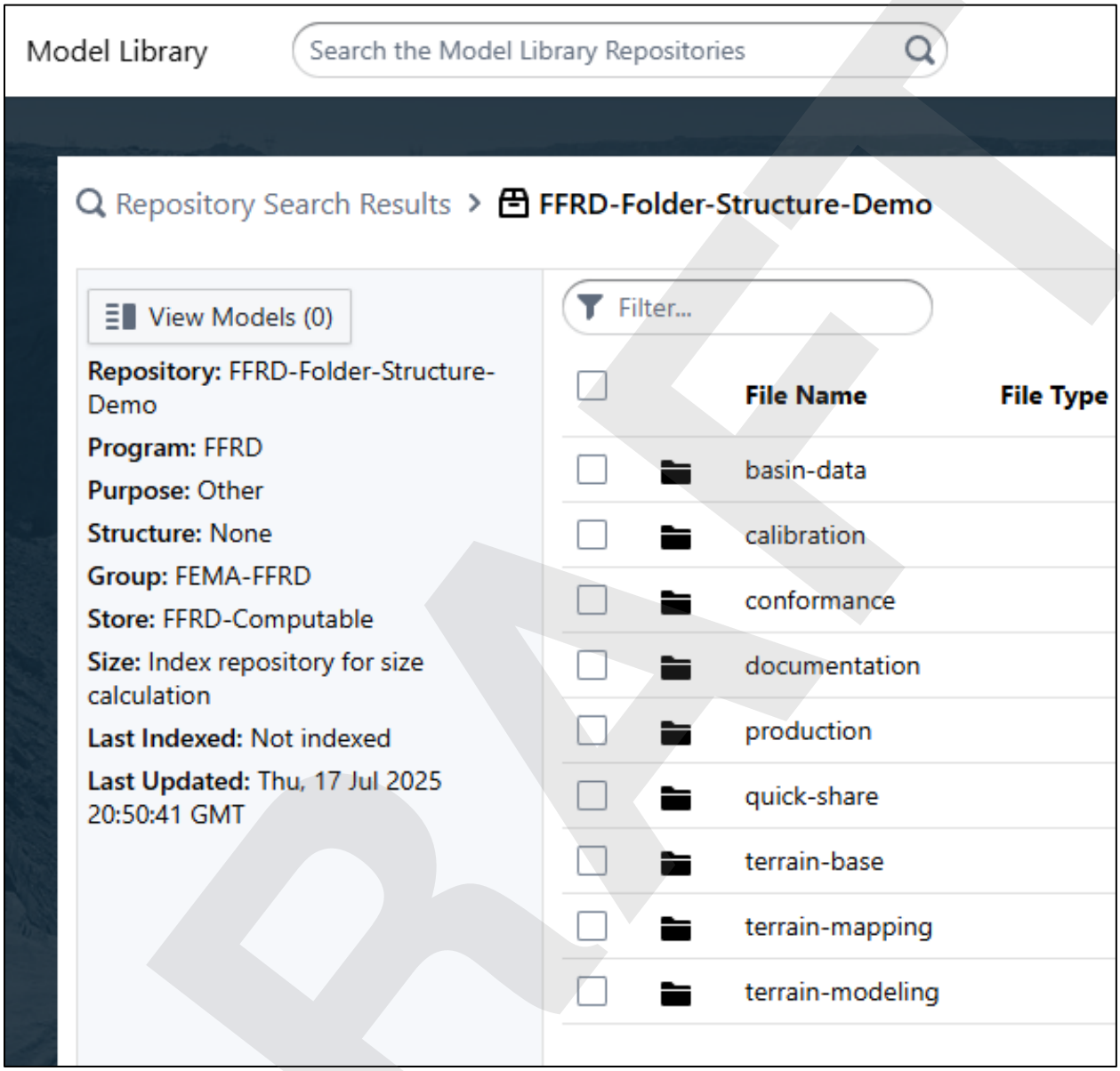
To maintain current model data in a stable location, user action to intentionally archive the current model prior to replacing it with an updated version is required. Within the *'calibration'* and *'conformance'* directories, each model has its own subdirectory. The *'archive'* directory will store the older models as they become out-of-date. Older models will be moved to the archive directory using the *rename* action on the subfolder level and entering "archive\model-name-YYYYMMDD" as the location to move to. This move process is described at the *Move a File or Directory*, Model Library user's manual page.

1.3 DIRECTORY STRUCTURE

The full formal *FFRD* directory structure is documented as a blank template on the FFRD Templates and Resources GitHub Site. The intended contents of each directory and subdirectory are included as read-me files within this directory.

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Figure 1. FFRD Folder Structure Demo Directory



1.4 USER ACCOUNT MANAGEMENT ACTIONS

The following subsections provide common user account management actions for FFRD.

1.4.1 TEAM MEMBERS: GAINING ACCESS TO MODEL LIBRARY

The process for gaining access to the Model Library is initiated by entering the URL (https://model-library.sec.usace.army.mil/) into a web browser. The general workflow for creating an account is included in the Model Library documentation, which can be accessed from the help button (Figure 2) in the upper right corner of the page.

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Figure 2. Model Library Help Button



Once access is gained, legitimate user verification by the Model Library administrators may be necessary. If access has not been granted, inquire with the help desk via email at mlhelp@usace.army.mil.

After new account verification is complete, users will need to be added to the FEMA-FFRD group in the Model Library. Reach out to the technical lead if access is needed.

1.4.2 TECHNICAL LEAD: GRANT GROUP ACCESS TO TEAM MEMBERS

The technical lead will grant access for team members to the project repository. The technical lead must already be added to the *FEMA-FFRD* group and assigned as a group administrator by the Model Library administrators.

To add a team member to the group:

1. Select the drop-down button in the upper right of the page
2. Select *Manage Groups*
3. Select the FEMA-FFRD card
4. Select the *Select new user* drop-down button and find the team member
5. Select the blue edit button next to the new team member
6. Apply the appropriate permissions (Read and Write for most team members)
7. Save
8. Verify user has access by communicating directly with them.

If any technical issues or questions arise, support can be obtained via email at mlhelp@usace.army.mil.

1.4.3 TECHNICAL LEAD: CREATE NEW FFRD REPOSITORY

At the beginning of an FFRD project, a new repository will be created in the Model Library by the technical lead, and empty folders will be created according to the standard directory structure.

An FFRD demo repository is available in the Model Library to aid in the creation of a new project repository. This demo repository is named ffrd-folder-structure-demo and includes all of the directories and subdirectories needed for a new repository. Note that additional basin-name-specific folders must be added, and read-me files are included in the demo directory to guide the format for those folders.

To create a new repository for a new project, follow these steps:

1. Download the demo directory to your computer
2. Unzip the downloaded directory
3. Add custom, basin-specific subdirectories as indicated in the read-me files
4. Click *Add a Model Repository* from the Model Library Splash Page

    a. Use the FFRD SOP standard naming for the repository (e.g., ‘ffrd_trinity’)
    b. Set Program field to FFRD
    c. Set Repository Group to FFRD
    d. Set Repository Location to FFRD-Computable
    e. Set Repository Purpose to Watershed

5. Open newly created repository
6. Click on Tools – Upload Files
7. Drag the local directory from your computer into the upload tray in the browser.

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JOB AID 8—DAM PHYSICAL REPRESENTATION

This job aid describes dam physical and operational model representation specifically for FFRD models. The guidelines for dam and reservoir modeling methods are included in SOP Volume II, and the guidelines for scoping specific methods to apply for each dam in a basin are included in Job Aid 01, Scoping Processes. This job aid provides the technical details and processes required for implementation.

1.1 COORDINATED DAM MODELING APPROACH ACROSS SYSTEM OF MODELS

Consistency of the model approach across the HEC-HMS, HEC-ResSim, and HEC-RAS models for each dam is important for FFRD projects. The SOP utilizes the Dam Scoping Table, Model Linking Register, and Name Table to ensure each model can be developed based on a coordinated approach.

A common dataset that contains the available physical dam information, reservoir capacity information, and curated historic observed data is provided prior to model development and calibration activities and will be leveraged throughout model development.

1.2 INITIAL POOL ELEVATIONS

For each storm event, initial pool elevations for reservoirs should be set consistently across each model (HEC-HMS, HEC-ResSim, and HEC-RAS). The initial pools should be set according to standard techniques available in user manuals for each software package.

For HEC-RAS models, initial condition (IC) points are only required for dams scoped for Type A, B, or C modeling. For all HEC-RAS IC points, the *fixed during warmup* option should be selected so that the pool level at the start of the simulation is consistent with the selected value.

Historic event modeling will leverage any available observed pool data as initial pool elevations. When no observed data is available, estimating the average pool level is acceptable, and light detection and ranging (LiDAR) elevations that capture the pool water surface may be used as an approximation.

Stochastic events will vary initial pool levels for each event for certain dams according to the Dam Scoping Table. The method for selecting pool elevations for stochastic simulations for each event is documented in other sections of the SOP.

1.3 RESERVOIR STORAGE CAPACITY

The reservoir storage capacity for each reservoir is an important aspect of the model representation for each modeled dam and must be consistent across the HEC-HMS, HEC-ResSim, and HEC-RAS models. All aspects of this representation are described in the sections below.

1.3.1 HEC-HMS STORAGE CAPACITY

A tabular Elevation-Storage Relationship represents the HEC-HMS storage capacity. It should be input such that it is based on the best available data and is consistent with the RAS Mapper Terrain.

1.3.2 HEC-RESSIM STORAGE CAPACITY

The HEC-ResSim storage capacity is represented by a tabular Elevation-Storage Relationship. It should be input such that it is based on the best available data and is consistent with the RAS Mapper Terrain.

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1.3.3 HEC-RAS STORAGE CAPACITY

The HEC-RAS storage capacity is represented by the RAS Mapper Terrain, which must account for underwater bathymetry. The RAS Mapper Terrain is generated by incorporating the best available data into the base terrain through GIS processes. Compare RAS Mapper Terrain-based storage with tabular estimates applied to HEC-HMS and HEC-ResSim to confirm that they are consistent.

1.3.4 STORAGE CAPACITY CONSISTENCY CHECK

After the RAS Mapper-based terrain has incorporated bathymetry to represent storage, it is critical that the terrain's elevation-volume relationship is validated against any known elevation-volume data. This could be a published elevation-volume curve, tabular data from historic documents, or National Inventory of Dams (NID) storage values. Elevation-volume curves can be computed for a given area within RAS Mapper or in the geometric schematic editor.

To compute a RAS Mapper terrain-based storage, a separate new geometry should be created for this purpose. A new 1D storage area should be created within this new geometry with a name corresponding to the reservoir of interest. The boundary of this storage area should be close to the perimeter of the water body as indicated by the LiDAR terrain digital elevation model (DEM) (see the note below regarding automatic generation of reservoir outline polygons). This geometry with a storage area over the reservoir needs to be associated with the terrain that has bathymetric terrain modifications applied. Right-clicking the storage area allows for computing and plotting elevation-volume data for the reservoir. To verify the computed elevation-volume curve has adequate resolution, the maximum elevation should be limited to near the top-of-dam elevation.

The elevation-volume curve computed by HEC-RAS from the terrain should be compared with available elevation-volume data. The terrain should be modified until there is a reasonable and satisfactory fit between the terrain surface's volume and available reference data.

The reservoir outline polygon needed to produce the storage capacity table can be created through an automated process. GIS geoprocessing tools, such as Contour, can be used to automatically generate an outline of the reservoir surface (assuming the reservoir is a flat surface in the LiDAR-based DEM). If using the Contour geoprocess, the base contour can be specified as slightly higher than the LiDAR-indicated water surface elevation (e.g., base contour of 0.1 feet higher than LiDAR water surface). A large contour interval can also be used by the Contour geoprocess so that only a single contour is created at the desired elevation near the LiDAR water surface. A contour type of *contour* is appropriate for this purpose (the output polyline feature can then be converted to a polygon using the *Feature to Polygon* geoprocess). The *Processing Extent* options of the Geoprocessing Environments can be used to limit the extent where contours are calculated. Some manual cleanup may be required to remove contours that do not correspond to the reservoir of interest. The *Simplify Polygon* geoprocess with a tolerance of approximately 1 to 2 feet can be used to reduce the number of vertices in a reservoir outline polygon.

1.3.5 STORAGE CAPACITY SOURCE DATA

There are several sources of reservoir bathymetry data, many of which are discussed below. Note that authoritative input data for the project will be collected during the scoping and data preparation phase and posted to the '*basin-data'* directory for the project. These datasets will not be available for all reservoirs, and therefore they have been roughly listed in order of their importance. Use engineering judgment to select the best available data to estimate reservoir storage for the project. For example, if recent bathymetric survey data is available, it should be utilized. If survey data are not available, historic documentation should be used to estimate bathymetry, and so on. It may be necessary to use information from multiple data sources to best approximate bathymetric data. The sections below describe the typically available data, followed by processes that may be used for application of the selected source data on an individual dam basis.

1.3.5.1 SURVEY DATA

Surveyed bathymetric data can be the most accurate representation of reservoir bathymetry. If this data is available and reliable, it should be used as the primary source of reservoir storage model inputs. The format of this survey data will vary and may come in the form of a surface dataset (e.g., triangular irregular networks [TIN], raster, etc.), individual survey points, surveyed transects or cross sections, or other formats.

1.3.5.2 HISTORIC DOCUMENTATION

Historic documents that describe the storage volume of reservoirs may be available. These documents may be available from the reservoir's construction, or more recent studies. Common examples of these documents include Water Control Manuals, Reservoir Regulation Plans, Reservoir Sedimentation Studies, As-Built documents, and other design documentation. Ideally, these documents will provide elevation-volume or elevation-area data (either tabularly or graphically), which describe how the reservoir's storage varies by elevation. If elevation-varied storage information is not available, storage volumes may be specified at given elevations of interest, such as conservation pool, spillway crest, top of dam, etc.

1.3.5.3 NATIONAL INVENTORY OF DAMS

The NID provides a national database of dams and reservoirs and includes some basic information on their storage volumes. Normal and maximum storage are provided as attributes for all reservoirs in the NID. According to the NID Data Dictionary, maximum storage is defined as, "the total storage space in the reservoir below the maximum attainable water surface elevation, including any surcharge storage." Normal storage is defined as, "the total storage space in a reservoir below the normal retention level, including dead and inactive storage and excluding any flood control or surcharge storage." For approximation purposes, it can generally be assumed that the reservoir storage below the LiDAR-indicated water surface elevation in the DEM is equivalent to the NID normal storage, unless available data indicates otherwise.

1.3.5.4 HISTORIC TOPOGRAPHIC MAPPING

Historic topographic mapping may be available from the period prior to the reservoir's impoundment, providing topographic contours for the terrain beneath the reservoir. For very old reservoirs, this information may not be available; however, historic topographic maps are available widely across the United States from the USGS beginning in the 1940s. The accuracy and detail of these maps vary and may be limited, but they do provide some estimate of the bathymetric surface. Data from these maps can be converted to DEMs and corrected to match a published elevation-volume curve where data is available.

1.3.5.5 TERRAIN INDICATIONS

When all other data discussed above is unavailable, LiDAR DEM terrain data may be useful for approximating bathymetry. For example, the storage volume of dry dams (dams that do not impound water permanently) may be accurately represented by the terrain DEM without additional information. For dams that do impound water but have no other information available regarding their storage, the elevation of the dam's toe on the downstream side is likely a reasonable approximation for the deepest portion of the reservoir. The reservoir's elevation is also well defined around the lake water's edge, where depths are zero. A rough approximation regarding the reservoir's volume can be estimated from this information alone. LiDAR DEMs may show locations where tributaries enter the reservoir body; these tributaries likely continue beneath the reservoir's surface, and any estimates of the lake's bathymetry should attempt to capture these features. The bathymetry beneath the lake's surface likely resembles the surrounding area,

which should be well described by available LiDAR DEMs. These surrounding areas can provide context information on things like valley wall slope, pre-impoundment river channel width and meander characteristics, valley shape, and many other geomorphic features that can inform bathymetric approximations.

1.3.6 METHODS TO INCORPORATE RESERVOIR BATHYMETRY

There are multiple methods available to incorporate reservoir bathymetry data into the RAS Mapper terrain to represent the best available reservoir capacity information. The sections below present typical workflows and processes for incorporating reservoir bathymetry. The selection of a specific method for each reservoir should be based on available source data and professional engineering judgment. Other approaches to represent bathymetry may also be viable; the methods described here are not exhaustive.

Regardless of the method applied, apply any required vertical datum conversions to the input data so that the reservoir storage representation meets the FFRD vertical datum standard.

1.3.6.1 SURVEY DATA APPROACH

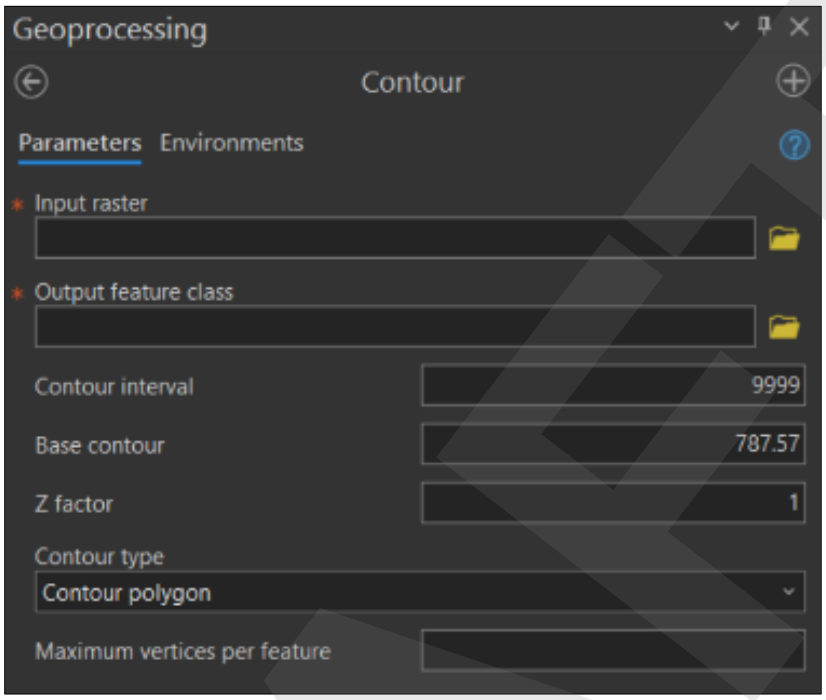
When an array of survey points is used as source data, a TIN is a simple method of interpolating elevations between known points, which preserves the exact elevation at the known points. This approach is appropriate for converting point-based bathymetric survey data into a raster format that can be merged with a DEM representing elevations above the reservoir pool. Compared to other interpolation methods, TIN creation has few required parameters and does not rely heavily on user judgment. However, the spatial distribution of input data points can cause errors in estimation, which will be discussed in the next section. This guide references tools in ArcGIS Pro; however, similar methods may be available in other GIS software, and alternative workflow processes are viable.

1.3.6.1.1 GIS PROCESSING STEPS

First, if necessary, a datum conversion should be applied such that the vertical datum of the bathymetric survey data matches the vertical datum of the DEM that it will be merged with. Next, the bathymetric survey data should be imported into ArcGIS Pro as a point feature class, using the XY Table to Point tool if the data was delivered in tabular format. Next, using a DEM as input, a polygon of the approximate reservoir extents at the time of the bathymetric survey should be generated. This can be done using the Contour tool. To create a single contour polygon, the base contour should be set to the desired pool elevation, and the contour interval should be set to a very large number beyond the range of DEM values.

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Figure 1. Creating a Single Contour Polygon



This reservoir extent polygon will be used as a constraint on the TIN creation. To ensure that no observed points are clipped out, the contour elevation should be set to several feet above the elevation at the time of data collection.

Optionally, to ensure that the slope of the TIN is logical, a thalweg line can be applied during TIN creation as a breakline. A breakline forces the process to consider areas on opposite sides of the breakline as independent from each other. Thalweg data may be provided with survey data or may be created using other methods. Breaklines are useful for constraining and refining interpolation between survey points and may be required to adequately reflect the reservoir's bathymetry.

With point data, a reservoir extents polygon, and (optionally) a polyline of the thalweg, the *Create TIN* geoprocessing tool can be used. The point data should be input as Mass Points with the height field set to match the elevation field in the Attribute Table. The reservoir extents polygon should be used as a Hard Clip. The thalweg should be used as a Hard Line. The output coordinate system should match the DEM, and the Constrained Delaunay option is up to the user.

Next, the TIN to Raster geoprocessing tool can be used to convert the TIN into raster format. The sampling distance should be Cell Size, and the Sampling Value should match the cell size of the DEM (this should be precise and may need to be copied and pasted from the DEM's Properties dialog box). Using the Natural Neighbor method versus the Linear method results in very similar storage capacity relationships, but the Natural Neighbor method is more computationally robust, producing a smoother raster surface. Finally, in the Environments tab, the Snap Raster should be set to the DEM to enable seamless merging of the bathymetry raster and the DEM.

Next, as an optional step, the resulting bathymetric surface can be merged with the LiDAR-based DEM to produce an elevation-storage relationship for all reservoir pools. This step should be done carefully to minimize any breaks in the data or large jumps in elevation, which can occur if there is a discrepancy between the bathymetric data and the DEM due to the uncertainty and slight estimation error that exists in all terrain data. The recommended approach is to create a contour at the exact pool elevation that existed during the bathymetric survey and use Extract by Mask to mask the TIN to Raster result to these extents. The Snap Raster should be the DEM when using Extract by Mask. Then, Mosaic to New Raster can be used to merge the result with the DEM, setting the Mosaic Operator such that the values from the

bathymetry raster will be carried forward wherever they exist, and the values from the DEM will be carried forward outside the extents of the bathymetry raster. In the Environments tab, the Processing Extent of the Mosaic to New Raster tool can be set to a display extent that covers the entire reservoir, which is necessary to reduce file size if the source DEM covers a much larger area than needed. If the result is unsatisfactory, a different reservoir pool extent could be used in Extract by Mask to mask the TIN to Raster result, and Mosaic to New Raster could be applied again using the modified TIN to Raster result.

1.3.6.1.2 SPECIAL CONSIDERATIONS AND POTENTIAL ERROR SOURCES

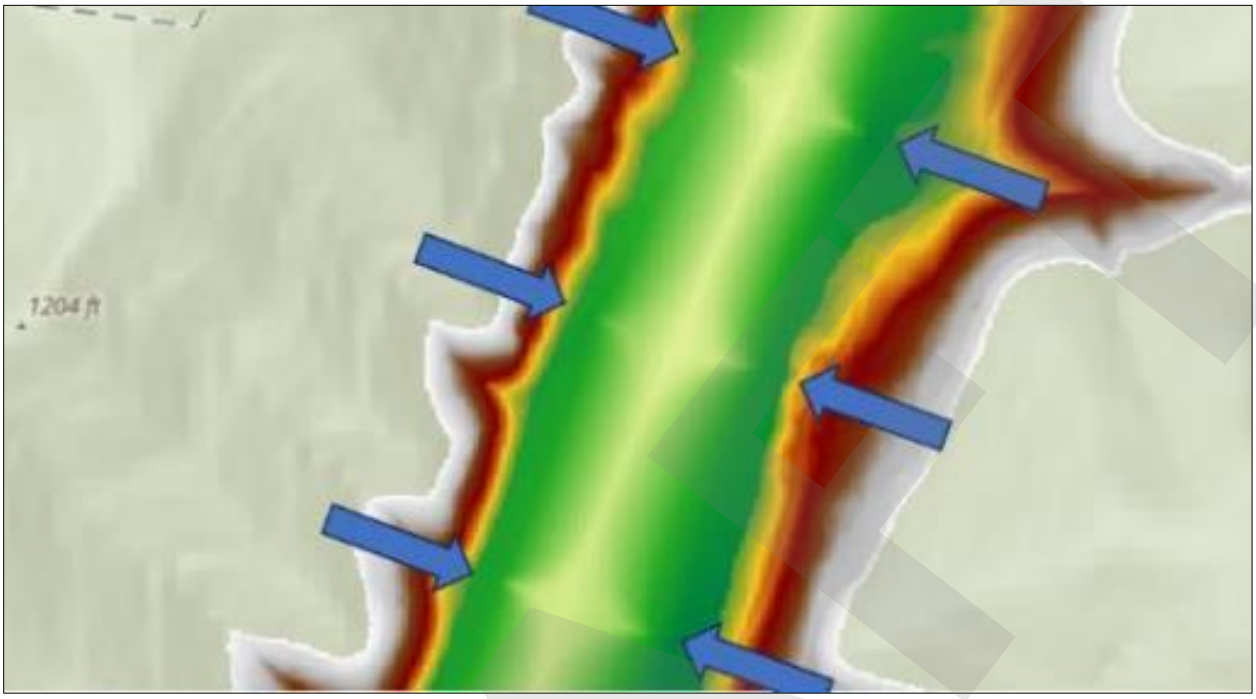
For the greatest accuracy in TIN creation, the ideal situation is for known data points to be plentiful and randomly distributed through space. This condition would require the use of multibeam survey equipment. However, bathymetric data is sometimes collected along defined paths in a reservoir using a single-beam echo sounder, which leaves large gaps between collected data. Figure 2 shows an example of this condition, which requires special considerations to be made during interpolation.

Figure 2. Large Gaps between Collected Data due to Single-Beam Echo Sounder Data Collection



In this example, data was collected in cross sections and transects, with very small point spacing along the cross section and very large point spacing between cross sections. This distribution of points creates error in the final result. This error occurs because, in a reservoir, elevations are typically changing very gradually parallel to the flow direction of the original stream channel, but elevations change rapidly perpendicular to flow. This feature of the data is called anisotropy. In TIN interpolation, there is no way to correct for this condition, so interpolated points are generated based on nearby values in all directions. After the TIN was converted to a raster, the result was that elevations were overestimated in areas far from observed points. Figure 3 shows the locations of cross sections where the elevations are correct, with overestimated elevations (in darker shades of green) in between the cross sections.

Figure 3. Cross Sections with Correct Elevations with Overestimated Elevations between Cross Sections



The overestimation in this example was more severe due to the narrow, steeply sloped valley in which this reservoir exists. A wider reservoir with flatter slopes would be less prone to this error. Other approaches, such as the Topo to Raster geoprocessing tool, can provide a more suitable interpolation between data points in some cases, but each approach has limitations. In general, the result of the TIN creation process should be carefully examined for reasonableness. If there is clear overestimation or underestimation of elevations, another approach should be considered. Additional breaklines can be used to guide interpolation and reduce errors. In some cases, if no one suitable approach can be found, engineering judgment may be necessary to combine the results of multiple approaches.

1.3.6.2 ELEVATION-STORAGE TABULAR DATA APPROACH

Many reservoirs have elevation-volume data available that describe how the reservoir's storage varies with water surface elevation. Figure 4 shows an example of such data. Occasionally, this volume data is provided in elevation-area format, describing how the reservoir's surface area varies with water surface elevation. Either of these formats can be converted to the other with some assumptions. Figure 5 shows an example calculation showing the conversion from surface area to storage volume, and Figure 6 shows the reverse process of converting storage volume to surface area. Note that if surface area is calculated from volume as shown in Figure 6, that surface area corresponds to the average elevation data shown in column D of the figure, and not the elevation data shown in column A.

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Figure 4. Example Elevation-Volume and Elevation-Area Curves

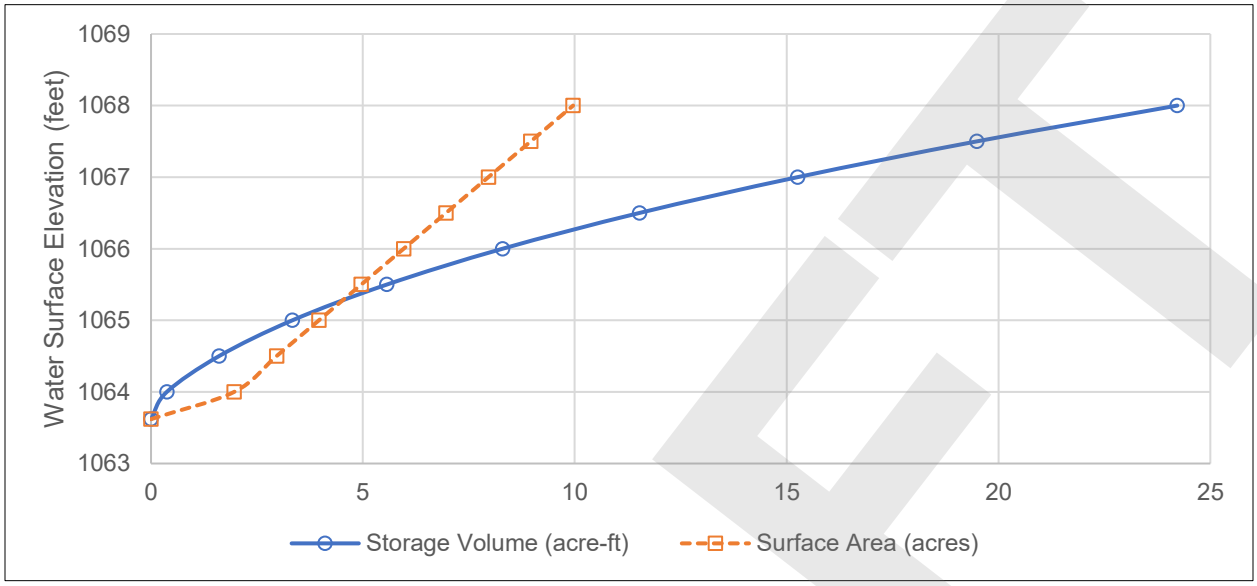


Figure 5. Example Conversion from Surface Area to Storage Volume

	A	B	C
1	**Elevation**	**Surface Area**	**Storage Volume**
2	*ft NAVD88*	*Acres*	*Acre-ft*
3	1063.6	0	0
4	1064.0	2.0	0.4
5	1064.5	3.0	1.6
6	1065.0	4.0	3.3
7	1065.5	5.0	=C6+AVERAGE(B6:B7)*(A7-A6)
8	1066.0	6.0	8.3
9	1066.5	7.0	11.5
10	1067.0	8.0	15.3
11	1067.5	9.0	19.5
12	1068.0	10.0	24.2

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Figure 6. Example Conversion from Storage Volume to Surface Area

	A	B	C	D	E	F	G
1	**Elevation**	**Storage Volume**		**Average Elevation**	**Incremental Elevation**	**Incremental Volume**	**Surface Area**
2	*ft*	*Acre-ft*		*ft*	*ft*	*Acre-ft*	*Acres*
3	1063.6	0		=A3	N/A	N/A	0
4	1064	0.4		=AVERAGE(A3:A4)	=A4-A3	=B4-B3	=F4/E4
5	1064.5	1.6		=AVERAGE(A4:A5)	=A5-A4	=B5-B4	=F5/E5
6	1065	3.3		=AVERAGE(A5:A6)	=A6-A5	=B6-B5	=F6/E6
7	1065.5	5.6		=AVERAGE(A6:A7)	=A7-A6	=B7-B6	=F7/E7
8	1066	8.3		=AVERAGE(A7:A8)	=A8-A7	=B8-B7	=F8/E8
9	1066.5	11.5		=AVERAGE(A8:A9)	=A9-A8	=B9-B8	=F9/E9
10	1067	15.3		=AVERAGE(A9:A10)	=A10-A9	=B10-B9	=F10/E10
11	1067.5	19.5		=AVERAGE(A10:A11)	=A11-A10	=B11-B10	=F11/E11
12	1068	24.2		=AVERAGE(A11:A12)	=A12-A11	=B12-B11	=F12/E12

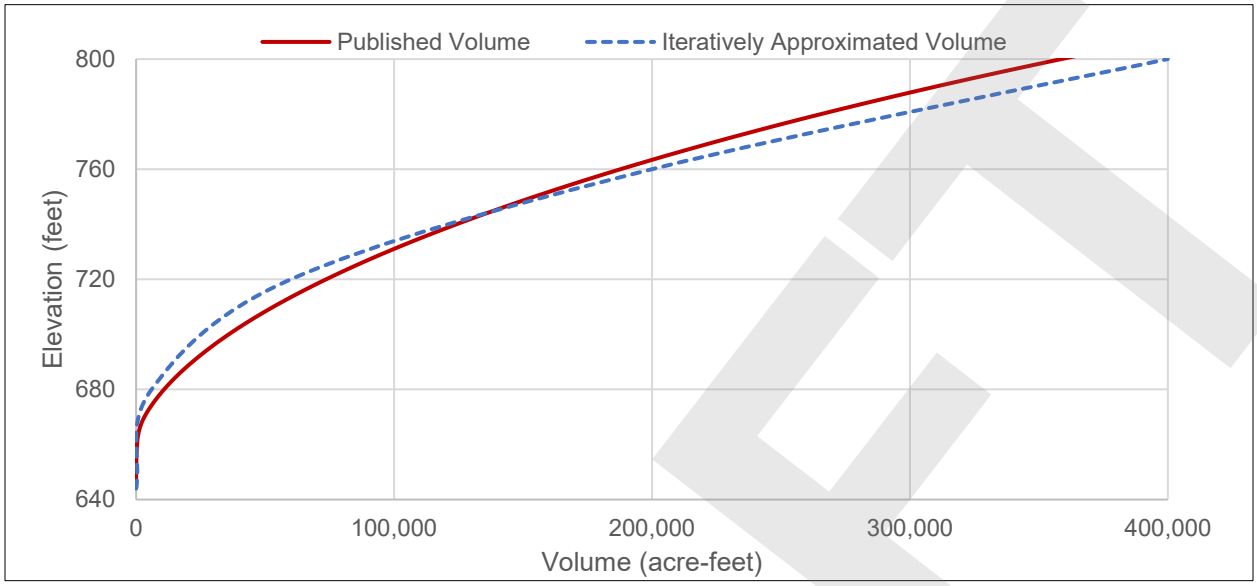
1.3.6.2.1 APPROXIMATE APPROACH

When elevation-volume data is available, but a bathymetric terrain surface is not, HEC-RAS terrain modifications can be used to modify the LiDAR DEM terrain surface and approximate the given elevation-volume curve. This approach involves using multipoint polygon terrain modifications and control points to approximate a bathymetric surface in RAS Mapper. This approximate bathymetry is iteratively modified until an acceptable match is achieved between the published elevation-volume data and the elevation-volume computed from the terrain surface. This process is similar to the one described in the Detailed Approach section below. Figure 7 displays an example comparison of what can be achieved by iteratively approximating an elevation-volume curve by using HEC-RAS terrain modifications. Additional refinement of terrain modifications could be performed to better improve the fit between the two curves. Refer to Section 1.3.4, Storage Capacity Consistency Check, for additional details on how to compute an elevation-volume curve from a terrain surface within HEC-RAS.

Historic contour data can be used to inform the shape of the bathymetric terrain but is not always necessary. This approximate approach differs from the approach discussed in Section 1.3.6.2.2, Detailed Approach, in which the terrain surface is corrected to closely match published elevation-volume data. While this approximate approach may not perfectly reproduce the curve, it is an efficient process and is appropriate for smaller impoundments that do not pose a high breach risk to downstream communities. Engineering judgment and team consensus will be required to determine which bathymetric estimation approach is appropriate for a given reservoir and when a given match between modeled and published volume estimates is acceptable.

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Figure 7. Example Comparison of Published and Approximated Elevation-Volume Curves



Terrain modifications in RAS Mapper take advantage of vector editing tools to create points, polylines, and polygons to modify an existing HEC-RAS terrain. These tools are useful for incorporating bathymetry into a terrain surface, as well as many other uses. Additional information regarding the use of terrain modifications is discussed in this document and in the RAS Mapper User Manual.

1.3.6.2.2 DETAILED APPROACH

For large reservoirs that pose significant breach risk to downstream communities, it may be warranted to pursue a detailed approach to estimate reservoir bathymetry that reproduces a published elevation-volume curve accurately. This detailed approach may also be required for reservoirs where HEC-ResSim operations are being simulated. This is required because HEC-ResSim specifies elevation-volume relationships using published tabular data, and any disagreement between reservoir storage in HEC-RAS and HEC-ResSim will result in disagreement of computed water surfaces in the reservoir.

The detailed approach consists of creating an approximate bathymetric terrain surface and then correcting that surface such that it reproduces a published elevation-volume curve. These two steps are described below.

1.3.6.2.2.1 Creating a Bathymetric Surface

There are several ways to create an approximate bathymetric terrain surface. One, which has been employed successfully on the FFRD pilot projects, is to generate a terrain surface from historic topographic maps from prior to the reservoir's impoundment, which are widely available across the U.S. Note that for historic topography to be useful in this process, the historic maps must include contour data in the reservoir pool, which is common in maps created prior to reservoir impoundment.

The process to convert historic topographic mapping to a terrain that can be utilized directly in HEC-RAS is described in Table 1. This process involves tracing topographic map features such as contour lines and stream centerlines. It may be possible to automate this procedure such that manual digitization of contour lines is not required; however, for smaller reservoirs where contour data is not particularly dense, manual digitization often seems to be the most efficient option.

The process is designed around the use of ArcGIS Pro and requires either the 3D Analyst or Spatial Analyst extension, specifically the Topo to Raster geoprocessing tool. It is recommended to review

available documentation for the Topo to Raster tool prior to use to understand its data requirements and limitations. Some resources regarding the Topo to Raster tool include:

- https://pro.arcgis.com/en/pro-app/latest/tool-reference/3d-analyst/how-topo-to-raster-works.htm
- https://pro.arcgis.com/en/pro-app/latest/tool-reference/3d-analyst/topo-to-raster.htm
- https://support.esri.com/en-us/knowledge-base/why-does-the-topo-to-raster-tool-generate-inaccurate-el-000012375

This process should produce a terrain surface DEM that can be directly incorporated into HEC-RAS; however, the terrain surface generated will likely not agree with other sources of elevation-volume data, such as that commonly available in water control manuals. The next section describes how a terrain surface can be corrected to improve agreement between a terrain surface and given elevation-volume data.

1.3.6.2.2.2 Correcting a Bathymetric Surface to Reproduce an Elevation-Volume Curve

Elevation-volume relationships, and specifically elevation-area relationships, can be used to modify a given terrain surface such that it reproduces the input elevation-volume relationship. It should also be noted that this volume correction process can be applied to a diverse range of initial terrain surfaces, and not only surfaces created from historic topographic mapping as discussed in the previous section. Initial terrain surfaces should resemble the true bathymetric surface as well as is practical for best results. The volumetric correction process can be performed in a script or spreadsheet using linear interpolation methods to return values from a given elevation-area curve corresponding to a specified contour surface area. If done in Excel, a custom linear interpolation function is very useful.

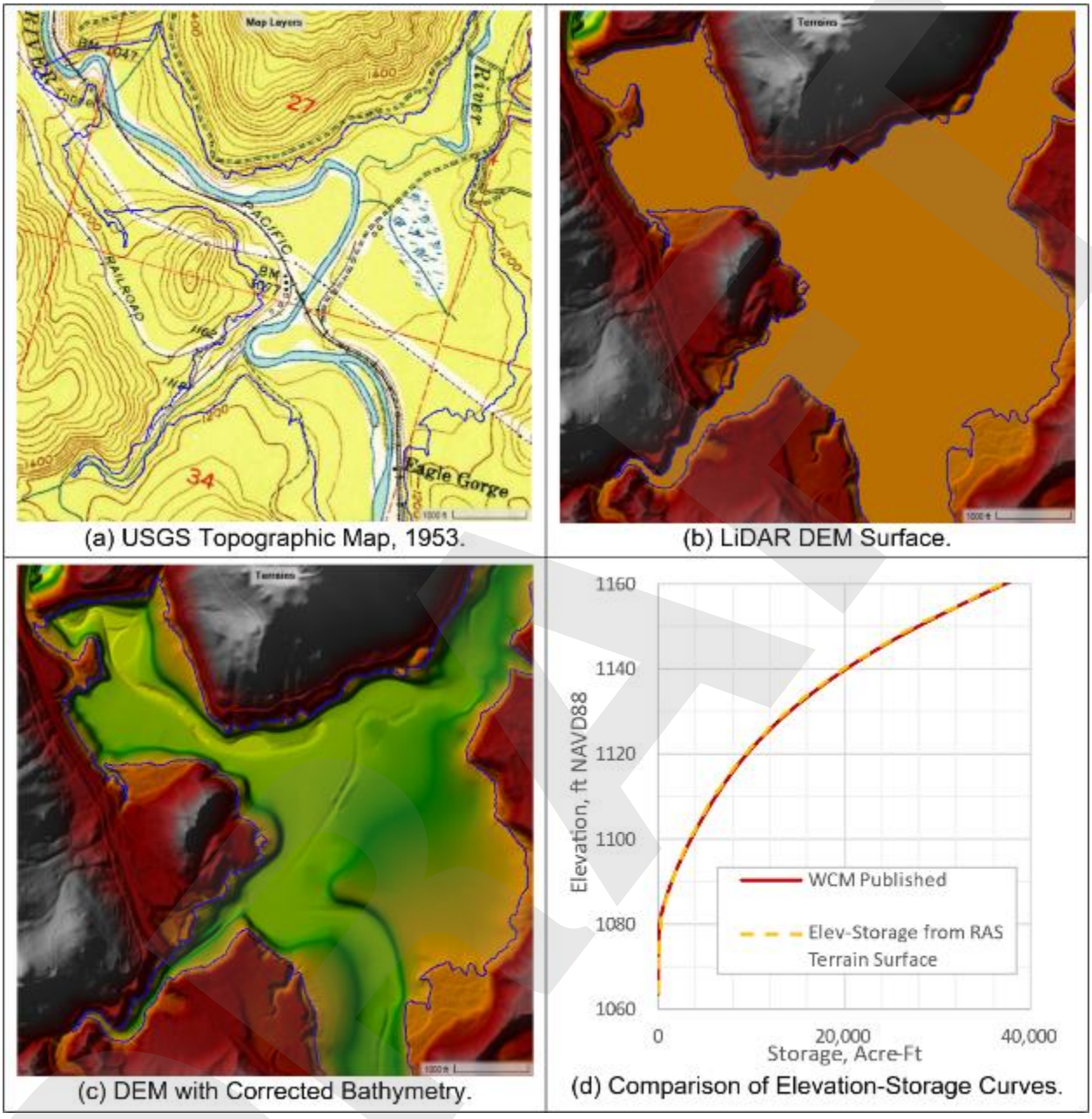
The steps to perform a volumetric correction process on a terrain are described in Table 2. This process generates elevation contour polygons from a terrain surface (step 5) and calculates the surface area associated with each contour (steps 6 through 8). Once the area of each terrain contour is known, the corresponding elevation can be looked up from published elevation-area data (step 9). A linear interpolation script is useful for this task; this is the corrected elevation attribute. Now each terrain contour has attributes of surface area, elevation (as computed from the historic bathymetry estimate), and a corrected elevation looked up from published data.

These elevation-corrected contours can be used as input data to generate a new terrain surface (steps 10 and 11). The elevation-volume relationship of the new surface should be verified (step 12). If needed, the process can be iteratively performed to improve the fit between published and modeled volume data (step 13). Finally, the process should create a terrain surface suitable for use in HEC-RAS, which reproduces elevation-volume relationships that will be used in the paired HEC-HMS and HEC-ResSim models (step 14).

As noted previously, this volumetric correction approach relies on elevation-surface-area relationships. Refer to Section 1.3.6.2, and specifically Figure 6, for an example calculation of how to convert an elevation-volume dataset to an elevation-area dataset. Note that the surface-area calculated using this process corresponds with the average elevation data shown in column D of Figure 6, not column A. Also refer to Section 1.3.4 for the process to verify a terrain surface's elevation-volume relationship in HEC-RAS (step 12).

Figure 8 displays several of the incremental steps of the detailed approach to approximating a bathymetric surface. Part (a) of this figure illustrates an example historic topographic map, including contour lines and stream centerlines, as well as a reservoir outline generated from LiDAR data. Part (b) illustrates the LiDAR DEM itself, which includes no bathymetric information. Part (c) shows the DEM resulting from this process, and Part (d) illustrates the excellent comparison that can be achieved between a terrain surface and a published elevation-volume dataset.

Figure 8. Example Components of Detailed Approach for Estimating a Bathymetric Surface.



(a) USGS Topographic Map, 1953.

(b) LiDAR DEM Surface.

(c) DEM with Corrected Bathymetry.

(d) Comparison of Elevation-Storage Curves.

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Table 1. Steps to Create a Terrain Surface from Historic Topographic Mapping using ArcGIS Pro

#	Steps and Process	Input Files	Output Files
1	Create outline of reservoir at LiDAR water level. Use the Simplify Polygon geoprocessing tool to remove excess vertices. The Contour geoprocess (with a very large contour interval and base contour slightly above the LiDAR-indicated water surface) can be useful in automatically generating a shapefile outline of the reservoir surface.	LiDAR Surface	SimplifiedReservoirOutline.shp
2	Digitize stream centerline through bathymetry. Upstream to downstream.	Historic Topo Map	StreamCenterline.shp
3	Digitize bathymetric contours into lines and assign elevations from contour map as an attribute (use historical topo or estimate from terrain). Additional contour lines can be estimated beyond what is shown in the historic mapping to add resolution for better interpolation.	Historic Topo Map	HistoricTopoContourLines.shp
4	Use Topo to Raster with inputs of the stream centerline (as a Stream), Contours (as Contours, assign the elevation field from attribute tables), and reservoir outline (as Boundary). Select appropriate cell size. Provide a value for the Smallest Z value to be used in interpolation; this should likely be the lowest value of the elevation-storage curve, but it must be lower than the lowest contour from above.	SimplifiedReservoirOutline.shp StreamCenterline.shp HistoricTopoContourLines.shp	HistoricTopoRasterSurface.tif

Note: This first output raster should resemble the topo map and bathymetric surface; however, it does not reflect the published elevation-storage curve. Steps below are used to correct the raster to match the storage curve.

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Table 2. Steps to Correct a Terrain Surface to Match Specified Elevation-Area Data

#	Steps and Process	Input Files	Output Files
5	Use the *Contour* geoprocessing tool to generate contour lines from the first *Topo to Raster* surface. For *contour type*, use the Contour Shell option. Contour at a relatively dense interval; an interval of around 1 foot should be adequate. Smaller contour intervals provide more definition of the terrain surface.	HistoricTopoRasterSurface.tif	HistoricTopo_Regenerated… Contours_1ft.shp
6	Create three fields: *NumParts* (long), *Area_Acres* (double), and *ElevCorrct* (double). Then for each contour polygon feature, calculate the number of parts for each contour.	HistoricTopo_Regenerated… Contours_1ft.shp	Same as left, but with number of parts calculated
7	Examine the number of parts for each contour interval. Use judgment to remove any contours that have a large number of parts. This is also the time to modify any contours to add definition to the bathymetry. Consider removing duplicate contours along the perimeter of the bathymetry.	HistoricTopo_Regenerated… Contours_1ft.shp	Same as left, but with unnecessary polygon parts and contours removed.
8	For each contour polygon feature, calculate the area within the contour (in acres).	HistoricTopo_Regenerated… Contours_1ft.shp	Same as left, but with surface areas calculated
9	For each contour polygon (that shares the same elevation), use the surface area to look up the corresponding elevation from the elevation-area curve. Copy this data into the *ElevCorrct* attribute field.	HistoricTopo_Regenerated… Contours_1ft.shp Elevation-Area curve	Same as left, with elevations from elevation-area curve
10	Use Create TIN geoprocessing tool to generate a surface from the corrected elevation contour data. Inputs are HistoricTopo_RegeneratedContours_1ft.shp as *Hard Line* type, input using *ElevCorrct* as Height field. Also SimplifiedReservoirOutline.shp as *Hard Clip* type with *Elev* as Height Field (assuming reservoir boundary was developed from contour data of equal elevation, if not, assign <None> for the height field).	HistoricTopo_Regenerated… Contours_1ft.shp SimplifiedReservoirOutline.shp	HistoricTopo_Corrected_TIN
11	Use *TIN to Raster* geoprocessing tool to convert TIN to a .tif raster. Use *Floating Point* data type, *Natural Neighbor method*, and *Cell Size* sampling distance with a sampling value (i.e., raster cell size) appropriate for your model.	HistoricTopo_Corrected_TIN	HistoricTopo_CorrectedElev-Storage.tif

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#	Steps and Process	Input Files	Output Files
12	Import corrected bathymetric surface into HEC-RAS and overlay with a storage area that has the same outline as the reservoir. Compute the storage-elevation curve and compare to the published storage curve. Verify that the curve matches the terrain. If not, review the previous steps to identify and resolve the issues.		
13	If elevation-storage curve matches published data, continue to next step. If match is not sufficient, steps 5 through 12 can be iterated using the corrected raster as an input. This should improve the fit with the elevation-volume curve.		
14	Combine into HEC-RAS terrain and model.		

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1.3.6.3 STORAGE APPROXIMATION WITH BASIC STORAGE ATTRIBUTES

In many cases, complete elevation-volume data may not be available for a given reservoir. In these instances, the NID may have some limited information on the reservoir's volume. One attribute of the NID database is Normal Storage. For approximation purposes, it can generally be assumed that the reservoir storage below the LiDAR-indicated water surface elevation in the DEM is equivalent to the NID Normal Storage, unless available data indicates otherwise. This single storage value can be used to estimate and validate a complete elevation-volume curve generated using HEC-RAS terrain modifications. Other terrain indications, discussed in Section 1.3.5.5, can be useful in approximating bathymetric terrain features.

The process involves iteratively creating HEC-RAS terrain modifications to estimate a bathymetric surface until the elevation-volume curve intersects the normal storage volume indicated in the NID at the water surface elevation indicated by the LiDAR terrain data. Figure 9 displays an elevation-volume curve computed in HEC-RAS using this approximation technique as well as the published normal storage point from the NID. Note that in this case the curve compares relatively well to the published value after only a single attempt to estimate bathymetry. The published NID value does indicate slightly more volume than estimated by the terrain (approximately 4 percent additional storage at LiDAR surface elevation of 519.3 feet). In this case, the terrain modifications could be refined to increase volume slightly to achieve better agreement between the terrain and NID volume.

One convenient way to achieve better agreement between the terrain and NID volume is to calculate the vertical difference in elevation between the elevation-volume curve and the NID storage value. In the example shown in Figure 9, the curve is approximately 0.2 feet above the published storage value. To adjust this curve, all the terrain modification control points can have their assigned elevations lowered by 0.2 feet and the curve should then intersect the published storage volume. This vertical adjustment assumes the overall shape and curvature of the elevation-volume data is reasonable.

Figure 9. Example Comparing Approximated Elevation-Volume Curve to Published National Inventory of Dams Normal Storage

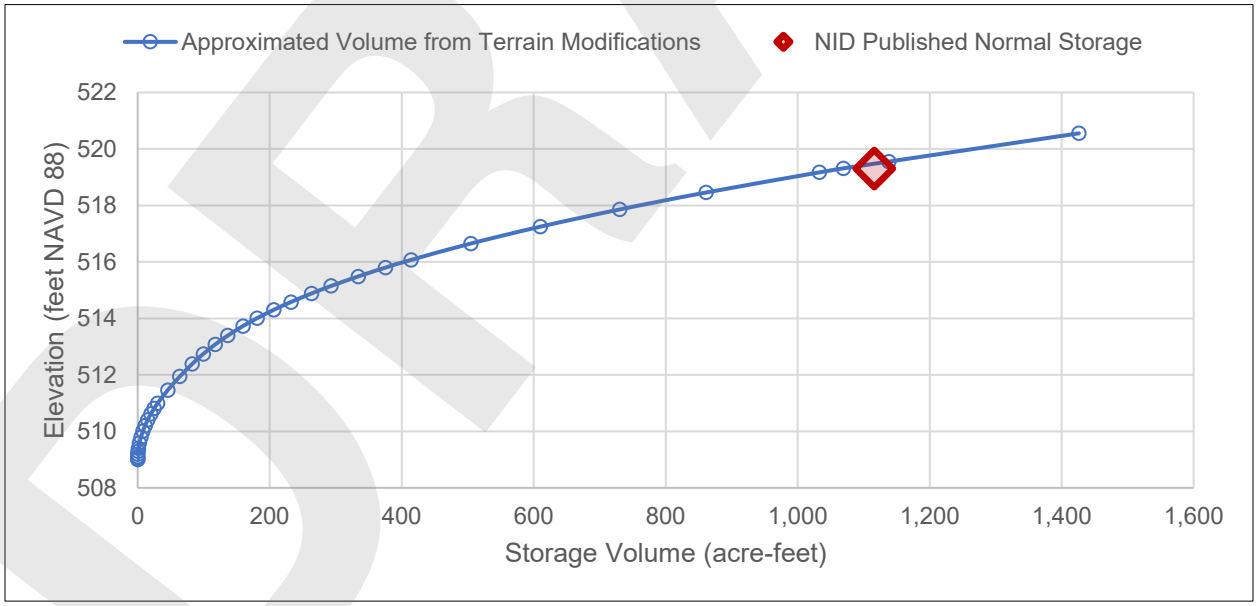
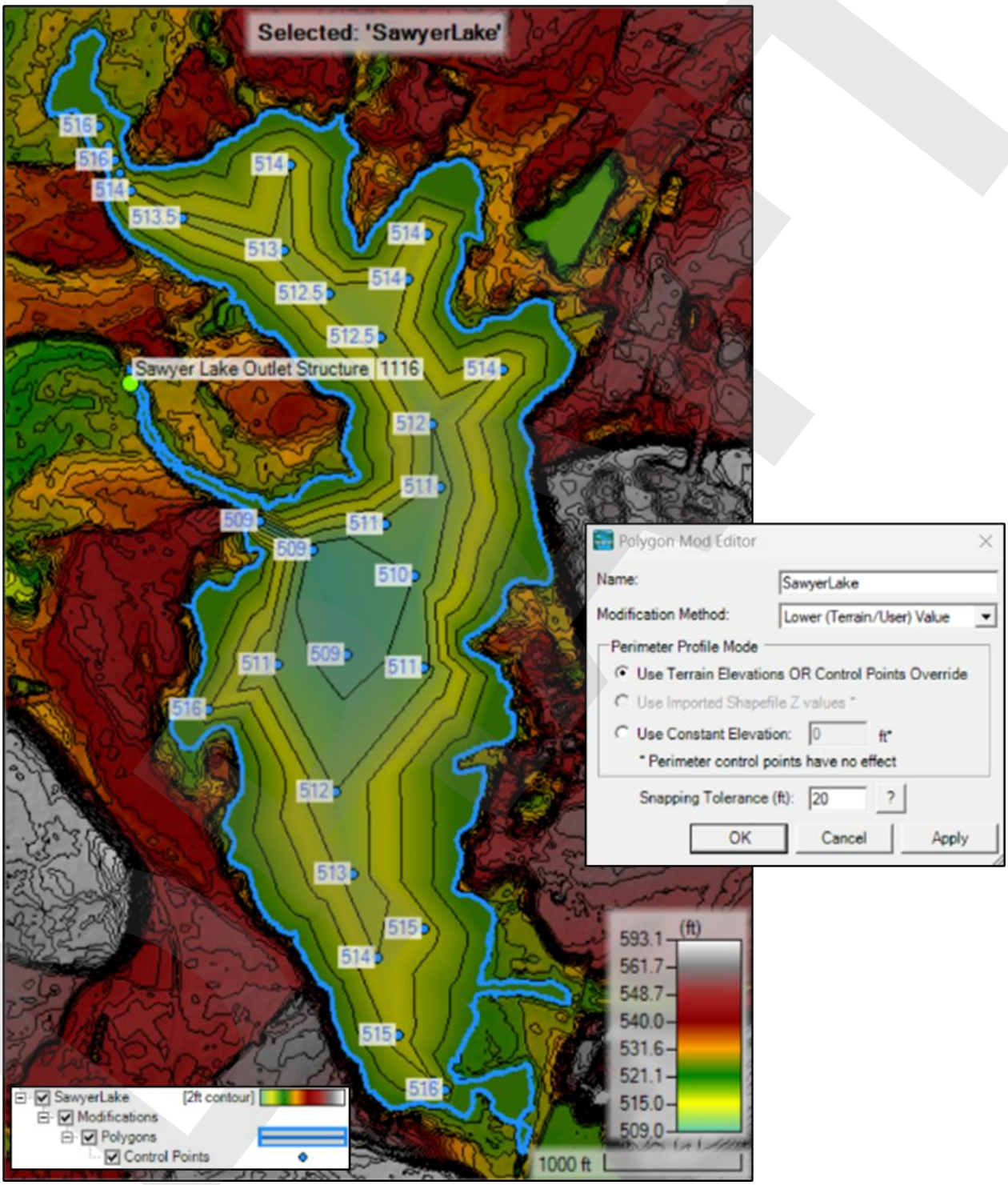


Figure 10 displays the estimated bathymetry and terrain modifications used to create this elevation-volume curve. A multipoint polygon terrain modification was added around the reservoir rim, shown in blue. The Lower (Terrain/User) Value modification method was selected, such that modifications could only reduce terrain elevations and never add to them. The Perimeter Profile Mode was set to Use Terrain Elevations OR Control Points Override option, so that individual control points (shown as blue circles)

could be added to define the lake's bathymetry. Each control point is labeled with the selected lake bottom elevation. Note that this same terrain modification approach is applicable to other approaches to bathymetry estimates discussed above.

Figure 10. Example Terrain Modifications Used to Approximate Bathymetry



1.3.6.4 STORAGE APPROXIMATION WITH NO SUPPORTING DATA

In cases where there is no known volume information on a reservoir, such as when the NID is missing a normal storage attribute, there is still a need to estimate bathymetry if breach analysis will be performed on that reservoir. In that case, the approach utilized in Section 1.3.6.3 should be followed, although there will be no known storage point to validate the bathymetry estimate against. Best judgment should be used to approximate bathymetry in these cases. Refer to Subsection 1.3.5.5 for more information on how features from a LiDAR-based DEM can inform bathymetric estimates.

1.4 RESERVOIR OUTFLOW

Operational releases from reservoirs are a key aspect of model representation. For FFRD models, the method applied for each dam in each of the models (HEC-HMS, HEC-ResSim, and HEC-RAS) will be guided by the selections made in the Dam Scoping Table. Modeling development decisions should always be made with an understanding of the interaction of other models in the FFRD system. The sections below provide additional modeling guidance for the implementation of the selected methods.

1.4.1 HEC-HMS RESERVOIR OUTFLOWS

For dams that are selected to represent a reservoir element in HEC-HMS, the following guidelines, techniques, and troubleshooting tips are offered. Note that the Dam Scoping Table will provide direction for HEC-HMS modeling using Simple Reservoir Operations, Complex Operations, or no inclusion in the HEC-HMS model.

1.4.1.1 UNMODELED RESERVOIRS

To account for the attenuation of flows through reservoirs not modeled in HEC-HMS, HEC-HMS may account for their presence with adjustments to the unit hydrograph parameters, increasing the time of concentration or storage coefficients. This approach can improve historic event calibration in HEC-HMS and improve the accuracy of HEC-HMS output data that passes to HEC-ResSim and HEC-RAS models.

1.4.1.2 SIMPLE RESERVOIR OPERATIONS

Simple reservoir operations include one-to-one storage and discharge relationships, uncontrolled outlets, and ungated spillways. Typically, if the outlets and spillways have simple rating curves not dependent on inflow or seasonality, the dam can be modeled with simple procedures in HEC-HMS. A single outflow curve is the simplest hydraulic routing method available in HEC-HMS. If outlets are gated or controlled, simple operations can still be used by analyzing observed data and estimating an average outflow for a given elevation or storage condition. Guidance on creating a single outflow curve can be found in the *Outflow Curve Option for an HEC-HMS Reservoir Element* guide. Storage-discharge and elevation-storage relationships should be uniform across the HEC-ResSim, HEC-HMS, and HEC-RAS models (to the extent possible).

1.4.1.3 COMPLEX RESERVOIR OPERATIONS

For more complex reservoir operations, the use of HEC-ResSim to model operating schedules, rules, and downstream constraints may be scoped for the FFRD project; however, in some cases rule-based operations in HEC-HMS may be scoped for the project to provide an accurate and more efficient approach. When HEC-HMS rule-based operations are scoped for use as the authoritative reservoir outflow model, the following guidelines should be used.

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1.4.1.3.1 STORAGE OBJECTIVE PATTERNS

Storage patterns can be used based on a published reservoir guide curve or developed based on observed historical elevations converted to storage. Taking the 50 percent exceedance value of a duration analysis of the observed period of record of daily elevations can serve as a guide curve of storage objectives. The relationship should be input to the HEC-HMS model as a Paired Data–Storage Pattern table of calendar days and corresponding volume targets.

1.4.1.3.2 ZONES

Zones should be established to implement rules and store objective patterns that may change throughout the range of elevations. A Paired Data–Daily Elevation Pattern is needed to define the maximum elevation of the zone. This can be a constant elevation, or the zone elevation limits can vary throughout the calendar year.

1.4.1.3.3 RELEASE RULES

Simple minimum and maximum release rules should be utilized first before adding release rate of change rules. Typically, this would include maximum release rules in a Zone to manage downstream capacity constraints or minimum release rules to better match gradual release patterns and smooth out spike releases. Release rate-of-change rules can then be added as needed. It is recommended to test shorter continuous simulations (approximately two water years) to see how the release rules perform and add complexity as needed.

1.4.1.3.4 DOWNSTREAM CONSTRAINTS

Downstream flow constraints are very computationally expensive, and using a downstream constraint rule should be avoided in HEC-HMS. If a downstream flow constraint is prescribed in reservoir operations, this constraint should be managed to the extent possible with a maximum flow release from the reservoir that considers additional downstream flows that will accumulate at the downstream constraint point.

1.4.1.3.5 GATED OUTLETS

Gated outlets should be added as General Outlets. The minimum curve should be set to zero, and the maximum curve will represent the maximum discharge possible or prescribed for a given elevation. The minimum time between adjustments should be sufficiently short so that the reservoir does not bottom out in between adjustments and not so short that the reservoir pool begins to oscillate unstably. It is recommended to start with a minimum adjustment time of 60 minutes and adjust as needed.

1.4.1.4 POTENTIAL MODELING OBSTACLES

Modeling reservoirs in HEC-HMS can be a trial-and-error process; the modeler should start with the simplest modeling method and add complexity as needed to best represent reservoir operations. During that process there are some common pitfalls.

1.4.1.4.1 RESERVOIR BOTTOMING OUT

The modeler should check that the minimum time between adjustments is not too long. If this is set too large, the gate settings may not be able to adjust to the prescribed storage pattern quickly enough. This is especially true for reservoirs with very low storage volumes near the bottom of the pool. Another potential workaround is to set a low minimum release near the bottom of the pool or a constant storage pattern to ensure minimal or no releases occur during times where available storage is very low.

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1.4.1.4.2 LONG RUNTIMES

Avoid using downstream constraints and, to the extent possible, any rules that necessitate a decision between two outlets on where to make releases. This would include adding main and auxiliary outlets and having a minimum or maximum release rule from auxiliary and main outlets.

1.4.1.4.3 ABRUPT SPIKES IN RELEASES OR LACK OF ATTENUATED RELEASES

Spikes in releases can be smoothed by applying a maximum rate of change release rule or by instituting a minimum release rule to allow releases to be more attenuated instead of, for instance, peaking at 1,500 cfs outflows and immediately dropping to 0 cfs outflows once the guide curve is matched.

1.4.2 HEC-RESSIM RESERVOIR OUTFLOWS

For dams that are selected for operations modeling in HEC-ResSim, the FFRD core standards are described in SOP Volume II and the guidelines, techniques, and troubleshooting tips that follow supplement those standards.

1.4.2.1 HEC-RESSIM SCOPING OF APPROACH

In the reservoir operations model, various levels of detail are available depending on the reservoir characteristics. As reservoirs and their operations increase in size and complexity, additional modeling techniques are required. Reservoir modeling decisions are made during the scoping and data preparation phase of the FFRD project. The following topics will be clearly scoped and communicated through a combination of the Dam Scoping Table, Name Table, Model Linking Register, and Scoping and Data Preparation Report.

- HEC-ResSim modeling approach for reservoir routing operations (detailed reservoir operations model, hydrologic model, or not at all)
- HEC-ResSim modeling approach for each unique outflow feature at each dam (e.g., are spillways modeled separately from other outlets?)
- HEC-ResSim modeling approach for overtopping surfaces (dams and ungated spillways)
- Determination of any locations where special treatment or coordination is needed between the hydrologic, hydraulic, and/or reservoir-operations model for consistency in modeled inflow, outflow, and reservoir elevation.

1.4.2.2 HEC-RESSIM MODEL DOMAIN

The HEC-ResSim model domain covers all reservoirs operating based on conditions elsewhere in the watershed. Even if each reservoir is operating independently of the others, a single HEC-ResSim model is preferred; however, in special cases it may be necessary to create several smaller headwater HEC-ResSim models for small reservoirs that cannot be modeled at the same timestep as larger reservoirs downstream in the system due to model instabilities and mass balance issues that occur when small reservoirs use too large a timestep.

1.4.2.3 MODELING OF OVERTOPPING SURFACES

Dam overtopping, including events larger than design storms, must be accounted for in the HEC-ResSim model and in the Model Linking Register.

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1.4.2.4 DIVERSIONS

When included in the scope, HEC-ResSim diversions should be added at major canal and diversion systems. These can either be connected to a downstream location or used to remove water from the system entirely. In heavily irrigated watersheds with many small diversions, they may be lumped to the nearest downstream junction.

1.4.3 HEC-RAS RESERVOIR OUTFLOWS

The guidelines for scoping and selection of hydraulic model treatment for dams is included in Job Aid 01, *Scoping Processes*, and full descriptions of each model treatment type are included in SOP Volume II. Specifics of modeling techniques for each type of model treatment are included in Section 8 of SOP Volume II. Table 3 provides a high-level summary of the treatment types. HEC-RAS techniques and troubleshooting tips for dam outflow representation are also offered below.

Table 3. Hydraulic Model Dam Modeling Treatment Types

Hydraulic Modeling
Type A **Operation Modeled outside of Hydraulic Model** HEC-RAS Element: 2D Connection +Outflow Blocked with high crest and no outlets (outflow sourced from another model or observations) +Outflow timeseries sourced from observed data or simulation from another model +Pool Bathymetry Terrain Modifications +Terrain Modifications for breaching
Type B **Operation Modeled by Hydraulic Model** HEC-RAS Element: 2D Connection +Outflow representing operation procedures for the dam. (i.e., may utilize features for gates, outflow curves, elevation-controlled gates, spillways, overtopping, etc.) +Pool Bathymetry Terrain Modifications
Type C **Simplified Outflows Modeled by Hydraulic Model** HEC-RAS Element: 2D Connection +Approximate representation of outflow features +Pool Bathymetry Terrain Modifications
Type D **Basic Representation in Hydraulic Model** HEC-RAS Element: Breakline +Approximate Low Flow Outlet Terrain Modification (notch)
Type E **Not Modeled** HEC-RAS Element: No modeling element

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1.4.3.1 RESERVOIR ROUTING CONSISTENCY

For dams modeled with the Type A treatment, consistency of the reservoir routing results between HEC-RAS and the model sourced for outflows (HEC-ResSim or HEC-HMS) should be carefully evaluated. The FFRD model approach includes an inherent difference between the modeling of inflows to the reservoir between the models as HEC-RAS routes HEC-HMS-generated excess precipitation and baseflows over a 2D grid while HEC-HMS applies unit hydrograph techniques to transform runoff. While the calibration process promotes close alignment between the model results, differences that persist may result in reservoir routing problems in HEC-RAS.

When the reservoir inflow from HEC-ResSim or HEC-HMS is out of sync with HEC-RAS simulated inflow, the HEC-RAS reservoir pool elevation hydrographs can diverge from the other models. If these differences are large, it is possible that the specified outflows for use in HEC-RAS could cause the reservoir to go dry and result in stability issues. A related issue that can arise is that a breach may be triggered based on the sampled trigger elevation for a stochastic event in the HEC-ResSim model, while the HEC-RAS model may not reach the trigger elevation resulting in an inaccurate routing of the breach outflows.

These pitfalls are avoided by verifying that HEC-HMS and HEC-RAS calibration efforts are coordinated such that flood routing for the historic events is well aligned. Other recommended checks during the conformance phase of the project include a comparison of simulated minimum pool and peak pool elevations between the models and spot check comparisons of event-level pool elevation hydrographs.

1.5 DAM BREACH REPRESENTATION

The workflows and other details regarding the dam breach approach are included in SOP Volume II and Job Aid 10, Dam System Response Development. However, the handling of reservoir breaching in HEC-ResSim for stochastic events requires special consideration, and an understanding of the interactions with the other models in the system is needed and described here. For dams modeled as Type A in the hydraulic model, the release of water through the breach is calculated by the hydraulic model based on the dam breach parameters (size and timing of breach opening); however, a separate specified outflow boundary condition is also directly linked to the simulated HEC-ResSim releases. If not addressed, the HEC-RAS model could release water based on breach outflow and large HEC-ResSim operational releases that do not account for the falling pool caused by the breach.

To address the issue, HEC-ResSim will breach by setting outflows to equal inflows when a pool above the trigger elevation is reached. The breach trigger elevation is sampled for each event and shared between HEC-ResSim and HEC-RAS. In the HEC-RAS model during that same event, the breach will be triggered at the same instant based on the HEC-RAS routed pool elevation and release a larger volume of water through the breach. The actual outflow in the HEC-RAS model is the sum of the HEC-ResSim releases and the HEC-RAS-computed breach flows.

There are a few imperfections in this method:

- HEC-RAS and HEC-ResSim Simulated Pool Hydrographs may not track perfectly due to the timing of inflows. HEC-ResSim receives inflows routed via HEC-HMS, while HEC-RAS simulates 2D flow over the mesh using HEC-HMS excess precipitation and baseflow outputs.
    - If pool hydrographs do not track well together, breaches will not trigger at the same time.
- Breaches at upstream dams may trigger breaches at downstream dams. HEC-ResSim will not be able to account for this fact since the large volume in a reservoir is not actually released after a breach.
- After a breach occurs, HEC-RAS may release too much flow, especially if inflows are high at the time of breach.

- It is possible HEC-RAS numerical stability issues will occur if HEC-ResSim inflows are forced out of a reservoir that is dry after a breach has released all water.

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JOB AID 9—LEVEE PHYSICAL REPRESENTATION

This job aid provides guidance for representation of physical levee structures within FFRD hydraulic models.

This job aid also discusses considerations related to levee breaches. The levee data lead also provides draft breach locations prior to the start of the calibration phase. Since HEC-RAS only allows a single breach per storage area/2D connection, the breach locations will impact the segmentation of levees that have more than one breach scoped.

1.1 LEVEE ALIGNMENT AND PROFILE DATA

The following subsections discuss levee alignment and profile data in the FFRD program.

1.1.1 NATIONAL LEVEE DATA REFINEMENT EFFORT

As described in Job Aid 01 and Job Aid 02, the National Levee Database (NLD) data is the official source for FFRD levee modeling. The NLD data for the study basin will undergo a data refinement effort prior to or as a part of the scoping and data preparation phase, to ensure the dataset is adequate for direct use in FFRD modeling. This NLD data refinement effort is described by reference documents stored directly in the NLD.

1.1.2 SOURCES OF LEVEE DATA

As stated previously, the NLD provides a near-comprehensive list of levee structures to be included in the hydraulic model and, in many cases, the NLD will be the best available data source. However, there are cases where NLD data may be missing, incomplete, outdated, or otherwise incorrect. Levees are unique and dynamic structures that can change or be modified over time. Levee raises, setbacks, erosion, or other modifications are examples where the NLD may not reflect the best understanding of the levee structure. For this reason, it is important for the hydraulic modeler to verify the accuracy of levee data. This verification can be done through discussion with the local USACE district, through comparison with historic and current aerial imagery, and through comparison with other available data. Other potential sources of levee data include risk assessments, district modeling and surveys, as-built drawings, and digital elevation models (DEMs) of the terrain, as well as other sources. If a discrepancy is found between NLD data and other sources, modeler judgment and discussion with the technical team lead may be needed to determine the best source of levee data.

One data source that is widely available for comparison with NLD data is a high-resolution DEM of the terrain. This terrain data can be used for verification of NLD crest profile data; however, it may not accurately depict specific portions of the levee, such as floodwall sections or closures. If issues are found with the NLD data and a decision is made to use terrain elevations to define the structure crest, it may be necessary to modify the NLD structure centerline to correspond with the high ground indicated by the terrain. Often the NLD centerline will not perfectly align with the terrain's high ground, and maintaining the NLD centerline while recutting the levee profile's crest data from the terrain can underestimate the true levee crest elevation.

1.1.3 COMMUNICATION WITH LEVEE DATA LEAD

If NLD data is not adequate and the modeling team finds an alternative improved data source, it shall be annotated in the calibration phase report in the levee system-by-system table. It will also be communicated to the levee data lead for possible NLD data refinement.

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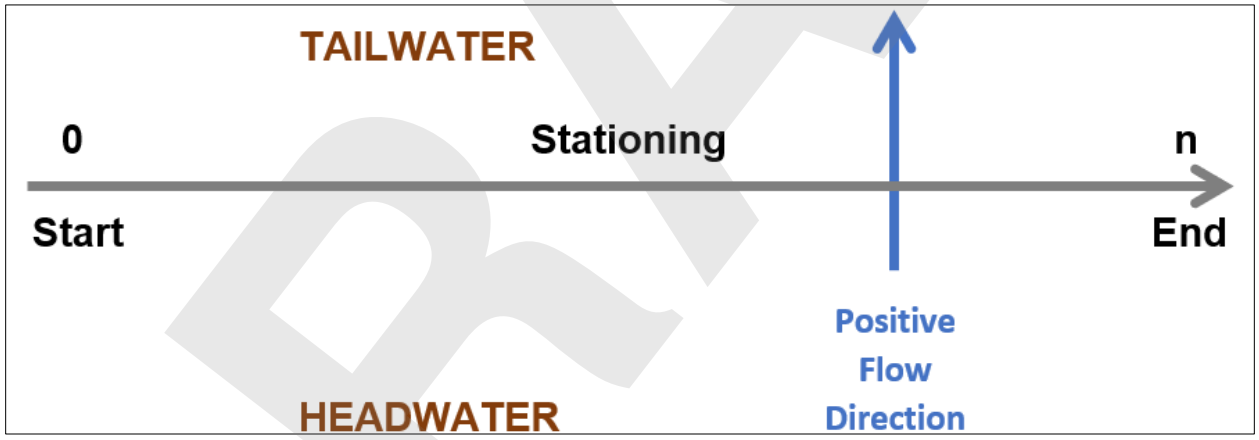
1.1.4 DIRECT USE OF NATIONAL LEVEE DATABASE PROFILE AND ALIGNMENT

Levee data should be adopted from the best available data source, which in many cases will be the NLD. Levee centerline data is available as shapefiles through the NLD, and this alignment should be adopted unless better alignment data is available. This NLD alignment may be drawn upstream-to-downstream or downstream-to-upstream. Unless there is a compelling reason to reverse this directionality, the direction of the NLD structure should be maintained. If structure direction is reversed, this and its rationale should be documented in both the hydraulic model report and the HEC-RAS Connection Data Editor description box.

The levee centerline data is provided by NLD for the full levee system. This centerline shapefile can be split at levee segments so that each segment is represented by its own connection within the hydraulic model. It is generally best practice to create a small gap between storage area/2D connections that would otherwise touch. This small gap (on the order of 1 foot wide) improves 2D face meshing and can improve stability in some circumstances. When these gaps are added between structure segments, it may also be necessary to add a corresponding gap to the structure crest profile data, as discussed below. An additional technique, which can improve 2D cell meshing, is to underlay a 2D area breakline beneath the storage area/2D connection structures representing a levee system. This breakline can particularly improve meshing in instances where it spans small gaps that can be placed between levee segments.

Note that depending on the direction the levee structure is drawn, and the side of the river the levee sits upon, the flow over the structure may be computed as either positive or negative. **Error! Reference source not found.** illustrates how HEC-RAS determines flow to be either positive or negative depending on the directionality of a levee structure.

Figure 1. Positive/Negative Flow Convention for Levee Structure Connections



In addition to levee centerline data, levee crest profile data is also available for many levees from the NLD. This data is accessed through the levee system's Profile tab and can be downloaded as a comma-separated values (.csv) file. These crest profile .csv files are typically formatted as x-coordinate, y-coordinate, distance (also known as stationing or m-value), and elevation (z-value). The distance (station, m) and elevation (z) values are of interest for defining the levee crest profile in HEC-RAS. By default, the stationing provided in the NLD .csv file should correspond to the direction of the centerline shapefile discussed previously. Once a connection structure is created in the HEC-RAS model, tabular XY coordinates and station-elevation data can be entered in the HEC-RAS weir/embankment data editor. An appropriate weir crest width and discharge coefficient should be entered in this editor; refer to the HEC-RAS User Manual for additional guidance on selecting weir discharge coefficients. If a structure's centerline data was modified (e.g., reversed or split), a corresponding modification should be made to the profile crest station-elevation data.

1.1.5 LEVEE MODEL CONFIGURATION DETAILS

Levee data should be adopted from the best available data source, which in many cases will be the NLD. Levees should typically be modeled as internal storage area/2D connections. Refer to Job Aid 06, Standard Naming Conventions, for guidance regarding naming conventions for levee structures. The HEC-RAS connection data editor includes a text description box, which should be utilized by the modeler to describe the source of levee data and any assumptions made regarding the levee structure. This text description should include the source of the levee centerline alignment, NLD ID number, NLD system and segment names, the source of crest profile data, and any other pertinent data or assumptions.

Levee structures should typically be incorporated into the HEC-RAS model on a segment-by-segment basis; this is especially true of long levee systems. Shorter hydraulic connections are more conducive to evaluation and troubleshooting of hydraulic model results compared with long connections. HEC-RAS is limited to a single breach location per connection, and therefore shorter segment-by-segment connections allow for potential breaching of each segment and multiple breaches in a multi-segment levee system.

1.1.5.1 OVERFLOW COMPUTATION METHOD

HEC-RAS prompts the hydraulic modeler to select the *overflow computational method* to calculate discharge over the structure, with the options of either the *Normal 2D Equation Domain* or *Use Weir Equation*.

2D equations more accurately calculate overflow for levees that do not have significant height above the surrounding terrain, are subject to full submergence during floods, and will generally be more stable than the weir equations but may overestimate overtopping discharge when there is a significant head difference over the levee structure. Weir equations better represent tall levees where the levee is expected to act as a non-submerged weir during flood conditions.

Weir equations should be used for FFRD models. 2D equations are not recommended for FFRD models because they are not fully compatible with the simplified physical breach method.

1.1.5.2 REFERENCE POINTS AND LEVEES

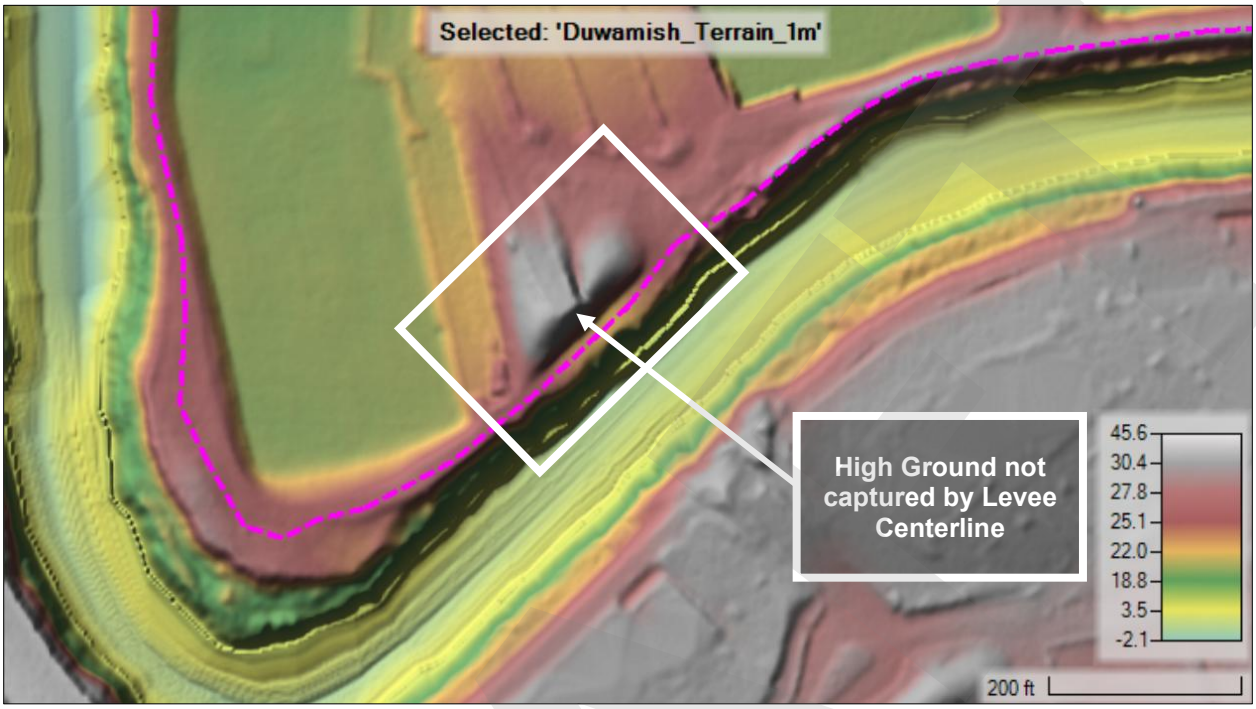
Reference points are required near levees to ensure adequate water surface elevation information is written out by the model in the vicinity of levees. The requirements for these saved output locations including reference points near levees are described in Section 8.3 of SOP Volume II. There are also specific naming convention requirements for these reference points defined within Job Aid 06, *Standard Naming Conventions*.

1.1.5.3 HIGH GROUND

In some cases, the levee centerline provided by the NLD may not capture high ground that is not technically part of the levee system but effectively acts as part of the levee structure. **Error! Reference source not found.** displays an example where the levee centerline, shown in pink, does not capture high ground created by a bridge abutment. Levee systems should be inspected for instances such as these, and if there is adjacent high ground that acts as a water-impounding surface, it should be included in the hydraulic model as part of the levee structure or as a separate breakline.

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Figure 2. Example of High Ground Adjacent to Levee Centerline



1.1.5.4 CLOSURES

Many levee systems include closure structures that are not permanently installed but are closed during flooding to prevent floodwater from intruding through railroad crossings or other features. These features are intentionally excluded from the crest profile data in the NLD but should be included in hydraulic models as part of the levee crest. Locations of closure structures are available within the NLD as part of the structure's Features tab. Without more detailed information, the crest of these closures should be interpolated from surrounding crest profile data.

1.1.5.5 INTERIOR DRAINAGE FEATURES

Levee systems frequently include interior drainage features such as gravity drain culverts, pump stations, gated pipes, and other features. Information on these features can be gathered from the NLD or other available sources. These features should be included in the hydraulic model with the best available data or approximated when detailed information is not readily available. Interior drainage culverts often utilize flap gates to prevent backflow from the river to the interior side of the levee. The flap-gate feature in HEC-RAS can be used to approximate these culverts' behavior and the diagram shown in **Error! Reference source not found.** may be useful for determining whether the *no positive flow* or *no negative flow* flap gate option should be used. Pump stations can also significantly impact inundation near levees, and where information is available, these pumps should be incorporated into the modeling. Detailed information regarding the pump stations may be unavailable, and in these cases, engineering judgment should be used to best approximate the impact of the pump station. Terrain modifications or other modeling techniques may be required to approximate some of the underground pipe networks that feed interior drainage systems.

1.2 LEVEE BREACH DATA

Levee breaching will be performed using the hydraulic model, and levees should be incorporated into the modeling with this use-case in mind. If breach locations are known, additional 2D cell refinement may be

required in areas where breach floodwaters are expected. Mesh refinement behind levees should take care to balance model runtime objectives with mesh resolution and detail.

Additional details regarding breach location, breach parameter, and system response curve selection are covered in Job Aid 11, Levee System Response Development, and SOP Volume II.

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JOB AID 10—DAM SYSTEM RESPONSE DEVELOPMENT

This job aid describes the procedures for collecting dam performance data, scoping of the dam breach modeling methodology, producing system response inputs, and communication with the team to incorporate dam breaches into the modeling system.

1.1 CONTEXT

To incorporate dam performance into FFRD, dams where the evaluation of breaches may have a significant impact on flood risk characterization must be identified, a system response curve must be estimated, and a breach scenario must be incorporated into the hydraulic and hydrologic models and cloud-compute inputs.

1.2 ROLES

The following key roles are required to incorporate dam breaches into an FFRD project:

1.2.1 DAM DATA LEAD

The dam data lead is the USACE expert responsible for the development and delivery of system response probability (SRP) curves for dams. They also contribute to dam database refinement efforts and the scoping of dams for breach assessment in the FFRD model. Additionally, they review and evaluate breach results from stochastic model simulations.

1.2.2 HYDRAULIC MODELER

The hydraulic modeler is responsible for developing the model that fully accommodates dam breaches as scoped.

1.2.3 RESERVOIR OPERATIONS MODELER

The reservoir operations modeler is responsible for ensuring that the operations model accounts for breach events that may occur for reservoirs included in the reservoir operations model by adjusting to appropriate post-breach operations releases.

1.2.4 CLOUD COMPUTE LEAD

The cloud compute lead incorporates hydraulic and hydrologic models and SRP relationships into cloud compute for the project and ensures that results are shared with the team for review.

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1.3 PROCEDURAL OVERVIEW

The following key tasks are performed to establish the required model elements, breach locations, and system response curves for FFRD dam breach modeling:

1. Prior to an FFRD Study: An effort to systematically refine and improve the quality of the National Inventory of Dams (NID) database within the basin is performed prior to the FFRD project. Guidelines for these separate efforts are under development.
2. Scoping and Data Preparation Phase:

    a. Confirm that NID data are adequate to support scoping of the model approach for dams, including the selection of a subset of dams to consider for breaching. FFRD project-specific assumptions may be necessary to support the study and will be documented in FFRD reporting and available as feedback to the NID data managers.
    b. Select dams within the study that will be considered for breach assessment and document the selection within the Dam Scoping Table. Criteria for selection of dams to breach are included in Job Aid 01, *Scoping Processes*.
    c. Consolidate all existing risk assessment data for dams scoped for breach.

3. Calibration Phase:

    a. Develop SRP curves for each dam included in the scope for breach. Development of dam SRP curves for the basin may include a combination of the sources as described in sections below.
    b. Establish crest elevation and toe elevation for each breach to be applied in SRP curves and hydraulic modeling.
    c. Upload SRP curve data to the Model Library in established directory location and format for incorporation into cloud compute.
    d. Develop hydraulic and reservoir operations models to accommodate sampled breach elevations during the conformance phase. This includes aspects of HEC-RAS modeling and HEC-ResSim modeling.

4. Conformance Phase: Review and evaluate resulting sampled breach elevations and number of breach events that occur within the single-realization conformance phase computation. Address any issues.
5. Production Phase: Review and evaluate breach results from multi-realization event computations.
6. Final Reporting Phase: Document breach inputs and results. Note that interim reporting is integrated into each FFRD project phase, and the end-of-phase reports will provide much of the technical documentation of dam breach inputs and results.

1.4 DAM BREACH SITE SELECTION

Dam breaches for FFRD should be consolidated into a single representative breach location, which is generally considered adequate for flood risk characterization at the basin scale.

Unique dam configurations or other circumstances may warrant a more complex treatment of certain dams, which must be coordinated with the full project team.

1.5 INTERVENTION

Intervention, sometimes referred to as flood fighting, will be accounted for in the system response probabilities, resulting in lower probabilities of breach.

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1.6 CONSOLIDATION OF EXISTING RISK ASSESSMENT INFORMATION

During the scoping and data preparation phase of an FFRD project, the dam data lead is responsible for collecting all existing dam risk assessment information.

1.7 SYSTEM RESPONSE PROBABILITY CURVE DEVELOPMENT FOR DAMS

An SRP curve will be provided for all dams scoped for inclusion in the FFRD study. It is recognized that relatively limited data exist to support SRP estimates. The sections below provide guidance for SRP development based on type of input data available.

1.7.1 DAMS WITH EXISTING QUANTITATIVE RISK ASSESSMENT STUDY

Dams with a current existing Quantitative Risk Assessment (QRA) will leverage the performance data directly. FFRD models typically include a single representative breach location and SRP for all potential failure modes, so consolidation of the existing data may be necessary.

1.7.2 DAMS WITH EXISTING SEMI-QUANTITATIVE RISK ASSESSMENT STUDY

Dams with a current existing Semi-Quantitative Risk Assessment (SQRA) will leverage the performance data directly. FFRD models typically include a single representative breach location and SRP for all potential failure modes, so consolidation of the existing data may be necessary.

1.7.3 DAMS WITH OTHER EXISTING RISK ASSESSMENT OR PERFORMANCE INFORMATION

Risk assessments or performance evaluations other than those described in the sections above should be evaluated on a case-by-case basis for the relevance of the existing data for FFRD modeling. When applicable, this existing information may be leveraged for FFRD SRP development.

1.7.4 DAMS WITH EXISTING DAM SCREENING TOOL RISK SCREENING

Performance assessments from finalized and approved Dam Screening Tool (DST) based risk screenings should be leveraged directly for FFRD SRP curves. FFRD models typically include a single representative breach location and SRP curve for all potential failure modes, so consolidation of the existing data may be necessary.

1.7.5 DAMS WITH NO EXISTING STUDY

Most dams in the United States do not currently have existing risk assessment studies to support SRP curve development for FFRD purposes. Additionally, no generalized, widely accepted default approach for approximating the performance of dams exists. For these dams, the two options described below will be applied at the judgment of the dam data lead for the FFRD project in coordination with the technical lead and project manager for the study. The time and cost to leverage more sophisticated approaches should be considered and weighed against other constraints.

1.7.5.1 RAPID SYSTEM RESPONSE PROBABILITY ASSESSMENT WITH DAM SCREENING TOOL APPROACHES

A rapid SRP assessment leveraging existing physical dam data based on methods documented within the DST may be applied for FFRD use. When applied, the dam data lead is responsible for reviewing and vetting the results.

1.7.5.2 SIMPLISTIC SYSTEM RESPONSE PROBABILITY APPROACH

A simplistic SRP based on a step function primarily driven by overtopping of the dam may be applied for FFRD use. This approach typically would include a 0 percent probability of breach for all loadings below an overtopping depth threshold (such as 2 feet) and a 100 percent probability of breach for any loading over the selected threshold.

1.8 HYDRAULIC MODEL BREACH PARAMETERS

Guidance for selection of hydraulic model breach parameters for FFRD is included in SOP Volume II, Section 8.

1.9 REVIEW AND EVALUATION OF BREACH RESULTS

A single realization of stochastic events is simulated by cloud computing during the conformance phase. Breach results from the conformance phase must be reviewed by the dam data lead to confirm that breach trigger elevations are being sampled appropriately and the number of breach events that occur at each breach location falls within the expected range. Similarly, the multi-realization results from the production phase must be reviewed and evaluated.

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JOB AID 11—LEVEE SYSTEM RESPONSE DEVELOPMENT

This job aid describes the procedures to collect levee performance data, scope the levee breach modeling methodology, produce system response inputs, and communicate with the project team to incorporate levee breaches into the modeling system.

1.1 CONTEXT

To incorporate levee performance into FFRD, one or more breach locations must be identified, system response probability (SRP) curve(s) must be estimated, and a breach scenario must be incorporated into the hydraulic and hydrologic models and cloud-compute inputs.

1.2 ROLES

The following key roles are required to incorporate levee breaches into an FFRD project.

1.2.1 LEVEE DATA LEAD

The levee data lead is the USACE expert responsible for the development and delivery of SRP curves for levees. They also contribute to levee database refinement efforts. During the scoping and data preparation phase, the levee data lead assembles existing levee risk assessment data, determines the breach quantity for each levee system, and creates the Levee Breach Locations file, including draft levee control locations (LCLs). During the calibration phase, they coordinate with the hydraulic modelers to verify and finalize breach locations and LCLs. Additionally, they review and evaluate breach results from stochastic model simulations and prepare the levee sections of FFRD reports.

1.2.2 HYDRAULIC MODELER

The hydraulic modeler is responsible for developing the model that fully accommodates levee breaches as scoped. Additionally, the hydraulic modeler confirms the levee control location (LCL) (initial overtopping locations) for each levee system based on modeled water surface profiles.

1.2.3 CLOUD COMPUTE LEAD

The cloud compute lead incorporates hydraulic and hydrologic models and SRP relationships into cloud compute for the project and ensures that results are shared with the team for review.

1.3 PROCEDURAL OVERVIEW

Establishing levee two-dimensional (2D) structures, breach locations, and SRP curves for FFRD modeling has the following main steps:

1. Prior to FFRD Study: Refine and improve the quality of the National Levee Database (NLD) within the basin prior to the FFRD project. Guidelines for these separate efforts are not included in this SOP. This effort includes improvement to the levee alignments, features, profiles, and leveed areas.
2. Scoping and Data Preparation Phase:

    a. Identify which levee systems/segments are included in the study area and what levee

data is available.
   b. Consolidate and evaluate existing levee risk assessment data
   c. Select the number of breach locations for each levee system
   d. Identify and document preliminary LCL breach locations
   e. Identify levee breach quantity and location(s) per guidance. These data are needed when the calibration phase starts. Note that the breach location—particularly the LCL—may be adjusted during the calibration phase.

3. Calibration Phase:

   a. Verify and finalize breach locations, including assessment of LCL locations with the support of hydraulic model results
   b. Define levee breach sampling scheme
   c. Develop SRP curves for each breach location and upload to project directory

4. Conformance Phase: Review and evaluate resulting sampled breach elevations and the number of breach events that occur within the single-realization conformance phase computation. Address any issues.
5. Production Phase: Review and evaluate breach results from multi-realization event computations.
6. Final Reporting Phase: Document breach inputs and results. Note that interim reporting is integrated into each FFRD project phase, and the end-of-phase reports will provide much of the technical documentation of levee breach inputs and results.

1.4 IDENTIFYING LEVEES IN THE STUDY AREA

The NLD will be used as the authoritative listing of levee systems in the study basin, and every system will include at least one breach location for the FFRD project. NLD data refinement efforts that occur prior to the start of the FFRD project will ensure that the levee inventory is of high quality for project use. The Levee Data Tracking Table will provide insight on the previously completed data refinement activities.

1.5 LEVEE BREACH QUANTITY AND LOCATION SELECTION

Breach location selection and quantity are a critically important but challenging aspect of levee modeling. The location and number of breaches impacts the potential inundation area in the FFRD results.

The levee data lead determines the breach quantity. The breach quantity will be set by a combination of many factors including levee mileage, number of levee segments, and risk assessment data availability. The breach quantity can be adjusted based on engineering judgment. The breach location will follow a general hierarchy for most critical areas to define but will also rely on engineering judgment in some cases. Documentation of applied judgment is required.

The following subsections outline the procedures. Selecting breach quantity and location can be expected to be iterative with the HEC-RAS modeling team, particularly when identifying an LCL, which is the location on the levee system expected to experience overtopping first. The initial breach locations are needed by the hydraulic modelers ahead of the calibration phase. The calibration model runs may yield a different understanding of the LCL location that will need to be adjusted.

1.5.1 LEVEE BREACH QUANTITY

Breach quantity is a balance of the level of detail required to display a representative picture of flooding and the inundation characteristics of the levee. Long levee reaches, levee systems with steep terrain, and/or short hydrograph floods present a theoretical need to increase the number of breach locations to provide a more comprehensive breach inundation. For example, a breach location at the downstream end of a long levee system will not cause inundation in the upper reaches of the leveed area. If available

information indicates this is the most likely scenario for levee performance to contribute to the system, it is acceptable. In the absence of risk assessment data, there is uncertainty in making the selection.

When using levee risk assessment information, the breach quantity will be scaled to accurately represent what is known about the levee and not mischaracterize levee performance by over-manipulating the input data. For example, combining multiple system responses from different Potential Failure Modes (PFMs) and locating them at an LCL or frequently loaded location will increase the annual probability more than intended. Similarly, arbitrarily splitting input data to apply to a larger number of breach locations may result in levee performance mischaracterization by the modeler and increase computation demands. Best practice is to add breach locations for the most significant contributors to the expected annual probability of failure; essentially where the levee was expected to have performance concerns. This approach will best yield results consistent with how risk assessors intended. PFMs with a lower probability of occurrence compared to the governing PFMs are not expected to trigger in the modeling and contribute to FFRD products. Levee-specific performance information of levee features including earthen embankments and floodwalls can be applied directly. The quantity of breaches will vary but typically will have a breach location for overtopping and at least one breach per feature (earthen embankment, floodwall, etc.). This approach increases the likelihood that FFRD results will be aligned with published risk assessment results.

When more detailed risk assessment information is not available, a minimum number of breaches should be implemented based on levee length to improve chances of characterizing inundation area from levee breaches.

Table 1 provides the recommended minimum number of levee breaches per levee length. The breach quantity can be adjusted based on engineering judgmentTable . The actual number of breaches may increase based on the underlying risk assessment data supporting the evaluation. When siting multiple breach locations, breach locations should be spaced somewhat evenly and prioritize another breach location in the upstream increment of the levee system. For example, if a levee is recommended to have two breach locations and the LCL is in the lower half of the levee, a representative, prior-to-overtopping location should be sought in the upper half of the levee system. Engineering judgment is required (See Section 1.5.2).

Table 1. Recommended Minimum Number of Breaches for Future of Flood Risk Data Modeling

Levee Length	Minimum Number of Breaches
<4 miles	1
4-8 miles	2
>8 miles	3

1.5.2 LEVEE BREACH LOCATION

Once the target quantity of breaches per levee system is determined, the breaches can be located. Breach location selection is generally based on current USACE Mapping, Modeling, and Consequences (MMC) Production Center's Technical Manual for Levees, which is available through the FFRD Templates and Resources GitHub page. Locations are identified using the following prioritized considerations:

1. Breach location at the incipient overtopping location, also known as the LCL, as understood at the time of model setup including initial without-breach model runs. Note that this will likely require hydraulic modeling team input from initial calibration runs and be finalized iteratively.
2. Known areas of poor expected performance based upon risk assessment results.
3. Locations of maximum levee height. Do not select areas with infrequent loading or those significantly lower than typical overtopping (OT) annual exceedance probability (AEP). Refer to loading function, as available. Taller levee sections have more potential susceptibility to various PFMs and would yield significant breach inflows.

4. Location in the upstream third of the levee system.
5. High population or structure density areas adjacent to the levee.

While this list is provided in priority order, combinations of factors may present a stronger case than a rigid hierarchy. For example, a representative breach location adjacent to a population center in the upstream third of the levee may be considered a more critical location than a maximum height levee at the downstream end of the levee. FFRD steering committee guidance stated that FFRD flood hazard data is as important for land use planning as it is for current structures; therefore, locating breaches to inform flooding to populated areas is a minor consideration overshadowed by the need to model most likely overtopping and breach locations. Engineering judgment is required. Location selection rationale must be documented.

1.5.2.1 MULTIPLE BREACH LOCATIONS FOR A LEVEE SYSTEM

The SRP curves should be representative of each unique breach location identified. For example, a single breach location at the LCL will have a simplified assumption that all critical prior-to-overtopping performance is combined and applies at the same location most likely to overtop. For levee systems with multiple breaches assigned, breaches will be assigned performance data that reflects the reason for selecting the location. For example, a levee system that has an LCL and two likely breach locations will have the breach locations represent the system response for those failure modes and the LCL will represent the remainder of the performance evaluated during the risk assessment.

1.5.2.2 BREACH LOCATIONS FROM RISK ASSESSMENTS

USACE risk assessments help inform levee performance during loading. The assessment may or may not document a likely breach location. Further, establishing breach locations from USACE risk assessment reports often requires manual searching through reports including Quantitative Risk Assessment (QRA), Semi-Quantitative Risk Assessment (SQRA), and Levee Screening Tool (LST) 1.0 risk assessments. Table 2

Table 2provides information on risk assessment breach location sources and their characteristics.

Table 2. Breach Location Sources

Risk Assessment Type	Breach Location Specified	In a Database
QRA	Yes	No
SQRA	Yes	No
LST (Current)	Yes	Yes
LST 1.0	Sometimes	No

The use of risk assessment information for locating breach locations is not without limitations. The information on likely breach locations contained in risk assessments may become out of date as levee improvements are made, or incipient overtopping locations may not match FFRD modeled locations. The former is a limitation generally accepted following coordination with the local USACE district. The latter should be adjusted iteratively as hydraulic modeling advances.

1.5.2.3 BREACH LOCATION SELECTION FOR LEVEES WITH LEVEE SCREENING TOOL 1.0 DATA

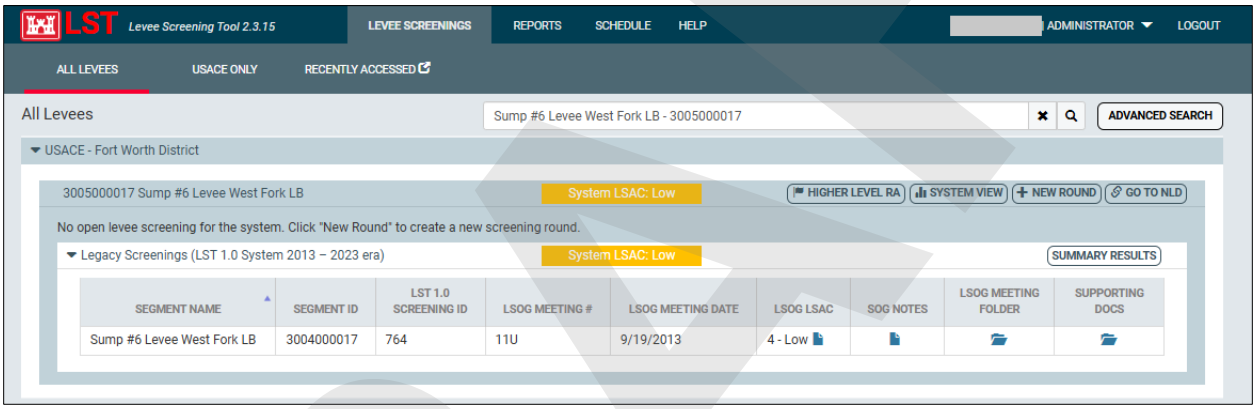
Screening-level risk assessment data is available in the LST. LST 1.0 data is available from the Legacy Screening round (Figure 1). The user is seeking information that will inform a breach location(s), and this

can be done by focusing on what scenario drove risk (overtopping or prior-to-overtopping breach). Reference the annual probability of breach field and the AEP top field for a comparison.

If overtopping is more likely, the LCL should be selected for single-breach levee systems. Information on the LCL may be discussed in the LST print-all found in the Levee Senior Oversight Group (LSOG) meeting folder. Additional supporting documents, including hydraulic and hydrologic memorandums or water surface profile versus crest comparison data may be available. Not all screenings will specifically identify an LCL; in those cases, tentatively select a localized low point to be validated by HEC-RAS modeler runs.

If the prior-to-overtopping condition is more likely, evaluate which performance mode was assessed as most likely to occur and locate the justification language in the print-all or PowerPoint presentation. Seek references to a specific location that was the primary concern during the risk assessment. Not all screenings will specifically identify a performance weak point; in those cases, tentatively select a breach location following general hierarchy guidance.

Figure 1. Location of Levee Screening Tool 1.0 Legacy Data in the Levee Screening Tool



1.5.2.4 TEAM COMMUNICATION

Selected levee breach locations for the project are stored in a *Levee Breach Locations* geospatial file. This file is established during the scoping and data preparation phase of the project and adjusted as necessary through the subsequent phases of the project. This file documents breach locations, names, key attributes, and rationale for the selection of each location.

A template for this Levee Breach Locations file is included in FFRD Templates and Resources.

The levee data lead is responsible for creation of this file during the scoping and data preparation phase, and the file will be available to the hydraulic modelers at the start of model development activities during the calibration phase. The hydraulic modelers must collaborate with the levee data lead to make necessary adjustments.

The team may also use a spreadsheet to manage interim communication of breach locations and attributes to gain a consistent understanding of breach elevations, selection criteria, and scenario-specific details.

Table 3 displays the recommended minimum attributes used for breach location documentation.

Table 3. Levee Breach Fields Used in Spreadsheet and Attribute Tables to Document Approach

Levee Breach Field	Description
FID	Feature ID in the geospatial shapefile
NLD System - Levee Name	Levee System Name in the National Levee Database
NLDID	Levee System ID in the National Levee Database
Length in Miles	Levee system length in miles. Used in identifying a target breach quantity for FFRD modeling.
Number of Segments	Number of segments in the system. A segment is a subcomponent of the levee system that in most cases designates a portion owned by a separate entity.
NLD - Segment Name	Levee Segment Name in the National Levee Database
Tier	Identifies the risk assessment data tier (1–4 integer)
SRP Source	Identifies the highest fidelity risk assessment type available to develop system response curves (QRA, SQRA, LST, LST 1.0, LLID)
Location Description & Rationale	User entered justification on breach quantity and location selection. In particular, documenting judgment applied in selecting the locations.
Location Summary	Short descriptor on the breach location selection criteria. (e.g., LCL, erosion, or Upstream 1/3).
Feature	Identify the levee feature at which the breach is located (embankment, floodwall, closure). This field informs the breach parameters..
FFRD Hydraulic Model Unit	Used to track which HEC-RAS modeling domain for each breach location.
FFRD Hydraulic Model Unit Lead	Used to identify which HEC-RAS modeling team to contact for information.
Location Short Name	Optional field for documenting team nomenclature for breach location.
Long	Longitude of breach location (decimal degrees, GCS North American 1983 datum)
Lat	Latitude of breach location (decimal degrees, GCS North American 1983 datum)
X (ft)	X coordinate of breach location (feet)
Y (ft)	Y coordinate of breach location (feet)
RAS 2D Connection Name	The RAS 2D connection name assigned. A key identifier used across the HEC-RAS model, elevation trigger, and coded results output data.
HW or TW	Identifies whether the RAS headwater or tailwater side corresponds to the riverside slope of the levee.
Station	Nearest station in the HEC-RAS model to the breach location.
Top	Elevation of the levee crest/crown/top at the breach location (feet, NAVD 88 datum). Used to define system response curve effective elevation range.

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Levee Breach Field	Description
Toe	Elevation of the levee toe at the breach location (feet, NAVD 88 datum). Used to define system response curve effective elevation range.
Height	Height of the levee at the breach location (feet)

1.6 CONSOLIDATION OF EXISTING RISK ASSESSMENT INFORMATION

The SRP curves developed for FFRD modeling are uniquely developed and dependent on the understanding of the subject levee's expected performance during flood loading as determined by a risk assessment. The NLD data can identify if a risk assessment was completed (*LSAC Rating* field) and if a higher-level risk assessment was completed (*Has QRA/SQRA (Yes/No)* field).

Table The levee data lead identifies all sources of levee performance data used to develop SRP curves.

Table 4. Risk Assessment Types and Description

Risk Assessment Type	Description
QRA	USACE-performed risk assessment with the lowest degree of uncertainty. System response curves are typically an output product.
SQRA	USACE-performed risk assessment with a moderate degree of uncertainty. System response curves are typically not an output product.
LST	USACE-performed screening-level risk assessment completed using the USACE-developed Levee Screening Tool. Risk assessments typically have a high degree of uncertainty. System response curves are an output product.
LST 1.0	USACE-performed screening-level risk assessment completed using the USACE-developed Levee Screening Tool's former model (pre-2023). Risk assessments typically have a high degree of uncertainty. System response curves are not an output product.
Levee Loading and Incident Data (LLID)	USACE dataset of levee performance during flood events for the USACE portfolio of levees and non-federal levees active in the PL 84-99 rehabilitation program. Levees without a levee-specific risk assessment are assigned "average" system response based on this data.

Higher-level risk assessments such as SQRA and QRA can be found on the Corps of Engineers Dam and Levee Safety (CEDALS) folder on ProjectWise. Only risk assessments that have been finalized and posted by the project's district with a signed memorandum will be used as a data source.

LST data is available from the Levee Screening Tool web application. LST 1.0 was decommissioned, and the export data is available from the NLD manager and others. Most data is accessible in the current LST.

The risk information datasets are managed and accessible by USACE. The data continues to expand and be updated through both routine and nonroutine Levee Safety Program activities.

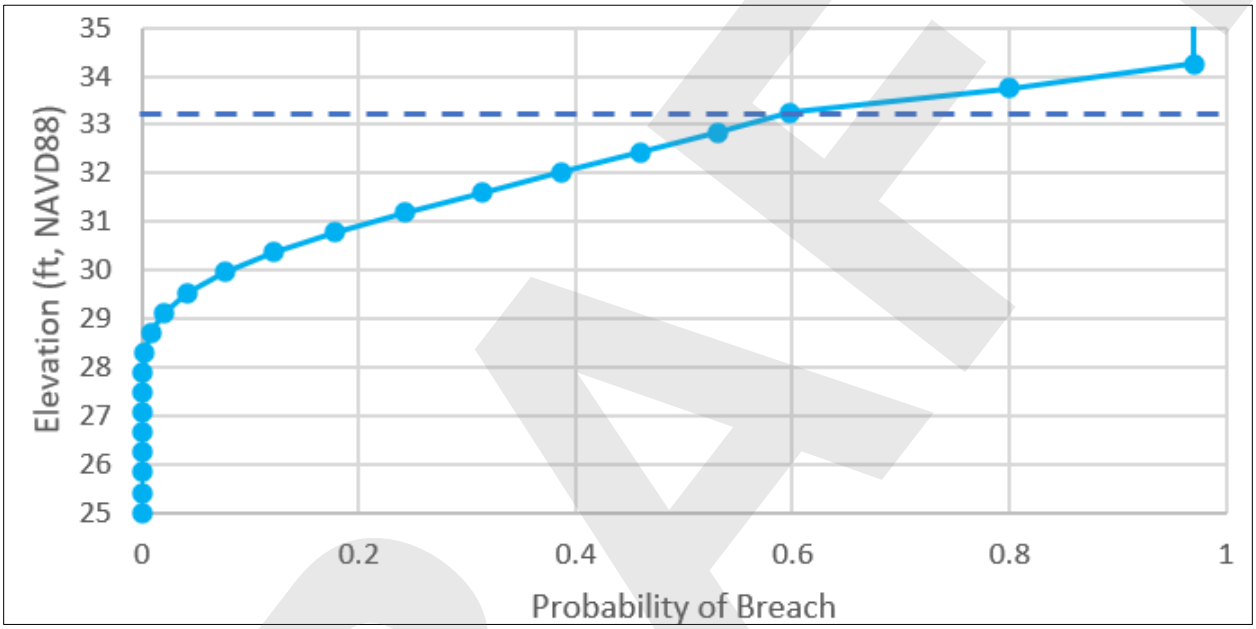
1.7 SYSTEM RESPONSE PROBABILITY CURVE DEVELOPMENT

An SRP curve represents the conditional probability of breach for a levee location. Figure 2 shows an example of a levee SRP curve plotted in terms of elevation versus probability of breach. The dashed line in the figure indicates the overtopping elevation for the breach location. The system response curve, a cumulative distribution function, provides a conditional probability of breach for prior-to-overtopping and overtopping events. The SRP curve begins at the origin (0,0) and increases monotonically.

The relationship of system response and annual probability is important to understand prior to developing SRP curves. Figure 3 provides an illustrative example of the area under an SRP curve and what that equates to in an annualized value. The curve is simplified to show p(f) of 1.0 at flood frequencies (AEP) exceeding the crest. The x-axis is the AEP of a flood loading event and the y-axis is the conditional probability of levee breach. In the figure, the area under the entire system response curve is the annual probability of inundation (API). The area under the portion of the curve prior to overtopping AEP is referred to as the annual probability of breach prior to overtopping (APF_{PTOT}) and the remaining portion of the curve represents the overtopping contribution to the API.

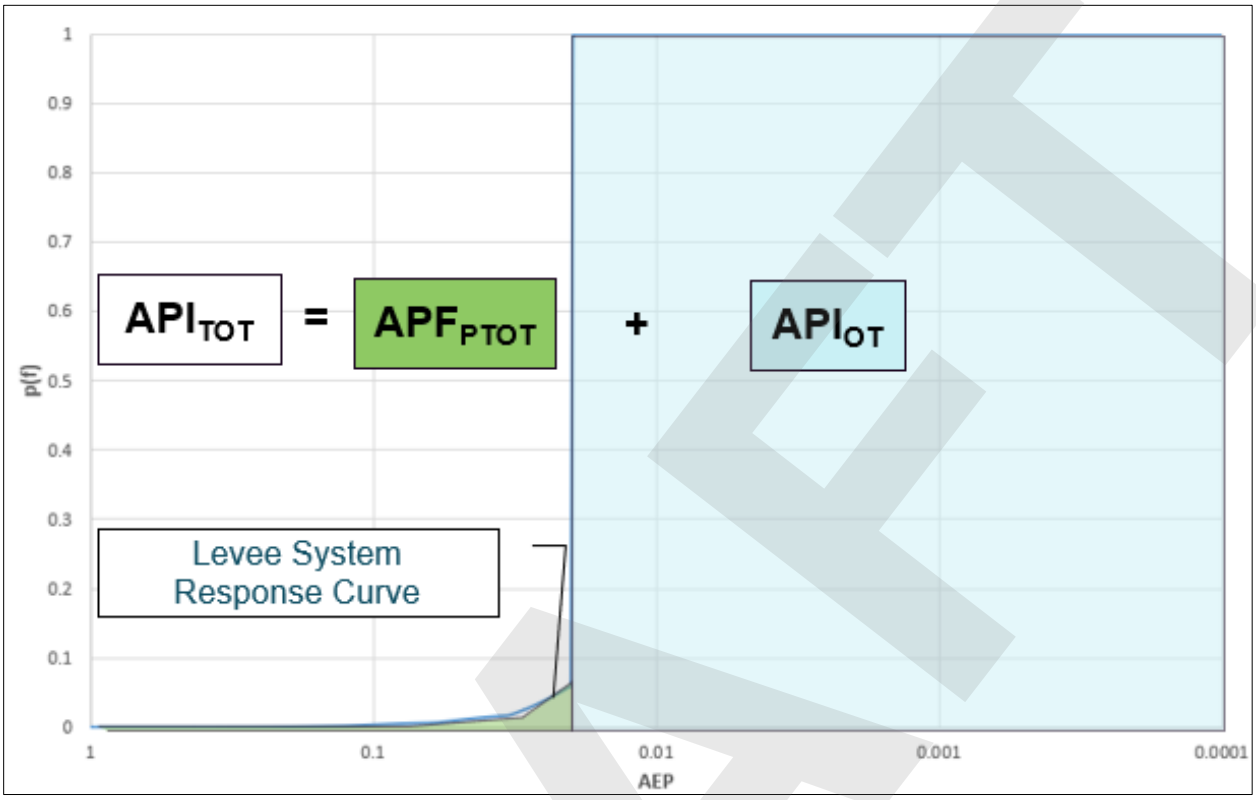
$$APF_{TOT} = APF_{PTOT} + API_{OT}$$

Figure 2. Example of a Levee System Response Curve: Elevation (feet) versus Probability of Breach.



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Figure 3. Example System Response Curve with Key Annualized Probability Terms Represented



A few key terms are important to define to build the understanding of how SRP curves are developed:

PFM: A PFM is a mechanism that once initiated potentially could progress to breach of a levee system or inundation of the leveed area.

Probability of Failure (p(f) or p(fail)): The probability of levee breach conditioned upon a given water surface elevation, percent levee height, or flood frequency.

Annual Probability of Failure Prior to Overtopping (APF_{PTOT}): The annual probability of failure (APF) represents the likelihood, in any given year, that a levee will breach considering the full range of potential loads to which it could be subjected (LST Technical Reference Manual). The APF_{PTOT} is a summation of only the likelihood of breach for loading below the levee crest.

API: The API is the APF_{PTOT} plus the annual probability of overtopping with and without breach.

OT AEP: The probability that levee overtopping will occur in any given year considering the full range of possible annual floods.

SRP curve development utilizes risk assessment information and/or USACE levee loading and incident data. The development process is dependent on whether the levee has been assessed with a risk assessment or not. Further, if the levee does have a risk assessment, the type of risk assessment informs SRP curve development.

1.7.1 TIERED APPROACH TO SYSTEM RESPONSE PROBABILITY CURVE DEVELOPMENT

SRP curve development uses existing risk assessment information and/or USACE LLID. The SRP curve development procedure varies depending on the type of performance data available. The following

represents the tiers of data that are available to characterize a levee's expected performance given flood loading (in order of highest fidelity).

1. QRA
2. SQRA
3. LST
4. Legacy LST (LST 1.0)
5. No Risk Assessment

QRA and LST risk assessments produce SRP curves for each PFM evaluated. FFRD SRP curves developed from these sources are combined, distributed, and/or down-scaled appropriately for the levee breach location scheme.

SRP curves developed from SQRA and LST 1.0 will use the risk assessment output values of APF for each PFM. An SRP curve is iteratively generated using the risk assessment APF as a target and adjusting log-normal shape parameters until the computed APF matches the target APF. FFRD SRPs developed are then combined, distributed, and/or down-scaled appropriately for the levee breach location scheme.

Levee systems without levee-specific performance data will be assigned a system response of an average levee based on USACE LLID. FFRD SRPs developed are then combined, distributed, and/or down-scaled appropriately for the levee breach location scheme.

1.7.2 INTERVENTION

In most modern risk assessment procedures, the ability to detect and intervene to arrest a breach mode during a flood is already included. For legacy levee performance data, including LST 1.0, an intervention correction is applied using LLID parameters. The LLID data for average levee performance including intervention will be applied for all levees without levee-specific data.

1.7.3 OVERTOPPING PERFORMANCE

For FFRD, the levee overtopping performance is considered using levee risk assessment for levee-specific assessment or USACE loading and incident data where it is not available.

Table provides a matrix of risk assessment type and the source of overtopping system response. Table 6 portrays the average overtopping system response based on the USACE levee loading and incident data. Levee performance during overtopping varies based on levee geometry, soil type, and erosion protection measures. Without readily available levee information, the average performance is considered an acceptable simplification. However, in some cases the assumption of active intervention may not be appropriate.

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Table 5. Overtopping Levee Performance Source

Risk Assessment Type	Overtopping Performance
QRA	Levee Specific
SQRA	Levee Specific
LST (Current)	Levee Specific
LST 1.0	Average Performance
No Risk Assessment	Average Performance

Table 6. Average Levee System Response during Overtopping

Loading	Probability of Breach
OT+0.5 foot	57%
OT+2 feet	68%
OT+3 feet	76%

Source: LST Technical Manual DRAFT. Figure 129. Probability of Breach during Overtopping for Different Depths and Soil Types. "All Levees" data.

1.7.4 MULTIPLE OVERTOPPING LOCATIONS ON THE SAME SYSTEM

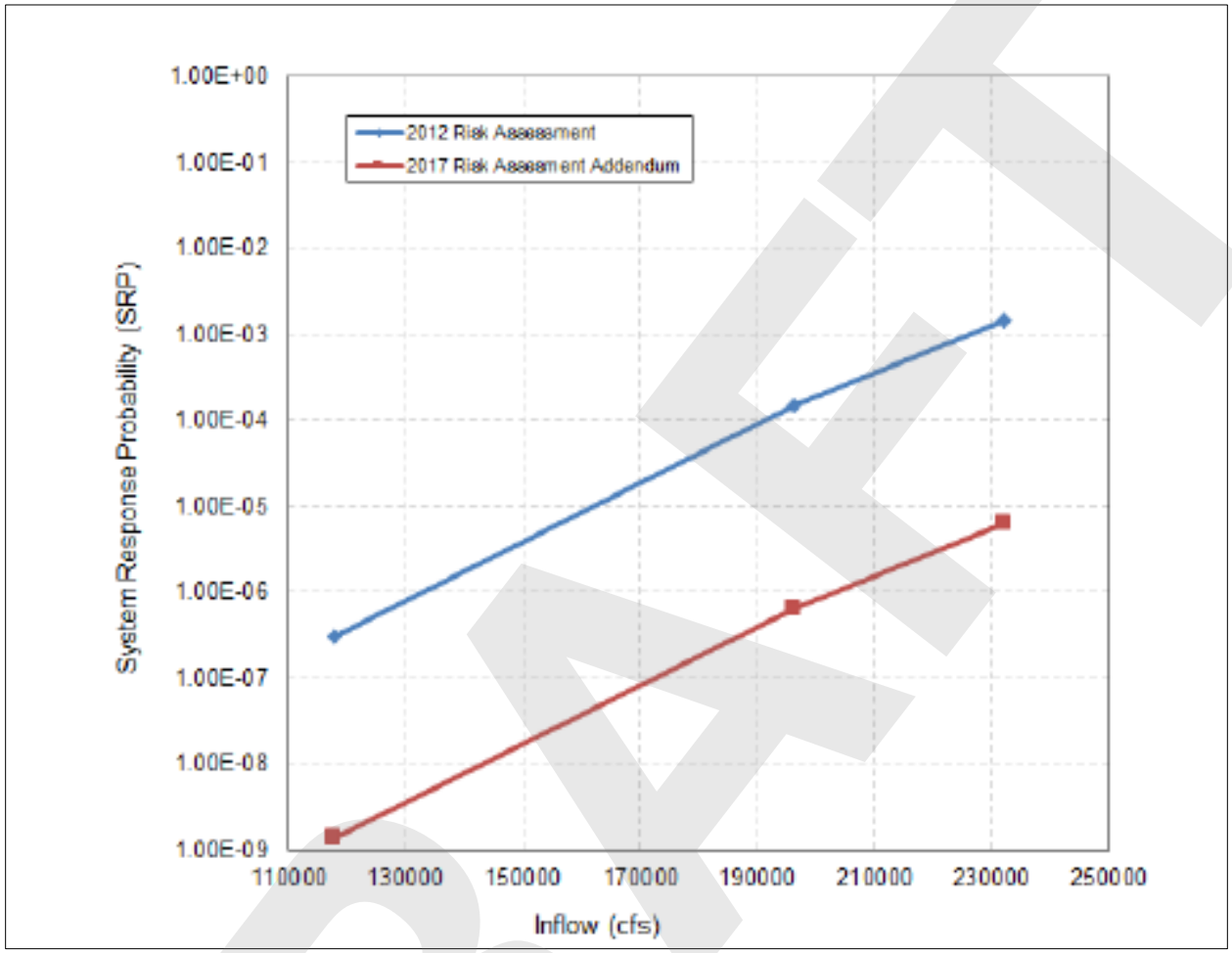
In consideration of multiple levee breach locations, every breach location assigned will include an overtopping system response. This will model a realistic scenario in which all potential areas of overtopping have a likelihood of breaching from overtopping. USACE subject matter experts (SMEs) concluded this approach creates a situation for the model to perform its own competing risk calculation, minimizing the length effect concern. Theoretically, as the initial overtopping location begins to overtop and possibly breach, the river stage at the other breach locations is less likely to achieve an elevation needed to trigger a breach. Thus, each of the individual breach locations of a particular levee system is competing against each other for the chance to breach. Once one is successful, the probability of another occurring drops so as not to introduce length effect issues of the probability of failure approach.

1.7.5 STARTING WITH A QUANTITATIVE RISK ASSESSMENT

From a QRA report, the probability of failure for a given percent loading, flow, or event AEP is typically available directly, but this information needs to be manually collected and then assigned to the breach location setup previously determined. Breach location assignment may involve a combination of one or more PFMs and the addition of expected overtopping performance. If loading function information is available at multiple locations and varies significantly, the PFM-specific system response should be converted to an AEP versus p(f) format before combining.

Figure 4 provides an example of input data collected from a USACE QRA report. In this example, the x-axis is provided in river flow and will need to be converted to percent levee height or AEP depending on whether the system response for this PFM is a standalone breach location or requires combination with other PFMs. To convert the axis, the user will collect information from the report that best provides an understanding of the flood loading function and coincident loading reference points on the levee.

Figure 4. Example System Response Probability Curves for a Single Performance Mode Included in a USACE Quantitative Risk Assessment Report



Combination of likelihood of breach for multiple PFMs into a single breach location is accomplished with simple addition using the formula below. This calculation assumes the PFMs are mutually exclusive. The combination also assumes that the frequency of flood loading is common at a given percent loading to arrive at the intended APF for the system. For example, combination of two SRP curves for different PFMs at 50 percent loading must mean the 50 percent loading at each location corresponds to approximately the same frequency of flood loading (e.g., 50 percent loading at both locations is equivalent to a 1-in-50 AEP).

$$P(fail)_{Total} = \sum p(fail)_i$$

1.7.6 STARTING WITH A SEMI-QUANTITATIVE RISK ASSESSMENT

In most cases, a semi-quantitative risk assessment will not include SRP curves for each PFM. The output users will be looking for from SQRA will be loading functions and the evaluated PFMs and their respective elicited APF with intervention. The general process for developing SRP curves from this data is to set the APF as a target and develop an SRP curve where the area under the curve matches that APF target. The shape of the curve is influenced by log-normal shape parameters (sigma) from USACE LLID. USACE maintains a spreadsheet for this computation. The detailed steps are:

1. Evaluate hydraulic and hydrologic data in the report. Locate and enter one or more flood loading functions required to represent the flood loading for the PFMs. Input data needed: Stage and AEP

for each loading curve. Enter the incipient overtopping point to document the cutoff point for APF calculation.

2. Locate and enter the APF with intervention for each elicited PFM as the target APF for the calculated SRP for each PFM. Use all credible breach modes from the SQRA report so that the total APF for the system is considered. Assign each PFM to a standard PFM type to associate the LLID shape parameters.
3. Perform log-linear interpolation of each percent loading function to a standardized set of nodes to normalize all PFM entries to a common AEP range so SRPs can eventually be combined.
4. Set an iterative computation of SRP at each AEP node by varying the shape parameter (σ, sigma ln) until the computed APF of the created SRP curve matches the target APF from the SQRA report.
5. Repeat for all credible PFMs.
6. Combine all or portions of the calculated SRP to match the desired distribution of PFMs at the breach locations. A system represented by only a single breach location will sum all calculated PFM SRP curves. A levee system represented by multiple breach locations will use specific PFM SRP curves directly and sum the remainder at the LCL, for example.

Figure 5 shows the calculation spreadsheet setup with orange cells indicating entry. Fail mode is selected from a static dropdown list. User selects the loading function that applies to this PFM. User enters the elicited PFM APF with intervention from the SQRA report. Finally, an iterative approach is used to change sigma ln until the computed APF matches the target APF. The prior-to-overtopping SRP curve is now created for this PFM. The expansion of percent levee height is a log-linear interpolation of the input flood loading curve. The calculation of SRP is done as a cumulative distribution function with a log-normal distribution for the given percent levee height ordinate with mean (μ) and standard deviation (σ). Finally, the computation of APF is computed as the area under the SRP curve using a Riemann sum trapezoidal approximation:

$$APF = \sum (x_i - x_{i+1}) \times (\mathrm{y_i} + \mathrm{y_{i+1}}) \div 2$$

Figure 6 and Figure 7 illustrate the computation of APF and the Riemann sums midpoint method.

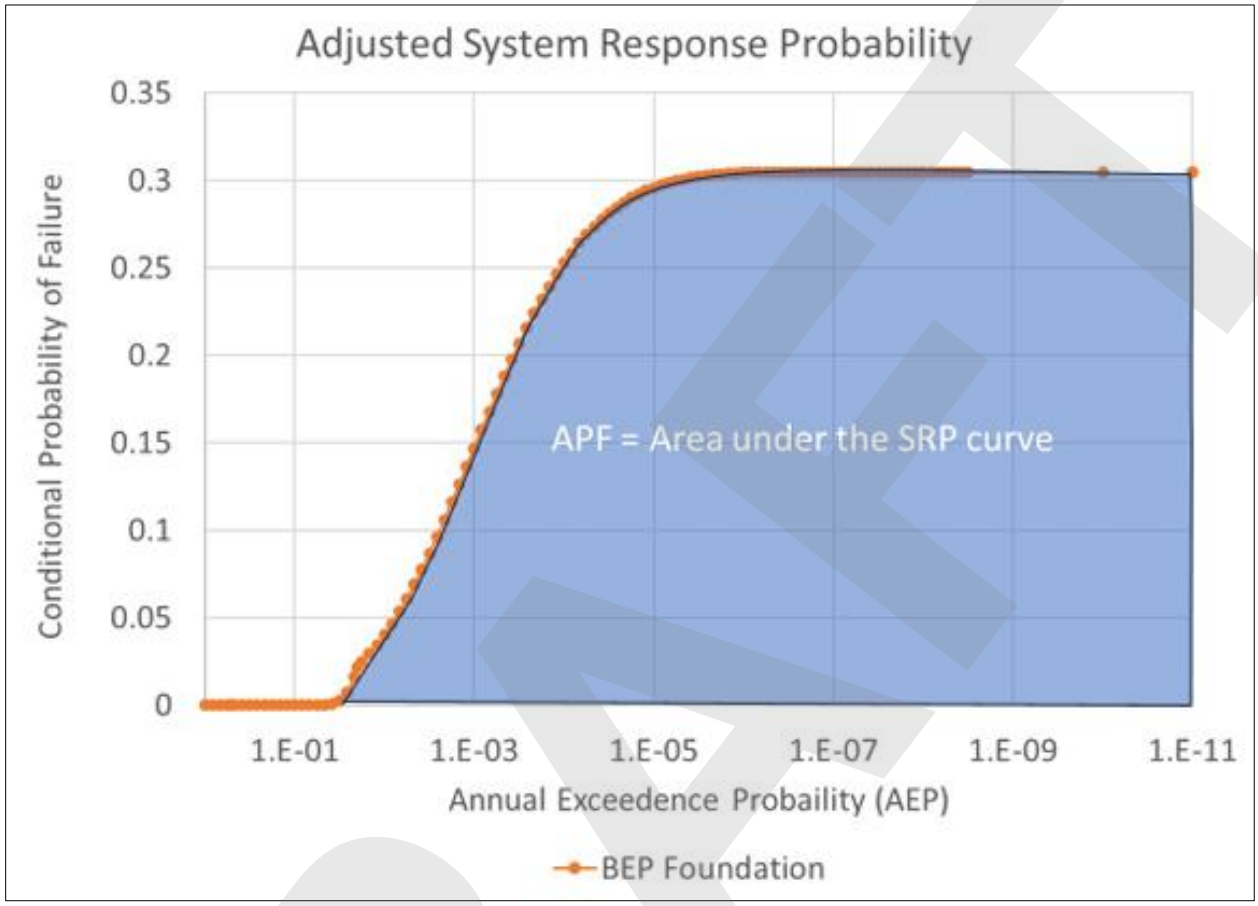
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Figure 5. Example System Response Curves Computation Sheet for a Single PFM.

Name	Embankment Erosion PFM 10		
Fail Mode	Embankment Erosion		
Use?	Yes		
Loading	Loading1		
Target APF	9.49E-05		
Computed APF	9.49E-05		
mu ln	3.217808671		
sigma ln	1.406530149		
mu	67.15290622		
sigma	167.622		
AEP	**% Levee Height**	**SRP**	**APF**
1.0000E+00	0	0	0
5.0000E-01	0.056636169	7.48826E-06	1.87E-06
3.3333E-01	0.089766204	3.14593E-05	3.25E-06
2.5000E-01	0.113272338	6.24695E-05	3.91E-06
2.0000E-01	0.131505112	9.56965E-05	3.95E-06
1.6667E-01	0.146402373	0.000129185	3.75E-06
1.4286E-01	0.158997827	0.000162108	3.47E-06
1.2500E-01	0.169908507	0.000194118	3.18E-06

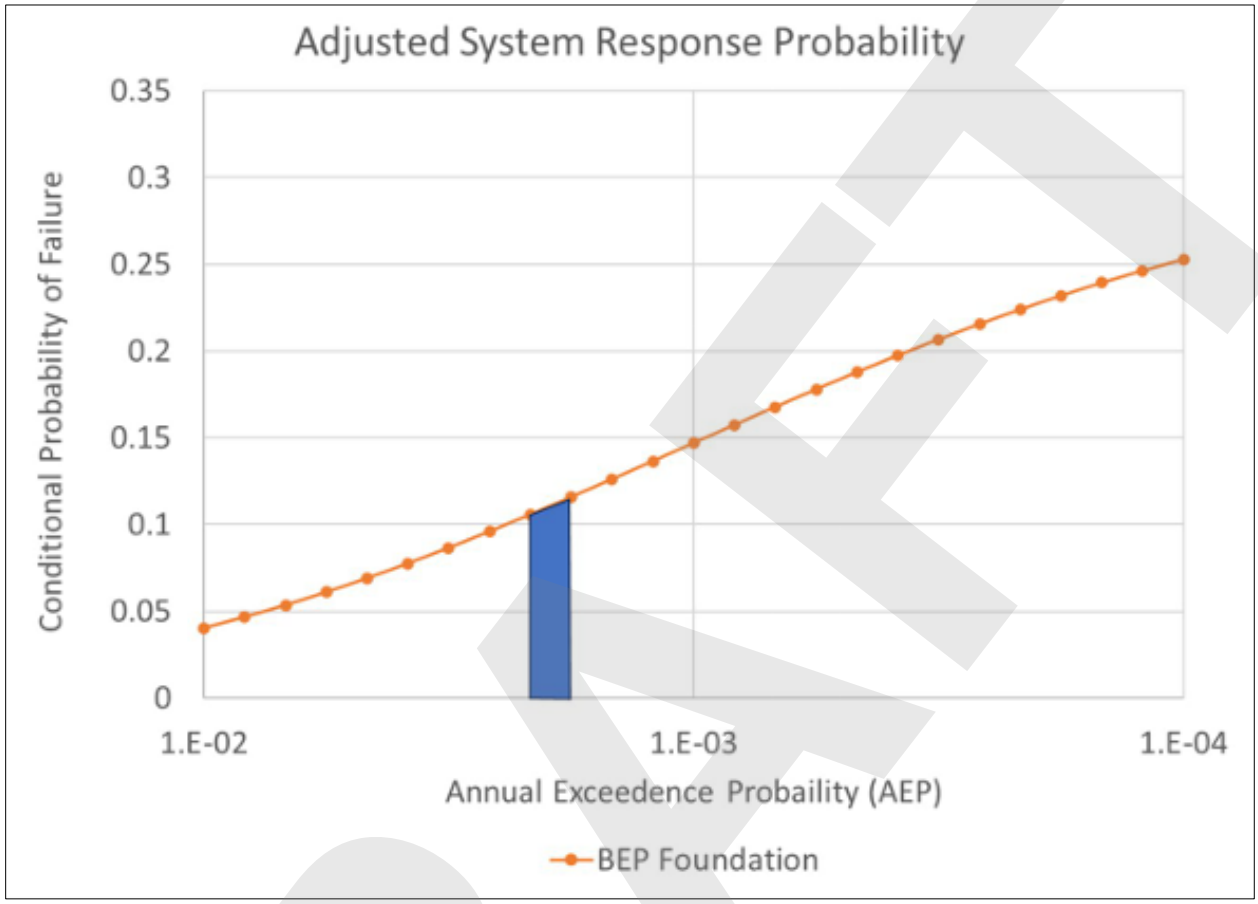
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Figure 6. Illustration of APF Computation from a System Response Curve (from Levee Screening Tool Application and Technical Manual)



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Figure 7. Illustration of Riemann Sum Trapezoidal Approximation (from Levee Screening Tool Application and Technical Manual)



1.7.7 STARTING WITH A SCREENING-LEVEL RISK ASSESSMENT

Screening-level risk assessment data is available in the LST. Once a multidisciplinary team completes a screening, the LST maintains database information on levee performance (PFMs), representative breach locations assigned to those PFMs, and representative loading functions assigned to those PFMs. The LST uses a standard set of PFMs that are assessed depending on the features of the levee (e.g., earthen embankment, floodwall, and/or closure structure). Overtopping with breach for the incipient overtopping location is also assessed.

Within the LST, a large database export can be accomplished with administrator privileges in the Reports tab on the main banner. SRP curves can be extracted on a levee-by-levee basis within the Computations tab of a levee segment. Figure 8 provides an example of the levee segment computation export with SRP curve in a consistent AEP format and provides percent levee height versus p(fail) curves for each PFM. In this example, the levee has an earthen embankment and floodwall backward erosion piping breach mode assessed with differing loading curves. The embankment reaches 100 percent of the levee height at a more frequent loading than the floodwall section. This is important to recognize when considering breach setup and combination of PFMs.

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Figure 8. Levee Screening Tool Levee System Response Curve Results for Two of the Evaluated Potential Failure Modes

AEP	BEP Foundation		FW BEP	
AEP	% Levee Height	BEP Foundation	% Levee Height	FW BEP
1	0	0	0	0
0.8254	0	0	0	0
0.6813	0	0	0	0
0.5623	0	0	0	0
0.4642	0	0	0	0
0.3831	0	0	0	0
0.3162	0	0	0	0
0.261	0	0	0	0
0.2154	0	0	0	0
0.1778	0	0	0	0
0.1468	0	0	0	0
0.1212	0	0	0	0
0.12	0	0	0	0
0.1	0	0	0.03367583	2.14143E-12
0.0825	0	0	0.069117546	1.94595E-10
0.0681	0	0	0.104559261	1.92506E-09
0.0562	0	0	0.140000977	8.55487E-09
0.0464	0	0	0.175442692	2.45256E-08
0.0383	0	0	0.210884408	5.34289E-08
0.0316	0	0	0.246326123	9.67223E-08
0.0261	0	0	0.279173034	1.4943E-07
0.0215	0	0	0.311719862	2.07341E-07
0.0178	0	0	0.34426669	2.68906E-07
0.0147	0	0	0.376813518	3.30941E-07
0.0121	0	0	0.409360345	3.90223E-07
0.01	0	0	0.441907173	4.44065E-07
0.0095	0	0	0.450607485	1.2582E-07
0.0083	0.047421851	2.7764E-81	0.474454001	3.62087E-07
0.0068	0.112145387	1.87145E-48	0.506523829	5.26921E-07
0.0056	0.176868922	6.94238E-35	0.536853086	5.49211E-07
0.0046	0.241592457	4.26477E-27	0.567182343	5.58801E-07
0.0038	0.306315993	5.42428E-22	0.5975116	5.5898E-07
0.0032	0.371039528	2.14621E-18	0.627840857	5.50104E-07
0.0026	0.435763063	9.64464E-16	0.658170115	5.33003E-07
0.0022	0.500486599	1.0029E-13	0.688499372	5.08926E-07
0.0018	0.565210134	3.70827E-12	0.718828629	4.79385E-07
0.0015	0.62993367	6.42581E-11	0.749157886	4.45976E-07
0.0012	0.694657205	6.27802E-10	0.775235795	4.07156E-07
0.001	0.75938074	3.92929E-09	0.801192291	3.65663E-07
0.0008	0.824104276	1.72495E-08	0.827148788	3.25786E-07
0.0007	0.888827811	5.67433E-08	0.853105284	2.88175E-07
0.0006	0.953551346	1.46944E-07	0.87906178	2.53179E-07
0.0005	1	1.95359E-07	0.897689383	1.61632E-07
0.0005	1.018550594	1.03893E-07	0.905018277	5.92637E-08

For each levee segment, the common-cause adjusted SRP results will be used. The LST will present SRP curves by PFM, all prior-to-overtopping PFMs combined, and overtopping with breach. Combining

and splitting the SRP curves are required based upon the breach quantity set up by the user for the subject levee system.

1.7.8 STARTING WITH A SCREENING-LEVEL RISK ASSESSMENT (LST 1.0)

The most prevalent levee performance data for levees in the United States comes from the first version of the LST (LST 1.0) with a screening level risk assessment completed for the entire USACE portfolio.

The input data needed to develop SRP curves from this dataset is the LST 1.0 database export file from October 2023. The file contains levee toe and crest loading frequency (AEP) and performance indexes for each PFM. The user should understand that it is typical that only a single generalized loading curve representing the most frequent loading-toe AEP and the incipient overtopping frequency are available. A segment may have a separate loading function for embankments, floodwalls, and/or closure structures if those features are present. Further, the levee performance was based on a relative risk Bayesian updating approach. Conversions of the performance data are included to correct the performance estimates to expected magnitudes and shape parameters from the USACE Levee Loading and Incident data as well as corrections to account for intervention. Screenings were performed on levees at the segment level. One or more segments may comprise a levee system. The following procedures are for a combined SRP curve at the segment level. Single breach setups for a multi-segment system will combine the curves as previously described in the p(fail) versus AEP format. Multiple breaches assigned to a single segment will require splitting relevant PFMs to each breach location as appropriate.

Given that all input data is retrieved from a single dataset, the process is mostly automated once computational setup is achieved. USACE maintains a spreadsheet for this computation. The detailed steps are:

1. Enter the NLD segment ID of the subject levee segment in the Loading tab
2. Extract the LST flood loading data. Toe and crest AEP estimates are available for the segment and additional entries for floodwall loading and closure structure loading, if applicable.
3. Select Yes for every applicable PFM. The LST used a standardized set of PFMs applicable to a feature type (embankment, floodwall, and closure structure).
4. Assign the correct loading function to each PFM.
5. On the Combined Curve tab, the target APF will be pulled from the LST 1.0 data. The intended LST 1.0 field is the *performance index*. There is a performance index associated with each evaluated PFM from the standardized suite of LST PFMs.
6. Iteratively adjust the sigma ln shape parameter until the calculated APF for the SRP curve you are creating matches the target APF.
7. Repeat for all credible PFMs.
8. The spreadsheet will perform the computation to account for intervention (Figure 9).
9. Combine all or portions of the calculated SRP (adjusted for intervention) to match the desired distribution of PFMs at the breach locations. A system represented by only a single breach location will sum all calculated PFM SRP curves. A levee system represented by multiple breach locations will use specific PFM SRP curves directly and sum the remainder at the LCL, for example.

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Figure 9. Unsuccessful Intervention Probabilities Used to Adjust LST 1.0 Performance Data to Account for Intervention.

Probability of Unsuccessful Intervention						
BEP Foundation	BEP Thru Embankment	Embankment Erosion	Embankment Stability	FW BEP Foundation	FW Instability	Closures
0.308665316	0.273506036	0.114042345	0.508579044	0.148523787	0.530448589	0.633144706

1.7.9 NO RISK ASSESSMENT IS AVAILABLE

When the levee system or segment performance has not been assessed with a risk assessment, average levee performance parameters are applied based on the USACE LLID (Levee Screening Tool Application and Technical Manual [September 2024] Appendix A: USACE Levee Loading and Incident Data). The LLID data uses historic performance during flood events to provide expected performance given flood loading.

The SRP curve is generated using methods and computations described above with the following LLID breach probabilities and log-normal shape parameters (Table 7). Each levee segment will have an SRP curve computed using these curve parameters with performance modes that apply to the levee features (embankment, floodwall, closure structure) as viewable in the NLD alignment line type.

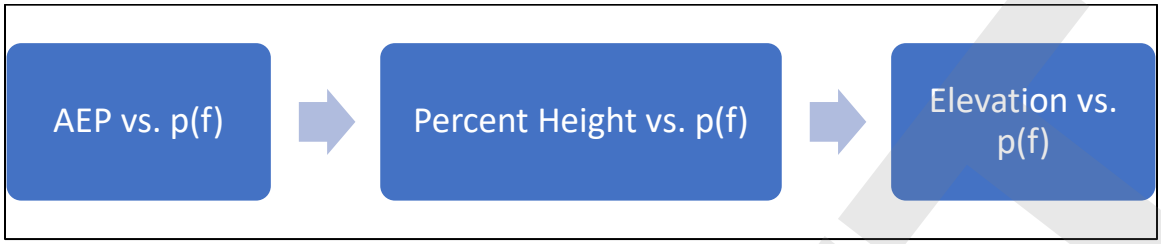
Table 7. Probability of Breach and Shape Parameters Derived from LLID (from Levee Screening Tool Application and Technical Manual)

Potential Failure Mode	Probability of Breach at 100% Loading	Mean Shape Parameters (Log-Normal) (μ)	Standard Deviation Shape Parameters (Log-Normal) (σ)
Backward Erosion and Piping (BEP) Embankment	0.0426	0.3536	0.2055
Internal Erosion (IE) Embankment	0.0426	0.3547	0.2060
Embankment Slope Instability	0.0088	0.5781	0.2435
Embankment Riverine Erosion	0.0090	3.2805	1.3858
Floodwall Backward Erosion and Piping (BEP)	0.0096	2.3064	0.9858
Floodwall Instability	0.0324	0.6203	0.3360
Closure Structure Failure or Misoperation	0.0419	0.9446	0.5464

1.7.10 SYSTEM RESPONSE PROBABILITY CURVE CONVERSION

During SRP curve creation, the axes will most likely need to be changed from the initial AEP versus p(f) format. In addition, the required standard number of ordinates on the FFRD curves may require a contraction or expansion of a calculated curve. Finally, the FFRD HEC-RAS model requires an elevation versus p(fail) format using the elevations at the breach location (North American Vertical Datum of 1988 [NAVD 88]). As Figure 10 illustrates, the AEP axis is then converted to a percent height and finally to a usable format for FFRD to the breach location selected in an elevation versus p(f) format.

Figure 10. System Response Probability Curve Format Conversion Process



1. The SRP curve generation process requires establishment of a related percent levee loading relationship to the flood loading. This step is to revert the AEP axis back to a percent levee height scale. This is available in all USACE-prepared computational spreadsheets. Verify that the scale represents the percent loading for the intended breach location.
2. The following step is to convert to the standardized number of ordinates. Currently, the SRP curve generation spreadsheets are set up to generate an SRP curve with 10,000 ordinates to conform to FEMA insurance pricing requirements; however, depending on the development method the raw curve may have a different number. Regardless, the user will convert the input SRP curve to a standard number of ordinates following Table 8. A USACE-prepared spreadsheet is available to perform this computation. Computationally, the input SRP curve is log-linearly interpolated into the final format.
3. The final step is to convert percent height into elevations at the breach location. The levee toe and crest elevation are documented in the breach location files (1.5.2).

$$Elevation = Percent\ Loading \times (Crest\ Elevation - Toe\ Elevation) + Toe\ Elevation$$

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Table 8. Prior-to-Overtopping Probability Curve Format Template.

Number	Percent Levee Height	Probability of Failure
1	0	#
2	0.05	#
3	0.1	#
4	0.15	#
5	0.2	#
6	0.25	#
7	0.3	#
8	0.35	#
9	0.4	#
10	0.45	#
11	0.5	#
12	0.55	#
13	0.6	#
14	0.65	#
15	0.7	#
16	0.75	#
17	0.8	#
18	0.85	#
19	0.9	#
20	0.95	#
21	1	#

1.7.11 SYSTEM RESPONSE ALLOCATION, COMBINATION, AND FORMATTING

SRPs representing performance for a specific PFM or levee segment are then combined to allocate system response to the defined breach location scheme. If the breach scheme identifies the need for a prior-to-overtopping breach location and the risk assessment source has identified a critical flaw at this specific location, the full system response for that breach mode is applied. If no critical location is identified, the combined performance modes evaluated in the risk assessment are downscaled and applied evenly to all breach locations. The combination computation applies De Morgan's rule for statistically independent and not mutually exclusive PFMs when the risk assessment data is unadjusted (Table 9).

$$Probability\ of\ Failure = 1 - (1 - p(fail))^N$$

Table 9. Summary of Adjusted and Unadjusted System Response by Risk Assessment Type

Risk Assessment Type	Adjusted or Unadjusted	Combination Method
QRA	Unadjusted	De Morgan's Rule
SQRA	Unadjusted	De Morgan's Rule
LST (Current)	Adjusted	Summation
LST (v1.0)	Adjusted	Summation

When levee SRPs need to be distributed to multiple identified breach locations, an adjustment to De Morgan's rule is applied to the conditional probability of failure for the system (P(fail)) to statistically downscale the system response by the number of breach locations ($N_{Locations}$).

$$Probability\ of\ Failure = 1 - (1 - P(fail))^{1/N_{\text{Locations}}}$$

Once the SRP is developed for the breach location scheme, the data is converted to the correct stage range from levee toe elevation to crest elevation. The overtopping response is added using De Morgan's rule to the prior-to-overtopping system response, if prior-to-overtopping at 100 percent of the levee height remains constant. Finally, if the probability of breach has not reached 1.0, the SRP curve is completed by interpolating from the overtopping response.

1.7.12 COMBINING OVERTOPPING PERFORMANCE WITH PRIOR-TO-OVERTOPPING PERFORMANCE

Risk assessments typically document overtopping differently and include a direct SRP curve that expresses probability of failure as a function of depth over the crest of the levee at that location. Therefore, this portion of the SRP curve will be added after the SRP curve is in the elevation versus p(f) format.

The elicited overtopping performance typically has a maximum range (i.e., only elicited to 2 feet over the levee crest height). For the purposes of FFRD, the overtopping performance will be extrapolated to 99.9 percent chance of breach and then extended as a straight line to 100 feet above the levee crest elevation to ensure that the sampled elevation range exceeds the potential FFRD hydraulic model elevation.

The final step when combining prior-to-overtopping and overtopping portions of the SRP is to consider that prior-to-overtopping PFMs have a non-zero chance of breach, even during overtopping. For example, a levee may still breach from internal erosion during overtopping, so it will be combined with likelihood of overtopping with breach; however, tailwater conditions as the leveed area fills ultimately cause the internal erosion SRP to plateau. To account for this, the prior-to-overtopping probability of failure is held constant during the addition of the overtopping performance probability of failure. This is not a constant, but this approach is used as a simplification to not ignore this contribution.

1.8 BREACH CHARACTERISTICS

Guidance for selection of breach parameters for FFRD is included in SOP Volume II, Section 8.

1.9 FORMAT FOR LEVEE SYSTEM RESPONSE PROBABILITY CURVE DELIVERY

The SRP curves are compiled, qualityFigure 11-control reviewed, and then uploaded to the Model Library as a single .csv file following the format shown in Figure 11. The file includes the 2D connection name in the HEC-RAS model to link the system response trigger elevation sampled from the system response curve to the water surface modeled in HEC-RAS. The remaining common data fields are data identifiers including the NLD system and segment ID numbers, and the basis for the system response curve (i.e.,

risk assessment source, potential failure mode, or default performance). The system response includes prior to overtopping and overtopping performance.

The provided SRPs formatted as .csv files will be used as an input for the cloud-compute actions that sample breach trigger elevations for the stochastic simulations.

A template for use in creation of the required SRP format for delivery is available on the FFRD GitHub Templates and Resources site, named *system-response*. Note that no commas are allowed in the header information fields.

The levee SRP .csv file should be initially delivered to the project directory in the following location.

```
Delivery Location: basin-name > conformance > system-response
```

Figure 11. Combined Levee System Response

	A	B	C	D	E	F	G	H	I	J
1	2d connection na	NLD_Fenster	2d connection na	NLD_Ft_Dent	2d connection na	NLD_HorseShBn	2d connection na	NLD_HorseShBn	2d connection na	NLD_LwrGreenRB_2
2	NLD System id	5505000094	NLD System id	2005000055	NLD System id	5505000009	NLD System id	5505000009	NLD System id	5505000043
3	NLD Segment id	5504000097	NLD Segment id	2004000055	NLD Segment id	5504000009	NLD Segment id	5504000009	NLD Segment id	5504000071
4	NIDID		NIDID		NIDID		NIDID		NIDID	
5	failure mode	default combine	failure mode	default combine	failure mode	BEP	failure mode	OT	failure mode	combined
6	source	default	source	default	source	HLRA - BEP PFM 3	source	HLRA - LCL	source	LST Legacy
7	stage	probability	stage	probability	stage	probability	stage	probability	stage	probability
8	69.1	0	19.75	0	39.9	0	48.5	0	38.5	0
9	69.2	0	19.9375	0	40.455	0	48.735	0	38.87	0
10	69.3	8.99E-17	20.125	8.99E-17	41.01	0	48.97	0	39.24	0.00E+00
11	69.4	4.49E-13	20.3125	4.49E-13	41.565	0	49.205	0	39.61	1.96E-05
12	69.5	6.11E-11	20.5	6.11E-11	42.12	0	49.44	0	39.98	6.82E-05
13	69.6	6.75E-09	20.6875	6.75E-09	42.675	0	49.675	0	40.35	2.02E-04
14	69.7	1.33E-07	20.875	1.33E-07	43.23	0	49.91	0	40.72	5.37E-04
15	69.8	1.36E-06	21.0625	1.36E-06	43.785	0	50.145	0	41.09	1.23E-03
16	69.9	9.96E-06	21.25	9.96E-06	44.34	0	50.38	0	41.46	2.49E-03
17	70	4.92E-05	21.4375	4.92E-05	44.895	0	50.615	0	41.83	4.58E-03
18	70.1	0.000184438	21.625	0.000184438	45.45	0	50.85	0	42.2	7.68E-03
19	70.2	0.000558024	21.8125	0.000558024	46.005	2.77E-08	51.085	0	42.57	1.20E-02
20	70.3	0.001421416	22	0.001421416	46.56	3.48E-07	51.32	0	42.94	0.017582186
21	70.4	0.003158266	22.1875	0.003158266	47.115	2.97E-06	51.555	0	43.31	0.024544141
22	70.5	0.006276578	22.375	0.006276578	47.67	1.88E-05	51.79	0	43.68	0.032845339
23	70.6	0.011372082	22.5625	0.011372082	48.225	9.17E-05	52.025	0	44.05	0.042416155
24	70.7	0.019072199	22.75	0.019072199	48.78	0.000359421	52.26	0	44.42	0.053147622
25	70.8	0.029967223	22.9375	0.029967223	49.335	0.001171411	52.495	0	44.79	0.064902475
26	70.9	0.044544889	23.125	0.044544889	49.89	0.003263916	52.73	0	45.16	0.077527517
27	71	0.063137062	23.3125	0.063137062	50.445	0.007951276	52.965	0	45.53	0.090863295
28	71.1	0.085893052	23.5	0.085893052	51	0.017251129	53.2	0	45.9	0.104752141
29	71.6	0.655893052	24	0.655893052	51.5	0.517251129	53.7	0.5	46.4	0.674752141
30	73.1	0.765893052	25.5	0.765893052	52	1	54.2	0.99	47.9	0.784752141
31	74.1	0.845893052	26.5	0.845893052	1000	1	1000	0.9900001	48.9	0.864752141
32	1000	0.845893152	1000	0.845893152	1000	1	1000	1	1000	0.864752241
33	1000	1	1000	1					1000	1

1.10 REVIEW AND EVALUATION OF BREACH RESULTS

A single realization of stochastic events is simulated by cloud computing during the conformance phase. Breach results from the conformance phase must be reviewed by the levee data lead to confirm that breach trigger elevations are being sampled appropriately and the number of breach events that occur at each breach location falls within the expected range. Similarly, the multi-realization results from the production phase must be reviewed and evaluated.

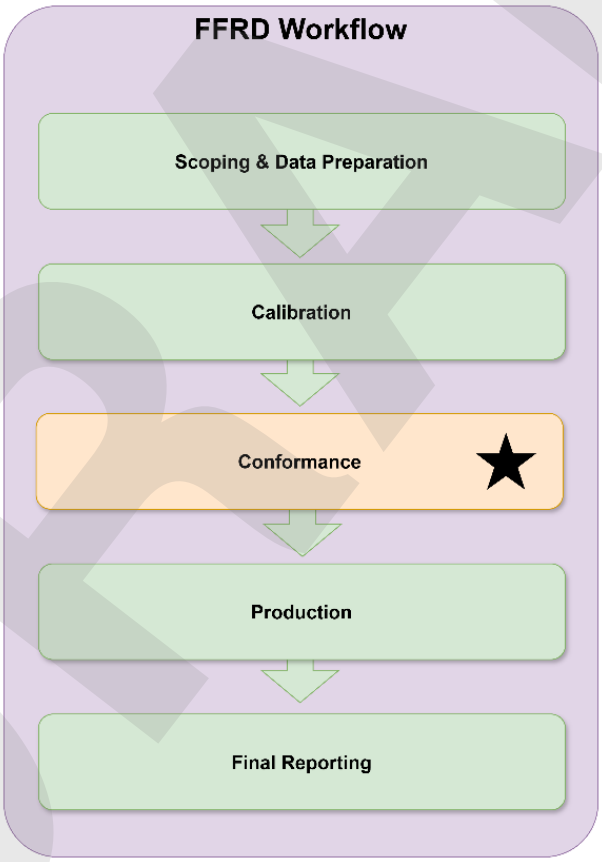
JOB AID 12—MODEL DELIVERY FOR CLOUD COMPUTE

1.1 WORKFLOW CONTEXT

This job aid describes specific requirements and steps for delivery of models and cloud compute of synthetic events during the conformance phase. The model integration lead is responsible for model delivery and verification quality control (QC), loading models to the cloud, and associating and updating the final Model Linking Register. This process validates model inputs required for successful computing, manifest updates, and execution in the cloud.

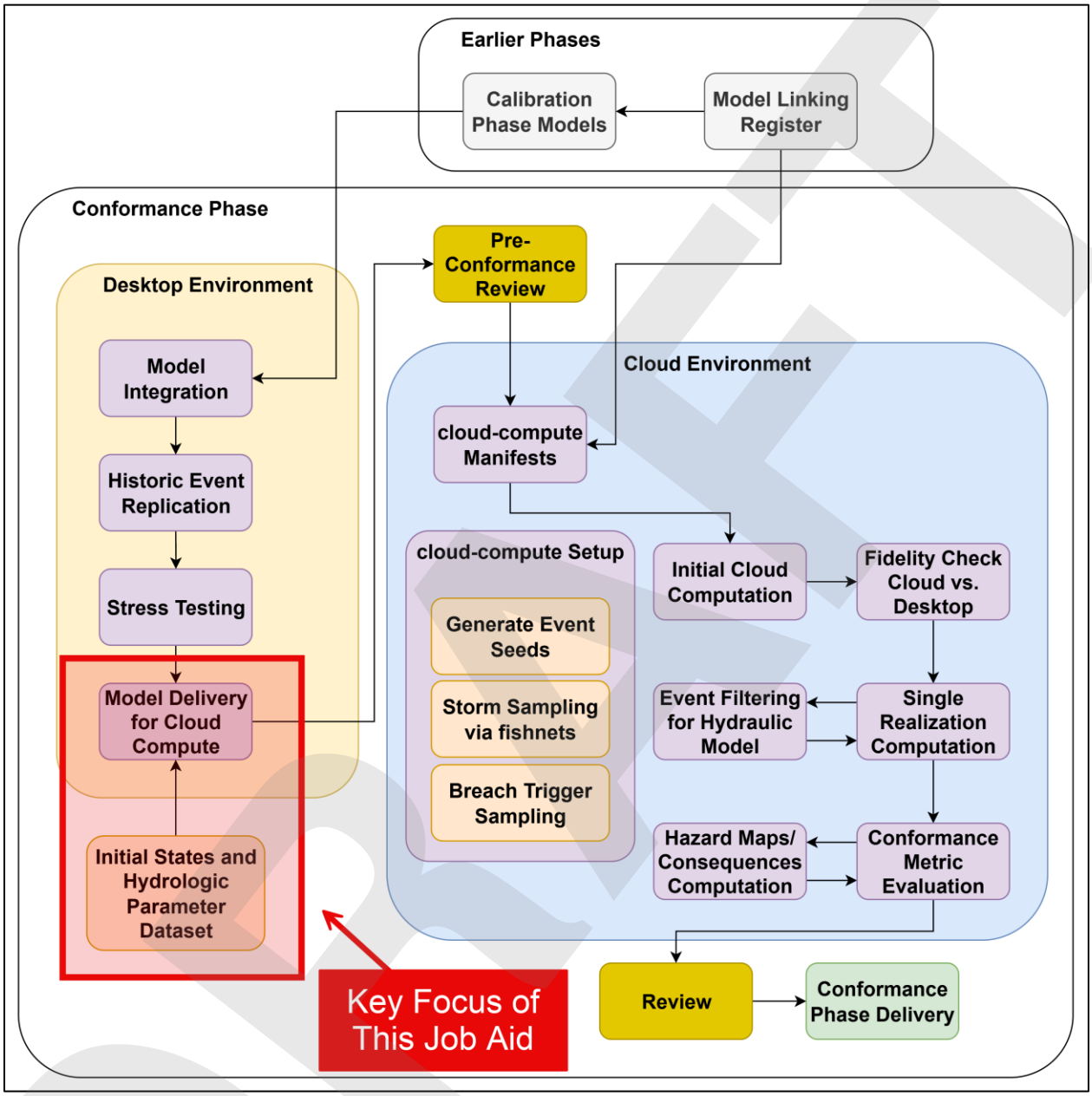
The model integration lead is also responsible for validating accurate model linking and confirming model performance in the desktop environment (these activities are not described in this job aid). To proceed with the conformance phase, the pre-conformance review checklist items must be complete, the linked system of models must be valid, and the cloud-preparation steps must be finished.

Figure 1. Models are Delivered for Cloud-Compute during the Conformance Phase of the Project, the Third Phase of the FFRD Workflow.



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Figure 2. Focus of this Job Aid in Relation to Steps within the Conformance Phase.



1.2 FINAL MODEL LINKING REGISTER DELIVERY

The Name Table, Model Linking Register, Dam Scoping Table, and Levee Breach Locations are fully developed during the scoping and data preparation phase. These products guide the model linking strategy throughout the FFRD project. Each may be updated during the calibration phase or conformance phase of the project when unique issues are identified.

The Model Linking Register is finalized early in the conformance phase when the full system of hydraulic models is linked in a desktop environment and tested. This register is an easily readable table containing information required to build cloud compute manifests. These cloud compute manifests control the actual linking of output data to input data that is summarized in the Model Linking Register with cloud compute. Several cloud compute manifests are required and will be included in various plug-ins (e.g., hms-mutator-

plugin, hms-runner, ressim-runner, fragility-curve-plugin, ras-runner, and ffrd-manifest-generator). Many of the plug-ins mentioned will become available in future SOP versions; the plug-ins are currently in development.

Scripting to automate certain steps in the production of the compute manifests has the potential to improve efficiency and ability to QC linking data.

1.3 DIRECTORY STRUCTURE

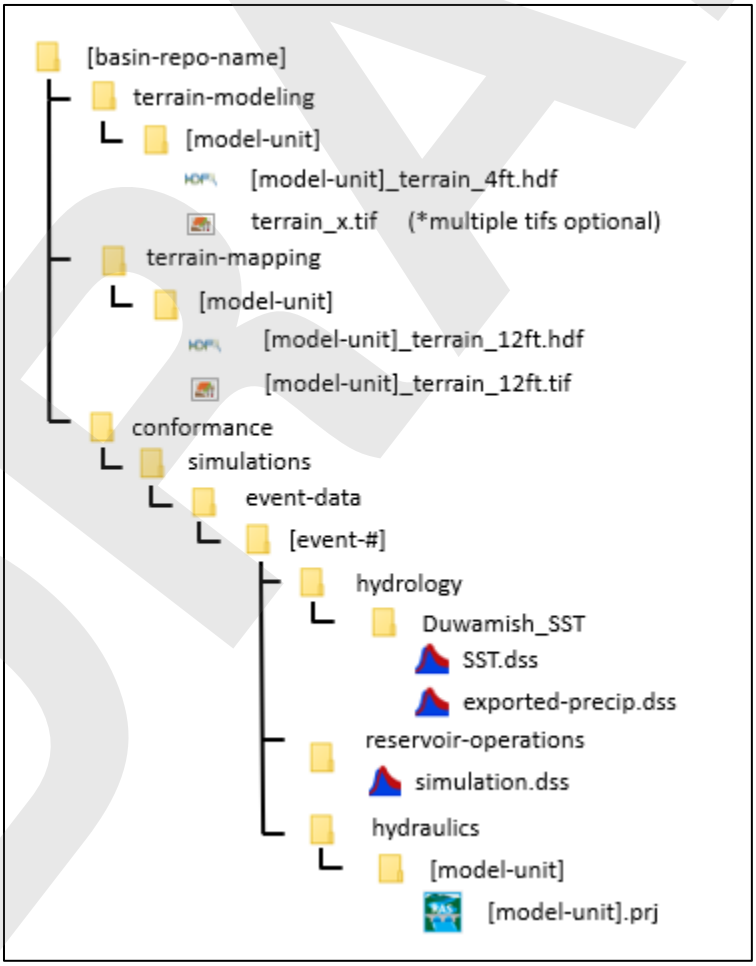
The desktop version of the model, prior to uploading to the cloud, should include internal referencing to input files as shown in Figure 3 below. This structure mimics the directory structure of how the output data from individual events are stored in Model Library by cloud-compute, which allows for efficient viewing and replication of cloud results in later steps.

Note

The terrain is stored at a higher-level directory where all models can share it.

The location to upload models for cloud compute is ***not*** shown in this figure. The actual model upload locations are described separately from this document as the locations will be unique for each model.

Figure 3. Model Library Output Data Directory Structure to be Mimicked on Desktop Prior to Model Upload. Not the Location for Model Upload for Cloud-Compute



1.4 NAMING CONVENTION CONTROL

The FFRD SOP includes robust naming convention criteria to apply to model folders, files, and internal model element names. The naming conventions are critical to control model linkages, perform various QC steps, and reliably access output data among other uses.

Uploading models for cloud computing is the most critical moment to ensure naming conventions follow the standards in the FFRD process.

Although the SOP includes various review steps and checklists to address naming convention issues, there is opportunity for human error. Automated processes to check for adherence to naming conventions using scripts have the potential to reliably catch issues and ensure consistency across the FFRD program.

1.5 TIME WINDOW FOR STOCHASTIC STORM TRANSPOSITION EVENTS

The model integration lead guides the event time window duration selection as described in SOP Volume II. The duration will be long enough to ensure rainfall in the upper basin has sufficient time to travel downstream for the model to reach peak level.

A specific starting date and time is selected for the stochastic storm transposition (SST) events and model preparation for cloud compute, but the date and time will vary for each event within cloud-compute.

1.6 HYDROLOGIC MODELING SYSTEM MODEL DELIVERY

The following sections serve as a checklist on how to transform a calibrated HEC-HMS project and prepare it for conformance and production simulations using cloud-compute.

The synthetic event sampling of the initial states and basin parameter sets are based on the large dataset, basin, and control files representing possible combinations of conditions and parameter sets on each day throughout the period of record used for SST.

1.6.1 SOFTWARE VERSION

The HEC-HMS software version used for model development must be verified and consistent with the version called for in Section 6 of SOP Volume II.

1.6.2 FIRST STEPS

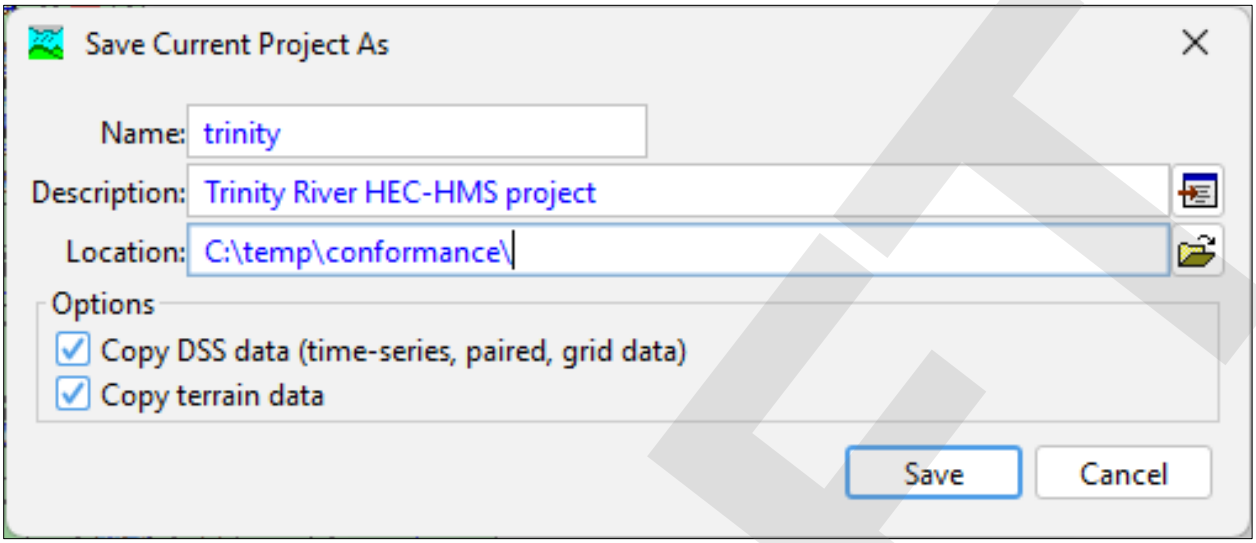
The HEC-HMS modeler or model integration lead will first use the *File>Save As...* option within the HEC-HMS user interface to create a copy of the project. Use the *Copy DSS data (time-series, paired, grid data)* option to internalize any externally referenced HEC-DSS data as shown in Figure 4.

Note

This process does not copy external data like .asc or .tif files.

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Figure 4. HEC-HMS Save Current Project As *Dialog*



1.6.3 MODEL SETTINGS

The following subsections describe modifications that must be made to HEC-HMS model files and project configurations to support stochastic simulations during the conformance and production phases.

1.6.3.1 BASIN MODELS

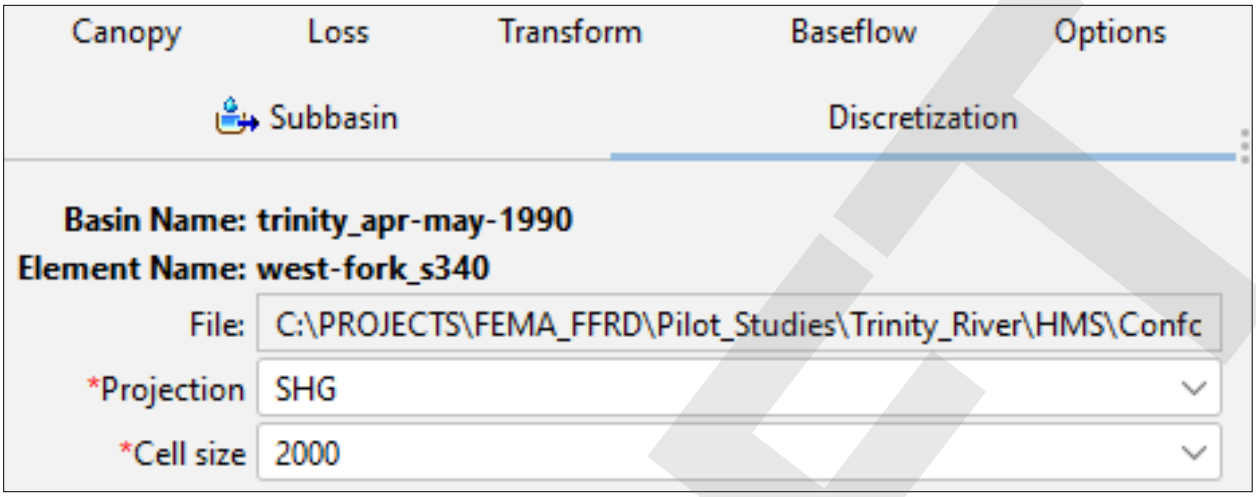
Several modifications to the basin models are required; these models were created during the historic event calibration phase. In general, initial conditions for all basin elements (e.g., subbasins, reaches, and reservoirs) must be specified in absolute, not relative, terms. This enables efficient stochastic sampling and specification of their values.

1.6.3.1.1 SUBBASINS

The Structured Discretization method must be used. The standard hydrologic grid (SHG) projection along with an appropriate cell size (e.g., 2,000 meters) is required as shown in Figure 5.

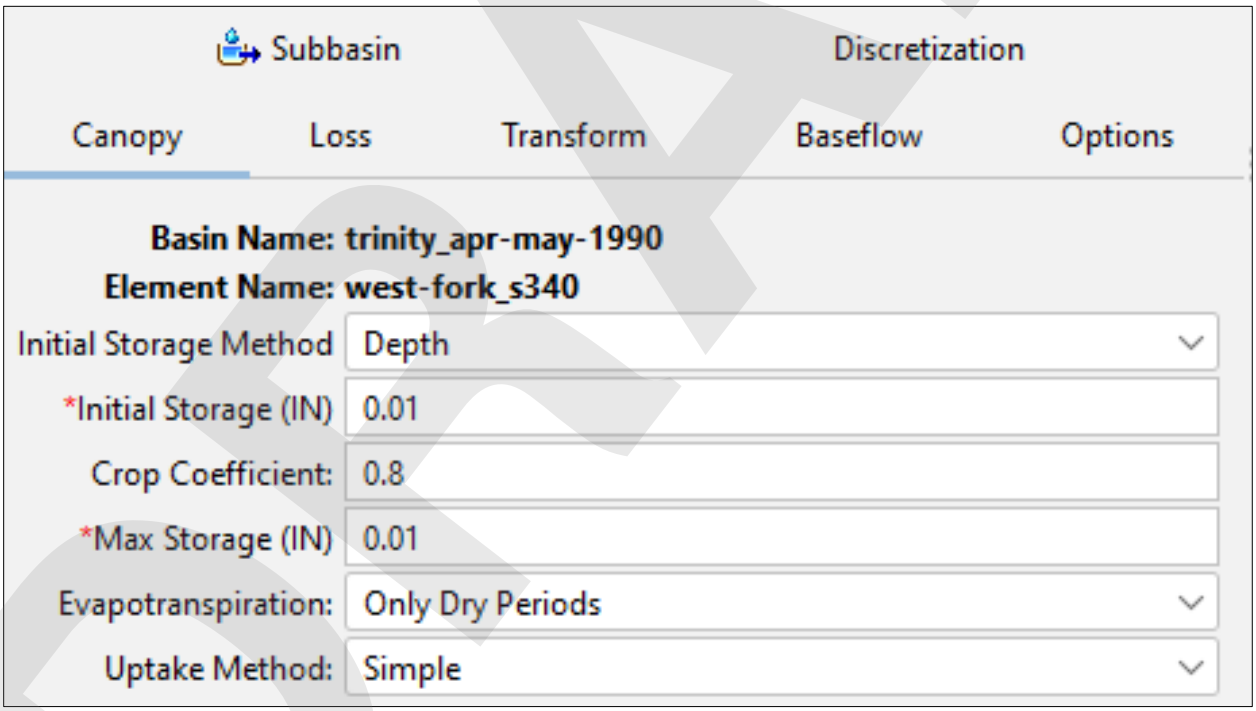
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Figure 5. HEC-HMS Subbasin, Discretization Component Editor



The *Simple Canopy* method and the *Depth Initial Storage Method* must be used. All other parameter selections/values achieved during model calibration should be left unchanged. Figure 6 shows an example of these options.

Figure 6. HEC-HMS Subbasin, Canopy Component Editor



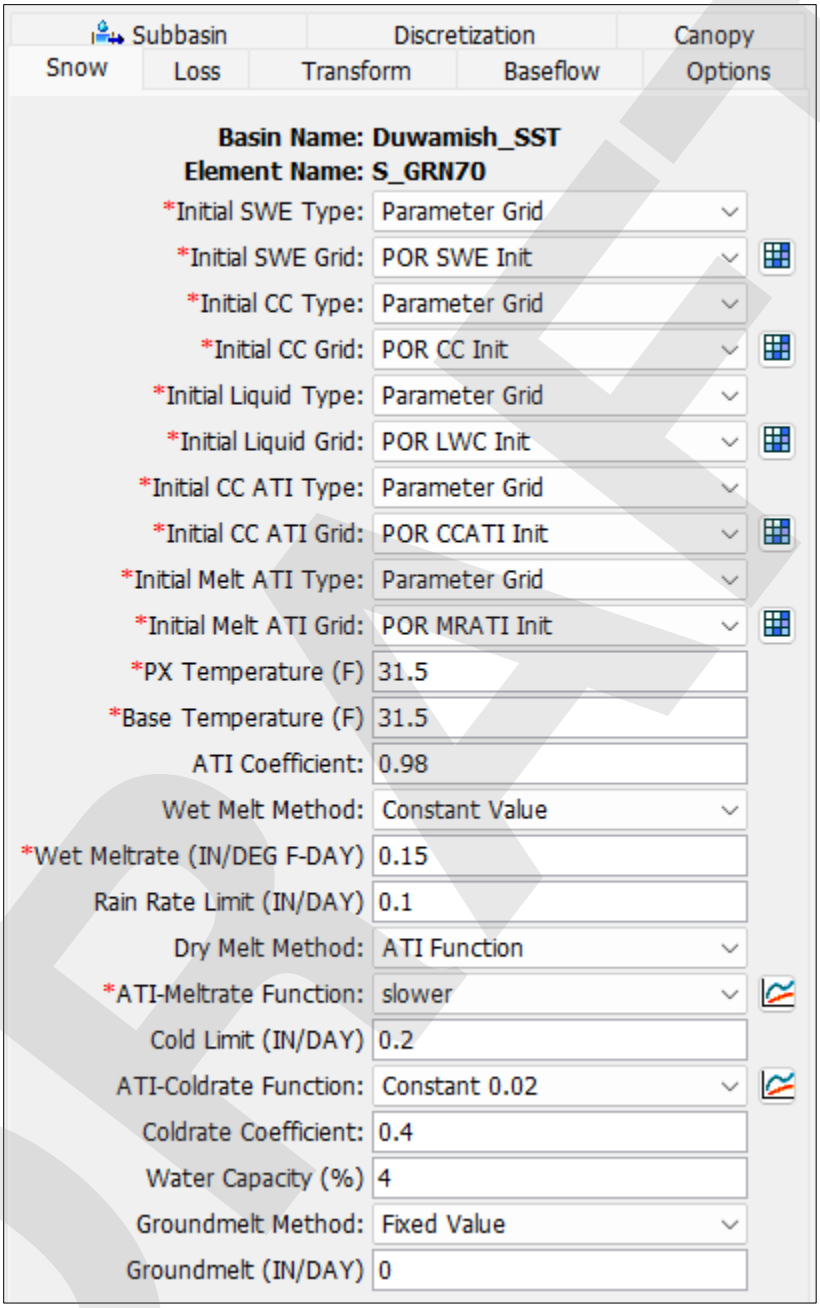
Note

The initial storage depth will be stochastically sampled and set during conformance and production simulations.

If snow accumulation and melting processes are relevant to the watershed of interest, the *Gridded Temperature Index Snow* method must be used (otherwise, this process should be set to *None*). For each

initial state variable, use the *Parameter Grid* option. All other parameter selections/values achieved during model calibration should be left unchanged. Figure 7 shows an example of these options.

Figure 7. HEC-HMS Subbasin, Snow Component Editor



Note

The parameter grids for each state variable will be stochastically sampled and set during conformance and production simulations.

If surface storage processes are relevant to the watershed of interest, the *Simple Surface* method is required (otherwise, this process should be set to *None*). The *Depth Initial Storage Method* must be used.

All other parameter selections/values achieved during model calibration should be left unchanged. Figure 8 shows an example of these options.

Note

The initial storage depth will be stochastically sampled and set during conformance and production simulations.

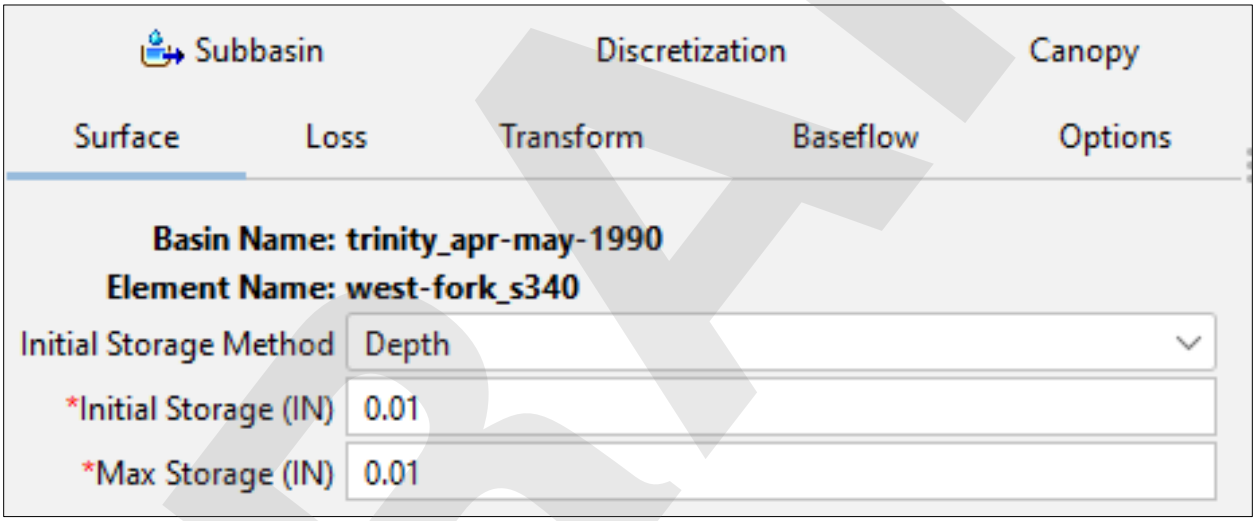
The *Deficit and Constant Loss* method is required. All parameter selections and values achieved during model calibration should be left unchanged. Figure 9 shows an example of these options.

Note

The initial deficit will be stochastically sampled and set during conformance and production simulations.

The *ModClark Transform* method must be used. All parameter selections/values achieved during model calibration should be left unchanged. Figure 10 shows an example of these options.

Figure 8. HEC-HMS Subbasin, Surface Component Editor



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Figure 9. HEC-HMS Subbasin, Loss Component Editor

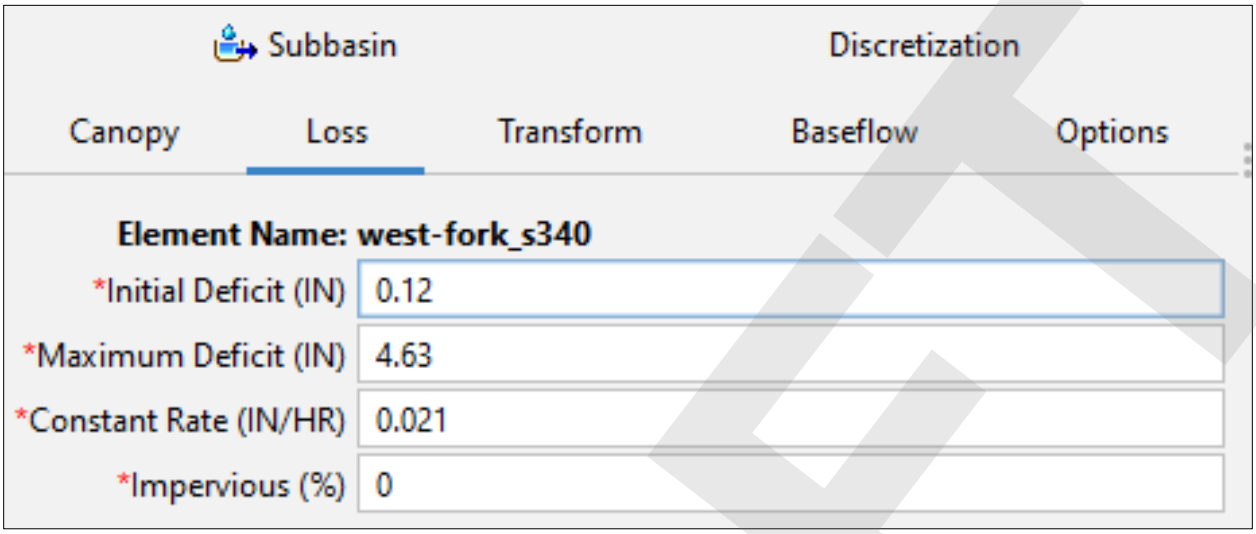
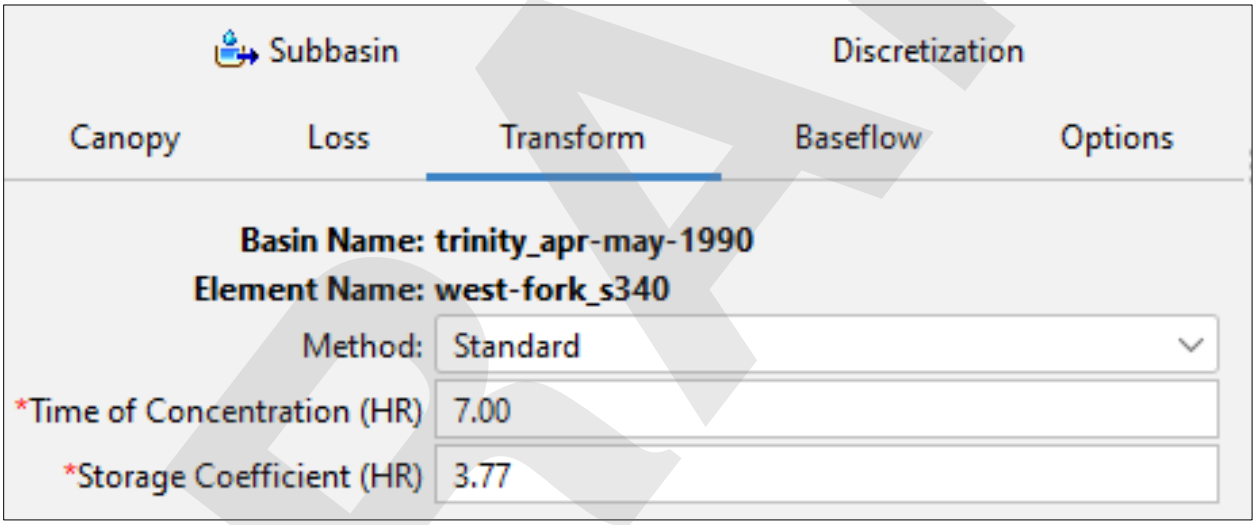


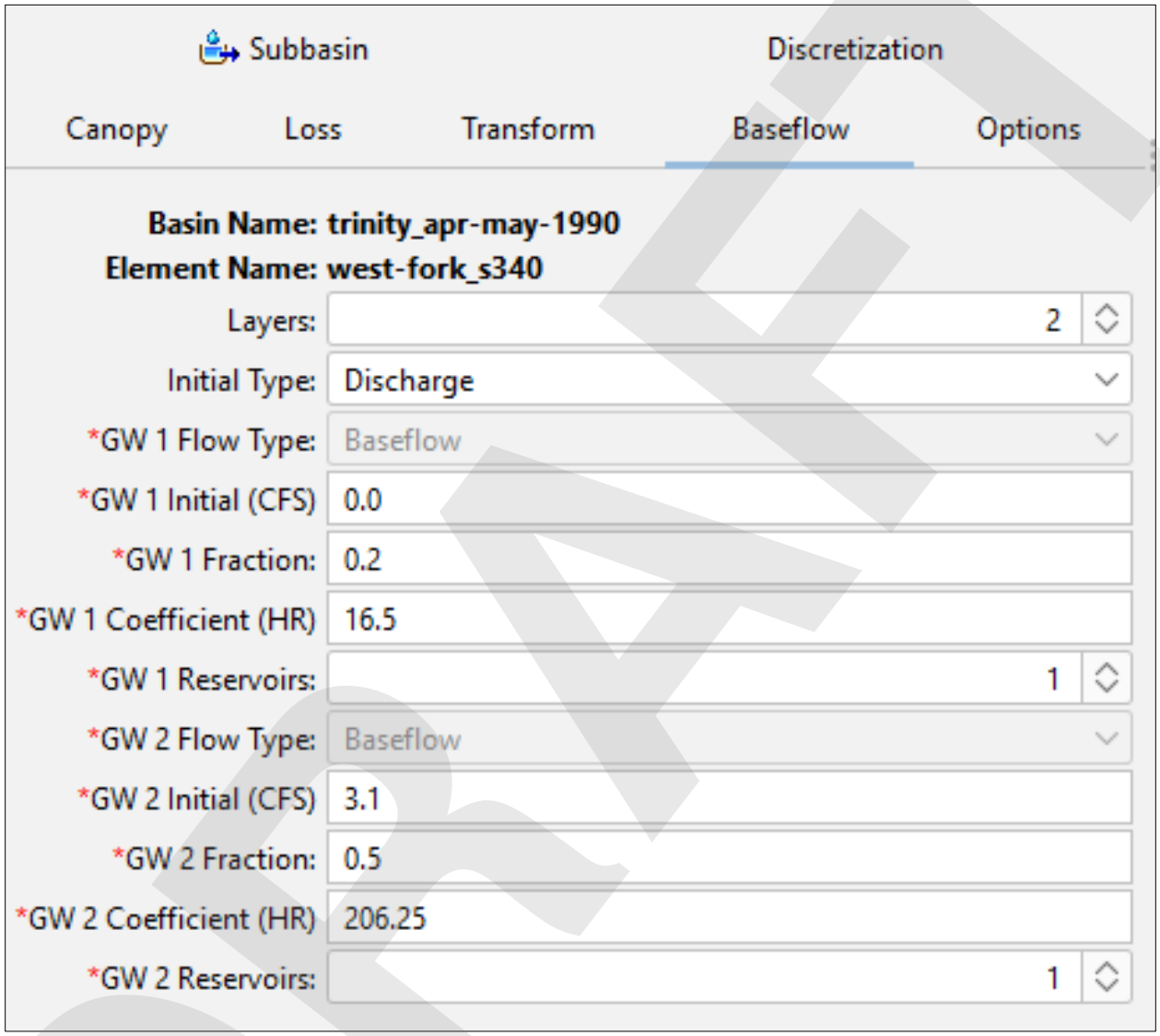
Figure 10. HEC-HMS Subbasin, Transform Component Editor



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The *Linear Reservoir Baseflow* method is required. The *Discharge Initial Type* must be used. All other parameter selections/values achieved during model calibration should be left unchanged. Figure 11 shows an example of these options.

Figure 11. HEC-HMS Subbasin, Baseflow Component Editor



Note

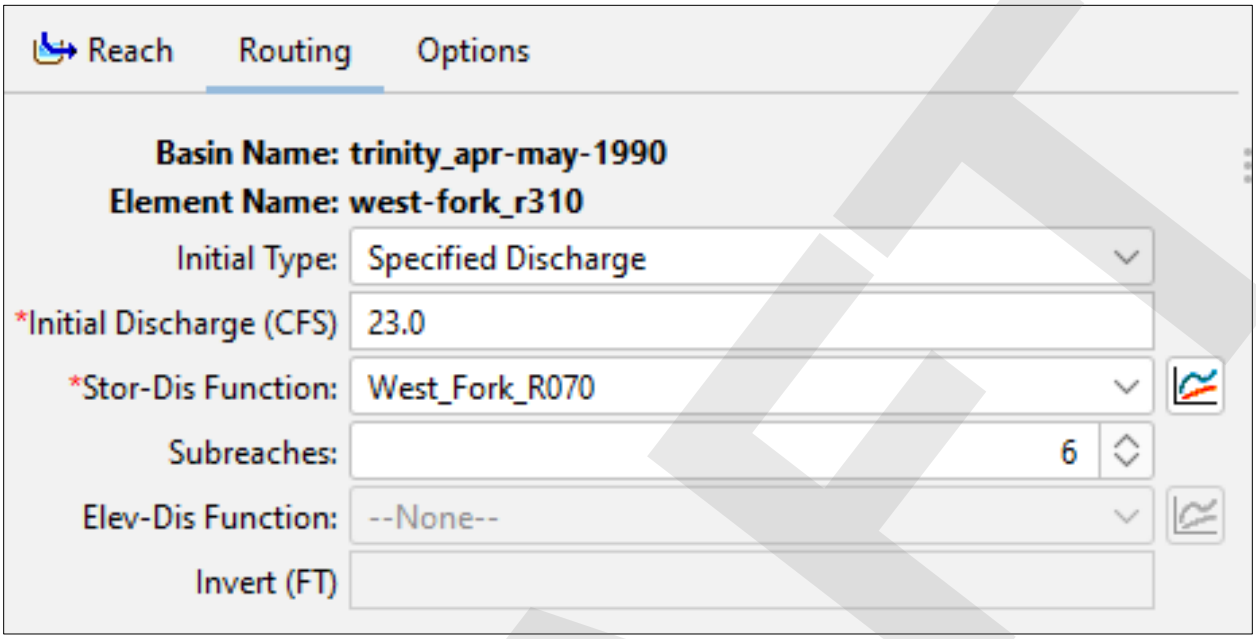
The GW1 and GW2 Initial discharge values will be stochastically sampled and set during conformance and production simulations.

1.6.3.1.2 REACHES

Depending upon the physical processes within the watershed of interest, multiple reach routing methods may be utilized within HEC-HMS FFRD applications. Regardless of the chosen routing method, the *Specified Discharge Initial Type* must be used. All other parameter selections/values achieved during model calibration should be left unchanged. Figure 12 shows an example of these options.

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Figure 12. HEC-HMS Routing Component Editor



Note

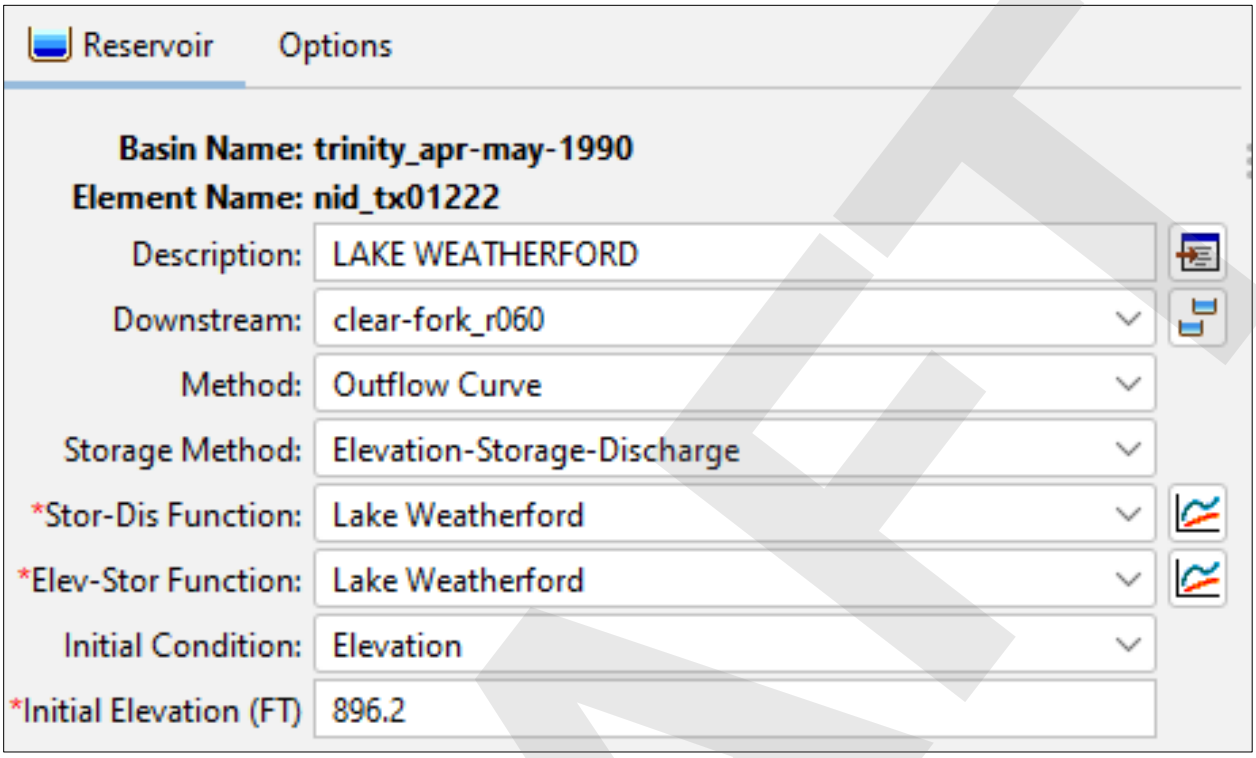
The initial discharge values will be stochastically sampled and set during conformance and production simulations.

1.6.3.1.3 RESERVOIRS

Depending upon the physical processes within the watershed of interest, multiple reservoir routing methods may be utilized within HEC-HMS FFRD applications. Regardless of the chosen routing method, the *Elevation Initial Condition* must be used. All other parameter selections/values achieved during model calibration should be left unchanged. Figure 13 shows an example of these options.

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Figure 13. HEC-HMS Reservoir Component Editor



Note

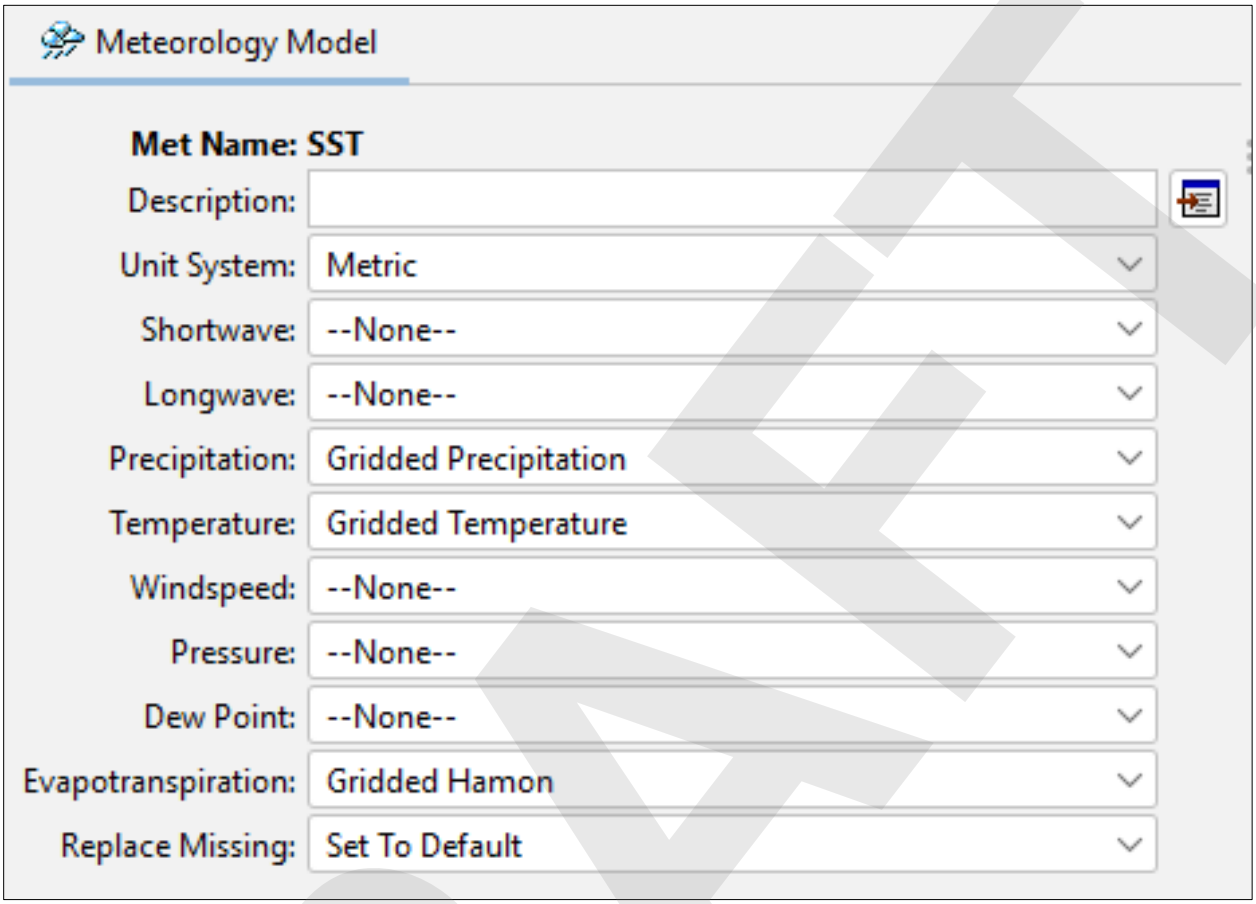
The initial elevation values will be stochastically sampled and set during conformance and production simulations.

1.6.3.2 METEOROLOGIC MODEL

Unlike the basin models, a brand-new meteorologic model named SST should be created for use during the conformance and production phases. The *Metric Unit System* must be specified (this is used when transposing the gridded boundary conditions). For all FFRD applications, the *Gridded Precipitation* method must be selected. If snow accumulation/melt or evapotranspiration processes are relevant to the watershed of interest, the *Gridded Temperature* method must be selected (otherwise, this process should be set to *None*). Similarly, if evapotranspiration processes are relevant to the watershed of interest, the *Gridded Hamon Evapotranspiration* method must be selected (otherwise, this process should be set to *None*). The *Replace Missing* option should be specified as *Set to Default*. Figure 14 shows an example of these options.

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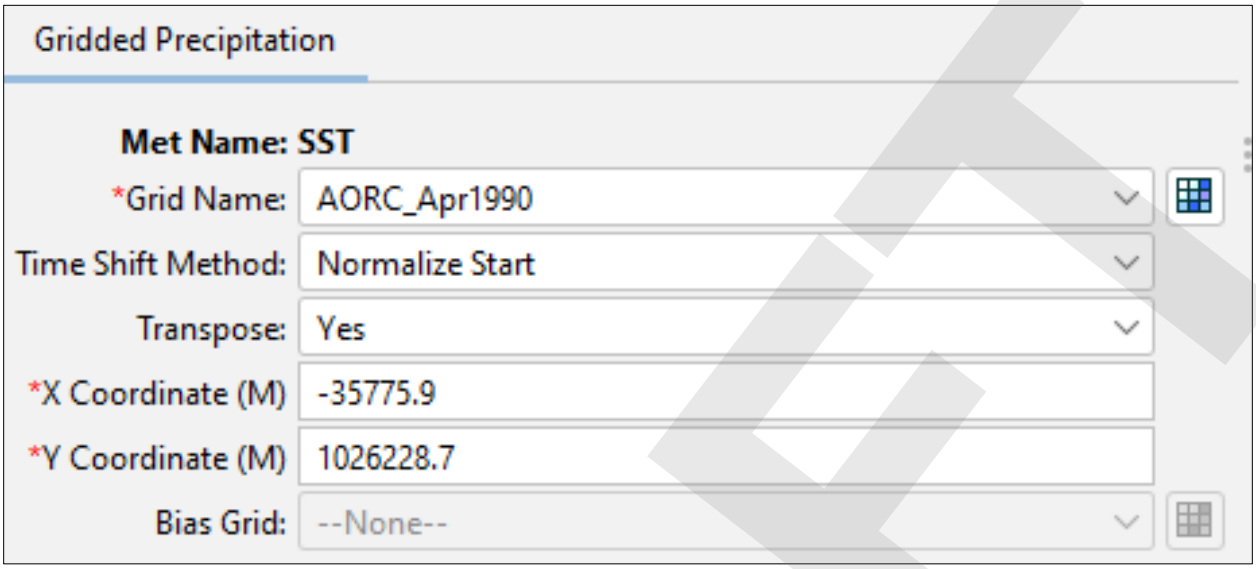
Figure 14. HEC-HMS Meteorology Model Component Editor



Within the Gridded Precipitation node, any valid precipitation grid name can be chosen. These datasets will be stochastically sampled and set during conformance and production simulations. The *Normalize Start Time Shift Method* must be selected. The *Transpose* option should be set to *Yes*. Default values near the middle of the watershed in the basin model's coordinate reference system should be specified within the *X* and *Y Coordinate* entry fields. Figure 15 shows an example of these options.

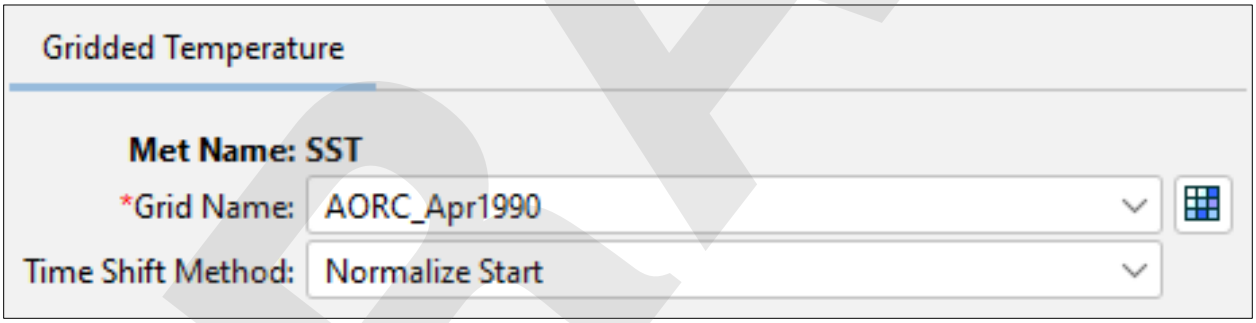
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Figure 15. HEC-HMS Meteorology Model, Gridded Precipitation Component Editor



If the Gridded Temperature method was selected, within the Gridded Temperature node, any valid temperature grid name can be chosen. These datasets will be stochastically sampled and set during conformance and production simulations. The *Normalize Start Time Shift Method* must be selected. Figure 16 shows an example of these options.

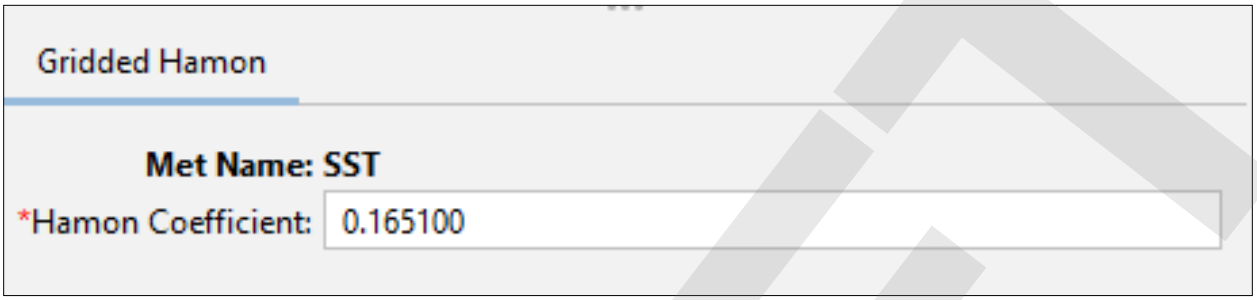
Figure 16. HEC-HMS Meteorology Model, Gridded Temperature Component Editor



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If the Gridded Hamon method was selected, all parameter selections/values achieved during model calibration should be left unchanged within the Gridded Hamon node. Figure 17 shows an example of these options.

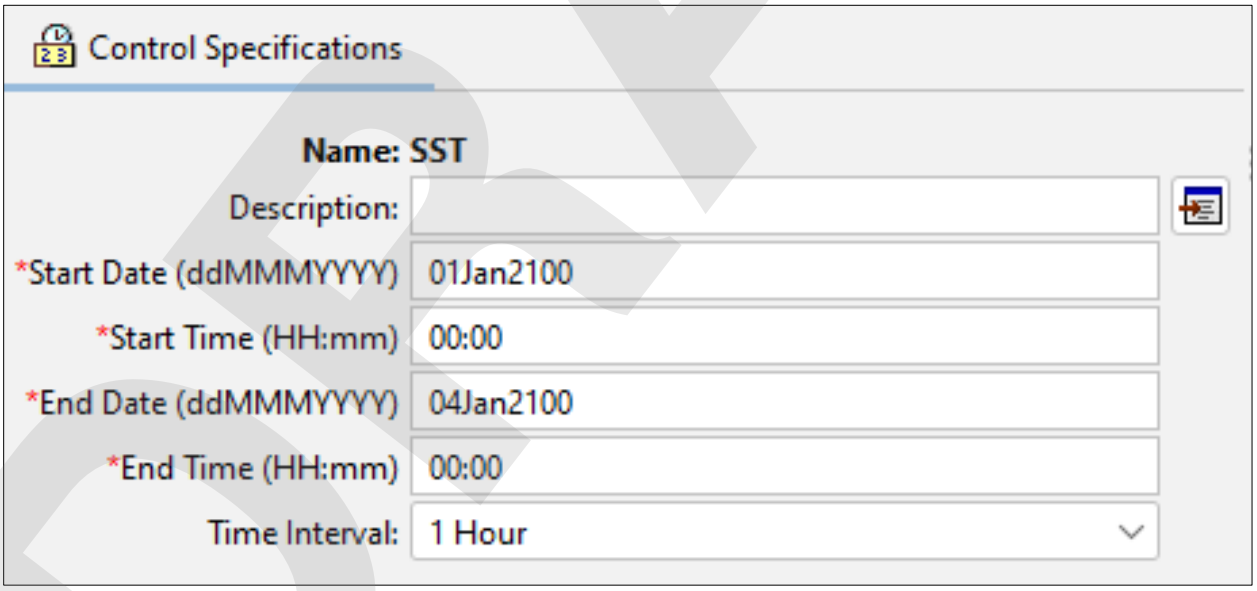
Figure 17. HEC-HMS Meteorology Model, Gridded Hamon Component Editor



1.6.3.3 CONTROL SPECIFICATIONS

Similar to the meteorologic model, a brand-new control specifications component named SST should be created for use during the conformance and production phases. *01Jan2100* should be entered within the *Start Date* entry field. *00:00* should be entered within the *Start Time* entry field. *04Jan2100* should be entered within the *End Date* entry field. *00:00* should be entered within the *End Time* entry field. Depending upon the physical processes within the watershed of interest, multiple computational timesteps may be utilized within HEC-HMS FFRD applications. Figure 18 shows an example of these options.

Figure 18. HEC-HMS Control Specifications Component Editor

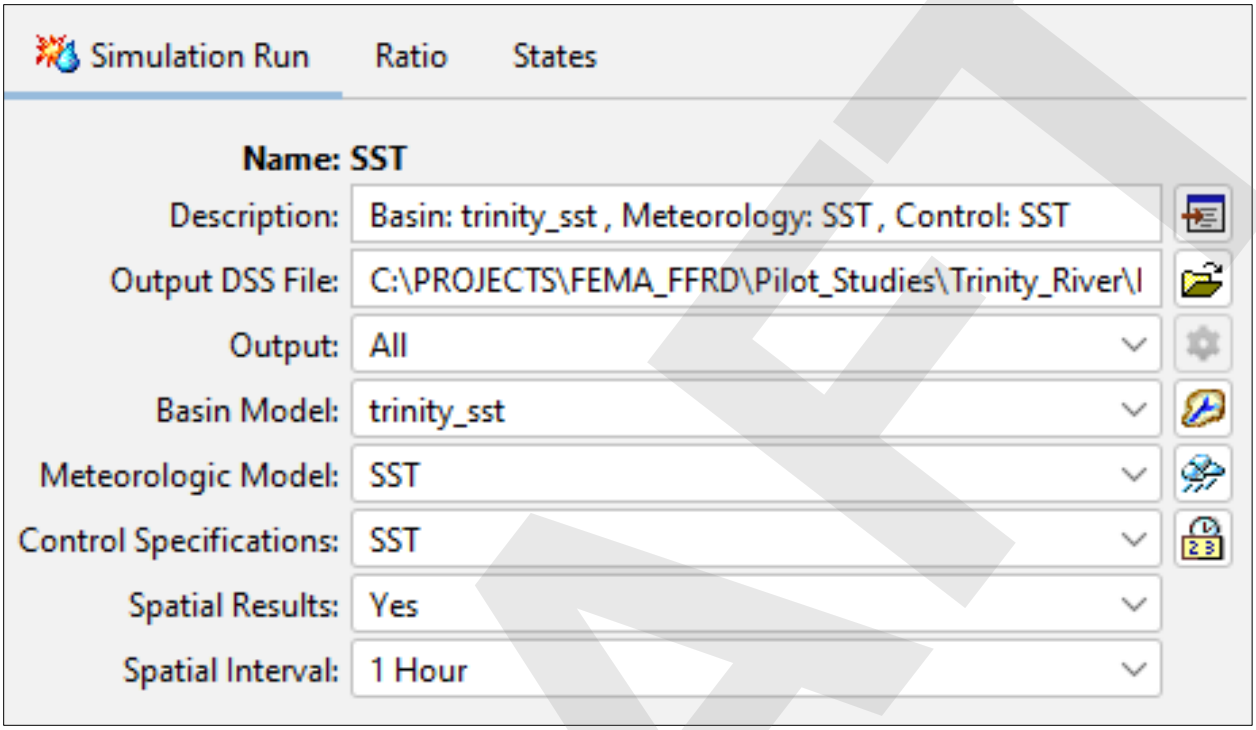


1.6.3.4 SIMULATION RUNS

Similar to the control specifications, a brand-new simulation run named SST should be created for use during the conformance and production phases. Any basin model can be selected within this simulation as they will be updated during the conformance and production simulations. The SST meteorologic model and SST control specifications (which were previously created) should be selected. The *Spatial Results*

option should be set to *Yes* and the *Spatial Interval* should be set to *1 Hour*. Figure 19 shows an example of these options.

Figure 19. HEC-HMS Simulation Run Component Editor



The *No Ratio* method should be selected on the *Ratio* tab. Additionally, the *No Save States* option should be selected on the *States* tab.

1.6.4 IMPORTING PRECIPITATION DATA (STORM CATALOG)

The tool used for generating storm catalogs produces a .grid file that follows the specification of a normal HEC-HMS .grid file that links gridded datasets to a project. This allows a list of entries for hundreds of storms to be quickly copied and pasted into the project, effectively importing them without needing to manually create hundreds of gridsets in HEC-HMS.

For the steps below, the *project.grid* file is the .grid file that exists in the HEC-HMS project and matches its name. It contains all the grid data used in calibration and POR simulations. The *catalog.grid* file was created by StormCloud/StormHub and only contains the precipitation data for storms in the storm catalog.

The instructions below assume use of Notepad++ to edit the files.

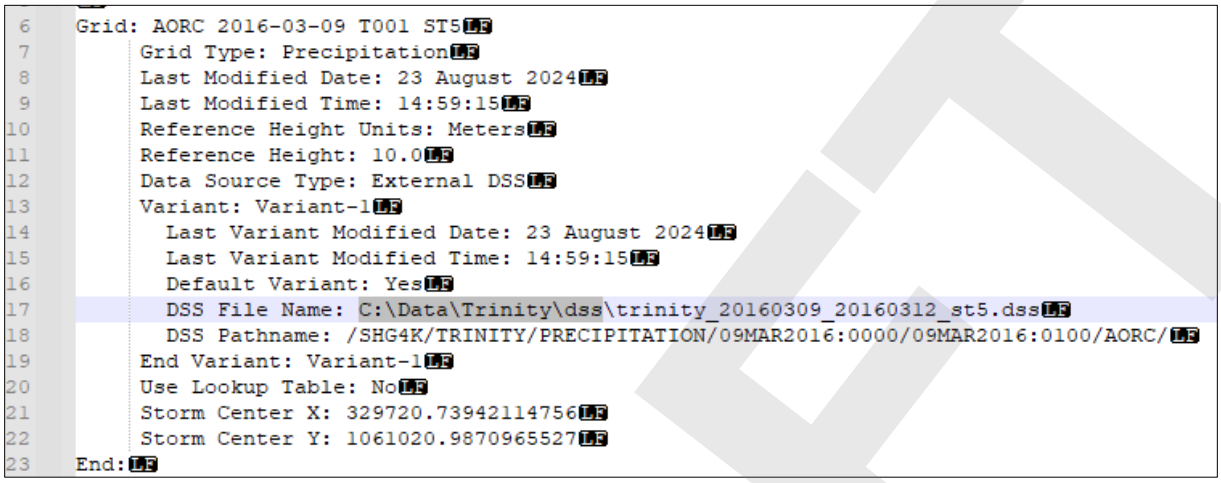
Make sure the HEC-HMS project is closed before editing any internal files.

1. Copy (or move) the storm data .dss files to the project *data* folder in a subdirectory called catalog.
2. Make a backup copy of the catalog.grid file and give it the file extension *.gridbak* with the same filename as the original catalog.grid file.
3. Open the original (not backup) catalog.grid file in a plaintext editor.
4. Find and replace file paths to point to the project's catalog folder.

    a. Scroll down to the first grid data entry in the catalog.grid file and find the *DSS File Name* entry.
    b. Highlight the portion of the file path up to the last backslash (just before the file name), as

shown in Figure 20.

Figure 20. Section of Text to Replace within .grid File



c. Press Ctrl+H to open the Replace dialog. Note that it automatically places the highlighted text in the *Find what* field.
d. In the *Replace with* field, enter *data\catalog*.
e. Click Replace All on the right side.

5. Save the catalog.grid file but leave it open.
6. Make a backup copy of the project.grid file and give it the file extension .gridbak with the same filename as the original project.grid file.
7. Open the original (not backup) project.grid file in a plaintext editor in a new window.
8. Scroll to the bottom of the project.grid file and find the last grid entry. Add a blank line after the *End:* tag for the last grid, and then press Enter again to go to the following line.

    a. Each entry must be separated by a blank line.

9. From the catalog.grid file, copy all the storm grid entries and paste them into the project.grid file below the blank line that was added.

    a. Make sure not to copy the catalog.grid file's header (the first four lines beginning with “Grid Manager: ...”)

10. Save the project.grid file. Close both .grid files.

When the HEC-HMS project is re-opened, hundreds of new gridsets are listed in the Watershed Explorer.

1.6.5 MODEL CLEANUP

The HEC-HMS project used during the conformance and production phases should be as lean as possible because it will be copied many times and reproducing unnecessary data is a waste. Below are several things that help reduce the size of the SST project. Most of the cleanup can happen within the HEC-HMS user interface (e.g., deleting basin models) but there are hints below for when deleting a large number of entries using the HEC-HMS input files is faster (e.g., within Notepad++). Always be careful when editing HEC-HMS input files and make a backup before doing so.

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1.6.5.1 BASIN MODELS

First, remove any basin models except for the finalized event calibration basin models. Also, remove any linkages to observed data (e.g., flow, stage, SWE) on the Options tab (set to none) of all basin elements (e.g., subbasins, junctions, reaches, etc.) as this information will not be utilized during stochastic simulations.

1.6.5.2 METEOROLOGIC MODELS

Next, remove any meteorologic models except for the SST meteorologic model created in the steps above.

1.6.5.3 CONTROL SPECIFICATIONS

Also, remove control specifications except for SST from the project.

1.6.5.4 TIMESERIES DATA

It is unlikely that any timeseries data will be necessary if all observed data is unlinked, gridded processes are used, and no specified discharge elements (reservoirs, sources, etc.) are in the model. If certain timeseries data is necessary, do not delete it.

The following process removes the links from the HEC-HMS project, but DSS files will need manual cleanup. The DSS files that contain the timeseries data were copied over during the initial Save As step. Use the File>References tool and filter to Time Series Data to see where these data are on disk and delete them manually. They should have been internalized to the copied project's data folder and are safe to delete because they are copies of the original.

1. Make a backup copy of the project.gage file that exists in the HEC-HMS project and matches its name. Give it the file extension .gagebak with the same filename as the original project.gage file.
2. Open the original (not backup) project.gage in a plaintext editor.
3. Delete everything below the file header (first 5 lines).

1.6.5.5 PAIRED DATA

Unlike timeseries data, it is very likely that several paired data objects are utilized within the basin models that will be used during the stochastic simulations. Thus, care needs to be taken to organize these paired data objects in such a way that any errors or compute issues can be efficiently investigated during the conformance and production phases.

Begin by culling any unused paired data objects (i.e., storage-discharge, elevation-storage, etc.) within the HEC-HMS project. Then, for the remaining paired data references that utilize external DSS files, combine all records into a single DSS file. This paired data DSS file should be placed within the project's data folder. For any paired data references that utilize the Manual Data Entry method, these records can remain unaltered.

1.6.5.6 SIMULATION RUNS

Remove any simulation runs or any other computes (e.g., Uncertainty Analysis, Optimization Trial, etc.) from the project except for the SST run created. Other compute types can be removed by deleting the corresponding directory within the base project directory (with the HEC-HMS project closed):

- Delete the montecarlo directory to delete all Uncertainty Analyses.
- Delete the optimizer directory to delete all Optimization Trials.

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- Delete the ensemble directory to delete all Ensemble Analyses.
- Delete the forecast directory to delete all Forecast Alternatives.
- Delete the frequency directory to delete all Frequency Analyses.

1.6.5.7 DSS FILES IN THE PROJECT DIRECTORY

There will be some straggler files associated with now-deleted simulations to clean up. Delete any DSS files with a name that matches a meteorologic model that was removed. These files contain a met data cache and can be removed. Verify this is the case by opening the DSS file in DSSVue and verifying:

- It only contains meteorologic data.
- The F-part for all paths contains "MET:xxxxx" where "xxxxx" is the name of the met model.

Delete any DSS files with a name that matches a simulation run that was removed. These files contain outdated results and can be deleted. Verify this is the case by opening the DSS file in DSSVue and verifying:

- It only contains simulation results.
- The F-part for all paths contains "RUN:xxxxx" where "xxxxx" is the name of the simulation run.

DSS files with names that match other simulation types (e.g., Uncertainty Analysis, Optimization Trial, etc.) can be deleted as well.

1.6.5.8 OTHER FILES IN THE PROJECT DIRECTORY

All files with the .out and .log extensions can be safely deleted. Additionally, all .sqlite files that do not match one of the basin models that remain within the HEC-HMS project can be deleted. Finally, the results directory can be safely deleted.

1.6.5.9 CLEANING UP GIS DATA

There will not be modification of the geospatial information for any basin element, so the source or processed terrain data will no longer be necessary. Thus, the gis, maps, and terrain directories should be deleted.

1.6.6 DEVELOPING CONTINUOUS INITIAL SNOW STATES

This process is only required for projects that utilize the Gridded Temperature Index snow method:

1. First, re-run the period of record with full spatial results turned on to get a full set of gridded state variables for snow.
2. Second, sanitize the output gridded datasets using the Vortex | Sanitizer tool.

1.6.7 HEC-HMS MODEL DELIVERY

The prepared and cleaned up model files must be delivered by posting them to the appropriate location on the project directory. There are two distinct deliveries associated with the HEC-HMS model delivery: the runnable model and template files for use by cloud-compute tools to generate input files that account for variability in antecedent conditions for each stochastic event.

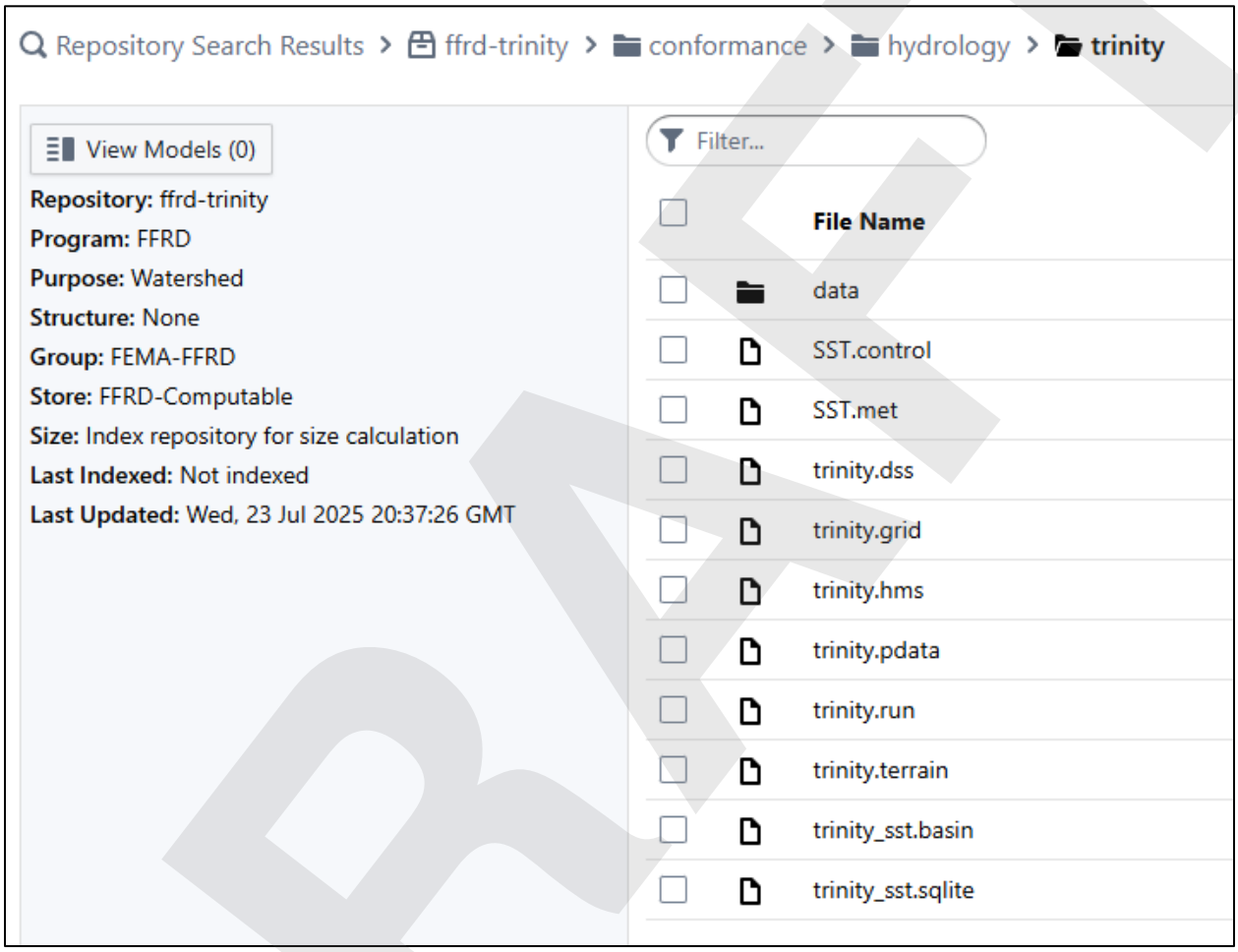
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1.6.7.1 BASE MODEL

Post the prepared and runnable HEC-HMS models to the following location within the project directory:

```
Delivery Location: conformance > hydrology > basin-name
```

Figure 21. Example Delivered HMS Model Dataset



1.6.7.2 ANTECEDENT CONDITION TEMPLATE FILES

The basin-record-gen-plugin cloud-compute actions take HMS template files to create an array of .basin and .control files to represent the variation in antecedent conditions for stochastic events. The input template files to be delivered are:

- Calibrated period-of-record HEC-HMS DSS output
- Individual .basin files from each calibration event
- Individual .control files from each calibration event

The template files that will be used by cloud compute will be posted to the location below.

```
Delivery Location: conformance > basin-generation
```

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1.7 HEC-RESSIM DELIVERY

The following sections serve to describe how to prepare and deliver the HEC-ResSim FFRD model for conformance phase and production phase runs using cloud compute.

1.7.1 SOFTWARE VERSION

The HEC-ResSim version used for model development must be verified and consistent with the version called for in Section 7 of SOP Volume II.

1.7.2 CREATE THE CONFORMANCE RUN TEMPLATE

A single alternative and a single simulation shall be the starting point for the conformance model:

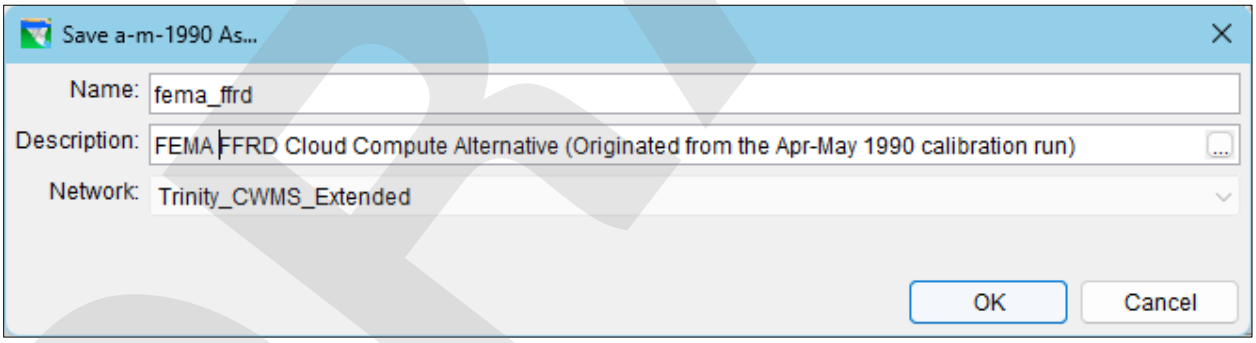
- Conformance alternative name="fema_ffrd"
- Conformance simulation name="fema_ffrd"

1.7.2.1 CONFORMANCE ALTERNATIVE

The simplest way to create the "fema_ffrd" alternative is to make a copy of one of the existing calibration alternatives.

1. From the Reservoir Network Module, select the Alternative Editor, click on one of the calibration runs, and use the Save As… option.
2. Save as *fema_ffrd* and provide an informative Description such as, "FEMA FFRD Cloud Compute Alternative – Originated as a copy of April 1990 calibration alternative for use with FFRD cloud computes".

Figure 22. Step 2 Window



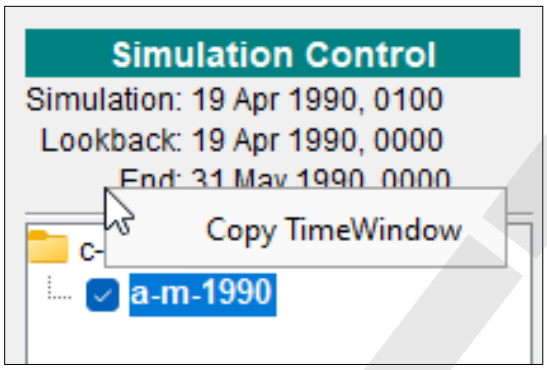
3. No other changes should be necessary, but verify that the appropriate timestep and flow computation method are selected. The inflow mapping must be consistent with the simulation date range and timestep.

1.7.2.2 CONFORMANCE SIMULATION

Next, move to the Simulation Module and create the new "fema_ffrd" simulation.

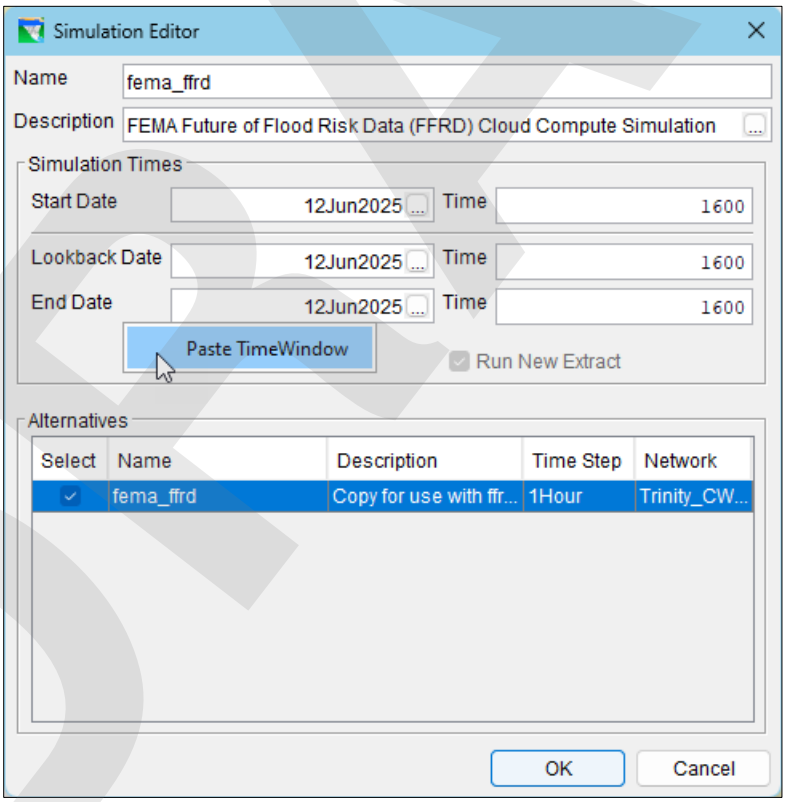
1. As previously stated, it is easiest to start with the calibration simulation that was paired with the alternative used to create the fema_ffrd alternative, because the simulation dates must match the available data. First, open the matching calibration simulation, and use the right-click shortcut menu to copy the Simulation Time Window.

Figure 23. Step 1 Window



2. Use the Simulation>Create a New Simulation option to create the fema_ffrd simulation, and give it an informative description such as, “FEMA Future of Flood Risk Data (FFRD) Cloud Compute Simulation.”
3. Use the right-click Paste TimeWindow option to make the simulation time window match the currently available data. The cloud compute modeler will automate the updates to these dates to match what is sampled during the production runs.
4. Run the simulation to confirm that nothing has broken, and note the total compute time for the report.

Figure 24. Step 4 Window



1.7.3 WITHIN-MODEL CLEANUP

As with HEC-HMS, the HEC-ResSim watershed used during the conformance and production phases should be as lean as possible because it will be copied many times and reproducing unnecessary data is a waste. While some model cleanup was done during the calibration modeling, the following steps shall be taken to ensure the most compact, streamlined model for conformance and production.

Note

Always be careful when directly editing HEC-ResSim model files and make a backup before doing so.

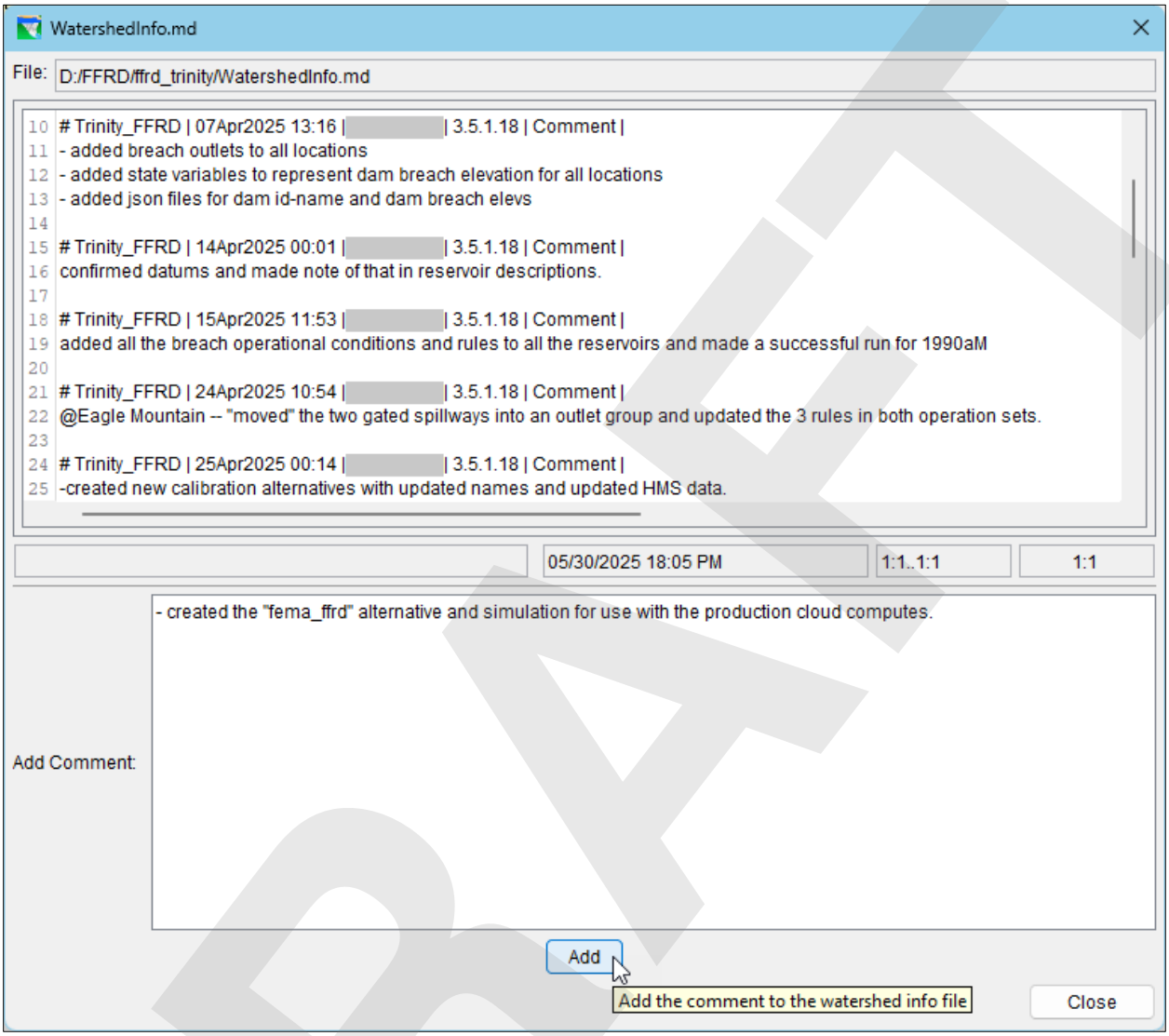
These instructions assume that some cleanup and thorough documentation occurred during earlier phases (e.g., all unnecessary networks, scripts, and particularly messy or disorganized pieces of the model were removed).

1.7.3.1 CREATE A ZIPPED BACKUP MODEL

Before zipping up the model, make sure the current state has been fully documented. Use the File>Watershed Info feature to log a comment about the current state of the model.

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Figure 25. Zipped Backup Model Window



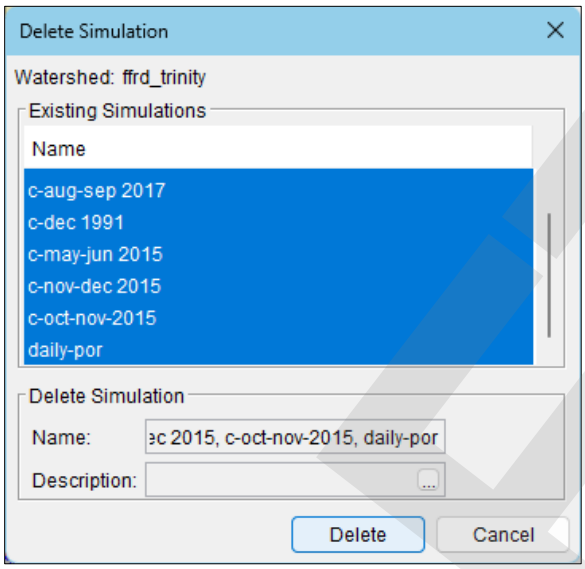
Then completely close the model and all associated files before creating a .zip file backup. Now that there is a backup of the model, delete as many superfluous files as possible.

1.7.3.2 REMOVE ALL SIMULATIONS EXCEPT *FEMA_FFRD*

Reopen the model and go to the Simulation Module. Delete all simulations except fema_ffrd, using the Simulation>Delete Simulation option.

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Figure 26. Delete Simulation Window



1.7.3.3 DELETE ALL ALTERNATIVES EXCEPT *FEMA_FFRD*

From the Network Module, Alternative Editor, delete all but the fema_ffrd alternative. There is no bulk delete option, but Ctrl+D can be used to make this process slightly faster.

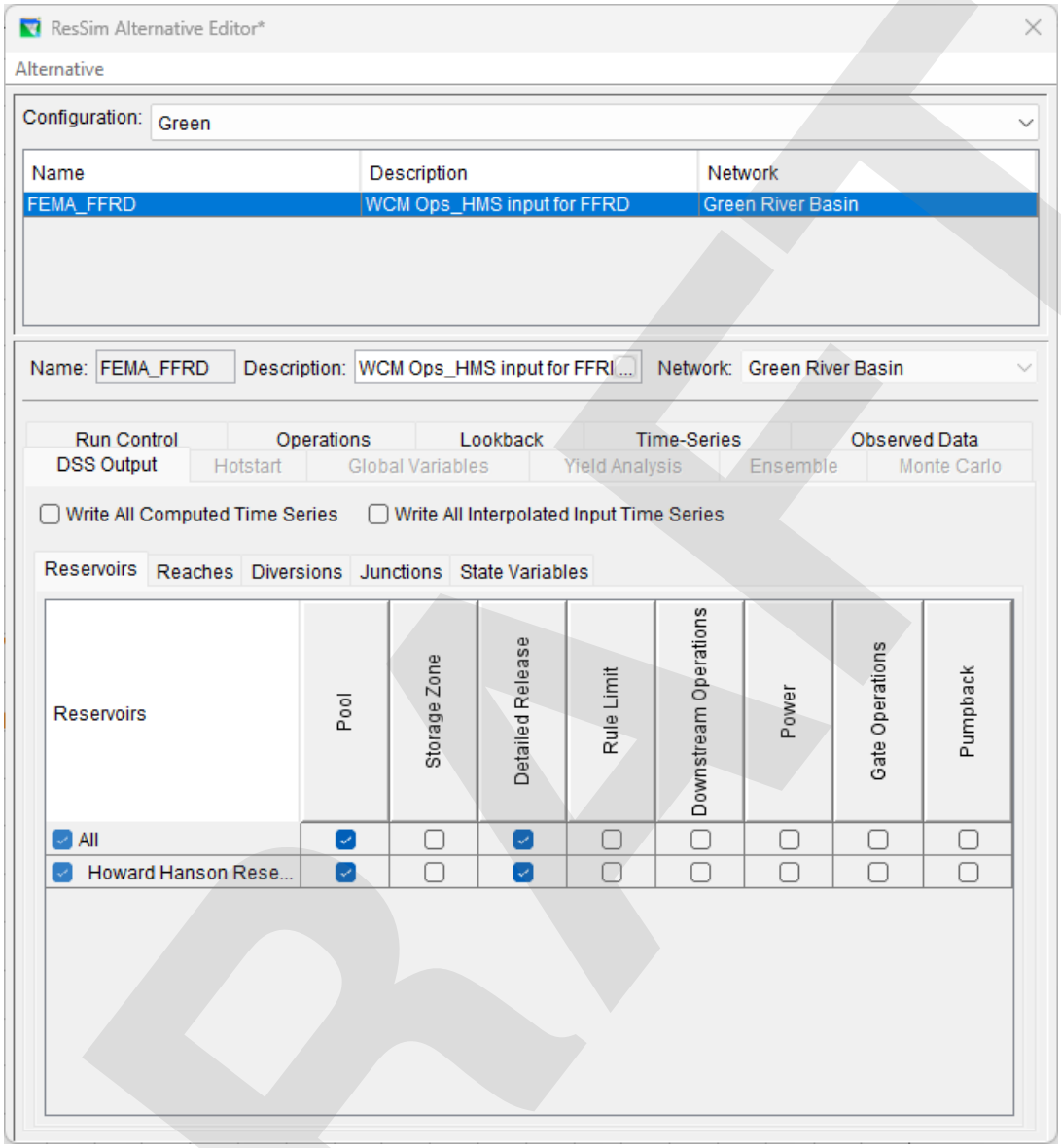
1.7.3.4 REMOVE ALL UNNECESSARY OUTPUTS IN THE FEMA_FFRD ALTERNATIVE

Generally, only simulated flow and elevation outputs are required to be saved. The outputs that are required for FFRD simulations either feed directly to a HEC-RAS model, aid in identification of interesting events (event filtering), or provide the required data to develop a relevant frequency curve for conformance metrics.

The DSS Output tab in the alternative writes all computed and interpolated timeseries by default. Prior to handoff, the user should uncheck both in the Alternative Editor. The user should also go a step further and uncheck additional items on the subtabs. On the Reservoir tab, the user should only need the Reservoir Pool and flows, so all other outputs can be unchecked. The user can also uncheck all outputs on the Reaches and Diversion tabs. For Junctions and State Variables, the user may want to uncheck some outputs; however, it is recommended that any record that may be referenced during validation (such as downstream control points or data handoff points between models) should remain selected.

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Figure 27. Alternative Editor Window

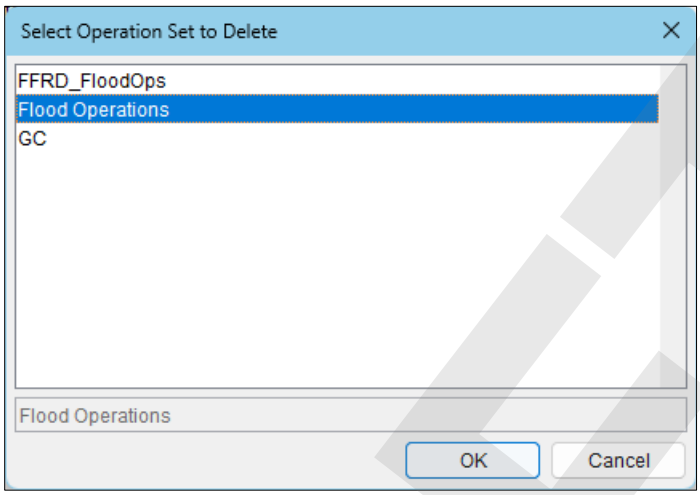


1.7.3.5 EXCESS RULES AND OPERATION SETS

Rules and operation sets create very little extra bulk to model file size but can create clutter and confusion. To alleviate this, confirm that the intent of each rule was included in its description during the calibration phase and that the operation sets used for FFRD runs are clearly marked. Cleaning up unneeded operation sets from a model with 16 reservoirs, each with 3 operation sets, required 208 clicks but did not result in any noticeable file size reduction, so the modeler should focus on tidiness in this case and may leave extra operation sets in the model without causing harm.

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Figure 28. Operation Set Deletion Window



1.7.3.6 DELETE UNNECESSARY MAP LAYERS

From the View>Layers menu in any module, remove any map layers that are unneeded. Do not remove the Stream Alignment or Study layers or associated subparts.

1.7.4 MANUAL MODEL CLEANUP

Next, remove remaining excess files via the Windows File Explorer or similar. Document changes in the WatershedInfo file and close the watershed and associated files. The final cleanup step is the most dangerous. As a precaution, zip a copy of the model again.

1.7.4.1 MAP FILES

Remove unneeded map shapefiles from the *maps* directory via the File Explorer. For example:

- highways
- project points

1.7.4.2 SHARED DIRECTORY FILES

The watershed *shared* directory holds inputs and other shared files. Remove the files that are not needed.

1.7.4.3 NETWORKS, ALTERNATIVES, SIMULATIONS

Remove remnant files associated with the networks, alternatives, and simulations already deleted from the HEC-ResSim GUI. Table 1 lists files to consider removing from the rss subdirectory.

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Table 1. RSS Subdirectory Removable Files

File Extension	Keep	Remove
bak	Any associated with fema_ffrd, ffrd_{basin-name}.*.bak, stream.align.bak, WatershedInfo.md.bak	Everything else
conf	rss.conf	
dbf	resop.dbf ressysop.dbf {network-name}_ExtendedOp.dbf {network-name}_ExtendedSysOp.dbf	Files associated with old networks
dbt	resop.dbt ressysop.dbt {network-name}_ExtendedOp.dbt {network-name}_ExtendedSysOp.dbt	Files associated with old networks
dss	{network-name}_Extended.dss {network-name}_Extended-GlobalVariablesTables.dss	Files associated with old networks
fits	Two files associated with each alternative	All other fits files
igv	one file associated with each alternative	All other igv files
malt	one file associated with each alternative	No other files existed
mcscript	one file associated with each alternative	All other mcscript files
ralt	one file associated with each alternative	No other files existed
rssrun	none	No rssrun files needed for ResSim standalone
simperiod	One file associated with each simulation	No other files existed
simrun	One file associated with each simulation	
xml	none	no xml reports were needed
Xsd, xlt	None	no AlternativeInput report files needed

1.7.5 HEC-RESSIM SCRIPT CONSIDERATIONS

HEC-ResSim models with scripts may be flagged as a security risk within the Civil Works Business Intelligence (CWBI) cloud, and no JAR files or Python modules will be accepted on CWBI. To address this concern, modelers will clean existing models of scripts by reviewing each script function and replacing it wherever possible with simpler operations. If scripted operations (rules or state variables) are necessary, the modeler will conduct a careful review and write justification for the CWBI security team to support the need for the script's inclusion on CWBI. Modelers should keep in mind that scripted rules and state variables will not be calculated during the HEC-ResSim lookback period, and this can impact overall model results. Scripts must be structured to only require the data available within the event.

1.7.6 HEC-RESSIM MODEL DELIVERY

Open the watershed and verify that each module still functions correctly. Make one last note in the *WatershedInfo.md* file of everything that was deleted. Post the model to the Model Library.

```
Delivery Location: conformance > reservoir-operations > [basin-name]
```

If the model is in zipped format prior to uploading to the Model Library, use the unzip option in the Model Library so that the model is posted in an unzipped format. Note that for HEC-ResSim, uploading in zip format may be required to avoid firewall filtering of specific files associated with the model.

Figure 29. Zip File Upload on Model Library



1.8 HEC-RAS DELIVERY

A final calibration phase HEC-RAS model for specific historic storm events is the starting point for the conformance phase.

Various changes are made to prepare the model for generalized synthetic storm events driven by SST before they are posted to the Model Library in preparation for cloud-compute, which are described in the sections below.

1.8.1 SOFTWARE VERSION

The HEC-RAS software version used for model development must be verified and consistent with the version stated in Section 8 of SOP Volume II.

1.8.2 PLANS AND NAMING CONVENTIONS

The HEC-RAS model is delivered with a single plan file and single flow file with generalized naming conventions to accept SST events that will not be confused for actual observed events. The following changes are made to the original calibrated model names:

- Project and Filenames: [model-unit]
- Geometry Name: [model-unit]
- Plan Name: SST
- Plan Short Name: SST

1.8.3 PLAN FILE

The following adjustments should be made on the desktop version of the model plan file.

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1.8.3.1 TIME WINDOW

The simulation time window selected for SST events, described in Section 1.5 above, will be used for the generalized SST event HEC-RAS time window. It will match the time window used in HEC-HMS and HEC-ResSim.

Note

The time window is somewhat imperfect in that initial conditions in the HEC-RAS model start during the look-back period prior to the 72-hour precipitation event. The processes for FFRD SST event time windows, start date, and initial conditions will be considered for improvement in future versions of the SOP.

1.8.3.2 WARM-UP TIME PERIOD

The plan file warm-up period settings should be set to balance accuracy and efficiency in terms of cost and computation time in the cloud-compute environment.

The use of a warm-up period allows for the channels in the model to be primed with water before the start of the simulation. Note that in the context of the FFRD HEC-RAS models, excess precipitation grids will not contribute to initial flows in channels, so the flow boundary conditions will be the primary contributor during the warm-up, if used.

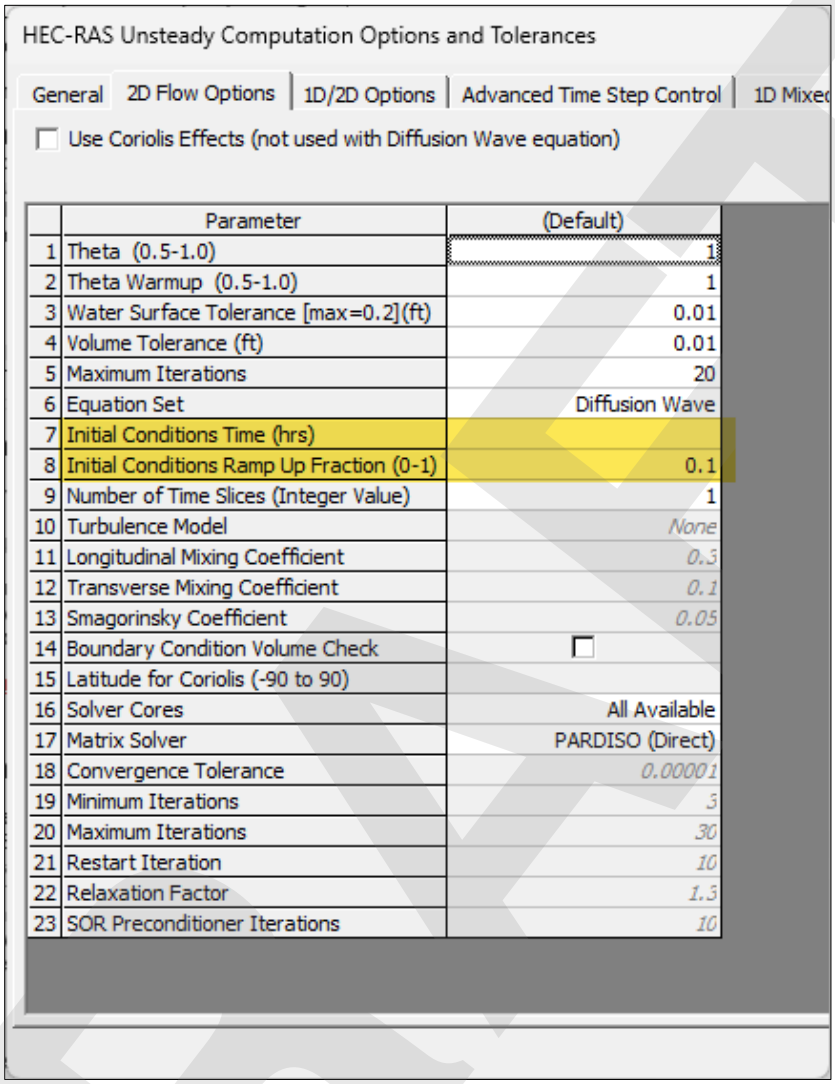
In some cases, the modeler may determine that warm-up time periods do not improve the ability of the HEC-RAS model to predict peak water surface elevations, and it may not be needed.

Note

Previous versions of HEC-RAS contain a bug that prevents the use of the *Fixed During Warmup* feature for Initial Conditions (IC) Points and also the 2D initial conditions time. If both a constant IC point elevation and initial conditions period are desired, consider using the *Number of warm up timesteps* option available under the *General* tab of the *Unsteady Computation Options and Tolerances.* This option can achieve a similar effect as the 2D Options' *Initial Conditions Time*.

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Figure 30. HEC-RAS Unsteady Computation Options and Tolerances Window



	Parameter	(Default)
1	Theta (0.5-1.0)	1
2	Theta Warmup (0.5-1.0)	1
3	Water Surface Tolerance [max=0.2](ft)	0.01
4	Volume Tolerance (ft)	0.01
5	Maximum Iterations	20
6	Equation Set	Diffusion Wave
7	Initial Conditions Time (hrs)	
8	Initial Conditions Ramp Up Fraction (0-1)	0.1
9	Number of Time Slices (Integer Value)	1
10	Turbulence Model	*None*
11	Longitudinal Mixing Coefficient	*0.3*
12	Transverse Mixing Coefficient	*0.1*
13	Smagorinsky Coefficient	*0.05*
14	Boundary Condition Volume Check	☐
15	Latitude for Coriolis (-90 to 90)	
16	Solver Cores	All Available
17	Matrix Solver	PARDISO (Direct)
18	Convergence Tolerance	*0.00001*
19	Minimum Iterations	*3*
20	Maximum Iterations	*30*
21	Restart Iteration	*10*
22	Relaxation Factor	*1.3*
23	SOR Preconditioner Iterations	*10*

1.8.3.3 INITIAL CONDITIONS POINTS

Set the *Keep initial elevations constant during warmup* option to Yes. This setting allows initial conditions (IC) points representing pool elevations that will be varied in SST events to remain constant when a warmup period is also used.

1.8.3.4 RESTART FILES

If restart files were used for the calibration runs, they must be removed for cloud-based SST events.

1.8.3.5 BREACHES

All breach locations with various breach parameters will be included in the final calibration phase model prior to the start of the conformance phase. For the model upload for cloud compute, all breaches should be turned on (*Breach This Structure* checkbox). The trigger breach elevation may be set to the toe of the

structure to be breached, but cloud-compute will handle random sampling of this value during computations.

1.8.3.6 EQUATION SET

The two-dimensional (2D) equation set selected in the plan file should be consistent with the selected equation set used for the calibration of the model. This should be verified prior to model upload for cloud computing.

1.8.3.7 MATRIX SOLVER

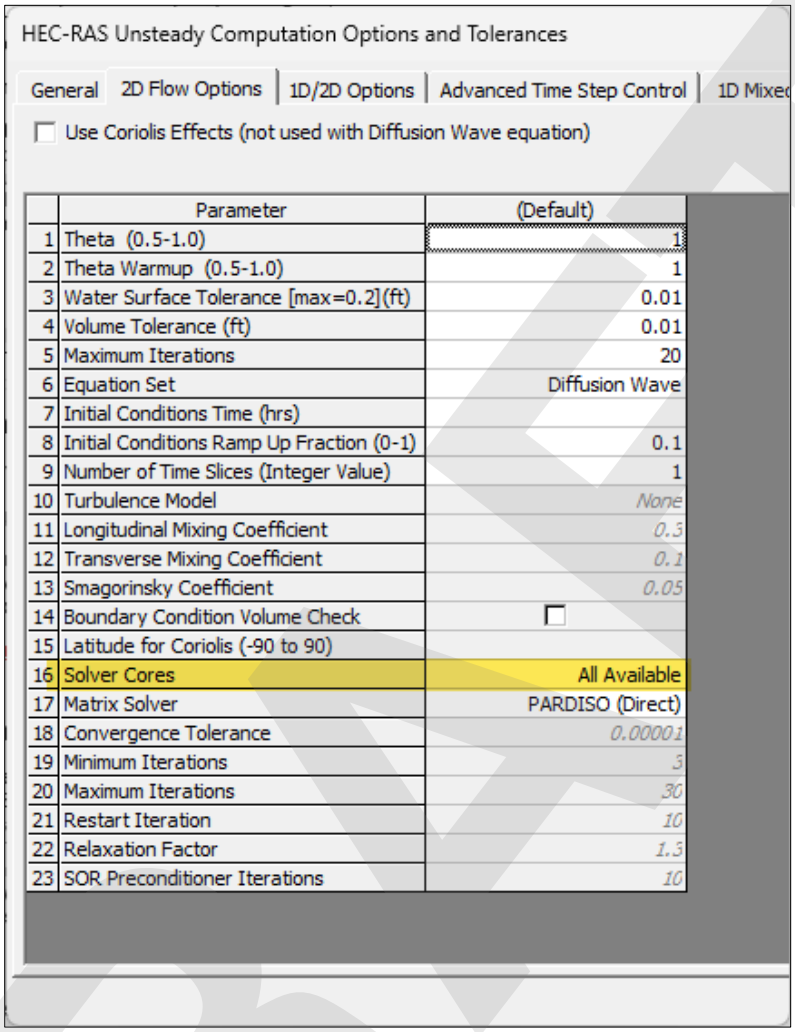
The plan file 2D *matrix solver* selected in HEC-RAS may be evaluated to balance computational cost and numerical stability prior to delivery to cloud compute.

1.8.3.8 SOLVER CORES SETTING

The plan file Solver Cores setting should be set to optimize efficiency in terms of cost and computation time in the cloud-compute environment. Coordination with the cloud environment support team is needed to determine the optimal Solver Core setting, which accounts for the type of compute resource to be used and computation time. Note that the *ras-runner* compute manifest should also call the appropriate compute resources that match this HEC-RAS solver core setting.

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Figure 31. View of Window that Defines Solver Core Settings

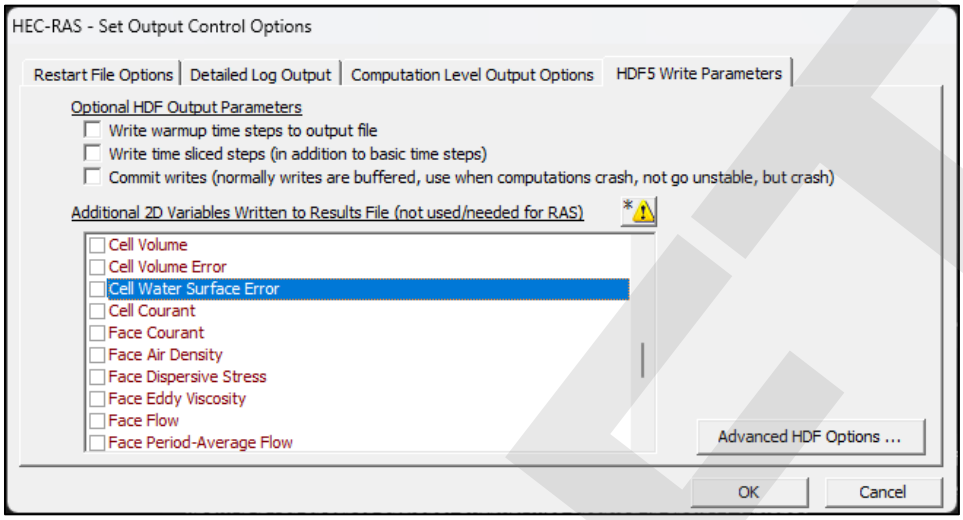


1.8.3.9 OUTPUT CONTROL OPTIONS

Review the Output Control Options and turn off any output that is not needed. It is common to turn on certain variables to be saved to the HDF5 output file during the model development and calibration process, but those should not be carried forward for the cloud-compute runs during the conformance or production phases of the project.

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Figure 32. Example Setting for Output Control Options



1.8.4 MAPPING OUTPUT INTERVAL

The mapping output interval can have a significant impact on HDF output file size. The hydraulic modeler should coordinate with the model integration lead and other team members to investigate whether the critical event-level outputs can be achieved with a longer mapping output interval. Note that the maximum water surface elevation and maximum velocity for each cell is recorded at the computation interval, which lessens the need for a short mapping output interval.

1.8.5 GEOMETRY

Although no special changes to the geometry file will be necessary to deliver the desktop model for cloud compute, it is possible that minor geometry adjustments will be needed in the conformance phase models due to unique issues that were not discovered during the calibration of individual models. Note if any updates to the models that affect linking take place, corresponding updates to the Model Linking Register are needed. The model integration lead will own the Model Linking Register throughout the process.

The geometry file (.g01) should be uploaded with the model to the compute directory. Note that the file is not required for cloud-compute, but it is retained to allow for a desktop modeler to work with the model locally in future steps if needed.

1.8.6 FLOW FILE

Several updates to the flow file are needed prior to upload for cloud compute to fully link to the input data sources for SST events. Relative path name references should be based on the directory structure.

Note

Cloud-compute ingests only the HDF files for most of the key inputs for the computation and does not use the flow file. However, the flow file itself (.f01) should be uploaded with the model to the compute directory to allow for a desktop modeler to work with the model locally in future steps if needed.

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1.8.6.1 OBSERVED DATA

Delete references to observed data hydrographs from the calibration events. Rating Curve tables may be maintained.

1.8.6.2 UPSTREAM INFLOWS

Any upstream inflow hydrographs must be updated to reference appropriate simulated outputs based on the Model Linking Register.

1.8.6.3 METEOROLOGICAL DATA

Excess Precipitation grids must be updated to reference HEC-HMS results from SST events. Reference the directory structure for relative pathnames for reference to the excess precipitation grids.

Filename: exported-precip.dss

1.8.6.4 HEC-HMS BASEFLOWS

Baseflow hydrographs must be updated to reference HEC-HMS results from SST events. Reference the directory structure for relative pathnames for reference to the baseflows.

Filename: SST.dss

1.8.6.5 DOWNSTREAM WATER SURFACE

Appropriate adjustments to the downstream water surface boundary condition must be made. This may be an adjustment from an observed timeseries to a rating curve, or from an observed tidal stage timeseries to a representative fixed water surface during synthetic rainfall events. SOP Volume II provides general guidance on selection.

1.8.6.6 DAM RELEASES FROM HEC-RESSIM OR HEC-HMS

Dam outflow hydrographs must be updated to reference HEC-ResSim or HEC-HMS results from SST events according to the Model Linking Register and Dam Scoping Table.

Reference the directory structure for relative pathnames for reference to the simulated dam outflows.

HEC-ResSim output filename: simulation.dss

HEC-HMS output filename: SST.dss

1.8.7 RAS MAPPER FILE

The RAS Mapper file must associate with the terrain-modeling file according to the directory structure. Reference to the default location for land cover should remain in place to allow modelers to download the model and recreate results. Links to plans and results from calibration efforts or any data other than the single delivered plan must be deleted.

The RAS Mapper file is also used in cloud-compute during depth and velocity grid generation. This process will refer to the separate path to a lower resolution grid. The requirements for this version of the RAS Mapper are under development.

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1.8.8 ADDITIONAL HEC-RAS DELIVERY STEPS

The following sections describe additional actions that are required to prepare the HEC-RAS models for cloud compute.

1.8.8.1 RUN PLAN ON DESKTOP

The HEC-RAS model plan must be executed on the desktop to build the intermediate files that are used directly by cloud compute. After all final changes to the model are made, all out-of-date intermediate files should be deleted, and the model should be run to create a clean and current set of files.

Note that the delivered .p01.HDF file will include results data that will be removed from the file during the cloud-compute.

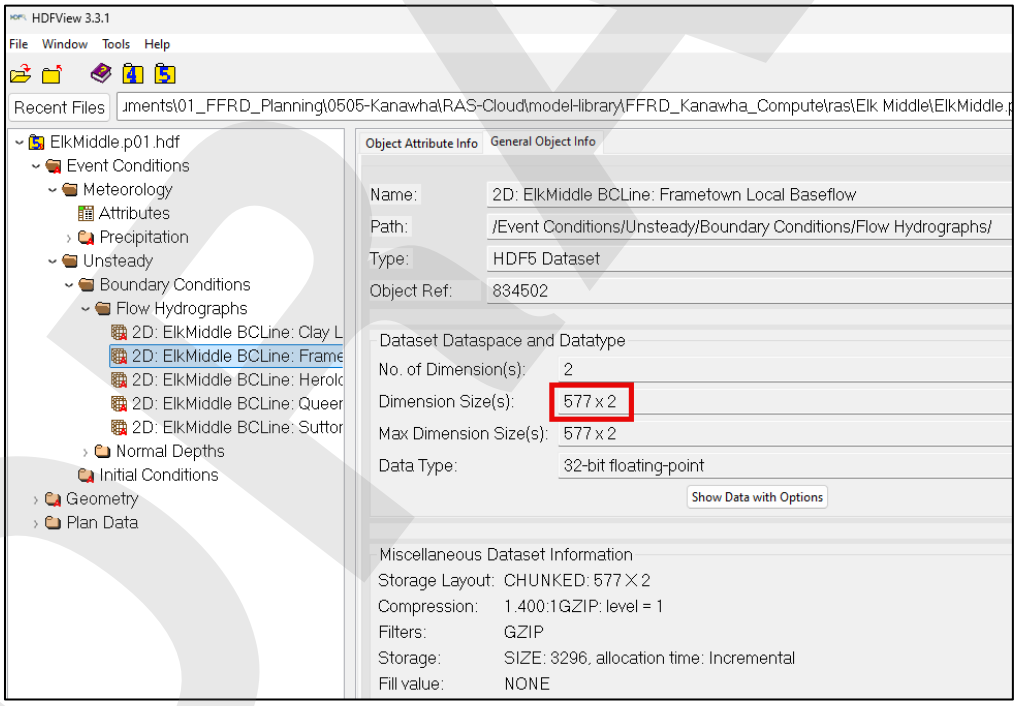
1.8.8.2 VERIFY DIMENSIONS OF INFLOWS IN HDF FILE

The HEC-RAS HDF file must have dimensions that match the number of timesteps for boundary condition inputs. For example, if the duration of the simulation for SST events is 10 days with hourly outputs, the HEC-HMS or HEC-ResSim outputs will have 240 ordinates.

Likewise, the HEC-RAS .p01.HDF file input tables should have 240 ordinates.

Path: Event Conditions>Unsteady>Boundary Conditions>Flow Hydrographs

Figure 33. HDF Input Dimensions



1.8.8.3 VERIFY DIMENSIONS OF EXCESS PRECIPITATION

The dimensions of the excess precipitation inputs must match the values expected by the HEC-RAS HDF file. The SST-generated excess precipitation grid dimensions can be checked via the Grid Info dialog in HEC-DSSVue, as shown in Figure 34. The HEC-RAS HDF file dimensions can be checked via the table size from the example path shown in Figure 35. Note that the HEC-RAS HDF table first dimension

includes a timestep component, while the second dimension value is a total number of values in a single grid (number of cells wide by number of cells tall).

Figure 34. Location to Check Excess Precipitation Dimensions

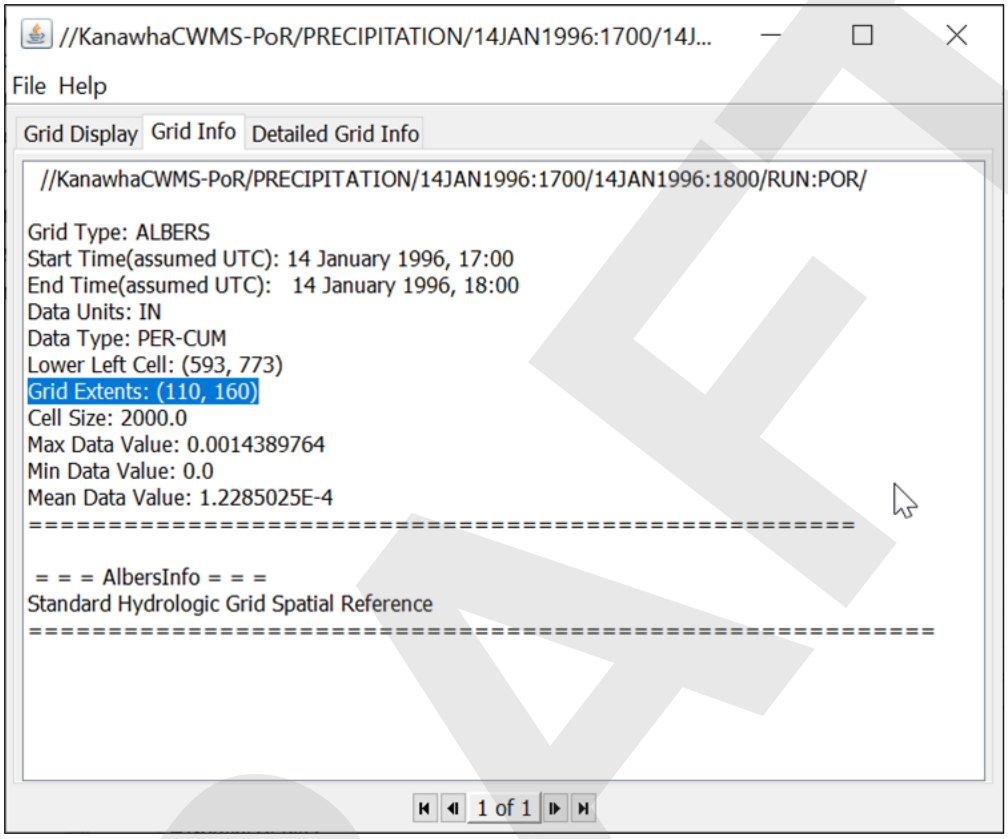
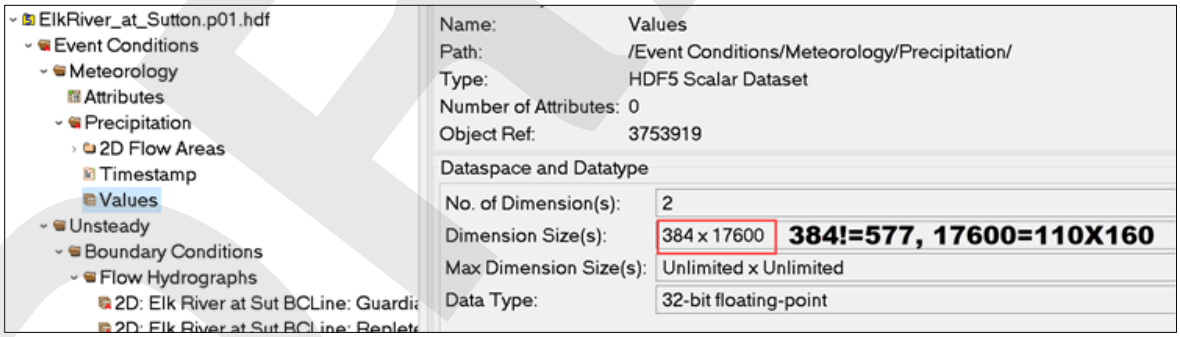


Figure 35. Precipitation Grid Dimensions in HEC-RAS HDF File



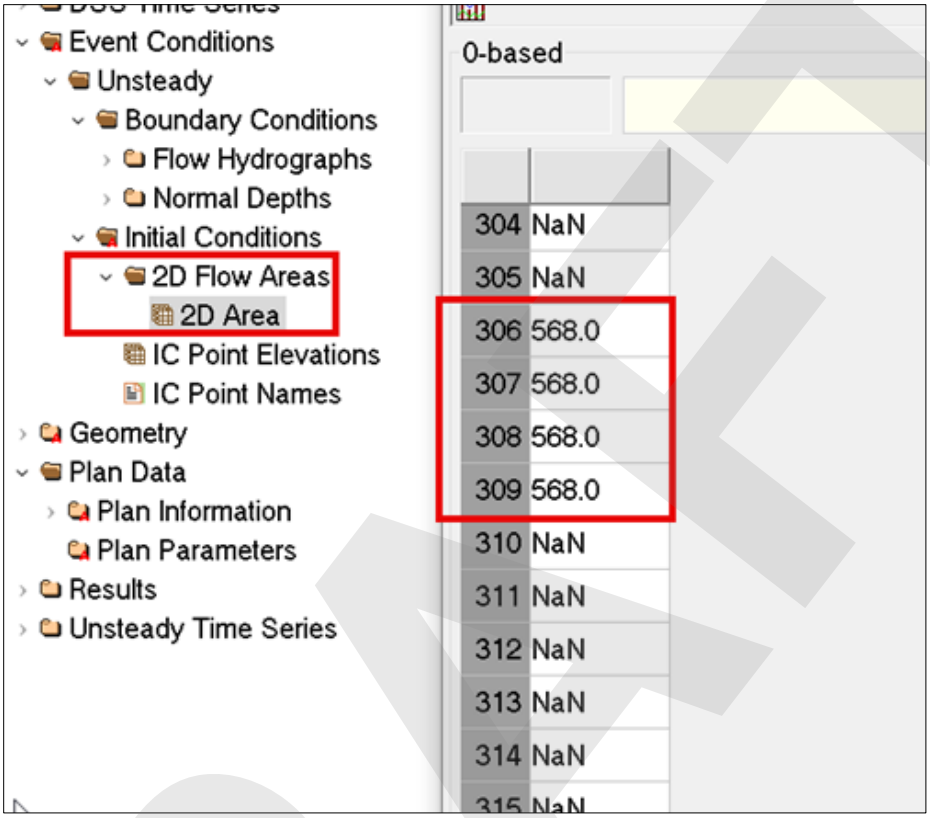
1.8.8.4 LIST CELLS AFFECTED BY INITIAL CONDITIONS POINTS

The cloud compute will vary initial reservoir elevations for each event for all reservoirs modeled as reservoirs in HEC-HMS or included in the HEC-ResSim model, as defined in the Dam Scoping Table. Each of these reservoirs will include an HEC-RAS initial condition point.

The control for the varied IC point elevation is not within the flow file that desktop HEC-RAS users are familiar with. Rather, the reservoir pool starting conditions are fed to the Linux/cloud compute for HEC-

RAS through the .p01.HDF file, from the Event Conditions/Unsteady/Initial Conditions/2D Flow Areas/Perimeter 1 path.

Figure 36. HDF File Path that Controls IC Point Starting Elevations



This path includes an array of IC elevations for each cell in the model, and must be overwritten prior to cloud compute. The HEC-RAS modeler or model integration lead must deliver a list of all cells that could potentially be wet for the highest possible sampled IC to the cloud compute team. To do this, a temporary model run using a very high IC point elevation for the reservoir should be computed. After the computation completes, the resulting HDF file must be opened, the previously mentioned path tabulated and exported to a table using the HDFView software. This list of cells will be used by cloud compute for overwriting pool levels in SST events.

Note

HEC-RAS will ignore an IC in cells with bottom elevations that are higher than the pre-set IC elevation. This is why it is appropriate to set IC elevations for all cells that are possibly wet for IC.

To generate this list of points, the following steps can be followed.

1. Open the .p01.hdf file and navigate to the array mentioned above.
2. Table>Select All>Copy. This data is the initial elevation applied to each cell in order of cell number.
3. To reformat these data into a list of cell numbers, paste these values into Excel. Insert a column to the left named Cell Num, number this column starting at 0.

Figure 37. Table Resulting from Step 3

	A	B
1	Cell Num	Initial Elev
2	0	NaN
3	1	NaN
4	2	NaN
5	3	NaN
6	4	NaN
7	5	NaN
8	6	NaN
9	7	NaN
10	8	NaN
11	9	NaN
12	10	NaN
13	11	NaN
14	12	NaN
15	13	NaN
16	14	NaN

4. Filter these values, remove all NaN rows. Also remove all rows referencing cells whose initial elevations may be specified but will not be varied in cloud compute.

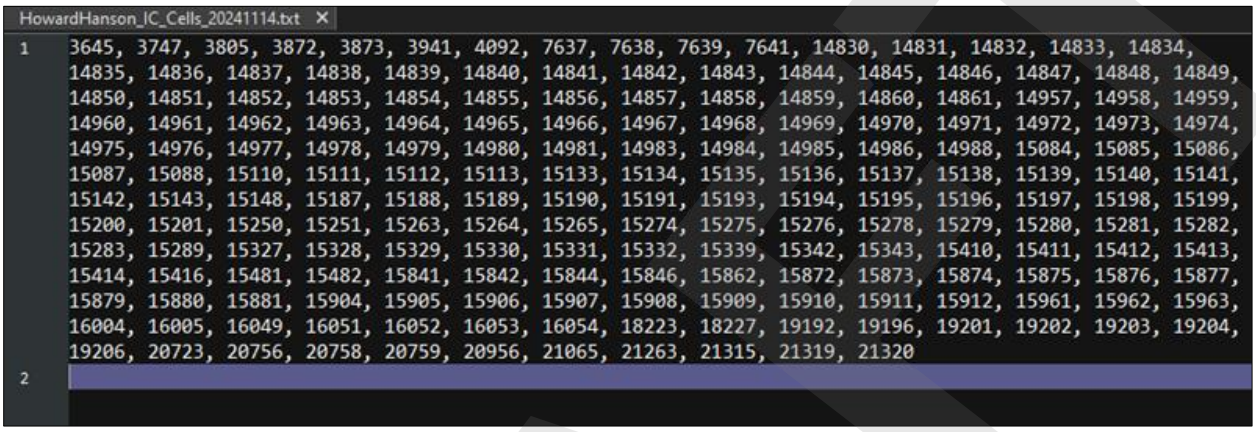
Figure 38. Table Resulting from Step 4

	A	B	C
1	Cell Num	Initial Elev	*Notes*
2	~~0~~	~~NaN~~	*Remove all 'NaN' values*
3	~~1~~	~~NaN~~	*Remove all 'NaN' values*
4	~~2~~	~~NaN~~	*Remove all 'NaN' values*
5	~~3~~	~~NaN~~	*Remove all 'NaN' values*
6	~~19567~~	~~500.79996~~	*This value will not be varied in cloud compute*
7	~~19577~~	~~500.79996~~	*This value will not be varied in cloud compute*
8	~~19574~~	~~500.8~~	*This value will not be varied in cloud compute*
9	~~22288~~	~~500.8~~	*This value will not be varied in cloud compute*
10	~~22289~~	~~500.8~~	*This value will not be varied in cloud compute*
11	3645	1220	*Keep these values…*
12	3747	1220	
13	3805	1220	
14	3872	1220	
15	3873	1220	
16	3941	1220	
17	4092	1220	
18	7637	1220	
19	7638	1220	
20	7639	1220	
21	7641	1220	
22	14831	1220	

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5. Sort remaining cells by smallest to largest cell number.
6. Copy these cell numbers (only the cell numbers) into a new sheet using Paste Special>Transpose. This will convert the column of cell numbers into a single row. Save the sheet as a comma-delimited file (.csv).
7. Cell numbers will be stored in a single row, separated by commas.

Figure 39. Text List Resulting from Step 7



1.8.8.5 DELIVERY OF CELL LIST FOR INITIAL CONDITIONS POINTS

This list of cells will be delivered as an additional field in the Model Linking Register, which will be utilized in the development of the HEC-RAS compute manifests described elsewhere.

When multiple IC points that will be used for reservoir pool initial elevations are present in a model unit, multiple cell lists will be prepared in a similar manner to that described above.

```
Delivery Location: basin-data > project-scoping > model-linking-register
```

1.8.9 HEC-RAS MODEL DELIVERY

Fully run the final model on desktop and perform all QC of results described in other sections.

Remove output files, unneeded subfolders, and backup files from the directory.

Post the full set of remaining files from desktop to the Model Library. Although not all remaining model files are required for cloud-compute, the full directory should be posted to the runnable directory to retain a reference and to allow the model to be downloaded for a desktop replication of a cloud event run.

```
Delivery Location: conformance > hydraulics > [model-unit]
```

1.8.10 LAND COVER

No special adjustments to the land cover layer are needed to deliver the model for cloud computing. The land cover layer should remain in its HEC-RAS default location and should be uploaded with the model to the compute directory. Note that the data is not required for cloud-compute, but it is retained to allow for a desktop modeler to work with the model locally in future steps if needed.

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1.8.11 MAPPING TERRAIN DELIVERY

As described in Section 3.5 of SOP Volume II, multiple versions of the HEC-RAS terrain will be managed by the hydraulic modeler.

The *terrain-modeling version is the full*-resolution terrain used for modeling, and the final version will be maintained in the following path.

```
Delivery Location: terrain-modeling > [model-unit]
```

The *terrain-mapping* version is the lower-resolution exported version of the terrain that will be used for results mapping and processing in cloud-compute. This terrain merges all vector-based terrain modifications and any layers or tiles associated with the terrain used for model development and will be posted to the following path.

```
Delivery Location: terrain-mapping > [model-unit]
```

1.8.11.1 CREATE EXPORTED TERRAIN FOR MAPPING

Merging and resampling of terrain to produce the *terrain-mapping* version is performed within HEC-RAS Mapper. This software provides adequate tools to perform this process seamlessly and minimizes opportunities for user error by using a consistent system. The merged/resampled output DEM should be generated directly from the HEC-RAS terrain that is associated with the HEC-RAS geometry file used for FFRD modeling. This terrain should include any vector terrain modifications used to alter the terrain, such as hydro-connectors, culverts, bathymetry estimates, levees, etc.

To merge and resample a terrain in RAS Mapper, right-click the terrain associated with the FFRD model geometry. Select Export Layer, and Export Raster as shown in Figure 40. The exported raster name should be descriptive and follow naming convention guidance as shown in Job Aid 06, *Standard Naming Conventions*. It should include the exported cell size as specified in Section 4 of SOP Volume II. The checkbox to *Export To Single Raster* should be enabled. This is required if the input HEC-RAS terrain .hdf file references several individual raster DEMs that may be of variable cell resolutions. The output cell size should be defined in feet.

Note that terrain modifications should be incorporated into the exported merged/resampled terrain. By default, if vector terrain modifications are included in the HEC-RAS terrain .hdf file, they will be burned into the exported terrain. If the width of a terrain modification is small (i.e., substantially less than the resolution of the exported raster file), the terrain modifications may be misrepresented in the output terrain surface as smoothing can occur when rasters are resampled. This is generally acceptable for cloud result mapping but is a reason for caution when using resampled terrains for hydraulic modeling.

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Figure 40. HEC-RAS Mapper Export Terrain Dialog

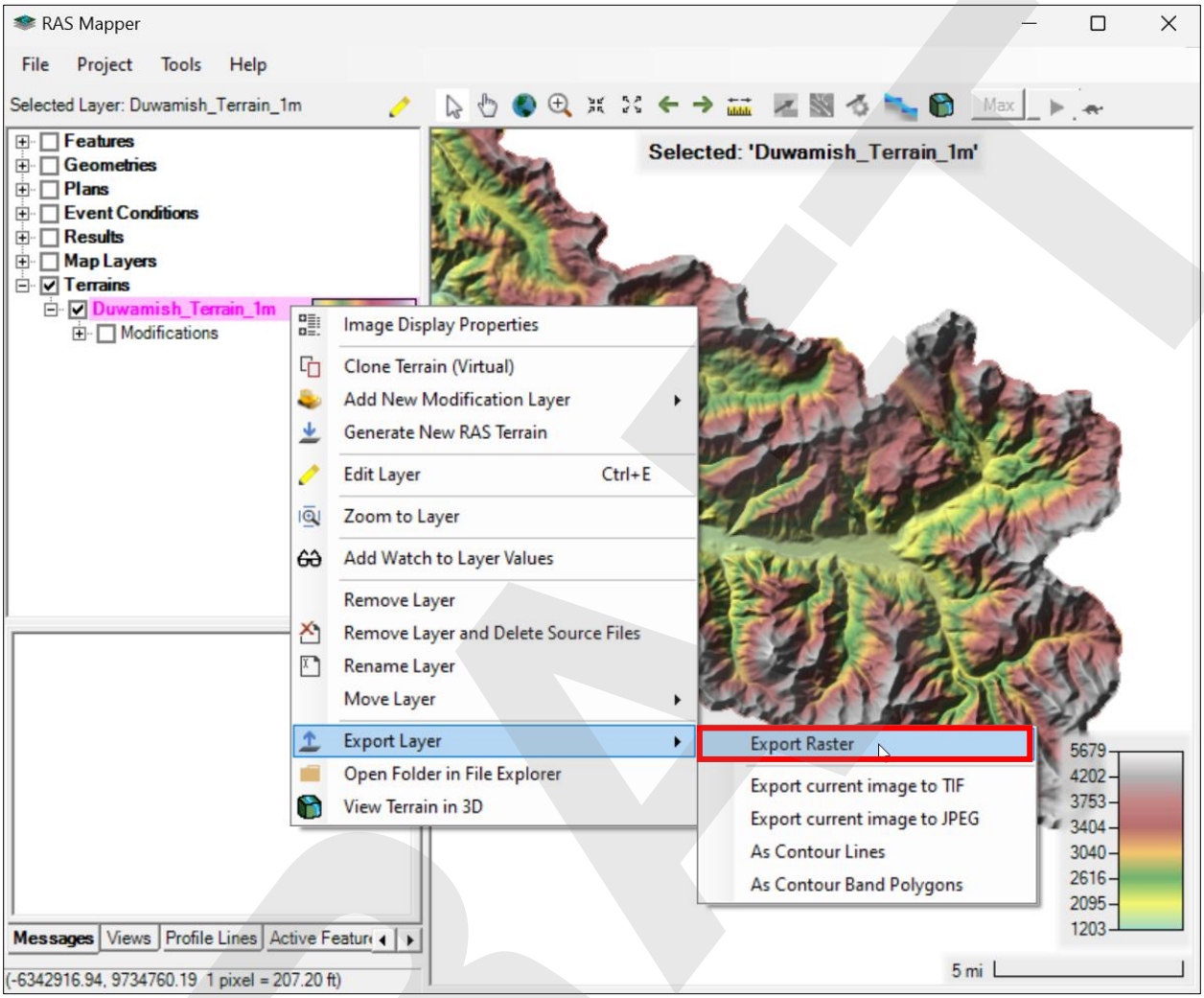
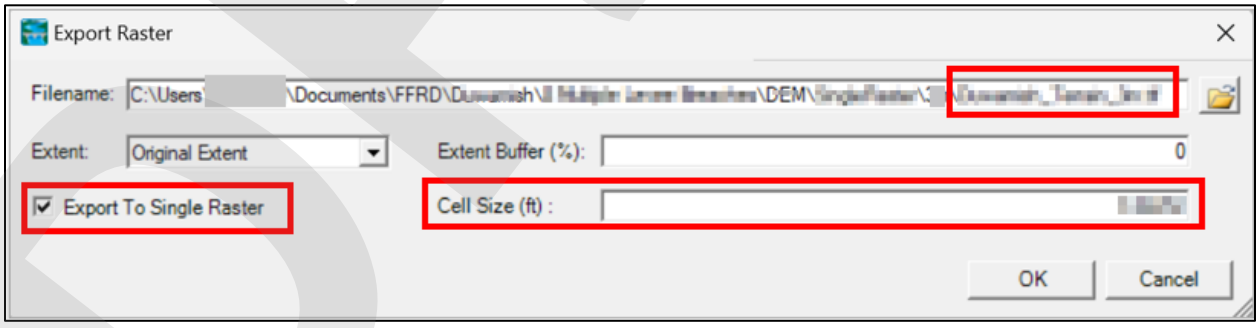


Figure 41. RAS Mapper Export Terrain Secondary Dialog



1.9 BREACH DATA DELIVERY

The system response probability tables for each breach location are produced by the dam and levee data leads as described in Job Aids 10 and 11.

The model integration lead is also responsible for ensuring that the dam and levee breach data is delivered in the appropriate format and directory location for use in the stochastic simulations via cloud computing. The System Response Template is provided on the FFRD Templates and Resources GitHub site. A single file is created for all breaches for the entire project. The template includes instructions for populating the data in an Excel table that includes data validation checks to avoid common errors and exporting to a .csv format for direct use in cloud-compute.

The system response table is delivered to the following location.

```
Delivery Location: conformance > system-response
```

Figure 42. View of System Response Template Format

	A	B	C	D	E
1	hydraulic model unit	peaceful-river	hydraulic model unit	peaceful-river	hydraulic mo
2	2d connection name	nld_1605557018	2d connection name	md_00523_happy-dam	2d connectio
3	Breach FID	1	Breach FID	2	Breach FID
4	NLD System id	1605557018	NLD System id		NLD System i
5	NLD System Name	Running Bear Levee	NLD System Name		NLD System N
6	NLD Segment id	1604557018	NLD Segment id		NLD Segment
7	NIDID		NIDID	NID_00523	NIDID
8	failure mode	LCL	failure mode	LCL	failure mode
9	source	LLID	source	SQRA	source
10	Top Elev	441.523526	Top Elev	570.8	Top Elev
11	Toe Elev	434.32	Toe Elev	553.2	Toe Elev
12	stage	probability	stage	probability	stage
13	434.32	0	553.2	0	
14	435.89	0.00056	556.895	0.0000	45
15	436.51	0.00142	558.44	0.0000	4
16	437.14	0.00316	559.985	0.0001	45
17	437.77	0.00628	561.53	0.0002	4
18	438.39	0.01137	563.075	0.0003	45
19	439.02	0.01907	564.62	0.0005	4
20	439.64	0.02997	566.165	0.0008	45

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JOB AID 13—EVENT SELECTION FOR HYDRAULIC MODELING

The process below aims to develop a list of specific events that may drive annual maximum hazards based on hydrologic, reservoir operations, and system response modeling results. This selected subset of events is then computed through the hydraulic models, which provide the modeled hazard information necessary to produce annual hazard and risk probability outputs.

Summary files that tabulate the maximums of many variables in many locations for each event are generated. Using the event maximum summary data, a list of annual maximum contributing events is generated.

1.1 EVENTS BASED ON HYDROLOGY AND RESERVOIR OPERATIONS MODEL RESULTS

Using summarized hydrology and reservoir operations event maximum outputs, apply the following steps to generate the list of annual maximum contributing events.

1. Read hydrology and reservoir operations summarized output event maximums following guidelines in Table 1.
2. Read the blocks.json file containing the relationship of each event number to a synthetic year.
3. Iterate over each synthetic year, over each parameter and location.

    a. Find the maximum value within the synthetic year.
    b. Store the event number and maximum value with the parameter and location.

4. Iterate over the stored maximums and event numbers for each parameter and location.

    a. Sort each parameter and location based on maximum value.
    b. Select the events within the target recurrence interval range (for FFRD, between 1/10 and 1/2,000 AEP).
    c. Store the event numbers for those selected events.

1.2 EVENTS BASED ON POSSIBLE BREACH EVENTS

The following steps identify events that may result in breaching, which could lead to annual maximum contributing events behind levees or downstream of dams.

1. Develop a rating curve relationship to estimate water surface loading at each specific breach location based on the most appropriate correlated output parameter from either the hydrology model or the reservoir operations model.

    a. Hydraulic model results from events identified in Section 1.1 are needed to support the development of these rating curves during the conformance phase of the project. For the production phase of the project, the previously established rating curves can be used directly.

2. Generate estimated water surface loading at each breach location for each event using the relationship from Step 1 and output summaries from other models.
3. Read the event-level maximum files and the breach trigger elevations, following guidelines in Table 1.

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4. Utilize the relationship developed in Step 1 and compare the breach trigger elevation at each breach location to determine if a breach may occur. If a breach may occur at any location, record the event number.
5. Store all identified event numbers that may include a breach of any structure.

Table 1. Key Parameters and Locations Identifying Annual Maximum Contributing Events

Parameter	Location	Model
Instantaneous Excess Precipitation	All subbasins	Hydrology
Flow (Instantaneous)	Subbasin outlets, reaches, and/or junction elements at primary gages	Hydrology
Flow (Regulated - Instantaneous)	At primary, secondary, and tertiary gages and reservoir outflows	Reservoir operations
Pool Elevation	At all dams with system response curves	Reservoir operations
Breach Trigger Elevations	At all dams and levees with system response curves	System response

1.3 FINAL EVENT LIST

After events are identified for hydrology, reservoir operations, and breaching, combine and deconflict the lists so that each event number is listed only once.

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JOB AID 14—CONSEQUENCE PROCESSES

1.1 BACKGROUND

This document describes the process for structure inventory acquisition, structure inventory adjustments, and consequences computation for a Future of Flood Risk Data (FFRD) project.

1.2 PREPARING STRUCTURE INVENTORIES (DATA PREPARATION PHASE)

The initial structure inventory delivery process for an FFRD project during the data preparation phase is described below.

1.2.1 STRUCTURE INVENTORY SCALES

Structure inventory points are represented by GeoPackage files that follow the U.S. Army Corps of Engineers (USACE) National Structure Inventory (NSI) data schema. There are two scales of structure inventory used in FFRD. The first is the Hydrologic Unit Code (HUC)-4 watershed scale, where all the structures in the basin are in one file. The second scale is at the modeling unit boundaries (the Hydrologic Engineering Center's River Analysis System [HEC-RAS] model boundaries), where the HUC-4 level file is split into one GeoPackage for each modeling unit. These smaller scale files cannot be created until the modeling unit boundaries have been finalized. This job aid describes how to create structure inventories using NSI data as the source.

1.2.2 STRUCTURE INVENTORY FILE TYPE AND NAMING CONVENTIONS

Each structure inventory file is a GeoPackage containing a point data layer named *nsi* as the only table. The name of the GeoPackage itself is defined in Job Aid 06, *Standard Naming Conventions*.

1.2.3 FULL WATERSHED STUDY STRUCTURE INVENTORY

During the scoping and data preparation phase, prior to the creation of the mesh refinement regions, the full study watershed structure inventory is developed. There are two primary methods to download NSI data.

The first is directly from the NSI website. This allows for downloading of NSI GeoPackages at the state level. Clip the state GeoPackage data to the study watershed boundary. If multiple states are required, clip each state file to the watershed. Merge the output files into a single GeoPackage for the entire watershed.

The second method is to use the county-level NSI Application Programming Interface (API) to download county-level data in the JSON format, convert the files to a GeoPackage, clip the data to the watershed boundary, and merge the outputs together to a single output GeoPackage. This can be accomplished using a Python script tool in a geographic information system (GIS) environment.

```
Delivery Location: basin-name > basin-data > structure-inventory
```

1.2.4 MODELING UNIT STRUCTURE INVENTORIES

Once the modeling unit boundaries are finalized, the full watershed structure inventory GeoPackage is split into separate GeoPackages for each hydraulic model unit. In Esri ArcGIS Pro, use the Split tool to create unique GeoPackage files for each modeling unit. This can be accomplished more efficiently

through a script that will perform the split and create the individual GeoPackage outputs with the appropriate naming conventions.

This dataset will remain without further refinements and serve as a reference to the original NSI dataset throughout the project.

```
Delivery Location: basin-name > basin-data > structure-inventory
```

1.3 IMPROVEMENTS TO STRUCTURE INVENTORY (CALIBRATION PHASE)

The purpose of this subsection is to identify potential issues in hydraulic modeling and the placement of the structure inventory points. This subsection applies to the calibration phase and occurs after the structure inventory data is prepared.

1.3.1 IDENTIFYING HYDRAULIC MODELING AND STRUCTURE PLACEMENT ISSUES

Improvements to the structure inventory can include resolving structure placement issues.

1.3.2 WORKFLOW AND TIMELINE

The following ordered steps occur during the calibration phase of the FFRD project.

1. Once the HEC-RAS model units are finalized and the HEC-RAS meshes are created, the consequences team member generates structure inventories for each modeling unit.
2. The hydraulic modeler utilizes the inventory by computing depths for the calibration events at the structures and reviews the depths for outliers.
3. The hydraulic modeler resolves hydraulic modeling issues identified and compiles observed issues with the structure inventory for the consequences team member to review.
4. Once the hydraulic model with computed depths at the structures for each calibration event is submitted for review, the consequences team member downloads the files locally for review.
5. The consequences team member reviews the depth estimates for each event and checks the issues identified by the hydraulic modeler that may be hydraulic in nature, and determines whether structures need to be relocated or removed. The consequences team member provides a list of potential hydraulic modeling issues back to the hydraulic modeler and updates the structure inventory as required to correct structure location issues.
6. The consequences team member delivers the calibrated structure inventory into the project directory.

```
Delivery Location: basin-name > calibration > consequences
```

7. As described in Job Aid 07, *Model Library Use*, the current dataset to be applied for computations must always maintain a consistent pathname and filename. When new iterations of the structure inventory improvements occur, outdated versions need to be archived by moving them to the archive subdirectory with an appended date stamp included in the filename.

1.3.3 IMPROVEMENTS TO HYDRAULIC MODEL OR STRUCTURE PLACEMENTS

Reviewing the hydraulic model outputs including the structure inventory helps focus reviews on high-risk areas that warrant greater scrutiny and provides an additional review perspective beyond gage-based metrics alone. The hydraulic team members and the consequences team members are responsible for completing the activities in this job aid.

The following subsections provide context for common issues in hydraulic modeling and structure inventory placements. The hydraulic modeling team member focus is on identifying hydraulic modeling issues during calibration of the hydraulic models; however, structure placement issues will likely be noticed and documented during the results review. The consequences team member is responsible for structure placement issues; however, it is likely the consequences team member may find hydraulic issues missed by the hydraulic modeler.

1.3.3.1 HYDRAULIC MODEL REFINEMENTS

Hydraulic model refinements are necessary when the hydraulic model results in depths at structures that are unreasonable for the modeled event. This may occur due to the lack of terrain modifications, poorly aligned cells, datum issues with the levee crest profiles, or other geometry or plan settings. Some common issues are illustrated below.

1.3.3.1.1 TERRAIN MODIFICATIONS

While defining river channels in hydraulic models, an island containing infrastructure may be improperly defined in the channel. Simply reviewing hydraulic model data without noticing the structures can yield hydraulic model results that look good but are inconsistent with reality. Figure 1 illustrates an example in which an industrial complex located on an island floods during every calibration event even though the complex is rarely ever flooded in the real world. The model terrain did not reflect the island adequately.

Figure 1. An Incorrectly Flooded Industrial Complex due to Terrain



1.3.3.1.2 STREAM ALIGNMENT ISSUES

If automated tools are used to define stream centerlines, or old orthophotography is utilized to review a model, a river may not align with the actual river geometry. Using the structure inventory can quickly identify these issues.

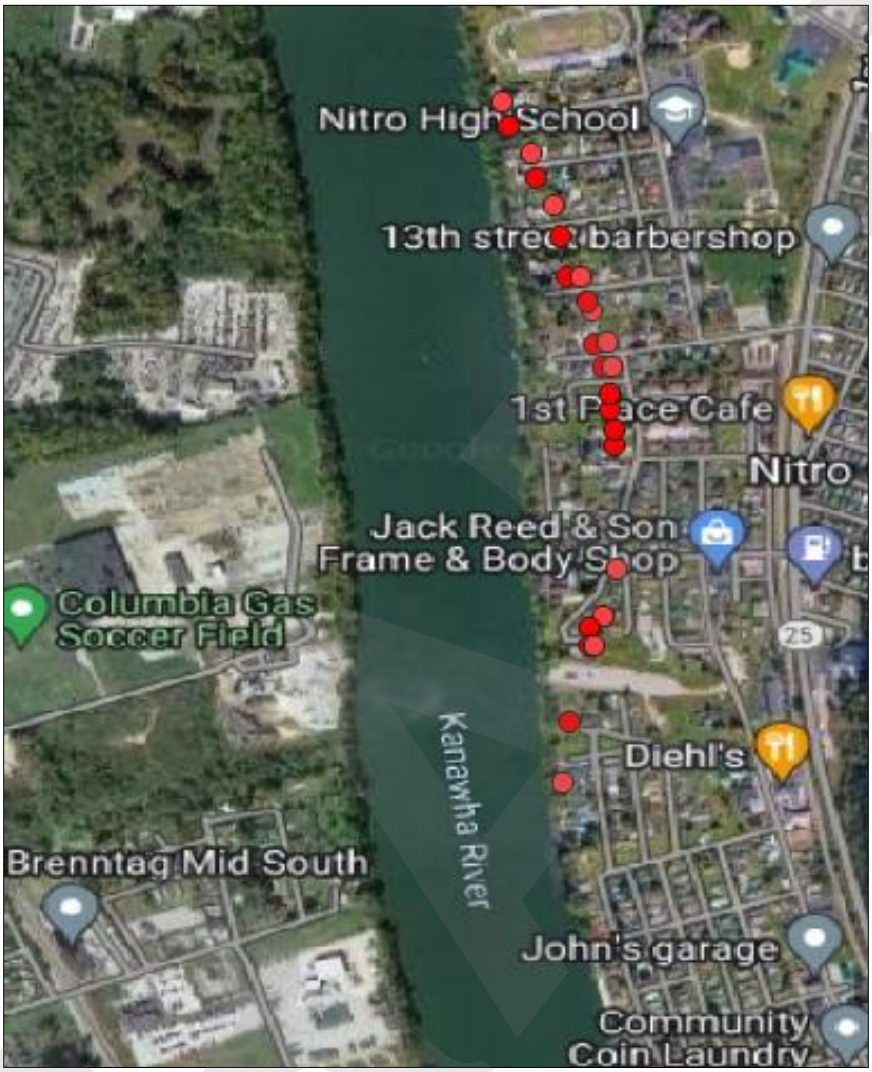
Figures 2 and 3 illustrate examples where simulated results do not align with the river channel and may cause unrealistic structural damage and consequences.

Figure 2. Misaligned Simulated Result Causing Unrealistic Structure Damage



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Figure 3. The Channel from Figure 2 without the Inundation Overlay



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1.3.3.1.3 POORLY ALIGNED CELLS

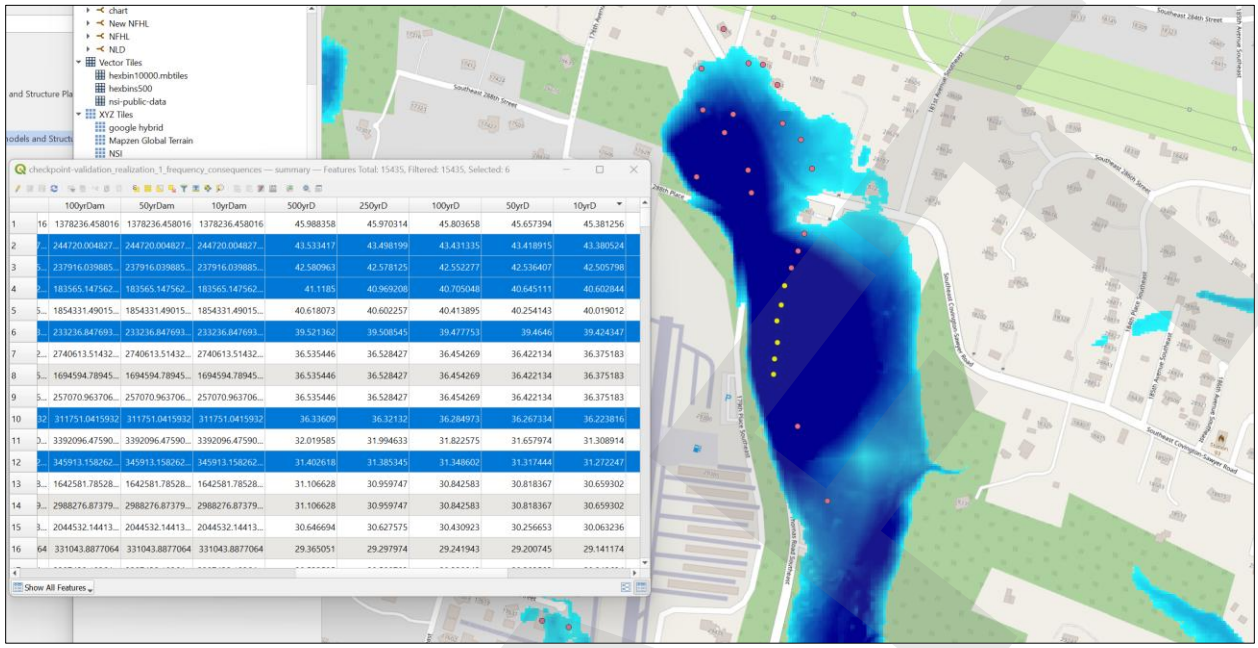
Utilizing large cell sizes and highly detailed terrains can yield results where terrain modifications should be used. Figure 4 illustrates the edge of a cell that coincided with a missing terrain modification resulting in significant pond flooding of structures that should not be flooded.

Figure 4. Incorrect Structure Flooding due to Missing Terrain Modification Coinciding with a Cell Edge



Sorting based on depth can support identification of problem areas in the hydraulic modeling quickly. See Figures 5 and 6 for examples, where 6 of the top 15 deepest depths for an event occurred in one area of the watershed. Figure 5 shows flooding of more than 45 feet at the structures. Figure 6 shows the structures on rooftops in OpenStreetMap. This indicates an issue with the hydraulic model and further investigation may be warranted.

Figure 5. Flooding in Excess of 45 Feet



		100yrDam	50yrDam	10yrDam	500yrD	250yrD	100yrD	50yrD	10yrD
1	16	1378236.458016	1378236.458016	1378236.458016	45.988358	45.970314	45.803658	45.657394	45.381256
2	7...	244720.004827...	244720.004827...	244720.004827...	43.533417	43.498199	43.431335	43.418915	43.380524
3	5...	237916.039885...	237916.039885...	237916.039885...	42.580963	42.578125	42.552277	42.536407	42.505798
4	2...	183565.147562...	183565.147562...	183565.147562...	41.1185	40.969208	40.705048	40.645111	40.602844
5	5...	1854331.49015...	1854331.49015...	1854331.49015...	40.618073	40.602257	40.413895	40.254143	40.019012
6	1...	233236.847693...	233236.847693...	233236.847693...	39.521362	39.508545	39.477753	39.4646	39.424347
7	2...	2740613.51432...	2740613.51432...	2740613.51432...	36.535446	36.528427	36.454269	36.422134	36.375183
8	5...	1694594.78945...	1694594.78945...	1694594.78945...	36.535446	36.528427	36.454269	36.422134	36.375183
9	5...	257070.963706...	257070.963706...	257070.963706...	36.535446	36.528427	36.454269	36.422134	36.375183
10	32	311751.0415932	311751.0415932	311751.0415932	36.33609	36.32132	36.284973	36.267334	36.223816
11	0...	3392096.47590...	3392096.47590...	3392096.47590...	32.019585	31.994633	31.822575	31.657974	31.308914
12	2...	345913.158262...	345913.158262...	345913.158262...	31.402618	31.385345	31.348602	31.317444	31.272247
13	3...	1642581.78528...	1642581.78528...	1642581.78528...	31.106628	30.959747	30.842583	30.818367	30.659302
14	9...	2988276.87379...	2988276.87379...	2988276.87379...	31.106628	30.959747	30.842583	30.818367	30.659302
15	3...	2044532.14413...	2044532.14413...	2044532.14413...	30.646694	30.627575	30.430923	30.256653	30.063236
16	64	331043.8877064	331043.8877064	331043.8877064	29.365051	29.297974	29.241943	29.200745	29.141174

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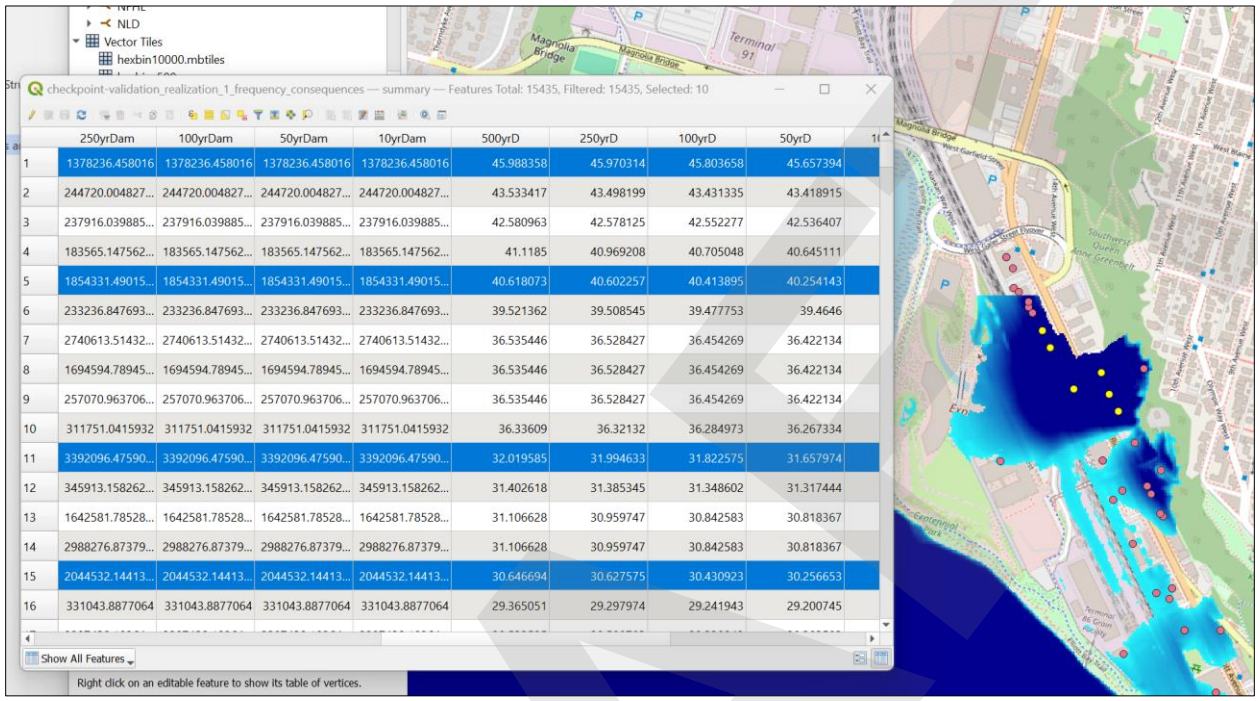
Figure 6. Structures on Rooftops



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In Figure 7, four more of the top 15 structures are flooded to depths over 30 feet in a single problem area.

Figure 77. Structures Flooded to Depths More than 30 Feet



	250yrDam	100yrDam	50yrDam	10yrDam	500yrD	250yrD	100yrD	50yrD
1	1378236.458016	1378236.458016	1378236.458016	1378236.458016	45.988358	45.970314	45.803658	45.657394
2	244720.004827...	244720.004827...	244720.004827...	244720.004827...	43.533417	43.498199	43.431335	43.418915
3	237916.039885...	237916.039885...	237916.039885...	237916.039885...	42.580963	42.578125	42.552277	42.536407
4	183565.147562...	183565.147562...	183565.147562...	183565.147562...	41.1185	40.969208	40.705048	40.645111
5	1854331.49015...	1854331.49015...	1854331.49015...	1854331.49015...	40.618073	40.602257	40.413895	40.254143
6	233236.847693...	233236.847693...	233236.847693...	233236.847693...	39.521362	39.508545	39.477753	39.4646
7	2740613.51432...	2740613.51432...	2740613.51432...	2740613.51432...	36.535446	36.528427	36.454269	36.422134
8	1694594.78945...	1694594.78945...	1694594.78945...	1694594.78945...	36.535446	36.528427	36.454269	36.422134
9	257070.963706...	257070.963706...	257070.963706...	257070.963706...	36.535446	36.528427	36.454269	36.422134
10	311751.0415932	311751.0415932	311751.0415932	311751.0415932	36.33609	36.32132	36.284973	36.267334
11	3392096.47590...	3392096.47590...	3392096.47590...	3392096.47590...	32.019585	31.994633	31.822575	31.657974
12	345913.158262...	345913.158262...	345913.158262...	345913.158262...	31.402618	31.385345	31.348602	31.317444
13	1642581.78528...	1642581.78528...	1642581.78528...	1642581.78528...	31.106628	30.959747	30.842583	30.818367
14	2988276.87379...	2988276.87379...	2988276.87379...	2988276.87379...	31.106628	30.959747	30.842583	30.818367
15	2044532.14413...	2044532.14413...	2044532.14413...	2044532.14413...	30.646694	30.627575	30.430923	30.256653
16	331043.8877064	331043.8877064	331043.8877064	331043.8877064	29.365051	29.297974	29.241943	29.200745

This is a low area in the terrain and a location where the mesh accumulates flooding inappropriately. Fixing the problems in Figure 5 and Figure 7 can likely address the most significant flooding in the watershed very quickly.

1.3.3.2 STRUCTURE INVENTORY REFINEMENTS

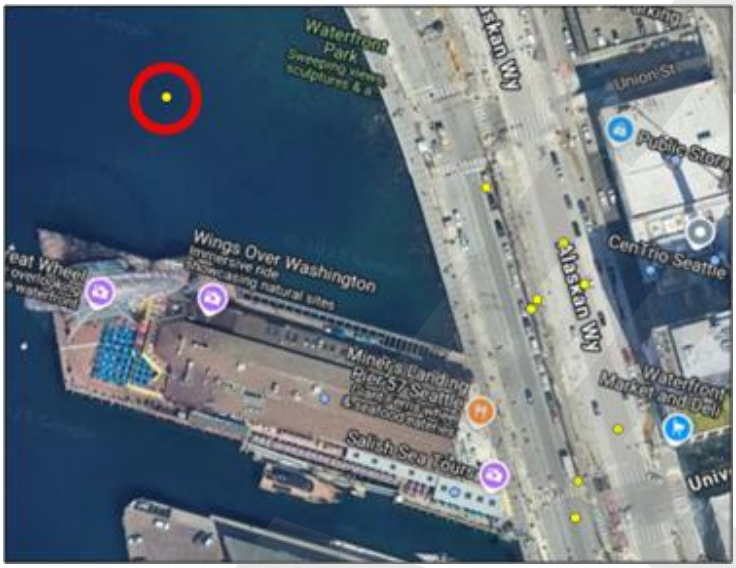
The base-level NSI utilizes parcel data and building footprints to locate structures. No single parcel data source in the United States is fully complete, and no footprint data source has yet achieved a perfect representation of all buildings in the United States. The generation of the NSI leverages other sources of information to fill in data gaps. An example is reliance on the Census data block-level housing units with other Census datasets to generate structures where no parcel or footprint data exists. Reconciliation of these diverse datasets can yield incorrect results occasionally. It is advisable that manual calibration of the inventory is performed to prevent overestimation of average annual damages. The consequence team member is responsible for calibrating the inventory used for production results. The following sections provide some low-effort solutions to improve significant issues quickly. The NSI technical documentation Frequently Asked Questions can provide background on common issues and how to address them. The primary objective of this job aid is structure placement improvements and correcting significant issues that impact consequence estimation.

1.3.3.2.1 INITIAL DEPTH OUTPUTS SORT BY DEPTH

Calibration using the computed results from the hydraulic model can quickly identify structures in areas previously flooded. In some cases, the structures are inappropriately placed and should be relocated. Common issues may occur near rivers where no building footprints exist, but a parcel record does. By default, the NSI drops a point in the center of the parcel. This may be good for a neighborhood, but some states extend parcels to the center of a river, and this can lead to structure centroids placed inside the river boundary. Sorting depth data and cross-checking the deepest depths against structures will identify

the structures inappropriately placed, and those structures can be quickly relocated. Figure 8 illustrates an example of deep depths on a nonexistent structure identified in a parcel for a potential harbor.

Figure 8. Nonexistent Structure in a Harbor



1.3.3.2.2 LOCATION BASED ON IMAGERY

Imagery resources are used in GIS software to adjust and verify structure locations. This process focuses on structures near water sources such as rivers, creeks, streams, and lakes. Existing data (e.g., flood maps and dam and levee breach events) are used to target water features for location calibration. A primary goal for water feature calibration is to ensure structure points are not located in areas subject to continuous inundation. Figure 9 shows a structure in a creek. Figure 10 shows imagery indicating a structure that does not exist in that location; this structure should likely be removed from the NSI.

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Figure 9. Structure in Inundation



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Figure 1010. No Structure Visible



1.4 IMPROVEMENTS TO STRUCTURE INVENTORY (CONFORMANCE PHASE)

Evaluating the quality of the hydraulic model and the placement of the structure inventory continues during the conformance phase of the project. The conformance phase starts with a calibrated structure inventory coming from the calibration phase. Hydraulic results spanning the frequency range from the 10-year to the 2,000-year event become available to aid in the review of the hydraulic model and structure placements.

1.4.1 CONFORMANCE PHASE STRUCTURE INVENTORY DATA MANAGEMENT

At the start of the conformance phase, a copy of the calibration phase structure inventory is promoted to the conformance phase directory location. Any updates to the inventory during this phase will be maintained with the same path and filename, and outdated versions will be archived.

```
Structure Inventory Location for Conformance phase:
basin-name > conformance > consequences
```

1.4.2 EVALUATION OF DAMAGES IN FREQUENT EVENTS

The hydraulic and consequence model results in the conformance phase are obtained as GeoPackages from the Model Library and reviewed to identify potential issues. Any structures flooded during the most frequent event (typically the 2-year event) are checked for accuracy using aerial imagery and other sources as described in Section 1.3. Any significant issues with hydraulic modeling are discussed by the team. Structure point locations are edited to reflect realistic results, as necessary. Examples include moving a point from the center of a structure to an edge to keep it from flooding at the most frequent events.

1.4.3 EVALUATION OF DAMAGES AND FLOOD FREQUENCIES IN LEVEED AREAS

Leveed area results are inspected to determine if flood frequencies and damages on the interior of the levee are higher than expected. Some water will likely pool in a levee interior during frequent events, but there should not be large amounts of structure flooding at frequent events (10-year and more frequent). The Annual Exceedance Probability (AEP) grid can be used in conjunction with the other results files to inspect levee system results for reasonableness.

Any significant issues with interior levee results are documented and reported. The results will be discussed with the team for further investigation. Any issues not addressed will be documented in the report, including any identified causes and potential solutions. An example of remaining issues could include levees that do not tie into high ground or lack of pumping and interior drainage systems in the hydraulic model.

1.5 COMPUTATION OF CONSEQUENCES

Technical procedures for computation of consequences based on hydraulic modeling results are included in cloud-compute plugin documentation. The inputs to consequence computation are a set of hydraulic model results describing the hazard, a structure inventory describing the exposure, and a set of damage functions describing the vulnerability. The FFRD directory structure described throughout this SOP supports the standardized delivery of input data for consequence compute.

1.6 PRODUCTION PHASE STRUCTURE INVENTORY

At the start of the production phase, a copy of the conformance phase structure inventory is promoted to the production phase directory location.

```
Structure Inventory Location for Production phase:
basin-name > production > consequences
```

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